

The Transportation/ Land Use Connection



Terry Moore and Paul Thorsnes, with Bruce Appleyard

New Edition



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Terry Moore, faicp, has been a vice president and planner at ECONorthwest since 1979. He has managed more than 500 projects in land-use and transportation planning, policy analysis, and market analysis for private and public clients. His current interests are regional planning for land use and transportation, transportation project evaluation and selection, congestion pricing, and sustainable urban development.



Paul Thorsnes is a senior lecturer in the Department of Economics at the University of Otago in Dunedin, New Zealand. Paul holds a Ph.D. in economics from the University of Oregon and specializes in urban economics. Paul worked with Terry Moore at ECONorthwest from 1989 to 1995. His recent research has focused on the economics of local public goods and amenities in housing markets.



Bruce Appleyard, aicp, has more than 15 years experience working with communities across the U.S. as regarding the transportation/land use connection. Bruce has been a project manager for work involving urban design options and consensus building. For his contributions, he was recently named one of the first 10 "Active Living Heroes" by the Robert Wood Johnson Foundation's Active Living Network. Currently, Bruce is working as a consultant as he pursues his Ph.D. in City and Regional Planning at the University of California, Berkeley.

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E-mail: pasreports@planning.org

APA headquarters office is at 1776 Massachusetts Ave., N.W., Washington, DC 20036.

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2007 Edition

TERRY MOORE, FAICP, AND PAUL THORSNES, WITH BRUCE APPELYARD, AICP

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Preface

We wrote our original PAS Report on the transportation/land use connection more than 10 years ago. Here's what we said in 1994:

Many planners view the coordinated implementation of such [land-use and transportation] policies as the key to solving a variety of urban problems: congestion, air pollution, the decay of central areas. As the concerns of residents of growing urban areas about this problem increase, so does their willingness to support public action to ameliorate them. They hope to preserve the benefits of growth and reduce its costs.

In this report, we synthesize the results of recent theoretical and empirical research to see whether that hope is realistic. We hope to provide a rigorous yet readable framework for thinking about land-use/transportation policy. We aim at an audience of practicing planners and public decision makers in state, regional, and local governments.

We decided to focus on regional policies despite the obvious importance of federal policies. The federal government strongly influences local and regional policy through its criteria for funding improvements in transportation facilities. Historically, the bulk of federal money has gone to highways. The Intermodal Surface Transportation Efficiency Act of 1991 (ISTEA), however, allows state and local governments considerable flexibility to build facilities that serve modes other than the auto. This flexibility puts greater responsibility for decision making on state and local governments, and opens many new opportunities for changes in regional and local policy.

We started this report three years ago for a practical reason. We were working on several projects for public agencies relating to transportation policies. In our review of the literature, we could find no single report aimed primarily at answering the questions of interest to us: What are the key interactions between land-use and transportation activities in theory? and, What measurements of those relationships exist that are relevant to the development of integrated policy for land use and transportation? We found the best answers to those questions in the writings of urban economists. Those writings bracketed what we were looking for: their textbooks provided the most compelling explanations of the key relationships; their professional literature added details for those willing to tackle the econometrics. But we did not find the piece in-between that would provide for professionals and policy makers already conversant in land-use and transportation policy a framework for evaluating the integrated land-use/transportation policies they were all talking about.

We were both reassured and disappointed to find that others agreed with our assessment. In the two years that we fussed with our manuscript, several articles with a focus on integrated land-use and transportation policies were published, several in the *Journal of the American Planning Association*. The articles rolled in faster than we could read them: we got stuck in the traffic. We might have abandoned the report except that (1) one of us remembered the comment of a former professor, a prolific publisher, who, when confronted by the fact that the article he had just published contained arguments identical to another that he had not read, said “Good ideas have to be repeated in about 10 good articles before they become part of the received wisdom,” and (2) the other one followed a spouse on a six-month sabbatical to New Zealand and had a lot of time (or thought he would) to rewrite the manuscript.

In any case, it is our judgment, apparently shared by the editor of the PAS Reports, that this report helps fill a gap that still exists, though others have made it smaller since we first noticed it. Our conclusions are similar to those of many other researchers who have viewed transportation from an economic perspective, a happy coincidence that similar methods and data do not inevitably produce. We are only slightly less pessimistic about the impacts that public policy can have, and the efficiency of planning, public involvement, and visioning (which, though soft in comparison to hard-headed economic prescriptions, may work ultimately by changing individual preferences and the rules for economic activity).

In short, many planners at all levels of government understood, conceptually and intuitively, that a transportation/land use connection existed and should be addressed at all levels of transportation and land-use planning. They lacked information, however, on exactly how that connection worked (directions and magnitudes of causes and effects), how it could be measured, and what the likely impacts would be of policies aimed at influencing it. Evidence about possible and partial answers was scattered throughout the professional literature of several fields. This area of research looked ready to grow. That state of affairs encouraged us to write that PAS Report. We targeted an audience of practicing planners working at a regional level on public-sector planning for the development of land uses and supporting transportation systems.

A lot has happened since then. The blizzard of white papers we soldiered through built to an avalanche that carries along all researchers and practitioners, only a few of whom appear to be riding on top. We all know a lot more, but it’s mixed in with a lot more debris.

The precipitating event for our current effort to sort through the mess is neither academic nor pecuniary. APA sold the last copy of our PAS Report in 2005 and was faced with the choice of either reprinting the old one or convincing us to create a new one.¹ Paul Thorsnes had left the country; Terry Moore left his senses; a deal was struck (“You write it and give it to us; we’ll print it and sell it.”)

We did have the sense, however, to realize that while seniority heads in the direction of senescence, if packaged properly, it can attract real talent. Bruce Appleyard found us, and had the blend of knowledge, curiosity, motivation, and naiveté that allowed him to be seduced into the deal.

Collectively, we assessed what we had and what it could become. We learned that the book had been used in planning schools for teaching.² We considered the book’s current organization and alternatives. We evaluated staying with a PAS Report or shifting to a real book. We really wanted the formatting changed (that was one dense layout). Here’s what we decided:

- Keep it as a PAS Report. That’s how it’s known and how it can get to a broader audience easier and cheaper.

- Keep the underlying theme of economics; individual and institutional decision making; and benefits, costs, and trade-offs.
- Keep the same general organization. This decision was the hardest one. We considered a lot of variations; they were clearly different, but we could never see that they were unambiguously better.
- Aim at a dual audience: planners who are (practitioners) and planners to be (graduate students).
- Make this report the one that planners can go to for an overview of most of what they need to know and for access to more than they are ever likely to care about. See the next point.
- Make the report more readable. At the most basic level, that meant reducing the text density and adding useful graphics.

We recognized the objectives as overly ambitious when we set them; the extent of the hubris is painfully clear now that we have a product that we call final.

PREFACE NOTES

1. That statement implies that the book was a best-seller. It is a comment on the public interest in the profession of planning that, by APA standards, it was: it sold about 5,000 copies in 10 years. We have high hopes, however, for Donald Shoup's new book, *The High Cost of Free Parking*, which, at 700 pages, got reviewed by a magazine in New York City as "beach reading." And it meant it, commenting on the wit, great writing, and thoughtfulness of the book.
2. Krizek and Levinson (2005).

Acknowledgments

There are always a lot of people to thank. We are not naming the family members and former friends that put up with our long absences and distracted presence. We are limiting ourselves to naming those people who contributed directly to the text of this book.

First, just to set the record straight, we did most of it. A lot of the text is from the first edition, which Terry Moore and Paul Thorsnes wrote. For the second edition, the primary authors/rewriters were:

- Front material; Chapters 1, 4, 5, 6, 7, 8; Appendices A, C, D, E, F, G; and Section Summaries (Terry Moore)
- Chapters 2, 3, 4, 5, 6; Appendices B, C (Paul Thorsnes)
- Chapters 5, 6; Appendix G (Bruce Appleyard)

Three people contributed substantially to the appendices. Professor Art O'Sullivan (Lewis and Clark College, Portland) wrote large parts of Appendix B; Joel Freedman (Parsons Brinkerhoff, PBWorld, Portland) wrote Appendix E; Brian McGuire (ECONorthwest intern) wrote Appendix F.

Several other staff at ECONorthwest did small or large amounts of research that went into the chapters and the appendices: Kate Coddington, Anne Fifield, Beth Goodman, Jake Holcombe, Jim Ebenhoh, and Becky Steckler. Randy Pozdena has been a source of inspiration and insight on transportation economics since before we published the first edition.

Many people contributed sidebars: Dan Burden, Robert Cervero, Reid Ewing, Larry Frank, Matthew Kitchen, Troy Russ, and Paul Zykofsky. The sidebars are a big improvement to this edition: they broaden the scope; add other perspectives; summarize expert opinion; and provide some visual interest absent in the first. In addition, Professor Susan Handy gave valuable advice on a draft of Chapter 5.

We also thank our respective firms and institutions for allowing us the time to work on the report. The views expressed in this report, however, are ours, and may not always be theirs.

We could not have started without the indirect financial and staff support of several public agencies that have hired us as consultants to develop some of the ideas contained in this book, especially the Oregon Department of Transportation, the Oregon Transportation and Growth

Management Program, Metro (Portland, Oregon), and the National Academy of Sciences (TRB: TCRP and NCHRP).

We could not have finished without the help of Jim Hecimovich, our editor, who added encouragement and discipline to editing and management. Jim says he believes in us, the urban economists, as the “compassionate capitalists” that planners need to become to provide market sense to their ideas. The success of the first edition and Don Shoup’s *The High Cost of Free Parking* indicate that, in fact, we and APA may be on the right path.

Readers Guide

Everybody should read everything in this PAS Report.

Somebody might. But most of you will skim and skip. That makes sense. Some readers with a lot of experience already know a lot of the material. Some students—the risk takers with more on their minds than studenting—will scan and highlight. Some planners may just use the book as an occasional reference. Different needs, different reads.

Rather than fighting with the skimmers and skimpers, who may be the biggest group of readers, we decided to help them, at least a little.

Every chapter starts with a very short summary of what's in the chapter. If you think you know that material, skip the chapter. The summary tells you what each section of the chapter covers: pick the ones you want. With the summary are two additional sets of materials:

- **The Basics.** This section is our attempt at the student's answer to the professor's question, Would somebody please summarize in two minutes the main themes and key ideas of this chapter?
- **The Shortcuts.** This section contains some ideas on what you might be able to skim or skip.

We have also done some things to try to encourage you to keep reading. Compared to the first edition, the paragraphs are shorter, the font is bigger, the examples are more colorful, and the colors are no longer limited to black and white. We dropped some of the heavier stuff into appendices, added graphics, and put sidelights into sidebars.

But our main contribution to readability is good logic and clear prose. We try to build logically from some basic ideas to more complicated ones; from theory to practice; and from practice to policy. For an overview of that logic, go to the last section of Chapter 1, "Different Needs, Different Reads," which provides additional suggestions to readers with different backgrounds about what to read.

CHAPTER 1

Orientation to This PAS Report

Organization: This chapter is orientation. It describes the purposes and organization of this PAS Report. It has four sections:

1. *Overview of the report.* Topic, purpose, audience, perspective.
2. *Premises before promises.* Key assumptions—ideological and testable—about planning, markets, land use, transportation, and the connection among them.
3. *A case for caring.* Why the transportation/land use connection matters.
4. *Orientation to the report.* How it is organized and shortcuts for different categories of readers.

Basics: The transportation/land use connection matters because highways shape the way America has grown and will continue to grow, but highways have problems with congestion. Building more roads will not solve these problems, but changes in the way we approach transportation and land use planning may.

- Public sector investment in land-use and transportation development is necessary.
- Highway-related transportation in urban, metropolitan areas is the focus of this report. The biggest problem with this type of transportation is peak-period congestion.
- Highway development has changed the way we live in America, but increasing investments in roadway capacity haven't slowed congestion.
- Because building more roads does not seem to solve congestion problems, this report investigates various mechanisms for using existing transportation facilities more efficiently, some of which may have the effect of limiting the number of cars on the road.
- Controlling congestion requires changing transportation policies and introducing mechanisms like congestion pricing, but it also requires complementary changes in land-use policies.

Shortcuts: Readers with broad experience in transportation and land-use planning can skim the rest of this chapter. Readers who are new to the field will find this chapter an easy read that will help them get oriented to the topic.

OVERVIEW OF THE REPORT

The Topic

Transportation planners now deal with policies that go well beyond more and better highways, the traditional engineering solutions to traffic congestion. They look at complementary changes in land-use and transportation policies coordinated at the regional level. Such policies include the following:

- Properly pricing auto travel
- Improving the quantity and quality of infrastructure that serves pedestrians, bicyclists, and high-occupancy vehicles
- Regulating the location and design of new development; for example:
 - Restricting urban expansion and encouraging or requiring suburban development at higher densities
 - Creating centers of new high-intensity development

Many planners view the coordinated implementation of such policies as critical to solving urban problems like congestion, air pollution, the decay of central areas, and a multitude of other ills they believe result from the pattern of development broadly defined as “sprawl.” As the concerns of residents of growing urban areas about these problems increase, so does their willingness to support public action to ameliorate them. They hope to preserve the benefits of growth and reduce its costs.

In this report we synthesize the results of recent theoretical and empirical research to give planners a better sense of the relationships between transportation and land use. We hope to provide a rigorous yet readable *framework* for thinking about transportation policy in the context of land use and development.

The Audience

The audience at the center of our target comprises technical people (planners, policy analysts, engineers, designers) in metropolitan areas who are already working on the integration of land-use and transportation policy. They will mainly be in the public sector, consulting firms, or advocacy groups. The majority will have a technical background and be familiar with most of the policies discussed in this report.

Not far from the bull’s eye are graduate students in land use, transportation, and public policy analysis, and the professors charged with their education. If the report works for practitioners, it should work for *future* practitioners as well.

A little farther from the center are policy makers and the citizens they serve. They would have to be dedicated and technically oriented to slug through this report. Our opinion is that policy in the area of land use and transportation would be improved if policy makers and citizens understood the points we make in this report, but we realize that (1) it’s not for everybody, and (2) skills other than technical ones are needed to get to agreement on good policy.

In some cases, we expand the idea of a region beyond the metropolitan area to discuss statewide policies. We do not, however, spend much time on federal policy, despite the fact that federal changes in spending, funding requirements for states and metropolitan planning organizations, tax policies, and regulations (e.g., on emissions and fuel efficiency) could have very significant effects on transportation problems.

We provide suggestions to all these readers about how to read this report in the Readers Guide at the beginning of the report and in the last section of this chapter.

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The Perspective

Given our intended audience, we had to make some decisions about what they wanted and what we thought they should have. We opted to focus on three issues:

1. Public sector transportation and land-use policies and practices in urban areas at the metropolitan level
2. Surface transportation policies in the context of land use more than the other way around
3. Economic explanations for travel behavior and how public policy can influence the economic incentives that influence that behavior.

The last point will be the hardest to swallow for some readers. Knowing that, we have tried to make the economics more digestible by cutting it into smaller pieces. Still, our evaluation framework is unabashedly economic. Our perspective has not changed since the first edition: underlying economic concepts—about benefits and costs, incentives, demand and supply, trade-offs and choices, incremental change—are essential to understanding why people make their travel and location decisions.

This perspective may be arbitrary, but it is not thoughtless, and it is ours. Our reasoning will become clearer in subsequent chapters, but we provide a summary in the next section.

PREMISES BEFORE PROMISES

Broad (Ideological) Premises

The future of urban land-use and transportation development in a region is uncertain, as are the contributions and likely future states of the factors determining that future. Thus, planners should be clear about the assumptions they are making (the premises) before getting into descriptions of end states and the policy prescriptions for achieving them (the promises).

Here, we give readers a necessarily brief explanation of the reasons for our perspective:

- Every country has a mixed economy made up of a private sector (business) and a public sector (government and its agencies). The U.S., as the reputed leader in free enterprise, has a huge government sector. Capitalism as we know it could not function without it. The centrally planned countries that remain have at least underground private sectors and usually much, much more. These assumptions might have been debatable 25 years ago, but not now (the Soviet Union has decentralized; the developing economies of Asia are considered market miracles).
- In a mixed economy, the private sector develops in the context of public policy choices (investments, rules, and regulatory constraints) the public sector makes on behalf of the larger society.
- The ideal mix of private/public is unknowable. Many combinations can work. Much of the debate about the proper mix (which is a continuing debate) is usually more ideological than technical.
- Currently waxing is the private side: capitalism.
- But that trajectory, influenced by so many forces, can change. Markets have tremendous advantages, but evidence—both theoretical and empirical—is clear that the overall welfare of a society depends on the activities of government, its investments in infrastructure, the rules private concerns must play by, and the regulatory constraints imposed. In other words, government policy. Of particular concern in the context of land-use and transportation policy are *externalities* and *public goods*.

Our perspective has not changed since the first edition: underlying economic concepts—about benefits and costs, incentives, demand and supply, trade-offs and choices, incremental change—are essential to understanding why people make their travel and location decisions.

In the context of this report, a key premise is: *given current realities, public-sector involvement in land-use and transportation development is clearly necessary in all countries. Exactly what that involvement should be is a question this PAS Report addresses.*

Another reason for our focus on the public sector: that's where the action is for regional surface transportation in urban areas. Even if the private sector *could* build, own, and manage our urban highway systems, it doesn't and isn't likely to soon in any significant way. Moreover, for reasons discussed in later chapters, some fundamental economic reasons exist for the public provision of inter- and intra-urban highway systems.

This set of premises for evaluating land-use and transportation development applies in most countries. But, as described above, *national* policy, despite its obvious influence, is not our focus; rather, it is *regional, metropolitan* policy. That decision derives in part from another: to focus on transportation issues *within urban areas*, which are, with a few exceptions,¹ issues about *surface (land) transportation*, primarily *highway* transportation.

The urban transportation problem is a classic example of what economists refer to as a problem of public goods and externalities. It requires a metropolitan view to create an efficient system. Isolated investment decisions by individual cities in the region are likely to lead to a suboptimal system, which is why, even with the flexibility in local transportation infrastructure decision making promised by the Intermodal Surface Transportation Efficiency Act of 1991, we opt for a metropolitan and not a local focus. At the other end of the continuum, the federal and state role in metropolitan transportation is much more one of funding than of system and policy design. Metropolitan and regional is our focus.

We focus on decisions and actions: we believe that good thinking and logical, inclusive discussion can move decisions and actions in the direction of doing better things and doing those things better. Thus, we focus on the practice of public sector planning in transportation, and on the policies (including regulations, programs, and investments) that give us the transportation system we have.

Our focus on transportation first and land use second needs explanation. It is not based on relative importance: we would argue that land-use planning is more important. It better captures a multidimensional vision of what a community wants to be; transportation planning should occur in the service of that vision. It is not based on causality: the debate about which causes what is usually resolved by saying it's simultaneous. Instead, our focus is based on (1) our observation that plenty of books on how to do land-use planning are available,² and (2) our assumption that what we can add is more information about how to incorporate transportation into that planning.

Thus, we view this report as a supplement to, not a substitute for, texts on land-use planning, growth management, and urban design. Similarly, it does not substitute for technical books and articles on all aspects of transportation planning. It fits a niche at the intersection of transportation and land use, with an emphasis on transportation.

Finally, we address all the issues about land use and transportation from an economic perspective. To track the theme of the report, we try to stay in the middle of the road. In the left lane are more radical critiques of the failure of the transportation market and the wishful paeans about the desirability of a carless, careless future. In the right lane are mainly free-market economists whose analysis starts from the assumption that public policy is often an ill-informed, inefficient, and undesirable intervention in a world where market forces yield superior results. The theme in our middle lane is: public policies uninformed by a knowledge of how markets work risk

In the context of this report, a key premise is: given current realities, public-sector involvement in land-use and transportation development is clearly necessary in all countries. Exactly what that involvement should be is a question this PAS Report addresses.

being inefficient and unfair (even when fairness is an explicit objective); and market decisions unconstrained by public policies that deal with externalities and public goods are likely to be inefficient and unfair.

Specific (Testable) Premises

To summarize: we have reasons good enough for us to think that a report about the connection between land use and transportation, written primarily for a U.S. audience of planners (practicing and preparing), should focus on highway-related transportation in urban, metropolitan areas. Other broad premises are possible, but this report does nothing more to describe them, much less justify them. The interested student is encouraged to write a different PAS Report.

Within that context, there are yet more premises that we start with. The logic goes like this:

- The biggest problem for surface transportation is congestion and the resulting increases in travel time for any given trip (accidents are in second place).
- Most congestion problems are urban problems.
- Urban congestion problems are primarily peak-period problems.
- Peak-period congestion problems are primarily a result of trying to move people, not freight:
 - Freight has become *relatively* less important over time. Moving people has become relatively more important (because of increasing values of time).
 - A lot of freight is more flexible in using off-peak transportation capacity.
 - Freight is more willing to pay for reduced congestion (one trip by a loaded truck is much more valuable than a trip by the equivalent congestion load of cars, but cars heavily outnumber trucks in urban congestion).
- The peak period is getting longer; the contribution of commuting to peak-period congestion is decreasing.
- Peak-period congestion is a result of too many people wanting to be on the same route, at the same time, in the same mode of travel, given the full cost (*to them, as they perceive it*) of that travel.
- Treating this problem requires improving the mechanisms for allocating space on the highway (the current mechanism is a combination of first-come/first-served and slow travel). That better mechanism has to both allow fewer cars (the relatively easy part) and ensure that the cars that remain are inhabited by the people who most highly value travel on that highway at that time (the not-so-easy part).
- One such mechanism is pricing. As we explain later in detail (excruciating and repetitive detail), peak-period travelers underpay for travel. Getting the pricing right both reduces the number of cars—reducing congestion to its efficient level (some congestion is efficient)—and allocates the seating in those cars efficiently.
- There are more reasons for public policy to pay attention to the form of urban development than just its impacts on urban transportation. The transportation/land use connection is about more than congestion. How communities organize their transportation systems can have big impacts (positive and negative) on other aspects of quality of life.

That string of bullets summarizes a basic argument of this entire PAS Report. But that string contains too many assumptions to ask readers to just accept them. Large parts of this report are dedicated to providing theoretical

Public policies uninformed by a knowledge of how markets work risk being inefficient and unfair (even when fairness is an explicit objective); and market decisions unconstrained by public policies that deal with externalities and public goods are likely to be inefficient and unfair.

and empirical reasons to support those assumptions: to promote them as being better than most others and useful as a basis for prescribing public policy to address urban transportation problems.

A CASE FOR CARING

Previous sections state briefly or imply our answer to the question, Why care about the transportation/land use connection? This section makes the case in more detail.³

Policy makers in the 1940s and 1950s didn't pay much attention to the connection. They had little trouble directing transportation policy: they built the roads and highways needed to accommodate the growing numbers of cars and trucks. As incomes rose, the demand for cars and highways increased. Congress responded with the Interstate Highway Act of 1956, which provided long-term federal support for road and highway construction. Most states also levied gas taxes to fund construction of roadways.

No single force has had a greater impact on the pattern of land development in American cities in this century than highways. Better highways decreased the cost (both time and money) of transportation within and between urban areas. Cars helped households relocate from housing near their workplaces to housing in the suburbs that provided better housing per dollar: larger lots, detached houses, cleaner environments, and more control over local public services. Retail firms followed their customers to the suburbs. Long-haul trucks freed manufacturing firms from relatively expensive sites near ports and railroad terminals. Service-oriented firms followed the retail and manufacturing firms they served to the suburbs. In short, highway transportation improvements abetted other market forces that encouraged increasing numbers of households and firms to pick suburban locations.

Land-use plans and policies guided and encouraged the new pattern of development that emerged at the urban fringe. Zoning took advantage of low-cost transportation to protect residential environments by segregating potentially conflicting uses—the car allowed easy travel between residential, commercial, and industrial areas despite the distances separating them. Zoning was also used to preserve environmental quality within residential areas. Land-development regulations required designs that facilitated access by car to new development; policies for public facilities subsidized extensions of water, sewer, and power to new developments.

The changes in the pattern of land development made possible by better roadways, and encouraged by land-use and public-facility policies, rapidly increased the demand for trips on those roadways. Whenever the demand pushed against capacity, the solution was to continue funding for construction of increased roadway capacity. Transportation planning and investment focused on increasing roadway capacity (though not only through new construction) through the 1980s.

Despite the tremendous investments made in roadway capacity, traffic congestion, the chief problem of intrametropolitan transportation, is virtually ubiquitous. This report provides more detail later, but Figure 1-1 gives some slap-and-dash evidence to illustrate the problem and some of its causes. It shows that factors that would reduce congestion have grown slowly; factors that would increase congestion have grown more rapidly; and actual congestion has grown more rapidly still.

Surveys of people across the country have consistently found that highway congestion is always near the top of the list of the problems of growth. It vexes local policy makers because it has no obvious solution. It doesn't affect just the very poor—it irritates the people with the most influence at city hall. It doesn't persist because of a lack of attention or

No single force has had a greater impact on the pattern of land development in American cities in this century than highways.

Would reduce congestion if growing

Highway system mileage (70-04)	0.2
Real price of gasoline (2004 cents/gallon) (70-04)	0.3
Average total cost per mile (2004 cents) (80-00)	0.9
Total government expenditures on transportation (80-00)	2.9

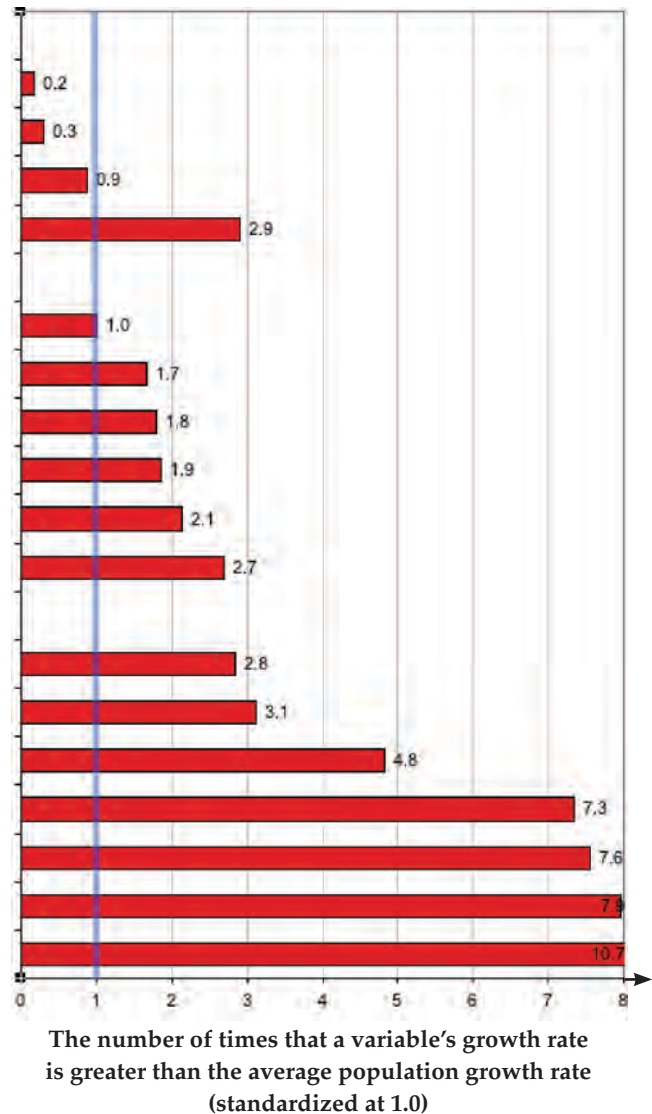
Would increase congestion if growing

Population (60-04)	1.0
Motor vehicle licensed drivers (70-00)	1.7
Motor fuel usage (70-00)	1.8
Per capita personal income (in 2004 dollars) (70-04)	1.9
Registered vehicles (70-03)	2.1
Annual vehicle miles traveled (70-04)	2.7

Measures of congestion

Annual person-miles of public transportation travel (billions) (82-03)	2.8
Daily vehicle-miles of travel on major roads (billions) (82-03)	3.1
Annual hours of delay per peak traveler (82-03)	4.8
Cost of congestion (billions of 2004 dollars) (annual) (82-03)	7.3
Total hours of delay (billions) (annual) (82-03)	7.6
Total gallons of "wasted" fuel (billions) (annual) (82-03)	7.9
Number of urban areas with more than 20 hours of delay per peak traveler (annual) (82-03)	10.7

■ The number of times that a variable's growth rate is greater than the average growth rate



Sources: Multiple. See note 4.

Figure 1-1. Historical trends in key factors relating to transportation cost and demand

money—governments at all levels have spent decades and billions of dollars improving urban transportation systems. Despite the prayers, promulgations, and programs, trips outstrip highway capacity, and most transit cannot attract enough riders to pay even a third of its operating cost, much less its capital cost. Why doesn't increasing transportation capacity, either highway or transit, reduce congestion?

Part of the answer is that land use changes in response to accessibility.⁵ New capacity encourages changes in land use that increase the demand for trips until congestion returns to its level prior to the new investment. New buildings appear along the new highway and create new trips. Their large parking lots make trips a trek for those otherwise trending toward transit. While land along freeways develops into buildings and parking lots, the anxiety of planners and politicians about vacant retail and office space in central areas manifests itself in schizophrenic policies to ban the auto and simultaneously woo businesses that depend on it. Most transportation planners now recognize this inevitable interaction between the transportation system and land-use pattern. They realize that controlling congestion requires not just changes in transportation policies, but probably complementary changes in land-use policies.⁶

This short history points to changes encouraging more attention to transportation policy and its connection to land-use and land-development policy. Moreover, concern about land use, independent of its connection to transportation, has increased. In summary, at least three things have changed since we wrote the first edition of this PAS Report (1994) that have led to more attention to the transportation/land use connection:

- **Increasing land-use planning.** At a minimum, there has been no abatement of land-development planning and regulation. All the needs for planning that planners accept as givens still exist. Environmental quality, water, risk assessment and disaster management, sprawl and Smart Growth: these and other issues now frequently meet the press. More planners and more local, regional, and state jurisdictions are involved in some kind of planning. More regulations requiring land-use planning and more groups fighting for and against those regulations exist.
- **Increasing understanding of the connections among of all aspects of planning.** Not only are land use and transportation connected to each other, they are also connected in complex ways to policies and outcomes for economic development, environmental quality, urban amenities, health, the quality and price of other urban infrastructure, fiscal stability, environmental justice, and so on. Planners have always known that—the connection is embedded in the idea of a comprehensive plan or an Environmental Impact Statement. But decision makers and the public are increasingly requiring a better evaluation of the broader effects of public policies and investments.
- **Increasing understanding of the transportation problem.** The reality is slowly sinking in. The easy roads have been built. Every unit of new transportation capacity costs more to build than it used to. The pricing of access to transportation facilities has problems unlike those of most other utilities. As the road system now functions, it does not even deny service to people who don't pay their bills, much less charge users of the service based on the costs they impose. Other utilities (e.g., electricity, water, solid waste collection and disposal) have the ability (admittedly, not always or even commonly used) to charge prices to recover cost or graduated to ration scarce supply during peak periods. Electric and water utilities, for example, have charged higher rates by season; electric utilities by time of day.

Lack of pricing by time of day and location is the major contributor to urban transportation problems (both use and funding). Funding is a huge constraint on the construction, operation, and maintenance of urban transportation systems at all levels. In most cases, government will not be able to reduce the growth of congestion—much less reduce it in absolute terms—with new facilities. Congestion is going to get worse without changes in policy. That is the context for considering land-use patterns: they have a lot to do with the location and type of transportation that will be demanded, and with the cost of supplying transportation systems.

DIFFERENT NEEDS, DIFFERENT READS

The more research we did, the more we mulled and molded that research into this report, the more difficult the organization became. The more knowledgeable we got absolutely, the more obvious our relative ignorance became. Answers we thought elegantly simple quickly became intolerably simpleminded when set next to the questions they generated. We believe that the problems we faced in trying to report our research reflect the complexity of the interrelationships we are trying to describe.

As long as this PAS Report is, it could be (and once was) much longer. The editors eventually replaced their good manners with good sense, rejecting

Congestion is going to get worse without changes in policy. That is the context for considering land-use patterns: they have a lot to do with the location and type of transportation that will be demanded, and with the cost of supplying transportation systems.

revisions and throwing out half of what was already bloating their hard disks. Despite their best efforts, this PAS Report takes some time to get through.

We kept our intended audience in mind as we pruned. We assumed most would have an acquaintance, but not a friendship, with the principles of policy evaluation, microeconomics, and land-use and transportation planning. That assumption makes the text more readable for people with that background. For students, that assumption should fit for most graduate students in planning schools, but probably not most undergraduates. For planning practitioners, that assumption probably works for most practitioners.

But the assumption will be incorrect for some. Even some of the ones who should and must profess to understand this stuff will be happy to have a refresher on and quick reference to the basic principles. To that end, we have added several appendices providing details the chapters do not and telling the story in a different way.

The **main text** tells the story from the perspective of transportation policy:

- Section I: historical and technical context for land and transportation development
- Section II: policy evaluation—framework and details
- Section III: policy process—packaging and adopted integrated transportation and land-use policies in a regional context.

The **first five appendices** are really the details of the evaluation component discussed in Section II of the main text:

- Appendix A: essential principles of rational decision making and policy evaluation applicable to *any* public choice about projects, programs, and investments
- Appendix B: microeconomic principles as they apply to public policy choices, in general and specifically for land use and transportation
- Appendix C: a primer for surface transportation planning
- Appendix D: benefit-cost analysis, project evaluation, and project selection
- Appendix E: transportation modeling.

Having material in appendices should also help instructors using this PAS Report in graduate planning courses: you can more easily tailor the readings to the background of your students. A final advantage is that the shift of material to appendices helped reduce for us the inherent problem of having to explain sequentially what needs to be known simultaneously; it freed us to tell a story without having to worry the story could not make sense without a prior understanding of all the technical tools the story was trying to gently introduce.

We hope that this overview has provided enough context for readers to understand where we are coming from and why. And, remember, both a Readers Guide and overview material exist at the beginning of each chapter to provide you with a more easy entry to the material.

For those planners and decision makers in metropolitan areas who have little familiarity with transportation, we think this *report can be skimmed and still provide some value*:

- Look at Chapter 2 and parts of Chapter 3 to get an overview of the forces shaping land use and transportation.
- Skim Chapters 5 and 6 to get an idea of the range of land-use and transportation policies available.
- Read Chapter 7 for an idea of how policies might be integrated into packages that address, each with increasing strength and difficulty, the problem of congestion.

- Read or skim Chapter 8 for ideas on how to incorporate all these ideas into an integrated, regional, and public evaluation of land-use and transportation futures.

For students and their professors, our advice is straightforward. Professors: some who have used this report previously teach the whole report and use it as the basis of their course; others use pieces of it in courses on transportation or growth management. Students: your choices, as far as we can tell, are limited to reading what your professor assigns, or not. Beyond that, any advice from us would be imprudent.

For planners and decision makers *not* in metropolitan areas (i.e., not in metropolitan areas of roughly 100,000 people or more), many parts of the report are probably not applicable. Though we believe the framework we provide for thinking about transportation policy still applies to small areas, some of your problems are not the same (and, we hope, not as bad), and the scale and intensity of your land uses and transportation infrastructure will not require or allow some of the solutions we discuss (e.g., congestion pricing, new transit systems).

For those wanting to get quickly to the punchline without going through all the theory, we recommend starting with Anthony Downs, *Still Stuck in Traffic* (2004).⁷ If you want a well-reasoned and clearly articulated overview of urban transportation problems from an economic point of view, read it. For an even shorter overview, read his summary chapter (Chapter 18, which would serve as a good introductory piece for planners trying to brief busy decision makers on the causes of urban traffic congestion and the effectiveness of potential policy responses). Though our report covers some of the same material Downs does, we believe it is complementary.

CHAPTER 1 NOTES

1. Seaports and airports (which this PAS Report does not address at all) and rail lines (this report discusses intrametropolitan rail as part of transit).
2. It would be silly to try a listing here: it is long. For the readers that have got this far and have no knowledge of the literature on land-use planning or growth management, try <http://www.planning.org> or <http://www.press.uillinois.edu/s97/kaiser.html>
3. Earlier sections of this chapter already gave a short answer to that question, and implied the points that would be made in a longer one. In the rest of this section, we make those points explicit. You can skip this section if you are already a believer; you might want to use some of these arguments if you are preaching to others.
4. Sources for Table 1-1:
 - **Real price of gasoline, United States, 1970–2004:** Energy Information Administration (http://www.eia.doe.gov/emeu/steo/pub/fsheets/PetroleumPrices_files/frame.html). The historical gasoline prices were taken from the Bureau of Labor Statistics (BLS) survey prices: 1919–1979 leaded regular; 1980–1981: unleaded regular; 1982–1994: Energy Information Administration (EIA) data extrapolated using BLS survey data; 1995–2005: EIA Survey. Real (inflation adjusted prices) price data is deflated by the CPIU, from the BLS, where year 2004 = 1.
 - **Highway system mileage, United States, 1960–2004:** 1960–95: U.S. Department of Transportation, Federal Highway Administration, Highway Statistics Summary to 1995, FHWA-PL-97-009 (Washington, D.C.: Annual issues), table HM-212. 1996–2004: Ibid., Highway Statistics, table HM-20, (Washington, D.C.: Annual issues), Internet site (<http://www.fhwa.dot.gov/policy/ohpi/hss/index.htm>) as of Feb. 2, 2006. As cited in the Bureau of Transportation Statistics web-site (http://www.bts.gov/publications/national_transportation_statistics/html/table_01_01.html). Calculations include all public road and street mileage in the 50 states and the District of Columbia. For years prior to 1980, some miles of nonpublic roadways are included. No consistent data on private road mileage are available. Beginning in 1998, approximately 43,000 miles of Bureau of Land Management Roads are excluded.
 - **Average total cost per mile (2004 cents):** American Automobile Association, Your Driving Costs (Heathrow, Fla.: Annual issues). As cited in the Bureau of

Transportation Statistics website (http://www.bts.gov/publications/national_transportation_statistics/html/table_03_14.html). All figures reflect the average cost of operating a vehicle 15,000 miles per year in stop and go conditions. Oil costs were included in gas prices. Post-2004, oil costs are covered by maintenance. Fixed costs (ownership costs) include insurance, license, registration, taxes, depreciation, and finance charges. Changes in the way costs were calculated make it difficult to compare pre-1985 data with more recent years. Prior to 1985, the cost figures are for a mid-sized, current model, American car equipped with a variety of standard and optional accessories. After 1985, the cost figures represent a composite of three current model American cars. The 2004 fuel costs are based on average late-2003 U.S. prices from AAA's Fuel Gauge Report (www.fuelgauge.com). Insurance figures are based on a full-coverage policy for a married 47-year-old male with a good driving record living in a small city and commuting three to 10 miles daily to work. The policy includes \$100,000/\$300,000 level coverage with a \$500 deductible for collision coverage and a \$100 deductible for comprehensive coverage. Depreciation costs are based on the difference between new-vehicle purchase price and its estimated trade-in value at the end of five years. American Automobile Association analysis covers vehicles equipped with standard and optional accessories including automatic transmission, air conditioning, power steering, power disc brakes, AM/FM stereo, driver- and passenger-side air bags, anti-lock brakes, cruise control, tilt steering wheel, tinted glass, emissions equipment, and rear-window defogger. Average total cost per mile deflated to 2004 cents using the BEA's "Price Indexes for Personal Consumption Expenditures by Major Type of Product."

- **Total government expenditures on transportation:** 1980: U.S. Department of Transportation, Bureau of Transportation Statistics, unpublished data. 1985–2002: U.S. Department of Transportation, Bureau of Transportation Statistics, Government Transportation Financial Statistics: 2003 (Washington, D.C.: forthcoming), tables 3-A, 13-A, and 14-A. BTS website: http://www.bts.gov/publications/national_transportation_statistics/. Transportation-related revenues and expenditures in 2004 dollars are calculated using the "Price Indexes for Personal Consumption Expenditures by Major Type of Product" from the BEA. Data for 1980, 1985 and 1990–2001 have been revised due to the Office of Management and Budget (OMB) revision to the source data on federal expenditure and federal grants. Federal grants to state and local governments for research and special programs, and emergency preparedness are included starting from 1985. Numbers may not add to totals due to rounding.
- **Population:** 1960: Selected Historical Decennial Census Population and Housing Counts. Available on the web at (<http://www.census.gov/population/www/censusdata/hiscendata.html>). 1970, 1980, 1990: U.S. Bureau of the Census, 1990 Census of Population and Housing (<http://www.census.gov/population/www/censusdata/hiscendata.html>) 2000: U.S. Census bureau, Census 2000 (http://factfinder.census.gov/servlet/SAFFPopulation?_sse=on). 2001, 2002, 2003, 2004: U.S. Census bureau, 2005 Population Estimates, GCT-T1 Population Estimates (http://factfinder.census.gov/servlet/GCTTable?_bm=y&-geo_id=01000US&-box_head_nbr=GCT-T1&-ds_name=PEP_2005_EST&-lang=en&-format=US-9&-_sse=on).
- **Motor vehicle licensed drivers:** BTS website: (http://www.bts.gov/publications/national_transportation_statistics/2002/html/table_automobile_profile.html). 1970, 1980, and 1990 numbers came from the U.S. Department of Transportation, Federal Highway Administration (FHWA), Highway Statistics Summary to 1995, FHWA-97-009 (Washington, D.C.: July 1997), table DL-201. 2000 number from U.S. Department of Transportation, Federal Highway Administration (FHWA), Highway Statistics (Washington, D.C.: Annual issues), table DL-22 (revised tables used when applicable).
- **Motor fuel usage, United States:** 1970, 1980, 1990: Federal Highway Administration website (<http://www.fhwa.dot.gov/ohim/summary95/section1.html>) 2000: Federal Highway Administration website (<http://www.fhwa.dot.gov/ohim/hs00/mf.htm>).
- **Per capita personal income, United States:** Nominal Per Capita Income from the BEA website: <http://www.bea.doc.gov/bea/regional/spi/default.cfm>. Per Capita Income in 2004 dollar calculated using the "Price Indexes for Personal Consumption Expenditures by Major Type of Product" from the BEA. Nominal dollars converted to 2004 dollars using the "Price Indexes for Personal Consumption Expenditures by Major Type of Product" from the BEA.

- **Registered vehicles, United States:** 1970: U.S. Census Bureau, Statistical Abstract of the United States: 1980. Table No. 1116. p. 649. (<http://www.census.gov/prod/www/abs/statab1951-1994.htm>) 1980, 1990: U.S. Federal Highway Administration, Highway Statistics, annual; and Selected Highway Statistics and Charts, annual. As cited in U.S. Census Bureau, Statistical Abstract of the United States: 2000. Table No. 1026. p. 627. (<http://www.census.gov/prod/www/statistical-abstract-us.html>). 2000: U.S. Federal Highway Administration, Highway Statistics, annual; and Selected Highway Statistics and Charts, annual. (<http://www.fhwa.dot.gov/policy/ohpi/hsspubs.htm>). As cited in U.S. Census Bureau, Statistical Abstract of the United States: 2004–2005. Table No. 1082. p. 692. (<http://www.census.gov/prod/www/abs/statab1951-1994.htm>). 2001, 2002, 2003: As cited in the U.S. Census Bureau, Statistical Abstract of the United States: 2006 (http://www.census.gov/compendia/statab/transportation/motor_vehicle_registrations/). Population data for per capita calculations from the U.S. Census Bureau: 1970, 1980, 1990: U.S. Bureau of the Census, 1990 Census of Population and Housing (<http://www.census.gov/population/www/censusdata/hiscendata.html>). 2000: U.S. Census Bureau, Census 2000 (<http://factfinder.census.gov/servlet/SAFFPopulation?sse=on>). 2001, 2002, 2003: U.S. Census Bureau, 2005 Population Estimates, GCT-T1 Population Estimates (http://factfinder.census.gov/servlet/GCTTable?_bm=y&-geo_id=01000US&-box_head_nbr=GCT-T1&-ds_name=PEP_2005_EST&-lang=en&-format=US-9&-sse=on). Motor vehicle registrations cover publicly, privately, and commercially owned vehicles. For uniformity, data have been adjusted to a calendar-year basis as registration years in states differ; figures represent net numbers where possible, excluding re-registrations and non-resident registrations. Figures exclude vehicles owned by military services. Estimates available for interim years in addition to those shown above.
- **Annual vehicle miles traveled, United States: 1970–2004:** The Federal Highway Administration website (<http://www.fhwa.dot.gov/policy/ohpi/qftravel.htm>).
- **Measures of congestions:** Texas Transportation Institute. 2005 Urban Mobility Report. p. 1 available online at (<http://mobility.tamu.edu/ums/report/>). 2004 dollars calculated using the “Price Indexes for Personal Consumption Expenditures by Major Type of Product” from the BEA. Pre-2000 data do not include effect of operational strategies and public transportation. Delay per Peak Traveler—The extra time spent traveling at congested speeds rather than free-flow speeds divided by the number of persons making a trip during the peak period.

Notes: The average annual population growth rate, used as the denominator in the above calculations, is taken over the same time period as a given variable is reported for. The numbers in parentheses indicate the time period over which a given variable is reported.

5. Other parts of the answer, especially what is referred to as *latent demand*, are addressed in later chapters.
6. We use the term land-use policies broadly in this PAS Report to include the range of policies local governments have used—typically as part of their land-use functions—to manage growth. This definition includes public infrastructure policies, which are critical to managing growth.
7. The first edition of our PAS Report acknowledged that we had been scooped by what became *Stuck in Traffic* by Anthony Downs (1992); this second edition acknowledges that we were scooped again, this time by Downs’s second edition.



*Thanks to the Interstate Highway System, it is now possible
to travel from coast to coast without seeing anything.*

—Charles Kuralt

*The only way to solve the traffic problems of the country is
to pass a law that only paid-for cars are allowed to use the
highways. That would make traffic so scarce, we could use our
boulevards for children's playgrounds.*

—Will Rogers

SECTION I.

FORCES SHAPING CITIES AND THEIR TRANSPORTATION SYSTEMS

This section describes and explains the evolution over the last two centuries of urban land development patterns and urban surface transportation policies. In 1800, less than 10 percent of the population in the U.S. lived in towns and cities. Depending on how one defines an urban area, 75 percent to 80 percent of the people in the U.S. and in other developed countries are now urbanites. This section draws from standard material in urban economics and regional science to explain the reasons for a huge shift in human living conditions in a short time.

That explanation hinges on advances in the technologies of goods production, communication, and transportation. Advances in production technologies have made possible and enforced the great movement from farm to city: it no longer takes so many of us to feed all of us. Now there are so many more of us because we can more easily feed ourselves and ward off other threats to a long life.

What are all of these additional people to do? In part, make more stuff. Chapter 2 describes how the advantages of proximity in producing and distributing many of the nonagricultural goods and services drive urban development.

Though the pace of urbanization in the U.S. and in other developed countries has itself slowed—there aren't as many farmers left to move to the city—the pace of change in urban land development has not. Technological advances drive changes in the variety of goods and services produced, and in where and how these goods and services can be produced most cost-effectively. Such changes obviously influence patterns of urban development. Change is unlikely to stop or even to slow given the continuing potential for innovation in production technologies and the rewards bestowed on innovators.

Advances in technology enrich us, but they also complicate our decision making: we have so many choices. A theme of this section is that advances in technology have complicated the setting for public transportation and land-use decisions because the range of choices available has increased.

This theme of evolution from constraint to choice is apparent in the organizations of Chapters 2 and 3. Cities form to exploit the cost advantages in the production and distribution of goods and services that proximity and specialization provide. Two hundred years ago, transportation and transportation infrastructure were both expensive. Rivers, lakes, and oceans offered natural infrastructure, and transportation by water, though expensive by today's standards, was much cheaper than the other modes available (i.e., foot and horse-and-wagon). So towns and cities arose in locations bestowed with favorable natural characteristics: falling water for energy, proximity to raw materials, or a good port providing access to low-cost water transport. Transportation improvements were limited to canal building and the construction of trails and rough wagon roads, which state and local governments worked to encourage.

The development of new technologies, especially steam power, led to the development of railroads that added a significant dimension of choice in intercity transportation. The new rail lines were private ventures and mostly connected existing cities, which reinforced original locational advantages of these cities and helped them grow. But new cities emerged at places the railroad itself made accessible. The development potential of rail prompted the first serious national transportation policy: encourage private, growth-inducing investment in rail.

A few developers took advantage of freight rail lines to build upscale exurban residential developments. But steam trains were too noisy and dirty to support much commuter travel. The growth of large cities had to await the development of electric streetcars. Streetcars made it possible for a large workforce to commute cost-effectively from low-density housing in suburban areas to employment that concentrated near the port or railway station. The streetcar lines were usually developed privately, but local governments exercised control over land development through control over the location of streetcar lines.

The next and more liberating technological innovation was the internal combustion engine. Cars and buses liberated commuters from the streetcar lines, and trucks liberated manufacturers from the port and from freight rail lines and stations. Trucks, buses, and cars

require good roads and highways, however, and building any highway system requires huge investment and acquisition of rights-of-way. Equally important, charging highway users is difficult. The difficulties in private investment and the success of large-scale public projects during the Great Depression and World War II led to national and state commitment to large-scale highway building.

The response of highway users to that investment has been so strong that as we build for economic development, we have had to develop widely varying public efforts to treat the negative effects of the car: traffic congestion, urban sprawl, and environmental pollution. The ultimate sources of these problems are in dispute, as are the means recommended to treat them.

The aim of our historical treatment of transportation development is to shed light on the sources of the problems. The problems are understandable but the treatments are vexing: each involves difficult trade-offs, and current technologies allow a great deal of choice. In the twenty-first century, the frontier is largely closed, and resources scarce; we can no longer simply build our way out of our problems. But technology gives us an impressive range of choice in how to deal with that scarcity. Making those choices is a challenge for public transportation and land-use policy. This section lays some of the groundwork for understanding those choices.

Section I has two chapters that provide both historical and technical context for thinking about transportation policy.

Chapter 2. Market Forces in the History of Urban Land Development. What causes cities and metropolitan areas to start and grow where they do? Understanding the history of urban land development is important because buildings and infrastructure are durable. What exists now influences decisions about investments that determine what will be in the future. We start with simple models and gradually add detail and complexity, showing how a small urban area with a single downtown center can grow into a polycentric metropolitan region. We describe how transportation facilities affect that growth, and what an efficient transportation system would look like at different stages of development. Our main purpose is to describe the kind of urban areas and transportation systems market forces would tend to create in the context of past and current land-use and transportation policies in urban areas. We think that description is fundamental to any evaluation of public policy, which is usually trying to get a different result than the one that current economic and policy forces are providing.

Chapter 3. The Evolution of Surface Transportation Policy. This chapter lumps the last 100 years of transportation development in the U.S. into three epochs, defined by a combination of technology and transportation policy. It looks at the pre-auto and early-auto era around 1900, the influence of the auto by 1940, and the growth of interstate highways after 1950. These technological changes were accompanied by changes in transportation regulation and investment. Both changes had huge effects on the transportation system and on land-use patterns.

Both chapters have an economic perspective. If you are already comfortable with your knowledge of urban economics, you can skip these chapters. If you are so uncomfortable that you plan to skip these chapters, you probably have the wrong book: Section II, Policy Evaluation, will be difficult without a basic knowledge of the principles in Section I and its supporting appendices.

Market Forces in the History of Urban Land Development

Organization: This chapter describes trends in urban land-development patterns over the last several hundred years and explains these trends from an economic perspective. It has three sections:

1. *Fundamentals of urban growth.* Why cities exist (from an economic perspective), and an explanation of the number and size distribution of the cities in a region.
2. *Urban development in a single-center city.* Most cities developed before cars and trucks did. These cities usually developed around a single employment center. The location of that center was usually determined by natural characteristics, such as the location of a natural harbor. The analysis of a simple, single-center city provides insights into urban land prices, development intensities, and the relative locations of land uses. It explains the broad patterns of land development in turn-of-the-twentieth-century (i.e., preautomobile) cities.
3. *Urban development in a multiple-center city.* Most employment in twenty-first century metropolitan areas in the U.S. is outside of the traditional central business district (CBD). The dispersion of employment is largely a result of the introduction of cars, trucks, and highways. Employers continue to cluster in especially accessible locations, but the location of new clusters is no longer strictly a market decision: policy is increasingly important. Suburban employment centers, for example, tend to form at the intersections of highways, the number and locations of which are decided through transportation policy. The analysis of the simpler, single-center city helps an understanding of the relationships among subcenters and the patterns of land development in their vicinity.

Basics: Understanding the history of land development is important because buildings and infrastructure are durable. What exists now influences decisions about investments that determine what will be in the future. This chapter uses a chronological description of changes in land development over time to introduce key concepts for evaluating policy.

- Cities formed and grew as businesses clustered near sources of raw materials, markets, or ports. Their locations optimize on the trade-off between the cost advantages of large-scale production and higher intercity transport costs.
- Before cars, trucks, and highways, most cities developed around a single high-density employment center near the port or freight-rail depot. Land prices fell with distance from the CBD to compensate for higher commuting costs. Development densities fell with distance in response to lower land prices; transportation costs cause land prices, which cause density.
- Improvements in transportation and building construction allowed monocentric cities to grow. Electric streetcar systems reduced commuting costs and steel-framed construction allowed higher-density development in the CBD.
- Well-located cities grew and urban land-development patterns changed as cars, trucks, and highways broke the strong link to the traditional CBD. The

flexibility of auto travel allows these well-located cities to grow until traffic congestion, air pollution, or land constraints counteract their locational advantage.

- There remain cost advantages to clustering: most cities are polycentric. Subcenters develop at highly accessible locations, such as at the intersections of major highways. This polycentric development efficiently trades off costly density for lower transportation costs.
- Changes in the characteristics of transportation may be the single most important determinant of changes in urban development patterns.
- Trucks and buses created the demand for land-development policies in the early twentieth century. Trucks and buses allow manufacturers and multifamily developers access to single-family areas. Zoning was instituted to keep them out.
- Land-development regulations contributed to the development of suburban areas to which cars, trucks, and highways gave access. The primary motivation for land-use and transportation policy, until recently, was the development of North America's vast land resources.

Shortcuts: This chapter lays part of the foundation of the economic analysis of land development. Readers with a good background in the history of land development and land-development policy can skim the chapter. People with a limited background in the economic history and theory of urban development should probably read the chapter because Chapters 3 and 4 will otherwise be hard to understand. Whatever you decide about Chapter 2 probably applies to Chapter 3 as well; if you are comfortable with or intolerant of the history and economics of urban development, skip both Chapters 2 and 3 and take your chances with Chapter 4. That's manageable, and you can refer to Appendices B and C for some help on economic principles.

This chapter examines the economic forces that shape urban land-development patterns. We start with relatively simple concepts and models, and then combine and build on them to provide a rich framework for understanding the complexities of land development in modern urban areas. Working from simple to complex corresponds nicely to the chronological development of cities. Nineteenth-century cities were smaller and their land-development patterns simpler.

We begin our discussion from the hypothesis and the hope that the complex environments found in large cities result from the interactions of only a few relatively simple forces—the same forces that create and shape small cities. The possibility that simple mechanisms cause complex systems drives scientific research. Researchers proceed by developing abstract, usually mathematical, models that reduce real-world complexity to a few key relationships. If successful, they gain important insights from the simple models, and these evolve into the more complex models needed to understand real-world systems.

FUNDAMENTALS OF URBAN GROWTH

A key reason cities exist, in an economic sense, is to economize on the transportation costs associated with conducting the activities essential and optional for daily life. Market forces similarly distribute urban areas around a region to distribute goods and services in ways that economize on transportation costs. The long-run trends in transportation technology have been to decrease the unit cost of transportation.

What if those trends continue? What if transportation in the future costs less than it does today?

A surprising implication is that the characteristics of land development in a distant future with very low transportation costs are similar to those in pre-industrial times when transportation costs were very high. Compare the characteristics of land development in medieval Europe—when transportation costs were very high and production technology primitive—to that in a future with a key component of Star Trek technology: matter “transporters.”

The results of that comparison, which we'll describe in just a minute, suggest that just as in medieval times, economic forces for urbanization are very weak in the Star Trekian future.

Consider first the drivers of land development in pre-industrial times. Less than 3 percent of the world's population lived in urban areas prior to the start of the Industrial Revolution. Given the limited technologies of the time, farming was a labor-intensive activity. So most people had to work on farms, and farm laborers couldn't commute far to work in the fields because transportation costs were extremely high (people traveled mostly by foot). So farm laborers and their households were spread relatively evenly across arable regions.¹

A key characteristic of the technologies for producing *nonagricultural* goods and services reinforced this pattern: large-scale production offered few cost advantages. Many goods were produced on the farm, either by the farmers or by itinerant specialists who traveled from farm to farm. Most households had little incentive or opportunity either to concentrate in towns or to travel.

The small proportion of people who did concentrate in towns in those times did so for two reasons. First, there were cost advantages. It was easier (less costly) to defend stores of food within the walls of a town. That defense usually required trained and armed (i.e., specialized) defenders who lived in the town: there are scale economies in defense. Similar cost advantages could be found in relatively large-scale trading operations, public administration, and religion.

Second, farming households benefited from the choice and competition in the markets the towns attracted. Some goods were imported from other regions or produced by specialists who worked in the town. Households could occasionally travel to the local market towns that offered one-stop and comparison shopping. Nevertheless, most such towns were small by modern standards, and most of the population was spread across agriculturally productive areas.

Jump from the middle of the second millennium to the middle of the third, a future with Star Trek technology. Think about what would happen to land use if, on a particular trip, traveling an additional mile took a very small amount of time and required a very small additional out-of-pocket expense; that is, suppose that matter transporters could be had for the price of an economy car. Suppose further that the cost of the energy expended in using the transporter is also relatively low. Finally, and most importantly, assume that the cost of a trip *does not vary with the distance covered*: given an upfront investment, one can move freely and cheaply from any place on the planet to any other place. The unit (i.e., per-mile) cost of travel is zero with this Star Trek technology.

How would the built environment develop under these conditions? If people got greater satisfaction from living on large lots and businesses could reduce production costs by using more land, households and firms would spread themselves widely. Constrained only by the number of people living in the world, everyone could live on a large lot with a view.² People who like warm weather could live in the tropics and commute to work in Canada. Businesses could use as much land as needed to minimize the cost of production, simply transporting intermediate goods and services and raw materials directly from their points of production. Face-to-face meetings with clients, suppliers, or consultants are possible regardless of location. Just as in high-cost medieval times, in this low-cost future, production and population would be spread out.

This vision of the future differs from the vision of most futurists as revealed by our content analysis of the covers of science-fiction novels. They

A key reason cities exist, in an economic sense, is to economize on the transportation costs associated with conducting the activities essential and optional for daily life.

envision gleaming, dome-covered, and intensively developed cities. These futurists apparently do not foresee dramatic decreases in the unit cost of transportation, or, more likely, they do not foresee the consequences of cheap transportation for land development (or their cities are responding to the high costs of providing things other than transportation, like breathable air). As we show throughout this chapter, however, the trend toward dispersal is already under way. Which begs the question:

Why Do Cities Exist?

This may seem a trivial question. People live in cities because that's where the jobs, shopping, entertainment, and social life are. But it is also where the crime, noise, air pollution, traffic, and, especially, high land and housing prices are. The question, Why do cities exist?, might be stated more precisely as, Why do households and businesses endure high prices and crowding in countries abundantly endowed with land?

The surprising answer is that achieving the same overall level of benefit, despite the high-priced land and crowding, costs less in cities. A central lesson in this chapter is that land is expensive in cities—very expensive in some cities—because the savings in transportation costs are so valuable.

Modern cities started to develop with the technological innovations that drove the Industrial Revolution. Prior to that time, people produced most of what they needed at home. They used human and animal power, along with simple tools. Each farmer was truly a jack of all trades. The invention of new tools, techniques, and sources of power made factory production of some goods more cost-effective than the old methods used at home. Workers within factories became specialists using specialized machinery. The productivity of the factory (i.e., output per factory worker) could exceed by many times the productivity of small-scale or home production. The rural Jacks and Jills could never achieve the quantity and quality of output achieved by factory Joes and Janes.

Most planners have heard the terms economies of scale, specialization, and division of labor as factors important in the Industrial Revolution. We discuss those and other terms under two broad headings that explain why cities exist:

- Internal economies of scale: the cost advantage an individual business gains with larger-scale production. A business experiences economies of scale when the business can reduce average production cost by producing higher volumes of output using specialized machinery and labor. Bigger and more specialized, though more expensive up front, is cheaper on average.
- External economies of scale: the cost advantage the businesses in an industry cluster or city enjoy as the cluster or city gets larger. These economies are variously referred to as economies of proximity, clustering, concentration, or (most often) agglomeration. Agglomeration economies are further divided into *localization* economies (that occur because of the size of an industry sector) and *urbanization* economies (that occur because of the size of an urban area). These economies arise from specialization, risk sharing, knowledge sharing, and reductions in the costs of transactions with other businesses (more below).

The economic advantage that comes from putting people with specialized training together with specialized machines under one roof is one of the two primary forces for urbanization. A business that exploits scale economies creates a destination to which a large number of workers must commute from their homes. This need for large numbers to travel from home to an employment center is a necessary condition for the formation of cities.

The question, Why do cities exist?, might be stated more precisely as, Why do households and businesses endure high prices and crowding in countries abundantly endowed with land?

But it is not a sufficient condition, as demonstrated in the Star Trek experiment. If the unit cost of transportation is zero, workers can live wherever they like. In the present, living farther from work means commutes that cost more in terms of both time and money. This cost of distance translates into willingness to pay (i.e., demand) to reduce the length of commutes; commuters compete for relatively accessible residential locations. That competition drives up land prices. People respond to the higher land prices by economizing on land: living on smaller lots and building vertically. The result is higher land prices and greater population densities in areas accessible to workplaces.

Cities exist because of production technologies with scale economies and transportation costs low enough to make commuting to work affordable, yet high enough to encourage people to compete to live close to their workplaces (and other destinations).

This concept applies to pre-industrial cities as well as to modern ones. In medieval times, cities arose due to scale economies in defense, government, and religion. The cities that grew relatively large also exploited scale economies in trade. Good, safe harbors accessible to agricultural areas are relatively rare. They are a form of specialized capital. The people who engaged in trade—sailors, ship builders, longshoremen, wholesalers, and so on—enjoyed cost advantages from locations on land proximate to a good harbor. They competed for the most proximate land, bidding up its price. The largest pre-industrial cities combined defense, public administration, and religion with trade.

Later, during the Industrial Revolution, the advantages of factory production gave factory owners the incentive and ability to pay wages high enough to pull people off the farm and into work in the factory. Because of scale economies, the owner of a factory could offer output at prices below the cost of home production, while at the same time offering wages attractive to farm laborers.

Factories often did not need to pull very hard. Improvements in farm technology and investments in labor-saving machinery concurrently reduced the demand for farm labor, and more farm workers became available to work in the factory. Advances in health care extended life spans and reduced child mortality. The dramatic increases in population contributed labor for the factories.

The high costs of commuting by foot help explain why millions of laborers chose to settle with their families in the deplorable conditions often found in late eighteenth- and nineteenth-century industrial cities.³ Manufacturers clustered near ports and energy sources to minimize shipping costs. High commuting costs forced workers and their families to live in crowded conditions near the factory. The high density, combined with industrial pollution and poor quality public services, made for a grim existence.

Cities also grow because of the cost advantages of proximity to other businesses. Economists refer to the cost advantages of clustering as *economies of agglomeration*, or also as *external economies of scale*—external because they do not result from the internal operation of a particular firm. Average production costs in some industries fall with the growth of the size of the cluster where they locate. Production costs fall for a variety of reasons:

- *Internal economies of scale in the production of intermediate inputs.* Businesses that supply key inputs to other businesses can invest in specialized training, equipment, and processes to reduce average production costs if their business customers demand sufficiently large quantities of their outputs. These customers seek locations near the input suppliers for the same reason that workers seek locations near their workplaces: to economize on transportation costs.

The economic advantage that comes from putting people with specialized training together with specialized machines under one roof is one of the two primary forces for urbanization.

- *Comparison and one-stop shopping.* Businesses can reduce transportation costs by shopping for intermediate inputs and business services in clusters of suppliers.
- *Lower-cost face-to-face contact with representatives of other businesses.* Some business transactions require more than telephone calls and emails. Proximity reduces the transportation costs associated with personal visits and facilitates coordination in the production and delivery of customized goods and services.
- *Labor-market pooling.* Employers benefit from the large pool of workers who live near a cluster of businesses; they can shop for experienced workers and gain insight through their workers into what other firms are doing. The regular interactions among a large number of relatively specialized workers encourage innovations: working in a cluster facilitates the spinning-off of new companies that more efficiently supply the inputs the original companies were producing in-house or importing from other cities or clusters. And workers benefit from the competition for their services and from the diverse range of employment opportunities.

The cost of producing a wide variety of goods and services decreases, other things (such as land price) being equal, when businesses locate close to other businesses. *The economic advantage of clustering is the second of the two primary forces that drive urbanization.* We emphasize that the primary advantage of clustering is the savings in transportation costs.

Modern cities, even fairly small ones, usually exist as a result of the combined forces of agglomeration and scale. Production (of widgets, education, government policies, and so on) forms the economic base of the city. Production activities attract the workers that attract the businesses that serve the producers, the workers, and their families—all to reap the advantages of proximity. Agglomeration economies attract other firms in industries that benefit from proximity to related firms. We discuss these issues in more detail in the sections that follow.

Before moving on, however, we note that advances in production technologies continue to influence modern urban development through the ways in which they affect the cost advantages of large-scale production:

- *Some changes diminish the extent of scale economies.* For example, advances in the ability of personal computers, monitors, and printers to produce professional-quality documents have dramatically affected the publishing industry. A firm with only a few employees and good computing and printing equipment can now compete effectively in some markets with publishing giants. Such changes allow decentralization of employment and affect land use.
- *Other changes increase the extent of scale economies.* For example, large manufacturing companies invest in computer-controlled robots to increase speed and efficiency in production. They need to produce and sell at high volumes, however, to spread the upfront cost of those sophisticated robots and the highly trained work force that supports them.

The advances in production technologies essential for improvements in living standards cause changes in land development through their effects on internal and external economies of scale.

The Size and Regional Distribution of Cities: Central-Place Theory

We can build on the analysis above to do something that sounds a bit grand: describe the forces that influence the number, size, composition, and location of the towns and cities in a region; that is, we can describe the theory of *Central Places* introduced by Walter Christaller and August Losch in the

Modern cities, even fairly small ones, usually exist as a result of the combined forces of agglomeration and scale.

1930s. Central-place theory attempts to explain the hierarchical system of cities and towns within a region.

Central-place theory focuses on the differences in demand and supply characteristics across the various types of sellers of retail goods and personal and business services. First, consider the determinants of the *number* of sellers of a particular product or service within a region:

- *The cost-minimizing scale of the business.* Some retailers and service providers enjoy cost advantages from operating at a large scale. Larger scale means fewer stores, all else the same.
- *The per-household demand for the product.* The higher the demand, the larger the number of stores, all else the same
- *The cost of traveling to the store.* Consumers care not just about the price at the store, but how much time and money they have to spend to make their purchases. High travel costs pull stores closer to their customers. There tends to be a larger number of smaller stores when or where transportation costs are relatively high.

Key ideas behind central-place theory are: (1) both consumer demand and the extent of scale economies vary across the products for sale, and (2) each retailer faces a trade-off between exploiting scale economies and reducing consumer transportation costs. The economically efficient distribution of retailers across the region minimizes the sum of production costs (the price at the store) and total consumer travel costs.

Central-place theory demonstrates that market incentives encourage this distribution of retail centers. To illustrate this, assume for simplicity that the economic base of the region is agriculture: agricultural workers and their households—the retailers' customers—are spread evenly across the region. Each retailer recognizes the trade-off between the size of its operation and its location. In the absence of competition, a monopoly retailer can simply choose a central location and enjoy low average production costs.

But the consequent high transportation costs for consumers far from the center present a profit opportunity for smaller operators to locate near the periphery. Competition and the lure of profits push retailers to make socially beneficial trade-offs between low-cost production on a large scale and at a convenient location.

The result is a market-driven hierarchical system of cities. The large central city contains all types of suppliers. The smaller intermediate centers lack the relatively specialized suppliers that can either exploit extensive scale economies (so prices at their central location make long shopping trips worthwhile) or offer products that customers purchase only occasionally. The small outlying centers include only the retailers that offer goods in high demand that can be supplied out of a relatively small operation. Households shop frequently in the small centers, less frequently at the intermediate locations, and only on special occasions at the large center.

Additional forces for agglomeration exist. The retailers attracted to any given part of the region have to locate close together. Consumers benefit from one-stop and comparison shopping. Moreover, public services can be provided more cost-effectively if businesses locate closer together. These forces increase the appeal of relatively larger centers.

These retail and service clusters have effects on urban development similar to those of a factory. All things being equal, the retailers' employees prefer shorter commutes, so they compete for land and housing closer to the employment center, which bids up the price of central land. The residents economize on the higher-priced land by living on smaller lots: residential density rises with proximity to the business district.

Key ideas behind central-place theory are: (1) both consumer demand and the extent of scale economies vary across the products for sale, and (2) each retailer faces a trade-off between exploiting scale economies and reducing consumer transportation costs.

The tidy distribution of retail and service centers predicted by the central-place model shows up clearly in regions that are primarily agricultural and in which the agricultural land is relatively uniform in its characteristics. The system of small towns and cities in, for example, Iowa and southern Minnesota, fits the theoretical prediction. The roadway system is essentially a grid that follows original survey lines. Villages and towns develop at roadway intersections.

The smallest centers provide goods and services from suppliers who enjoy relatively high per-capita demand but for whom large-scale production offers little cost advantage: convenience stores, personal services, branch banks, gas stations, elementary schools, agricultural supply stores, and so on. Larger centers contain these same types of sellers plus businesses that can exploit scale economies, such as supermarkets and high schools, and sellers who experience lower per-capita demand, such as ethnic restaurants and jewelry stores. The pattern fits the central place model because the conditions in Iowa match the assumptions.

From Rural To Urban: Changes Over Time in Regional Development

Conditions that vary from the assumptions distort the tidy pattern of development predicted by the model, yet the forces at work in the model still operate. For example, the central-place hierarchy established in medieval times was distorted over time as the changes in technology that occurred over the course of the Industrial Revolution influenced the location decisions of large-scale businesses.

Natural harbors attracted urban development as improvements in the technology of shipping freight by sea gave port cities good access to distant markets. During the Renaissance in Europe, goods could be shipped most cost-effectively by firms that invested in specialized ships, equipment to load them, and people trained to sail them. Economies of scale in shipping encouraged the growth of port cities.

As shipping companies attracted workers to ports, other business followed—businesses that served shipping companies (ship builders, financiers, accountants) and retail firms that sold goods to workers. The concentration of people in the port cities attracted retailers and service providers away from agricultural centers.

Later improvements in production technology caused the growth of urban areas around manufacturing firms, some of which found port locations desirable, others not. Relatively large-scale manufacturers produce for markets often larger than the region in which they locate. They find themselves in a tug-of-war between locations with good access to markets (such as ports) and locations near sources of energy, raw materials, and labor. This tug-of-war was often won in the 1800s by areas proximate to difficult-to-transport energy and raw materials. These manufacturing centers attracted their share of retailers and service providers.

Continuing improvements in both production and transportation technologies since the Industrial Revolution have profoundly affected the regional distribution of centers. The trend has been toward a decrease in the number of small centers and an increase in the size of larger centers. There are fewer towns and villages, and the larger centers are much larger.

To see why, consider the effect on agricultural service centers of improvements in the technologies of farming and transportation. Improvements in farm technology have dramatically reduced population densities in agricultural areas. Per-capita demand for various products has, however, increased dramatically as incomes have increased. The net effect on retail centers is hard to predict, though the population effect probably dominates.



Stephen Phraner, istockphoto.com

Natural harbors attracted urban development as improvements in the technology of shipping freight by sea gave port cities good access to distant markets. Here is the harbor in Boston.

Predicting the effect of reductions in transportation costs is easier. A householder who can travel comfortably, safely, and cost effectively at 60 miles per hour is in a much better position to take advantage of internal and external scale economies in retail than is a householder who has to travel at low speeds in the heat or cold on rough roads. Another innovation, the household refrigerator, reduced the need for frequent trips to the grocery store. It is no longer a big deal for a farm householder to travel 60 miles each way once a week for a major shopping expedition. Many of the small towns that once provided retail goods and a variety of services to agricultural areas have shrunk or disappeared into the rural landscape, while larger centers have grown and now provide a larger variety of goods and services.

Next, consider the effects of improvement in transportation technology and infrastructure on the size and locational preferences of manufacturing firms. Lower transportation costs give manufacturing firms access to larger market areas. Given access to more customers, they can increase the scale of the factory, often reducing the cost of producing each unit. Households in developed countries, for example, can afford household appliances (e.g., refrigerators) in part because low transportation costs allow manufacturers to spread the cost of sophisticated production machinery over large volumes of output. Improvements in shipping freight, first by ocean and river, then by canal, and eventually by train, truck, and plane, allow a smaller number of manufacturers each to more fully exploit scale economies. A glance at a map of the U.S. suggests the locational advantages for manufacturers that helped cities like New York, Boston, Chicago, St. Louis, San Francisco, and Seattle to grow.

URBAN DEVELOPMENT PRIOR TO CARS AND TRUCKS: CITIES WITH A SINGLE EMPLOYMENT CENTER

The patterns of land development in contemporary urban areas are complex. In particular, employment tends to be distributed across the metropolitan area, much of it in centers, but much also distributed at low intensities along arterial streets and highways. As recently as the 1920s, or even the 1950s, the pattern of development was simpler because most businesses located near the center of the city. Office and retail land users were in the central business district (CBD), and most manufacturing was outside but still near the CBD. Businesses in some small towns still cluster in or near the CBD, though that pattern is far from ubiquitous.

In this section we describe the forces that determine land use in a single-centered (monocentric) city. We use many of the key ideas later as a basis for explaining the more dispersed development in contemporary cities.

The Causes of Monocentric Urban Development

The economic model we describe in this section, which is known as the monocentric-city model, speaks to a key common characteristic of cities before the development of high-capacity intercity highways: most cities of size had as their prime locational advantage some form of a *central import/export node*, typically a port or railroad station.

Cities grew as advances in technology reduced the cost of mechanical substitutes for farm labor and increased the cost advantages of large-scale factory production. They grew near ports, canals, and railroad stations as manufacturers sought locations with good access to markets and to sources of raw materials. Locations near the import/export node became intensively developed employment centers.

Prior to the development of railroads in the mid-1800s, nature played the dominant role in the location of the import/export node. Transportation by water was far less expensive than land transport (and still is, by weight), so



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As recently as the 1920s and even into the 1950s, office and retail land users were still concentrated in CBDs. Here is Chicago's CBD skyline, looking in across Lake Michigan.

many manufacturers were attracted to areas near good ports. Though the quality of a port can be improved (such as by dredging, building breakwaters, etc.), making more than marginal improvements was expensive, so the location of port cities was usually determined by the natural characteristics of potential ports. Port cities developed on natural harbors along sea coasts, the Great Lakes, and navigable rivers.

Manufacturers also sought out locations on rivers that provided energy as well as relatively low-cost transportation. The first rapids upstream from a river's outlet into the ocean or into a larger river was attractive because a mill race could be built around the rapids to power machinery, and raw materials and finished goods could be transported on the river. Again, nature determined the location of the rapids. For the most part, people had little control over the places most suitable for urban industrial development.

That changed over the course of the nineteenth century. Early in the century, one type of man-made transportation infrastructure affected the locations of urban development: the canal. Even so, canals were expensive to build, could not be built just anywhere, and generally had to follow natural contours.

Later in the century, the developers of railroads faced fewer constraints. Railroads were cheaper to build and could negotiate irregular terrain more easily than could canals. Railroads expanded the locational possibilities for import/export nodes. But existing cities, and ports more generally, were still attractive targets for railroad developers. Just as in eastern parts of the U.S. the location of a canal could determine which harbor would ultimately develop into a large port city, later and in the West the lucky harbor would be that to which the railroad was extended.

People still had relatively little control, however, over the location of employment within the city. The reason was the high cost of shipping freight from the central import/export node (the wharf or the railroad depot) to manufacturing plants, and the high cost of face-to-face meetings between the suppliers of business services and their clients. Prior to the development of the truck, freight was shipped within the city by horse and wagon, so manufacturers had little choice but to seek locations near the import/export node. The suppliers of business services had little choice but to locate near their clients. Cities were, therefore, monocentric.

Most of these monocentric cities shared two characteristics:

1. *Land uses tended to be segregated.* Services and retail concentrated in the central parts of the CBD. Workers lived mostly beyond the CBD or in the upper stories of CBD buildings. Manufacturing firms mostly located outside of the CBD and often between the residential areas and CBD.
2. *Land values and the intensity of urban development fell with distance from the center of the CBD.* Buildings tended to be taller and closer together toward the city center, and employment and residential densities correspondingly rose with proximity to the center. Land price and land density spiked at the center and dropped with distance from the center.

The Economics of Land Development in Monocentric Cities

This section explains the key characteristics of land development using an economic model of a monocentric city formally articulated by William Alonso, Richard Muth, and Edwin Mills in the 1960s.⁴

Modeling the monocentric city starts with a long list of simplifying assumptions. While these assumptions appear to render the model unrealistically simple, they serve the purpose of modeling well: they allow us to focus on a few key drivers of urban land-development patterns. The modeling follows the common practice of attempting to isolate the most important



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Railroads were cheaper to build and could more easily negotiate terrain than canals. Consequently, they expanded the locational possibilities for import/export nodes.

forces at work in all monocentric cities and then exploring their implications with the aid of mathematics. We can explore the implications of relaxing the assumptions once we have developed the basic model.

Here are the assumptions:

1. The city develops on a flat, featureless plain; no location has better natural amenities (such as a view) than any other. This allows a sharp focus on the effects on land development of the characteristics of the transportation system.
2. All travel occurs radially toward or away from a point at the center of the city: all freight is shipped to and from the single import/export node, and all workers commute on a streetcar system with its terminus at the import/export node. And there are no natural impediments to development. These assumptions provide the symmetry needed to analyze a three-dimensional reality in only two dimensions.
3. The per-mile cost of traveling from anywhere in the city to the center of the city does not vary with distance. This highly unrealistic assumption allows us to focus on the effect of differences in the relative costs of travel across land users.
4. Infrastructure and buildings are malleable: buildings can be torn down and replaced without significant cost, so existing structures have no effect on new development. This, again, is very unrealistic, especially over relatively short time periods. But it allows us to focus on the economic forces at work in the urban economy.
5. No land or transportation user imposes “spillover” costs or benefits on other land or transportation users. Here we are assuming away all of the traffic noise, industrial pollutants, and congestion that plague urban areas. Don’t worry, much of the remainder of this book deals with these issues.
6. There are three principal types of land users—households, manufacturing firms, and providers of business services (office firms)—distinguishable by two key characteristics: (1) the per-mile cost of transportation to and from the central import/export node, and (2) the ease with which each land user can economize on high-priced land.⁵
7. Retailers are a fourth type of land user. Retail customers travel to a retailer’s location, so visibility and accessibility are important to retailers.
8. Freight arrives by ship or train at the central import/export node. From there the freight travels by horse and wagon, which is expensive, to the factory. Manufacturers dislike multistory plants because of the high cost of moving material vertically.
9. Workers from each household commute on the streetcar to and from their place of employment.⁶ Their per-mile cost of travel is significantly lower than the manufacturers’ costs, and their willingness to economize on high-priced floor space is not much greater.
10. Providers of services have to interact face-to-face with their clients, requiring relatively frequent travel by professionals whose time is valuable. They do not mind riding the elevators in tall buildings and can economize on floor space, but it is very costly for them to spend time traveling to speak with clients.

There are four distinct implications for land use, land prices, and the intensity of land use under these restrictive but relevant conditions.

1. The land users segregate themselves into concentric rings around the central import/export node. Office firms locate in the innermost ring, nearest the import/export node, even though they are not importers or exporters of freight. Manufacturing

The economics of land development in monocentric cities suggests the following broad pattern of location decisions:

- 1. The land users segregate themselves into concentric rings around the central import/export node.*
- 2. Land price falls with distance from the city center.*
- 3. The intensity of land use falls with density from the city center.*
- 4. Retailers join office firms in the center of the CBD. Industry locates around them, and the bulk of residences even farther.*

firms also prefer locations close to the import/export node to reduce the cost of transporting freight, but office firms can outbid them for that land for two reasons:

- The cost per mile of transporting the highly paid professionals that provide services to manufacturers is high relative to the cost per mile of moving freight.
- Office firms can use land intensively relative to manufacturers; they can operate cost-effectively in tall buildings.

The owners of office firms, through their willingness to pay for floor space in tall buildings, allow the developers of these tall buildings to outbid the developers of manufacturing space for land close to the central import/export node. The greater concentration of office firms allows them as a group to outbid both manufacturers and households for space in the most central locations.

Manufacturers locate in the second ring. And households locate farther away in the outermost ring because they have the lowest per-mile transportation costs; they need only transport themselves to and from their workplace twice each day.

2. Land price falls with distance from the city center.

To illustrate the determinants of land price, consider the development of residential land. To isolate land price, suppose all households are essentially the same: all have the same number of members with similar tastes and incomes. To be satisfied with their living arrangements, similar households have to get similar amounts of satisfaction from their lifestyles regardless of where they live. Some locations have more desirable characteristics than others.

In the monocentric city, locations closer to the city center have the advantage that commuting costs less. People who live farther from the city center have to get something to compensate for their longer commutes. In the simple monocentric-city model, they get lower-priced land. Land price rises with better accessibility to the center. The same logic applies to office firms and to manufacturers. Greater distance from the central import/export node (which is also the location of the hub of the radial streetcar system) means higher transport costs, which capitalize into the price of land.

Office firms capture the most central locations because they can outbid the other land users. Manufacturers locate in the next ring because they can outbid households.

3. The intensity of land use falls with distance from the city center.

Again, consider the development of residential land. Land prices affect the density of development in residential areas because both builders and households respond to relatively high land and floor-space prices.

The builder of housing can use varying amounts of land to produce housing that provides the same level of satisfaction to a given household. As land becomes more expensive, the builder uses less of it by, for example, building two-story houses and placing houses closer together (i.e., building on smaller lots).

By themselves, these adjustments decrease the attractiveness of the house to consumers. Builders compensate by improving other features of the house, such as by installing better quality appliances, carpeting, or trim. These features add to the builder's costs. But the additional revenues from the higher density (i.e., the increase in the number of houses per unit land area) compensate for these higher per-house costs. Otherwise, similar households can get similar amounts of satisfaction from their house even though the density of development is higher.

Homeowners and renters also make tradeoffs that work to increase the density of development. Though builders can economize on land, higher land prices still translate into higher floor-space prices. As the price of floor space increases, households trade-off the quantity of floor space for more of the other goods and services they desire. Though these householders live in smaller houses than similar households who live in less accessible areas, they are content with the smaller house because of the superior accessibility of the location.

Again, the same logic applies to office firms and manufacturers.

To summarize, one might think that high-intensity development causes high land prices. In fact, causality generally works in the other direction: *high-priced land causes high-intensity development*. The price of the land is high because it has good access, and people compete for that access through their bids. But because they have to pay more for land, each tries to economize on it, which results in higher-intensity development.

4. Retailers join office firms in the center of the CBD.

The CBD has value because it provides retailers access to the maximum number of businesses and households (potential customers), given the strictly radial transportation system. Retailers can outbid manufacturers for central locations. Indeed, the benefits of easy access and visibility allow retailers to outbid office users for street-level space. Retailers in early twentieth-century cities tended to occupy the street-level space, and office dwellers behind and above them.

Relaxing Some of Our Assumptions

This description of the land-use pattern conforms only roughly to the pattern found in most cities before 1930. We can add more realism by relaxing assumptions.

We start by relaxing assumptions about the characteristics of the transportation system. Suppose that the transportation system remains radially oriented, but that most travel occurs on a limited number of main streetcar lines. If the streetcar follows an arterial street, land on the street between stations offers better access to the city center than locations the same distance from the city center but off the arterial. Some businesses would find locations along the arterial and near streetcar stops attractive, outbidding households. Urban uses develop along the arterials farther in miles from the city center, looking somewhat like a starfish with the streetcar lines operating in the arms of the starfish.⁷

Next suppose that freight could be shipped not just from a point (like a single railway station) but instead from anywhere along a line (e.g., from anywhere on the shore of a bay). Manufacturers would each seek a location on the waterfront, and office firms would seek locations that provide generally good access to their clients. In this case, office firms probably would not outbid manufacturing firms for all of the land next to the bay. Office firms instead would exploit economies of agglomeration by clustering around the point with the best access to other areas along the shore. Manufacturing firms could take advantage of the large supply of land with direct access to shipping. Workers would seek locations that provide good access both to their employer and to the central shopping area.

Now relax the assumption that all households are relatively unwilling to economize on high-priced land. Households generally are willing to reduce the costs associated with high-priced floor-space in accessible areas by making do with less housing (smaller lots and dwelling units). Some households are willing to do with no private outside space (i.e., a yard). These households can occupy a relatively small area of land if they live in tall buildings. As



Threejays, istockphoto.com

Retailers in early twentieth-century cities tended to occupy the street-level space, with office dwellers behind and above them (as well as residents).

long as the costs of building taller do not eat up all the value of proximity, the apartment developer can bid more than manufacturers for accessible land. In principle (and in practice), this intensity effect can outweigh the transportation-cost effect, and householders could occupy the middle ring. We see this effect in large, space-constrained cities.

Next suppose that land characteristics and natural amenities vary across the residential areas of the city. Households, for example, care about the view *and* accessibility. So they bid up house and land prices in areas with a good view. All else the same, households economize on high-priced floor space and developers economize on high-priced land, so the intensity of development increases.

Readers might object that not all high-priced residential land is intensely developed. Most cities contain residential areas of large single-family homes on relatively large lots developed on expensive land. Though it may seem otherwise, that land is intensely developed in relative terms. The people who live in these extravagant neighborhoods have high incomes (remember, we assumed in the simple model that everyone has the same income). People with similar incomes who live in areas with poorer CBD access or amenities typically have even larger estates. Though the lots in the accessible wealthy subdivision seem large to the average person, those lots seem small to their high-income consumers.

We can relax more of the assumptions, but you get the idea. The simple monocentric city model reveals the key economic forces at work in land development. The characteristics of the transportation system determine patterns of accessibility. Businesses and households compete for accessible locations, bidding up the land prices. Land users economize on high-priced land, increasing development densities. Competition for land and price adjustment continues until similar households and businesses are satisfied with their locations. Economic considerations of businesses and households shape the city and the region.

The standard for achieving efficiency is surprisingly simple. Resources are allocated efficiently to production of a particular good until the value of an additional unit of resource is equal to that resource unit's value in the next best productive activity.

Are the Characteristics of Monocentric Development “Efficient”?

“Efficiency” has a special meaning in economics. The key objective of an economic system is to make the best use of the productive resources (raw materials, plant and equipment, and labor) available; or, to put this objective more in the language of economists, to maximize the net social benefits from production. Meeting this objective is complex in practice: allocating hundreds of types of resources to production of thousands of goods and services, then allocating these end products across millions of consumers is a big, complex job.

Despite this complexity, the standard for achieving efficiency is surprisingly simple. Resources are allocated efficiently to production of a particular good until the value of an additional unit of resource is equal to that resource unit's value in the next best productive activity. For example, allocate more land to manufacturing as long as the value manufacturers capture from use of the additional (i.e., marginal) land exceeds the value other land users (e.g., service providers, retailers, or households) would get from use of the land.

Similarly, a particular good is produced in an efficient quantity if output increases until its value to consumers—net of production, distribution, and marketing costs—is equal to the net value consumers place on the marginal unit of the other goods and services produced. Production and consumption are efficient if, on the margin, benefit equals cost across all of the numerous goods and services produced.

Is this standard met by the allocation of land and other resources under the conditions in the monocentric-city model? Yes. Land goes to the land

user willing to bid the most. That user's willingness to bid comes from the value of land in the production of the goods and services that the land user sells or consumes.

Central land is most valuable to service providers, even though each of these service providers uses very little of it. Land near the center is allocated to service providers until manufacturers are willing and able to outbid them. Land is allocated to manufacturers until residential land users are able to outbid them. And urban development stops where rural users are able to outbid urban households.

Continuity in the land-value surface or distance gradient is a key characteristic of this efficient market equilibrium development pattern. Land values fall with distance from the city center. Where land use transitions from services to manufacturing, the land value must be similar for both types of land user. A jump in land values (i.e., a discontinuity in the land-value gradient) will quickly be bid away. Similarly, the value of land is roughly the same for manufacturers and for residential users at the land-use boundary. This important aspect of efficiency is also of practical importance when we look below at the effects of land-use regulation.

Of interest are the incentives that land prices in the monocentric city generate. Land prices rise, sometimes to spectacular levels, with proximity to the central import-export node. Spectacularly high prices speak to the scarcity of land with good access to nodes with desirable characteristics. Not much land is close to very good ports because nature failed to supply very many harbors with highly desirable characteristics.

While land owners like high land prices, the high prices encourage land users to look for alternatives. High land prices at good ports may make costly improvements to naturally poorer harbors worthwhile, increasing the supply of good ports. High land prices also encourage land users to economize on land by building higher, increasing floor space without using more land. The value in improving harbors and building higher provides incentive to develop skills and machinery for making those improvements more cost effectively. High prices that result from scarcity drive innovation that reduces that scarcity.

Characteristics of Efficient Transportation in a Monocentric City

To this point, our discussion of urban development has focused on land use. Nonetheless, we had to make some important assumptions about the transportation system that allowed the region and the monocentric city to develop.

We made these assumptions impossibly simple to allow us to focus on explaining land-use patterns in a monocentric city. We (implicitly) assumed that (1) roadways occupied no space; (2) in essence, everyone could travel unimpeded to the city center (but only at speeds similar to those actually found in turn-of-the-century cities); (3) no one could travel in a circumferential direction; and (4) congestion did not exist. These unrealistic assumptions, which yield *land-use patterns* with important similarities to the patterns found in real cities, provide little insight into the *travel patterns* in real cities.

To gain that insight, one must ease the restrictive assumptions of the monocentric city model only a little. Suppose that the basic assumptions still hold: four types of land users, the city develops on a flat plain, radial transportation, etc. Now suppose, however, that transportation requires space: travelers must compete with land users for land on which to build streetcar lines and roadways.

The fundamental factors that influence the characteristics of an efficient urban transportation system are: (1) the value travelers place on the service the road or rail provides (i.e., the value of being able to get from an origin

The fundamental factors that influence the characteristics of an efficient urban transportation system are:

(1) the value travelers place on the service the road or rail provides (i.e., the value of being able to get from an origin to a destination), (2) the value of the land that the road or rail occupies, and (3) the technologies that exist to transport people and freight.

to a destination), (2) the value of the land that the road or rail occupies, and (3) the technologies that exist to transport people and freight.⁸

The price of the land used for transportation is determined by the competing demands of other valued uses—factories, office buildings, stores, and houses. In the monocentric city, the price of land increases as one approaches the center of the city. Because roads occupy land, the cost of building a road with given characteristics increases as the price of land increases, just as the cost of building a house with given characteristics increases as the price of land increases.

Efficiency requires that the builders and users of the transportation system respond to higher land prices in ways analogous to the response of the builders and users of housing or office buildings. The design of the efficient transportation system takes advantage of ways to use less land as the price of land increases. Mass transit, such as streetcars, allows many more “person trips” using the same amount of space. Less frequent entrances to a highway allow more car trips on the same amount of asphalt by speeding up traffic.

Transport developers, like land developers, can economize on high-priced land by building vertically (e.g., double-decker highways).⁹ Given the characteristics of the infrastructure, travelers economize on high-priced space on roadways and transit by using less of that space. For example, people stand in crowded streetcars, commuters carpool, and drivers slow down to increase the density of vehicles: *the system gets congested*.

We cannot predict a priori how the proportion of land used for travel changes with distance from the city center. The scarcity of land in the CBD causes high land prices, but the activities in the CBD make travel there valuable. On the one hand, the high price of land encourages a supplier of transportation infrastructure to economize on land by devoting less of it to roadways. On the other hand, the increasing volume of traffic encourages a supplier to increase the capacity of the infrastructure, which generally requires the use of more land. The proportion of land devoted to travel depends on the relative sizes of these effects.

The characteristics of an efficient transportation system vary across cities; these characteristics depend on the size of the city, the goods and services produced in the city, and the technologies available with which to move people and goods within the city. But, as we demonstrate later, one characteristic remains common: *an efficient transportation system experiences congestion during peak periods*.

Efficiency requires that the builders and users of the transportation system respond to higher land prices in ways analogous to the response of the builders and users of housing or office buildings.

The Evolution of Land Use and Transportation in Monocentric Cities

We can demonstrate the power of the monocentric-city model by using it to explain the relationships between the innovations in transportation and the changes in urban development that occurred between about 1850 and 1950. That century witnessed phenomenal growth in the population and land area of cities. Despite that growth, most cities remained dominated by a single center of employment and looked like our monocentric-city model.

But virtually everything else about cities *did* change between 1850 and 1930. The population of many cities grew dramatically. For example, the population of New York increased nearly tenfold (from 696,000 in 1850 to 6,930,000 in 1930), and that of Philadelphia increased more than tenfold (from 121,000 in 1850 to 1,950,000 in 1930).

As urban populations grew, so did the land area occupied by the city. For example, the radius of Boston grew from about two miles in 1850 to about six miles in the 1890s. Most other major cities experienced similar rates of growth.

Though the population of the country as a whole grew rapidly over the same time period, cities grew faster. In the 200 years from 1800 to 2000, the

share of the U.S. population living in cities grew from 6 percent to almost 80 percent (O'Sullivan 2007). From 1900 to 2000, the share doubled, from about 40 percent to 80 percent.

Changes in transportation technology are key to understanding these changes in urban size.¹⁰ Nineteenth-century innovations in transportation decreased the cost of commuting to work and of moving both people and freight between cities. Prior to 1850, most urban dwellers commuted to work by foot. In the 1850s, rails were laid in the streets of most large urban areas to accommodate streetcars powered by a horse. These small streetcars roughly doubled travel speeds and relieved commuters of walking in the muddy streets.¹¹ Larger and faster electric streetcars began to replace horse cars in the 1890s. Cable car lines were built to climb hills too steep for electric or horse-drawn cars. The streetcar and cable-car lines emanated radially from the CBD because manufacturing and office firms remained tied to the import/export center by the high transportation costs they faced.

These improvements combined to increase the land area of many cities. Streetcars significantly decreased the cost of commuting. The monocentric-city model predicts that, even if the population of a city remained unchanged, the lower commuting costs would increase the radius of the city. Households would take advantage of the streetcar to access lower-priced land at the urban fringe, causing the land area of the city to grow. They would develop that lower-priced land less intensively, generally decreasing residential densities. In other words, suburbanization would occur even without growth in population.

But streetcars encouraged growth and stimulated even greater suburbanization. The cost advantages of business agglomeration encouraged urban employment growth. Urban economic growth was constrained, however, by the number of workers that could reasonably commute to the CBD from residential areas. The streetcar made it possible for far more workers to travel to the CBD at a reasonable cost. The development of the streetcar thereby encouraged employers in less well-located areas to move to take advantage of the economies available in cities with especially good access to raw materials, manufactured inputs, a variety of services, and a large workforce.

The population of many cities grew also because of improvements in the technology of transporting freight between cities. Some cities grew in part because of scale economies in transportation and the growth of trade between cities and regions; big transportation firms located in port cities.

Lower transportation costs that allowed manufacturing firms to more fully exploit scale economies also spurred growth. Given access to a large market area and extensive scale economies, a single large factory could underprice smaller factories spread throughout the market area. Cities with especially good access grew, often at the expense of smaller cities with poorer access to large markets.¹²

Empirical research on the effects of the streetcar on land development supports the theories we have been discussing, though it is more descriptive than statistical. (See the sidebar for a summary of a few of the more rigorous studies.)

Researchers have used other data gleaned from historical records to test statistically the monocentric-city model's predictions about the effects of changes in transportation costs on patterns of urban development. They have focused on changes over time in land-price and population-density gradients (i.e., on changes in the rate at which land price and population density/land-use intensity fall with distance from the city center).

In a seminal paper, Mills (1972) reports the rate of change in population density as a function of distance from the CBD in four metropolitan areas (Baltimore, Milwaukee, Philadelphia, and Rochester) at 11 points in time

HOW THE STREETCAR AFFECTED LAND DEVELOPMENT

Foster (1981) reports that civic leaders in most large urban areas in the 1890s eagerly supported private construction of electric streetcar lines. They saw the streetcar as an effective way to provide access to undeveloped land near the urban fringe, which would relieve the crowding in central areas of the city and promote economic development. Land developers typically built the streetcar lines to provide access to residential land at the urban fringe. Cities became less circular as streetcar lines extended unevenly into the urban fringe. Warner (1978) and Wachs (1984) document the development of suburbs that developed near streetcar lines built near Boston and Los Angeles, respectively.

The streetcar failed to reduce crowding and congestion in central areas: the radially oriented streetcar system made locations in the CBD even more attractive to business. As the costs of commuting and shipping freight between regions fell, the CBDs grew. Foster (1981) reports that in the early 1900s, traffic congestion (streetcar, horse and wagon, and pedestrian) was a major problem in the central areas of large cities in large part because more workers could commute to the CBD. ■

between 1880 and 1963. In 1880, an average of 88 percent of the population in those cities lived within three miles of the city center, and the population density fell by about two-thirds with each additional mile from the city center.

Those numbers fell over time. By 1930, only 56 percent of the population lived within three miles of the city center (the land area of cities had increased), and the population density fell more slowly (though still quickly), falling by about half with each additional mile from the city center (the density gradient had become flatter). These observations support the hypothesis that land prices (and, therefore, population densities) respond to drops in per-mile transportation costs.

Mills (1972) and Kau and Sirmans (1979) looked directly at changes in land rents in Chicago at six points in time from 1836 to 1928 using data published by Homer Hoyt in 1933. Researchers have studied Chicago intensively both because Chicago has developed on a flat plain (relatively few topographical features exist to complicate the analysis), and because data exist that show land development and land values at different times in Chicago's history. Hoyt, for example, sifted through historical records to produce maps of Chicago that show land values at six dates between 1836 and 1928.

During that period, Chicago grew dramatically both in population and in land area. Statistical analysis reveals the expected decline over time in the rate at which residential land values and population densities fall with distance from the center of the city.

This pattern is both worldwide and continuing. Anas, Arnott, and Small (1998) report similarly dramatic decreases in the population density gradients in London and Paris since the early 1800s. Almost 90 percent of London's population in 1801 lived within three miles of the city center. Jordan, Ross, and Usowski (1998) looked at more recent trends in density gradients, from 1970 to 1990, in 79 U.S. metro areas. In all but one (San Francisco) density gradients continued to fall, most likely in response to on-going improvements to urban highway systems.

The basic implications of the monocentric model still apply. We show in the next section, however, that land-use patterns in modern metropolitan areas have become much more complex because employers are no longer tied to a single central import/export node.

The development of the truck and the passenger car in the early twentieth century created the potential for continued growth and for a dramatic change in urban land-use and travel patterns because it decreased the per-mile costs of moving both freight and commuters.

URBAN DEVELOPMENT WITH CARS AND TRUCKS: CITIES WITH MULTIPLE EMPLOYMENT CENTERS

The development of the truck and the passenger car in the early twentieth century created the potential for continued growth and for a dramatic change in urban land-use and travel patterns because it decreased the per-mile costs of moving both freight and commuters. Streetcars remained important to commuters through the 1920s, but the drop in the price of a car due to mass production allowed increasing numbers of households to purchase one. The introduction of the car continued the trend toward lower commuting costs per mile and encouraged residential development on land between the spokes of the streetcar system (Muller 2004).¹³

Trucks and highways similarly continued the decrease in the cost of shipping raw materials to the city and the city's product to markets: well-located cities continued to get bigger.

Black and Henderson (2003) report that, within the set of cities in which about 90 percent of the urban U.S. population resided, the mean city size increased about fourfold from 1900 to 1990 (from about 130,000 to about 530,000 people).¹⁴

Cars, trucks, and highways provided new flexibility for business location decisions: there no longer needed to be a single import/export node.

Manufacturing could move out of central areas, eliminating the natural fit of a radially oriented transport system.

Achieving the liberating potential of cars and trucks, however, required the construction and expansion of roadway systems. Local governments worked to improve urban roadway systems as cars and trucks began competing for space in the 1910s and 1920s. But the economic depression of the 1930s and the conversion to a war economy in the 1940s inhibited large-scale improvement of roadways and construction of intercity highways.

It was not until the second half of the twentieth century that governments at all levels were able to devote significant proportions of their budgets to road building. Indeed, concerns about defense provided some of the motivation for the construction of a key piece of transportation infrastructure: the interstate highway system. The patterns of urban development have changed irreversibly as a consequence. In particular, the relative importance of the traditional CBD has diminished dramatically.

Major cities in North America are no long monocentric. Commercial and industrial land uses are now scattered around most metropolitan areas. The monocentric-city model does not apply as well, or at least as directly. But is there still something systematic going on? Or is the pattern of development not much of a pattern at all?

We argue in the rest of this section that patterns still exist, that they depend on transportation costs and the characteristics of the transportation system, and that key aspects of the monocentric-city model still have relevance. Businesses get benefits from clustering. Business clusters develop at relatively accessible places in the larger urbanized area, such as near the intersections of major highways. And these clusters have some of the characteristics of monocentric cities; our simple model still helps us understand land prices, land allocation across uses, and the intensity of development in and around these commercial subcenters. We also call on central-place theory to shed some light on the size and distribution of metropolitan employment clusters.

The interstate highway system changed the patterns of urban development irreversibly. In particular, the relative economic importance of the traditional CBD diminished.

The Causes of Polycentric Urban Development

Consider a simple model that includes the same activities as in the monocentric-city model (manufacturers, service providers, retailers, and households), but that has a transportation system with a key similarity to the one we specified to explain the locations of medieval villages. In the medieval system, people could walk everywhere; transportation was expensive per mile, but each location was equally well-served by infrastructure (e.g., walking paths). Suppose that the urban transportation system is similarly ubiquitous, consisting of a grid of arterial streets or highways that extends throughout the flat plain that the population inhabits. Cars and trucks can travel at relatively high speeds on these arterial streets.

As usual, we have to make some assumptions to make some points:

- Every point on the grid has equal access to inputs, to manufacturing processes, and to markets in other regions.
- Every commuter has access to a car.
- Every manufacturer ships freight by truck.
- Manufacturers and other businesses can locate anywhere along any of the arterial streets.

In this world, if there were no benefits to being closer to other firms and cost advantages exist to using land more extensively, manufacturers respond to land prices by spacing themselves as widely as possible. In other words, no other reasons (such as proximity to a port) exist to cause competition for land, and businesses reduce cost by occupying lower-priced land.

But the cost advantages of agglomeration *do* exist. Manufacturers are willing to trade low land prices for proximity to other manufacturers and to the businesses that provide inputs to the manufacturing firms. Retailers cluster to take advantage of the traffic that each of them generates. Clustering reduces the *non-land* cost of doing business.

Agglomerative economies provide a sort of gravitational force attracting businesses toward each other.¹⁵ This force is weaker than the need to ship freight to and from a central import/export node, but by itself it still suggests that businesses would be attracted to a single central area. *Why, then, are there multiple centers?*

The answer is that the costs of transportation (and, therefore, of compensating wages) and floor space rise with the size of the employment center. Land prices rise as more firms compete for the most accessible locations, encouraging more intensive development, which costs more per unit of floor space. Average transportation cost rises for two reasons:

1. As the number of employees in the cluster rises, the length of the average commute increases; the employers in the cluster have to attract workers who live farther from the cluster.
2. Transportation infrastructure competes with businesses for scarce land in the cluster, further bidding up land prices.

Alternative locations start to look attractive as transportation and land costs rise in a large central cluster. Given other accessible locations on the transportation grid, subcenters form to effectively trade-off the cost advantages of agglomeration for the land and transport cost advantages of locations away from the central cluster.¹⁶

To summarize, the causes of polycentric urban development are:

- Cars, trucks, and highways break the tie to the central import/export node. Freight can be delivered to anywhere a highway (and other supporting infrastructure) can be built.
- But transportation still involves time and out-of-pocket costs.
- Firms reduce these costs by clustering.
- Other costs rise as a cluster grows.
- Another separate cluster forms when these other costs get high enough to overcome the transportation-cost advantages of the existing cluster.
- The result: development of multiple employment subcenters.

Is There an “Efficient” Pattern of Polycentric Land Development?

We argued that, given the constraints in production and transportation technologies, the characteristics of transportation and land development in the monocentric-city model—radial transportation, segregation of land uses, and land prices and development densities that rise with proximity to the center—are economically efficient.

Actual monocentric cities varied in their details but tended to have the general characteristics predicted in the model. So, a case can be made that land development patterns in turn-of-the-twentieth-century cities, warts and all, represented reasonably efficient uses of the resources and technologies of the time (we give the counterargument, based on external costs, later).

Can one identify a set of similarly efficient characteristics in the case of polycentric cities? The answer is a limited yes. Subcenters have many of the characteristics of the CBD in a monocentric city, so the characteristics of a monocentric city generally carry over to subcenters. Though we can call on central-place theory to help evaluate the relative locations and sizes of centers, we cannot say as much about the general configuration of the

Clustering reduces the non-land cost of doing business.

transportation system as we could in the monocentric city. The good news is more choice about policy; the bad news is the same.

We begin our evaluation with the claim that urban economies benefit from decentralized development. Some sprawl is good. Large turn-of-the-century cities were dense, dirty, congested, and high-priced in part because of the high per-mile cost of shipping freight within the city and the need to ship that freight to the central import/export node. The truck and the highway opened up low-priced suburban land for manufacturing, and manufacturers could save significantly by spreading out on cheaper land. Competitive markets encourage these cost savings to be passed on to consumers.

As their employers moved, workers could similarly spread out without significantly increasing the time cost of their commutes. Residential developers responded to the demand for suburban-style living. Retailers and service providers followed their customers to suburban areas. Breaking the tie to the central import/export node has greatly reduced the costs of doing business in urban areas.

Yet suburbanization also has its costs in practice. In most cities, highway development reduces the demand for locations in the traditional CBD. A key assumption in our simple model is that buildings and infrastructure are “malleable,” so the CBD can adapt to the fall in its relative status. Well, they are malleable given enough time and money. In most cities, however, redevelopment of the traditional CBD has been neither cheap nor easy.

Suburbanization also has costs at the urban fringe. As we discuss later, too much decentralization may have occurred due to a variety of subsidies to suburban land and transportation development. Suburbanization has in practice probably pushed agricultural and natural land uses farther from the city center than is optimal. Decentralization can be efficient, but it has its costs, and may well have gone too far.¹⁷

Central-place theory provides insight into the efficient pattern, especially of service and retail subcenters. Because some businesses value a location central to the entire region (e.g., because they sell to a regional market or need access to specialized suppliers or facilities), the largest cluster would, all else being equal (e.g., environmental and public services quality), form at the center of the urban area.

Many office and retail firms follow their customers as both manufacturing firms and households move into suburban areas. Smaller subcenters, containing subsets of the activities found in the CBD, form at accessible points in suburban areas (e.g., at intersections of major arterials that provide access both to the CBD and to other subcenters). Those firms that can attract enough local customers to operate at an efficient scale move to suburban centers but still benefit from relatively good access to the large central cluster (e.g., to obtain specialized services). An overall centralizing tendency remains, but it is not as strong as the centralizing forces at work in the monocentric-city model.¹⁸ The system of subcenters optimizes the trade-off between agglomeration economies and travel costs.

In practice, the number, sizes, and locations of clusters depend on the characteristics of the urban area (e.g., population and type of industries), the characteristics of different sites (e.g., physical constraints and public policies and services), the characteristics of the transportation system, and, of course, on chance (e.g., the idiosyncratic preferences of business owners).¹⁹

The location decisions of manufacturers do not have to follow the central-place hierarchy. Manufacturing subcenters may develop, for example, in areas in which the industry has a history and workers therefore have specialized knowledge and skills. The gravitational attraction of this manufacturing center distorts the neat theoretical hierarchy of central places as suppliers and service providers respond to the development of this manufacturing

Decentralization can be efficient, but it has its costs, and may well have gone too far.

subcenter. Relatively large service and retail centers (edge cities) may develop in suburban areas because conditions for development are poor in the traditional CBD and the transportation system renders those suburban locations especially attractive. These development patterns might reasonably be considered economically efficient given the historically imposed constraints (of the location of specialty workers or of conditions in the traditional CBD).

Many of the key insights of the monocentric city model apply to development within and around employment subcenters. Subcenters usually consist mostly of nonresidential land users; the market tends to segregate land users. Both land prices and, consequently, land-use intensities, tend to fall with distance from subcenters. We can fruitfully view individual subcenters through the lens of the monocentric model.

Polycentric cities are connected collections of specialized centers. Metropolitan areas have expanded as transportation costs have dropped. Subcenters, which at one time might have been individual cities, allow firms to exploit agglomerative economies. The interactions of these subcenters have allowed metropolitan areas to take on the characteristics of regions. We view these generally decentralized patterns of development as reasonably efficient and therefore as significant contributors to the growth of income and quality of life in developed countries.

Characteristics of Efficient Transportation in a Polycentric City

Even in our simple model, an efficient transportation system results from complex trade-offs between land and transportation capital. In real cities, transportation planning becomes considerably more difficult: several modes of travel can occur on the same roadway, people can travel in directions other than directly to and from an activity center, and urban areas typically contain multiple activity centers.

We cannot describe in any detail the efficient transportation system in a polycentric city. But we can use our simple models to identify some of the important characteristics of that system.

Remember: *a cluster grows until growth-induced costs, especially commuting costs, overwhelm the cost savings from agglomeration.* As clusters reach their optimal size, continued growth in the metropolitan area leads to the formation of additional clusters. Because we assumed a grid system of arterial streets, clusters form at the intersections of arterial streets or highways that offer especially good access to other clusters and to residential areas. The result is a metropolitan hierarchy of urban subcenters similar to the regional hierarchy of urban centers in central-place theory.

Once started at an arterial intersection, a business cluster generates demand for greater transportation capacity into and out of the cluster. In particular, the connections between one cluster and the other clusters in the system are important. The increasing amounts of travel on the links between clusters encourage increases in capacity that lead to the formation of a hierarchy of highways, with the highways that connect clusters becoming relatively large.

That transportation hierarchy, in turn, influences the location of new firms and residences and, hence, the growth of existing and new clusters. *One can start with land use, or one can start with transportation—in either case, the basic feedbacks lead to a hierarchy of central places and transportation links connecting them.*

The growth of a cluster at a highway intersection has effects on the transportation system similar to the effects in a monocentric city. The growth in the cluster increases the demand for travel in that part of the local street grid. The value travelers place on access to the cluster encourages improvements

One can start with land use, or one can start with transportation—in either case, the basic feedbacks lead to a hierarchy of central places and transportation links connecting them.

to the local roadway system that increase capacity in and near the cluster, improvements like those we discussed in the context of the monocentric city. Those changes improve access to the cluster, which encourages the further growth of the cluster, which encourages further improvement to the transportation system. At some point, of course, the cost of adding capacity to the cluster's transportation system outweighs the advantage of that capacity to travelers—the cluster and the local transportation system have reached their efficient sizes.

The growth of a cluster also encourages improvements to roadways that provide access to the less intensively developed areas nearby. The cluster serves the households and businesses located nearby, and the cluster's employees generally locate within reasonable commuting distance of the cluster. That encourages the improvement of the streets in the regional grid that feed into the cluster's grid.

An efficient transportation system experiences congestion. The design of the transportation system economizes on high-priced land, which increases the cost of space on the highway. Travelers economize on the high-priced space by using less of it. This economizing implies congestion, even on a transportation system efficient in size and use.

The reason is simple: at some point the cost of additional capacity exceeds the value of the increased travel speeds the new capacity allows. In short, building enough capacity to eliminate congestion is inconsistent with the objective of maximizing the net benefit generated by the transportation system.

Efforts to eliminate congestion fail because rising transportation costs (especially slower travel) is a key brake on the growth of the cluster. Reducing congestion allows a well-located cluster to continue growing, which increases travel demands, eventually driving the costs of capacity expansion to uneconomic levels. Economic forces simply do not allow congestion-free travel.

More generally, when we took this tour of a hypothetical urban area, we were struck by the complexity of the processes that shape it: thousands of decision makers, feedback upon feedback. *We can describe in general the forces that shape cities, but neither we nor any planning model we can conceive of can describe with accuracy the optimal size or configuration of land use and transportation systems in a real urban area.* We will need to return to this point in subsequent chapters.

We can describe in general the forces that shape cities, but neither we nor any planning model we can conceive of can describe with accuracy the optimal size or configuration of land use and transportation systems in a real urban area.

The Reality of Land Use and Transportation in Polycentric Cities

Employment has, as theory suggests, moved into the suburbs. Over two-thirds of U.S. employment in 1948 was in central cities. That dominance fell steadily over time; by 2000, the suburbs contained the majority of the jobs. Most of metropolitan area jobs are dispersed outside of both traditional CBDs and large employment subcenters.

O'Sullivan (2007) calculates from information from Anderson and Borgia (2001) that about half the employment in several medium-size metro areas—Cleveland, Indianapolis, Portland, and St. Louis—is dispersed along highways, arterial streets, or in small subcenters. The other half is on average equally distributed between the CBD and in one or more large subcenters.²⁰ In other words, only about 25 percent of employment even in medium-size metro areas is in the traditional CBD. O'Sullivan also reports that in large cities, almost two-thirds of office employment is in the CBD (38 percent) and large subcenters (26 percent).

Population has similarly suburbanized. In 2000, the central-city share of metropolitan population was 36 percent. Census journey-to-work data show that most commuting trips (63 percent) correspondingly originate in the suburbs, over two-thirds of which also end in suburban destinations. Only 8 percent of all commutes originate in the central city and terminate in the suburbs.

O'Sullivan (2007) summarizes several key empirical findings:

- Subcenters are common in metropolitan areas.
- Subcenters vary in composition, but the sources of cost advantages of agglomeration are usually identifiable.
- Traditional CBDs continue to be important centers of employment.
- Firms in subcenters interact with the firms in the CBD and in other subcenters as central-place theory suggests.

Some representative findings about the characteristics of polycentric development include the following:

- Mieskowski and Smith (1991) look at development in Houston. Houston provides a good opportunity to study recent changes in location decisions because so much of its development is relatively new: over half of the city's commercial and residential development, and two-thirds of the city's office development, occurred since 1960. Seven radial and three circumferential freeways serve the city. Mieskowski and Smith find that distance from the CBD has little value as a predictor of residential densities in Houston and that areas with poor access to freeways tend to remain vacant longer than areas with better access. Between 1969 and 1989, about 75 percent of office space (117 million square feet) "was built in clearly defined office employment centers at or near the intersections of major freeways or thoroughfares" (p. 193).
- Shukla and Waddell (1991) study the determinants of firm location (and, therefore, of land use) in the Dallas/Fort Worth urban area. This area, like Houston, has experienced rapid growth. They find that manufacturing firms value proximity to the CBD, the airport, freeway interchanges, and other subcenters. Wholesale firms locate near both areas of manufacturing and retail sales, as one would expect given their transport orientation. Smaller retail firms (firms with fewer than 50 employees) are the most decentralized, consistent with the predictions of central-place theory. Finance, insurance, and real estate (office-oriented firms) are the most centrally oriented group.
- Giuliano and Small (1991) study the distribution of employment in Los Angeles subcenters. They find that the distribution of subcenters in the Los Angeles region fits the central-place concept surprisingly well considering Los Angeles's reputation for sprawl. They identify 32 subcenters in the Los Angeles area that vary considerably in size and composition. They are able to fit 27 of the 32 subcenters into three groups: (1) specialized manufacturing, (2) mixed industrial, and (3) mixed service. Subcenters near the center of the region tend to be bigger and more densely developed than those farther from central Los Angeles, and central subcenters contain more service-oriented firms. Downtown Los Angeles, at the center of the region, contains 10 percent of all employment in the region, and 31 percent of the employment is in subcenters.

Given these and other papers that describe subcenter development and characteristics in individual metropolitan areas, the logical next step is explaining aspects of the variation in subcenter characteristics across metro areas. McMillen and Smith (2003) study subcenters across 62 metro areas. According to their research, the number of subcenters in a metropolitan area increases with metro-area population and with levels of traffic congestion, consistent with theoretical prediction. The traffic congestion raises transportation costs that counteract the cost advantages of agglomerating in centers, and thereby encourages the formation of additional centers.

McMillen (2004) looks at the spatial variation in employment densities in the same set of metropolitan areas. He finds, consistent with predictions from the monocentric-city model, that employment densities fall with distance from both the CBD and subcenters in all of the metro areas studied. These results provide good evidence that the economic explanation of polycentric development has some merit.

Many researchers (Hamilton (1982), White (1988a and 1988b), Cropper and Gordon (1991), Richardson, Gordon, and Choi (1992)) have studied commuting patterns to investigate how strongly proximity to work influences the location decisions of households. One-way commute time still averages well under 30 minutes, though the overall average increased from 19 minutes to more than 22 minutes between 1990 and 2001.²¹ Buried in these averages are the four-hour, 100-mile commutes that make headlines, but they make up a small percentage of total trips.

These findings suggest that most households tend to locate relatively close (in terms of travel time) to their workplaces. It also supports the intuitively reasonable (if not obvious) idea that trip makers are constrained by a time budget for travel; given a 24-hour daily budget, most people don't want to spend more than about an hour commuting.

But the wide distribution of employment in suburban areas combined with the flexibility of cars, trucks, and highways allow households to consider other factors in their location decisions. Researchers have undertaken scores of empirical studies to estimate statistically how different attributes of a property (e.g., regional location, site characteristics) affect its price.²² They find that residential land values tend to fall slowly with distance from the centers of contemporary urban areas. The value of residential land depends on access not just to the city centers, but also to less intensively developed employment centers, a variety of urban amenities (e.g., recreation, shopping, and culture), and a range of characteristics of the residential neighborhood (e.g., quality of schools, air quality, age of other buildings).²³

The body of empirical work supports our argument that major changes in urban form have resulted primarily from changes in the characteristics of transportation. The car, truck, and highway have given both businesses and households greater locational flexibility. That flexibility has changed dramatically the character of urban development. The change has not come without cost: some parts of most urban areas have experienced increases in traffic congestion, declining environmental quality, and a lack of maintenance and redevelopment. In particular, most of the older central cities have suffered from the relative decline in the importance of centrality.

Policy makers hope that changes in policies can address these problems. To provide insight into the potential of policy to evoke desirable change, we discuss in Chapter 3 the effects that transportation policies in the past have had on the changing pattern of urban development.

The body of empirical work supports our argument that major changes in urban form have resulted primarily from changes in the characteristics of transportation.

The Causes and Effects of Urban Land-Development Policy

Local government regulation of urban land development is virtually ubiquitous in the U.S. and other developed countries. Local governments influence land development directly through zoning, subdivision codes, and building codes. They influence land development indirectly through decisions about the characteristics of transportation systems and other infrastructure.

In this section, we take an economic approach to explaining the historical evolution of land-development regulation and policy and its broad effects on land-development patterns. We start by arguing that innovations in transportation technology—trucks and buses, in particular—prompted the implementation of modern land-use controls; transportation innovation caused regulatory innovation.²⁴ We then look at how land-use regulation

evolved in the context of the next big transportation innovation: high-speed urban highways.

Relatively comprehensive land-use policy, such as zoning, is a phenomenon of the era of cars and trucks. The first citywide zoning ordinances were adopted in the 1910s: eight cities had adopted citywide zoning by 1916. Sixty-eight more cities had adopted zoning by 1926, the year that the U.S. Supreme Court upheld the constitutionality of zoning. By 1936, another 1,246 municipalities had zoning ordinances. The shift to zoning as standard public policy was abetted by the fact that central cities were able to annex most developing areas on the urban fringe. Zoning took off at roughly the same time as did cars and trucks.

Why didn't zoning take off earlier? Most zoning is hierarchical, with single-family residential uses at the top of the hierarchy. The central concern of zoning is that nonconforming uses may hurt residential house and land values. For many households, their house is their primary asset. It is difficult to diversify this asset, and insurance policies are not generally available to protect a household against a drop in house value due to the effects of nearby development. Some form of regulatory protection makes sense. Single-family development began in earnest in the late 1800s when electric streetcar systems were built. So why didn't zoning become an issue in the late 1800s?

Fischel (2004) argues that the answer is that homeowners and developers could adequately protect residential areas by influencing the location of streetcar lines. Streetcar lines were not designed to carry heavy freight, so they did not encourage manufacturers to leave central areas. Indeed, they helped workers commute to their CBD work places. Apartment and neighborhood commercial developers competed for land close to the streetcar lines. Homeowners and single-family developers needed control only over the location of new streetcar lines to prevent nonconforming uses in single-family developments more than a block or two from existing streetcar lines. They did not need citywide zoning in the era of streetcar commuting and suburbs.

The use of trucks for hauling freight within the city and of buses for transporting people changed things. Trucks are able to move freight at significantly lower cost than are horses and wagons, so trucks allowed manufacturers to consider locations distant from their traditional ones. Buses, in contrast to streetcars, could transport apartment-dwelling commuters on any street. Beginning in the 1910s, the owners of homes in single-family residential areas could legitimately be concerned about factories or apartment complexes going up next door, which would result in a big drop in the market value of these households' biggest asset.

The result was a push, not only by the relatively well-heeled owners of single-family dwellings, but also by realtors, lenders, and developers, for citywide zoning. A good case could be made that transportation innovation—trucks and buses—initiated land-development regulation.

What was the effect of this early land-use regulation on land-development patterns?

It tended to preserve the patterns that the market produced in the streetcar era, especially the segregation of single-family residential land uses from any others.²⁵ High-density residential and commercial uses were restricted to locations near streetcar lines and major arterial streets, and manufacturing was restricted to areas with railroad and highway access away from and usually downwind of single-family residential areas. As cars became more affordable through the 1950s, more of the areas in between the spokes of the radially oriented commuter transportation system filled in with more single-family housing, which was protected from nonconforming uses.

The shift to zoning as standard public policy was abetted by the fact that central cities were able to annex most developing areas on the urban fringe. Zoning took off at roughly the same time as did cars and trucks.

The zoning tended to preserve existing land-use patterns, especially those outside of the CBD.

Cities remained largely monocentric into the 1950s. That pattern changed with the construction over several decades of the system of high-speed urban and interurban freeways. Urban areas have become larger, with employment distributed widely across the metropolitan area, both in clusters and along arterial streets, highways, and freight rail lines.

How have high-speed highways and the associated land development patterns influenced land-development policies?

The construction of high-speed, high-capacity highways opened up new frontiers for land development. Prior to highways, most urban dwellers lived rather tightly packed on high-priced land in dirty and congested cities that frequently looked out upon miles and miles of lovely land suitable for housing. While cattle and corn cavorted in sunshine and spaciousness, city dwellers suffered air pollution and crowding. Both households and businesses took advantage of the access highways provided.

Land-use policy evolved to abet that process. The main objectives of conventional land-development policy have been to encourage suburban development, to move families into single-family housing, and to protect the value of that single-family housing. This explicitly prodevelopment attitude may seem a little strong now, but promoting economic development, in general, and escaping polluted and crowded urban environments, seemed eminently sensible at the time. Policy aimed to accomplish the following:

- *Supply urban services cost-effectively to urban fringe areas.* Just as federal, state, and local governments supported the highway improvements that made suburban land development possible, so too did they support and subsidize extensions of water, sewer, and drainage systems, development of schools and recreation facilities, and so on. Governments were committed to extending public services to support land development.
- *Promote homeownership.* The rationale is that homeownership provides incentive for households to look after their property and to participate in local policy development. Investing in a home is also a form of savings. The federal government has promoted homeownership by insuring home loans, implicitly allowing homeowners to deduct imputed rental income from taxable income,²⁶ explicitly allowing homeowners to deduct mortgage interest payments and local property taxes from taxable income, and not taxing the capital gain upon sale of a house.
- *Ensure quality construction.* A home is often a household's biggest investment. Householders would understandably be reluctant to purchase a house if they are uncertain about the quality of the structure and of local infrastructure. Given limited consumer information, unscrupulous residential developers might give in to the temptation to cut corners. Local governments establish, administer, and enforce subdivision and building codes in an attempt to ensure that housing meets or exceeds basic construction standards.
- *Segregate land uses.* The market tends to segregate land uses. But the desire was strong to protect single-family homes from activities that could hurt home values. Cheap travel by car made it possible to restrict commercial and industrial activities to areas well away from residential areas.
- *Limit development densities.* The market tends to provide low-density housing because low-cost transportation gives access to large amounts of undeveloped land. The consequently low land prices encourage extensive use of land. Zoning reinforces that tendency because households had found crowding unpleasant.

The explicit prodevelopment attitude of conventional land-development policy may seem a little strong now, but promoting economic development in general, and escaping polluted and crowded urban environments, seemed eminently sensible at the time.

Clearly, local governments are active players in land development. What are the effects of these policies?

Let's start with the positives. We have already described what we consider to be the enormous economic advantages of large-scale polycentric urban development. Land-development policies contributed to these economic gains. And the quality of the environments in which most urban households live is much higher now than it was in the 1950s.

Less obvious is that land-development policy has probably contributed to the efficiency of the provision of local public goods and services. This somewhat controversial idea was first developed by Charles Tiebout in 1956, before the construction of the highway system. Tiebout's idea is this. The many households looking to purchase a home vary in their demands not only for housing characteristics but also for the characteristics of local public goods and services. Cheap cars and gas, and high-capacity highways give households unprecedented mobility. They can take advantage of this mobility to shop for service districts that offer public services that fit their preferences given their budgets.

This shopping puts market pressures on service-district officials. Homebuyers bid up the prices of houses in areas that offer especially attractive packages of services and taxes. Homeowners are keen to receive high bids if they decide to sell their house. So they have incentive to put pressure on their local elected officials to implement popular services and to do a good job of providing those services.

Cheap travel forces public service providers to think like restaurant or grocery store owners. Households differ in their preferences for *private* consumption goods. They "vote with their dollars" (i.e., their purchases), which determines the market's allocation of resources to the production of private consumption goods and services. Suppliers have to be sensitive to consumer wants to make a living in a competitive environment, which is good for consumers.

Analogously, households "vote with their feet" when shopping for local public goods and services. Service districts compete for households just as retailers compete for customers. The result is segregation across service districts by household preferences for public goods and services. The situation is analogous to the segmentation of markets for consumer goods. It generates substantial benefits and is made possible by low-cost highway travel.

Land-development policies come into play in Hamilton's extension of Tiebout's analysis to include the effects of the property tax (Hamilton 1975). Local public services were mostly paid for with local taxes, especially property taxes. Relatively wealthy households could get better public services for given property tax payments if they could band together in their own communities to prevent the entry of lower-income households living in cheaper housing.

The problem is that households can cheat on the system. A household can get a good deal on public services by moving into a relatively cheap house in a district that offers high-quality public services. This household pays less than most other households for access to the same public services.

Zoning and development regulations can prevent this form of "free-riding" by preventing the availability of relatively cheap housing. The usual regulatory tools are minimum lot size (i.e., density) constraints that increase lot prices, and subdivision, building, and design codes that increase the cost of building the structure. Conveniently, these regulatory constraints can be justified on the grounds of preserving environmental amenity and preventing shoddy construction, which they succeed in doing. And they succeed in preventing free-riding, which improves the efficiency of public-services provision. These advantages are significant.

The disadvantages of this “fragmented system” of public-service districts and local governments are several:

- *It encourages income segregation.* Demand for public services tends to vary with income, so households tend to sort themselves across service districts by income, which may be economically efficient but has its drawbacks.
- *It complicates decision making about issues that spill over to other service districts or that have a metropolitan scale.* It may, for example, reduce opportunities to exploit economies of scale in production of public services. And it allows one service district to engage in activities that negatively affect households in neighboring districts.
- *It tends to increase commuting distances.* Households have to travel to choose among competing suburban service districts. This travel tends to increase traffic congestion, air pollution, and traffic accidents.

Zoning as an exclusionary device probably also contributes to excessive sprawl. The problem stems from the use of the property tax to pay for public services. As noted above, the variation across service districts in the characteristics of public goods and services is typically paid for with variation in property taxes.²⁷

The property tax has advantages for funding public goods and services. In particular, expenditure on real property is a pretty good proxy for demand for local public services: most of the people who are willing and able to pay for expensive housing probably also want expensive and high-quality public services. The property tax allows local service districts to pay for their activities by charging each household a price that reflects that household’s willingness to pay for the service.

Large-lot zoning contributes to sprawl and longer commutes, however, if the households who are willing to pay their share of the high-quality public services would prefer to live on smaller lots. Demand for housing is in practice not a perfect proxy for demand for local public services. The intensity of suburban land development would likely be higher if local public services were not funded by property taxes because the incentive to impose large minimum-lot-size requirements and other similar regulations would diminish.²⁸

This rationale for minimum-lot-size constraints sheds light on empirical observations of zoning. For example, a series of case studies to evaluate Oregon’s growth management policies (ECONorthwest 1991c) found that cities around Oregon were experiencing, on average, actual built densities considerably below allowable densities. A subsequent study of residential development in the Portland, Oregon, region found lots in new residential subdivisions were larger than the minimum lot size allowed in the local jurisdictions’ zoning ordinance.

Some of this lower-than-required density could result from constraints on developers imposed by parcel geometries and topographical variation: if every lot has to be bigger than some minimum, it may often not be possible to get all lots at the minimum. But interviews with developers indicated that market conditions played the major role. One interpretation is that zoning and building codes put a floor on home values. This floor guarantees a tax base high enough to attract households willing to pay more than the minimum for the public services offered in the jurisdiction.

Another contributor to excessive sprawl is the seemingly sensible effort to put distance between manufacturing/commercial activities and residential areas. When highways were first being built, simply segregating uses must have seemed a reasonable and cost-effective way both to allow industry to continue to function cost-effectively given prevailing prices and policy (in many cases, by continuing to use the ambient environment as a sump for

A fragmented system of public service districts and local governments can have the following disadvantages:

1. *Encouraging income segregation.*
2. *Complicating decision making about issues that spill over to other service districts or that have a metropolitan scale.*
3. *Increasing commuting distances.*

industrial waste) and to protect residential areas from those industrial externalities. Now, however, that separation contributes to traffic congestion. It also discourages manufacturers who would be willing to spend more on reducing emissions if they then could locate closer to their employees.

Finally, the subsidies for single-family residential development encourage sprawl. The various subsidies to homeownership and public-services extensions reduce the market price of land and housing, which not only encourages householders to buy rather than rent, but also fuels demand for bigger lots and houses, which implies excessive suburbanization (sprawl).

In short, land-development policies have had their desired effects. Tremendous value has been created. But after five decades, the basic objective of the economic development of suburban areas has probably been more than achieved.

CHAPTER 2 NOTES

1. This is still the case in agricultural areas of less-developed countries.
2. This simple world seems to ignore the high costs not only of moving people to areas with nice amenities, but also of supplying this dispersed development with public services. If you allow us the transporter assumption, however, you should be able to allow another for the sake of this discussion: that the costs of extending public services could be no greater than they are now (perhaps lower densities allow general use of wells and septic tanks, and power is generated by the same clean, cheap fusion that powers the transporter).
3. Farm laborers continue to move from agricultural areas to work in factories in dirty and crowded cities in developing countries. Factory pay and working conditions may compare poorly to those in developed countries, but they may look good to people familiar with life on subsistence farms.
4. The basic ideas in the monocentric-city model were first laid out by Johann Heinrich von Thunen in 1826 in his effort to understand the variation in agricultural land rents and land uses with distance from urban markets. Every textbook on urban economics provides a more detailed description of the monocentric-city model than we can provide here. See, for example, O'Sullivan (2007) or Mills and Hamilton (1989).
5. By these criteria any number of distinctions could be made. The added complexity, however, adds little to the present analysis.
6. Householders obviously need to travel to places in addition to work. In monocentric cities, other destinations, such as retail firms, are typically in the CBD. This reduces, for the moment, the need to consider nonwork trips. Nonwork trips become much more important, however, when a variety of firms locate in suburban areas.
7. The city would remain roughly circular, however, if distance were measured in terms of travel time from the city center rather than in terms of straight-line miles. Indeed, analysts of retail markets often use *isochron* (equal time) maps to define a market area.
8. Remember, in this section we are *assuming functioning markets and ignoring the effects of public policy*. In Chapter 3 we will add the complications of public policy, demonstrating that it is both necessary and dominating in the context of efficient urban transportation.
9. In real cities, factors other than the price of land clearly influence decisions about design (e.g., displacement of existing development). But these factors have the same kinds of effects on decisions as higher land prices.
10. Muller (2004) describes in detail how historical patterns of urban land development responded to changes in transportation technology.
11. In earlier industrial cities, rain and unpaved streets significantly increased the cost of transportation.
12. Note that the urban populations could not have grown without improvements in the technology of producing food: cities require an agricultural surplus. Until recently, most people lived on farms (in 1800 only about 3 percent of the world population lived in urban areas) not just because there were few economies of scale in production, but because low agricultural productivity required it. The improvements in the technologies of manufacturing and transportation that occurred in the nineteenth century coincided with improvements in farm technology. Improved farm productivity made it possible to take advantage of the other technological improvements.

13. Readers might object that car travel requires greater out-of-pocket expense than travel by streetcar. True enough, but a big part of that expense is the purchase of the car itself, which is an expense that is sunk, and therefore irrelevant, when deciding on a given day whether to commute by car or by streetcar. Another big expense is the value of time spent commuting. Car travel reduces or eliminates the time spent walking to the streetcar stop and waiting for the next streetcar to come along. On decent highways, cars can travel faster, covering more miles per time period. Commuters with relatively high values of time would find car travel cheaper per mile, even though the out-of-pocket expense is higher. We provide more on this in subsequent chapters.
14. The *median* city size increased about fivefold, probably reflecting the growth in employment in services relative to employment in manufacturing: services firms are more strongly attracted to locations in bigger cities. Interestingly, the distribution of relative city sizes was remarkably stable over this long period of rapid technological change; cities got bigger on average, but the distribution of city sizes around that average did not change much at all.
15. Other researchers have referred to clusters as subcenters or activity centers. We use the terms interchangeably in this report.
16. The theory of subcenter formation has been developing for about 25 years. See Eberts and McMillen (1999) for a good review with an emphasis on the role of public infrastructure.
17. At this level of generality we don't see much cause for debate. Trying to fit all the activity of a New York City onto a single square mile or spreading it uniformly over the state of New York seems obviously less efficient than the current pattern. There can be too much density; there can be too little. The tricky question is about a range for optimum levels of density under different sets of conditions.
18. Schwartz (1993) provides empirical evidence in support of central-place theory, showing that despite the undisputed predominance of growth companies in suburban rather than central-city locations, these new suburban companies rely on central-city firms for professional and financial services.
19. A model does not yet exist that explains modern development patterns as well as the monocentric-city model explains urban development prior to 1930. The key to the simplicity of monocentric cities is the connection with the import/export node. Trucks and highways break that connection. The identification and estimation of prime causes in contemporary urban development relies on piecing together partial analyses.
20. The subcenters are large and dense: at least 10,000 workers and at least 5,000 workers per acre.
21. O'Sullivan (2007), taken from the USDOT "Summary of Travel Trends, 2001 National Household Travel Survey."
22. Economists refer to these as hedonic price studies. Hedonic prices are the unobservable prices of the characteristics of goods (such as housing) that possess more than one desirable characteristic.
23. See, for example, Heikkila et al. (1989).
24. The development of the argument we present follows that in Fischel (2004), a must read for those interested in the economics of zoning.
25. See Levine (2006) for an extended discussion of this issue and an important conclusion for planners concerned about urban form, density, and transit-oriented development: single-family zoning is what inhibits the market from providing the kind of urban development those planners want.
26. Imputed rental income is essentially the net income the householder would receive if he rented his house to someone else at market rates. Switzerland is the only developed country that taxes imputed rental returns.
27. The inequities that arise from purely local funding of public services have encouraged most states to centralize the funding base of key public services, such as public education. That funding provides a floor on local spending. Spending above that floor, which varies across service districts, is usually paid for with property taxes.
28. Again, for much more detail, see Levine (2006).

The Evolution of Surface Transportation Policy

Organization: This chapter lumps the last 100 years of transportation development in the U.S. into three epochs, defined by a combination of technology and transportation policy. It has three sections:

1. *Transportation policy before cars and trucks.* Rail provided low-cost travel both between and within cities after the mid-1800s. Policy supported private rail development to encourage economic development. But rail, especially an intra-urban streetcar system, has the characteristics of a natural monopoly: a single company can supply a product more cost-effectively than can multiple competitive suppliers. Policy evolved to treat the excesses of monopoly.
2. *Transportation policy during the rise of cars and trucks.* After 1900, cars and trucks had the potential to revolutionize transportation. But they required good roads, and roads and highways have the characteristics of a public good. In particular, access to roads was difficult to charge for, so private roads tended to be inadequate. Transportation policy turned toward improving roads and highways.
3. *Transportation policy during and after the highway boom.* Cars and trucks generate a variety of costs not borne by car and truck suppliers and operators. These unpriced costs are called negative externalities and generally lead to too much of the unpriced activity. Transportation policy since the 1970s has sought to curb the excesses of the auto: traffic congestion, air pollution, as well as traffic accidents and injuries.

Basics: The analysis in Chapter 2 demonstrates that changes in land development over the last 200 years were caused primarily by transportation and technical innovations that revolutionized the production of goods and services. Changes in transportation caused changes in land use, and transportation policy has been vital to implementing these transportation innovations. Understanding the evolution of transportation policy helps explain the determinants of both land-development patterns and transportation system characteristics.

- The economic rationale for public policy involves market “failure.” Poorly defined or enforced property rights, barriers to entering a market, and imperfect information are primary sources of market failure.
- Urban commuter rail systems have the characteristics of a natural monopoly: a single supplier enjoys lower production costs than two or more suppliers operating independently. This cost advantage provides a barrier to entering the market. Given only walking to employment centers as an alternative to rail transit prior to cars and highways, a monopoly transit operator is able to raise fares substantially above production costs.
- Legislation allowing governments to inhibit monopoly or regulate their pricing was passed in the late 1800s and early 1900s. But by the 1930s competition from cars and trucks reduced the profitability of both commuter and freight rail to the point that regulatory bodies had to acquire or otherwise prop up rail.

- Uncongested highways have the characteristics of a “public good”: many travelers can use the highway simultaneously without much negative effect on other travelers, and charging for access to the highway system is difficult. Because charging is difficult, the private market undersupplies highways. So, governments fund the construction and maintenance of highways, paying for them largely through taxes on fuel, tires, and truck axle weight.
- The funding of highway improvements indirectly through taxes on travel inputs had effects on decision making concerning infrastructure improvements. Decisions are made bureaucratically, mostly by state departments of transportation (DOTs). The objectives of DOTs often differ from those of local or metropolitan governments.
- The interstate highway system also has had strong effects on travel, land use, and land-use policy. The system provides low-cost access to North America’s vast land resources. Households and businesses responded by moving to the suburbs. Land-use policy abetted that process. The system provides huge economic benefits.
- But highway construction and travel, and the way they are funded, have resulted in a variety of excessive “external” costs. An external cost of an activity is a cost imposed on people not directly involved in the activity. Transportation-related externalities include the additional time costs imposed by an additional vehicle entering a congested highway, the costs of auto emissions into the atmosphere, and the costs vehicle traffic imposes on the safety of pedestrians and other vehicle travelers.
- External costs cause inefficiently large amounts of an activity to the extent that decision makers can ignore them. A driver entering a congested freeway ignores the fact that he is slowing everyone else down a little. He can ignore the cost because access to the highway is unpriced. Access to the highway is unpriced as a matter of policy.
- Conventional policy toward congestion has been to try to eliminate it by providing more capacity. The problem is that the faster travel times the new capacity provides encourage additional trips. It fails to treat the source of the problem: unpriced access.
- Conventional policy toward air emissions, noise, and safety is regulation. Regulation typically consists of requirements for equipment, such as catalytic converters, mufflers, and seatbelts. These have positive effects, but they provide little incentive for drivers or automobile manufacturers to find more effective methods to reduce external costs.

Shortcuts: This chapter provides additional fundamental aspects of the economic analysis of transportation and land-use policy. Those readers unfamiliar with these concepts should read the chapter (and perhaps Appendix B) carefully. Readers familiar with the economic concepts can more quickly skim each section to pick up the historical context of policy development.

In Chapter 2 we argue that market forces, not land-development policies, play the dominant role in shaping urban form. In analyzing those market forces, however, we took the characteristics of the transportation system and transportation policy as given. In this chapter, we apply economic analysis to explain these characteristics, again in the context of their historical evolution.

Transportation policy has played key roles in determining the characteristics of the transportation system. State, regional, and local governments have wide-ranging legal and financial powers to influence transportation in urban areas, including:

- directly supplying or regulating the supply of most transportation infrastructure (roadways, transit, sidewalks, bike paths, and parking);¹
- controlling access to roadways, influencing the price of parking and fuel, and the price of purchasing and licensing privately owned vehicles; and
- affecting the design of cars and trucks through regulations intended to make vehicles safer, cleaner, or more fuel-efficient.

Exercise of these powers has profoundly affected the characteristics of the transportation system, which are key determinants of urban form.

Economists often explain public policy as a response to “market failure.” The idea is that efficient market performance requires certain conditions. An absence of one or more of these conditions may motivate calls for government “intervention” in the market. Three key conditions for functioning markets are provided here:

1. *The rules of the game, especially property rights, are well defined and enforced.* Property rights are rights of control over the use of a resource and are thereby the ultimate source of the strong individual *incentives* in market economies. Land and real estate are a huge class of property, and one that, by definition, affects urban form. It should be no surprise that a great deal of public policy pertains to the exercise, enforcement, and transfer of property rights.
2. *Resources are mobile; that is, they are not prevented from entering and being traded in a market.* Markets work efficiently when prices attract resources to their most highly valued activities. Critical to that function is the mobility of resources; that is, the *ability* of resources to gain access to the market. For example, women or ethnic minorities suffer lack of access to some labor markets. Similarly, some firms can successfully prevent the entry of potential competitors to some markets. A range of public policies treat barriers to participation in markets.
3. *Market participants have good information about product prices and characteristics.* Prices are the critical signals that market participants use to decide how to allocate the resources under their control. Good decision making (and efficient market transactions) requires that participants know both the market prices and characteristics of the goods and services they are buying.

There are several additional motivations for government policy. First, the performance of the economy tends to cycle: periods of generally low unemployment and prosperity tend not to last indefinitely. Second, household incomes and quality of life vary in market economies. Some of this variation is essential to the efficient functioning of the system: the prospect of higher incomes motivates hard work and costly investment in valuable training. But some households suffer unreasonably low incomes through no fault of their own, and public policies attempt to treat this inequity. We do not focus in this book on these motivators for government policies, though we do note where they play a role in land-use and transportation policy.

A third motivator of government activity in markets is the risk involved in some very large projects. Large projects are risky for a variety of reasons. Some projects, for example, require acquisition of large amounts of land for rights of way. Some land owners will be reluctant to sell at reasonable prices. Many large projects have unique characteristics, which hinders accurate forecasting of future costs and revenues. Large projects also tend to attract the attention of regulatory bodies. Unforeseen regulations affect streams of costs and revenues.

High risk makes it difficult to raise in private financial markets the financial capital necessary to build large projects. Though this situation is improving over time, the private market likely undersupplies large infrastructure projects. Governments at varying levels have been given powers of condemnation and taxation that help overcome these financial obstacles.

We tie our analysis of transportation policies to three sources of market failure, each of which dominated an epoch in the evolution of policy:

- *Natural monopoly.* A natural monopoly occurs when a single supplier can produce at lower average production cost than can multiple suppliers operating in the same market. The natural monopolist’s low operating cost gives them the ability to prevent the entry of other suppliers to the market

There are three conditions for a functioning market:

1. *The rules of the game, especially property rights, are well defined and enforced.*
2. *Resources are mobile; that is, they are not prevented from entering and being traded in a market.*
3. *Market participants have good information about product prices and characteristics.*

(i.e., resources are immobile): when a competitor enters, the larger firm can in concept drop price below the entrant's cost of production without suffering a loss itself. In the absence of competition, the monopoly supplier has the incentive to charge prices above that which provides a competitive market return. Turn-of-the-twentieth-century streetcar systems had the characteristics of a natural monopoly.

- *Public goods.* An uncongested roadway system has the characteristics of a public good: many people can use the system simultaneously, and it is difficult to deny access to people who refuse to pay. The property rights to the local street system cannot effectively be enforced. From the 1920s through the 1960s, governments have used their powers of taxation to fund provision of a number of important public goods, especially of roads and highways.
- *Externalities.* Since the 1960s, policy makers have become increasingly concerned with reducing the negative "spillover" effects of a variety of commercial activities, traffic congestion, car and truck pollution, and urban sprawl. Economists refer to these as negative *externalities* because the effects are external to the private production process. They arise because market participants can avoid some costs, and the market price of the activity is therefore too low. Market participants can avoid the cost because property rights to some resource, such as a highway, the neighborhood environment, or the atmosphere, are poorly defined or enforced.

We recognize that policy has rarely focused exclusively on a single form of market failure. The federal government clearly recognized the need for public participation in the provision of roadways at least as early as 1916 when Congress passed the Federal Aid Road Act. Before then, local governments supported road construction and installed public sewer and water systems. The development of zoning ordinances in the 1920s indicates that local governments were concerned about reducing the effects of negative externalities long before 1970. But breaking the evolution of policy into epochs makes our exposition of the economics of market failure more interesting and accessible: we can explain the economic analysis of each type of market failure in the context of historical policy making. Moreover, the breakdown of policy into epochs corresponds with the evolving thrust of public policy.

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TRANSPORTATION POLICY BEFORE CARS AND TRUCKS: ECONOMIC DEVELOPMENT AND NATURAL MONOPOLY

Government policy in the U.S. evolved as the country developed. Prior to the mid-1800s, most intercity or interregional shipping of freight was by water, and most personal travel was by water or on foot or horse. We noted in Chapter 2 that large cities developed along natural harbors. The development of the steamboat in the early 1800s allowed upstream travel on rivers and was critical to the development of river towns. Overland roads and canals were built privately and paid for over time through the collection of tolls.

In principle, little government activity was needed in intercity or interregional transportation because private suppliers could collect tolls to pay for their efforts. The construction of a road or canal, however, involves a large upfront expenditure and a large risk. States, in the interest of economic development, subsidized private construction of roads and canals. With the exception of postal roads, which are explicitly mentioned in the Constitution, the federal government played a relatively small role in the early development of transportation, in large part because this was not regarded as a federal concern.²

The federal government played a larger role in the development of railroads in the second half of the nineteenth century. The government declared

all railroads postal roads, which allowed federal subsidies of private railroads through postal contracts. The large upfront expenditure required for construction of interregional and interstate railroads inhibited private development.

The federal government responded by granting federal land to the states, which then sold the land and used the proceeds to subsidize the construction of railroads. Rolling stock, maintenance, and operating costs could be paid for with shipping charges and passenger fares. Railroads were privately operated—more than 1,000 railroad companies were in operation in 1890—but federal and state assistance played an important role in their development.

The large traffic volumes and generally desirable characteristics of railroads encouraged little government attention to the wagon roads in rural areas. Many were impassable during bad weather. Local governments imposed property and head taxes in rural areas to improve and maintain rural roads, but the proceeds tended to be small. Much of the maintenance was carried out piecemeal by the farmers who needed to use the roads.

Local government policy was more important in urban areas. Rail transport and innovations in production technologies encouraged the growth of well-located cities. Private land developers built local streets, which were then deeded to the public, and private companies or the land developers themselves extended streetcar lines to newly developing areas. Local governments helped coordinate the construction of these private transportation investments and raised money through property taxes to maintain the urban public street system.

Though rail transportation allowed cities to grow and brought commuters out of the mud in the streets, it introduced another problem. The owner of a rail or streetcar line had a competitive advantage the suppliers of most goods and services can only dream about: a *natural monopoly*. The monopoly is deemed “natural” because it derives not from illegal business practices or special legal status (e.g., patents), but from the characteristics of technology: once someone (a business or a government) supplies the good or service, it is impossible for anyone else to do so at the same cost, so they do not enter the market. A natural monopoly allows the supplier of rail services to raise fares above cost because of a lack of the competition necessary to keep market-determined fares closer to cost.

Urban water, sewer, electricity, natural gas, and telecommunications also operated (and continue to operate) under the conditions of natural monopoly. Local government policy had to deal with natural monopoly before the age of cars and trucks. In the rest of this section we describe how natural monopoly allows prices to diverge from costs, and how government can intervene to improve efficiency.

What Are Natural Monopolies and How Do They Distort Prices?

A natural monopoly results from the cost advantages of large-scale production. Most production processes exhibit at least some economies of scale: large production runs allow investment in specialized labor and machinery that, while expensive, push down the *average* cost of the large volume of output. The cost advantage of using highly specialized equipment combined with low-cost shipping has brought many technically sophisticated consumer goods into the reach of the average household.

Indeed, low-cost transportation plays a key role in preventing natural monopoly. Many production processes exhibit extensive scale economies. A big, specialized plant can produce large numbers of washing machines, for example, at costs much lower than can be achieved in a small plant with unspecialized tools and labor. It still requires a significant number of



Chicago Historical Society

Local government policy was important in urban areas. Rail transportation, like streetcars, and innovations in production technologies encouraged the growth of well-located cities. Here is downtown Chicago.

plants, however, to satisfy the large national and global demand for washing machines. Fortunately, the low cost of shipping washing machines, even across long distances, enforces competition among the owners of very large production plants. Extensive scale economies do not necessarily result in monopoly prices.

Transportation costs and scale economies are the critical components of central-place theory. A single grocer, for example, may be able to supply groceries at lower cost in a small town than could two or more grocers. Scale economies are not terribly extensive, but they are extensive relative to the local demand. Stores in the next town, however, discourage the small-town monopolist from charging prices significantly above average production cost as long as traveling to the next town is not too expensive. Because the small-town store is small, prices may be relatively high to cover its high average costs.

Importantly, as long as the set-up costs of starting a grocery business are not too high (i.e., barriers to entering the market are low), grocers will tend to distribute themselves over space so as to minimize the *combination* of production and transportation costs. Some small grocers will charge relatively high prices, but low transportation costs will encourage fewer stores operating at a more efficient scale with more competitive pricing.

Problems result when barriers to the entry of competitors exist. Consider the case of an urban streetcar system. A particular line or branch line can be built and operated by a private company, which then has control over its rails.³ There may not be enough demand along that line to warrant installation of a competing set of rails and purchase of the streetcars to run on them. It may be most cost-effective simply to run larger streetcars more frequently along the one set of rails. The company that owns the rails enjoys a natural monopoly along that line.

Like the small-town grocer, the apparent monopolist will not necessarily be able to charge monopoly prices. Another company could build a competing line on a neighboring radial street. A commuter living between the lines could then choose between them, enforcing competition and competitive fares. So far, a competitive system of private streetcar lines could be feasible. But there are at least three factors working against this arrangement:

1. As we described in Chapter 2, in the absence of zoning, homeowners tended to resist construction of new streetcar lines due to their concern about the negative impacts of the consequent changes in land use.
2. A single company enjoys cost and service advantages from operating the entire network rather than individual lines. For example, the large stock of streetcars and their drivers can be reallocated around the system as demand varies throughout the day or year, meaning less down time for labor and equipment. Handling tickets and transfers can be done more cost effectively, as can scheduling along various lines. And more specialized equipment and labor can be used for administration and maintenance.
3. Transportation drives land use. A rail line is a major capital investment, and land development requires additional major infrastructure investments. Once infrastructure is provided, land proximate to these investments gains value, and more valuable land gets used more intensively. More intensive land use means that the existing rail line works even better: the marginal costs of adding new capacity are small and average costs keep falling. Unless a competing streetcar system lays a parallel track in the same right of way (unlikely), it is relegated to competing in that subarea by moving to an adjacent, less developed, corridor.

Extensive scale economies and other barriers prevent entry in the provision of streetcar commuter service. Conditions in a turn-of-the-century city

Problems result when barriers to the entry of competitors exist.

may have been good for a streetcar monopolist: workers had to commute to work and no good substitutes for the streetcar existed. In the jargon of economics, demand was *inelastic* at prices near average production cost: a monopoly supplier could raise prices without much harm to ridership. The profit-incentive to form a monopoly and raise price was strong.

Similar arguments can be made for the tendency toward natural monopoly in intercity freight rail services. Mergers of companies in the late 1800s, not only in rail service, but also in other manufacturing industries, raised concerns about monopoly and monopoly pricing.

Government Response to Monopoly Pricing of Rail Transportation

States attempted to regulate railroads in the 1880s, but the railroads argued successfully that state regulations are unconstitutional because railroads engaged in interstate trade. Congress responded by passing the Interstate Commerce Commission Act in 1887 and the Sherman Antitrust Act in 1890. This antimonopoly legislation was strengthened in 1914 with the Clayton Antitrust Act and the Federal Trade Commission Act. Public policy aimed at encouraging competition and otherwise discouraging monopoly pricing.

Given the necessary legislative support, governments can take any of three approaches to combat monopolistic pricing:

1. *Promote competition among suppliers.* Monopolies are often created by mergers of two or more companies. Government can discourage monopoly by preventing merger. Denying a merger may prevent attainment of some of the cost advantages associated with large-scale production, but it encourages competition that pushes firms to operate as cost-effectively as they can and charge prices near average production cost. Preventing mergers encourages competitive behavior.
2. *Regulate natural monopolies.* The details of this kind of regulation can get complicated (which, of course, makes it a challenge to do well). A public utility commission is generally given the power to evaluate pricing schedules proposed by the private monopoly. A common practice is to grant rates that cover operating costs and provide a more-or-less market return on shareholder investment in fixed capital.

In principle, the regulated monopolist earns only a market return, and the government itself does not have to supply the good or service. A private company may in some circumstances be in a better position than the government to operate the service cost-effectively. Most private producers of electricity, natural gas, and even garbage service continue to be regulated by state or local governments.⁴

3. *Produce the good or service itself.* Government production of transit services has an advantage inasmuch as government has the legal authority to acquire land for rights-of-way without having to go through lengthy negotiations with land owners. Another advantage of public production is that the books are open to public scrutiny. Regulating a private monopolist requires information about production costs the private company may be less than eager to reveal. This is also a disadvantage of public ownership. Management is open to public scrutiny, which may hinder innovative decision making and investments.

In practice, local governments tend to supply basic public services, such as water, sewer, and drainage services, which are natural monopolies, the management of which tends not to require large amounts of entrepreneurial acumen. Sometimes, more complex organizations, such as ports, are owned by the local government but are organized like a private corporation; the managers operate relatively independently from the local government itself.

Government can take any of three approaches to combat monopolistic pricing:

1. *Promote competition among suppliers.*
2. *Regulate natural monopolies.*
3. *Produce the good or service itself.*

As it turns out, the need for monopoly regulation of streetcars and railroads dissipated fairly quickly. Many commuter rail systems and some intercity rail operations were eventually acquired by the public, but not in the interest of preventing high pricing. Instead, local governments took over the systems in an effort to prevent their demise. Cars and trucks in the 1910s were similar to computers and cell phones in the 1990s inasmuch as prices were continuously falling with advances in production technologies and techniques. The Ford Model T was priced at \$300 in 1917. Cheap cars provided a substitute to streetcars for commuters and by the 1930s were becoming the dominant mode for commuting.

The public began acquiring streetcar systems in the 1920s and 1930s—not to control monopoly pricing, but to subsidize them sufficiently to keep them running. Not everyone could afford a car, but there were too few streetcar commuters to cover both operating and capital cost.

Public agencies created in the late 1800s to prevent *high* prices started working in the 1930s to prevent *low* prices. The truck, for example, started to compete effectively against intercity rail. The Interstate Commerce Commission in 1935 was empowered to impose regulations on trucking that included restrictions on the entry of new trucking companies and on the routes trucks were allowed to use. Prices were evaluated not on trucking costs, but on their effect on railroads. The commission tried to protect railroads sufficiently to keep them in business and their workers employed. As we discuss in the next section, other federal, state, and local agencies were at the same time working aggressively to improve the roads that made trucking profitable.



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Cars and trucks in the 1910s were similar to computers and cell phones in the 1990s inasmuch as prices were continuously falling with advances in production technologies and techniques. The Ford Model T was priced at \$300 in 1917.

The Effects of Policy to Prevent Monopoly Pricing

The concern with private monopoly prices is that they are too high. The price of a streetcar trip at rush hour (i.e., the peak period) should roughly equal the full cost of supplying that trip. That cost includes the short-run costs of energy and labor, the medium-run costs of rolling stock, and the long-run opportunity cost of the investment in rights-of-way, rails, and system management. A monopoly supplier can increase profit by raising price above this cost. The policy objective is to keep prices low enough so that they accurately reflect long-run production costs.

Lower rail prices encourage more transport by rail. Within a monocentric city, that means more travel on the urban streetcar network, a network that extends farther from the CBD, more land development at the urban fringe, and, perhaps surprisingly, more intensive land development in the CBD. Cheaper travel leads to spatially larger cities.

To see this, it helps to work through how the financing of a streetcar extension affects land values, and vice versa. The line of reasoning is a bit involved, but worth the effort.

Start with a well-located monocentric streetcar city initially in equilibrium—wages, land prices, and transport costs are such that there is no tendency for the city to grow or shrink. The streetcar lines extend to the edge of the residential districts that form the boundary of the city. The city boundaries are not expanding because land values fall with distance from the CBD (to compensate households for the longer commutes) and an urban land developer values land at the boundary about the same as does a farmer. Farmers are able to outbid developers for land beyond the current urban edge.

Now suppose an innovation in construction techniques occurs that reduces the cost of building relatively tall office buildings (e.g., steel-frame construction). Space in CBD office buildings becomes cheaper. Existing service firms respond by expanding, and some service firms in less well-located cities move into the CBD. This business expansion increases the demand for workers, which bids up wages—the new and larger firms can

increase wages because of the savings on rents. The demand for housing increases as workers from other cities are attracted by the higher wages, and incumbent households have higher incomes to spend on housing (and other goods). Housing prices are bid up, which flows through to bid up the price of residential land. In particular, land at the original city boundary becomes more valuable in residential use than in farming—a discontinuity appears in the land-rent gradient.

Streetcar lines need to be extended to develop land at the urban fringe, and this extension could be financed in several ways.

- The developer of a sufficiently large tract of land could borrow money to extend the line and then pay off the construction loan with the proceeds from sales of building lots.
- The private owner of the streetcar line can borrow the money to extend the line and pay off this longer-term loan through fare revenues.
- The public owner of the streetcar line could borrow money to extend the line and pay off the longer-term loan with the increase in property tax revenues the new development generates.

Each of these seemingly different financing methods has a similar impact on market land values. The residential developer estimates the revenues from the sale of building lots and subtracts the costs of development, including that of extending the street car line, to determine his maximum willingness to pay for the undeveloped land. The cost of extending the line thereby “capitalizes” into land values. The competitive streetcar operator charges fares sufficiently high to cover the costs of the extension. The purchasers of building lots factor the transportation costs into their willingness to pay for the lot, so overall land-developer revenues fall by the cost of constructing the streetcar line. If property taxes are used to pay for the extension of the line, lot buyers similarly factor in the higher property taxes into their willingness to pay for lots.

In short, no matter the mechanism with which the extension is financed, the effects on the land market are the same as if a competitive private streetcar operator charged fares equal to the full long-term cost of providing the streetcar service.⁵

In any event, the city would grow. The savings in the cost of constructing office space induces growth in population and income, and land values increase throughout the city. Land-use intensity increases in the CBD (due to the lower cost of tall buildings). The effect on the intensity of land use in residential areas depends on the effect of higher land prices (which increase intensity) relative to the effect of higher incomes (which increases lot sizes, decreasing density). Streetcar lines are extended, and farmland at the urban fringe converts to residential uses. In the new equilibrium, the value of land at the new urban edge is again about the same in urban as in agricultural uses.

How would things be different with a profit-maximizing monopoly supplier of streetcar services? The process would look similar, but the city both before and after the construction innovation would be smaller. The monopoly supplier maximizes profit by raising fares above long-run production cost. The higher commuting costs discourage some of the growth in employment and population. The higher fares also capitalize into land values: urban land values are everywhere lower. Residential developers are able to outbid farmers only at shorter distances from the CBD. The city both before and after is smaller than is economically efficient. The streetcar owner benefits at the expense of the land owners in the city.

Note that competitive pricing of transportation results in economically efficient transportation and land-use patterns. That efficiency has a number

The public began acquiring streetcar systems in the 1920s and 1930s—not to control monopoly pricing, but to subsidize them sufficiently to keep them running. Not everyone could afford a car, but there were too few streetcar commuters to cover both operating and capital cost.

of dimensions. To travel, commuters have to pay the fare. That fare influences their willingness to pay for housing. Residential development spreads into agricultural areas only if commuters who have to pay the full cost of their transportation are willing to pay enough for building lots to allow the residential developer to outbid farmers for undeveloped land. The fares collected by the transit operator pay for the cost of maintaining the system, and the prospect of increasing revenues guides the operator to increase capacity. The value of the resulting economic efficiency capitalizes into the value of land, maximizing its aggregate value. *The full-cost pricing of transportation is a key driver of the efficiency of the metropolitan transportation system and of the land-use pattern it influences.*

TRANSPORTATION POLICY DURING THE RISE OF CARS AND TRUCKS: PUBLIC HIGHWAYS

Attitudes about the role of government in American society changed rapidly as the twentieth century progressed. The abuses of monopolists provided one indication that free enterprise does not always produce socially favorable outcomes. But the contemporary commitment of government to big public construction projects was born in the Great Depression of the 1930s.

The Depression was just the deepest of a long series of economic recessions and depressions that plagued the nation in the nineteenth and early twentieth centuries. Research into the causes of business cycles has motivated fundamental change in the structure of financial institutions and in beliefs about the ability and obligation of government to act in ways that reduce the effects of economic downturns. The relatively stable record of economic growth since the late 1930s indicates the effectiveness of these changes in public and private policies.

The public works projects initiated during the Depression not only contributed to the immediate objective of reducing unemployment, but also suggest a response to a major source of failure in a market economy. Analysts realized that many of the projects undertaken mainly to make work for the unemployed (1) turned out to have considerable social value, and (2) were unlikely to have been undertaken by private firms even in a healthy economy. Examples include improvements to recreational facilities on public lands (e.g., trails and buildings in national forests and parks), improvements to the system of rural roadways, and construction of dams for flood control and recreation in addition to irrigation and generation of electricity.

In each of these cases, a single good provides benefits to all of the members of large groups of people, each of whom benefits in differing amounts difficult to estimate. Private firms would probably not have undertaken these projects because the difficulty (even impossibility) of charging many of the beneficiaries of the project would prevent them from recouping costs. Goods with these characteristics form a special class that economists refer to as *public goods*.

Since the 1930s, governments at all levels have committed themselves to providing a wide range of public goods. At the metropolitan level, public goods include the system of city streets, police and fire protection, parks and recreational facilities, and a wide range of urban environmental characteristics. Public goods with federal support include rural roads, national defense, national parks and wilderness recreation areas, and clean air and waterways. Government provision of these goods and regulations that protect naturally supplied public goods (and support for private provision of other public goods, such as scientific research) have played major roles in the sustained growth in incomes and quality of life since the end of World War II.

In the rest of this section, we first describe in the abstract the problems public goods pose for efficient market allocation of resources, and then look at the policies governments have used to relieve these problems in the context



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Since the 1930s, governments at all levels have committed themselves to providing a wide range of public goods, including city streets, fire and police protection, and, of course, parks. Here, one of the most famous: Central Park in New York.

of transportation policy. Though we do not address environmental quality in this section, it is not because we fail to recognize it as an important public good. We treat it in detail in a later section in the context of externalities.⁶

What Are Public Goods and How Do They Distort Prices?

A good or service is termed a public good if its users are *both* “nonrivalous” and “nonexcludable.”

Nonrivalry in consumption means that the good or service can be used or consumed by more than one person simultaneously. Many people, for example, can travel unimpeded on an uncongested highway or enjoy the ambiance of a well-designed residential neighborhood. Many people can watch the same movie, football game, fireworks, or an over-the-air TV broadcast, or listen to the same lecture or radio show.

It is the other characteristic of a public good—nonexcludability—that motivates and justifies government provision. The market supplies a wide variety of goods that are nonrival in consumption, but those goods are “exclusive” inasmuch as people who don’t want to or cannot pay for them don’t have them (e.g., cable TV). On the other hand, if suppliers cannot collect payment (i.e., if the price is effectively zero), they obviously have little incentive to supply or maintain the facility or service and exclude themselves from the market for supplying such goods or services. The price of and consequent revenues from some public goods are too low to generate private provision of anything approaching an efficient quantity or quality of the public good.

Private suppliers have managed to find clever ways to provide goods that appear to have both of the characteristics of public goods. For example, over-the-air television and radio broadcasts are paid for through sales of advertisements, and satellite television providers require customers to pay for decoders. Residential and commercial development pays for construction and maintenance of their highly localized public goods through lot or house prices and space rents. Some private organizations, such as land conservancies and community radio stations, are able to operate through donations of time and money. The value that consumers place on public goods encourages private suppliers to find ways to collect enough of that value to supply the goods.

Nevertheless, a variety of important public goods remain undersupplied by the private sector. The government is in a unique position from which to direct the supply of public goods because it has the legal authority to force people to pay. Governments at all levels can impose a wide range of taxes.⁷

But what should each of the many consumers of the nonrival good pay? In contrast to private goods, everyone consumes (as a rough approximation or by assumption) the same level of public goods and services. So an apparently reasonable choice is to charge everyone the same amount; that is, to impose a head (a.k.a. poll) tax.

But, as is true for paying for private goods, individuals vary in their valuation or willingness to pay for public goods and services. Households with children tend to place more value on parks and playgrounds. People who like to walk or cycle to work tend to place relatively low value on the characteristics of the road system valuable to drivers. And, of course, *willingness* to pay for public goods and services varies with *ability* to pay (i.e., with income and wealth). In short, a head tax high enough to cover the cost of provision of an efficient quantity of the public good or service may overcharge a high proportion of its consumers.

The alternative is to charge individuals and households differing prices for the same quantity of the public good. The price varies with customer valuation. Private goods sell for a single price, and consumers vary in how much they purchase and consume: one price, differing quantities. The quantity of a

A good or service is a public good if its use is both “nonrivalous” and “nonexcludable.” For instance, many people can watch the same movie, football game, etc., if they can tap the public airwaves with an antenna. This public good gets paid for privately by advertisers.

public good is the same for all, but each pays a different price: one quantity, differing prices.

Estimating payment by asking people about value will not work well. No individual can be excluded from consumption of however much of the public good is supplied. It makes sense from an individual's perspective, then, to underreport his valuation of the public good. He pays less for his consumption of the public good and has more to spend on consumption of private goods. If everyone does this, of course, revenues are lower than optimal and the government undersupplies the public good. Economists refer to this as the "free-rider" problem. It arises because individual preferences and willingness to pay are not directly observable.

The method typically used in practice is to find some observable *proxy* for preferences for the public good; that is, the government looks for some observable behavior probably closely correlated with preferences for public goods and services. These observable quantities include:

- *Income.* Willingness to pay for most public goods and services probably varies positively with income. The federal government and many state and local governments require individuals to report their incomes accurately. Tax payments then vary with reported income—every household pays a different amount of tax for similar public goods and services.
- *Expenditure on private goods.* A goods and services tax, or sales tax, for example, matches variation of expenditure on public goods with that on private consumption goods as a whole. Most states and some cities impose a sales tax.
- *House value.* Expenditure on housing is probably positively correlated with demand (willingness to pay) for local public goods and services. The value of a house, even one that has not sold for a long time, can be estimated from sales of similar houses. Most local governments fund local services partly through a property tax.⁸
- *Gasoline consumption.* Consumption of gasoline is positively correlated with demand for use of the roadway system. Both the states and the federal government tax gasoline; revenues are earmarked for expenditure on public roads and highways.

None of these measures is perfectly correlated with individual or household demand for local public services. Taken together, however, overall tax payments may correlate reasonably well with willingness-to-pay for public services. As we described in Chapter 2, households can and almost certainly do take advantage of their mobility to "vote with their feet" to find service districts, cities, or even states and metro areas, that better match their preferences for taxes and public goods and services.

In addition to determining the variation in payments for public goods and services across individuals or households, each public provider of goods and services also must decide the total budget. How big should the sales, income, gasoline, or property tax rates be? This is typically decided politically by elected representatives: Congress, state legislatures, or city councils. It may also be decided by a direct vote of the people, as in school budget elections.⁹ A concern is that the free-rider problem reappears: each individual has an incentive to vote against a proposal to raise taxes or against a candidate who supports higher public spending.

Currently, about 75 percent of gross domestic product (GDP) consists of market exchange of private consumption goods (including housing), and governments at all levels spend about 19 percent of GDP on provision of goods and services (the remainder of government spending consists of transfer payments, such as social security). It is difficult to evaluate the extent to

Individual preferences and willingness to pay are not directly observable.

Proxies need to be used. Examples of proxies governments use include:

1. *income;*
2. *expenditure on private goods;*
3. *house value; and*
4. *gasoline consumption.*

which political decisions are efficient, except to the extent that households vote with their feet. Their ability to do that depends on the localness of the public good and the extent of household mobility (transportation, again).

Government Response to Market Underprovision of Roads and Highways

The first significant federal support for road construction corresponded in time with the legalization of the federal income tax through passage of the Sixteenth Amendment to the Constitution (1913). The Federal Road Act of 1916 provided funds for improvements to rural roads on a 50-50 matching basis. The states administered expenditure of the funds and were required to set up state highway departments to oversee program operation and to decide funding priorities. The departments were required to provide progress reports, audits, and verification of the quality of finished work. The 1916 Act laid the groundwork for continuing federal support for road improvements and signaled the shift from local control over roads to more centralized state and federal control.

Publicly funded road construction boomed after World War I. The federal government temporarily took over control of the railroads to try to reduce delays caused by congestion. After the war, Congress focused its road spending on creating an integrated interstate road network, in part to provide an alternative to railroad freight transport. Most of the new road system in rural areas was gravel, but by the end of the 1920s 60 percent of U.S. households had a car, and more than 3.5 million trucks were in operation.

Federal, state, and local governments had risen to the challenge of providing unlimited access roadways as a public good, and these roads provided the infrastructure base for significant competition to rail transportation in both rural and urban areas. This competition transformed the regulatory bodies created to control rail monopolies into regulators of trucking to protect employment in the now less cost-competitive railroads.

Gas taxes helped fund roadway improvements. By 1929, every state imposed a tax on gas of between two and six cents per gallon. The first national tax on gasoline consumption was imposed in 1932. The tax was one cent per gallon of gas, and revenues went into the general fund. The tax was implemented primarily to prop up government revenues dwindling with the onset of the Great Depression. The federal gas tax increased to 1.5 cents per gallon in 1940 to increase revenues for expenses related to World War II.

By 1950, taxes on gasoline covered most federal and state expenditures on roads. National expenditure on roads was roughly equivalent to revenues from the national gas tax, an outcome supported by highway-user lobbies. Expenditure by the states was at least double federal expenditure on roads. Common practice was to issue bonds to pay for construction of highways, and then pay those bonds off, in part, with funds from gas tax revenues. Many states passed legislation to dedicate gas taxes to fund roadway construction and maintenance.

Conditions in the 1950s were ripe for a major federal road construction project. The success of large-scale, federally supported public infrastructure projects in the 1930s (e.g., rural electrification, flood control, rural roadways) had shown the economic value of large federally funded projects. Demographic trends were increasing the demand for highways. Incomes were rising in the post-war boom, car ownership was continuing to increase, and the baby boom was in its infancy. Householders coveted single-family houses with room for kids in clean and secure suburbs. Businesses could see the advantages of highway access to relatively low-priced land in suburban areas. And earlier federal and state road-building campaigns had laid the conceptual, physical, and administrative groundwork for a very large construction program.



Tim McCaig, istockphoto.com

By 1929, every state imposed a tax on gas. The first national tax was imposed in 1932. By 1950, taxes on gasoline covered most federal and state expenditures on roads.

A key issue in undertaking such a large construction program is funding. Congress had already authorized the construction of a roughly 40,000-mile interstate highway system in the Federal Aid Highway Act of 1944. But it did not provide funding for the construction, operation, and maintenance of the system.

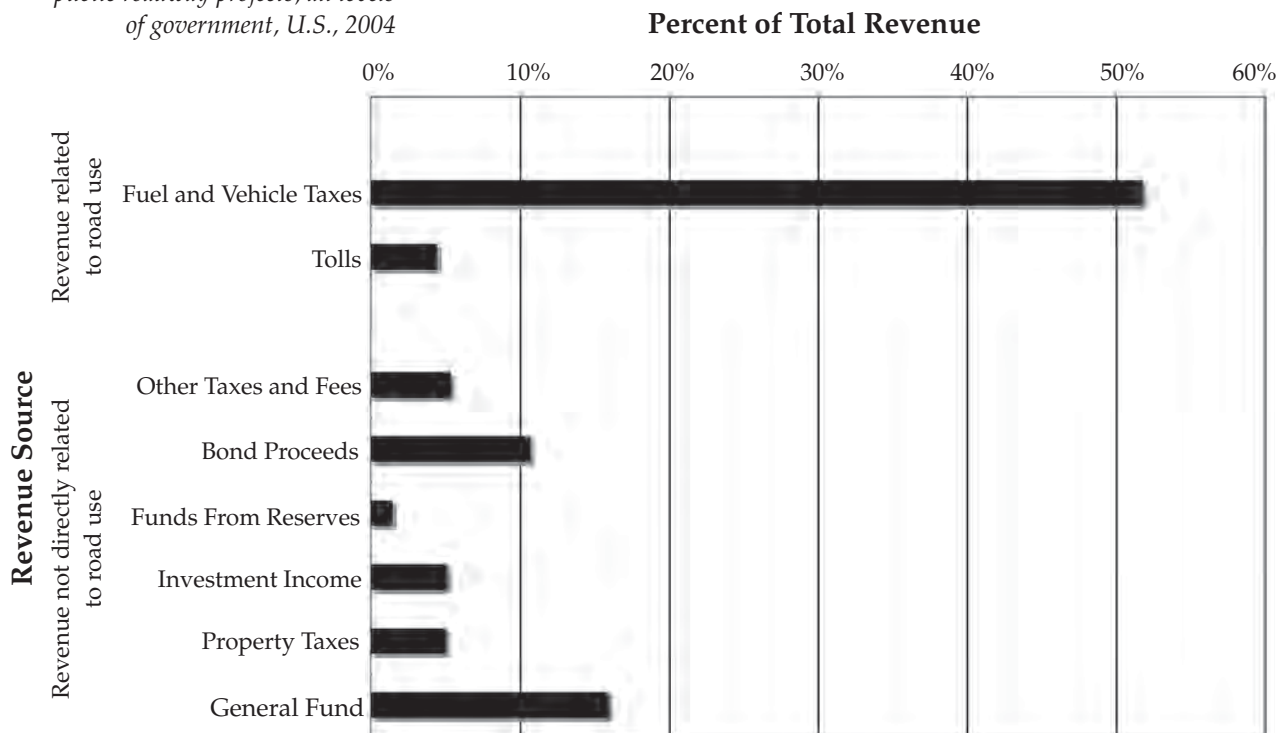
The debate on funding was renewed in the mid-1950s. One option is tolls. Truck and car associations opposed tolls for two reasons: (1) toll collection given the technology available at the time was inconvenient and time consuming, and (2) tolls on top of gas taxes appear to represent double taxation.

In the end, the government struck a deal to fund the federal share of construction costs through an increase in the federal gas and diesel tax to three cents per gallon, taxes on tires and tire components, a manufacturing tax on trucks, buses, and truck trailers, and a tax on the weight of trucks (because large trucks cause tens of times more wear and tear on a highway than does a passenger car). The revenues raised from these taxes would go into a newly created Highway Trust Fund, which was dedicated to funding transportation improvements. The result was the Federal-Aid Highway Act of 1956, also known as the National Interstate and Defense Highways Act.

This basic funding system has remained in place since then. Figure 3-1 shows the sources of revenues explicitly devoted to transportation infrastructure and services by all levels of government as of 2004.¹⁰

Users of the roadway system paid for about 60 percent of the cost of providing infrastructure through fuel taxes, motor-carrier taxes, and highway and bridge tolls. This arguably pays most of the direct expenditure on the state and federal highway system. Highway users, as users,¹¹ probably pay for the construction and maintenance of the national and state highway system. Most of the remainder, used principally to fund construction and maintenance of local roads and streets, comes from general fund or property tax revenues.

Figure 3-1. Revenue sources for public roadway projects, all levels of government, U.S., 2004



Source: Federal Highway Administration, *Highway Statistics*, 2004.

This policy for funding roadway improvements is a combination of “indirect user pays” and “get the money where you can.” That funding system can make sense, however, if the highway system can reasonably be considered a pure public good: tax payments vary across individuals and may correlate reasonably well with the variation in their willingness to pay for use of the highway infrastructure.

The problem is that the highway system is a public good only when it is uncongested. As we argue later in this chapter, the decision to fund construction and maintenance of the public highway system using mainly taxes on purchase of inputs to travel was an unfortunate, though difficult-to-foresee, misstep in transportation policy.

The Effects of Post-War Highway Transportation Policy

Two aspects of post-1950s surface transportation policy have had important effects on transportation and land use, and on transportation and land-use policy.

The first aspect is the highway system itself. In 1956, the federal government committed itself and the states to the construction over several decades of a more than 40,000-mile system of interstate highways. The objective of such a system is primarily the speed (and potentially the safety) that results from unimpeded travel. Lanes and curves are wide, grades are gentle. There are no stop lights. Access is limited—neither households nor businesses have direct access to the interstate highway. Access and exit points are cleverly, and sometimes elaborately, designed so that cars and trucks can enter or exit the highway with minimal disruption to travel speeds on the highway.

The second aspect of transportation policy is the way the construction and maintenance of the system is funded. The primary funding for the interstate system comes from taxes on fuel, and tires, and weight charges on trucks. Though it collects its share of tax revenues, the federal government does not plan, build, or maintain roadways. Rather, it allocates funds to the states for the planning, construction, and maintenance of interstate freeways and major urban and rural routes. State departments of transportation (DOTs) carry out these projects with additional funds from state taxes on fuel. Local governments pay for roadway improvements through intergovernmental grants, property taxes, general revenues, and sometimes through developer contributions.

The problem is that the highway system is a public good only when it is uncongested. . . . [T]he decision to fund construction and maintenance of the public highway system using mainly taxes on purchase of inputs to travel was an unfortunate, though difficult-to-foresee, misstep in transportation policy.

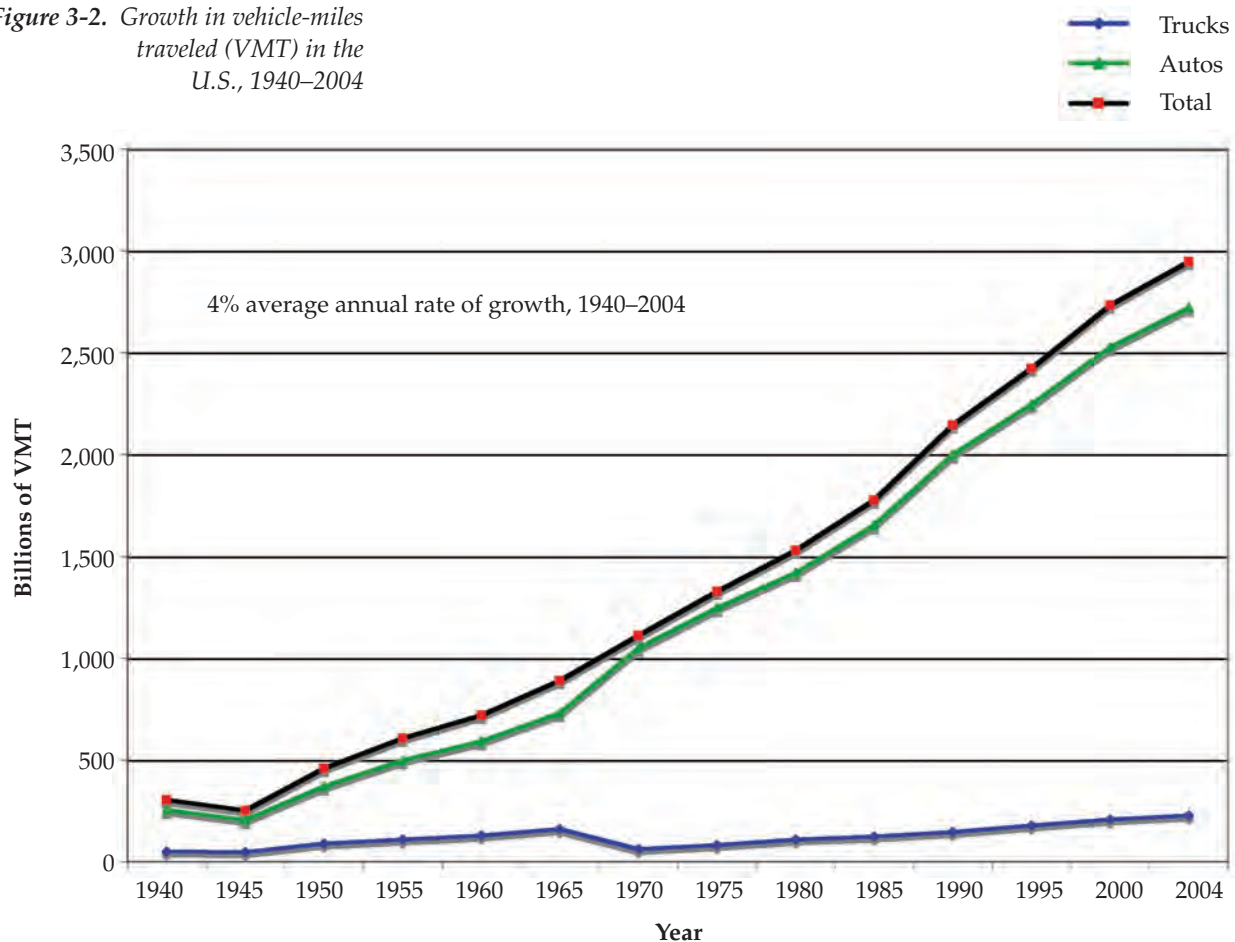
The Effects of New Highway Capacity

The extensive highway improvements yielded big reductions in the per-mile time costs of travel by car and truck. Time savings encouraged more travel, and more dispersed, low-density, polycentric urban land development.

Figure 3-2 shows the stunning growth in car and truck vehicle-miles traveled (VMT) over the period of construction of the interstate system. The volume of travel increased more than sixfold between 1950 and 2004, increasing even in the 1970s when the price of fuel increased dramatically. Part of this growth in travel is a result of a doubling of the population over the same time period. The rest is a result of growth in income and the decrease in the per-mile cost of travel made possible by construction of the high-speed highway system.

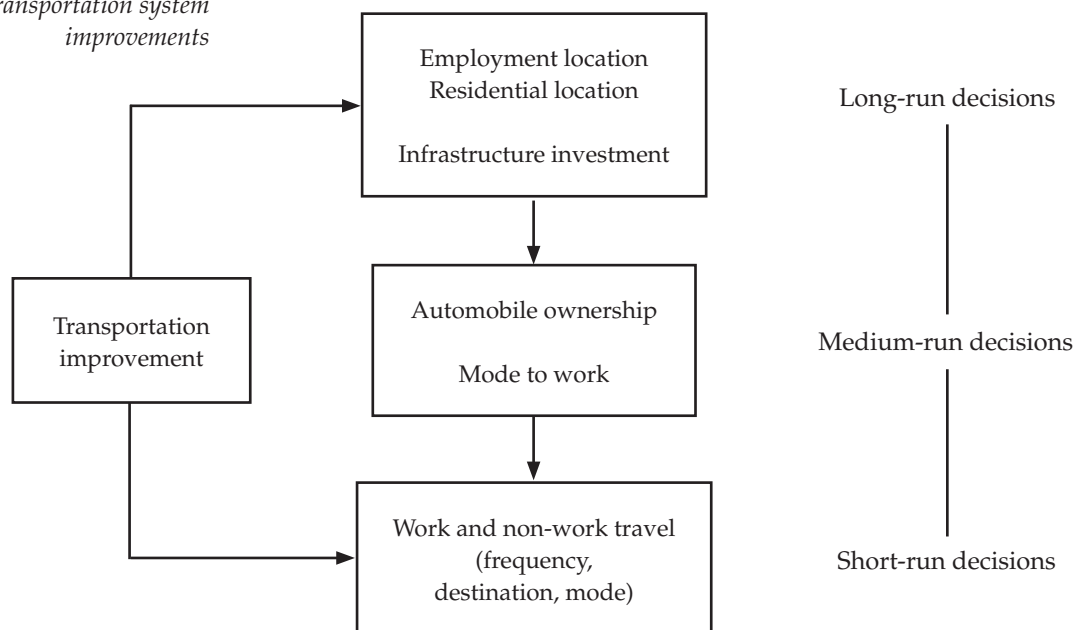
Figure 3-3 provides a useful framework with which to understand the determinants of the long-run growth in VMT. It starts on the left with a transportation improvement that decreases per-mile travel costs, usually by decreasing travel times. For example, a new freeway link could double average travel speeds. The figure shows three time frames and the decisions that affect travel that are made in each of those time frames: the short, medium, and long term.

Figure 3-2. Growth in vehicle-miles traveled (VMT) in the U.S., 1940–2004



Source: Federal Highway Administration, *Highway Statistics*, Summary to 1995, and 1996–2004.

Figure 3-3. Short-, medium-, and long-run response to transportation system improvements



Source: Adapted from Ben-Akiva 1985

In the short term, the pattern of car ownership and the locations of origins and destinations are fixed. One can imagine the effects of opening a new freeway link in an urban area in the 1960s. Travelers (1) change routes from arterial streets to the freeway; (2) change modes from mass transit to cars; (3) change the timing of their trips from off-peak to peak; (4) make additional trips; and (5) make more longer trips. As travelers switch to the improved route, congestion decreases on other routes.

Within a short period (days or weeks, not years) travelers establish a new equilibrium traffic pattern. The new trips increase the total volume of traffic on the system¹² but not by enough to bury all of the new capacity in congestion. In the short run, the new urban freeway decreases traffic congestion and travel times, increasing the total number of trips, and decreasing trips on nonhighway modes.

In the medium term, probably up to a year or two, a large proportion of householders can alter their access to a car. Faster travel makes commuting to work by car more desirable relative to other modes, such as bus or commuter rail. A higher proportion of commuters will choose to buy a car, or a second car, and drive to work. The new capacity also decreases the cost of nonwork car trips, especially during off-peak hours. A second car offers the flexibility to take advantage of those cheaper trips. Over a year or two, the new capacity encourages increases in the number of cars and trips per household, resulting in a substantial increase in VMT.

In the long term, the improvements in transportation infrastructure lead to changes in land development patterns that increase the volume of traffic on the system because: (1) both households and businesses already in the urban area can relocate to lower-priced but accessible areas that require longer-distance trips, and (2) the population of the urban area grows in part because people from other areas take advantage of the lower-cost transportation.

Travel time is an important criterion in household and business location decisions. High-speed freeways allow longer trips in the same amount of time. Households can consider a move to new developments at the urban fringe or to suburbs farther from work but with local public services more to the householders' liking. Land in suburban locations is cheaper, which encourages householders to purchase larger lots. Residential densities generally fall, which means that the same number of householders travel longer distances to reach the same destinations.

Businesses also take advantage of new highway capacity to locate in the suburbs. Retailers move to be closer to their customers and look for locations with good visibility. Manufacturers take advantage of relatively low-priced land with good access to suburban freeways and the ability to locate closer to their workforce. Service providers move to the suburbs to take advantage of lower-priced space and proximity to both customers and workforce. This by itself tends to reduce trip lengths.

But households, given further improvements in suburban freeways, take advantage of their suburban workplaces to move even farther into the suburban fringe. The improvements in the transportation system in well-located metropolitan areas encourage (or, at least, allow) in-migration from other areas.¹³ In the new equilibrium, most trip times can be reasonably short, most not exceeding 30 minutes, but the tremendous increase in highway capacity allows many more relatively long trips.

The big increase in VMT coincides with what must be the largest and widest-ranging increase in the quality of urban life in history. New highway systems and urban roadway improvements played a crucial role in the post-war economic boom. Though many other factors fueled the economic growth in that period (such as technical innovations that decreased the costs of production of many goods), the highway system provided the access to

The big increase in VMT coincides with what must be the largest and widest-ranging increase in the quality of urban life in history. New highway systems and urban roadway improvements played a crucial role in the post-war economic boom.

the relatively low-priced land that both firms and households desired. These and many other improvements in production fed back to increase incomes for workers, which households used, along with access to lower-priced land, to create comfortable suburban lifestyles.

Those big improvements had big costs. The costs to the public of construction and maintenance of the highways themselves are considerable, as are the private capital and operating costs of cars and trucks. In addition, the cost of urban development is the lost value of land in farm and natural uses.

So what are the *net* effects? Car and truck operators are, at least on average, clearly willing to bear construction and maintenance costs, including the costs of purchasing land at market prices. Urban land values usually far exceed the value of the land in agricultural uses.¹⁴ That simple analysis suggests that the enormous public investment in transportation infrastructure has generated large net benefits. Put another way, all this travel by car and truck costs a lot, and on average it is well worth it.

But the net benefit is not as big as it could be. A variety of external costs reduce the efficiency of the system: time wasted sitting in gridlock, car and truck emissions, and unsightly and environmentally damaging urban sprawl. We take up these issues in the next major section. That analysis does not change our conclusion that the highway system generates net benefit, but it does reveal that the net benefits are smaller than the simple analysis suggests.

Conventional funding practices have two main effects on the highway system:

- 1. Decisions about the allocation of revenues for construction and maintenance have to be made bureaucratically.*
- 2. Indirect funding fails to generate the price signals that the users of congested highways need to use highway capacity efficiently.*

The Effects of Highway Funding Practices

Conventional funding practices have two main effects on the highway system. Both of these effects result from the failure to directly price (i.e., toll) access to the system itself. First, decisions about the allocation of revenues for construction and maintenance have to be made bureaucratically: institutional arrangements affect the characteristics of the transportation system. Second, indirect funding fails to generate the price signals that the users of congested highways need to use highway capacity efficiently. We discuss the first of these effects in the remainder of this section and the second effect in the following section in the context of negative externalities.

The indirect funding of highways means that federal, state, and local transportation agencies face a more difficult allocation problem than that faced by private businesses. Revenues guide the decisions of private producers of goods and services: they increase production of the goods and services that sell well and distribute them to the places where they sell well. Because the government collects fuel taxes from the sellers of fuel, the revenues themselves at best give only a rough idea of where the fuel gets burned. Even if fuel use could be tracked precisely, fuel use corresponds only roughly with the value travelers place on the journey.

Most planning agencies instead use some form of a priority index to rank proposed projects. The index typically consists of the sum of points assigned to various criteria that indicate the benefit of the project in terms of the objectives the agency sees as valuable. For example, transportation planners might assign from 0 to 10 points to each of several conditions—level of congestion, safety, condition of the pavement, and construction cost—to arrive at a priority index that can range in value from 0 to 40. The higher the project's index, the higher the priority for the project. A project receives funding based on its priority index, the size of the initial planning agencies improvements budget, and, of course, political factors.¹⁵

The way that federal funding comes back to states for allocation to projects affects how transportation projects are identified and ranked. The funding system established over the course of federal highway legislation from 1916 to 1956 elevated the role of state DOTs relative to that of local or metropolitan planning organizations. The objectives of the state DOTs often differ from

those of local planning organizations. For example, state DOTs tend to be concerned about the performance of the highway itself—speed, safety, durability, etc.—whereas local government agencies are often also concerned about local economic development and land-use impacts. Differences in objectives affect project ranking.

Planners would, of course, like to rank projects as rigorously and consistently as possible. Large projects require hundreds of millions of dollars to complete, contribute to travel woes during the construction period, and have significant long-run effects on the quality of urban life. Thus, since the mid-1960s, researchers, funded by the same federal programs that fund highway construction, have worked to develop computer programs to simulate traffic flows in metropolitan areas. Transportation planners in most urban areas use some form of a four-step urban transportation modeling system, described here.

The standard four-step approach consists of a series of computer models that yield a simulation of *equilibrium* traffic flows on different routes and modes at various times of the day, week, or year. Sophisticated variants of these models yield more than a general description of the effects of changes in system characteristics: they give a detailed accounting of changes in traffic flows by route and mode. Planners use these models to simulate the effects and to estimate the benefits to drivers of proposed changes in the system.¹⁶

But even the best ranking and modeling systems yield estimates of the net benefits from alternative projects that are relatively crude and sometimes (because of poor theory, data, and analysis) worse than the informed guesses of traffic experts.¹⁷ In the absence of tolls, one cannot directly observe travelers' willingness to pay for travel on the highway. As we show in Appendices C and D, and in Chapter 4, willingness to pay is a measure of the benefits produced by transportation improvements: any estimate of benefits is crude without that measure.

For these reasons, and others we note in the next section, the bureaucratic allocation system has resulted in a highway system that at this point is probably overbuilt in most metropolitan areas: capacity in suburban areas may be too high (it has cost more than travelers would have been willing to pay for it) even where congestion is severe. We explore that topic more in Chapter 4.

But even the best ranking and modeling systems yield estimates of the net benefits from alternative projects that are relatively crude and sometimes (because of poor theory, data, and analysis) worse than the informed guesses of traffic experts. In the absence of tolls, one cannot directly observe travelers' willingness to pay for travel on the highway.

TRANSPORTATION POLICY DURING AND AFTER THE HIGHWAY BOOM: TREATING NEGATIVE EXTERNALITIES

The vast improvement in mobility made possible by roadway policies in the 1950s and 1960s did not last, and it came with some unanticipated costs. Many transportation planners in the 1950s expected that added capacity would eliminate congestion for a long time. Had the populations of most metropolitan areas remained unchanged over time and businesses spread instead of clustered, that expectation might have matched reality.

But economic forces thwarted engineering visions of speedy travel. As pavement increased, so did congestion. The Texas Transportation Institute, in its *2005 Urban Mobility Report*, found that from 1993 to 2003:

- Peak-period¹⁸ trips took an average of about 7 percent longer.
- Travelers spend 47 extra hours per year in travel compared to 40 hours in 1993.
- The percent of congested freeway mileage has grown from 51 percent to 60 percent over the period.

As VMT increased, so did the impact of transportation on other activities. In many cities, trucks and cars contribute the bulk of urban noise and air pollution. A heavy truck causes many times the damage to highways that a passenger vehicle does. Automobile accidents increase the demand for

a wide range of services, such as police, emergency services, attorneys, and courts. Traffic increases the hazards of walking and cycling, while diminishing the comfort, convenience, and amenity of those modes; about 17 percent of the fatalities that result from auto accidents are cyclists and pedestrians. Where a driver parks can be a nuisance to others. And the sprawling, low-density development that suburban freeways encourage has a variety of negative environmental effects.

Most of the current debate about transportation policies (and, to a large extent, land-use policy) is about how to reduce both congestion and the various other spillover costs (also called external costs) of transportation and urban sprawl.¹⁹ In this section we begin with a brief economic analysis of externalities, then apply that analysis to the transportation and land-use policies of the last 30 years. Refer to Appendix B for a more technical discussion of externalities and their impacts on market performance.

What Are Externalities and How Do They Distort Prices?

A variety of activities generate costs or benefits that spill over onto people not directly involved with the activity. Cars and factories generate noise and air pollution. The traffic generated by large retailers provides benefits to other nearby retailers but imposes costs on nearby households. Drivers on a congested highway slow other drivers down by forcing them to reduce their speed to avoid accidents as following distances decrease. Pedestrians and cyclists risk injury traveling along busy streets. Quality building design not only contributes to the satisfaction of a building's occupants, but also to the quality of the entire neighborhood. Urban areas teem with spillover costs and benefits.

Economists call these spillovers externalities. The reason is that their effects tend to be "external" to decisions being made about the activities that generate them. A driver entering a congested highway, for example, tends not to think much about the impact of his trip on everyone else's travel speeds. He makes his decision to travel based on the costs and benefits he experiences, largely ignoring the effects his trip has on others.

External costs and benefits reduce the efficiency of the market system. To see this, consider the textbook case of a factory that pollutes the air. A factory owner has to pay market prices for most of its inputs: labor, materials, and machinery. Those prices reflect the opportunity costs of using those resources. Prices provide the factory's managers with strong financial incentive to use resources as cost effectively as possible; production costs are as low as technically possible. Competition forces suppliers to price their products close to their full production costs. So goods are produced cost-effectively and priced near cost, which is good for consumers. So far, things are operating as they should.

Inefficiency arises if some inputs (i.e., resources) are not priced at their opportunity cost. Many factories continue to dispose of waste into the atmosphere at no charge (though regulations currently limit these emissions). These emissions impose costs on everyone in the air shed, many of whom have no direct ties to the factory. Because there is no charge on emissions to the atmosphere, the factory owners have no incentive to economize on emissions, so they emit too much. Moreover, their total production costs are lower than the total costs to society of that production, so product prices are inefficiently low. Factories produce and consumers buy too much of the good.

This discussion suggests an alternative definition of an externality: a cost or benefit not priced in the market. Because it's not priced, decision makers can ignore it or take advantage of it. The result is too much production when there are external costs and too little when there are external benefits.



Stephen Strathdee, istockphoto.com

A variety of activities generate costs or benefits that spill over onto people not directly involved with the activity. Cars and factories generate noise and air pollution.

Why is it that some resources are available at no charge (i.e., are not priced)? The problem in essence is usually one of poorly defined or enforced property rights. The atmosphere is a scarce resource: Who “owns” it? Maybe no one. Or maybe we all own it collectively. Lacking the rights of ownership, no individual has incentive to look after the resource or charge for its use. The atmosphere has historically been an “open-access” resource: anyone can use it for any purpose he likes at no charge.²⁰ One use is breathing. Another use is waste disposal. Those uses conflict.

Most of the highway system is similarly open access. Yes, transportation agencies are charged with keeping highways in good condition. And, yes, freeways are “limited” access in that one can enter the highway only at a limited number of interchanges. But they are, nevertheless, *freeways*: drivers are not charged directly for their use of the highway. That would be OK from an efficiency standpoint if highway space was not scarce. But space on congested highways is scarce: each additional driver imposes time costs on other drivers. This cost is an external cost that individual drivers tend not to consider. So, too many drivers enter a congested highway. And though DOTs attend to maintenance, they do not do anything directly on a day-to-day basis to limit the number of vehicles entering the highway. The highway is allocated strictly first come, first served.

What would happen if external costs (i.e., access to scarce resources) were priced appropriately? Then the users of those resources would have strong financial incentive to economize on the use of those resources. The cost of that use would capitalize into the price of the goods and services brought to market. These higher product prices would generate lower sales. Would there still be pollution and traffic congestion? Yes! But not as much.

Note that the efficient amount of production can vary city by city based on people’s preferences. It can vary in the same city over time as preferences change, which can happen with changes in income, education levels, and activities.

The same conclusions follow for other urban externalities, such as noise, auto pollution, or accident risk. The rule of thumb is that, if all the costs are internalized, a harmful activity will be increased only if the marginal social benefit exceeds the marginal social cost. This is efficient in that it maximizes the total *net* benefits from economic activity.

What is the magnitude of the external costs of car and truck travel? Parry and Small (2005) constructed a sophisticated economic model to estimate the gasoline tax needed to cover the social costs of infrastructure, congestion, auto air pollution (global and local), and safety. Their model takes into account the long-run effects a higher gas tax would have on both VMT and fleet fuel efficiency. Using the best estimates of various costs, such as time costs, they estimated the optimal tax at about \$1 per gallon in the U.S., or about two-and-half times current state and federal gas taxes combined.

Put another way, the *average* social cost that car and truck traffic imposes is equivalent to about \$1 per gallon of gas consumed. The largest component of this cost is the time cost of congestion. By comparison, the optimal tax is about \$1.35 per gallon in the U.K., about half the current tax rate there. The difference in optimal tax rates across the two countries is mostly a result of differing congestion levels.²¹

Parry and Small note, however, that increasing the gas tax is not an effective means with which to internalize most of the external costs. A higher gas tax encourages households to purchase and consume less gas—probably a good way to reduce air emissions. The higher gas tax reduces congestion and accident risk indirectly through its effect on VMT and on the size and power of cars. But Parry and Small estimate that taxing VMT directly would have about *four times* the benefits of the recommended increase in the gas tax



Daniel Stein, istockphoto.com

The atmosphere has historically been an “open access” resource: anyone can use it for any purpose he likes at no charge. One use is breathing. Another use is waste disposal (like fuel emissions). Those uses conflict. Ask Los Angelinos.

because many external costs are more directly related to VMT than to fuel consumption. We explore the reasoning behind this in the next section.

Government Response to Traffic and Emissions Externalities

The obvious policy response to the inefficiency caused by an unpriced resource is to price it. This means: pricing access to the atmosphere for waste disposal through a charge per unit of emissions (e.g., a carbon tax); pricing access to a highway with congestion tolls and/or a VMT tax. As we describe in detail in Chapter 4, the technological constraints to pricing transportation externalities have dropped significantly in the last decade. The political difficulties appear also to be diminishing, probably due to the growing recognition of the shortcomings of conventional policies.

Conventional transportation policy toward traffic congestion and other auto-related external costs has mostly been to try to eliminate them. North America's resources are vast. The primary economic objective since Europeans began to access the Americas in the 1500s until essentially the 1960s and 1970s has been to gain access to and develop these resources. Rather than find ways to ration scarce resources, the approach—for centuries—has been to find more of them or to find technologies to get more out of them. While this probably should be the first course of action, it has (or should have) limits.

In the case of traffic congestion, efforts to create more capacity or to get more from existing capacity have taken three forms:

1. *Inventing clever methods for getting more VMT from the same highway capacity.* These methods include on-ramp signals that reduce the “turbulence” caused by cars entering a congested highway, lanes on which the direction of travel can be reversed between morning and evening rushes, onboard computers that use GPS systems to warn drivers of congested areas and identify alternative routes, and advanced systems for quickly removing stalled vehicles from travel lanes.²²
2. *Building more highway capacity.* The interstate highway system was being expanded in most growing metropolitan areas even before the entire system was completed in the early 1990s. Expansion takes primarily two forms: (1) adding lanes to existing highways, and (2) building more highways, usually bypasses and beltways in less-intensively developed suburban and exurban areas.
3. *Expanding the capacity of alternative modes of travel.* A seemingly obvious way to reduce traffic congestion is to encourage the use of mass transit, walking, and cycling. The Intermodal Surface Transportation Efficiency Act (ISTEA) of 1991 and its reauthorization in 1998 and 2004 increased funding for mass transit and provided the flexibility to shift federal transportation funds from highway construction and maintenance to funding of infrastructure for other modes. By 1999, about 7 percent of eligible funds were shifted, 82 percent of which was spent in five states: New York, California, Pennsylvania, Oregon, and Virginia (Dilger 2003, Chapter 3). The general trend of spending on highway expansion and maintenance continued, but mass transit ridership ended its decline in 1996 and started upwards in 1997.

The other conventional approach to the treatment of externalities is regulation. Regulation involves bureaucratic, as opposed to market, allocation of resources. A regulatory approach to combating traffic congestion, for example, might include banning cars with odd-numbered license plates from driving on certain days. Such a policy will reduce traffic congestion (at least for awhile) but allocates valuable highway space crudely.²³ As with pricing, regulation has not generally been used to combat traffic congestion. The primary mechanism for allocating cars to highways has

The obvious policy response to the inefficiency caused by an unpriced resource is to price it. This means: pricing access to the atmosphere for waste disposal through a charge per unit of emissions (e.g., a carbon tax); pricing access to a highway with congestion tolls and/or a VMT tax.

been and continues to be “first-come, first-served” with no regulatory limit on the number of trips.

Regulation is ubiquitous, however, in land use. Control of land-use externalities provides the main rationale for land-use regulation. Traffic congestion could be, and often is, included among these externalities. Indeed, some state DOT officials see traffic congestion primarily as a failure of land-use regulation rather than as a failure to somehow limit access to highways (e.g., through pricing or regulation). As Figure 3-3 shows, households and businesses respond to improvements in the transportation system over time by changing locations. These changes in location tend to increase VMT on the improved highway. Regulations that prohibit more intensive land development discourage a major source of traffic congestion.

The problem is that controlling congestion with land use requires strong regulation. Consider three examples:

1. *Low-density zoning.* Low-density zoning serves a variety of positive purposes. It reduces the negative (and positive) spillovers associated with density, inhibits free-riding on local public goods and services, prevents the higher-density development that accessibility encourages and that generates traffic congestion on open-access highways. Thus, a common objective of low-density zoning is to limit traffic congestion.

That policy also has some negative consequences. Businesses enjoy significant cost advantages when located in business clusters. Low-density residential development implies longer average trips for workers and customers to these clusters. While inhibiting growth in the number of trips, low-density zoning increases the average length of trips, which increases traffic congestion on the highways that serve the business clusters. In addition, low-density zoning directly encourages urban sprawl by encouraging businesses to locate closer to their workers and consumers in low-density suburban areas.

2. *Limits on the spatial extent of land development.* An explicit urban growth boundary (UGB), or simply strong agricultural or natural resource zoning outside of urban areas, prohibits or discourages exurban development in response to highway expansion. This type of policy addresses urban sprawl and the “problem” of the long-run land-development response to a highway improvement.

It also creates a land-value differential at the zoning boundary: land owners inside the boundary enjoy an economic windfall as developable land values rise with the constraint on land supply; land outside the boundary suffers a corresponding economic “wipeout.”²⁴ Longer term, the constraint on land supply encourages higher-density urban development, which limits sprawl, but increases conflicts among the now more-proximate land users, including increasing congestion on local streets.

An advantage to this higher-density development, however, is that mass transit thrives on density. Metropolitan areas that have adopted a UGB, such as Portland, Oregon, show that the UGB combines well with a commitment to relatively high-quality mass transit. Highways remain congested because access to highways remains open, but sprawl is reduced and more trips are completed on mass transit.

3. *Limits on floor space or dwelling units.*²⁵ In the extreme, the local government can limit the effect of a highway improvement on land development simply by limiting development. A combination of ceilings on dwelling units and on the amount and location of developable land could provide the conditions necessary to build a congestion-free transportation system. Indeed, it is in this sense that traffic congestion can be considered a failure of land-use policy.

Some state DOT officials see traffic congestion primarily as a failure of land-use regulation rather than as a failure to somehow limit access to highways (e.g., through pricing or regulation).

A disadvantage of this system is that it makes the per-capita cost of transportation infrastructure inefficiently high: the cost of “marginal” infrastructure exceeds its value in time savings because some congestion is efficient. Another disadvantage is the lost cost savings that some businesses achieve by locating in a relatively densely developed cluster.

More fundamentally, the tight control on land development represents a shift from a market to an administrative allocation of resources. Rather than affecting the allocation of resources directly, market forces must channel through a central decision-making body, such as a zoning commission or city council. Essentially, a significant cost of treating traffic congestion with land-use regulation is the loss of the advantages a market system offers relative to an administrative system.

Regulation is also the dominant public policy response to auto-related environmental and safety externalities. The first step typically involves the determination of target improvements in environmental or safety outcomes. The cost and revenue data available to private business managers are usually not available to regulators, so some other criteria are typically used to set the targets. For example, the Environmental Protection Agency sets ambient air quality standards primarily by assessing each pollutant’s effect on health.²⁶ In practice, the rationale for criteria varies, and politics play a role.

There are at least three general ways regulations are implemented to pursue the chosen targets:

1. *Impose equipment requirements.* For example, all new cars are required to come equipped with specified emissions-control technologies, such as catalytic converters. Similarly, all cars are required to come equipped with seat belts and driver- and passenger-side airbags that meet minimum specified requirements.
2. *Let the market meet the standards.* For example, the Corporate Average Fuel Economy (CAFE) standards first adopted in 1975 specify the minimum average fuel economy for an automobile manufacturer’s fleet of passenger cars and light trucks. The company decides how it will meet the standard. In general, performance standards put the onus on the car maker or car owner to ensure that the car meets the specified standards. This approach has an advantage inasmuch as it creates incentives for car makers and automotive service suppliers to compete to find cost-effective and consumer-friendly ways to meet the standard. It brings a far greater number of minds to the problem. The challenge for the public agency shifts from specifying equipment to setting up testing facilities to ensure that the car maker or user has in fact taken steps adequate to meet the standard.
3. *Consumer education about externality-related performance characteristics of a car or car component.* For example, the EPA provides fuel economy data posted in the windows of new cars for sale, and it and the U.S. Department of Energy list fuel economy and air emissions statistics for most cars. Similarly, the Insurance Institute for Highway Safety and the National Highway Traffic Safety Administration crash test cars and light trucks and publish crash-test ratings. Car makers include favorable ratings in their marketing campaigns.

An important source of market failure is limits in the information readily available to consumers about important product characteristics. As Ralph Nader pointed out in his 1965 classic *Unsafe at Any Speed*, car makers trying to minimize production costs have incentive from a marketing perspective to at least not emphasize the vehicle shortcomings made necessary to achieve cost savings. The market therefore may supply less information than that necessary for consumers to make well-informed decisions.

An important source of market failure is limits in the information readily available to consumers about important product characteristics. As Ralph Nader pointed out in his 1965 classic Unsafe at Any Speed, car makers trying to minimize production costs have incentive from a marketing perspective to at least not emphasize the vehicle shortcomings made necessary to achieve cost savings.

Car buyers have over time revealed themselves to be ready consumers of crash-test ratings and to a lesser extent environmental ratings. In both cases, however, the externalities problem is still there. Consumers tend to worry more about their own safety than that of others, and emissions mostly affect other people. Nevertheless, good information is critical for good market outcomes.

The Effects of Policies that Treat Traffic Externalities

What are the likely effects of the transportation policies that combat congestion by adding transportation capacity or trying to get more out of existing capacity?

Increasingly efficient use of transportation infrastructure. The pressure to reduce congestion quickly and cheaply has led to the implementation of some clever and effective ways to make better use of existing highway infrastructure. These efforts generally make good economic sense because their benefits in terms of numbers of trips and travel times are large relative to their costs. The problem is that businesses and households respond to the reduced congestion in ways that create more congestion. Benefits go up, but congestion may not be affected much (see Appendix C).

Too many vehicle trips. Access to most highways remains essentially open (e.g., unpriced), so highway space is allocated on a first-come, first-served basis. Too many people drive and open-access highways (especially those that serve business clusters) suffer the “tragedy of the commons.”

Too much pavement? This effect is hard to pin down. On the one hand, one should expect the implicit policy of building our way out of congestion to generate too much pavement. Pavement in congested areas is usually expensive, however, and the budget is not unlimited: most highway construction is funded with gas taxes. Parry and Small’s (2005) estimates indicate that gas taxes are too low to cover the costs of auto-related externalities, especially the time cost of congestion.

As we demonstrate in the next chapter, an optimal congestion toll pays for the optimal highway. If drivers paid for travel on a highway like they pay through higher peak-time prices for travel on an airplane, the highways that serve major employment centers could well be bigger than they already are. So, a lot of pavement may not be too much in some places.

But much of the new capacity in congested metropolitan areas is being built in suburban areas where land is cheaper and construction costs are lower. Maybe there is too little pavement in congested centers and too much in beltways and bypasses. One can easily imagine transportation officials in the absence of price signals focusing more on the low costs of additional capacity in suburban or exurban areas than on the high net benefits in serving employment clusters.

Too much urban sprawl? Bypasses, despite their logical objective of separating through traffic from local traffic, encourage businesses and households to move from established centers to locations near the bypass.²⁷ This encourages inefficient sprawl if money would be better spent improving infrastructure to supply more trips to existing centers.

This is especially the case if more space-efficient transportation modes (e.g., mass transit) could serve the existing centers effectively. Bypasses encourage sprawl because they reduce transport costs to suburban areas. They also encourage sprawl simply because they encourage travel by car, and cars require a lot of space for travel and parking. Congestion on suburban highways is unpriced, which encourages more trips and more sprawl.

On the other hand, bypasses and beltways evolve over time into essentially a grid of high-speed highways. This grid provides the transportation basis for the efficiency of a polycentric urban area—as described in Chapter 2, multiple centers connected by good transportation infrastructure provide greater net

The likely effects of transportation policies that combat congestion by adding capacity or trying to get more out of existing capacity are:

1. *increasingly efficient use of the infrastructure;*
2. *too many vehicle trips;*
3. *too much pavement (maybe);*
4. *too much urban sprawl (again maybe);*
5. *too few trips by transit and other modes; and*
6. *perhaps too much travel.*

benefits than a single center. Bypasses impose significant adjustment costs on existing centers and allow and encourage low-density development, but low-density development is not necessarily bad.

Too few trips by mass transit or other nonauto modes. The federal government offers subsidies to transit to reduce congestion for highway users; to improve the mobility of people who cannot afford or are otherwise unable to drive; and to internalize the space-saving, fuel-efficiency, and safety of transit. Transit does not, however, have significant long-term effects on traffic congestion because access to highways remains open.²⁸ Even with below-cost transit, transit ridership is probably inefficiently low because car travel is underpriced and because the characteristics of suburban land development and transportation infrastructure inhibit access to transit.

These same characteristics—low development densities, wide busy arterial streets, and large parking lots—discourage trips by foot and bicycle.

Too much travel? Unpriced congestion clearly generates too many vehicle trips. Pricing trips on congested highways would encourage ride-sharing and travel on transit. So whether there are too many person-trips on highways is unclear. Subsidizing mass transit encourages more person trips. On the one hand, unpriced highways and underpriced transit suggest too much urban travel. On the other hand, there are likely too few trips by foot or bike.

What are the likely effects of the land-use policies used to combat congestion?

More congestion. As we argue in Chapter 2, it seems highly likely that standard land-development regulations that separate land uses, limit densities, and mandate wide streets and large parking lots contribute to traffic generation and congestion. Comprehensive growth management policies—strict limits on density combined with a tight UGB—could work to reduce congestion; the combination reduces both the number of people and the length of trips. Any less-comprehensive changes in land-use regulations do not reduce traffic congestion because the fundamental cause of traffic congestion is open access to the highway system rather than land-development patterns.

More urban sprawl. Standard residential zoning contributes significantly to sprawl. Changes in a variety of these regulations—density limits, street and parking characteristics, building codes—could reduce sprawl itself and treat its most unpalatable characteristics. A main cause of urban sprawl is the low-cost travel made possible by suburban highways. Proper highway pricing would reduce both congestion and sprawl.

What are the likely effects of the policies used to combat auto pollution and injuries?

Fewer injuries and deaths. This conclusion is not as obvious as it appears. One would expect drivers to be concerned about their own safety and express that concern through willingness to pay for safer cars. Today, safety features help sell cars, but that was much less the case in the 1960s. The reasons must be limited consumer information about the risks and a tendency for consumers to skimp on expensive safety measures they are unlikely to need. Regulations and public information campaigns have contributed to the considerable improvements in auto safety features.

Still too many auto-related injuries and deaths. There are several reasons the number of injuries and deaths are inefficiently high. First, too much VMT generates too many traffic accidents and injuries because the number of accidents is positively correlated with VMT. Second, drivers' care about their own safety reduces accidents, but they do not consider additional external costs. For example, the size and weight of a big new SUV may protect its occupants in a crash but increase the risk to those in other cars. Third, insurance may reduce the incentive to drive safely. Insured consumers bear far less than the full out-of-pocket cost an accident imposes both on themselves

The likely effects of land-use policies used to combat congestion by separating uses and limiting densities are:

1. more congestion; and
2. more urban sprawl.

and on the other drivers involved. Insurance thereby reduces the “price” of risky behavior: driving too fast, passing too often, and so on. But that effect is not unambiguous. Insurance premiums are actuarially calculated as a probability times the cost of an accident. That probability, and therefore an individual’s insurance premium, rises with speeding tickets and accidents. The threat of rising premiums means less and safer driving.

Still too much auto pollution. The case for a positive impact of pollution regulations and information campaigns is somewhat stronger. Car owners have little individual incentive to pay extra for cleaner cars, so regulations and standards matter, but car companies and consumers find ways around the regulatory constraints. For example, light trucks face lower fuel-consumption standards than cars. The car companies have responded by designing light trucks (i.e., SUVs) that get poor mileage but have strong appeal to suburban consumers. Market-driven technological innovations that increase engine efficiencies and lengthen maintenance cycles have been the main contributors to cleaner cars.

CHAPTER 3 NOTES

1. Clearly, some land-development policies influence transportation system characteristics. Subdivision codes, for example, usually include street and parking standards. We recognize that a clean distinction between land-use and transportation policy simply does not exist.
2. See, for example, Chapter 1 of Dilger (2003) for a more detailed description of early transportation policy.
3. Labbe (1982) provides an interesting account of the development of public transit in Portland, Oregon, a story similar to that in many American cities. Typically, land developers would obtain a franchise from the city government to build a transit line from the city center (or from existing transit lines) to land they planned to develop.
4. A recent regulatory innovation is to decompose the components of service provision and regulate only the component with the characteristics of natural monopoly. For example, it’s the transmission of electricity, and not its generation, that has the characteristics of a natural monopoly. So electricity generation has become deregulated, while the pricing of access to the electricity grid remains regulated. Analogously, a local government could decouple ownership and regulation of the rails from ownership of the streetcars themselves. Currently, however, monopoly pricing is usually not a concern in rail mass transit because of competition from other modes.
5. One method of financing might be preferred over another if the costs associated with originating the loan or collecting the money to pay off the loan differs across methods. Economists refer to these administrative costs as “transactions” costs. Our analysis above assumes that transaction costs are similar across methods of finance.
6. As we show later in the context of traffic congestion, externalities and public goods are closely related problems.
7. Clearly, the government does not have to supply public goods directly. It just needs to collect the revenues necessary to pay for either public or private supply of the goods and services.
8. There is another interesting rationale for paying for local public goods and services with a tax on the value of housing: the net benefits from provision of the public goods and services capitalize into house prices. As described in Chapter 2, mobile households compete for houses in service districts that offer a desirable package of public goods, services, and amenities. That competition bids up house prices.
9. Private businesses decide how much to produce based on the revenues generated from sale of the product in the market. As revenues increase, so does production. Thus, people reveal their willingness to pay for the product by purchasing it at the price established in the market. Since people cannot be excluded from using or consuming some public goods, the government has difficulty using price as a signal to guide production decisions.
10. Receipts in 2004 totaled \$107 billion. Governments actually spend more than this on transportation-related services. Some public agencies—local police, fire, attorney, courts, and other social services—provide services related to transportation. The Federal Highway Administration does not attempt to estimate the proportion of the cost of providing these services that result from transportation.

11. That language, though cumbersome, is intentional. Even though highway users may not pay the full cost through taxes and fees directly related to highways and streets, those additional costs get paid somehow and probably by people that also use the highways. But they are collected in other ways.
12. Referred to as induced demand and explained in Appendix C.
13. Though the information is only anecdotal, many relocations by households and firms are strongly influenced by quality-of-life issues: congestion is always at or near the top of the list of the problems they hope to escape. Appendix A (Figure A-2) discusses in more detail the factors that drive regional growth.
14. Complicating this are the subsidies that both urban and rural land owners receive.
15. See Appendix D for a more detailed discussion of project evaluation and selection.
16. See Appendix E for a more detailed description of transportation models.
17. In that context the usually extensive process of model calibration can be seen as an exercise in changing the model until it finally produces the results that common sense and expert judgment deem reasonable.
18. In most metropolitan areas, the idea of “rush hour” is obsolete—congestion happens for multiple hours on both morning and evening weekdays.
19. As we noted in Chapter 2, cars and highways also reduce the negative external effects of some activities by allowing people to put distance between the source of the externality and their homes.
20. Regulations clearly limit access to the atmosphere. Air-quality regulation takes a variety of forms, is complicated, and a complete discussion is beyond the scope of this book. We do, however, give a sense of the regulatory approach shortly in our discussion of policies to control auto air emissions.
21. Interestingly, the gas tax in the UK is much higher than the estimated optimal amount and the gas tax in the U.S. is much lower.
22. We discuss these Transportation System Management (TSM) techniques in more detail in Chapter 6.
23. Of course, over time many households may be able to take costly measures to overcome the regulatory constraint: diversify license-plate numbering across vehicles and drive to work in the car that has the correct license plate.
24. Or, if the boundary has been established for a long time, land values outside the boundary are not wiped out, they are just kept from increasing in value as rapidly as they would have otherwise.
25. Shoup (2005a) points out that limiting the number of parking spaces, and pricing them accordingly, would also have large effects on highway travel; every vehicle trip requires a parking space. Conventional land-development policy, in contrast, requires sufficient parking to accommodate peak demand at zero price. More on this in Chapter 4.
26. The EPA is forbidden by statute to consider the costs of attaining the ambient air quality standards.
27. This process not infrequently leaves blight in its wake. See Edwards, (1991) for interesting case studies of the negative effects of bypasses on land development.
28. That’s not to say that transit cannot reduce congestion. Giuliano (1992) reports that transit proved effective, for example, in reducing congestion during the summer Olympics in Los Angeles. The effects, however, were short run. Giuliano reports that even before the Olympics ended, traffic congestion had increased as people returned to the highways in response to the news that the Olympics had not caused the expected increase in congestion.



Transportation planning in the 20th century was getting people from here to there. In the 21st century, the challenge will be about getting there here.

—Carl Hosticka, Councilor, Metro, Portland's regional government

If we want to show our commitment to good transportation policy, going the extra mile won't help.

—Terry Moore, FAICP

SECTION II.

EVALUATING URBAN TRANSPORTATION AND LAND-USE POLICIES

Section I uses a historical description of urban development and policy to introduce important principles for understanding how transportation and land-use development interact in a metropolitan area. Those principles are the basis for a more thorough treatment of transportation and land-use policies in Section II. These policies fall along a continuum from direct (tell people what to do) to indirect (change prices and then let people do what they want).

Chapter 4 sets the stage for a more detailed evaluation of public policies that can change the structure and performance of a metropolitan transportation system. It explains why urban transportation problems derive in large part from a failure to fully estimate and charge users the cost of providing transportation facilities, and how the efficiency of the transportation system could be increased by getting the prices right. As economists, we tend to think of pricing policies as *ideal policies*: they directly attack the roots of the problem.

Despite the likely effectiveness of pricing policies, no federal or regional agency with responsibility for land use, environmental quality, or transportation has implemented pricing in a comprehensive manner. The barriers to pricing are more political than technical. Policy makers have been more likely to deal with the efficiency issues through policies that directly address the perceived resource misallocation (e.g., with zoning, growth boundaries, pollution standards, requirements for pollution control equipment, light rail transit, auto mileage requirements) than with correcting the improper pricing that is the root cause of the misallocation (e.g., with emissions fees, congestion pricing, parking meters, and development fees).

Chapters 5 and 6 accept that direct pricing is not a mainstream idea, and that it has some technical and political hurdles to clear that make other, traditional policies a lot easier for regions to adopt now. Chapter 5 focuses on land-use policies; Chapter 6 on transportation policies.

Because most of these policies do not, in general, aim directly at bringing prices into line with costs, economists tend to classify them as *second-best policies*, which clearly shows their professional bias. But they have a point: (1) resources are allocated inefficiently when prices are wrong (e.g., because some of the costs of production are not internalized into the price charged to consumers); (2) fixing the prices is the obvious, cost-effective, and efficient way to fix the misallocation of resources; (3) any other types of policies, which are not as cost-effective and efficient, should be trying to approximate outcomes that ideal pricing solutions would deliver more or less automatically; thus, (4) those policies are (at best) second-best.

Our view is not that pricing policies are always better in all ways, and politicians are ignorant or weak when they choose not to adopt them. Rather, it is that thinking of other policies in terms of and relative to pricing policies can give a lot of insight into the efficiency and fairness of those other policies. For that reason, we start our discussion of policy in Chapter 4 with an evaluation framework that is based on trying to get transportation prices to reflect full transportation costs, a necessary condition if transportation consumers (travelers) and providers (highway and transit agencies) are going to make efficient choices.

We cannot in the abstract distinguish which of these policies will better approximate the outcome of a system of efficient prices: the effects of each policy depend on how they are implemented. For example, given enough information and good fortune, a zoning board might designate an efficient mix of types and locations of land uses. Analysts can, however, assist policy makers in evaluating these second-best policies by describing how the policies work, and how they might help or hinder a shift toward proper pricing. Some of the policies we describe improve the prospects for future implementation of pricing more than others. We intend the analyses in Chapters 5 and 6 to help policy makers foresee the long-term consequences of implementing a variety of policies.

We do not pretend to cover every land-use or transportation policy, or even every category of policy types. Our discussion is illustrative, not comprehensive. We are more concerned with describing broad concepts and a way of thinking about the relative merits of transportation and land-use policies than with passing final judgment on policies. We hope to

provide tools that policy analysts and decision makers can use in their own evaluation of these policies.

Section II has three chapters that evaluate many different ways that public agencies might address the performance of urban surface transportation systems:

Chapter 4. Framework for Evaluation. Addressing the problems of planning from the perspective of economics, Chapters 2 and 3 argue that market forces are strong determinants of land-use and transportation patterns, but that they operate within constraints set by public policy and investment. Chapter 4 builds from that logic. It starts with a description of the components of travel cost for two reasons: (1) those components of cost are what matter to travelers, and any policies aimed at affecting travel behavior should be adopted based on consideration of which cost components they will affect, and how sensitive travel demand is to changes in those cost components, and (2) that analysis helps make the case that travelers do not consider the full costs of their travel.

Chapter 4 then describes the principles of economics and policy evaluation as they apply to transportation (see Appendices A through D for more detail). It makes a case for viewing transportation and land-use problems as failures in pricing, and then shows how public policy could directly address those pricing failures. Examples of public policies that address transportation through pricing are toll roads, congestion pricing, parking meters, parking cash-out, and emission charges.

It is not our claim that deviations from pricing policies are always wrong, or that ubiquitous congestion pricing is technologically and politically ready for immediate implementation. Rather, our point is that nonpricing policies are likely to be more effective if they are built on an understanding of what the land-use pattern and transportation system would look like in a world where prices better reflected the full social benefits and costs.

Chapter 5. Achieving a Balanced Transportation/Land Use Connection. This chapter starts with an explanation of our point of view in this book about land-use policies: that many factors suggest a need to shift metropolitan transportation policy toward one in which transportation investments are subordinate to a long-run vision (usually in the form of a land-use plan) for a livable metropolitan area. The chapter then provides an economic framework for evaluating land-use policy, a description of what some planners might prescribe as strong land-use policies for influencing the structure of land use and transportation systems in metropolitan areas, and the reasons such strong policies are difficult to implement comprehensively. We conclude that land-use policies are more likely to be adopted incrementally in most metropolitan areas and devote the last half of Chapter 5 to a discussion of some specific land-use policies that seem most promising for their ability to integrate with and affect transportation. It divides those policies into two categories: (1) land-use policies that affect development patterns at an urban/metropolitan level, and (2) urban design policies that affect development patterns at a neighborhood level. The chapter does not attempt to discuss all land-use policies, but instead focuses on ones that, by intention or chance, have potentially significant effects on transportation.

Chapter 6. Transportation Policy Evaluation. Chapter 4 acknowledges that pure pricing will not always be the policy that local decision makers want to use to improve land-use patterns and transportation system performance. Chapter 5 sets a broader policy and institutional context for evaluating transportation policies. Chapter 6 then evaluates transportation policies in that context. It organizes policies into three categories: (1) policies

to improve transportation infrastructure: adding more supply has been the obvious first response to the problem of congested or unsafe surface transportation in urban areas; (2) policies to improve the use of vehicles: how to get more person-trips out of fewer vehicle-trips; (3) policies to mitigate traffic externalities: when streets get “improved” to facilitate the flow of vehicles, those improvements can have negative effects on the social and economic environment along those streets that policy can mitigate.

Together, Chapters 5 and 6 provide a relatively complete categorization and a cursory but consistent evaluation of transportation and land-use policies. They provide the policy pieces that Section III then assembles into complementary packages to address regional transportation performance.

CHAPTER 4

Framework for Evaluation

Organization: When market prices reflect marginal social costs, markets allocate resources efficiently—the right amount of goods and services are produced cost-effectively and allocated to the consumers who value them most highly. But prices do not reflect marginal social costs in highway transportation or in land development. The reasons for the poor pricing are more political than practical, and the number of political barriers that need to be lowered or torn down are getting fewer as the shortcomings of conventional policy become more apparent.

This chapter provides a framework for thinking about changes in policy that will get the prices right: how they could be implemented, and their likely effects on highway transportation and land use. The chapter has three sections:

1. *The costs versus the pricing of highway travel.* Estimates of how the large total costs of transportation break down into components. The objective is to get a feel for the extent to which drivers fail to perceive the full social cost of the trips. This “underpricing” generates too many of certain types of trips, which have significant impacts on land use.
2. *Efficient pricing of highway transportation.* The case for pricing: why it is needed, how it might be done, and the potential effects on transportation and land use. Key categories for improved pricing are access to congested highways, parking, and auto emissions of pollutants into the environment.
3. *Efficient pricing of land development.* Pricing externalities and public-service extensions at marginal cost has benefits similar to those in transportation.

Basics:

- Pricing an activity at its marginal social cost provides the signals and incentives needed for economically efficient allocation of resources: the right amount of the activity takes place, by those who most value that activity, supplied by those who can do so most cost-effectively.
- Surface transportation is badly priced. Fuel and other input taxes price roadways roughly at average, rather than marginal cost. This underprices access to congested corridors due to the time costs drivers impose on other drivers, and overprices access to uncongested corridors. Drivers also underpay for a variety of other resources: the natural environment, administration of the highway system, and the health and safety of other road users.
- Efficient pricing of highway access varies with demand over the course of the day. A driver who enters a congested highway imposes time costs on all other drivers. The price of access should reflect that cost, which varies with travel volumes and speed.
- Good pricing reduces but does not eliminate peak-period congestion (or air pollution or traffic accidents). Highway infrastructure is expensive in congested areas; traffic volumes high enough to require some reductions in travel speeds are necessary to get the most out of the investment in infrastructure.

- Congestion pricing encourages more intensive use of highways. Vehicle trips, not person trips are priced. Drivers respond by increasing their demand for travel in higher-occupancy vehicles, such as car- and van-pools, private jitney service, or public transit.
- Congestion pricing increases the intensity of land development. By increasing the per-mile cost of driving, congestion prices encourage fewer and shorter vehicle trips. Fewer trips reduces congestion, allowing higher-intensity development in commercial centers. Shorter trips require more intense residential development proximate to commercial centers.
- Congestion pricing affects the process of transportation infrastructure decision making. Revenues from congestion tolls signal the value travelers place on space in each corridor, indeed on each tolled travel mode in each corridor. Revenues in excess of the annualized cost of infrastructure and related services recommend capacity additions *in that corridor*. Decision making becomes local.
- The barriers to efficient highway pricing are political, not technical. Efficient methods for collecting highway tolls are in operation in the U.S. and in cities in other countries. The main political hurdle does not involve privacy: some toll-collection methods, such as prepaid “smart cards”, do not reveal drivers’ identities. The political barriers have more to do with raising prices on basic public facilities and potential adverse impacts on the poor.
- Parking is even more badly underpriced than highway space. For every car, there are three to four parking spaces, all but one of which are empty most of the time. Though expensive, parking is usually offered for free to drivers and is paid for indirectly through lower wages and higher retail prices. Pricing parking at market rates that vary appropriately by time of day would reduce the number of auto trips.
- Parking spaces are badly oversupplied in part due to minimum parking requirements. The objective is to require sufficient numbers of parking spaces during peak periods at zero price. Eliminating these requirements and pricing parking at cost would have significant impacts on the intensity of suburban land development.
- Pricing other externalities would further decrease travel and increase land-development intensities. These externalities include auto emissions, industrial emissions, and a variety of localized commercial and residential externalities.

Shortcuts: This chapter is essential reading in that it lays the groundwork for the discussion and evaluation of the variety of commonly implemented land-use and transportation policies in Chapters 5 and 6. Readers with little background in economics should read Appendices B and C. Readers with little background in transportation planning should read Appendix C. Readers with a good background in economics might prefer to read the more detailed material in Appendix C and then skim this chapter.

Since at least the 1950s, the bulk of transportation policy in the U.S. has denied, ignored, or dismissed scarcity. North America is abundantly endowed with developable land. Highways, commuter rail lines, railroads, and airports provide access to that land. Building more capacity avoids the conflicts that come with decisions about how to allocate the capacity that exists.

As sensible as that policy might have seemed a half century ago, its futility becomes increasingly apparent. Highway space in large and growing cities and along a variety of heavily traveled intercity corridors is far from abundant. Land supplies are dwindling in well-located but topographically constrained metropolitan areas. Congestion, air pollution, and urban sprawl are serious problems. Environmental, land, and transportation resources are undeniably scarce.

The thrust of policy needs to shift to how to allocate these scarce resources—and how to do so efficiently and fairly. Prices do much of that allocation in market economies. When prices approximate marginal social costs, resources get put only to uses that exceed that cost. Pricing access to congested highways, for example, (1) encourages travelers to economize on their use of the highway by traveling at other times, car-pooling, using other modes,

and living closer to their destinations; (2) allocates scarce highway space to those who value it most highly; and (3) generates revenue that signals the value of and pays for additional highway capacity.

Current funding mechanisms either badly underprice access to resources or set prices at average, rather than marginal, cost. Taxes on fuel, tires, and so on pay for most of highway infrastructure and maintenance, but they fail to signal the costs associated with wasted time on congested highways. Though roughly right on average, they are wrong in most places most of the time. Space for parking is expensive, yet much of that cost is paid through higher product prices or lower wages. And drivers fail to cover the full cost their travel has on air and water quality and on a variety of transportation-related services (e.g., police, fire, emergency medical services, attorneys, courts) paid for by taxpayers in general rather than directly by roadway users. Drivers impose costs on other drivers, on people using other modes of travel, and on land development—costs they can largely ignore under current policy.

Imposing efficient prices is difficult, especially from a political standpoint. Imposing higher charges on access to any resource is unpopular; doing so on an essential resource, such as transportation infrastructure, is especially hard on lower-income households.

But all forms of government action involve technical and political difficulties. Pricing is technically no more difficult, and in many cases easier, than the more familiar types of regulation. Because pricing is so much more efficient, raising the total value of transportation and land resources, pricing policies should serve as the basis for any analysis of policy. *That is the main point of this chapter.* We are not idealistic theorists hoping for immediate policy changes. A practical first step toward pricing is to understand it, so that the true costs of our current system for selecting and paying for transportation capacity are more apparent to technicians, the public, and policy makers. We postpone until Chapters 5 and 6 the discussion of alternatives to pricing.

We are not idealistic theorists hoping for immediate policy changes. A practical first step toward pricing is to understand it, so that the true costs of our current system for selecting and paying for transportation capacity are more apparent to technicians, the public, and policy makers.

THE COSTS VERSUS THE PRICING OF HIGHWAY TRAVEL

Driving is expensive. The Bureau of Transportation Statistics (BTS), for example, estimates that in 2005 the average annual cost of owning and operating a car driven 15,000 miles a year was about \$7,800: 52 cents per mile.¹ That is a big number relative to a median, after-tax, household income of less than \$40,000 per year. It suggests that a two-car household spends almost 40 percent of its disposable income on car travel.

At 52 cents per mile, the out-of-pocket expense of a 20-mile, round-trip commute is about \$10.40, assuming free parking at the work place, or about \$230 per month (assuming 22 work days per month). This high cost potentially doubles for a two-worker household.

There are, of course, some caveats. First, the actual expenditure on car travel varies a great deal. The cost is especially high for expensive, new vehicles with low mileage (Sport-Utility Vehicles (SUVs), for example). But there are also a lot of smaller, older, and more fuel-efficient cars. It is unlikely that many of the households earning near the median income really spend 40 percent of their disposable income on car travel.

Second, the BTS estimates that about 71 percent of these total out-of-pocket costs are fixed; that is, they do not vary, over the course of a particular trip. Such costs include insurance premiums, license and registration fees, annual depreciation, and finance charges. The remaining 29 percent, or about 15 cents per mile, represent the average out-of-pocket costs of a particular trip. The average *variable* cost of a 20-mile, round-trip commute is thus about \$3 out-of-pocket, or about \$66 per month (assuming 22 work days per month). For economists, these variable (or additional, or marginal) costs are the ones to look at to understand travel behavior: they are the ones people

consider when they make decisions about where, when, and how to make a particular trip.

Third, significant costs of car travel are not included in owning and operating expenses, including the following:

- The time costs associated with car travel. This includes the time spent driving, which could have been spent doing something else, including working for pay. It also includes time spent looking for, buying, and maintaining the car.
- Parking. The total cost of providing parking probably exceeds the total cost of providing travel lanes. But most of these costs are paid indirectly through higher prices paid for housing or retail products, or lower wages.
- Some of the costs of infrastructure and transportation-related services. These include highway law enforcement, emergency services, legislative and judicial services, prisons, and so on. A big chunk of these costs is paid for with taxes on fuel, tires, and so on, but some of it is paid for with revenues from other taxes, such as those on property, retail sales, and income.
- Traffic-related deaths and injuries. More than 40,000 traffic-related fatalities occur in the U.S. each year; the number of injuries is many times higher. Much of the accident cost is paid out-of-pocket through insurance premiums, but some is not; nor are additional costs in terms of pain, suffering, and time.
- Environmental damages. These include local and global air pollution, water pollution, flood control and drainage costs due to impervious roadway and parking surfaces, noise, and visual blight.

There are more costs of owning and operating an automobile than those of buying, operating, and maintaining an automobile. They include:

- *the time costs associated with car travel;*
- *parking (in most cases);*
- *some of the costs of infrastructure and transportation-related services (e.g., police);*
- *traffic deaths and injuries; and*
- *environmental damage.*

How big are all these costs? What is the full annual social cost of travel by car? And what proportion of these costs is paid directly by those doing the driving? Answering these questions is difficult, and the available answers generate controversy. The best estimates of the full social costs of car travel of which we are aware were compiled and reported in a series of technical reports by Mark Delucchi from the Institute of Transportation Studies at the University of California, Davis.²

Estimating the *opportunity costs* of the resources used in transportation can get complicated. It is often not sufficient, for example, simply to multiply an observed market price by the quantity traded. Oil may sell for \$60 per barrel, which results in gasoline retailing for \$2.50 per gallon, on average. But gasoline is subject to taxation and a variety of regulations; the price at the pump is higher than its cost of production and distribution given the price of oil. Moreover, only the *marginal* barrel of oil costs \$60 to produce. Almost all of the millions of *infra*-marginal barrels cost less, some of them much less than the market price. Yes, users all pay \$60 per barrel of oil, but only part of that \$60 represents the value of resources foregone: the remainder is a transfer from oil consumers to oil producers; it is not a social cost.

We cannot reproduce or critique here all of Delucchi's estimates. Instead, we focus on the findings in his summary report, which has a number of advantages. First, the summary is unusually comprehensive. Second, it builds explicitly from an economic framework and is sensitive to distinctions between monetary (explicit) and nonmonetary (implicit) costs as well as internal and external costs. Third, it rates, on a scale of A to D, the quality of the analyses behind the estimates it reports. Fourth, it provides ranges of estimates rather than a single point estimate. In many cases, these ranges are big.

Delucchi's report also has disadvantages. First, some of the estimates, even in ranges, are controversial. Shoup (2005a), for example, while acknowledging

the general quality of Delucchi's report, disagrees with how it handles the estimate of parking costs. Second, the estimates are for travel in 1990–1991; we are not aware of more recent estimates of the same quality.

We deal with this second shortcoming by reporting percentages. The dollar value of the costs of car travel has risen since 1990, if for no other reason than general inflation.³ Some costs, such as the time costs associated with congestion and with construction, have probably risen faster than inflation. But given the large uncertainties in the estimates of many of the components of travel costs, the proportions reported by Delucchi probably remain fairly accurate.

Table 4-1 summarizes Delucchi's summary. It shows six general categories of costs, each of which appears in bold, and low and high estimates of the costs in each category. We report the average of these low and high estimates in the first column of percentages, and then indicate the range by the $\pm$ percentage shown in the second column. Indented and not in bold below each general category are subcategories that are especially large or otherwise of interest. Their percentages do not always add to category totals because we left out some subcategories to simplify the table.

Some costs, such as the time costs associated with congestion and with construction, have probably risen faster than inflation.

TABLE 4-1. SOCIAL OPPORTUNITY COSTS OF ROADWAY TRAVEL

	ESTIMATED COST (PERCENTAGE OF TOTAL COST)	ERROR RANGE (PERCENTAGE OF TOTAL COST)
Personal (internal) nonmonetary costs	30.0%	$\pm 8.9\%$
Travel time	20.8%	$\pm 4.5\%$
Injuries (e.g., pain and suffering)	6.0%	$\pm 3.1\%$
Purchase and maintenance time	3.2%	$\pm 1.2\%$
Goods and services privately supplied and priced	36.2%	3.1%
Annualized cost of vehicle fleet	12.4%	$\pm 1.6\%$
Parts, supplies, maintenance	7.0%	$\pm 0.6\%$
Fuel and oil	3.2%	$\pm 0.1\%$
Travel time that displaces paid work	8.4%	$\pm 0.8\%$
Priced private parking	0.1%	$\pm 0.0\%$
Goods and services publicly supplied	7.8%	$\pm 2.7\%$
Roads and highways	6.3%	$\pm 2.1\%$
Off-street parking	0.6%	$\pm 0.2\%$
Police and emergency services	0.4%	$\pm 0.1\%$
Judicial and correction system services	0.4%	$\pm 0.1\%$
Goods and services bundled privately	7.1%	$\pm 4.1\%$
Nonresidential off-street parking	4.2%	$\pm 2.3\%$
Residential off-street parking	1.1%	$\pm 0.5\%$
Privately supplied roads	1.8%	$\pm 1.3\%$
Monetary externalities	3.0%	$\pm 1.2\%$
Travel-time delay (during paid work)	1.0%	$\pm 0.5\%$
Accident costs	1.1%	$\pm 0.0\%$
Nonmonetary externalities	15.9%	$\pm 13.3\%$
Injuries (e.g., pain and suffering)	2.2%	$\pm 1.8\%$
Travel-time delay (off-work)	2.4%	$\pm 1.5\%$
Air pollution	10.8%	$\pm 9.4\%$
Other pollution	0.4%	$\pm 0.3\%$

The midpoint estimate of the absolute annual social cost of transportation-related goods and services in the U.S. in 1990-91 is \$2.49 trillion. So, for example, personal, nonmonetary costs represent 30 percent of this total.

Source: Delucchi 2004.

What is the base (the denominator) for these percentages? The average of the high and low estimates of the total annualized social cost of transportation: \$2.49 trillion in 1991. At that time, gross domestic product (GDP) was about \$6 trillion. The numbers are not directly comparable, however, because GDP understates the full value of production in an economy because not all that is produced is traded in formal markets.

Consider first the personal (i.e., *internal*) nonmonetary costs of travel—costs that the travelers impose on themselves and so do not involve a transfer of cash. These costs make up between 21 percent (30 percent *minus* 8.9 percent) and 39 percent (30 percent *plus* 8.9 percent) of the total cost of travel, with a midpoint estimate of 30 percent. Travel time is the largest component, making up almost 21 percent (± 4.5 percent) of total travel costs. Note that these travel-time costs *do not include* the costs of delay imposed by other drivers on congested highways (which are nonmonetary *external* costs). Note also that these costs are not included in GDP because the government agencies that produce the national accounts do not count them. Including these costs in GDP would increase it, decreasing travel costs as a percent of GDP.

The next cost category is goods and services produced privately and priced in markets. These costs also are internalized; one can think of these as personal monetary costs. These costs in total are large, but not much larger than personal nonmonetary costs. Time costs associated with travel are more than double the annualized cost of the entire motor-vehicle fleet. Note that the cost estimates exclude taxes (which are transfers rather than opportunity costs; i.e., costs that account for using up real resources) and resource “rents” (e.g., the difference between the market price and the production cost of oil). The contribution to the total cost of private parking that is priced (i.e., that the driver pays for in cash) is small (more on this below).

Next consider the opportunity cost of transportation-related goods and services supplied by the public sector. Building and maintaining roads and highways—now we’re into the big money, right? Wrong, if Table 4-1 is right. The entire category amounts to about only 8 percent ($\pm$ about 3 percent) of total transportation-related costs. Note that the cost of streets includes the cost of on-street parking.

Do drivers, as drivers, pay for the goods and services supplied by the public sector? The answer is: Yes, for the most part. In 1989, federal, state, and local governments collected about \$44.5 billion in fuel taxes, registration fees, and highway and bridge tolls. They spent about \$65 billion on roadway planning, construction, and repair (\$33.3 billion), maintenance (\$19.7 billion), and related services (\$12 billion).⁴ Thus, approximately \$20 billion in expenditures were not covered directly by fees paid by drivers. About \$18.5 billion of the revenue came from general fund revenues, mostly by state and local governments to fund improvements to local roads and streets. The remaining \$2 billion came from net gains in interest on roadway funds invested and through local bond issues.⁵ Thus, user fees covered about 70 percent of funds spent explicitly on roadway improvements and services in 1989. (More later on why this is *not* a funding scheme that produces an efficient transportation system.)

Next in Table 4-1 is the category of travel-related goods and services supplied privately, but not charged for directly. These goods and services are paid for indirectly through purchase of other goods. Private land developers, for example, pay for construction of many local streets. The costs of these streets, which include room for on-street parking, are included in the prices of houses or building lots. The costs of residential off-street parking—driveways, garages, and parking structures—are also included in house prices or rents. Finally, the lion’s share of nonresidential parking—at retail centers and work places—is provided at no charge to the driver, as driver. This parking is paid



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The lion’s share of nonresidential parking—at retail centers and work places—is provided at no charge to the driver, as driver. The cost is passed on to all consumers through higher prices and lower wages.

for through higher retail prices and lower worker wages (so drivers may still pay a lot of these costs, but as consumers and employees).

Next are monetary externalities: costs measurable in monetary terms and imposed by drivers on other travelers. There are two main subcategories. First are the time delays that drivers on congested highways impose on other drivers who are on the work clock. The market prices that time (the driver's wage plus the cost of vehicle time), but other drivers fail to perceive this cost (i.e., the cost is *external* to these drivers). Accident costs include those incurred by uninsured drivers—someone pays those costs, but not the driver who caused them. Note that these monetary externalities are small in percentage terms (3 percent of total travel costs), but this small percentage represents about \$75 billion in 1991, more than \$100 billion in 2006.

The final category is nonmonetary externalities. These are large: 16 percent of total travel costs (twice the costs of building and maintaining the road system). But note that the estimates of these costs are highly uncertain: the error range of 13 percent is almost as big as the estimate itself. These costs are hard to measure. The largest category is travel-generated air pollution, which plagues many metropolitan areas and is a significant source of greenhouse gases. The next largest category is travel-time delay for people not explicitly on the work clock, mainly commuters. The total estimated value of external travel-time delay is about 3.5 percent (± 2 percent) of total travel costs, or about \$90 billion in 1991, \$135 billion in 2006.

Delucchi lists additional costs for which estimates are unavailable. These include the aesthetic costs of highways and associated development, the costs that roads and highways impose due to the difficulty and danger in crossing them, and environmental damage and habitat loss due to the space occupied by roadway infrastructure. These costs are likely significant, and there certainly are others not included in the study.

What can we say in summary about the extent to which travelers pay for the costs they impose?

- Travelers pay about two-thirds (plus or minus quite a bit) of the annualized, marginal social cost of their trips; these costs are properly internal to the trip.
- Publicly supplied infrastructure and services are badly mispriced. Travelers pay directly, through a variety of travel-related taxes and fees, for at least two-thirds of the costs of these publicly provided goods and services. But these charges rarely reflect marginal costs (more on this in the next section).
- Travelers fail to pay directly for about 25 percent (plus or minus a lot) of travel-related costs. Almost 75 percent of these costs consist of non-monetary external costs, including that of pollution, time on congested highways, and accidents. The remainder consists of costs that private suppliers bundle into house prices, retail prices, or employee compensation packages.
- Time costs are big. Added across all categories, lost time accounts for more than one-third of the total costs in Table 4-1—about five times more than all the money the government spends on building and maintaining highways and public parking! Even if these numbers are only rough approximations, the policy implications (which, in our opinion, are underappreciated by policy makers) are important: time matters, a lot.
- Taken together, these estimates indicate that travelers on congested highways fail to pay perhaps a third of the social costs of their trip. But in fact the proportion of costs not paid is higher because some costs, such as fixed vehicle expenses, should not be counted in the cost of a particular trip. Car travel is underpriced.

Time costs are big. Added across all categories, lost time accounts for more than one-third of the total costs in Table 4-1—about five times more than all the money the government spends on building and maintaining highways and public parking!

Underpricing a resource leads to its overuse (i.e., its “exploitation”) unless other means, such as regulation, are used to limit access. Pricing has advantages over regulation. The next section explores the opportunities for and consequences of improved pricing in transportation.

EFFICIENT PRICING OF HIGHWAY TRANSPORTATION

Traffic engineers tend to think of traffic flows in physical terms, adapting concepts from the study of the flow of fluids, such as water through pipes or drainage channels. That analogy offers insights into traffic flow on a congested highway: if too much volume tries to push through a constrained space, the flow gets turbulent, speeds drop, and fewer cars pass a given point in a given amount of time. Taking steps to reduce the causes of turbulence increases both the amount of water flowing through the pipe system and the amount of cars flowing through the highway system. Appendix C provides more detail.

Water molecules do not react consciously or rationally to changes in their environment. Reducing turbulence in a particular watershed drainage system, for example, won’t attract water molecules from neighboring watersheds. Travelers, however, do react to incentives. One has to add economics to the physics to understand traffic.

Because people “buy” similar trips day after day, they tend to be well-informed consumers. They do a good job of processing information about their travel destinations, modes, routes, and travel times, and comparing them to transportation options and conditions to make optimum travel decisions. The principles of microeconomics capture the most significant determinants of these decisions: travelers’ considerations of benefits and costs.

While people consider their *personal* costs and benefits of travel, they tend not to consider the costs of the effects their travel or location choices have on others. These external costs are important sources of inefficiency in the transportation system; too many trips of relatively low net value are made. The full social costs of those trips exceed the benefits they provide.

Increasing roadway capacity can not be assumed to cure this problem. Smoother, faster flow encourages more trips of lower value. Downs (2004a) uses the term *triple convergence*: new trips converge on the new capacity as drivers change routes, modes, and travel times. These changes can restore congestion remarkably quickly. In the long run, drivers change the locations of origins and destinations, contributing to further congestion. The only real solution is to somehow limit traffic volumes.

In this section we make the case for pricing as the most appropriate device with which to limit (i.e., ration) trips. Setting the price of access to scarce resources, such as highway capacity, at marginal social cost has several advantages. First, it limits the number of car trips. Second, it allocates these trips according to those of highest net value. Third, it provides accurate signals about where additional capacity is most highly valued. Fourth, it provides the revenue to pay for that capacity.

We discuss pricing of access to three types of transportation-related resources: congested highways, on- and off-street parking, and the natural environment as a sump for automobile emissions. Access to these three resources is badly underpriced under current policy. We discuss how and why current pricing is wrong, how pricing could be improved, the effects of better pricing on transportation and land use, and the impediments (mostly political) to better pricing.

Pricing Congested Highways

The basics

Highway capacity currently is paid for in large part through taxes on fuel. The fuel tax is easy to collect, and the amount drivers pay varies with how

Underpricing a resource leads to its overuse (i.e., its “exploitation”) unless other means, such as regulation, are used to limit access. Pricing has advantages over regulation.

much they drive. On its face this seems a sensible way to fund highway construction and maintenance, especially compared to taxes on property, retail sales, or income.

One of several problems with the fuel tax is that it underprices travel on congested highways. That has to be the case. A fuel tax implicitly assumes that any vehicle with about the same fuel efficiency consumes transportation resources at the same rate no matter where or when it is driven. For several reasons (e.g., the high value of lost time shown in Table 4-1), that average expenditure on fuel taxes bears little relationship to the cost of the transportation services most drivers consume.

As shown in detail in Appendix B, the price for the use of a scarce resource (e.g., space on a congested highway) allocates the resource efficiently when it equals the *marginal social cost*. A driver entering a congested highway tends to think about the cost of the trip in terms of his out-of-pocket costs (fuel and vehicle wear-and-tear) plus his time cost. He ignores the time costs that his trip imposes on other drivers (because everyone has to slow down a bit to make space for his vehicle). These *external* time costs are larger per mile than the fuel tax during peak periods. Thus the fuel tax underprices access to the highway even if fuel-tax revenues cover the costs of highway construction, maintenance, and operation.

Figure 4-1 illustrates the conditions on a segment of a highway that experiences peak-period congestion. The vertical axis measures dollar amounts—costs, benefits, and prices—while a measure of traffic volume (e.g., vehicles per unit distance on the highway) appears on the horizontal axis.⁶ The figure consists of five curves. We explain each in turn.

We begin with the two curves labeled PTC for *Private Trip Cost*. These curves show how the dollar value of the typical driver's perceived cost of traveling on the highway varies with traffic volume.⁷ It consists of the sum of the monetary (i.e., out-of-pocket) costs of vehicle operation and the non-monetary costs of the time spent driving. The horizontal portion of the curve on the left corresponds to uncongested travel. At low traffic volumes, most drivers travel at the speed limit or better.

Trip cost depends on travel speed. PTC rises as the highway gets congested because cars slow down to maintain safe travel distances as traffic density

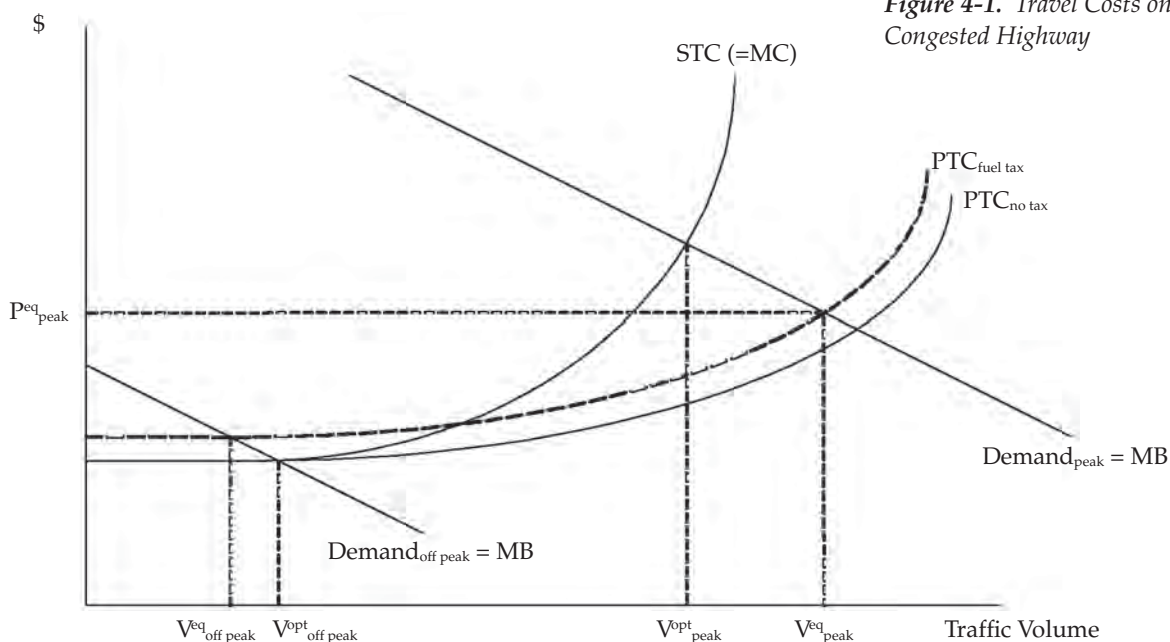


Figure 4-1. Travel Costs on a Congested Highway

risers. Slower travel translates to higher time costs; the PTC curve begins to slope upward. It gets steeper as more cars enter the already-congested roadway because each additional vehicle slows traffic more than the vehicle that entered before it. This curve gets its shape from the speed/flow relationship (Figure C-1 in Appendix C). In fact, the PTC curve is simply the speed/flow curve inverted because lower speeds translate directly into higher time costs.

There are two PTC curves in the figure. The lower solid curve reflects the costs of vehicle operation and time *excluding any fuel tax*. The higher dashed curve includes the fuel tax in PTC—this curve reflects the costs drivers experience on most highways. The PTC with fuel tax is parallel everywhere to the PTC without tax because the fuel tax is assumed to not vary as traffic volume rises and travel speed falls.

Figure 4-1 has two demand curves: one corresponding to off-peak travel (left) and the other corresponding to peak travel. In practice, the demand curve shifts continuously from the far left in the wee hours of the morning to the far right at peak demand at the height of the morning commute. The curve then shifts gradually to the left until demand rises again during the evening commute. Demand during the evening commute differs to some extent from that in the morning because commuters tend to have more flexibility in their evening travel schedules. This daily shifting in demand for travel on the highway is a key characteristic of the analysis.

The demand curves show the marginal benefit of travel—the monetary equivalent of the value or benefit drivers derive from a trip on this highway segment at this time, all else constant. The following thought experiment might help. Assume that drivers are given access to the highway in order of their bids at an auction. The highest bidder gets on first; presumably she makes her high bid because the trip has high value for her; she is at the far left of the graph. The next highest bidder gets in just to the right of her on the graph, and so on. Thus, the demand curve arranges trips from high to low benefit. The high-benefit trips on the left side of the graph are priority trips with few or no substitutes. As the time or out-of-pocket expense of the trip declines, the quantity of travel increases. The additional trips are those of lower priority, with more substitutes, or that were simply too expensive.⁸

The PTC and demand curves allow one to identify the peak and off-peak traffic-volume *equilibriums*. Drivers enter the highway as long as the benefit each perceives from the trip, as shown by the demand curve, exceeds the perceived cost. The cost each driver experiences is shown by the PTC curve. Thus, during peak demand the equilibrium traffic volume is $V^{\text{eq}}_{\text{peak}}$; peak travel volumes tend to settle near this level.⁹

The market “price” of a peak-period trip is $P^{\text{eq}}_{\text{peak}}$.¹⁰ This price is paid in the form of expenditure on fuel (including fuel taxes) and other vehicle operating costs, and the time spent on the highway segment. The off-peak travel volume is labeled $V^{\text{eq}}_{\text{off-peak}}$. The “price” of the peak-period trip is higher than that at off-peak due to the slower congested travel.

Is the peak-period equilibrium traffic volume (i.e., the congestion level) economically optimal? In other words, has the scarce highway capacity been allocated well? No! The values of the marginal trips are less than their full social cost. Marginal trips are not worth their cost.

The marginal *social* cost of a trip is shown by the *Social Trip Cost* (STC) curve. When the highway flows freely at low traffic volumes, social costs equal private costs: the social and private cost curves coincide. But social costs rise faster than private costs as congestion builds because an additional car entering the highway slows everyone else down. That slowdown has costs in time (primarily), operation (more fuel per mile), accidents, and environmental externalities (air pollution). A driver entering a congested

highway pays only a small fraction of these *external* costs. For now, to keep it simple, we assume these external costs are just time. Then, the social cost of an additional trip (i.e., the marginal cost of the trip) is the PTC *plus* the value of the additional time everyone else has to spend on the highway when the additional car enters.

Pozdena (1990) reports that an additional car entering a congested high-capacity highway can, in aggregate, cause an hour of delay. That one extra car may slow everyone down by only a few seconds.¹¹ But if one car slows each of 600 cars enough to increase travel time by six seconds each, the total effect on other cars is one hour. The value of that external time cost at any traffic volume is shown by the vertical distance between the STC and PTC curves.

The *optimal* traffic volume is $V_{\text{peak}}^{\text{opt}}$. Each of the relatively low-value trips between $V_{\text{peak}}^{\text{opt}}$ and $V_{\text{peak}}^{\text{eq}}$ has net social cost. These trips provide net benefits for the individuals making them, but only because they can ignore the costs their trips impose on others. If the objective is to get the maximum net benefit from the highway, travel volume should equal the optimal traffic volume, $V_{\text{peak}}^{\text{opt}}$, and only the relatively high-value trips should be made (it's not just a matter of quantity, but also of quality).

Note that the *highway suffers congestion at the optimal traffic volume*. The congestion is less and travel speeds higher than that at the equilibrium peak traffic volume, but traffic is still not free flowing. Eliminating congestion does not maximize net benefits. The reason is that *eliminating* congestion costs more than it's worth. Getting the most out of expensive highway infrastructure means allowing an optimal amount of peak-period congestion.

The fuel tax raises the private cost of a peak-period trip, but not by enough to reduce travel to its optimal level. The external time cost of marginal trips exceeds the tax. Too many cars enter the highway during peak periods even though each pays the average cost of infrastructure. Conversely, the fuel tax overprices travel during off-peak periods, which discourages drivers from taking trips that provide net benefit during those periods.

The fuel tax is a form of *average-cost* pricing. It, at best, covers cost *on average*. That might be considered good enough if the highway never or rarely experienced congestion. But because costs rise significantly during peak periods, so should price. Indeed, *marginal-cost* pricing varies over the course of the day from not much to potentially quite a bit.

The sidebar on the following page, The Private and Social Costs of Travel, gives a numerical example consistent with the relationships illustrated in Figure 4-1.

Improving the pricing of highways

Answering the question of how to improve the pricing of highways is simple in concept: charge tolls that vary by location and time of day (i.e., as demand changes). A congestion toll adds to the out-of-pocket cost of a trip, which shifts the PTC curve upward. For the equilibrium volume with the toll to coincide with the optimal volume, the PTC curve must shift up until it crosses the demand curve at the optimal volume, $V_{\text{peak}}^{\text{opt}}$. The vertical distance between the PTC and STC curves at any volume equals the external cost of the trip at that volume. The optimal congestion toll, therefore, equals the external time cost imposed by an additional driver at the optimal traffic volume.

This simple analysis (simple, at least, relative to a complete analysis of an entire transportation system) illustrates the basic concept. But it leaves important practical questions unanswered. Can analysts determine the optimal congestion toll? Can the toll be collected efficiently?

We take the second question first. One technical barrier to implementing a congestion toll involves setting up a collection system that does not itself



Stephen Walls, istockphoto.com

An additional car entering a congested highway can, in aggregate, cause an hour of delay.

THE PRIVATE AND SOCIAL COSTS OF TRAVEL

The table below shows the effects of congestion on the cost of a hypothetical 10-mile trip segment. The estimates are based on the following formula:

$$\text{Trip time per vehicle} = 12 \text{ min} + [0.001 * (\text{Volume} - 400)] \\ + [0.000015 * (\text{Volume} - 400)]^2$$

Intuitively, the formula says: (1) the free-flow travel time is 12 minutes; since the distance covered is 10 miles, the average free-flow speed is 50 mph; (2) free-flow is unperturbed up to volumes of 400 vehicles per hour (assume one lane in the direction of interest); (3) as the number of vehicles per hour increases beyond 400, travel time increases primarily linearly at first, but grows at a faster rate as volumes get bigger. All that is consistent with the physics of traffic described in Appendix C.

this implies an average wage of \$12/hour). The total value of the congestion externality (of the lost time on the roadway) caused by one more driver is the amount estimated in (E). Assume that the *operating* cost of the trip is \$2.00 (about 20¢/mile). The base time cost (no congestion) is \$1.20 (10¢/minute * 12 minutes), and the base cost is the sum of operating and time cost: \$3.20/trip. Column (F) shows the private costs increase beyond the base cost because the travel time (B) increases ($F = \$2 + (B * 10\text{¢}/\text{minute})$). But the social costs imposed by one new driver ($G = F + E$) increase even faster because it counts the value of the delays to *all* drivers. As the cost of the trip goes up, it exceeds the benefits of the trip (H, whose value are derived by reading the demand curve at every volume)—after the 1,400th driver, the cost of traveling is greater than its benefit and the trip no longer becomes worthwhile.

TRAFFIC VOLUME, TRAVEL TIME, TRAVEL COST, AND CONGESTION EXTERNALITIES

A	B	C	D	E	F	G	H
TRAFFIC VOLUME (VEHICLES)	TRIP TIME (MINUTES)	INCREASE IN TRAVEL TIME PER DRIVER (MINUTES)	INCREASE IN TOTAL TRAVEL TIME (MINUTES)	EXTERNAL TRIP COST (\$)	PRIVATE TRIP COST (\$)	SOCIAL TRIP COST (\$)	MARGINAL BENEFIT (DEMAND)
200	12.0	0	0.0	0.00	3.20	3.20	31.10
400	12.0	0	0.0	0.00	3.20	3.20	27.44
600	12.8	0.007	4.2	0.42	3.28	3.70	23.78
800	14.8	0.013	10.4	1.04	3.48	4.52	20.12
1,000	18.0	0.019	19.0	1.90	3.80	5.70	16.46
1,200	22.4	0.025	30.0	3.00	4.24	7.24	12.80
1,400	28.0	0.031	43.4	4.34	4.80	9.14	9.14
1,600	34.8	0.037	59.2	5.92	5.48	11.40	5.48
1,800	42.8	0.043	77.4	7.74	6.28	14.02	1.82
2,000	52.0	0.049	98.0	9.80	7.20	17.00	

Source: O'Sullivan, Arthur. *Urban Economics, Fourth Edition*. Boston: McGraw Hill, 2000, p. 548.

As traffic volume (A) increases, trip time (B) in minutes per vehicle increases as well. Every additional car entering the roadway causes a small increase in travel time for every other car (C), so the increased time grows down the column. For example, when the 600th vehicle enters the highway, it slows the travel time of all other vehicles from 12.793 minutes to 12.800 minutes: it causes an increase in travel time of 0.007 minutes for each vehicle. Because one new driver (after 400) affects all the other drivers, the total affect of one new driver is $D = (C * A - 1)$; this is the congestion externality of one new driver (for the 600th driver, for example, $0.007 * 600 = 4.2$ minutes of total delay). The amounts in column D vary because the impact per driver depends on how many drivers are already on the road (Column A).

Assume an opportunity cost of 10 cents per minute (\$6 per hour; if the value of lost time is on average one-half the wage rate,

The table illustrates the point made in theory and with the graph. The *actual* travel, based on drivers' consideration of *private* trip cost, will be 1,600 vehicles (private trip cost (F) equals marginal benefit (H)). The *optimal* travel, based on social trip cost, will be 1,400 vehicles (social trip cost (G) equals marginal benefit (H)).

The table shows the impact of *one* driver (the *marginal* cost) on all other drivers at various volumes. To get the *total* costs (e.g., private (F) or social (G)) one would have to multiply the marginal costs in the table by the number of new drivers above 400. For example, at 1,000 vehicles, 600 drivers (beyond the 400 that cause no affect) are causing an externality cost of \$1.90 on each of 1,000 drivers, for a total loss of \$1,900.

This example used with permission from O'Sullivan 2000 (4th edition), 548.

contribute to congestion. The old method for collecting road tolls uses toll booths that force all vehicles to stop to pay the toll. Having to stop at toll booths obviously works against reducing travel times.

Recent technical innovations offer less cumbersome methods for collecting tolls. A variety of methods are in use. Video cameras in London and Stockholm capture license plate numbers and check them against accounts that can be paid on-line, via telephone, and at retail shops. In Singapore, tolls are deducted electronically from a prepaid “smart card” as a car passes under any of 45 gantries. The system differs little in concept from that used with prepaid mobile phones or photocopying cards. Cars that access the toll lane on I-15 in southern California have transponders installed that communicate with electronic receivers. A variety of electronic systems have been used around the U.S. to collect standard tolls on roads and bridges that do not vary with congestion (Fielding and Klein 1993). A recent (2006) demonstration project in Seattle showed that areawide pricing using satellite tracking of vehicles is technologically feasible (see the sidebar on the following page).

So, yes, tolls can be collected efficiently with smart cards and transponders now, and those technologies are crossing political boundaries. States in the Midwest are working to make an “I-pass” transponder compatible for paying tolls on interstate highways across state lines. Going farther, the technical obstacles to GPS-based tolling are more or less solved for small areas. A big challenge remaining is making the local applications scalable to a large area at a reasonable cost, and getting public and political support.

How can the appropriate tolls be determined? The data needed for setting optimal congestion tolls are traffic volume, associated travel speed, and the dollar value of travel time. Probably the most uncertain of these is the value of travel time. Brownstone and Small (2005) summarize the existing evidence and report estimates from observations on traveler decision making in two congestion-pricing demonstration projects in southern California (SR91 and I-15). Though the two projects differ in their characteristics, they both yield estimates of the value of travel time of about \$20 per hour for a typical traveler. This corresponds to a little more than 50 percent of the average wage in that area.

The value of travel time varies considerably, however, across travelers. Part of this variation derives from ability to pay—low-income travelers are understandably more reluctant to part with any given amount of cash than are high-income households. Part relates to the traveler’s situation. For example, commuters who are especially concerned about getting to their destination on time are willing to pay more for a faster trip. In general, the vertical position of the PTC curve varies across travelers.

That variation suggests that a single optimal congestion toll or travel speed does not exist. Similar variation does exist, however, in demand for most consumer goods. The private suppliers of these consumer goods respond by tailoring prices and product characteristics to market segments or niches. The same can, in concept, be done with travel times, prices, and travel characteristics: congested, but toll-free, lanes or routes; toll lanes; toll-free high-occupancy vehicle (HOV) lanes; high-occupancy toll (HOT) lanes; combinations (e.g., lanes free to HOV next to single-occupant toll (SOT) lanes), and so on. Demonstration projects explore these issues.

Optimal tolls should vary over time with changes in traffic volumes. Singapore, for example, sets tolls to achieve target speeds on expressways of between 45 and 65 km/h. In August 2004, the toll for cars was S\$3.00 (\$1.74 in U.S. currency at the time) during the peak half hour, S\$2.50 (\$1.45) during the half-hour on either side of the peak, and S\$1.00 (58 cents) an additional half-hour before and after the peak.¹² The toll was higher for trucks. Travel speeds are reviewed every three months, and tolls adjusted as necessary



Slobo Mitic, istockphoto.com

Toll booths are inefficient. Recent technical innovations, including video cameras, “smart cards,” and transponders, make collecting tolls technically feasible and more efficient.

AREAWIDE TOLLING USING GPS

By Matthew Kitchen, Puget Sound Regional Council

The Seattle region has a lot of congestion on key roadways during peak travel periods. Having looked at the high costs of supply-side improvements to highways and transit, the Puget Sound Regional Council (the region's Metropolitan Planning Organization) decided to investigate a largely unused demand-side policy: congestion pricing. With funding from the U.S. Federal Highway Administration, the Council designed and conducted a demonstration project.

The project compared the travel behavior of volunteer participants before and after congestion pricing. The project required a toll system that would accommodate the complexity of participant trip characteristics, simulate efficient systemwide pricing, and accurately measure pricing response. But how could a group of 500 volunteers be charged for their driving by location and time of day (essential for true congestion pricing, as Chapter 4 and Appendix C explain) throughout a metropolitan region, and why would anyone volunteer for that punishment? Read on.

The Council chose a high-tech vehicle-positioning and tolling system. An electronic device for sending, receiving, and displaying information about position and tolls was installed in each participating vehicle. While a car's engine was running, the device communicated periodically via a cell phone network with a central computer. The system (1) pinpointed a vehicle's location every few seconds using information beamed from satellites (GPS), (2) matched the vehicle location with a digital map of tolled facilities, (3) logged a corresponding toll in the accounting system, and (4) communicated to a display panel on the transponder the toll rate per mile and the trip costs.



Researchers gave no details of the experimental design to the participants during an initial period of data collection, during which they collected baseline household-level travel information.

They used that information to develop unique travel budgets for each participating household. They calibrated the budgets so that if a household were to continue its measured travel patterns over the course of the experiment, it would spend most of its travel budgets on the tolls that were being charged, but would have \$100 to \$300 left as compensation for having participated in the experiment. Researchers loaded these travel budgets into accounts, and the experimental toll prices were deducted from these accounts when participants drove in tolled locations during tolled times.

Participants got to keep all the money still in their accounts at the end of the demonstration project. This "hold-harmless" design gave participants the opportunity to participate without risking their own funds or limiting their travel options, yet provided them with an incentive to adjust their driving behavior so as to enjoy the surplus remaining in the account at the end of the experiment.

The data-collection portion of the project ended in the spring of 2006, after 18 months of operation. The project fielded hundreds of customer service calls, issued more than 4,000 billing invoices, logged more than 100,000 data transactions to the central computer system via cell phone, and recorded more than 750,000 trips.

Preliminary analysis estimates short-run demand elasticities during the peak periods in the range of between -0.06 to -0.19 , (approximately 7–10 percent reduction in vehicle use during peak travel times resulting from a 60–75 percent increase in the variable costs of auto use), with considerably more traffic reduction on specific facilities.

The project is the first large-scale operational test showing the feasibility of *areawide* (as opposed to cordon or corridor) congestion-based charging. It demonstrated the technical and behavioral possibility of reducing wasted time and converting it to significant revenues for investment. Detailed analysis of the behavior data will continue over 2007 and beyond.

This approach to road tolling has many technical and policy issues that require additional analysis. Key lessons for future installations of GPS-based tolling systems include:

- Broad-based tolling of highways can result in significant changes in driver behavior.
- The selected technology for area-wide pricing is mature; off-the-shelf technology was very successful.
- The costs of employing a dedicated device for GPS toll applications are currently high, but the costs are rapidly decreasing with technical advances.
- Privacy will continue to be a notable area of policy concern, but one that can be managed through both technical and institutional approaches.
- An operational program where GPS-based tolling is at its core will require robust verification and enforcement components. ■

to achieve target speeds. All autos of similar type on the highway segment pay the same toll.

Another complicating factor is the variability of travel times around the typical and expected travel times. Travel times vary around their equilibrium values due to variation in weather conditions, construction or congestion on alternative routes, special events, and traffic “incidents,” such as breakdowns, crashes, and so on. Variability tends to rise with congestion. So the question of how much travelers value reliability arises.

Drivers have the option of choosing to travel on toll or untolled lanes on I-15 in southern California. The less-congested toll lanes offer greater reliability. The toll varies in real time depending on travel conditions (dynamic pricing) and is displayed on a message board before the entrance to the tolled lanes. Brownstone and Small (2005) note that the dynamic toll provides important information to travelers: a high toll indicates especially severe congestion on the untolled lanes. Drivers can choose the quicker and more reliable toll lanes, but only for a relatively high price.

The value of travel-time reliability has so far been less accurately measured than the value of travel time. An interesting aspect of reliability is that travelers are probably more concerned about arriving late than early—they are willing to pay perhaps quite a bit more to avoid a delay than to enjoy especially smooth sailing. Small, Winston, and Yan (2005) report evidence that about two-thirds of the value of express lanes relative to standard lanes come in terms of lower travel times, and about one-third in terms of better reliability. Lam and Small (2001) report that women appear to value reliability more highly than men, perhaps because childcare responsibilities restrict their travel-time flexibility.

In sum, congestion pricing is technically feasible, and optimal tolls can be sensibly if not precisely determined. The basic problem differs little from that faced by the private suppliers of goods and services that experience periodic fluctuation in demand: restaurants, theaters, golf courses, air travel, telephone and Internet systems, and so on. Suppliers of these goods and services have worked out sensible and accepted systems for varying prices with demand.

Peak-period pricing appears mysterious mainly to the public sector. Electricity, for example, is another system that experiences significant variation in demand. Like roadway pricing, electricity is usually (not always) priced at average, rather than marginal, cost, though there are no serious technical barriers to implementing peak-period pricing. While electricity is supplied by private companies, public utility commissions regulate prices. Apparently, political barriers to peak-period pricing exist. We discuss these immediately below in the context of the likely effects of congestion pricing on transportation and land use.

In sum, congestion pricing is technically feasible, and optimal tolls can be sensibly if not precisely determined.

Effects of congestion pricing on transportation and land use. We start with a discussion of the effects on transportation in two parts: 1) on traveler choices, and 2) on the supply of transportation infrastructure. A key to understanding traveler choices is that vehicle-trips, not person-trips, are priced. Commuters will try to economize on vehicle trips much more than on person-trips, which has effects on the demand for high-occupancy modes. A key to understanding the effects on the supply of transportation infrastructure is how congestion tolls price travel not only by time of day but also by location; tolls are higher on more congested routes or corridors. The resulting revenues signal the value of capacity on those corridors, which has important implications for decision-making processes.

Effects on traveler decisions. Congestion tolls increase the price of a vehicle-trip during peak travel periods. Returning to Figure 4-1, $P^{\text{opt}}_{\text{peak}}$ is greater

than $P^{\text{eq}}_{\text{peak}}$. The higher price discourages the vehicle trips of lower value, those between $V^{\text{opt}}_{\text{peak}}$ and $V^{\text{eq}}_{\text{peak}}$. These vehicles are “tolled off” the highway. As a result, travel speeds increase and trip times decrease for the vehicles that remain.

What becomes of the person-trips tolled off the highway? Some are simply not taken at all. But what one might also think of as “triple divergence” happens: these people change travel modes, routes, or times to economize on use of the tolled facility. Some people with flexibility in their travel schedules will take their trip at off-peak times. Some people will shift to alternative, usually unpriced, routes, which will increase congestion on these routes. This *trip diversion* to alternative routes (especially through neighborhoods) is a big concern when congestion pricing is considered. Some people will shift toward higher-occupancy vehicles: car pools, van pools, jitneys, buses, or commuter rail. The strength of this effect depends, of course, on the availability and characteristics of these other modes.

That these alternative modes be available is important for reasons of both efficiency and equity. From an efficiency standpoint, simply pricing highway capacity is not sufficient. The objective is not to restrict the number of person trips to attractive centers, but just the opposite—the purpose of the transportation system is to help people get to those places. But given the increasing marginal costs of adding new highway capacity, that purpose is best achieved by allocating the available capacity effectively.

An interesting option is van-pool or jitney service. Unlike standard public buses, these vans are smaller and more nimble in terms of timing and route. The service could be simple—in high-density travel corridors, hail a jitney as it travels down the street.¹³ Or it could take advantage of sophisticated logistics software to efficiently pick up commuters at home and deliver them to workplaces or transit stations, eliminating much of the costly walking, waiting, discomfort, and uncertainty associated with standard public bus service. Each jitney pays an economically appropriate congestion toll but splits that toll across the relatively large number of passengers in the vehicle.

Unfortunately, municipalities usually restrict or prohibit jitney service. The rationale is often public safety, but Downs (2004a, 147–49) points out underlying political reasons, including protection of existing public transit. The efficiency rationale for public (as opposed to private) bus service seems weak. There may be some network economies, especially in large markets, but little evidence exists that private suppliers could not compete to provide better service at lower cost. In fact, that’s why jitney service is discouraged: it competes too effectively against public bus service.¹⁴

For an example of an equity rationale, consider high-occupancy toll (HOT) alternatives. If a highway is tolled, many of the lower-value trips tolled off the highway are a lower value because the travelers wanting to make them have relatively low incomes. Indeed, this might be the strongest argument against toll lanes: they become “Lexus lanes.” But they also become HOT and mass transit lanes, and lower-income travelers are more likely to use mass transit and benefit from that improved service. Improving the efficiency of the transportation system is no doubt good for the wealthy, but it can also be good for the less wealthy, who may then have a system efficient enough to allow them to forego the cost of owning and driving multiple cars.

Equity issues could also be dealt with through subsidies for low-income commuters. Food stamps could be a model. The low-income household purchases highway or HOT access at prices that vary with income. A low-income household may pay 50 percent of the congestion toll, for example, rather than some fixed rate that does not vary with congestion. Important is that these prices remain highly correlated with marginal cost to maintain the incentive to economize on high-cost roadway space.¹⁵

What becomes of the person-trips tolled off the highway? Some are simply not taken at all. But what one might also think of as “triple divergence” happens: these people change travel modes, routes, or times to economize on use of the tolled facility.

THE CONGESTION TAX

Pricing gets more capacity on the highways by “tolling off” many drivers: doesn’t that mean that the rich benefit at the expense of the poor? In theory, it does not have to be that way. This example illustrates the possibility that the drivers tolled off could be better off if the money collected from the remaining drivers is used to reduce the taxes of all drivers. [Alternatively, the money could be used to provide services that the drivers value at least as much as what they cost. That is a slippery slope: the public sector argues that its services are very valuable; conservatives argue that consumers can get more value for the money by spending it themselves instead of having government spend it.]

Art decides to pay the tax and continue to use the highway. Driving takes Art less time than before the congestion tax was implemented; assume he saves \$0.33 for the trip. Because the

government can use the revenue from the congestion tax to cut other local taxes (e.g., the rate for the fuel tax could now, in theory, be reduced), Art’s other taxes drop by \$1.84. Despite paying the \$2.10 congestion tax, Art realizes a net benefit of \$0.07 for the trip.

Betty decides not to use the roadway after the congestion tax is imposed. She is “tolled off.” Betty also receives the benefit of having lower taxes. But she loses the consumer surplus that she previously got from the trip she is no longer taking. Before the tax, Betty was willing to pay \$5.21 for the trip, and that exceeded the previous cost of the trip (\$4.33), so her consumer surplus was \$0.88 per trip. But even though she is no longer driving, she is better off: her tax savings exceed the lost surplus by \$0.96.

This example used with permission from O’Sullivan 2006, 213–14.

BENEFITS AND COSTS OF THE CONGESTION TAX

	COST		BENEFIT		NET BENEFIT
	Tax Paid	Lost Consumer Surplus	Decrease in Time Cost	Lower Other Tax	
Art	\$2.10	—	\$0.33	\$1.84	\$0.07
Betty	—	\$0.88	—	\$1.84	\$0.96

In sum, congestion pricing, and marginal-cost pricing generally, improves the efficiency of the allocation of travelers to transportation infrastructure. Congestion pricing reduces the number of vehicle trips during peak periods, which reduces travel times, but has less effect on the number of person trips. More of these person trips will be made at other times, on other routes, and using higher-occupancy vehicles.

Effects on infrastructure decisions. Obviously, the design capacity of a segment of highway should depend on the demand for trips. Increasing congestion signals increasing demand, but that signal is distorted to the extent that drivers ignore the costs to society of their decision to use the highway. Moreover, congestion itself fails to show the value of the marginal trips taken: the level of traffic congestion tells nothing about the slope of the peak demand curve in Figure 4-1. Thus, congestion is an unreliable indicator of the value of additional capacity. Congestion suggests it may be sufficient to add more capacity, but it is not sufficient evidence that such capacity would be worth its cost.

A well-designed congestion toll (one that approximates the difference between STC and PTC) provides a much better indicator. A congestion toll amounts to a fee for the use of scarce (and, therefore, expensive) space on urban highways.¹⁶ In the short run, as demand and congestion increase, so does the user fee. Just as in the market for private goods, as price and revenues go up, so should the quantity supplied. If revenues rise above the annualized cost of a larger highway, new capacity should be added. Pricing properly in the short run encourages proper investment in the long run.

A common misunderstanding about congestion pricing is that it will generate revenues to fund all sorts of transportation improvements. It won't, and it shouldn't. Drivers are paying the toll for access to vehicle space on the highway. Revenue from the congestion toll pays for the full annualized cost of providing the highway of optimal capacity.

But highway capacity is already there, most of it "paid for" through fuel tax revenues. If drivers demonstrate a willingness to pay tolls that generate revenues in excess of what is needed simply to operate and maintain the existing capacity, what should be done with those excess revenues? In rough order of likelihood, here are the situations tolling might create for a transportation authority:

- Tolls pay less than or up to the full annualized cost of operating and maintaining the toll facility. ("Operating cost" is used broadly here to include payments to the private sector if tolling is done through a public-private partnership. To keep it simple, assume that the toll facility will be maintained at a level that will give it a very long life and that operations and maintenance (O&M) thus includes a component of capital amortization.) This is the likely case in most areas simply because the conventional policy is to provide sufficient capacity *at zero price to drivers*. In this case, there is no excess toll revenue after paying for O&M (which, for efficiency should be fully funded).
- Tolls pay more than the annualized cost of operating and maintaining the toll facility. This case will be less common: it will occur only in places where congestion and travel are both very high for relatively long periods, and in places where few good alternative routes (e.g., the bridges of San Francisco and New York) exist. The options:
 - Use the excess revenues for additional improvements or O&M in the corridor. The high toll revenue signals the high value of capacity in this corridor. The toll encourages travelers at the margin to switch to higher-occupancy vehicles or alternative routes. A high priority should be to make improvements in the corridor that facilitate HOV travel (though HOVs should still be tolled), reducing the incentive travelers have to switch to congested but untolled routes and improving the efficiency of the use of valuable highway space. Depending on local circumstances, both technical and political justification may exist for using some of the excess revenue to make improvements to the alternative routes in the corridor to which some of the tolled-off drivers are diverted. The key point is that the revenue should be spent in the tolled corridor so that the people who paid the tolls, or suffer from being tolled off, gain the benefits.
 - Give the excess revenues back to all drivers. The tolls ration demand and, in concept, pay the full annualized cost of the facility. Construction costs may already have been paid. Revenues in excess of those needed for efficient (i.e., benefits greater than costs) new transportation investments in the corridor (especially for HOVs) should ideally be returned in a lump sum to drivers. This is impractical if the identity of those who pay the tolls is unknown (as it would be with the use of prepaid smart cards). A more likely way to handle the reimbursement is to reduce the local taxes directly or indirectly linked to transportation expenditures. The sidebar on the previous page gives a numeric example illustrating why giving back the excess revenues (or equivalent benefits) to the drivers in the tolled area is important for equity reasons.

A common misunderstanding about congestion pricing is that it will generate revenues to fund all sorts of transportation improvements. It won't, and it shouldn't.

On the one hand, politicians fear tolls because many voters see highway access as a right—one for which they already pay through gas and other

taxes. On the other hand, the increasing concern of voters about the external costs of congestion (with climate change being the most recent and ubiquitous), the crunch on public funds for capital construction, and a shift toward user-pay policies all argue in favor of pricing policies. User fees make sense to most people, which suggests the government could use the revenues from the congestion toll to improve mobility within the tolled corridor, either by adding highway capacity when appropriate or improving the quality of service offered by HOVs.

Switching to a fee-based system has important ramifications for state and metropolitan transportation agencies. Current policy provides relatively little insight into the costs and benefits of increasing capacity. Gas tax revenues are collected at the state and federal levels, and federal taxes are reallocated back to the states. State Departments of Transportation (DOTs) play a big role in deciding what gets built where (see Chapters 3 and 5, and Appendix G in this report), and state objectives often conflict with local objectives.

Congestion toll revenues are clearly attached to specific highways or corridors. Suddenly, the costs and benefits of expanding capacity in a given corridor become easier to see. Rather than state DOTs going through a complex rating process that politicians can influence, local and regional planning organizations can spend locally generated revenues on local projects. A shift from state to regional or metropolitan decision making would have significant, though idiosyncratic, effects (e.g., Would all metropolitan planning organizations adopt similar transportation policies?) on urban transportation systems and the allocation of funds across the state (which might generate political opposition to tolls from representatives of areas that benefit from the current system).

Effects on land use. Discussing the effects of reasonably comprehensive congestion pricing on land use requires some speculation since no extant system is truly comprehensive. We need to simply focus on the main effects.

Suppose that everyone commutes to the Central Business District (CBD) in cars on radial highways. The critical aspect of congestion tolls is that they increase the *per-mile* cost of a vehicle trip, which encourages commuters to economize on the number and length of vehicle trips during peak periods. Tolls also have the long-run effect of encouraging changes in locations (trip origins and destinations) that decrease trip lengths. For the sake of the example, assume that buildings can readily be modified and replaced (which is true in the long run). Then the analysis goes like this:

1. A congestion toll, all else equal (including other taxes), increases the per-mile cost of driving, which encourages travelers to move closer to work. But more after-tax income spent on transportation means less is available to spend on other goods, including housing. Thus house prices fall generally and especially at greater distances from the CBD.
2. But toll revenues allow other taxes to fall or expenditures on public goods and services to rise. Some of this rise in after-tax income capitalizes into house prices, somewhat offsetting the effect of the toll. If the size of the drop in taxes or value of improvement in public goods and services is unrelated to distance from the CBD, the price of housing close to the CBD could increase to more than it was before congestion tolls. Households respond by economizing on floor space, and builders respond by building taller. Residential densities increase near the CBD.
3. More importantly, tolls improve the efficiency of the transportation system. Diminished congestion allows businesses in the CBD to more fully exploit economies of agglomeration, increasing profits and wages, which attract additional businesses and households from rural and other

Discussing the effects of reasonably comprehensive congestion pricing on land use requires some speculation since no extant system is truly comprehensive.

metropolitan areas. More business increases the size and density of the CBD and more people with higher incomes push house prices up generally, encouraging further density.

Computer simulations (e.g., Sullivan 1983) indicate that optimal congestion tolls would in the long run decrease average trip lengths, increase the average density of land development, and decrease the amount of space devoted to highways (i.e., highways are currently overbuilt). Higher per-mile vehicle costs and the increased density of development encourage the use of higher-density travel modes. Lee (1992) concludes that underpricing of transportation induces inefficient land development at the urban fringe and inefficiently high transportation costs to bring products to market.

Extending this kind of reasoning to land development in modern polycentric cities is difficult. But we can speculate. Congestion tolls allow faster access for higher-value trips to employment centers, which encourages more high-value trips to those centers. More trips on transit means less space devoted to parking. Thus, commercial centers would tend to get bigger and denser. Increased demand for trips on transit encourage the construction of housing oriented to transit, especially in larger metro areas with rail transit systems. This higher-intensity development, combined with more expensive highway travel, discourages sprawling development at the urban fringe. Lee (1992) comes to similar conclusions: better transport pricing would cause subcenters (including the CBD) to be more intense, closer together, and (possibly) fewer; the overall spatial extent of urbanization would be less; and transportation between subcenters would take some type of high-capacity form. Chapter 5 builds on this logic when discussing the importance of transit-oriented development patterns.

Shoup (2005a) notes that if only three parking spaces exist per vehicle (one at home and two elsewhere) and the average cost of a parking space is \$4,000, the total capital value of parking is roughly equal to the capital value of the entire vehicle fleet plus all roads.

Pricing Parking

The basics

Every vehicle trip launches from and lands in a parking space. The vast majority of trips are intermediate goods: they are a means to acquiring some other good, service, and desired activity. If you can't stop the car and get out, the trip has no value. (For the great majority of drivers. We acknowledge the existence of cruisers: not just the young with the chest-crushing bass, but the old with the wispy comb-over flogging the passengers bundled up in the backseat of his cool convertible. Drive on.)

Suburban residential areas include garages, driveways, and on-street parking. Suburban shopping malls are surrounded by parking lots sufficiently massive to provide free parking on the busiest shopping days. Municipalities provide parking on city streets and in parking structures. Businesses routinely supply parking to their employees. Shoup (2005a) estimates there are at least four parking spaces per car in the U.S: thus, at least three-quarters of those spaces are empty all of the time. But every car has to be parked somewhere during the 90+ percent of the time it's not moving.

All that parking is expensive. Table 4-1 shows the midpoint of high and low estimates of the total annualized cost of off-street parking is about 6 percent (± 3 percent) of the total cost of driving, almost double the cost of fuel. Shoup (2005a)¹⁷ notes that if only three parking spaces exist per vehicle (one at home and two elsewhere) and the average cost of a parking space is \$4,000, the total capital value of parking is roughly equal to the capital value of the entire vehicle fleet *plus* all roads.

Despite the high cost of free parking (the title of Shoup's book), drivers pay directly for less than 10 percent of the cost. Most people park at no daily or hourly charge at home, work, and shopping. Of course, most urban dwellers

pay for on-street parking at parking meters and occasionally pay for hourly parking in parking structures. Some commuters pay for parking in structures by the month. But the great majority of parking is unpriced.

How is all this parking paid for? The land developer and home builder provide on- and off-street residential parking. Its cost is bundled into the prices of houses, lots, and land.¹⁸ The cost of commercial parking is bundled into the retail prices of the goods and services the businesses in the commercial development sell. Employee parking is paid for by employers who provide free parking as part of the employee compensation package. Strictly speaking, employees pay for parking through lower wages, but because of federal and state income-tax policy, the government pays some of the cost too. We all pay for parking (drivers or not), just not explicitly.

Why is most parking not priced? First, part of the answer is that pricing parking has its own costs. Pricing parking on city streets requires parking meters and enforcement staff and equipment. Pricing parking lots and structures requires limiting access and staffing either toll booths or enforcement staff. These investments may not pay for themselves in many suburban areas. Development patterns contribute to the costs of charging for parking in two ways.

1. *Many small parking lots make pricing costly.* A significant proportion of retail development occurs along suburban arterial streets, each with its own parking lot. Building a fee-collection station that serves 500 parking spaces is more cost effective than building 10 toll stations to serve 50 spaces each.
2. *Parking requirements make the “market-clearing” price of parking very low.* Minimum parking requirements are virtually ubiquitous. The rationale is that each developer should supply sufficient parking for her development to prevent “free-riding” on neighboring parking lots. Sufficient parking typically means enough spaces to accommodate peak demand at zero price. It seems a bit odd to charge for parking when ample parking spaces are available—parking spaces the developer was required to put in. Low prices are unlikely to generate revenues sufficient to cover the costs of the people and capital required to collect the charges.

Second, the federal tax code encourages businesses to supply free parking for their employees and customers. Businesses can write off as a business expense the cost of providing parking, whereas commuters and shoppers must pay for parking out of *after-tax* income. The result is that consumers who drive to shop pay less in terms of higher prices, and employees who drive to work enjoy higher disposable incomes if parking is supplied (and written off) by businesses. Drivers pay only the after-tax cost of parking indirectly through lower wages and higher retail prices.

Third, retailers (and employers) may fear that charging for parking will hurt their business:

- Many people visit suburban retailers as much for the entertainment value as for the shopping convenience. People enjoy window shopping and browsing. Browsers often become buyers. Retailers may reasonably fear that parking charges will discourage browsing, even with the lower retail prices that accompany parking charges.¹⁹
- Stores (or employers) who charge for parking may be at a competitive disadvantage. Efficient parking charges generate net social benefits, but one retailer or employer can often gain an advantage over others by charging less for parking or offering free parking, given that everyone else charges. Since everyone feels the same incentive to offer low-cost or free parking, everyone eventually does.²⁰

Development patterns contribute to the costs of charging for parking in two ways.

1. *Many small parking lots make pricing costly.*
2. *Parking requirements make the “market-clearing” price of parking very low.*

All of these forces conspire to make parking a dominant urban land use. By itself free parking encourages car travel by reducing the out-of-pocket cost of a trip. In addition, the vast amounts of parking needed to accommodate these trips discourage any other form of travel: walking from a bus stop across a parking lot to a mall is not just time consuming, it is also inconvenient (in bad weather, carrying packages), risky, and uninspiring.

Improving the pricing of parking

The issue for policy makers is how best to encourage private businesses to charge market rates for parking. Optimal parking pricing is similar to congestion pricing or the pricing of restaurant meals. Off-peak users pay for variable costs: maintenance and toll collection. Peak users pay for the parking infrastructure—the capacity that must be built to meet their needs. So price should vary by time of day.

The obvious first step, a step beyond the power of local planning agencies, is to change the federal tax code. Without the current tax break, employers would have less incentive to offer free parking as a benefit separate from an employee's wage. They could make the employee equally well or better off by instead explicitly raising wages by the cost of parking. The employee would then have the choice of whether to spend those additional wages on parking or to try to save some of the money by using lower-price modes to commute to work. This change in tax code has received some attention at the federal level.

A more immediate step, but still beyond *local* planning agencies, is adoption of regional parking policies. When many businesses compete for customers, free parking becomes a necessary enticement. The lesson is that policy to encourage market pricing for parking must be regionwide if not nationwide. To garner sufficient support for changes in policy, businesses in one part of the region cannot feel they suffer a competitive disadvantage because of the policy. This may be difficult to achieve given the fragmented system of local governments found in most metropolitan areas. It requires a regional authority.²¹

Regional parking policy has to include at least three components:

1. Dropping minimum parking requirements. Standard planning practice is to require enough on-site parking to accommodate *peak* parking demand at *zero* price. Pricing reduces the quantity of parking demanded. Minimum parking requirements should be reduced, and design standards implemented with an eye toward facilitating parking charges and accommodating non-auto modes of travel.
2. Charging for publicly owned parking at market rates. The pricing of most metered on-street parking is too low. The evidence is that vacant spaces are too rare during peak periods. This forces drivers to “cruise” for parking, which wastes time and contributes to downtown traffic congestion. Individual public jurisdictions feel the same incentive to keep parking prices low as do their business constituents.
3. Encouraging private owners of parking to charge market rates. A regional tax on all private parking spaces negates the federal tax loophole but does not address the prisoner's dilemma. Unbundling the ownership of parking from the businesses it serves improves the incentive to price effectively. In most of the urban centers developed before the general implementation of parking requirements (i.e., in most CBDs), private businesses own and operate surface parking and parking structures. They charge time-varying, market-clearing tolls.

The regional transportation authority would also serve a coordinating function. Construction of a single system for collecting parking taxes, coordinated by the regional transportation authority, would reduce the cost



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The pricing of most metered on-street parking is too low. Drivers are forced to cruise for parking, which wastes time and contributes to congestion.

of administering the tax. The regional transportation authority could also research efficient designs for parking facilities and assist the owners of existing development to convert free parking to priced parking.

Getting the full benefits of pricing requires access to alternatives to car travel. People can carpool to work or shopping, or simply shop less, but other alternatives are often missing or of poor quality. A jitney service appears to have merit, as do improvements to public mass transit.

Locally, many cities are torn between a desire to stimulate economic development in a downtown and to price parking (for both efficiency and fiscal purposes). Some justify (not without reason) parking subsidies as necessary for density, which in turn is necessary to make transit work. Most have come to accept the idea of metered on-street parking as compatible with successful downtown retail (prices can be used to ensure that a street space is usually available for someone who wants to make a purchase). The bigger problem for a downtown is the free parking in shopping and employment centers in the rest of the city or in neighboring cities.

The big problem in implementing parking pricing is the vast supply of parking infrastructure in many urban and suburban areas. Highway capacity is probably also overbuilt, but not to the same extent. The good news is that surface parking can be redeveloped.

Effects of parking pricing on transportation and land use

Parking prices differ from congestion prices. Congestion prices are charges per vehicle mile, parking prices are charges per vehicle trip. Congestion pricing discourages vehicle miles traveled (VMT) during peak periods; parking pricing discourages vehicle trips during peak periods.

But parking prices are related to congestion prices. Congestion prices are higher on highways that serve desirable locations and that occupy land with high opportunity costs (because they are in desirable areas). Parking prices are similar. Shoup (2005a) estimates that optimal parking charges will be similar to, though likely higher than, congestion tolls for the typical driver. Payments for parking do not, however, vary with distance traveled, as congestion tolls do.

Parking charges reinforce the effects on transportation of congestion charges. They discourage vehicle trips, and so encourage travel by high-occupancy vehicle. Research into the factors that influence the success of transportation demand management (TDM) programs suggests that drivers respond strongly to increases in the price of parking.²²

The effects of parking charges on land use differ somewhat from those on congestion. First, parking charges do nothing directly to discourage long-distance commutes. Second, they probably give more of a boost to retailers and employers in central areas. Parking in suburbia is usually unpriced. Thus, travelers are pulled to central areas by lower congestion and easy-to-find parking, but they are also pushed from low-density strip development that no longer offers free parking. Third, the big oversupply of parking lots means opportunities for infill, especially in well-located suburban areas.

Parking charges are not a substitute for congestion pricing, though implementing them would help reduce congestion, at least in the short run. Parking charges will reduce the number of vehicle trips. Reducing the number of small parking lots along arterials would also increase the capacity of the arterial by increasing travel speeds.²³ Parking charges do nothing, however, to internalize the cost an additional driver imposes on other drivers when he enters a congested roadway. In the absence of congestion pricing, most roadways in attractive areas will continue to suffer from inefficiently high levels of congestion. For parking pricing to change that outcome, it would have to be ubiquitously (regionally) applied and to vary with time of day.

Research into the factors that influence the success of transportation demand management (TDM) programs suggests that drivers respond strongly to increases in the price of parking.

A DOZEN REASONS FOR RAISING MOTOR-FUEL TAXES

1. Motor fuel taxes are lower now than in the past.
2. U.S. fuel taxes are well below levels in other countries.
3. Fuel taxes are well below their theoretical optimum.
4. Drivers show remarkable tolerance for fuel price changes.
5. The cost of transportation projects continues to rise faster than revenue.
6. Congestion is growing in part because we are not spending enough on new capacity.
7. Relative declines in fuel tax revenues increase reliance on non-transportation related taxes to support transportation projects.
8. The relative decline in fuel tax revenues is increasing borrowing for transportation projects and programs.
9. Fuel taxes have low collection costs and are relatively fraud proof.
10. Fuel taxes are user fees that send "price signals" to motorists to use the transportation system more efficiently.
11. Fuel taxes are fairer toward the poor than the alternatives currently available.
12. Fuel taxes make it easier to transition to better user fees in coming years.

Until electronic tolling is widespread, the alternatives most often put forward to avoid increasing fuel taxes are less effective, less efficient, and less equitable. ■

Source: Wachs 2003

Pricing Auto Emissions

Cars generate noise, spew chemicals into the atmosphere, and leak oil, gas, and coolant onto the pavement, which find their way into streams, rivers, and lakes. These emissions generate a variety of costs that are hard to estimate with precision, but are likely big. The midpoint estimate in Table 4-1 puts the cost of auto emissions at more than 10 percent of the total cost of car travel, almost double the cost of roads and highways.

Drivers, as drivers, do not pay the full cost of their use of the environment as a waste dump. The fuel tax could, in theory, be a relatively crude fee on emitted wastes from burning gas in car engines: air pollution is roughly correlated with gallons of fuel consumed. The gas tax is, however, explicitly viewed as a fee for use of highways. In that case, drivers are using scarce environmental resources at no explicit out-of-pocket cost to themselves.

Drivers do pay some costs implicitly. The conventional approach for reducing environmental pollution is to set ambient air and water quality standards, which cannot be exceeded more than some specified number of times per year. To achieve these standards, regulators impose requirements on equipment and materials. All cars, for example, are required to have mufflers and catalytic converters. Gasoline cannot contain lead and other chemicals that vary by state. These requirements cost drivers in the form of higher car prices or poorer performance.

The concern with the "command-and-control" approach to emissions is that its costs are unnecessarily high. Some of the costs are bureaucratic: regulators have to define a wide variety of equipment and materials standards, then monitor and enforce compliance. Some of the costs arise from inefficiencies in their application—a standard set of requirements fails to fit all drivers and all situations well. Probably most costly is the failure of regulation to generate innovation. What would the world's competitive car makers have come up with in the last several decades if they and drivers had felt strong incentives to reduce auto emissions?

Pricing generates these incentives. The most efficient policy imposes an emissions tax per unit of emissions by type of emission. These fees encourage drivers to drive less and to look to buy lower-emissions cars. Manufacturers then have incentive to find ways to meet demand as cost-effectively as possible. Those drivers for whom the value of polluting exceeds the tax would continue to pollute, but the government could use the revenues from tax collections to make other improvements to the transportation system, compensate those affected by emissions, or mitigate the effects of the emissions on other activities.

Of course, practical problems arise in implementing an efficient system of emissions charges. First, the government must be able to measure the emissions of each vehicle. In principle, instruments could be mounted on each vehicle that measure and record emissions at regular intervals. Alternatively, the government could require all owners to take their vehicle into an inspection station in which average emissions are measured. Such requirements already exist in many urban areas troubled by air pollution. The government could collect the emissions fee at the inspection station by multiplying the per-mile emissions tax appropriate to the vehicle by the number of miles traveled since the previous inspection.

This procedure is rough because it fails to recognize that the damage done to the environment by a particular vehicle varies by location. For example, a vehicle typically does greater damage the higher the intensity of land development simply because more people are affected by the emissions and probably because the impact of another unit of pollution is greater in dense corridors where levels of pollution are already high. A person who drives a

given car in downtown areas should be charged more than another person driving the same car in rural areas.

One solution is to use technology similar to that used to price congestion. When a driver purchases fuel, for example, he could also pay his congestion and emissions tax. The amount of the tax can be recorded on a magnetic card inserted into a computerized receiver mounted on every car. During periodic inspection, a technician measures the characteristics of the vehicles emissions and records that information on the card or in the vehicle's receiver. Transponders on the roadside send information to each vehicle both about congestion charges and emissions fees on that corridor at that time. The receiver displays the total externality price of driving on each section of each highway.

Of course, setting the emissions charges is difficult given the uncertainty in the values of the damages emissions cause. Baumol and Oates (1988) suggest a "standards-charges" approach. The idea is analogous to the setting of congestion prices in Singapore: define an ambient environmental quality target, then set an emissions fee that might achieve it. Monitor the actual effect of the emissions fee on ambient environmental quality to obtain information about the ability of drivers and car manufacturers to reduce emissions. Given that information, revise targets and fees as necessary.

The effect on travel behavior of a reasonably efficient emissions fee system depends on how easily (1) manufacturers can improve the environmental performance of cars and trucks, and (2) drivers can reduce their travel by auto. Given the necessary incentives (primarily from the expected increase in consumer demand in response to fees), automakers probably can develop the technology to produce significantly quieter and cleaner cars and trucks (e.g., by adapting vehicles to the use of cleaner fuels, such as electricity or natural gas). The lower impact of such improvements on the environment reduces emissions fees and drivers' incentive to change travel behavior. To the extent those incentives exist, drivers will change their behavior to avoid driving in areas with relatively high fees.

The effect of this hypothetical fee on urban form depends on changes in travel behavior and on improvements in the quality of urban environments. On the one hand, higher fees in intensively developed areas encourage firms and households to relocate to areas where the fumes and noise their autos generate disturb fewer people (thus, reducing their cost in fees). On the other hand, cleaner environments encourage more development in central areas. The net effect on urban form of efficient emissions fees is uncertain.

In the past, the damage caused by auto traffic in central areas has undoubtedly contributed to household and businesses relocating to suburban areas. Chapter 2 explains that a cause of polycentricity is the eventual dominance of the diseconomies of congestion (including pollution) over the economies of concentration. Environmental quality matters. Our guess is that cleaner environments, combined with improvements in nonauto modes, would in net increase the attraction of central areas to both firms and households.

EFFICIENT PRICING OF LAND DEVELOPMENT

We conclude in Chapters 2 and 3 that public decisions about the siting and pricing of transportation facilities have had stronger impacts on land development patterns than land-use regulations have had. Nevertheless, as our discussion of parking requirements demonstrates, land-use regulations matter. At least part of the rationale for conventional land-development policy in North America has been similar to that of transportation policy: there is abundance, not scarcity; therefore, address increasing demand by adding capacity (e.g., use more land at the urban fringe).



Ian Francis, istockphoto.com

Manufacturers can improve the environmental efficiency of cars to address emissions externalities. The plant in the foreground is rape seed, oil from which can be used to make biofuels.

Scarcity, however, has been hard to banish. Households have responded aggressively to the availability of accessible land in especially desirable locations. As with highway capacity, a policy of simply building bigger parking lots and separating conflicting uses is no longer adequate in these popular areas. Instead, land-development policies have to provide land users with the incentives to use land more efficiently.

One method is to price land development at marginal cost. In this section we briefly outline opportunities for doing that. We start with a discussion of pricing land-use externalities and then extend that discussion to alternatives to setting prices that nevertheless generate pricelike incentives. We then look at pricing extensions of public services and at the property tax.

Pricing Land-use Externalities

Policy can help internalize the external costs that a variety of land users impose on their neighbors. Local governments could price the external effects of industrial land uses. The resulting changes in land use, especially changes in the relative locations of trip origins and destinations, feed back to influence travel patterns. Similar reasoning applies to externalities generated by retail firms and to externalities that households impose on other households.

Pricing industrial emissions

The U.S. Environmental Protection Agency (EPA) and some states and metropolitan areas have implemented programs that price industrial emissions. The most common approach involves some form of emissions trading. Rather than setting a price directly, a pollution-control agency sets the maximum total quantity of emissions to meet a performance standard. Rights to these emissions are distributed among polluters, who then are free to trade them. Trading establishes a price for emissions and allocates emissions to the polluters with high abatement costs.

This sort of program works less than efficiently when the effect on society of industrial emissions varies with the location of the emitter. A manufacturing plant far removed from residential development does less direct damage to human populations than does the same plant located within an urbanized area. That fact motivates zoning's segregation of land uses. But zoning is a blunt regulatory instrument relative to pricing: some manufacturing firms can reduce emissions much more easily than others and would be willing to reduce emissions in exchange for a more accessible location.

Efficient pricing makes that trade-off possible. The system is analogous to an efficient system of congestion pricing in that emissions prices vary by location. The closer a manufacturing firm's location to resources and development harmed by its emissions, the higher the emissions fee. The difficulty is in setting the fees. To set fees accurately, the government must have an accurate measure of the damages done to society of a wide array of pollutants. Moreover, the government must be able to measure with accuracy the amount of each pollutant emitted by each firm.

There are alternatives to pricing given the ability to monitor emissions: performance zoning is one. In this case, the zoning authority sets ceilings on the quantity of emissions allowed rather than setting prices. These ceilings are lower (i.e., emissions standards tougher) in areas where emissions do more damage, such as near residential areas. Nevertheless, firms that can meet these standards cost-effectively gain more flexibility in location. The "price" of that flexibility is bearing the costs associated with meeting the performance standard.²⁴

Reasonably efficient performance zoning could have significant effects on transportation and land use. Travel volumes decrease as manufacturing firms move closer to residential areas and business centers, improving the efficiency of the transportation system and local economy.

Zoning is a blunt regulatory instrument relative to pricing.

Pricing retail externalities

In principle, retail externalities differ little from industrial emissions. Successful retailers create traffic in their vicinity. Noisy trucks make early morning deliveries. Vehicle traffic harms nearby environments throughout the day. Some neighbors will consider the large number of people in the vicinity as a threat to safety. The damage inflicted to nearby environments decreases rapidly with distance. Segregating retail activities to areas distant from their customers, however, increases auto travel, which increases the costs of access and has its own negative effects on the environment.

The problems a zoning authority faces in defining commercial zones are identical to the problems it encounters when defining industrial zones. The inefficiencies of commercial zoning, however, may not be as large. Though retail firms impose negative externalities on residential areas, they provide positive externalities for other firms. Retailers benefit from clustering. Because the negative externalities typically diminish quickly with distance, there is no need to send retailers miles downwind. Finally, retailers benefit from locations close to heavily traveled arterial streets, while households prefer quieter locations. Zoning land along arterial streets and near major intersections for commercial uses probably just follows the market.

Potential benefits arise, however, from pricing certain commercial externalities. As explained in our discussion of pricing parking, parking requirements eat up land and the design of parking lots and their entrances can contribute to congestion on arterial streets. Imposing a tax or standard based on the impact on traffic of the design of the parking lot could encourage retailers to adopt more efficient designs. The conventional focus on minimum parking requirements gives the retail developer little incentive to develop innovative designs.

Pricing or performance standards would also encourage the location of retail firms in what are now exclusively residential zones. Central Place theory (Chapter 2) predicts that retailers locate to establish their own market areas. The size of an individual retailer's market area depends on the local demand for its product and the extent of the firm's scale economies. Agglomerative economies encourage retailers to cluster. But some retailers, such as small grocers, would probably benefit from locations within residential neighborhoods.

Zoning in most newer residential areas simply excludes all retailers. Given proper incentives, some retailers could probably adopt building designs that limit external costs and may be willing to pay either in cash or in kind (e.g., by providing a local neighborhood amenity) to locate within the neighborhood. These adjustments require at least flexibility in the zoning rules and usually costly negotiations between the planning department and developers. But these extra costs could have long-term payoffs in reduced car travel.

Pricing neighborhood externalities

Residential neighborhoods teem with minor externalities. The noise from a neighbor's lawnmower may be irritating on a peaceful weekend. The poor condition of a neighbor's yard may cause more than minor irritation when it comes time to sell a house.

Many of these minor externalities cannot readily be priced. Though the economic theory remains solid, the effectiveness of pricing is reduced by transaction costs. It is too expensive, relative to benefits, to set up a permit system for lawn mowing in which the price of the permit increases with the noise generated by the mower. In many cases, the existing system of give-and-take—combined on occasion with polite complaints—works pretty well, especially in neighborhoods consisting of fairly homogeneous households. Most people realize that irritating a neighbor can cause problems.

Pricing or performance standards would also encourage the location of retail firms in what are now exclusively residential zones.

But efforts, through zoning in particular, to keep residential neighborhoods homogeneous may be costly on a larger scale. We discussed Tiebout sorting in Chapter 2. Households are willing to bear relatively long commutes to live in neighborhoods in which others have similar preferences for local public goods and services. The most important of these is school quality, but other community services and the extent to which households maintain their yards and houses also matter. Commutes are longer and neighbors are often willing to fight vigorously to prevent nuisances and to live in desirable neighborhoods.

Funding local public goods and services with property taxes plays a role. A household that purchases a relatively small and lower-valued house enjoys the same level of public services as its neighbors but pays less in property taxes. Turnbull, Dombrow, and Sirmans (2006) find that otherwise similar houses tend to sell for higher prices when most of the other houses are larger and more expensive—homebuyers compete for the good deal on local public goods and services. Because larger houses generate more tax revenues, the owners of large houses have incentive to try to prevent the construction of smaller houses. Funding local services, especially schooling, using sources other than property tax reduces these incentives.

Improved financial incentives might also contribute to control of home maintenance externalities. Landlords, for example, encourage renters to better maintain their properties by requiring a deposit from which damages are subtracted. Something similar might work at the neighborhood level. Or property taxes could be used to provide subsidies for basic maintenance, such as yard maintenance, trash removal, or painting. These kinds of financial incentives internalize the externalities involved in home maintenance.

We recognize that implementing any system of prices in residential neighborhoods would be a big step in any city. Change requires more justification and detail than we can provide here. Yet dealing effectively with the more severe problems found in residential areas virtually requires pricing. Courts in New Jersey, for example, have required that residential zoning be revised so that all neighborhoods allow their share of low-income housing. Other cities are considering or have adopted policies to encourage such inclusionary zoning. A reasonable system of taxes and subsidies could ease the concerns of people in neighborhoods most resistant to these changes in zoning.

Governments historically have not priced extensions of urban services at cost. Conventional land-development policy is pro-development: simply extend urban services to new development.

Pricing Extensions of Urban Services

Public agencies (cities, counties, special service districts, and public utility commissions) supply or regulate the private supply of a range of urban services: sewer, water, and drainage; electricity and natural gas; police, fire, and emergency services; schools and recreation facilities. The provision of many of these services, especially public utilities, requires large investment in physical infrastructure; production of these services exhibits extensive network and scale economies. Public agencies provide or regulate private production in large part to avoid the potential abuses of private monopolies.

How service extensions are priced influences the prices land developers are willing to pay for undeveloped land. To determine their willingness to pay for land, developers work backward from the revenue they expect to generate from lot or house sales, subtracting the expected costs of development, construction, permitting, financing, and so on. The land seller essentially gets the leftovers. A larger charge for extending urban services means less left over for land, which influences whether and how land gets developed.

Governments historically have not priced extensions of urban services at cost. Conventional land-development policy is pro-development: simply extend urban services to new development (Frank and Downing 1988). The developer constructs and pays for infrastructure within the development,

and the local jurisdiction extends services to the development. Until the 1970s, local jurisdictions funded extensions of public services through general tax revenues (usually raised through property taxes) or through general obligation bonds eventually paid for with general tax revenues. This is essentially average-cost pricing.

The problem is that the cost of supplying infrastructure varies by the type and location of development (Frank 1989). Costs vary with the density of development, distance from existing facilities, slopes, and soil conditions. Average-cost pricing encourages too much development in high-cost areas and too little in the areas closer to central facilities and developed at higher densities. Better-located and higher-density development essentially subsidizes higher-cost development. And the lower-density development feeds back to affect the transportation system by increasing trip lengths and discouraging the use of modes other than the auto.

Voter resistance to higher taxes has led to an increasing number of communities attempting to make developers pay not only for the infrastructure within the development, but also for their share of the cost of extending infrastructure to or improving infrastructure in that area. Burge and Ihlandfeldt (2006) report that roughly a quarter of local jurisdictions charge development impact fees. Many others require developers to extend urban services or infrastructure at their own expense. These practices have two effects: increasing the funding available for infrastructure extension or improvements, and improving the locational efficiency of development.

Burge and Ihlandfeldt estimate the effects of impact fees using a unique data set of land development in counties in Florida. They find that impact fees tend to increase the amount of residential development. One reason is that the impact fees simply increase funding for services extensions. Another is that impact fees reduce local resistance to the additional development. Neighbors appear sensibly less prone to oppose a development proposal if it pays for itself. This has resulted in an increase in development of relatively small houses in inner suburban areas, improving the use of existing infrastructure and providing more conveniently located housing for moderate-income households.

Improving the Efficiency of the Property Tax

The property tax has long been used by local governments to finance local public services. Until World War II, receipts from local property taxes exceeded the expenditure of the federal government. With the growth in state and local governments during and after the Great Depression of the 1930s, however, the relative importance of the property tax fell; property tax collections fell to about 2.6 percent of gross national product in 1950 from about 5 percent of GDP in the 1920s. Property tax collection rose to about 3.9 percent of GDP by 1970 to fund the increase in demand for local public services after World War II (especially primary and secondary education).²⁵ Since then, property-tax collections have decreased to about 3 percent of GDP, similar in magnitude to receipts from sales taxes.²⁶

Most jurisdictions impose the same tax rate on the values of both land and improvements. The tax rate, for example, may be \$20 per \$1,000 of market value, or 2 percent of capital value per year. If the assessed value of the structure is \$200,000 and that of the lot is \$100,000, the total tax payment would be \$6,000 per year, \$4,000 on the value of the house plus \$2,000 on the value of the land.

How do market participants react to an increase in the local property tax rate, all else held constant? Holding the per-square-foot prices of land and housing constant, homebuyers can reduce their tax liability by consuming smaller lots and houses. An increase in the tax on a good usually has this



dsharpie, istockphoto.com

Impact fees increase the funding available for infrastructure extensions (e.g., for sewer drains) and improve the locational efficiency of development.

effect: higher prices and lower quantities produced and traded. In the absence of market failure (such as externalities), taxes reduce the efficiency of the allocation of resources.

An example makes the argument less abstract. Many people consider building additions to their homes. Most homeowners recognize that the cost of a new addition is more than the cost of plans, permits, materials, and labor: the addition increases the market value of the house, which increases the homeowner's property tax obligation. Increasing that obligation by increasing the tax rate on improvements increases the total cost of building an addition. At the margin, fewer additions will be built if the tax on improvements is increased.

Is the assumption of constant land and structure prices reasonable? Yes, in the case new structures. Building materials are mobile; that is, they are easily and routinely shipped from one location to another. The result is that base materials prices are determined regionally or nationally with adjustments for transportation costs. A change in local tax rates does not affect materials prices in the long run (i.e., the local supply of building materials is highly price elastic). Enough workers are similarly mobile over the long run that changes in tax rates on structures have little effect on local wages. Thus, higher property tax rates induce consumers to economize on structures.

The assumption of constant land prices, in contrast, is not valid. Land prices respond to changes in the property tax rate. The reason is that land is immobile. The higher property tax rate decreases consumer willingness and ability to pay for land, putting downward pressure on prices. While labor and materials can seek higher prices elsewhere, land is stuck where it is (i.e., the supply of land in any one area is highly *inelastic*). The land owner has little choice but to accept the lower prices on offer, and land prices fall enough to compensate for the higher tax payments; that is, tax payments *capitalize* into land prices.

The effects on land of an increase in the property tax rate differ from those on structures. A higher property tax discourages improvements while having no effect on land consumption. Thus, reducing the tax rate on structures while increasing the rate on land encourages more intensive use of land. That kind of shift increases the efficiency of the allocation of resources to land development and could affect the efficiency of the transportation system as well.

Reformers, at least since Henry George proposed the single tax (a tax only on the value of land) in the 1880s, have argued that land, not improvements, should be taxed. To our knowledge, no jurisdiction has completely exempted the value of improvements from the property tax. Anderson (2005) points out that two-rate tax systems are in use in Australia, Canada, Jamaica, New Zealand, and 17 municipalities in Pennsylvania, including Pittsburgh. Severn (1992) reports that many jurisdictions grant tax abatements on improvements in new developments as part of their economic development strategy. Though tax abatements are rarely permanent, they do decrease the overall cost of improvements.

A higher property tax discourages improvements while having no effect on land consumption. Thus, reducing the tax rate on structures while increasing the rate on land encourages more intensive use of land.

CHAPTER 4 NOTES

1. http://www.bts.gov/publications/national_transportation_statistics/html/table_03_14.html.
2. Delucchi, Mark A. (2004) Summary of Theory, Data, Methods, and Results: Report #1 in the series: *The Annualized Social Cost of Motor-Vehicle Use in the United States, based on 1990–1991 Data*. Institute of Transportation Studies, University of California, Davis, Research Report UCD-ITS-RR-96-03(01)_rev1. Available online at http://pubs.its.ucdavis.edu/publication_detail.php?id=153.
3. The Bureau of Labor Statistics, the keepers of the Consumer Price Index, report that prices rose generally by about 56 percent between January 1990 and January 2006.

4. Examples of maintenance are patching, bridge painting, snow and ice removal, and pavement marking. Examples of related services are research, law enforcement, and safety.
5. Roadway-related income and expenditure totaled about \$71 billion in 1989. About \$65 billion was spent on services, the remainder to paying interest on and retiring debt. About \$63 billion was raised through taxes (user fees and general fund); the remaining \$8 billion was raised through bond issues and interest on invested funds.
6. Other expositions of these relationships use the ratio of volume to capacity on the horizontal axis, which gives a dimensionless measure that approaches 1.0 as the volume of a highway segment reaches its calculated maximum capacity for throughput.
7. In reality, the PTC curves vary across individuals because different individuals value time differently and drive cars that vary in operating costs. Focusing on the typical driver simplifies the analysis without altering the results.
8. We assume the *value* (benefit) of the trip to society coincides with the value to the individual; we have not identified external *benefits* of trips.
9. It's possible, of course, that on a given day some additional drivers might underestimate the level of congestion on the highway and contribute to a higher traffic volume. They would, however, regret their decision and would avoid entering the highway in the future. Thus, on a day-to-day basis, the traffic volume should vary near the equilibrium volume.
10. The term 'price' refers to the amount an individual pays for a good or service, and the term 'cost' refers to the value to society of the resources expended to produce the good or service.
11. The entry of one truck can cause a total of two to five hours delay.
12. Olszewski and Xie (2005). At the time, S\$1.00 = US\$0.58.
13. This idea, novel and almost unused in the U.S., was working fine for the three months one author lived in Ponce, Puerto Rico, 35 years ago.
14. Downs (2004a, 149) cites Polhill and Edgar who describe an inadvertently created opportunity in Florida 15 years ago when jitney's flourished and beat the public bus system on performance and cost before the loophole allowing them was closed.
15. A general admonition of economists about public policy is relevant to transportation as well: the multiple social problems that are correlated with low income (housing, health care, education, transportation, and more) are dealt with most efficiently by getting more income to low-income households and then letting them choose the goods and services they find most beneficial. That said, direct income-redistribution policies are nonexistent and inefficient at a local level, and politically difficult at the state and federal level. Advocates for social justice have to take their wins where they can get them, which is usually as an add-on to the primary missions of executive federal and state agencies.
16. See Appendix C and O'Sullivan (2006, 221–23) for more detail on this somewhat complex but important argument.
17. Shoup's book, at over 600 pages, is *the* definitive treatment of parking. It is written, as is this report, from an economic perspective.
18. A land developer works backward from the revenue he expects from sales of houses or lots, subtracting development costs, to determine his bid for undeveloped land. Households value parking, and so some, probably most, of parking costs are recovered through higher house prices. But local regulations may require the developer to provide more parking than householders are willing to pay the full cost of. These costs translate into lower offers for land. Thus house buyers and land owners share the cost of parking. The same argument applies to non-residential development.
19. Retailers may fear that people are simply more sensitive to the direct costs of parking than to the price savings on purchases that the parking revenues would allow retailers to offer.
20. This an example of what economists refer to as a "prisoner's dilemma." The situation is one in which everyone benefits from cooperation, but each individual does even better if he cheats while everyone else cooperates. See any introductory economics text for a description. The problem was initially framed in 1950 by Merrill Flood and Melvin Dresher. Albert Tucker formalized the problem using as an example reductions in prison sentences designed to encourage prisoners to "rat" on their colleagues—ergo, the prisoner's dilemma.
21. See Downs (2004a, 298–320).
22. Kuzmyak (1990) and Wachs (1991) conclude that parking charges have been the most effective TDM policy. The strength of the response of drivers to increases in

the price of parking depends on the magnitude of the increase and on the availability and price of alternatives to driving alone.

23. In an effort to reduce congestion on urban arterials, local governments in Washington County, Oregon, the fastest-growing part of the Portland metropolitan area, have imposed limits on the number of entrances to urban arterials in newly developing areas. That kind of regulation will help encourage parking designs amenable to parking charges. But taxing free parking would reduce the need to regulate directly the number of entrances to streets.
24. Standard zoning has already moved in this direction by distinguishing, for example, between "light" and "heavy" manufacturing.
25. Aaron (1975).
26. Statistical Abstract of the United States, 2004.

CHAPTER 5

Achieving a Balanced Transportation/Land Use Connection

Organization: The first edition of this PAS Report (1994) moved from the history and economics lessons of chapters 2, 3, and 4 directly into the details of how to evaluate transportation and land-use policies in the context of objectives for urban transportation. We knew that regions had broader objectives than just those for transportation, but we gave them a shorter discussion and an even shorter evaluation. In this edition, we provide additional material about transportation's connection to land use to provide a context for our evaluation of transportation policies in Chapter 6.

This chapter has three sections:

1. *Section 1: The balance of transportation policy relative to land-use policy.* Chapter 2 noted that while land-use policy is almost ubiquitous, private market decisions have been the prime movers in land development. Public agency decisions, in contrast, have been the drivers of transportation system development. This section makes the case for a stronger role for regional land-development policy in addressing metropolitan transportation problems.
2. *Section 2: Planning instead of pricing.* Chapter 3 makes the case for pricing. But proper pricing is not yet at the top of political agendas, and it is difficult to do comprehensively. Could comprehensive changes in land-development policy get similar results more easily? This section describes the types of land-use policies many planners feel are essential to improving transportation and quality of life.
3. *Section 3: Local land-use policies that can affect transportation.* As with pricing, comprehensive regional land-development policy faces significant technical and political hurdles. In practice, changes in land-development policies will be incremental and implemented by local planning agencies. This section describes and evaluates the most important of these policies.

Basics: One can argue that transportation and land-use policy making are out of balance for the following reasons:

- Largely because of the need for a seamless network, transportation planning is done at large geographic scales (national, state, and metropolitan) and has a large, dedicated revenue source.
- Largely because most of the impacts of land development are local, most land-use planning is done by the numerous smaller units of local government whose perspective tends to be small scale and independent.
- Though land development affects transportation, and transportation planning requires land-development forecasts, the numerous small agencies charged with influencing land development have little influence over transportation. Instead, transportation agencies tend to think in terms of immediate transportation objectives and not about the objectives for land development (and related objectives for quality of life) typically treated in local land-use plans.

- The direction for change seems clear: More attention to land-development planning when doing transportation planning (and especially in the existing metropolitan process for transportation planning) is likely to yield more efficient transportation plans and net improvements in regional quality of life.

What changes in land-development policy would integrated regional planning entail? Most planners agree on (1) more attention to the design of buildings and neighborhoods to allow transit, cycling, and walking to provide alternatives to travel by auto, which probably requires (2) higher densities and mixing of uses, which probably require (3) dealing with land-use externalities in ways other than physical separation.

Transportation and land use affect each other in many ways, and vice versa. A framework for thinking about those impacts focuses on five dimensions: density, diversity, distance, design, and destination.

The scope of transportation considerations has widened over time: from the pavement, to the right of way, to the corridor, to an integrated evaluation of the broader effects of a single transportation project on the transportation system and other aspects of quality of life.

Many land-use policies have impacts on the transportation system. This chapter discusses those policies in five broad categories:

1. Creating boundaries: zoning and urban growth containment
2. Creating centers: transit-oriented development (TOD)
3. Creating corridors: retrofitting major urban arterials
4. Pricing public infrastructure and services: concurrency and impacts fees
5. Playing the market: incentives and public investment.

An understanding of land use—market forces, public goals, policies, and the likely effects of those policies—is crucial for a thorough evaluation of transportation policies. All public actions, and their collective effects, should be evaluated simultaneously in an integrated process. The objective of such planning must be to achieve multiple objectives, not just one (e.g., an improvement in the level of transportation service).

Shortcuts: This chapter lays the foundation for the importance of the transportation/land use connection, the core topic of this report. If you bought the report based on the title, you should read at least the first half of this chapter, which describes why the relationship is important and documents some of the recent research supporting that proposition. If you are a transportation planner trying to figure out what land-use planners are talking about, you should read the second half of the chapter, which discusses typical land-use policies in more detail. If you are a land-use planner confident about your knowledge of the policies described, skim or skip the second half.

Chapter 4 describes ideal policy from the economist's perspective: get the prices right. But getting prices right is hard: not only technically, but equally important, politically. There are, of course, alternatives to pricing. What they are and how they work is the focus of this chapter and the next one.

An investigation of public policy implies a need for government action. Chapters 3 and 4 make the case that good reasons exist for such action: policy may improve the efficiency and fairness with which resources get allocated and used. At the top level, an evaluation of such policy is about whether to do nothing (let the private sector handle it) or to do something (government action). If government action is deemed a reasonable option, an evaluation of transportation policy must consider many questions about the type and scope of that policy, including:

- Incentives or regulations?
- Regional or local?
- Incremental or comprehensive?
- Transportation or land use?

These questions overlap. They need to be considered simultaneously, but the limits of language require they be discussed sequentially: we have to pick an order for our exposition. The one we chose derives from these considerations:

- Because this report is about transportation in the context of land use, we discuss land-use issues and policies first (in this chapter) to set the context for a discussion of transportation policies in Chapter 6.
- Because this report emphasizes an economic perspective on transportation, the discussion of land-use policies in this chapter continues with that emphasis.
- Given that the economic emphasis strongly supports directing policy toward getting the prices right, we often use the likely effects of pricing policies as a touchstone for evaluating other policies. How well do nonpricing policies approximate the results one would expect to get if transportation was priced more efficiently (i.e., along the lines described in Chapter 4)? In other words, which nonpricing policies might be considered *second-best policies*.

We start by arguing that land-use impacts and policies have been underrepresented in decisions about transportation policy and investment. Improving the balance requires raising the stature of land-development planning at the metropolitan or regional level.

We then describe and evaluate the kinds of changes in land-development policies that could positively influence both transportation and the quality of urban life generally. That discussion starts what might be considered the land-use planner's ideal, analogous to the economist's ideal described in Chapter 4.

Recognizing the technical and political difficulties in implementing the ideal, we then describe and evaluate a range of more incremental changes in land-development policy.

THE BALANCE OF TRANSPORTATION AND LAND-USE POLICY

Why the balance might need adjustment

We view as idle any debate about whether a connection between land use and transportation exists, and whether it is strong enough to be worth caring about. Chapters 2 and 3 give a historical perspective: the connection clearly exists and *has* mattered. Land development, at a large scale, has responded to changes in transportation technologies and investments in transportation infrastructure. That land development feeds back to influence the performance of transportation infrastructure, which feeds back to influence (usually negatively) the environmental characteristics of land development.

A legitimate academic question is whether the connection matters as much as it once did because the pace of transportation development is slowing. Downs (2004a) and Giuliano (1995b) point out that in today's metropolitan areas, most of the transportation infrastructure that one will see 20 years from now already exists. The big roads—and the easiest and cheapest ones—have already been built. Cars and trucks will continue to improve, but their basic functionality is pretty well set. In economic terms, one should expect diminishing returns (reduced bang for the buck) from highway expansions in developed metropolitan areas.

That conclusion supports the case that, whereas transportation considerations have dominated policy at the metropolitan and regional level, land-use considerations should play a larger role. Cities in the U.S. have been more likely to build an urban form around a transportation network rather than the other way around. The central areas of older eastern cities developed around subways and streetcars. Their suburbs and the newer metropolitan areas in the western U.S. grew primarily in the twentieth century and around a transportation technology of highways and automobiles. Transportation set urban form at the metropolitan and even neighborhood levels more than a vision of urban form set the goals for a supporting transportation network.

Land-use impacts and policies have been underrepresented in decisions about transportation policy and investment. Improving the balance requires raising the stature of land-development planning at the metropolitan or regional level.

A fundamental reason for the dominance of transportation is the need to network. Local transportation infrastructure must connect to the metropolitan transportation system, which has to connect to the state and national system. This networking requires planning at a large geographic scale, so federal, state, and metropolitan transportation planning organizations have been given a great deal of authority and funding.

In contrast to transportation most land-development planning is done at a local level:¹

- Land-development policy, though almost ubiquitous, is conducted mostly at small scale: tens or hundreds of municipal governments and service districts make land-development policy in medium and large metropolitan areas.
- Most individual land developments affect locally both neighboring land uses and transportation.
- Private households and businesses, not large governments, are the primary creators, owners, operators, and users of the buildings that compose and define urban areas.

This “fragmented system” of local land-development planning and policy making suggests a possibility for improvements from a larger-scale perspective on land development—one better in balance with the regional and metropolitan perspective taken in transportation.

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Fragmented local land-use policy: the promise and the problems

What explains the “the fragmented system of local government?” Part of the answer is history and inertia: the small and separate municipal service districts of 100 years ago have grown together over time, and political interests and arguments for local control make consolidations difficult. Though those arguments for local control may be self-serving for local elected officials and employees, many academicians, practitioners, and citizens feel strongly and for good reason that local government is highly desirable (as did, by the way, our Revolutionary founders of the system of federal government). We explore the reasons in this section.

Two key characteristics of local governments are important to our discussion:

- *There are a lot of them.* The 2002 Census of Government counts 87,525 units of local government: counties (3,034), cities and towns (19,429), towns and townships (16,504), school districts (13,506), and other special districts (35,052). Most metropolitan areas contain tens to hundreds of local units of governments. Fisher (1996), using the 1987 Census of Governments, found that the Chicago metropolitan area consisted of 789 local governments, 341 of which were school districts. The Seattle metropolitan area contained 86 local units of government, and relatively small Madison, Wisconsin (population 345,000), had 75 local units.²
- *Their boundaries overlap.* Municipality boundaries do not, of course, overlap, but they frequently coincide across the contiguously developed areas in metropolitan areas—there are often no undeveloped or unincorporated areas between cities in a metropolitan area. The boundaries of school districts and other special service districts do not generally coincide with municipal boundaries, even when school districts have the same name as a municipality. Local government boundaries create a tangled web in most metro areas.

To begin to understand these characteristics, look at what local governments do. Local governments primarily treat market *inefficiencies*, and to a lesser extent market *inequities*. For our purposes we limit our discussion to treatment of market inefficiencies.³

Local governments step in to provide goods and services the private sector fails to supply efficiently. To treat the effects of market inefficiencies, local governments perform four major functions for the market:

1. *Supply or regulate the supply of basic utilities*, which have the characteristics of a natural monopoly. A “natural” monopoly arises when a large supplier has cost advantages over smaller ones over the full range of demand. The highway and arterial street system has the characteristics of a natural monopoly. Public utility districts typically provide sewer, water, and drainage systems. Regulated monopolies typically supply electricity, natural gas, telephone, and cable TV/Internet. Garbage collection is sometimes provided publicly, sometimes privately.
2. *Supply or regulate the supply of local public goods and services*. Public goods and services are *non-rival* in consumption (i.e., one person’s consumption does not limit others’ consumption of the same good or service) and *nonexcludable* (i.e., it is difficult to charge for access to the good or service). Examples include police and fire protection, environmental quality, public streets and infrastructure, recreation areas, and cultural amenities. These goods and services are usually provided by municipalities (cities and townships).

Note that the local street system has the characteristics of a public good. Neighborhood streets are usually built by private land developers; the costs are included in the prices of building lots and structures. Once built, local streets are usually deeded over the local municipality, which maintains them. Sometimes, neighborhood associations maintain ownership of the streets and collect dues to pay for maintenance.

3. *Control externalities among land users*. Some land uses tend to generate negative externalities: noise, pollution, traffic, and so on. An apparently easy solution is land-use zoning that separates these uses. Local (and regional, state, and federal) governments also treat externalities through regulatory limits, time-of-day restrictions, or requirements for equipment or other mitigation measures.

Other land users generate positive externalities. Public schools are an example. Better educated people tend to be better citizens, or education at least helps citizens participate more effectively in the democratic process. Reducing or eliminating the out-of-pocket cost of education encourages more education. Importantly, from a land-development perspective, however, is the common practice of public school zoning; only children who live in a given school’s catchment area or zone can attend the school. Schools vary in their characteristics. Households compete aggressively for homes in the zones around public schools with desirable characteristics.

4. *Monitor and regulate the characteristics of private building and infrastructure*. One rationale for land-development policy is that construction quality is difficult to observe. The purchaser of a new property may have difficulty selling the property in the future if potential buyers have difficulty discerning quality. Development and building codes help to ensure a minimum level of quality. Subdivision plans provide information about the location of buried utility lines.

Another rationale is that building codes help, along with minimum lot size zoning, to prevent free riding on public services, especially public schools. Most local public services, including schools, are paid for (to a significant extent) with property taxes. Minimum input (i.e., land and structure) requirements prevent low-value and therefore low-tax houses.

Given these various roles, are there efficient sizes for local governments (and, consequently, an appropriate number of governmental units)?

Controlling externalities among land users is one of local governments four major market functions. Some land uses (e.g., public schools) funded by government generate positive externalities.

Assume municipal and service-district boundaries are flexible. Assume local governments trade-off (1) the relative simplicity of providing customized and limited services for a smaller and more homogeneous group of households and businesses, against (2) better control over larger-scale externalities and the ability to exploit scale and scope economies in the production of government services.

Then consider the diversity of household and business demand for public services. Think about suppliers of *public* goods and services as if they were suppliers of *private* goods and services. Assume that each municipality, like each restaurant, caters to a niche in the spectrum of consumer preferences, and simultaneously benefits from larger-scale production. A town's police force and fire service, for example, benefit from more highly specialized training and equipment. But that specialization has costs spread only by serving a larger area. On the one hand, local businesses and households would like goods and services highly customized to their own needs and desires. On the other hand, those goods and services cost less per consumer if they can be spread over more consumers.

Businesses and households can, to some extent, have things both ways. The households in some neighborhoods, for example, form neighborhood associations that contribute to the services provided by the municipality (e.g., through neighborhood crime watch, beautification projects, events). Or businesses and households can buy into developments with more formal neighborhood associations (e.g., at the extreme, gated communities with private recreation facilities).

Neighboring municipalities can also reach agreements to share the costs of special training and equipment. For example, municipal police or fire departments can agree to share the special services they develop with neighboring municipalities in exchange for quid-pro-quo sharing of costs.

Municipalities can also choose not to supply goods or services with extensive scale economies. Instead, special districts can be formed. Water, sewer, drainage, garbage, and school districts often serve multiple municipalities. In principle, these service districts can be much larger than the municipalities they serve to exploit network and scale economies.

Though a smaller jurisdiction can provide more customized services, the smaller the jurisdiction the more likely an impact from negative externalities; public and private actions taken in surrounding areas are more likely to have negative effects on the surrounded municipality. One municipality may put its manufacturing zone in an area negatively affecting residential development in a neighboring municipality. A municipality that hires more police to crack down on teen-age cruisers may shift the problem to another municipality.

In practice, metropolitan areas consist of many municipalities, special service districts, private and quasi-governmental neighborhood bodies, and intergovernmental and interagency agreements. Conceptual studies in economics and public finance and administration do not converge on an optimal or market-enforced pattern of jurisdiction sizes and boundaries.

Indeed, history appears to play a major role in determining the number and boundaries of service providers. The large and older metropolitan areas in the north and east of the U.S. tend to consist of a larger number of smaller municipalities and service districts than do the large but newer metropolitan areas in the west. A big part of the reason is that land development, thanks largely to reduced transportation costs, has over time filled in the land that once separated small towns in what are now large metropolitan areas. These municipalities have, however, often maintained their boundaries and independence. The web of service districts can get pretty tangled.

This tangle of boundaries and agreements may seem inefficient. Within that context, getting things right requires costly negotiation, contracts, and

Though a smaller jurisdiction can provide more customized services, the smaller the jurisdiction the more likely an impact from negative externalities; public and private actions taken in surrounding areas are more likely to have negative effects on the surrounded municipality.

vigilance. Unifying service delivery under a single larger unit of government reduces or eliminates these transactions costs.

But efficiency means maximizing net benefits, not minimizing a particular type of cost. Enduring high transactions costs can be worthwhile. Nobel laureate Ronald Coase (1960) focused thinking in economics about the importance of transaction costs and property rights in understanding the size and organization of businesses and government.⁴ A key idea is that organizations evolve to make the best trade-offs between production costs, transactions costs, administrative costs, and product characteristics.

The attention to transactions costs sheds light on important changes over time in the ways businesses are organized. For example, manufacturing firms 40 years ago tended to be large and vertically integrated, some directly controlling production from raw materials to retail. Many of these businesses have since vertically disintegrated, preferring market exchange, negotiation, and short- and long-term contracting with independent suppliers to direct ownership and control of supply. A large vertically integrated business has low transactions costs but relatively high administrative and production costs. Changes in various technologies have reduced transactions costs; they still can be high, but bearing them allows even bigger savings in production and administrative costs.

The same ideas appear to apply in the production of local public goods and services. The size and scope of public and quasi-public agencies evolve in attempts to optimize on scale economies, administrative costs, transactions costs, and the diversity in business and household demand.

Zoning may be seen as a way to reduce transactions costs. A residential developer wants to maximize the return from sales of building lots and houses. Sale prices depend on customers' expectations about the characteristics of land development nearby (i.e., they are concerned about externalities). The land developer could negotiate and contract with all neighboring land owners to ensure nothing bad will happen. This is costly. Zoning eliminates these transactions costs.

A cost of zoning (i.e., one of its trade-offs) is rigidity in subdivision design, resulting in less consumer choice in the characteristics of development and needing some design changes that consumers may not consider worth the cost. Sometimes, however, neighbors might not oppose designs inconsistent with zoning or would be willing to accept the designs with some kind of compensation from the developer (such as land dedicated to a community park). Meetings of the local zoning board or planning commission can be seen as a way for the developer to conduct these negotiations cost-effectively. The negotiations are still costly, but they may be worth it.

In summary:

- Public facilities and services in most metropolitan areas are typically provided by multiple and overlapping general purpose and special service districts, each of which has effects on land development.
- The strongest entanglements are the closest. Any given jurisdiction tends to worry more about what its neighbors are doing than what jurisdictions on the other side of the metropolitan area are doing. Interests tend toward the local rather than the regional.
- Governance arrangements evolve over time to deal with changes from new technologies, new infrastructure, and changes in household and business demands. Transactions costs, production costs, and administrative costs are all important in this evolution.

Land-development policy has evolved in a context of historical boundaries in ways that trade-off the advantages of large scale for those of local control. But most of the institutional focus remains local. As incomes have

The size and scope of public and quasi-public agencies evolve in attempts to optimize on scale economies, administrative costs, transactions costs, and the diversity in business and household demand.

increased and basic transportation networks have been completed, the pressure for a larger-scale focus on land development has grown. Chapter 8 describes the increase in the number, scope, and sophistication of efforts to introduce land-use planning at the metropolitan level and to integrate it with regional transportation planning. The next section provides more information about this shift.

Adjusting the balance of transportation and land-use policy

A metropolitan transportation system must serve the businesses and households that the region comprises. What do those businesses and households want? A higher quality of life. Is faster, cheaper, safer, more convenient transportation consistent with that objective? Yes, all being equal.

The complication is that providing faster, cheaper, safer, more convenient transportation does not leave everything equal.

The higher incomes that better transportation helps generate increase demands for a higher quality of urban life in the context of increased dissatisfaction with traffic congestion and sprawling suburban development patterns. Increasingly limited transportation capacity is likely to get used more intensively. (Pricing helps do that; we explore other mechanisms in Chapter 6.) Increasing amounts of new development will be in higher-density and mixed-use development patterns, which imply changes in land-use patterns, designs, and policy, which will in turn influence transportation facilities and policy.

In this context, a case can be made that the attention paid to land use in the context of transportation planning is inadequate. The strength of transportation policy as a determinant of urban form, relative to the strength of land-use policy, has been reinforced by institutional arrangements. Specifically, government transportation agencies operate (1) separately from land-development agencies, (2) at larger scales (federal, state, and regional), and (3) with much better funding:

A case can be made that the attention paid to land use in the context of transportation planning is inadequate. The strength of transportation policy as a determinant of urban form, relative to the strength of land-use policy, has been reinforced by institutional arrangements.

- Transportation projects are large in scale, and executed in a coordinated fashion by public agencies operating at the state and regional levels—Departments of Transportation (DOTs) and Metropolitan Planning Organizations (MPOs). Land-use projects are mostly small private developments overseen by subregional governments, such as cities and townships.
- Most transportation projects are paid for using public funds generated from gas tax revenue (roughly \$60 billion per year, federal and state) and other taxes and fees. Land development relies mostly on private funding and financing involving investors, banks, and space users. And public land-development policy relies primarily on local taxes, especially property taxes.
- DOTs and MPOs focus mainly on transportation objectives: vehicle-miles-traveled, level of service, and so on. The objective is to provide the transportation capacity necessary to serve planned or forecast land development. These organizations tend to view traffic congestion as a failure of local land-use authorities to regulate land development according to plans and forecasts: they built the highway to serve projected land use, then local governments allowed land to develop more intensively than planned or forecast, which resulted in traffic congestion.
- Local authorities focus mainly on providing local services, promoting local economic development, and treating local externalities. They tend to take transportation as given and are close to the local businesses and households who benefit from the access transportation infrastructure provides.
- Local land-use authorities are constrained by standards set by federal and state agencies. The formulation of street standards, for example,

may lack consideration of local community or environmental contexts, emphasizing vehicle throughput and safety.

Thus, in U.S. metropolitan areas, through much of the twentieth century, transportation led land use. Surface transportation has been run by large, well-funded state and regional agencies that have a clear mission and strong advocacy groups; land-use responsibility has been decentralized across many local governments. That arrangement makes some sense if the dominant objectives are opening the vast land resources of the U.S. to development and providing businesses and households choice in the range of local public goods and services. But does it still make sense—should transportation's leadership continue?

Probably not. The broad objectives of accessing and developing suburban areas have been achieved. The objectives are now shifting toward improving the quality of urban environments. Pricing could be a powerful tool, but it still lacks political support and could never be comprehensive. Land-development regulations, though virtually ubiquitous, are mostly oriented toward aiding and abetting the old development objective. Pressure is mounting to change land-development policies.

Transportation planners are looking more to land use as a way of ranking transportation investments: build the transportation capacity where the growth is desired and expected. The sidebar on the following page shows how the scope of transportation considerations has widened over time: from the pavement, to the right-of-way, to the corridor, to an integrated evaluation of the broader effects of a single transportation project on the transportation system and other aspects of quality of life.

Some argue land use should *lead* transportation. Citizens involved in urban and regional planning are engaged and inspired by discussions of natural areas and urban form. Land use and urban design seem central to the public debate about quality of life and community development, while transportation is essential but derivative.⁵ The demand for transportation derives from the creation of desirable destinations: residences, work places, shopping areas, and cultural and recreational amenities. Thus, the primary focus should perhaps be on the characteristics of these destinations.

Cervero (2001) provides guidance that reflects this strong view:

- Embrace urban planning and articulate a vision for the future that can guide subsequent planning decisions.
- Use infrastructure investments to leverage, focus, and shape the location and intensity of development, including water and sewer, as well as transportation.
- Engage in regional cooperation, including local land use as well as transportation decisions.

Cervero and others believe land-use planning at the regional/metropolitan level should be leading decisions about transportation infrastructure. Such leadership would be abetted by, and may require, a regional land-use authority with the powers to create and enforce a regional land-development plan. That plan would be guided by a regional political process and then implemented by the many local land-development agencies in the region. Those conditions represent a big change from the status quo.

We do not expect metropolitan areas to move quickly and uniformly to adopting regional land-use plans with strong implementing measures. We want to emphasize only the change in direction. The correct solution—the one that most planners would probably endorse even without having slogged through this report—is that land-use and transportation planning have to happen together. They play off one another; there are trade-offs.



Scott Leigh, istockphoto.com

The broad objectives of accessing and developing suburban areas have been achieved. The objectives are now shifting toward improving the quality of urban environments.

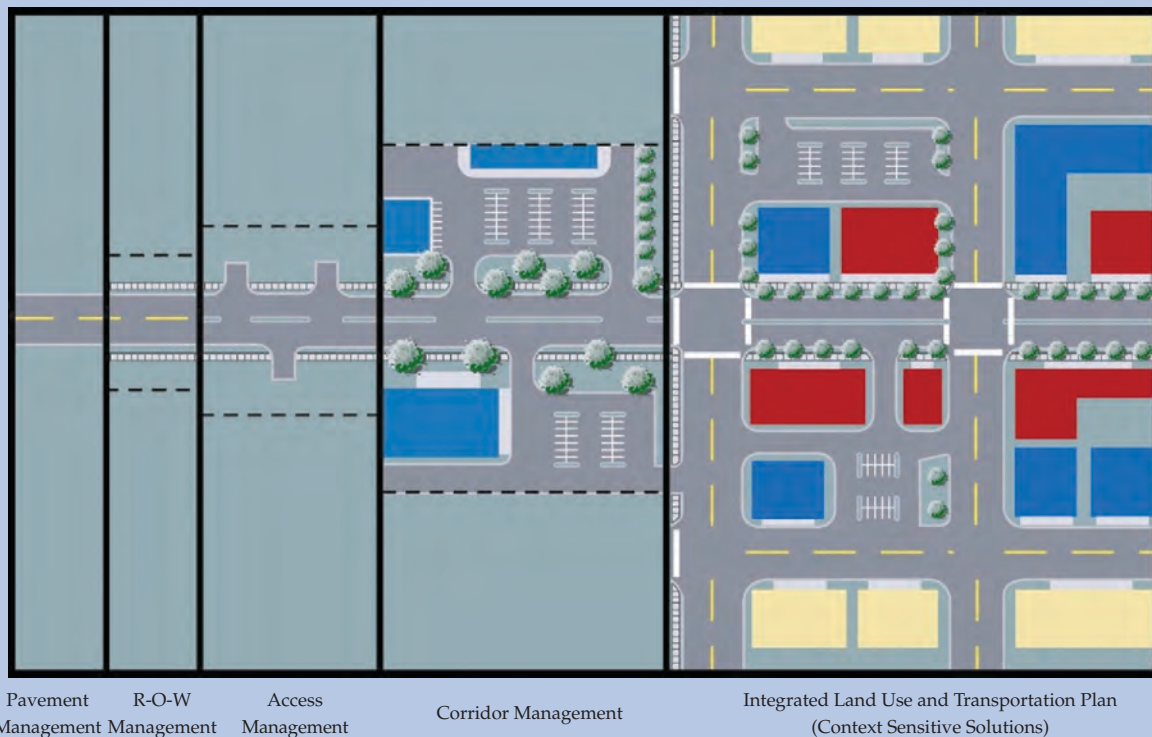
THE EVOLUTION OF TRANSPORTATION PLANNING AND ITS INTEGRATION WITH LAND USE

By Troy Russ, AICP, Principal, Glatting Jackson Kercher Anglin, Inc.

Transportation planning has evolved as its relationship with land use has been better understood. The figure illustrates that evolution.

Early road engineers limited their concerns to travel lanes themselves: to construction for speed and safety, and to pavement management. Their scope expanded to a larger right-of-way to deal with encroachments on safety and the likely need for future travel lanes. When road widening became more expensive, they developed access management plans, limiting the number of driveways and regulating the spacing of intersections to increase the efficiency of corridors. The evolution of transportation planning led planners and engineers to work together to create corridor plans for both transportation investments and land-use allocations.

Now planning, and the thrust of this report, is moving to the right of the figure: to transportation planning that looks beyond the corridor to include parallel roadways and planning complete communities through an engaged visioning process. Some of the most effective transportation solutions come from land-use decisions outside a roadway's corridor. Effective subdivision regulations that require connectivity can create alternative routing for motorists, reducing needs for additional capacity directly on the main arterial in the corridor. Integrated land-use and transportation plans provide a framework for context-sensitive transportation investments and policies that fit the character of the community.



The transportation system creates a structure to which land use must conform, so transportation planning should be done with consideration of the implications for land use (and economic development, environmental quality, urban amenity, and more) as well as for transportation when it sets that structure.

MPOs are a logical place to look for regional coordination of multiple governments, nongovernmental organizations, and private interests. Given the right powers, MPOs could act as a transactions-cost reducing broker for land-use decision making. As currently structured, however, they are in a coordinating rather than decision-making role. They are conveners of local government interests in the MPO region (cities, townships, and counties) that determine regional transportation priorities. With a few notable

exceptions (Metro in Portland, Oregon), they have virtually no direct control over land-use decisions.

Nonetheless, the role of land use in metropolitan transportation planning is so obvious as to be almost inevitable. In conducting transportation planning, MPOs allocate population and employment to the small-area zones in their transportation models. These allocations are exogenous to the models. So who decides what is going to get developed in those zones and how? Is that development inevitable (historical growth trends, when extrapolated, dictate development?), or can its type, intensity, and location be influenced by public policy?

Our answer, of course, is that urban development can and is influenced by public policy. Thus, MPOs must, by choice or by force, consider a long-run development pattern to do their transportation planning, and they are finding themselves involved, at some level, with regional land-use planning (the visioning at least, but little on the regulation). Tying land-use policy into the MPO long-range transportation planning process is probably the simplest route to the coordination and planning of land use and transportation across a metropolitan area.

Chapters 7 and 8 describe in more detail the ways those ties might be made: how land-use and transportation policies can be combined, technically and in a public process, to get better results for both. But we can't combine policies without first describing and evaluating them individually. The rest of this chapter does that.

PLANNING INSTEAD OF PRICING

In Chapter 4 we pushed the economist's policy prescription: pricing. Get the prices right, and the largely independent decisions of hundreds of land developers and thousands of households and businesses will work together. Getting some important prices right is possible technologically, but a hard sell politically.

Though full pricing is hard, is full regional planning any easier? This section examines broadly what would be entailed in controlling highway congestion and addressing quality-of-life issues through land-use planning.

We conduct the examination in two steps. The first step describes private land development in the context of local, rather than regional, development controls. The exercise sheds further light on how conventional land-development policy contributes to less-than-ideal urban development. But it also shows how the developer's perspective changes as the size of a development increases. A developer's objective is to maximize net return from land development, which means maximizing (net) quality of life for residents and business owners. The developer has more control as the size of the development increases. It is unlikely, however, that the primary focus is on vehicle throughput.

The second step looks at land-use and transportation planning from a metropolitan or regional perspective. The idea is that the regional agency has objectives very similar to those of a private land developer. Public agencies rather than a private developer are addressing these objectives because metropolitan development is typically too big for the private sector. What might be the key elements of regional planning?

Private land development with local development controls

Most commercial and residential development is planned, paid for, and built by private-sector developers. This development occurs in the context of the local public regulatory system described above. This section illustrates issues that arise and the trade-offs made by describing briefly how the development process changes as the geographic scale (i.e., the size of the development

Tying land-use policy into the MPO long-range transportation planning process is probably the simplest route to the coordination and planning of land use and transportation across a metropolitan area.

parcel) increases. Size matters because developers can exploit scale economies and internalize more externalities by developing larger parcels.

Start with a small, subdividable parcel, anywhere from two to 20 acres. These parcels are common in the many metropolitan areas growing at a modest pace. They come to market when a farmer splits some land to raise money for a capital improvement, or when a family who has enjoyed living on a large semi-rural lot decides to move, or when a speculator who has been holding the land decides to sell. The key characteristic of these small, but frequently numerous, parcels is that their developer can financially be concerned only about the most highly localized amenities.

The most common objective in developing these small parcels is simply to maximize the number of building lots given the local zoning and development codes. Development codes specify minimum infrastructure requirements, including: street and sidewalk sizes, geometries, and connectivity; water, sewer, and drainage; street lighting; and other utility requirements. Zoning codes specify maximum residential densities, minimum lot sizes, building setbacks, and so on. The cheapest, easiest, and therefore most common approach is to submit to the local authority a subdivision design that meets the minimum legal requirements and otherwise maximizes the number of building lots. This subdivision can often be approved “by right”; it does not require review by the zoning board, planning commission, or public.

No surprise that these small developments tend toward blandness. They promote development by reducing its cost. Some jurisdictions implement more detailed development codes in an effort to promote better neighborhood designs. But complex regulations are difficult to decide on, codify, implement, enforce, and otherwise support over time. The tendency is to keep things simple.

Small parcels do not prevent developers from being creative about design or providing more amenities. Some developers who follow development and zoning codes increase neighborhood amenities by imposing requirements on the designs of houses. One method is to specify a restrictive list of design characteristics. These requirements may be included as restrictions, known as covenants, in the deed to the building lot, so that they “run with the land.”⁶

Another method is to specify design principles or standards, and to establish a process by which proposed house designs are reviewed and approved. The developer may initially supervise this process and then hand over the approval process to a board of elected homeowners. These quasi-governmental homeowner associations are common and essentially provide regulatory authority beyond that of the local jurisdiction.

Other developers may want to create amenities using less conventional designs. Some of these amenities are modest: retaining mature trees or clustering houses on somewhat smaller building lots to provide communal open space for recreation. But even simple amenities may conflict with zoning or development codes: smaller lots violate minimum lot size regulations. Developers not infrequently have to fight for approval of designs that appear to depart little from the standards.

The reason for the fight is that the developer reaps the benefits of higher sale prices while the neighboring properties of the jurisdiction itself bear the (perceived and possible real) higher costs. Neighbors believe that smaller lots will change neighborhood character, create more traffic, and attract households with different socioeconomic and demographic characteristics. Service providers might worry that the semipublic open space provided will eventually become a maintenance cost for them. The costs of unusual designs come in the form of thinking through their consequences and changing regulations or otherwise dealing with the precedent.



Rachel Dewis, istockphoto.com

The cheapest, easiest, and therefore most common approach to development is to submit to a local authority a design that meets the minimum legal requirements and otherwise maximizes the number of building lots.

Finally, some developers perceive that a market exists for more unconventional designs with broader positive social consequences. “New urbanist” or “neotraditional” design is a prominent example.⁷

Neighborhoods in pre-auto era cities provide the inspiration for neotraditional designs. They include elements inconsistent with standard suburban zoning and development codes: a mix of lot sizes, but generally relatively small lots (so relatively high density) with small building setbacks; narrow streets and alley access to garages; street trees and public spaces; and mixes of dwelling types and even land uses. The idea is to take advantage of a relatively large development parcel to build a well-integrated, walkable, diverse, and generally high-amenity neighborhood. These (and other unconventional designs) may have strong market appeal but don’t fit the suburban mold and are often opposed by zoning commissions and neighbors.

Some suburban jurisdictions take steps to accommodate unconventional development. For example, some areas may be zoned for higher-density or mixed-use development, the details of which are negotiated on a case-by-case basis. The land-use plans that underlie zoning and development codes may allow for unconventional designs, even if the codes do not. Developers can use the language in the plans to either argue for approval or seek approval through legal processes.

The problem is that the developers of small parcels lack the financial incentives to look at the development’s impact on the larger community. Incentives to look broader improve as the size of the parcel increases. Recognizing this, larger-scale professional developers look for larger parcels, often from 50 to 200 acres. These larger parcels occasionally come on the market whole (a rural-fringe farmer sells off the entire farm), or the developer assembles a number of separate parcels. Assembly of parcels, which requires negotiations with multiple land owners, is time consuming and expensive.

The developer of the larger parcel could still submit a simple plan consistent with development and zoning codes that maximizes the number building lots. In concept, this plan could be approved by right and would deliver neighborhoods that differ little from those built up from smaller subdivisions.⁸ But the larger parcel gives the developer more to work with. Moreover, larger parcels are more likely to be found in unincorporated areas with weaker regulations. In these cases, the developer has the option of establishing a new suburban jurisdiction. The zoning and development codes essentially become endogenous to the development.

There are examples. One of the first New Urbanist developments, Seaside, Florida, was developed in Walton County, Florida, which had no zoning at the time. Suburban development around fast-growing Houston occurred on large, unzoned parcels. The characteristics of development there continue to be controlled by restrictive covenants. Foster City in the San Francisco Bay area is a relatively large area (3.8 square miles of land) that now houses about 29,000 people developed in the 1960s by T. Jack Foster and colleagues on land reclaimed from the bay.⁹

The private developer of a large parcel potentially has a great deal of control over land use. That control, in principle, could be used to design a built environment and a transportation system that accommodates the trips generated. A high-capacity roadway system could accommodate the relatively long and frequent car trips associated with low-density development and segregated uses. Or a combination of roads, mass transit, bike lanes, and foot paths could accommodate travel in high-density, mixed-use development.¹⁰ That development could be designed carefully to avoid many of the negative externalities associated with density.

Such a land developer is unlikely to think of the transportation system simply in terms of throughput. He or she would probably think about the

The problem is that the developers of small parcels lack the financial incentives to look at the development's impact on the larger community. Incentives to look broader improve as the size of the parcel increases.

characteristics of land use first, then think about the transportation system required to serve it well. At a minimum, a private developer would think about land use and transportation in an integrated way, and would consider how the transportation system affected amenity and property values.

A (benevolent) land developer might, in principle, design a metropolitan area with a transportation system that works, but a private land developer is unlikely to be able to gain control of a significant portion of an entire metropolitan area. The theory has some practical problems: most congested metropolitan areas are too big for a private developer to control. A mid-size metropolitan area might cover 100 square miles. Metropolitan areas take decades to develop, and they develop on land owned by hundreds or thousands of households and businesses. Purchasing and assembling large tracts of land is expensive.

The lack of such control and the high transactions costs associated with thousands of property owners provide the rationale for government involvement in land use: to control external effects and to exploit economies of scale in service provision. Public policy is typically necessary to coordinate, over decades, the development of large numbers of relatively small private parcels.

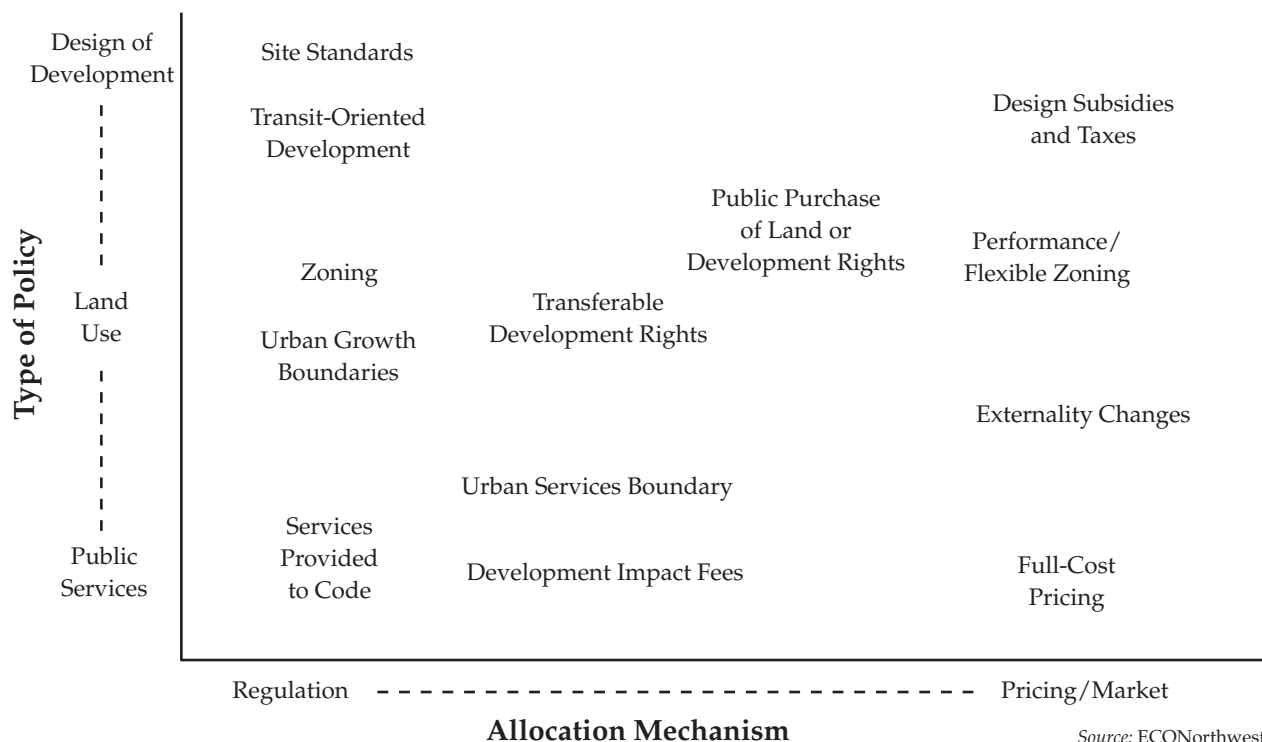
What should that policy look like?

A framework for thinking about land-use policies

Scores of land-use policies affect transportation, at least in theory. Figure 5-1 gives a sampling, classifying the policies on two dimensions that relate to the way the policies try to affect development: (1) through the element of development they address: design, use, or public facilities; and (2) through the tools they use to affect those elements: regulations or incentives.

On the horizontal axis, policies farther to the right involve greater reliance on pricing. The vertical axis is not strictly a continuum, but we treat it that way. For example, flexible/performance zoning has more of a market orientation than traditional zoning and can affect both the use of land and the design of development—we place it to the right and between land use and development design.

Figure 5-1. *Classification of land-use policies*



Source: ECONorthwest

TABLE 5-1. ANOTHER CLASSIFICATION OF LAND-USE POLICIES LIKELY TO AFFECT TRANSPORTATION

DEFINING A REGIONAL DEVELOPMENT PATTERN	CREATING LIVABLE PLACES
Regional planning	Zoning: mix of use
Transit-oriented development	Zoning: density
Corridor planning	Performance zoning
Jobs-housing balance	Designing for density/amenity
Urban growth boundaries	Street connectivity
	Streetscape
PROVIDING PUBLIC INFRASTRUCTURE AND SERVICES	MISCELLANEOUS POLICIES
Provision: where and how	Pricing externalities
Pricing	Changes in property taxation
Concurrency/impact fees	Purchase or transfer of development rights
	Location-efficient mortgages

Source: ECONorthwest

Table 5-1 shows a slightly different way to categorize land-use policies, focusing on ones likely to have stronger impacts on transportation.

We use “land use” to mean the arrangement and regulation of development patterns at a city or metropolitan level: land-use plans and zoning work at this level. We use “urban design” to apply at a neighborhood or site level; at the level of human and environmental interaction. We organize our discussion of land-use policies under these two headings.

That organization fits with yet another way to organize land-use policies in the context of their connections with transportation. Since the first edition of this report, some of the literature on the transportation/land use connection has agreed on several *dimensions* (the “Ds”) for discussing the influence of the built environment on travel behavior.¹¹ There were originally three Ds; we have seen as many as seven; we discuss five: *destinations*, *density*, *diversity*, *distance*, and *design*. Destinations address regional-scale issues (land use); the other four Ds address neighborhood-scale issues (urban design). We provide some background on these elements of the land use/transportation connection before providing more detail about specific land-use or urban design policies.

Destinations

Destinations are relevant at the city, regional, and corridor level. The accessibility of important regional activities (e.g., jobs and homes) influences driving rates. In the terminology of land-use planners, a large part of what is important about destinations is the balance of jobs and housing.

Ewing and Cervero (2001) find that this dimension probably has the most potential for big effects on transportation in the long run. The logic is hard to refute: in theory, if people lived near where they work, shop, learn, and recreate, they would make shorter trips and could make fewer trips by automobile. Thus, a land-use strategy at the regional level that might have a big impact on surface transportation is to plan and to provide incentives for employers to locate closer to their employees (recall that the conventional policy is to reduce externalities by separating commercial from residential land uses).

We discuss five dimensions of the influence of the built environment on travel behavior: destinations, density, diversity, distance, and design.

There is evidence that jobs-housing balance has effects on transportation choices. Cervero and Duncan (2006) report that the number of jobs within a four-mile distance to housing leads to a decrease in vehicle-miles-traveled (VMT) and vehicle-hours-traveled (VHT); Ewing and Cervero (2001) find lower auto use.

The question is whether the market will respond to zoning that allows a mix of uses. Zoning does not typically restrict the development of housing in areas zoned for commercial and industrial development (conventional zoning protects houses from other uses, not the other way around). People could live nearer their work if they so desired and expressed those desires through market mechanisms (i.e., by bidding up the price of housing available in these areas).

But little residential development today occurs in industrial or commercial areas. Is that because no strong market exists for that kind of development or because people have no examples of good mixed-use to select?

Advocates of balancing jobs and housing take a more direct approach to encouraging land-development patterns that reduce the amount of travel by auto. Rather than simply relaxing zoning regulations to allow mixed-use development, they recommend that zoning require it.

Rather than simply relaxing zoning regulations to allow mixed-use development, advocates of balancing jobs and housing recommend that zoning require it.

Density

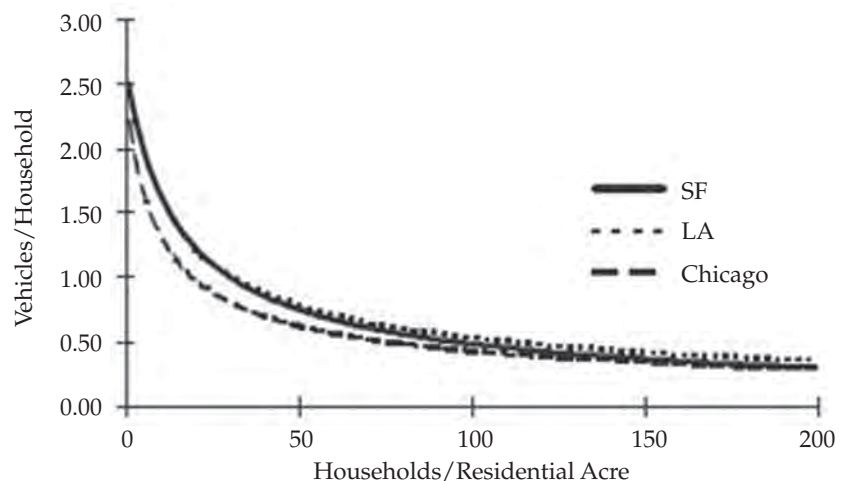
Density is a measure of the intensity of development. Density is often correlated with factors that are in turn correlated (causally) to lower auto use: parking scarcity and higher parking prices, congestion, better transit service, and walkability.

While many empirical studies of density thresholds exist, few have formally controlled for factors that would tend to increase driving (e.g., household income, household size) or would decrease auto use (e.g., land-use mix (diversity) and pedestrian amenities (design)).

Holtzclaw et al. (2002) studied density and transportation in Chicago, Los Angeles, and San Francisco, and found that doubling the density of an area can reduce driving by more than 30 percent. Figure 5-2 shows the expected results: the average drops quickly to about one car per household as residential densities rise to about 30 households per residential acre; at higher densities there will be an average of 0.5 cars per household.

In the U.S., only modest increases in density are needed to make a difference. Bernick and Cervero (1997) point to a significant decrease in auto use at the lower range of the density spectrum. Figure 5-2 shows, for example, a greater decrease in auto ownership when going from just five to 10 dwelling

Figure 5-2. Density and auto ownership



Source: Adapted from Holtzclaw et al. 2002

TABLE 5-2. RECOMMENDED RESIDENTIAL DENSITIES FOR TRANSIT SERVICE

SERVICE LEVELS	RESIDENTIAL DENSITY THRESHOLDS
Bus: Minimum service (20 buses/day)	4 dwelling units/acre
Bus: Intermediate service (40 buses/day)	7 dwelling units/acre
Bus: Frequent service (120 buses/day)	15 dwelling units/acre
Light Rail: 5 minute peak headways	9 dwelling units/acre (25–100 sq. mile corridor)
Rapid Rail: 5 minute peak headways	12 dwelling units/acre (100–150 sq. mile corridor)
Commuter Rail: 20 trains/day	1-2 dwelling units/acre (existing track)

Source: Pushkarev and Zupan 1977, 1982

TABLE 5-3. RECOMMENDED RESIDENTIAL DENSITIES AND EMPLOYMENT CENTER SIZES FOR TRANSIT SERVICE

MINIMUM SERVICE LEVEL	RESIDENTIAL DENSITY	COMMERCIAL/OFFICE SPACE
1 bus/hour	4–6 dwelling units/acre	5–8 million sq. ft.
1 bus/30 minutes	7–8 dwelling units/acre	8–20 million sq. ft.
Light rail and feeder buses	9 dwelling units/acre	35–50 million sq. ft.

Source: ITE 1989

units per acre (a five-unit-per-acre increase), compared to going from 50 to 100 dwelling units per acre (a 50-unit-per-acre increase).

Pushkarev and Zupan (1977, 1982) found that residential densities in transit corridors, along with the size of the downtown and the distance of the transit stations from downtown destinations, explained demand for a variety of transit modes. The effects of density are interrelated with regional employment center size, the land-use structure within the transit corridor, transit service characteristics, and public policies, such as the price of parking. Their findings are in Table 5.2. Table 5-3 shows similar recommendations from another study.

Frank and Pivo (1994) report that the modal share of transit work trips increases as residential densities rise from nine to 13 residents per gross acre and commercial densities rise from 50 to 75 employees per acre. The threshold for finding a nontrivial number of shopping trips on transit was 75 employees per acre and more than 18 residents per gross acre.

Diversity

Diversity is the dimension that covers mixed use.¹² Mix relates to concerns about a balance of jobs and housing.

Mix clearly depends on scale. At the metropolitan scale, something very close to a jobs/housing balance and an appropriate mix of uses may exist; at the level of a single lot, there is very rarely a job/housing balance or a mix of uses. Cervero (see his sidebar below) suggests that the key is to think in terms of mixed uses and balancing jobs and housing at the corridor level (six to eight miles linearly) so that centers of dense development can be linked by high-capacity transit.

Distance

Density and Diversity tend to reduce Distance. In their study of travel characteristics of TOD in California, Lund, Cervero, and Willson (2004) found that residents living and working within one-half mile of transit stations are

MEASURING DESIGN

By Reid Ewing, National Center for Smart Growth, University of Maryland

Building on the work of urban design pioneers (Lynch, Whyte, Appleyard, and Jacobs), Ewing et al. (2005) worked with a team of urban design and planning experts to conduct the most rigorous analysis to date to demonstrate which qualities of urban form correlated highly with walkability. As of this writing, field tests of the relationship of these urban form qualities (listed below) with walking were getting underway at Columbia University. Each definition is followed by our examples for enhancing these qualities:

- **Imageability** is the quality of a place that makes it distinct, recognizable, and memorable: it is what defines sense of place. One way to enhance this is to preserve and celebrate a community's urban history and environmental authenticity. Another is for new construction to incorporate either timeless or valued historic design elements.
- **Enclosure** is the degree to which streets and other public spaces are visually defined by buildings, walls, trees, and other vertical elements. One measure of this is the building height to width ratio. Several prominent urban designers recommend a range from 1:1 to 1:3.
- **Human Scale** is about having the size, texture, and articulation of physical elements match the size and proportions of humans and, equally important, correspond to the speed at which humans walk.
- **Transparency** is the degree which people can see or perceive what lies beyond the edge of a street or other public space. This quality can be enhanced by ground-floor windows, entrances, and retail activities.
- **Complexity** is the visual richness of a place, which depends on the numbers and kinds of buildings, architectural diversity and ornamentation, landscape elements, street furniture, signage, and human activity.

almost four times more likely to take transit than those who live one-half mile to three miles out. Compared with other areas in the region, those who live and work within a one-half mile are five times more likely to ride transit.

Design

The details of design at the scale that humans interact with their environment are important. Dealing with the balance of form and function, thoughtful design (form) can make high-density urban development aesthetically appealing to live in, inviting for pedestrians, and politically acceptable to the adjacent community.

For instance, adequately proportioned sidewalks landscaped to provide shade and a buffer from the noise, exhaust, and threats of traffic help create inviting, livable urban spaces. Dumbaugh (2005) found that design elements he termed "livable streetscape features" caused drivers to be more careful, creating a safer street environment. Functional design in this example means sidewalks and bikeways should be well proportioned and part of an interconnected network that includes such things as well-designed, safe street crossings.

Attractive and comfortable buses and bus stop shelters help the performance of transit, though they are not nearly as important as transit's operational attributes, like frequency of service (see end of Appendix C for elasticities).

Design is the hardest dimension to uniformly quantify, but efforts are being made (see sidebar).

Regional planning with strong development controls

The argument so far goes like this:

1. The private sector does most of the land development in a metropolitan area.
2. Private developers are not big enough to be able to develop any more than a very small piece of that metropolitan area. Thus:
 - a. They cannot deal with external effects (negative or positive) of their developments or fully exploit scale economies; and
 - b. They will not have strong incentives to spend a lot of money to make sure their development meshes well with a current and future regional transportation system.
3. Local governments may also have more incentive for local gain than for optimum regional efficiency.
4. There is probably a role for:
 - a. a greater consideration of the way land use interacts with transportation (it does so on several dimensions—the five Ds); and
 - b. a long-term, regional view.

What would a regional land-use authority with the control and perspective of a private land developer do about transportation in the context of land use? To help think about this question, assume a regional authority with enforceable control over land-use and transportation infrastructure for an entire metropolitan area. Assume further that it cannot or will not move toward more, better, and direct pricing of access to highways. In contrast to our extreme pricing solution (ubiquitous congestion pricing) in Chapter 4, this situation provides a basis for thinking about an extreme land-use planning solution.

The planning literature, while not unanimous, generally supports the need for a consistent vision of the structure of a region and to the centers and neighborhoods that compose it. This is, after all, how most land developers

and most businesses generally operate. They define the central themes and characteristics of their products, which guide business policy and employee behavior.

The vision and policies for achieving it that this literature typically advocates are consistent with the tenets of “Smart Growth” and ideas about “New Urbanism” found in the works of Calthorpe (1993), Calthorpe and Fulton (2001), and many others. A fundamental principle is that more attention should be paid to the overall quality and sustainability of urban life and neighborhoods. The more specific principles focus on more attention to and better design of:

- buildings in terms of access and integration with the neighborhood;
- neighborhoods in terms of walkability, livability, open space, environmental protection, and efficient use of infrastructure;
- transportation in terms especially of mode choice (i.e., land use and infrastructure design to allow transit, cycling, and walking to compete effectively with the car); and
- social and environmental justice in terms of affordable housing, design for special needs, and who gets what.

All the regional planning efforts we are familiar with (see Chapter 8) espouse more or less the same objectives. Some general agreement exists among planners and advocates of urban planning about the directions of changes in land-use and transportation needed to pursue those objectives in metropolitan areas, including:

- a shift in trips from cars to high-occupancy vehicles and nonauto modes;
- greater density, in part to facilitate the shift in modes and to reduce trip lengths;
- a greater mix of uses, in part to facilitate the shift in modes and to reduce trip lengths; and
- a hierarchy of centers and corridors.

The ideal planning solution typically espoused is comprehensive, long-run, and top-down. It looks out 20 to 50 years to create a structure of denser centers (the polycentric city) connected by denser corridors along high-volume transportation corridors. The density allows transit and other modes to function better (for reasons already given and then expanded on in Chapter 6)—so much better that people shift to transit rather than continuing to drive a car on congested highways.

Urban designers involved in regional planning might accept calling that level of planning the *macro-architecture of the region*. It is necessary but not sufficient. It is not enough to designate centers and corridors. Policy must ensure that those centers and corridors develop consistently with the principles above: they must have a mix of uses designed in such a way to control negative externalities and to facilitate walking and the use of high-occupancy vehicles.

Such regional planning sets a framework for development and must be accompanied by standards and regulations for *neighborhood design* and, ultimately, *building design*. The regional planning framework may exert considerable control over the policies set by local units of government.

The sidebar on the following two pages gives one expert’s view of how these elements of macro and neighborhood architecture interact.

Could heavy-duty regional planning deliver a better, more efficient metropolitan area? In principle, yes. But like pricing, there are high technical and political hurdles. The technical hurdles are not obviously any easier to surmount than those for pricing. Regulating land development at the macro, neighborhood, and building levels requires vast amounts of information to

It is not enough to designate centers and corridors: they must have a mix of uses designed in such a way to control negative externalities and to facilitate walking and the use of high-occupancy vehicles.

TRANSIT-ORIENTED CORRIDORS

By Robert Cervero, University of California, Berkeley

Transit-oriented development (TOD) has not produced many of its hoped-for benefits—traffic congestion relief, improved air quality, land conservation—in the U.S. I have long been an advocate of TOD as a model for sustainable urbanism and transport. TOD is the most sensible form of smart growth: the logical place to target compact, mixed-use, pedestrian-friendly growth is in and around transit stations. It seems to resonate with people of all backgrounds and political persuasions.

TOD has failed to “deliver the goods” in the U.S. in large part for institutional reasons: it largely takes an atomized form: designed, planned, and built from a narrow municipal perspective. What one typically finds in the U.S. is a few nodes, or islands, of TOD in a sea of auto-oriented development. Unless enough TODs are linearly aligned along natural travel corridors to create a critical mass, changes in transit ridership and auto use will be small. In order to draw enough motorists into the trains and buses so as to put an appreciable dent in car traffic, enough origin-destination combinations must be formed by TODs in non-downtown settings. Otherwise, stand-alone TODs can backfire, creating dense nodes in otherwise auto-oriented settings that increase traffic congestion on roads feeding into transit stations. Many TOD proposals encounter NIMBY (Not In My Backyard) resistance because local residents justifiably argue that they will increase, not decrease, traffic congestion.

The solution is to plan TODs at a corridor scale to create an axis of what might be thought of as “codependent” nodes that rely on each other and, in doing so, create trips conducive to transit riding. Land uses typically found scattered throughout suburbia—housing, offices, shops, restaurants—would be mixed along transit-served corridors. Think “corridors,” not “nodes.”

I suggest we drop the term TOD in favor of TOC: transit-oriented corridors. Corridors are natural subregional travel sheds: six- to eight-mile axes along which significant shares of household trips occur. The aim is to enable many activities, such as for shopping, recreating, and perhaps even working, to occur within these somewhat self-contained travel sheds. People will make trips outside of local travel sheds for regional destinations, but TOC design would allow a bigger share of trips to be made within corridors—and by transit.

TOCs reduce the bane of public transit—cross-haul trip making. Once trip origins and destinations are scattered, travelers switch in droves to private cars. Traffic funnels to the same limited set of tangential highways, beltways, and crosstown connectors. This merging of helter-skelter, zig-zag travel patterns throughout a region gives rise to suburban gridlock (Cervero 1986a).

The benefits of organizing TODs along linear corridors go beyond simply increasing transit ridership. In contrast to a monocentric pattern of urbanization, mixed-use TOCs allow trains and buses to get filled in both directions not only during peak hours, but also in the midday and throughout the week.

Each TOD within a TOC does not have to contain a balance of jobs, housing, retail, and services. Individual TODs can be

predominantly single-use in nature—some might be principally job centers, others bedroom communities, and others a mix of activities. What matters is that a good balance of urban activities occurs along the six-to-eight-mile travel shed. At a corridor scale, balanced development leads to balanced travel flows, which in the case of pricey rail-transit investments means efficient use of expensive fixed-guideway infrastructure.

There are more benefits. Putting offices and entertainment uses near each other, for example, can allow for shared parking. Though some might argue that parking itself is antithetical to transit-oriented communities, in America more than 95 percent of households own cars: TODs or TOCs cannot be “anti-car.” For evening activities catering to the middle class, some provisions of parking will always be necessary.

One reason that TODs in the U.S. do not sum up to TOCs is the focus on building housing—specifically, housing targeted at downtown workers who, for lifestyle reasons, want to commute via transit. California has relied on redevelopment laws to acquire land and underwrite the costs of TODs. State law requires that affordable housing be provided in all redevelopment districts, thus placing a focus on housing production. For the commute trip, however, housing is only one end of the journey. As jobs leave downtowns for the suburbs and beyond, the absence of destinations served by transit results in even TOD residents driving for the majority of trips. Recent surveys of TOD residents in California found transit captured between 3 percent (in the case of LA’s Blue Line LRT) and 44 percent (in the case of the Bay Area’s Pleasant Hill BART station) of all work trips made by TOD residents, largely a product of “self selection”—people opting to live near transit for the very reason of economizing on their travel to work (Lund et al. 2004). But for nonwork trips, fewer than 10 percent of all journeys by TOD residents were by transit.

The best example of TOC is in Stockholm, Sweden. There, the last half-century of strategic spatial planning has given rise to a regional settlement and commuting pattern that has substantially lowered car dependency in middle-income suburbs. Stockholm planners used a “necklace of pearls” as a metaphor for how growth should unfold beyond the central city. Stockholm officials built the Tunnelbana railway in advance of demand, using it to guide the siting and development of new towns, most interspersed by open space and green wedges, very much akin to Ebenezer Howard’s celebrated Garden Cities of Tomorrow. While some stations have a balance of land uses, others are more specialized, functioning as either employment centers or residential communities. But within the six-to-10-mile linear corridors served by Tunnelbana, one finds a balance of jobs, housing, retail, and population services. Consequently, 55 percent of peak-period commuters travel in one direction on trains while 45 percent travel in the other.

Stockholm is a fairly wealthy city by global standards, meaning most households own cars. The city and surroundings are so oriented toward transit riding, however, that for many trips

transit is far more convenient than driving. Stockholmers drive, though judiciously—such as when grocery shopping or taking weekend excursions to the countryside. For the daily grind of going to work and many other trips, however, stockbrokers and day laborers alike often take transit. Stockholm’s transit modal share is nearly twice that found in bigger rail-served European cities like Berlin and even higher than inner-London’s market share. Between 1980 and 1990, it was the only city in a sample of 37 global cities that registered a per capita decline in car use—a drop of 229 annual kilometers of travel per person (Kenworthy and Laube 1999).

In the U.S., the best example of TOC is found in Arlington County, Virginia. The Rosslyn-Ballston mixed-use corridor mimics Stockholm’s necklace-of-pearls approach. During peak hours, Washington Metrorail subway stations are both trip origins and destinations: trains and buses are full in both directions. From 4 p.m. to 6 p.m. on weekdays, the count of station entries and exits are within 10 percent of each other along the Rosslyn-Ballston axis. The presence of so much retail-entertainment-hotel activities has further filled trains and buses during the midday and on weekends. Arlington County averages higher shares of transit boardings and alightings at its stations in off-peak hours

than any other jurisdiction in the Washington metro area except for downtown D.C.

The San Francisco Bay Area is trying to replicate the Stockholm and Rosslyn-Ballston experiences. Despite the best of efforts to create transit-oriented communities along the BART rapid transit system over the past 30 years, only a handful of moderately dense nodes around stations like Pleasant Hill and Fruitvale exist outside of San Francisco, and these are flanked by large parking structures and surrounded by sprawl. The Metropolitan Transportation Commission, the regional transportation planning body, recently adopted Resolution 3434 that sets “corridor-level thresholds” for housing production and establishes “Corridor Working Groups” to “create a more coordinated approach to planning for TOD” (Metropolitan Transportation Commission 2006).

The serendipitous thing about pursuing TOCs is that over time, they can give rise to a Transit-Oriented Metropolis (TOM) (Cervero 1998). When development patterns friendly to transit take form throughout a region, matched by a network of fixed-guideway transit that mimics the interconnectivity of freeway networks, the U.S. will truly be on a pathway to sustainable urbanism. ■

do well. Here planners and policy makers run head on into a huge literature about the efficiencies of decentralized market decisions and the failures of central planning. It will be hard and potentially inefficient to devise, implement, and enforce regulatory policies to mimic the incentives of pricing in the absence of any changes to pricing.

Trying to solve regional land use/transportation problems with purely or primarily regulatory policies is also politically difficult. The private sector will react to what it will characterize as heavy-handed, inefficient, naïve, and unfair restrictions on private development. The local public sector is unlikely to react much differently. Policy changes in practice will probably be more incremental. They are discussed in the next section.

LOCAL LAND-USE POLICIES THAT CAN AFFECT TRANSPORTATION

Full-blown regional planning has a different set of implementation problems than full-blown transportation pricing, but it is not clear they are any easier to solve. Should regions use market-based incentives or regulatory standards? Neither this report, nor any other, will make a case for one over the other to the satisfaction of all parties. We simply assert that, as a practical matter, metropolitan policy will combine both, just as it will combine long-run and short-run perspectives, and will establish some level of a comprehensive framework (an informal vision; adopted goals; an enforceable plan) to guide incremental policy.

The rest of this chapter looks at those incremental policies in the area of land use. The best of these we refer to as “second best” because our economic perspective means that the best policies are the ones that aim directly at getting the prices right. Thus, a measure of the effectiveness of these policies is how well they approximate the land-use and transportation results that one would expect from proper pricing, and how much they cost to implement (directly, and with external costs) compared to pricing.

Should regions use market-based incentives or regulatory standards?

We simply assert that, as a practical matter, metropolitan policy will combine both.

Our discussion of policies divides land-use policies into two main categories:

1. Policies at a regional scale (land use). In the context of the five Ds, the focus here is on Destinations
2. Policies at a neighborhood scale (urban design). In the context of the five Ds, the focus here is on Density, Diversity, Distance, and Design.

Land use (regional form)

What policies are available to address regional form, and what are they likely to accomplish?

Some of the policies will require state or, preferably, regional requirements and enforcement. Absent those requirements, there are strong incentives for local governments to capture the benefits of growth and deflect its costs. For example, local governments have strong incentives to zone for retail uses in states with a sales tax (45 out of 50); let other jurisdictions worry about industrial land; and zone out multifamily uses (which typically provide sales and property tax revenues less than the costs of the services they require).

Chapter 8 discusses in more detail the politics of regional policy. In the rest of this section, we focus on the policies themselves: how are they intended and likely to work? We organize the policies into four categories:

1. Creating boundaries: zoning and urban growth containment
2. Creating centers and corridors: TOD and retrofitting corridors
3. Pricing public infrastructure and services: concurrency and impacts fees
4. Playing the market: incentives and public investment

We have noted elsewhere that this report is about transportation policy in the context of land use and development. We try to be comprehensive on surface transportation planning in metropolitan areas; we do *not* try to be comprehensive about land-use policies. Thus, this section gives some examples of how land-use policies affect transportation; it does not come close to a full treatment of land-use policies. Many of the policy headings addressed below are the subject of books on land use and growth management that identify, describe, and evaluate those policies; other PAS reports provide detail and direct readers to even more detail.¹³

Land development is one of the most regulated sectors of the U.S. economy, and conventional zoning actually gets in the way of market forces.

Creating boundaries: zoning and urban growth containment

Conventional zoning segregates uses and limits densities. The rationale is the control of externalities by separating potentially conflicting activities. The greater the potential conflict, the greater the distance separating the activities.

The actual impact of zoning on land development is difficult to estimate. The market tends to segregate land uses because land users vary in how they value various aspects of accessibility and public amenity. Many have argued that low-density, sprawling development occurs solely because of consumer preference and market forces.

But Levine (2005) argues that land development is one of the most regulated sectors of the U.S. economy, and conventional zoning actually gets in the way of market forces: “the design template for urban sprawl is written into the land-use regulations of thousands of municipalities nationwide.” Conventional zoning, subdivision codes, and parking requirements almost certainly aid and abet sprawl.

Relaxing zoning regulations would allow improved mobility in urban areas by allowing higher residential densities (thereby improving transit service and the shift to transit ridership) and decreasing distances between origins and destinations by allowing mixed-use development. Relaxing minimum parking space requirements would also allow development at higher densities.

Somewhat stronger are *flexible and performance-based zoning standards*. Standard residential zoning prohibits commercial or industrial development of any kind. Flexible zoning allows planners to negotiate with developers who want to construct some buildings for retail or other commercial use (e.g., in a planned unit development; see Mandelker 2007). Performance zoning reduces the need to negotiate by specifying standards that non-conforming land uses must meet, without specifying actual designs or building materials. The idea behind both flexible and performance zoning is to allow developers to innovate to control external effects using design rather than distance.

Zoning can also function almost as a flexible fee system. Planners bargain with a developer on behalf of neighboring residents and property owners. This bargaining may include requirements for on- and off-site improvements: recreational space, intersection improvements, and so on. If the nonconforming designs the developer proposes have value to the developer's customers, money will be available to compensate neighbors for the risk of negative externalities generated by the development. The in-kind payment substitutes for monetary development impact fees.

The main lesson here is that standard zoning may, in a real sense, leave money on the table. Neighboring land users are concerned about negative impacts. Those concerns can be treated by segregating land uses. But this may not be the most cost-effective way to treat these concerns, especially given the increasingly high cost of transportation. Flexible and performance zoning allow developers to propose alternative methods to control externalities, methods that have positive impacts on transportation.

Urban containment strategies are a way to, well, contain urban development. The idea is to discourage sprawl by drawing a line around the metro area and outlawing urban development beyond it. The line may be an urban services boundary (no extension of urban services beyond the line) or an urban growth boundary (no urban development beyond the boundary).¹⁴

An urban growth boundary (UGB) has potentially strong effects on land-use and travel patterns by affecting two of the 4Ds: density and destination. Urban areas grow as economic activity increases, either because the area offers better access to raw materials and business services, or because the area offers a relatively high quality of life. Growth increases the demand for developable land, but a growth boundary restricts its supply at the urban fringe.

Theory is unambiguous about the effect of a restriction on land supply: higher land and house prices. Existing homeowners benefit, of course, which provides political support for the program. But many people are concerned about the equity effects.

There is an outlet for the demand for land and housing. Rising suburban land prices encourage more intensive development in suburban areas and redevelopment in more central urban areas. Thus, urban containment not only reduces the extent of urban expansion, but also promotes development at higher intensities throughout the urban area. Achieving these higher intensities, however, usually requires and encourages the relaxation of zoning.

Nelson et al. (2002) consequently find that local planners play a significant role in determining the severity of housing price inflation attributable to growth management policies. The urban containment boundary pushes development inward, giving planners leverage with which to negotiate with land developers to provide externality-inhibiting building and neighborhood designs. Local authorities can also promote infill and redevelopment by coordinating and helping pay for improvements to aging infrastructure.

Flexible and performance zoning allow developers to propose alternative methods to control externalities, methods that have positive impacts on transportation.

Stronger steps are possible. Planners can push for measures to encourage or require developers to include affordable housing in their new projects. Incentives and mandates together—along with supportive land-use policies—are likely to make the most positive contribution to housing opportunity.

These changes in land development potentially reduce the length of trips and encourage use of alternatives to the automobile. The degree to which an urban containment boundary reduces auto use and vehicle trips depends on the employment of additional strategies for the other Ds: design, diversity, and the regional accessibility and importance of destinations.

Nelson and Dawkins (2004) evaluated more than 100 urban containment plans, exploring urban containment as a framework that regions and communities can use for managing growth. They categorized the urban containment strategies into those that represent a combination of either strong or weak boundaries and those that either restrict or accommodate development:

- *Weak Restrictive* (grow out, not up) strategies use weak containment measures, with restrictive policies for accommodating growth in desirable areas. Mostly found in California and Colorado.
- *Strong Restrictive* (grow somewhere else) strategies use strong growth containment measures, but ineffective measures to capture their fair share of the regions' growth. A well-known example is Boulder, Colorado, but there are others, mostly in California.
- *Weak Accommodating* (grow downtown, please) strategies use containment measures ineffective at controlling growth in rural areas, while weakly accommodating development pressures within the urban area. This type of containment strategy is common, with perhaps the best example found in the Minneapolis-St. Paul region.
- *Strong Accommodating* (grow up, not out) strategies use effective containment measures characterized by aggressive open space preservation, combined with strong measures to attract the region's growth into the urban area. The Portland, Oregon, UGB and accommodating zoning is a good example. Nelson and Dawkins also point to examples along the west coast of Florida.

Nelson and Dawkins looked at the effects of the first round of urban containment programs (adopted prior to 1985) on the amount of development activity taking place in central cities and on the ratio of central-city to metropolitan-area development activity. They found that central cities in contained metropolitan areas were attracting more development activity than central cities in uncontained areas. Suburban areas in both contained and uncontained metropolitan areas continued to grow.

Drawing an effective boundary, however, requires a great deal of information—information planners do not always have.¹⁵ Realistically, the positioning of the UGB is influenced at least as much by political concerns as technical ones. The political aspect of the UGB encourages great expenditures on lobbying and speculation. The difficulty of enforcing unpopular land-use restrictions encourages great expenditures to avoid them.

Creating centers and corridors: TOD and retrofitting corridors

Urban containment can encourage density. Density encourages the use of modes other than automobiles. An obvious place to allow and encourage density, then, is near high-quality transit stations.

Handy (2005) finds little evidence that suburban transit stations by themselves attract high-intensity development. Our observation is that there are two typical reasons for that. First, conventional zoning may discourage it. But most jurisdictions far enough along to be building transit stations know

Drawing an effective urban growth boundary requires a great deal of information—information planners do not always have.

that supportive zoning is a necessary condition for development; if they have not come to that conclusion on their own, it is very likely to be imposed on them by state or federal funders. Thus, we view the second reason as more critical and widespread: the transit stations are ahead of the market. As Handy notes, they cannot make development happen by themselves.

But density may come, quickly or eventually, with zoning that allows it and with designs that are both attractive generally and that facilitate access to transit. These designs are known as TOD.

TOD development guidelines address all four neighborhood-scale dimensions of land-development policy and may result in development both at sufficiently high density to encourage transit use and sufficiently livable to attract those who could otherwise choose a more typical suburban lifestyle. And maybe the livable designs are more attractive than the access to transit. Lund, Cervero, and Willson (2004) report that TOD residents are *more likely* to use transit if:

- transit competes effectively with highway travel in terms of time (i.e., highways are congested and slow, and transit travel is quick);
- good pedestrian connectivity exists at the destination (i.e., transit users don't have to walk too far or wait too long after getting off the rail car);
- employees are allowed flexible work hours; and
- employees have limited vehicle availability.

TOD residents are less likely to use transit if:

- trips involve multiple stops (or "trip chaining");
- good access to jobs is provided via highways;
- they can park for free at their workplace; and
- their employer helps to pay vehicle expenses (e.g., tolls, fuel, etc.).

These findings are all consistent with the basic theory of travel choice described in Chapter 4 and Appendix C.

Cervero and Duncan (2002a) find that the probabilities of a TOD resident commuting by rail vary with:

- residential location (within one-half mile of a station);
- workplace location (within one-half mile of a station); and
- household car-ownership levels (0 or 1).

They find that commuters who work and live within a one-half mile of transit and do not own a car have a high likelihood (greater than 80 percent) of commuting by rail. Of course, it is likely that self-selection occurs: households with economic or other reasons for not owning a car are those most likely to take advantage of this type of housing. It's hard to generalize on the basis of limited experience to all households in a metropolitan area.

In practice, making TOD work often involves a public-private partnership (sometimes referred to as joint development). But private developers may not find relaxed zoning requirements and allowances for higher density enough of an inducement. They may not perceive sufficient demand or are unwilling to bear the risks associated with large-scale, unconventional development.

Local governments can provide support in a variety of ways. They can help with land assembly and provision of infrastructure. Or they can simply help fund construction of buildings (more below in the subsection, "Playing the Market"). Between 1970 and 2002, for example, the Washington, D.C., Metropolitan Transit Authority (WMATA) formally entered into

LAND VALUES AND PROXIMITY TO TRANSIT

Land-use values of TOD have been shown to increase with the increasing indirect cost of driving associated with congestion (tolls and congestion pricing may yield similar effects), the presence of well-designed TOD, and the improvement and maturation of the transit system itself.

Cervero (2004) investigated several cases of TOD in the San Francisco Bay Area and found that a robust economy, supportive zoning, and transit system network expansion and maturation are important to maintaining high land values. In the early 1990s during an economic downturn, the Santa Clara County light rail system had no measurable effects on land values (Landis et al. 1994). However, by the late 1990s, when the economy was booming, and real estate values and traffic congestion were at an all-time high, Cervero (2004) found appreciable land-use value benefits accrued by the proximity to transit. Land-value benefits of TOD sometime take time to accrue and therefore point to the need to bridge the gap between current and future markets by assisting TOD with regulatory and financial incentives that we will discuss further below.

38 joint development projects in the D.C. region. The market value of these ventures has exceeded \$2.5 billion. Collectively, these projects—everything from air-rights leases and land rents to station connection fees—yield the agency about \$6 million in annual revenues.

Nevertheless, investment in TODs should not be counted on to yield good financial returns to the public sector in the short run. The idea is that TOD yields less tangible but longer-term benefits in the form of less suburban sprawl, experience with higher-density and mixed-use development, higher transit ridership, and more choice for metro area households.

The literature about the transportation/land use connection is extensive regarding TOD. But as Robert Cervero argues in his sidebar above, focusing on a single TOD can miss the point: one must address centers in the context of corridors; nodes in the context of noodles. Doing so helps prevent thinking that a desirable development pattern consists of similar centers, each with a complete mix of uses dotted around the metropolitan area. An emphasis on transit-oriented *corridors* probably results in a more efficient and practical pattern (in the sense that the private sector might actually see value in building to the planning specifications).

The typical issue for corridor planning in the U.S. is not, however, about rail transit—it's about highways. A rail corridor provides special access not along its length, but only at its stations. Thus, land use for rail is first about getting the stations in the right place (Destination) and then about station area design (all the other Ds, bundled under the heading of TOD).

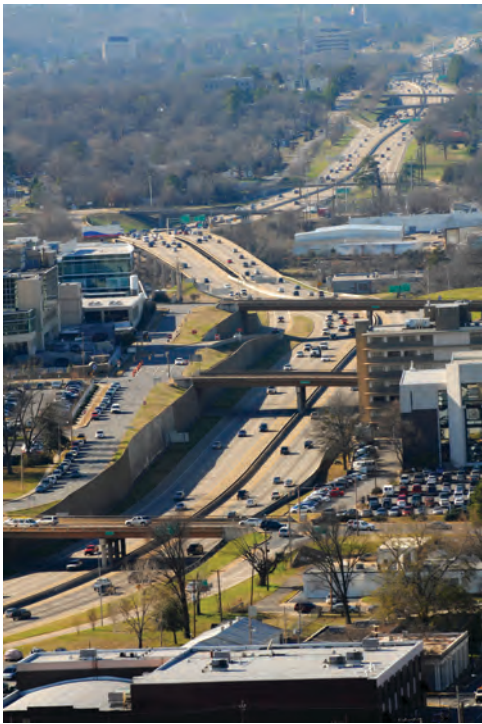
In contrast, the typical corridor in metropolitan areas is a highway arterial. They may have the length and linear form that Cervero wants from TOCs, but they are the antithesis of a string of pearls: they are just a string and usually a frayed one. They are, however, great examples of what can happen when markets and transportation create places without a long-run vision supported by enforceable standards.

Highway corridors are the predominant way that metropolitan areas offer the balance of good access and relatively cheap land that many businesses (especially land-intensive, auto-dependent retailers) want. They allow those retailers to give retail customers what they want: value (the right mix of choice, quality, and cost).

The early and few arterial connections between cities were the logical places for retail expansion as auto transportation became prevalent in the first half of the twentieth century. Downtowns were crowded; land was expensive; retail development was shifting to a land-extensive automobile orientation: land along the arterials fit the bill.

But times change, and retail strips are probably destined to decline because their efficiency is declining. Freeways have bypassed some retail strips; malls at interchanges have supplanted others. But market forces for both retail and residential uses are changing, presenting opportunities for strips to reinvent themselves as centers on a boulevard. Developers are starting to find a market demand for community and for places that have a “there” there.

A sensible retrofit for some of these corridors might (1) consolidate the dispersed, oversupplied, and underperforming retail activity in mixed-use or specialized centers along the strip; (2) make public investments in streetscape improvements and other public amenity to help that development succeed; (3) shift retail entitlements along the entire length of the corridor to fewer centers, and allow and encourage higher-density residential uses; and (4) increase transit just ahead of the market when it looks like it could tip the transition. That is probably not a bad description of how redevelopment occurred along the Orange Line of DC Metro in Arlington County, Virginia.



David H. Lewis, istockphoto.com

The typical corridor in metropolitan areas is a highway arterial. They may have the length and linear form that Cervero wants from TOCs, but they are the antithesis of a string of pearls: they are just a string and usually a frayed one.

Pricing public infrastructure and services: impact fees, concurrency, and parking

Urban land cannot develop without basic urban infrastructure: electricity, drinking water, wastewater collection, and transportation are basic at the site level; schools and recreational facilities should be basic and the neighborhood level. Consistent with the theme of this report:

- If those facilities are not provided, external costs are likely; external costs mean improper pricing; improper pricing means inefficiency.
- The facilities may be provided but priced in a way that gives the wrong signals to consumers and producers about the true cost; again, improper pricing means inefficiency.

The obvious solution, in concept, is to make sure all the necessary facilities and services get priced properly. There is a huge professional literature in public finance on that topic, and it is as contentious as all professional literature. We cannot cover it here. We are just going to comment on the hot topics in planning of impact fees and concurrency.

Impact fees and concurrency are related, and overlap. To oversimplify greatly, an impact fee aims, ideally, at charging development for the cost of facilities the public sector needs to build to adequately support development, while concurrency says developers must provide adequate facilities—in other words, two different ways to try to ensure the necessary facilities get built. One might think of impact fees directly trying to get the prices right, and concurrency as a second-best regulatory policy to get the facilities right.

Transportation planners and local government finance officers typically estimate the occupants' use of infrastructure capacity (in the case of transportation, the road or public transport network) through trip generation models. While model results can vary widely according to the assumptions used in the model, a connection can often be drawn between the benefits that commuters and freight haulers receive from the availability of nearby public infrastructure. For our purposes we will focus on roadway infrastructure, although the case has been made for trails benefiting commercial development.

Impact fees are usually calculated to take into account the average (not marginal) capital (not operational) costs that hypothetical new road users cause. Capital investment in transportation facilities is typically “lumpy,” occurring through large projects rather than small incremental improvements that add exactly the amount of extra capacity required. In other words, a given development may sometimes necessitate a large, expensive improvement of which the development would not have exclusive use. Rather than charging the “straw that broke the camel’s back” for the full cost of a new improvement, impact fees typically charge developers for their proportional cost on the entire network. They pay for a share of any new road immediately required, as well some proportion of the cost of existing roads and the roads needed at some point in the future. When new investment is required, the impact fees collected from all developments in a jurisdiction are used to fund the infrastructure.¹⁶

Not all developer contributions are based on the development’s proportional impact on networkwide costs. Some are based on the fixed cost of any improvements immediately necessitated by the development. For example, a development may be required to pay the full cost of the improvement of an adjacent intersection or the addition of a lane on an adjacent collector road leading from the development to the nearest arterial road. They may not be required to pay for their share of the use of existing or planned infrastructure.

Concurrency, usually implemented through Adequate Public Facilities Ordinances (APFOs), is an example of a land-use and transportation pricing

Times change, and retail strips are probably destined to decline because their efficiency is declining. Freeways have bypassed some, and malls at interchanges have replaced others.

policy. In theory, concurrency allows land development to occur only where the infrastructure necessary to support that development is provided concurrently (at the same time).¹⁷

Concurrency can be related to “urban service boundaries” if the local government (1) has already made a decision where to provide infrastructure and (2) allows development only where adequate infrastructure exists and developers are not allowed to provide the required infrastructure themselves. If developers are allowed to pay for the infrastructure, concurrency and impact fees are similar, but with one clear distinction: concurrency emphasizes the timing and location of where infrastructure investments are made, rather than who pays.

In practice, this concurrency is not exact; usually the required infrastructure is provided before land development occurs, and sometimes communities allow it to be installed within a set amount of time after development occurs. But the basic principle is that demand for infrastructure from new development is not allowed to outstrip supply.¹⁸

Proponents of concurrency in the context of transportation facilities hope it has the following benefits:

- *Less congestion.* New development should not generate traffic that exceeds the capacity of the transportation infrastructure that serves it. Congestion levels arising from development are kept below a maximum threshold, so that a minimum level of service (LOS) is always met.
- *Effective use of existing infrastructure.* Development is directed to those areas where public investment has already occurred or where government has decided it is worthwhile to invest in additional infrastructure.
- *Spatial and temporal connection with developer impact fees.* The benefits of concurrency are not as much about who pays, but rather about the when and where of costs in relation to the timing and location of development.
- *Integration of transportation and land-use planning and development.* Concurrency guides development to where sufficient transportation infrastructure exists. Without concurrency, impact fees and developer contributions may be collected but spent in areas other than where the development occurs that is causing the impact. Furthermore, the revenue may be insufficient to pay for the necessary improvements. Without concurrency, in both cases, the development could proceed without adequate transportation infrastructure. With concurrency, the development could proceed only when the government or developer had made the necessary improvements to the required standard.
- *Streamlines the political process.* Initial decisions on transportation investment are made with the explicit consideration of where growth should occur, thereby potentially achieving a more balanced transportation/land use connection. As a result, subsequent land-use decisions (comprehensive planning, rezonings, etc.) should be easier and quicker to make—benefiting the community and developers alike.

Concurrency policies often require adjustment after initial implementation. For example, concurrency policies have exacerbated sprawl by pushing development farther into exurban areas where minimum LOS standards are more easily met. This is inconsistent with the intent of concurrency: policy makers believed it would curb sprawl by directing growth towards underused infrastructure in central urban areas.

Many jurisdictions have found that older, central parts of a metro area are more spatially constrained than they originally envisioned. One modification in the APFO process is to accept higher congestion levels (lower LOS ratings) in these areas. Palm Beach County, Florida, for example, recognized

Concurrency policies often require adjustment after initial implementation. For example, concurrency policies have exacerbated sprawl by pushing development farther into exurban areas where minimum LOS standards are more easily met.

that many central-city arterials and collectors were constrained not by funding limitations but by physical constraints, such as narrow rights-of-way between historic streetscapes, and established neighborhoods, rivers, railways, etc. In response, it classified these as “constrained facilities” and allowed a lower LOS standard (i.e., allowed more congestion).

In the case of central-city streets, accepting a lower LOS can encourage desired land-use types. High-density development around designated centers can be promoted through lowering the minimum LOS and acknowledging the service of other modes, such as walking, bicycling, and public transport.

A lower LOS is not the only way to encourage desirable land uses—certain types of projects can also be exempted entirely from concurrency requirements. Florida’s statutes allow local governments to apply to the state for approval to create Transportation Concurrency Exception Areas, which must promote public transportation or be designated in local comprehensive plans for urban infill, urban redevelopment, or downtown revitalization.

Particular land-use activities can be exempt. Some Maryland counties with adequate public facilities requirements exempt housing for the elderly, while Baltimore County exempts industrial development and hospitals, and Montgomery County exempts affordable housing, some small health care facilities, and places of worship. Clark County, Washington, allows exceptions for schools, fire and police services, and public transport facilities.

Another policy area at the intersections of land-use and transportation is *parking*: it is a transportation use that is also a land use. In downtowns, it is a transportation use that occurs in buildings (parking structures). Its amount and design is controlled more by land-use regulations than by highway engineering standards.

Parking requirements have a direct influence on urban density. As a rough rule of thumb, except in dense downtowns, conventional parking requirements translate into a 1:1 ratio of parking area to commercial space. Those standards typically require three to four parking spaces for every 1,000 square feet of gross leasable area (three for office; four for retail). A typical parking stall consumes about 300 to 350 square feet of space, including aisles, landscaping: about three stalls per 1,000 square feet. Thus, every 1,000 feet of commercial space requires at least 1,000 square feet of parking. That is a 1-to-1 ratio, or a floor area ratio (FAR) of 0.5.

Those parking requirements constrain the form and massing. Assume a one-acre parcel (the size of a block in downtown Portland), and a standard 10,000-square-foot plate for a commercial office building. To meet on-site parking requirements using surface parking, the building can go no higher than three stories. To go more than three stories, either (1) the floorplate must be smaller and a larger percentage of the parcel must be dedicated to parking, or (2) structured parking must be built.

Both means of increasing density have economic problems. Going to a smaller, nonstandard floorplate size and a taller building will increase costs per square foot. Structured parking is, economically, even more of a problem. The average cost of a parking stall (not counting land cost) is five to eight times greater in a parking structure than in a surface parking lot. Our calculations are that structured parking cannot work without a subsidy until the price of land reaches \$40 to \$50 per square foot. Those costs are found in downtowns, but not in the suburban centers of most midsize metro areas.

It may be that the market demands these spaces because they are required for the success of some businesses. But developers and their banks are increasingly willing to build with less parking. Thus, one policy is right at the transportation/land use connection: relax or remove minimum parking requirements. Alternatives are shared parking, parking maximums,¹⁹ incentives for car-sharing, and “park-it-once” districts.

One policy is right at the transportation/land use connection: relax or remove minimum parking requirements. Alternatives are shared parking, parking maximums, incentives for car-sharing, and “park-it-once” districts.

In the language we introduced earlier in this chapter, these policies are at best “second best” to pricing. The direct policy is to price the parking properly: “free” parking (free in the sense that the user of parking does not directly pay for the full cost of that parking), like unpriced highway capacity, means inefficient levels of congestion.

Playing the market: incentives and public investment

The section above on TOD discusses the importance of the private sector (developers, builders, financiers, and others) in creating urban places. Cervero et al. (2005) studied what local government and transit providers offer to developers and what developers truly value. They interviewed 35 TOD developers across the U.S.

They found that public agencies viewed the adoption of better procedural strategies (e.g., expediting entitlement reviews and excluding TODs from concurrency requirements) as less effective than other measures for jump-starting TODs. In contrast, developers said they valued what locals dismissed as less important. Specifically, they found that increased certainty, reduced turn-around time, and upgraded transit services are generally preferred. One of the most cost-effective measures a local planning agency can offer is to streamline the project review process. Developers also generally agreed that supportive zoning, help with land assembly, funding set-asides for streetscape improvements, and other tools under public-sector control can substantially help make TOD work.

Incentives are usually thought of as either reducing inefficient regulatory burdens or reducing development costs by paying for things that developers might otherwise have to pay for. Clearly, providing key public infrastructure, including highway and transit facilities, is in that second category.

So are rarer techniques like the purchase of development rights. Rather than simply lobbying government to preserve wild areas, for example, the Nature Conservancy purchases the land it wants to conserve at market prices. Boulder, Colorado, has used a program of green space acquisition and focused service extension to limit growth at its fringe. Local governments similarly can purchase the land they want to control. Purchasing the land along a bypass highway allows the government to control the characteristics of development along the bypass, as in King County, Washington.

Urban design (neighborhood form)

Many of the policies related to urban design are already covered in the discussion of density, diversity, design, and distance (four of the Ds) above. Other aspects of urban design as they relate to transportation are covered in Chapter 6. This section gives a sampling of some of the main policies.

We use the term *urban design* to apply at a neighborhood or site level; at the level of human and environmental interaction. Urban design has at two different elements that together define a third:

- *Form.* Form addresses arrangement and appearance. A trip has an origin, a link, and a destination: point, line, point. A drawing of the physical form of the transportation/land use connection can be primarily characterized by “paths” and “nodes” (Lynch 1960).²⁰ Paths facilitate the movement of people and goods (i.e., streets, rail lines, etc.). Nodes are places where a trip ends, often including the exchange of passengers or freight. We can update the language to match current issues and thinking about the broader reach of the transportation/land use connection: paths are *corridors* and nodes are *centers*.
- *Function.* The function of a transportation system is movement (mobility); the function of a land-use system is activity. Linking the two is the

The direct policy is to price the parking properly: “free” parking (free in the sense that the user of parking does not directly pay for the full cost of that parking), like unpriced highway capacity, means inefficient levels of congestion.

function of accessibility. Urban design cannot stop at appearance (form). It must consider function by thinking holistically about environment, economy, community, and so on. Both the seen and unseen components of urban design matter, which leads to the next urban design element.

- *Experiential quality.* If an urban area is designed well for form and function, it should provide a good experience for inhabitants as well as travelers. That experience might be called Quality of Life or Livability, which is the overarching goal of urban public policy (see Appendix A). In the context of economics and the discussion in Chapter 6, a lot of the land-use planner and designer concerns about livability are about reducing negative externalities.

According to Handy (2005), the local neighborhood dimensions of density, diversity, distance, and design—the dimensions at the level of human interaction with the urban environment—have a bigger effect on walking than on driving, and especially on “purposeful walking” (e.g., going to work, as opposed to recreational walking). An increase in purposeful walking has also been associated with other quality-of-life benefits as well (see sidebar).

WALK AND ROLL: HEALTH, ENVIRONMENTAL, AND ECONOMIC BENEFITS

By Lawrence Frank, University of British Columbia

The 1996 Surgeon General’s Report on Physical Activity and Health documents the benefits of achieving moderate regular activity 30 minutes per day most days of the week, even through short trips (e.g., a 10-minute walk to and from transit). Walking is the most readily available physical activity to nearly all people, offering additional transportation and environmental benefits.

Physical activity is a key predictor of healthy body weight, and hence they are both important predictors of morbidity. The primary measure used is the body mass index (BMI), which is the ratio between one’s height and weight. While the presence of a causal relationship between the built environment and health is still being evaluated, evidence shows significant associations between the neighborhood design, physical activity, and obesity.

Providing opportunities for physical activity, with sidewalks, trails and livable streets, are shown to increase the level of physical activity, as well as cognitive and creative health (Appleyard 2003). Several studies show that increased levels of walkability are associated with higher physical activity and lower BMI. For example, residents in Atlanta’s most walkable areas are 2.4 times more likely to get the recommended 30 minutes of moderate activity (Frank and Engelke 2005). Another study found that each additional kilometer walked was associated with a 4.8 percent decrease in the likelihood of being obese, and each additional hour spent per day in a car was associated with a 6 percent increase in the likelihood of being obese (Frank, Anderson, and Schmid 2004). Research also shows that providing such physical activity opportunities provides important economic benefits—helping the bottom line as well as the waistline. For example, Wang et al. (2004) found that for every dollar invested in trail development, nearly \$3 (\$2.94) of public health benefits are produced.

Importance of Street Connectivity

The SMARTRAQ Project analysis in Atlanta, Georgia, found that doubling the current regional average intersection density,

from 8.3 to 16.6 intersections per square kilometer was associated with a reduction in average per capita vehicle mileage of about 1.6 percent. Furthermore, the Frank et al. (2006) study of King County, Washington, found that per-household VMT declines with increased street connectivity, all else held constant.

The Importance of Mixed Land Uses and Human Scale

The King County, Washington, *HealthScape* Study showed that the number of retail uses (not simply retail square footage) led to increases in walking trips. This result and evidence from other studies suggest zoning should allow and perhaps encourage more small to modest-size neighborhood stores and restaurants.

Findings from another SMARTRAQ study shows that teens too young to drive are the most sensitive youth age group to “built environment” factors. For example, they are several times more likely to walk if they live in neighborhoods with a mix of uses, have parks, and a connected street network (Frank et al. 2007). This study of 3,100 children in the Atlanta region found that the presence of recreational spaces within a kilometer walking distance of a child’s home was the most consistent predictor of walking for youth of all age groups. This same study, however, showed that additional household vehicles meant that young people walked less often and, when they did walk, they walked shorter distances.

Economic Benefits

A rigorous 2004 survey of park and trail use in Eugene, Oregon, found that city businesses and stores benefited directly from biking and walking trails that attracted customers from out of town, as well as facilitating their travel. The website, <http://www.bicyclinginfo.org/bikecost/>, provides a tool for estimating trail costs and benefits, such as the number of new trail users, the measured economic benefits, time savings, decreased health costs, etc.

DONALD APPLEYARD (1981) ON LIVABLE STREETS

Nearly everyone in the world lives on a street. People have always lived on streets. They have been places where children first learned about the world, where neighbors met, the social centers of towns and cities, the rallying points of revolts, the scenes of repression. But they have also been the channels for transportation and access: noisy with the clatter of horses' hooves and the shouts of their drivers, putrid with dung, garbage and mud, the places where strangers intruded and criminals lurked. The street has always been the scene of this conflict, between living and access, between resident and traveler, between street life and the threat of death. ■

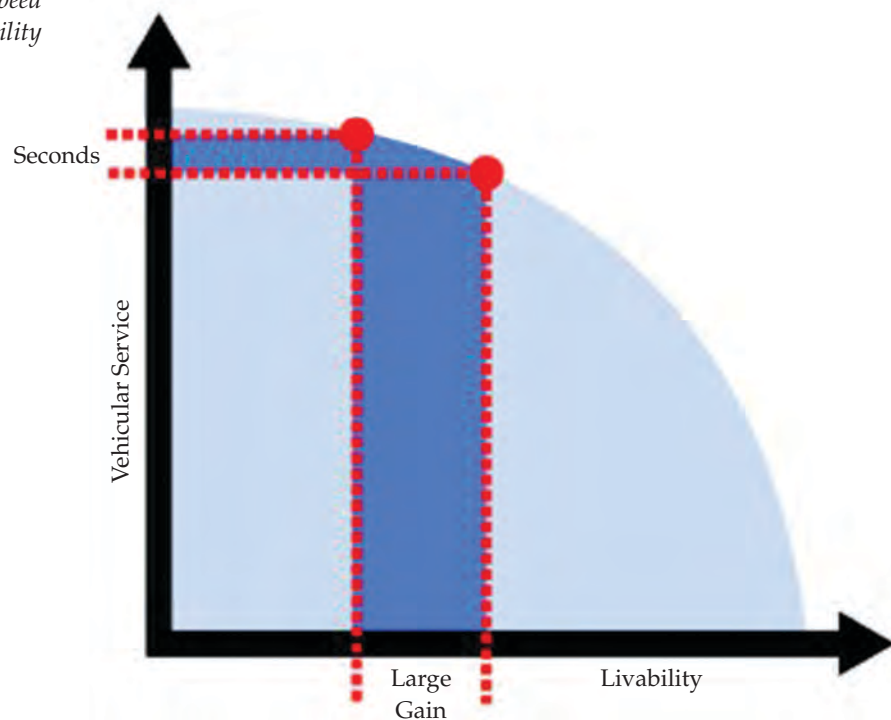
We can characterize the relationship between four of the five Ds and walking as follows:

1. *Distance* has a direct relationship with walking.
2. *Design*
 - a. Function (such as adequate and safe paths and crosswalks) helps overcome distance and has a strong but indirect relationship with walking.
 - b. Form adds to the quality of the walking experience and has an indirect and weaker relationship with walking.
3. *Diversity* has a direct relationship with distance—placing important destinations closer together—and has an indirect relationship with walking.
4. *Density* is usually associated with both diversity and distance, and has an indirect relationship with walking. Density, however, is often used as a measure of walkability and lower auto use because it is:
 - a. often accompanied by diversity and walkable distances functionally well designed, as well as other factors that make walking competitive with driving (good transit service, parking scarcity, etc.); and
 - b. relatively easy to gather, calculate, and compare across urban areas

As traffic volumes and speeds on city streets increase, quality-of-life factors that lead to healthy and strong communities are weakened by making them inhospitable places for pedestrians and neighbors who want to create stronger social ties. That, in turn, affects other aspects of community life and safety, which affects public services and costs. Donald Appleyard (1981) first articulated the notion of the ecology of the street and the unequal distribution of power often held by drivers over residents, pedestrians, and cyclists.

Streets compose, on average, more than one-third of urban areas and the majority of their public space. They provide more than access, and certainly more than conduits to desired destinations: they contain many of those destinations.

Figure 5-3. Trading speed for livability



Source: Glatting Jackson Kercher Anglin, Inc.

In the jargon of economics, one can describe the issue as a trade-off along a production-possibility curve. Figure 5-3 illustrates the possibility that, for certain streets and traffic volumes, a relatively small reduction in speed may yield a relatively large increase in livability. Moreover, as described in Appendix C, it may even be that the reduction in speed *increases* throughput.

INTEGRATING LAND-USE AND TRANSPORTATION PLANNING

Table 5-4 cross-tabulates some of the policies discussed in this chapter with the 5Ds to talk about the likely effects of the policies along those dimensions. A plus sign (+) suggests the potential to affect the dimension in a beneficial way (in the context of improving the connection between land use and transportation); a minus sign (–) suggests the opposite; an equal sign (=) suggests neutrality or uncertainty.

TABLE 5-4. GENERAL EFFECTS OF SAMPLE LAND-USE POLICIES ON THE 5DS

LAND-USE TOOL OR STRATEGY	NEIGHBORHOOD DENSITY	NEIGHBORHOOD DIVERSITY	NEIGHBORHOOD DISTANCE	NEIGHBORHOOD DESIGN	(REGIONAL ACCESSIBILITY TO) DESTINATIONS
Euclidian zoning	–	–	–	–	+ ^a
Flexible zoning	+	+	+	+	=
Design guidelines	+ ^c	=	+ ^b	+	=
Density bonuses	+	+	+	+	+ ^a
Tax incentives to attract growth and density	+	+	+	+	+ ^a
Growth boundaries	+	=	+	=	=
Transferable development rights	+ ^a	=	+ ^a	=	+ ^a
Parking pricing	+	=	=	=	=
Regional land-use planning	+	+	+	+	+
Transit-oriented planning/incentives	+	+	+	+	+ ^a

Source: Authors' opinion

Note: A plus sign suggests the potential to affect the dimension in a beneficial way (in the context of improving the connection between land use and transportation); a negative sign suggests the opposite; an equal sign suggests neutrality or uncertainty.

a. Possibly effective if coordinated at regional level.

b. Distance may be overcome via construction of inviting, direct paths.

c. Design guidelines can make higher densities attractive and appealing to the surrounding communities.

The professional literature in land use and transportation consistently supports the view that land-use and transportation development have had effects on one another and probably will continue to. The debate is about the degree, importance, and predictability of those effects. Noted researchers in the field line up on both sides.

On one side, Boarnet and Crane (2001) are skeptical about the ability of planners to use urban form to influence travel behavior, arguing that we just don't know enough. Giuliano and O'Brien (2004, 269) quotes and supports their argument, saying that "Empirical evidence on the land use impacts of both highways and transit indicates that transportation investments do not have consistent or predictable impact on land use. The evidence clearly

shows that land use change does not necessarily follow transportation investment. . . .” On the other side (e.g., Cervero, Handy, Ewing) are numerous articles on the theoretical importance of the relationship and empirical results consistent with, in specific cases, the theory.²¹ The sidebar shows one study’s list of potential effects.

We, as is obvious by now, find the theory compelling. For us, land use and transportation have to affect one another: it cannot be otherwise. We view everything else through that lens. Urban form must affect transportation at both the large scale (look at different metropolitan forms around the country and you’ll find some different transportation systems) and the small scale (transit is much more likely to be found serving nodes of density along dense corridors). Similarly, transportation must affect urban form at the metropolitan scale: it certainly has, and though the effects may not be as great as they once were, where and how we build our new transportation infrastructure to accommodate growth will surely affect where, and probably

HOW LAND USE FORM INFLUENCES TRANSIT USE

Summary of Research Findings from TCRP Report 16

How do characteristics of land-use form (e.g., residential density, CBD employment size and density) influence the demand for light rail and commuter rail transit and the cost of providing such service?

- Residential densities have significant influence on rail transit station boardings.
- Residential densities have more influence on light rail ridership and costs than on commuter rail.
- Both the size and the density of the CBD influence light rail ridership.
- CBD density is more important for supporting commuter rail ridership than light rail ridership.
- Other factors within the control of transit agencies, such as the availability of feeder bus service and park-and-ride lots, also influence ridership.

How does the built environment around new rail transit stations affect the mode of access and the size of the catchment area?

- Residents in higher-density residential areas are more likely to walk to transit.
- Nearly all commuters walk to their destinations in CBDs, but 25 to 50 percent ride buses at other destinations.
- Use of feeder bus service depends mainly on the level of service and parking available, not on the built environment
- Catchment areas are larger in more suburban areas and where transit station parking is ample

Does neighborhood land-use mix and urban design influence the demand for transit?

- The types and mix of land uses influence the demand for transit as well as the use of nonmotorized modes.

- Residents of “traditional” neighborhoods are more likely to use nonautomotive modes for nonwork trips than residents of “suburban” neighborhoods.
- It is difficult to sort out the effects of land use mix and urban design because they are strongly correlated with density.

How Transit Influences Land Use

What public policies and institutions are needed for transit-supportive development to occur near transit stations?

- Regions with successful transit focused development have the following characteristics:
 - Commitment to a regional vision of high-capacity transit connections between regional centers or in development corridors;
 - Strong, respected institutions that people trust to deliver services;
 - Political cultures that value transit;
 - High-quality transit service that attracts riders;
 - Regional growth that provides the development to channel to station areas;
 - Transit stations in areas where the market supports development;
 - Regional policies that focus growth in transit corridors and limit it elsewhere;
 - Station area policies and programs to support private-sector investments and transit-friendly development; and
 - Long-term commitment.

Source: TCRP Report 16, 1996.

how, we build housing and offices. Finally, we find it easy to conclude that many transportation and land-use problems in today's metropolitan areas would be less acute if public agencies had chosen different land-use and transportation plans, policies, and investments 30 to 50 years ago.

CHAPTER 5 NOTES

1. Local land-development policy—zoning, subdivision codes, and building codes—is ubiquitous. In Chapter 2 we explained why local government regulation of land development has, over time, become so common. Zoning protects households against uninsurable risks. It takes advantage of low-cost auto travel by separating potentially incompatible uses. A variety of land-development policies, such as minimum parking requirements and street standards for subdivisions, aid travel by automobile. Large-lot zoning and tough building codes have been used to prevent free-riding on local public services.
2. Metropolitan areas consist of entire counties. In some states and regions of the country, counties by default are divided into six-mile townships that provide general local public services. In other states or regions, the county supplies general services to all unincorporated areas.
3. Most economists agree that addressing inequities with redistributive policies works better at higher levels of government: state or federal. Among the reasons (and relevant to the discussion in this chapter) is that in a mobile society people can move to places where redistributive policies will make them better off. If a local government were to implement, for example, a significant income tax to build public housing to give to the homeless, the policy will encourage more homeless people to move to the city and more high-income people to move to another city. That is not a comment on the need for redistributive policies, but on the need to implement them at a higher level of government with greater geographic coverage.
4. The classics are Coase (1937; 1960); many more recent expansions of his ideas exist (see, for example, Williamson (1985)).
5. The demand for travel is primarily an intermediate demand: the real demand is to be in some place to conduct some activity. Thus, if land-use planning can find ways to make desirable origins and destinations closer and more connected, transportation can operate more efficiently.
6. Deed restrictions are common, but land development in Houston represents an extreme case in that standard zoning does not underlie restrictive covenants: private covenants replace public zoning. See Berry (2001) for description and analysis.
7. Those design concepts were pioneered in the 1980s and 1990s by, for example, Andres Duany and Elizabeth Plater-Zyberk. Perhaps ironic is that some of the earliest developments with what have now become conventional suburban characteristics (e.g., minimum lot sizes, prohibition of commercial uses, and curvilinear streets) were inspired by the disamenities in many pre-auto era urban neighborhoods (e.g., Llewellyn Park, New Jersey, and Riverside, Illinois). New Urbanism demonstrates perhaps that the problem is less from density and use, and more from uncontrolled externalities.
8. Large developments rarely get off so easy in practice. A common concern is that a relatively large development will have significant impacts on local infrastructure, especially the local street system. Large developers often negotiate off-site improvements, usually to streets, to obtain development approval.
9. See Fischel (1994) provides an interesting of land-use externalities and their control in the context of Foster City.
10. As we describe in more detail in Chapter 6, eliminating congestion on unpriced roads would be tough in densely developed areas even with strict control over building and roadway design. Alternative modes of high enough quality and low enough costs, combined with control over land development, could do the trick.
11. Cervero and Kockelmann (1997) were the first to formally discuss these dimensions as a framework to analyze the built environment's effect on travel behavior. An additional neighborhood, station-level dimension we use is Distance. The fifth dimension, related to Jobs/Housing Balance, is a measure of "regional accessibility" between home and important Destinations. This concept was first published by Ewing and Cervero (2001).
12. Sometimes "designation" is used because "diversity" expresses the land-use designation (in a plan or zoning ordinance) that determines the mixing of uses (horizontally and vertically) allowed or required.

13. Please see the subject index of PAS Reports at www.planning.org. With this report, there are now 547 reports in the series, which began in 1949, so readers can get both historical and contemporary perspectives on many, many land-use policy issues.
14. Strong agricultural zoning has similar effects.
15. Even with a sophisticated Geographic Information System, planners will probably still lack the information they need to draw an efficient UGB. The information simply is unavailable to determine the limits of urban development in an environment of efficient prices.
16. For those interested in a detailed discussion of impact fees, see Nelson 1988b.
17. For those interested in a detailed discussion of concurrency and APFOs, see S. Mark White (1996).
18. Concurrency policies exist in New York, Washington State, Florida, and Maryland. In 1972, a New York court held in *Ramapo* that governments can use timing and sequencing of infrastructure as a tool to regulate growth. Washington State has mandated concurrency policies since the Growth Management Act of 1990. Florida adopted a statewide growth management law in 1985 that requires all local governments to adopt and implement concurrency requirements for roads, sewers, parks, schools, and other public services. In 1978, Maryland authorized (but did not require) local governments to adopt ordinances requiring “adequate public facilities” as a condition of development.
19. In downtown Portland, Oregon, the city allows a maximum of 0.25–0.75 parking spaces per residential unit and 1.0 parking spaces for every 2,500–4,500 square feet of nonresidential space.
20. Lynch defines the five elements of urban “image” people use to characterize the urban environment: paths, nodes, edges, districts, and landmarks.
21. For just one example, Handy (2005) concludes that Ewing and Cervero (2001) showed a statistically significant, but limited, link between built environment character and travel behavior. A 10 percent increase in local (neighborhood) density, for example, correlated with only a 0.5 percent decline in vehicle trips and VMT. The highest elasticity was for regional accessibility (what we refer to as destination): a 10 percent increase in regional accessibility correlated with a 2 percent decline in VMT. When these neighborhood dimensions are combined with destinations, the built environment has its greatest influence on lowering VMT. Ewing and Cervero were looking at the affects of these policies in the absence of proper pricing of auto use, or parking, as discussed in Chapter 4. They recognize proper pricing would probably cause changes in these four land-use dimensions and have a greater effect on lowering auto use.

Transportation Policy Evaluation

Organization: Chapter 5 set the land-use context: this chapter focuses on transportation policies. It starts with a section that explains why we organized those policies into three categories, and then proceeds with a section on each:

1. *Policies to improve transportation infrastructure.* Adding more supply has been the obvious first response to the problem of congested or unsafe surface transportation in urban areas. For highways, the choices are primarily (1) new roads or more lanes, and (2) getting more out of existing capacity. As this gets more expensive, more attention is paid to adding capacity for modes alternative to the car: trains, bikes, and legs.
2. *Policies to improve the use of vehicles.* The lion's share of the vehicles travelers use to get around are purchased and owned privately. Nevertheless, many are not used with high efficiency; the passenger and back seats of cars often go unoccupied. Publicly owned vehicles, such as buses and train cars, are often also underused. This section investigates policies to get more person-trips out of fewer vehicle-trips.
3. *Policies to mitigate traffic externalities.* When streets get "improved" to facilitate the flow of vehicles, those improvements have negative effects on the social and economic environment along those streets. More and faster traffic on a street increases noise, inconvenience to pedestrians and bicyclists, accident risk, and, ironically, congestion. Economic theory and desires for an integration of land-use and livability considerations with transportation planning require that those effects be dealt with, primarily through better design.

A short section at the end of the chapter provides tables that summarize the characteristics of many of the policies.

Basics: Policies, investments, and other actions in all three categories are essential for an effective and balanced transportation system.

- New roads or more lanes are part of the package. New roads will obviously be necessary in expanding urban areas. New lanes might be cost-effective in areas of heavy congestion and where right-of-way can accommodate those lanes or is not abutted by dense development. Such new lanes can make sense as HOV lanes.
- Getting more out of existing capacity clearly makes sense. Transportation System Management (TSM), including Intelligent Transportation Systems (ITS), does that. Among the many possibilities for policy and investment:
 - Better engineering to fix turns that are too tight, bottlenecks, or lanes that are too narrow.
 - Better coordination of traffic flow (e.g., signalization; controlling flow with ramp metering).
 - Better management of incidents and accidents: possibly the most cost-effective thing that can be done to get more out of the existing system.

- Better information for travelers: ITS solutions (e.g., 511 numbers), and more.
- Education and enforcement. The bad driving habits of a small percentage of drivers can cause a lot of accidents and congestion for the majority.
- TSM by itself cannot bring congestion to efficient levels. TSM effectively works like adding lanes; it increases the capacity of the highway. But triple convergence almost guarantees that highway space remains scarce.
- Thus, Transportation Demand Management (TDM) is required for more efficient urban transportation.
 - The ideal policy is pricing road use and parking by time and location: congestion pricing. Achieving that ideal will happen in some metropolitan areas, but it will take 10 to 20 years. In the meantime, other less direct mechanisms can influence trip costs in ways that improve efficiency: for example, parking cash-out, priority transit, and HOV lanes.
 - Yet other policies ration demand and influence behavior through regulation (e.g., trip-reduction ordinances, or limits on parking spaces).
- Local mitigation techniques, such as traffic calming, are also probably essential because pricing access to the local street system is very difficult, and pricing of highways (or other effective highway demand management) will encourage drivers to look for unpriced alternatives, such as travel on arterial streets.
- In general, discouraging highway vehicle trips will increase the demand for substitutes—high-occupancy vehicles (HOVs), mass transit, bicycle, and walking—so policy must be coordinated: if prices or regulations encourage a shift in the demand for auto travel, they will work better if other policies are simultaneously encouraging a supply-side response in other modes of travel.

Shortcuts: Why are you reading this report? Right, because you care about solutions to urban transportation problems. If you are a technician, read this chapter. If you are a policy maker with limited time looking for an overview, Chapter 7 packages these policies: you could skip the details here and jump there.

A FRAMEWORK FOR THINKING ABOUT TRANSPORTATION POLICIES

The fundamental principle of transportation planning, evaluation, and project selection should be to invest in transportation projects for the *transportation* benefits that they provide. The reason is that if there are other, more important objectives than transportation (e.g., land use, economic development) they will (1) mainly be achieved through the transportation impacts and be double-counts of those impacts, and (2) probably be better achieved by having agencies specializing in those objectives aiming policies at them directly. The benefits of transportation projects and policy accrue directly and primarily in two ways:

- Better access. That could mean more destinations, but practically, in urban areas, it means (1) that the transportation policy reduces some or various measures of travel time from what they would have been in the absence of the policy, and (2) that, given that the major contributor to measures of urban travel time is urban congestion, a transportation policy should be aimed at reducing congestion—especially peak-period congestion—from what it would have been in the absence of the policy. The objective here should focus on travel time for *people and freight*, which is not always the same as travel time for *vehicles*.
- Less accident damage (property and personal).

Any actions that attempt to achieve those two benefits should simultaneously pursue a third and conflicting one:

- Controlling (to an often unspecified politically acceptable level) negative environmental, land-use, social, and economic externalities. We include in that constraint the ones about fairness and the distribution of impacts (also referred to as “environmental justice”).

Conventional transportation policy already focuses on these first two objectives. Urban freeways, for example, with their wide lanes, limited access points, and gentle curves allow, *in uncongested conditions*, relatively high-speed and safe travel.

Freeways, like all transportation projects, have effects that go beyond transportation: that point is clearly a theme of this report. Most of those other impacts (especially land-development impacts) are, in an economic sense, just another way of describing what is fundamentally an access/travel-time benefit. Reducing the amount or variability of travel time improves access, which makes households and business put a greater value on land that enjoys those benefits, which raises land values, which intensifies land development. The increase in land values helps put a dollar value on the transportation benefits. Focusing on travel times and safety, in principle, covers most of the benefits of surface transportation improvements or policies.

But the travel that transportation systems facilitate generates external costs. Air, noise, and visual pollution impose external costs on nontravelers. Congestion is another big one: highway congestion reduces the transportation benefits the highways were specifically designed to provide. The high costs associated with these externalities motivate policy response.

In short, our point: transportation policies aim primarily at reducing the amount and variability of travel time and accident cost, subject to the constraints of dealing with the external costs that those policies impose.

The conventional way to organize transportation policies is under two broad headings: policies that affect supply (capacity), and policies that affect demand (behavior). Table 6-1 shows that organization. It further divides the policies, as we did for land-use policies in Table 5-4 on page 149, as to whether they are primarily regulatory (requirements) or market-based (incentives).

TABLE 6-1. SUPPLY-SIDE AND DEMAND-SIDE TRANSPORTATION POLICIES

	SUPPLY SIDE	DEMAND SIDE
PRIMARYLY REGULATORY	Building more roads or expanding existing ones Building more transit facilities and increasing service and amenities in existing transit systems Improving highway maintenance Adding roving response teams to remove accidents Traffic management centers ITS mechanisms for speeding traffic flows Deregulating public transit activities Upgrading existing city streets Staggering work hours for more workers Developing means of transit feasible in low-density areas Building special roads for trucks only	Prohibiting certain license numbers from driving on specific days Changing federal work laws that discourage people from working at home Ramp metering on expressways Encouraging transportation management associations Encouraging more people to work at home Keeping minimum residential densities higher Clustering high-density housing around transit stops Limiting growth and development in local communities Improving the jobs/housing balance Using traffic-calming devices to slow flows Concentrating jobs in a few suburban clusters Making some lanes HOV lanes
PRIMARYLY MARKET-ORIENTED	Converting free HOV lanes to HOT lanes	Road pricing with tolls set to raise peak-hour flows Commuting allowance for employees Charging high taxes on gasoline Charging high taxes on parking during peak hours Eliminating tax deductibility for employers for providing free parking Increasing automobile license fees "Cashing-out" free parking provided by employers

Source: Downs 2004a, 78.

There is a lot to recommend that type of organization. Engineers started with supply policies, focusing on new supply. As it has become harder and more expensive to build new roads and rails in developed areas, they have shifted to transportation system management. As supply solutions came up short and as planners and economists got more involved, demand-side solutions get more attention. As the land use/transportation connection gets more attention, the emphasis on design and mitigating the negative effects of highway projects gets more attention also. The sidebar illustrates the shift that has occurred.

Another advantage of the supply/demand organization for us as authors is that was the one we had been using up to two days before the deadline for delivery of the final version of this chapter to APA for typesetting.

But categorizing policies as supply-side or demand-side has some problems. Chief among them is that demand and supply are not independent: those forces operate on each other, iterate, and ultimately equilibrate.

To make that less abstract, consider two examples:

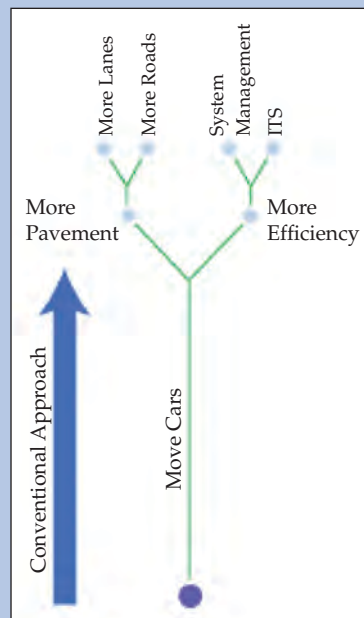
- Access management. Access management is supply-side. It makes more efficient use of capacity by changing highway design so that turbulence is reduced. It usually means reducing curb cuts and creating medians: fewer left-turning movements off the highway and fewer entries onto the highway. Ramp metering is a variation of access. But access management

EXPANDING THE TRANSPORTATION POLICY PALETTE

By Troy Russ, AICP, Principal, Glattig Jackson Kercher Anglin, Inc.

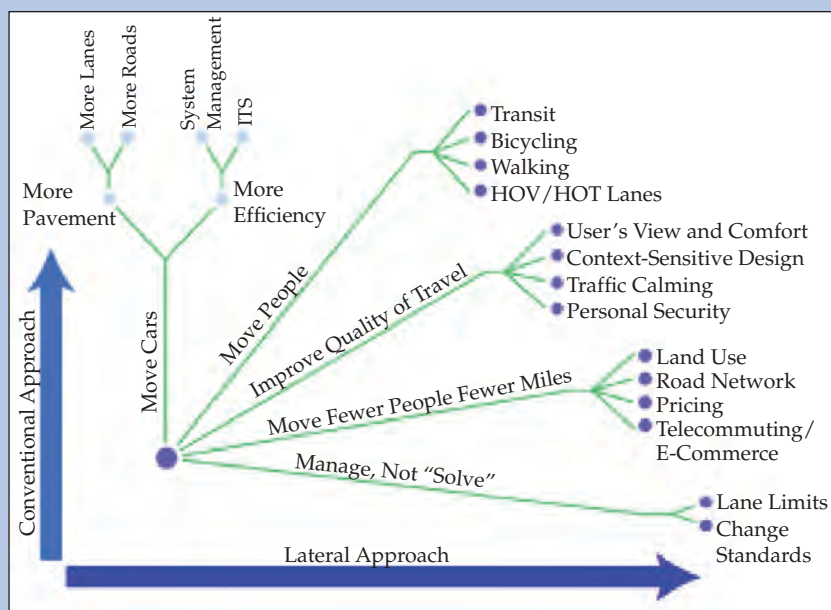
Conventional Approach to Transportation Planning

Since the 1950s, transportation professionals have been directed to move more vehicles. Thus, they focused making more, bigger, or more efficient roads. Their efforts coincided with growth in vehicle miles traveled (VMT) greater than population growth, declining population densities in cities, and an obesity epidemic partially caused by a poor walking environment.



A Balanced Approach to Transportation Planning

An alternative transportation approach seeks to make streets, highways, and transit systems more livable by striking a new balance between the goals of moving traffic and building livable communities, and by supporting broader community land use goals. Such an approach looks more at the movement of people (not just vehicles), the quality of a trip, and land-use actions to reduce transportation problems.



is also demand-side: it influences behavior. It changes where and how people can drive and how land-uses develop.

- HOV lanes. Adding a lane to a highway and restricting it to HOVs can clearly be thought of as supply-side. But the motivation for doing so might be demand-side: the faster travel in the HOV lane encourages some single-occupant-vehicle (SOV) drivers into carpools and buses, reducing the demand for SOV travel (at least temporarily) in the unrestricted lanes.

For this chapter, we use a different organization. It derives from a simple observation: travel requires paths and vehicles:

- Paths for urban surface transportation are roads, parking lots and structures, rails, bike lanes, and sidewalks.
- Vehicles are, correspondingly, cars, motorcycles, trucks, and buses; trolleys and light- and heavy-rail cars; bicycles, skateboards, roller blades, strollers, wheelchairs; and legs.¹

Transportation policies are about improving the performance (management) of a transportation system composed of paths and vehicles. Paths and vehicles deliver the ultimate transportation products: mobility and access. Transportation policies deliver access by aiming for (1) more, or more efficient use of, paths; or (2) more, or more efficient use of, vehicles.

That framework helped us clarify issues we found muddy and allowed us to take a razor to the long list of things we could talk about in this chapter:

- Paths are mostly provided publicly. The reasons are not an arbitrary historical artifact: transportation systems cut over a lot of property and political boundaries, and require big risk and investment. The paths that are provided privately are a small part of the regional transportation system, have little effect in the larger scheme of regional transportation, or are subject to heavy public regulation (e.g., privately built and operated toll roads). Thus, public policy is very concerned with paths, which are almost exclusively public.
- Vehicles are primarily provided privately. Most of the cars, vans, trucks, bikes, wheelchairs, and legs are privately owned and operated. The public sector may address standards for design and operation of those vehicles (that's public policy), but it does not provide or finance them, and it allows a lot of consumer choice. Urban passenger-rail cars and most buses are provided publicly (primarily), but there is no strong economic reason that they have to be.
- Getting more motor vehicles on the road does *not* have a logical connection to the broad goals of reducing travel time or accidents. We consider it no further as a potential public policy. Using vehicles more efficiently (HOV) or increasing alternatives (legs and bikes) is logically related to both of those goals.
- Accident prevention is in practice handled primarily as a constraint on adding or fixing paths or HOVs so that they can handle more people. The amount of transportation money that gets spent on improvements that are exclusively or primarily about traffic safety is relatively small.² Thus, we focus our discussion on benefits relating to congestion relief.
- The negative derivative effects of transportation improvements (externalities) can be caused by the construction of paths or the operation of vehicles. People are aware of these problems and use state agencies, local governments, and interest groups to try to mitigate them by changing policy—by getting governments to take action.

Given all these points, we organize the policy discussion in this chapter as follows:

Transportation policies are about improving the performance (management) of a transportation system composed of paths and vehicles. Paths and vehicles deliver the ultimate transportation products: mobility and access.

- **Policies affecting paths (i.e., the transportation infrastructure that vehicles use).** Most “path policies” involve increasing transportation capacity: more lane miles, more throughput per lane, more parking, more rail transit, more transit ridership, more bike paths, and so on. There are two major subheadings: more paths and more efficient paths.
- **Policies affecting vehicles.** Aside from public buses and rail cars, most vehicles are privately supplied. The public policy emphasis is on encouraging more use of legs and bikes relative to motor vehicles, and more efficient use of motor vehicles.
- **Policies to reduce negative effects on community livability.** Local streets require trading quick and easy travel by motor vehicle for neighborhood environmental amenity. This improvement to amenity often benefits travel by bike or foot.

POLICIES TO IMPROVE TRANSPORTATION INFRASTRUCTURE

Surface transportation policy focuses on providing the infrastructure needed for swift, safe, and reliable travel. Reducing the growth of traffic congestion on urban freeways and arterial streets (i.e., reducing deterioration of Level of Service) has become the main object of that policy.

Conventional policy views this as a problem of capacity, so the objective has been to build more and use what we have more efficiently, which reflects the organization of this section.

Though we are talking here about highways, the notion of trip components and times is critical in understanding ridership on transit. Consider that, for a park-and-ride trip by transit with a transfer, the components are: walk, drive, park, walk, wait, ride, wait for transfer, ride, walk.

Increasing infrastructure capacity

Physically increasing highway capacity means more pavement. This comes in the form of new lanes on existing highways or new highways.

The increasing cost of providing more pavement has motivated attention to other types of paths. In particular, an increasing number of growing metropolitan areas are building, or thinking about building, rail transit systems. The most common is light-rail. More attention is also being paid to improving bike and foot paths.

Conventional transportation policy focuses on improving limited access highways and major arterials. That focus could be broader. Taylor (2002) gives a simple example that makes the point. He divides an SOV commute trip into eight separate components related to vehicle (walk to car; drive to arterial) and path (collector, arterial, freeway, parking structure), and shows time spent on each path. For his hypothetical but reasonable trip, the length is about 11 miles and the total time (door to door) is 36 minutes. But only 40 percent of that time is spent on the congested freeway. Almost another 40 percent is spent on collectors and arterials getting to and from the congested freeway. Walking takes 17 percent of the time, almost all of it getting from the parking lot or structure to the final space at the place of work.

The example makes several points: (1) if the main benefit is time savings, policy and investment ought to focus on where those savings are possible, (2) the returns to investment in congestion relief on freeways will be smaller than most people think because travel time on freeways accounts for less than half the travel time for a typical trip, and that percentage will drop if congestion is relieved, and (3) many paths and modes are important to even a single trip. Though we are talking here about highways, the notion of trip components and times is critical in understanding ridership on transit. Consider that, for a park-and-ride trip by transit with a transfer, the components are: walk, drive, park, walk, wait, ride, wait for transfer, ride, walk.

In this section we analyze and evaluate various policies for increasing the capacity of surface transportation infrastructure. While each varies in its

effects overall, none will cure the problem of traffic congestion in growing areas. The reason is triple convergence: new capacity reduces travel times in the short run, which travelers take advantage of in the long run.

More lanes for congested highways

Transportation authorities commonly respond to traffic congestion on highways and arterial streets by expanding, usually by adding lanes. That response makes sense only to the extent that congestion is a good indicator of the need for the additional capacity. Congestion is a good indicator in a relative sense; improvements to congested urban freeways are more likely to yield larger net benefits than similar improvements to less-congested highways. But congestion is not necessarily a good indicator in an absolute sense: the proportion of trips of relatively low value varies across similarly congested highways.

Different from the issue of net benefits are the issues of triple convergence and latent demand, but they are frequently addressed as if they were the same thing. Cervero (2003), Downs (2004a), and Chapter 4 and Appendix C point out that new travelers and new development take advantage of new capacity, which can accelerate the rate of traffic growth until congestion returns, perhaps in only a year or two. The expensive new capacity appears at that point to provide no benefit to a commuter who used the corridor before and after the improvement; travel to work is as slow and unreliable as it was before the improvement.

Overlooked in this lamentation is that there are now more commuters using the facility, and some are living in locations and housing that they find better. These new travelers gain value from the new facility. It is possible that the added capacity was a good investment even though it was quickly used up. It is also possible that the congestion relief and added capacity for new trips was largely offset by increases in congestion during construction (see Appendix D for an overview of how to evaluate transportation improvements in terms of full benefits and costs). In the absence of tolls, however, calculating net benefits is difficult and uncertain.

The details in Chapter 4 about congestion, underpricing, and external costs cover most of the issues relevant to an evaluation of new lanes as a policy. Transportation investment decisions of this type would be improved if the revenues from congestion tolls were there to provide information about user values. In the absence of congestion tolls, evaluations should consider, hypothetically, what people would be willing to pay in tolls *if* the new capacity were to be tolled. The answer might often be: not much, and not anywhere near the amount it would take to pay the debt service on the amortized cost of the new capacity.

Why do highway expansions that probably could not pass a cost-benefit test get built? The main reasons: political pressure; the difficulties in measuring costs and benefits; an impatience with having to deal with the real complexities of the type of thinking (never mind the numbers) that benefit-cost analysis introduces; and the availability of funding. Large numbers of people traveling every day on congested highways means a lot of frustration, and that frustration rains on elected officials: (1) somebody needs to do something, (2) it's too crowded, (3) build more roads, (4) now, (5) or else. . . . And state DOTs have fuel-tax revenues to spend (though never enough to cover all of the projects on their lists). New lanes, highways, and interchanges are going to get built.

If the capacity added is large, if the area to which it is added is not very congested and not a likely location for significant new development, and if regional growth is slow, triple convergence may be slow: excess capacity may remain. For all the reasons explained in earlier chapters, that excess

Large numbers of people traveling every day on congested highways means a lot of frustration, and that frustration rains on elected officials: (1) somebody needs to do something, (2) it's too crowded, (3) build more roads, (4) now, (5) or else.

capacity (better access; reduced travel time) is an inducement to development. Cervero (2003) found that the quality, quantity, and location of new development in the vicinity of a new road project are both a result of the road project and a determinant of how quickly growth in traffic volumes will accelerate the degradation of the new capacity. Road improvements and faster travel times encourage building along new highway corridors, mostly by redistributing regional economic activity away from other areas in the region (as opposed to by generating new economic growth for the region) (Boarnet 1995, 1997).

New capacity will always be part—and probably always the biggest single part—of any integrated program for management of urban surface transportation systems. As urban areas grow, new road capacity (especially for growth at the urban edge) and fixes to old capacity will be needed. Our point is that there are good reasons to consider shifts in emphasis toward other policies for managing the transportation system.

More highways and bypasses

As an urban area grows, both limited and unlimited access highways can evolve from serving mainly through traffic to serving local traffic. Shopping malls near freeway interchanges and poor system management along commercial strips (too many driveways and unsynchronized stoplights) reduce traffic flow, reliability, and safety. Vehicular movements inhibit transit, cycling, and walking. A solution is to build a bypass, ostensibly to serve through traffic.

The most common resolution is a suburban (often exurban) bypass. The advantage of a bypass from a state DOT's perspective is its *relatively* low (though still high) construction costs. Adding lanes or new arterial streets and highways in urban corridors is expensive. Trucking interests and through travelers support the bypass. Many metropolitan residents support the bypass for the access it provides to lower-cost residential areas with a preferred bundle of public facilities, services, and prices (taxes). And many exurban land owners support the bypass for the windfall increase in property values the new access generates.

But bypasses are controversial. The controversy arises from the potential land-use impacts of bypasses. A bypass built in the urban fringe encourages decentralized development (and reduces pressure for centralized development and infill) by improving access to developable land. Much of this development consists of low-density residential development. A bypass can also hurt businesses in the bypassed corridor, especially if conventional land-use policies guide the development of land near the new bypass.

Retailers follow their customers and employers follow their workers to areas near the new bypass. Edwards (1991) reports that the central business districts (CBDs) of many small and medium-size cities, for example, suffer decay as businesses move to more accessible locations along bypass highways or near freeway interchanges. Thus, in addition to having to deal with the problems of growth and change along the bypass, the local jurisdiction must deal with the problems associated with the decay of the bypassed cluster.

A good example of these effects are found along a portion of Michigan state highway 11, built originally as a bypass south of the Grand Rapids, Michigan CBD. Over several decades, the unlimited-access bypass has become a local shopping magnet. Regional shopping malls anchor each end of the nearly 10-mile-long retail strip. Big-box retail and strip development fill in between and beyond the malls. Bypass capacity has expanded over time to three lanes in each direction, plus a turning lane. A large commercial and industrial area has developed south of the bypass. Peak congestion on the former bypass is sufficiently high that a new freeway bypass is nearing completion on exurban land several miles to the south.

Bypasses are controversial. The controversy arises from the potential land-use impacts of bypasses

The disadvantages get a lot of press. The now seven-lane arterial is congested, ugly, and virtually unusable by people traveling by nonauto modes. The congestion prevents the corridor from functioning as a bypass for most of the day. Parking lots cover most of the land near the street, unoccupied much of the time because of parking requirements or desires. As the retail grew along the bypass, it fell to nearly nothing in the bypassed CBD. The costs of land development along the bypass are high.

Less salient are the advantages. Remarkable amounts of trade occur in this 10-mile strip. Though ugly and long, from a regional perspective the retail is, in fact, clustered along this thin linear corridor. The corridor itself is congested, but access to it is relatively easy all along its length. Though arguably less than optimal, the large, congested linear cluster serves relatively well a variety of important local economic functions. And the retail concentrated there discourages strip development elsewhere. The benefits of land development along the bypass are also high.

Another potential disadvantage is the “need” for the new freeway bypass farther to the south. This new bypass, though limited access, could have many of the impacts of the old one. Does it make sense simply to keep building bypasses?

The answer is not an unequivocal “no.” Multiple layers of bypasses in relatively large metropolitan areas can evolve to form a grid of high-speed, high-volume urban arterials. Large commercial subcenters form at especially accessible locations, such as near freeway intersections. These subcenters provide substantial economic efficiencies. Though urban freeway systems frequently suffer congestion, congestion pricing could, theoretically, treat that problem and encourage more intensive land development.

Nevertheless, the disadvantages of bypasses have led to consideration of alternatives.

One alternative to a conventional suburban bypass is to build express lanes—lanes with no access to the adjacent development—in existing corridors. Constrained by the harbor, two large lakes, and steep slopes, Washington state officials added express lanes to serve drivers traveling through Seattle on Interstate 5. These lanes are in the same right-of-way as the other lanes that provide access to the intensely developed parts of Seattle. The construction costs of this option are high relative to those of standard bypasses, an option that was simply unavailable in Seattle.

A perceived advantage of the express lanes is their relatively low impact on land use. The express lanes do, however, speed commuting into and out of the downtown, with several likely land-use effects: (1) more low-density residential development farther from the CBD, but near the existing corridor (to the north and south in the case of Seattle), and (2) more development in the CBD. The result is more and longer commutes on the express lanes, which contributes to congestion (that pesky triple convergence again). The additional capacity will not in the long run eliminate congestion; some form of rationing is also required. But the express lanes effectively protect the CBD as a commercial center by not just maintaining, but improving, access.

The express-lane option requires care in design. Rather than bypassing the entire city of Medford in southern Oregon, for example, Interstate 5 was elevated to pass directly above the CBD (again, topography and development limited the feasibility of the construction of a low-cost suburban bypass). Rather than building a relatively expensive interchange to provide direct access to the CBD, lower-cost interchanges were constructed on the northern and southern edges of town. The bulk of new commercial development since then has occurred near those interchanges. Two new clusters have developed while the former CBD, located between them, has deteriorated markedly, an effect many citizens did not envision or desire.



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Interstate 5 in Seattle. Washington state officials added express lanes to serve drivers traveling through the city in the same right-of-way as the other lanes. Where topography restricts building bypasses, the high costs may still provide great benefits.

A second alternative is to build an exurban bypass and use restrictions on land development to prevent sprawl. Though that option is not often exercised, it can work. Interstate 205 in the metropolitan area of Portland, Oregon, and I-280 connecting San Francisco with San Jose provide anecdotal evidence that this policy can work as long as decision makers can tolerate it.³ After the I-205 bypass was built, the expected increases in development (including the inevitable regional mall at an interchange) occurred in the urban area it traverses. But along the segment of the highway to the south of the metropolitan area, such development has not occurred: I-205 still looks like a parkway. The reason: lack of full urban services or the zoning that would allow them to be built (specifically, in Oregon, the areas in question are still outside the metropolitan urban growth boundary) has prevented the development of land with otherwise good access.

The land-use-regulation option is not often exercised. Many of the owners of land near a bypass will oppose restrictions on development because the access provided by the bypass increases land values. These capital gains are windfalls because the landowners typically pay little of the cost of building the bypass.⁴ Windfalls or not, they motivate aggressive lobbying. That lobbying often succeeds.

An option for the government is to purchase land near interchanges when it purchases the highway rights-of-way. That strategy, however, raises project costs and raises issues about property rights: should the land be appraised for its value as vacant land (e.g., farmland) without access, or as commercial land with access? At this writing (2007), issues of property rights, takings, and givings are a hot topic in many areas of the country.⁵ There are likely to be strong political forces working against limiting land-use impacts near a bypass.

Sometimes, however, politics and topography work as planners hope. Cannon Beach, Oregon, grew along a section of Highway 101 built near a scenic stretch of Pacific Ocean beach. As the town grew, so did congestion on the highway. The congestion in town motivated construction of a bypass in the 1950s farther from the beach. In the short run, a lack of both demand and good buildable land (much of the land along the new bypass was wetland) kept development in check. In the longer run, municipal ownership of large amounts of the property, restrictive zoning supported by the Oregon land-use program, and the happy convergence of aesthetics and tourist-related business interest have prevented development along the bypass, while the downtown area continues to thrive.

A third alternative is to build a bypass with no interchanges. The bypass then improves travel conditions for through traffic without improving access to exurban areas. Though simple in concept, the solution has the same political problems as land-use restrictions. Being near a freeway or rail line provides no access benefits to properties that cannot access it: in fact, such facilities would diminish land value with noise, emissions, visual blight, and *decreased* access (because they are barriers to local travel). The owners of land near the bypass have incentive to lobby for interchanges. The technical debate turns on the question of efficiency: if we build this expensive bypass for through traffic, are we willing to forego the economic advantages of access to cheap undeveloped land to protect and stimulate the economic advantages of the downtown and other existing subcenters?

What can we conclude from our analysis?

- Expensive construction in existing corridors may in the long run provide more net benefits than are revealed by comparing construction-cost estimates. The objective should be to maximize net long-run benefits, not minimize short-run costs.
- Bypasses even with their consequent land-development impacts may be desirable if they truly serve their intended function of relieving downtown



Jeremy Edwards, istockphoto.com

Cannon Beach, Oregon presents a case study of the benefits of restricting land development near bypass interchanges. In this case, topography, municipal land purchases, and land-use restrictions all played roles. An alignment that would have brought the bypass much nearer to the beach was ultimately not chosen, despite and because of the beach's scenic attributes.

areas of traffic that really does not want to be there. That congestion relief may facilitate other streetscape improvements that improve livability.

- Having the best of all worlds—cheap construction, free-flowing traffic, and limited impact on land development—does not come easily. It requires coordinated policy: limiting access to the highway (preferably with congestion tolls), and binding, long-term restrictions on land development.

More parking

Chapters 4 and 5 discuss parking in detail. We add a just a few points here to put parking in context.

On the transportation side of the equation, flying down a path with no ability to land does not often make for a useful trip: a stopping point at the end of a path—parking—is essential for all wheeled travel. That simple logic suggests that expanding parking capacity where it is constrained can have the same effects on congestion that expanding lane capacity can have.

Since travel depends on cost, the direct and indirect (time) cost of parking is quite relevant to the demand for trips. The logic should be clear by now: (1) the effects of adding parking capacity are similar to those of adding lanes, as are the effects of underpricing it, and (2) since flying without landing has little value for most travel, the pricing of parking can be used as a demand-side policy to ration scarce highway capacity. These points are expanded a little later in this chapter, and greatly in Shoup (2005a).

More priority mass transit

An alternative to building more highway capacity is to build mass transit capacity on separate rights-of-way. Examples include commuter rail, heavy rail (subways, els, and monorails), light rail transit (LRT), and bus rapid transit (BRT). Advantages of grade separation are that these HOVs do not get caught up in traffic congestion, and they do not contribute to it directly by taking up scarce highway space.

The disadvantage of grade separation is its cost. Construction of new commuter rail lines, for example, requires large upfront capital expenditure, and fares rarely cover half of operating costs.

The arguments in support of subsidies are numerous, but three stand out. The first is to provide mobility for the poor, which mass transit accomplishes, though probably not cost-effectively. Recent trends toward the suburbanization of lower-income households increases the challenge. The second is to provide highway congestion relief. The idea is that good performance characteristics will encourage some people off congested highways. The third is to change the pattern of future urban development to one that is oriented toward transit, with all the purported benefits.

Can subsidies to rail transit positively impact highway congestion? The travel demand elasticities reported in Appendix C provide some insight. The elasticity of transit ridership with respect to transit fares is low: a 10 percent decrease in fares almost everywhere results in a less-than-10 percent increase in transit ridership. Commuter ridership in all but the most densely developed U.S. cities is a very small proportion of total commuting. Thus, a big drop in transit fares has only modest proportionate effects on transit ridership and almost negligible effects on highway congestion even in the short run.

The effects of lower fares on congestion in the long run are even smaller. The reason is that the elasticity of peak VMT on congested highways with respect to travel time is very high: triple convergence again. The fundamental problem is that the transit subsidy does not address the cause of inefficiently high traffic congestion: unpriced access to the highway.

What are the effects on land use? Most commuter rail systems are oriented radially, so they improve access to both the CBD and to areas in suburbia



Slobo Mitic, istockphoto.com

Light rail transit (LRT), as shown here in Sacramento, California, operates in a separate right-of-way, which prevents delays by bypassing congestion.

beyond the end of the train lines. Thus, commuting on rail is especially attractive to commuters to the CBD because it gives them the option of living in the urban fringe. They can drive to the suburban train station, then make relatively good use of the half-hour transit ride by preparing for the work day or reading the newspaper. The walk to work in the CBD is often less time consuming than getting into and out of a parking structure.

The effect on land use is similar to that of a new lane in a congested highway. The rail transit line improves access to both the CBD and exurbia, so the intensity of development increases in both places. In exurbia, this pattern means primarily more low-density development, though the train line could support the development of suburban subcenters and higher-density development.

Transit tends to fare better, as it were, at the other end of the line, usually the CBD. Indeed, part of the appeal of light rail systems are the benefits to a CBD ailing from suburbanization. Radially oriented rail improves access to the CBD from suburbia. As usual, that better access translates into development.

Handy (2005) finds that LRT alone does not automatically create higher densities around stations. Other conditions must exist to increase the likelihood of higher-density development:

- A region is experiencing significant growth.
- The light-rail system adds significantly to the accessibility of the locations it serves.
- Land-use regulations (zoning, comprehensive planning, public investments) are conducive to development.
- Public-sector support in the form of capital investments and land-use policies, such as high-density zoning allowances, restrictions on parking, and financial incentives, are in place.

LRT alone does not automatically create higher densities around stations. Other conditions must exist to increase the likelihood of higher-density development.

Landis and Cervero (1999) find that BART had impacts around stations in downtown San Francisco but not around stations in the suburbs. Examples of high-density development around suburban Metro stations are evident, however, in certain counties in the Washington, D.C., metropolitan area, specifically Arlington County, Virginia, and Montgomery County, Maryland. In other words, such development doesn't happen automatically: the form, density, and timing of the development around stations are highly dependent on demand conditions (markets: demographics, land and building prices, density) and how land-use and design tools are applied.

LRT is special when it comes to the CBD. First, it can serve somewhat like a streetcar on congested CBD streets (but with less frequent stops) because the rail lines are laid in the city streets. Second, installing light rail allows and encourages upgrading and updating of CBD infrastructure. Finding the money to upgrade dilapidated infrastructure in many CBDs is a big problem. Light rail is a source of funds. Done well, a light-rail installation can give a CBD a significant facelift whose short-run benefits may be as important as the accessibility it provides. From a development perspective, street-cars (trolleys) may work even better: they are less expensive, stop more frequently, and are perceived by developers to carry a more local clientele that they can more directly design for.

More bike and walking facilities

Bike and walking paths are important—we make the case in the later section of this chapter on externalities. But bike paths are often controversial. The lowest-cost bike path is one built on the shoulders of roadways. Car drivers who are not avid or commuter bicyclists believe that bike lanes take something away from drivers: that they take up space that could be used for

lanes, that the presence of cyclists complicates driving and increases the risk of fatal accidents, and that cyclists don't pay the fuel taxes that pay for roads. For cyclists such lanes are clearly better than the nothing they have typically received, and better than sidewalks, but they lack both safety and amenity.

An alternative is separated bike paths. Bicycle groups have had some success in lobbying for consideration of bike paths in transportation planning. Old rail lines have been converted to high-amenity bike paths. Neighborhood streets, away from arterials, are frequently designated as bikeways (more on this in the last section of the chapter). Separated bike paths are being built along new suburban arterial streets. And efforts are being made to build connected bike paths through new residential developments.

Bike paths in practice serve also as foot paths. Foot paths, either sidewalks or paths separated from streets, have potential to save car trips. The trick, of course, is to get origins close enough to destinations. Conventional zoning and parking requirements often prevent this, putting distance between commercial centers and residential neighborhoods, and requiring a long hike across a dismal parking lot. With higher residential densities and some thought to the design of commercial development, walking trips could become more important, especially for shopping.

The effects on overall urban transportation in the U.S. of these efforts probably are not large in relative terms by any common measure (e.g., person- or vehicle-miles traveled), but there are several arguments for giving these paths more attention:

- They give a subset of travelers a relatively safe, healthy, and enjoyable alternative to driving.
- Positive livability externalities (see a later section of this chapter).
- One has to start somewhere. In Holland bikes are a significant mode for commuting in urban areas. Yes, Holland is small, densely settled, and flat, but there are many metro areas in North America where bikes have potential. This general paradox of short-run/long-run is one inherent in transportation infrastructure: in the short-run—before people, technology, and the built environment adapt—the new transportation infrastructure looks very inefficient; but if they would adapt in the long run, and if that adaptation provides significant transportation benefits, then one cannot get to those benefits without starting to build in the short run.
- They provide a lot of recreational amenity not only for cyclists, but also for walkers, strollers, and skaters.
- Though users of those pathways do not pay for them directly with user fees, there are benefits to the motorized vehicle drivers that are paying for them with fuel taxes: the absence of walkers and cyclists from the roadway at least reduces driver stress and may eventually reduce in measurable ways the rate at which congestion would have otherwise increased (especially in the long run with different land-use designs and patterns for centers and corridors, which require better ability to make more nonmotorized trips for their success). And they don't pollute the environment (for which drivers don't pay).
- From an equity standpoint, it seems hard to justify the priority that a single human in a high-speed steel box typically gets over single humans on two legs or two wheels sans motor.

Improving the efficiency with which infrastructure is used

Building more capacity is expensive and getting more so. Using congestion as a guide, the most needed capacity is in the highest cost areas. Most business owners work relentlessly to get more productivity out of their investment



Car drivers who are not avid or commuter bicyclists believe that bike lanes take something away from drivers: they take up space that could be used for lanes, the presence of cyclists complicates driving, and cyclists don't pay the fuel taxes that pay for roads. On the other hand, cyclists appreciate such lanes, but they lack both safety and amenity.

in production infrastructure (i.e., plant and equipment); so should public transportation authorities. In this subsection we look at methods for getting more vehicle trips out of existing highway and rail transit capacity (we look at methods for getting more person trips out of existing vehicle miles in the next section).

Better highway efficiency: transportation system management (TSM)

The objective of TSM is to get more out of existing capacity (i.e., faster, more reliable, and safer travel). The basic problem is turbulence: even a little disrupts the smooth, linear flow of traffic when congestion is heavy, reducing travel speeds dramatically and encouraging driver behavior that increases the chances of an accident, which further reduces travel speeds and the reliability of travel time.

TSM strategies therefore target a variety of sources of turbulence:

- **Poor highway geometries.** For limited-access highways, this means turns that are too tight, bottlenecks, or lanes that are too narrow. The fixes are primarily engineering ones: lanes are not added, but the roadway is physically reconfigured to reduce or eliminate the source of turbulence. That could include unbending curves, separating flows with barriers, and widening lanes and shoulders. Many of these types of fixes reduce the risk of accident and injury if speed is held constant, but there is now good evidence, which we discuss later in this chapter, that such improvements will, in many cases, increase speed and, with it, the number and severity of accidents. Trade-offs, again.
- **Inefficient traffic coordination or use of system capacity.** The best example is signal timing. If the flow is a jerky go and stop, throughput drops. Optimized signal timing and signals on demand improve flow. Ramp metering and expedited toll payment, discussed in other categories below, could also fit here. Many urban areas temporarily reverse the direction of travel on some lanes on an hourly basis to deal with uneven bidirectional flows (e.g., the Golden Gate Bridge in San Francisco; Rock Creek Parkway in Washington D.C.; Seattle I-5 express lanes).
- **Inefficient merging of vehicles entering a busy road.** This happens at highway onramps. The main strategy involves smoothing the entry of vehicles both through improvement in highway geometries near the intersection (e.g., merging lanes) and ramp metering to help the entering traffic merge smoothly. This also happens on arterials: the problem there is too many vehicles trying to get on the road, and the problems caused when they slow down to get off in too many places (e.g., unprotected left turns into driveways on a busy arterial). The problem has motivated attention to access management.
- **Incomplete information about congestion and alternatives.** If drivers know about traffic problems before they're part of them, they can take steps to avoid them. Radio station traffic helicopters treat this problem. Sensors installed along highways allow transportation managers to monitor highways in real time. Information about highway conditions can be communicated by radio, overhead displays, or through GPS navigation systems. Systems are on the horizon that will recommend routes around the unusual traffic backup.
- **Too many cars on the highway at the same time.** An efficiently designed system used by well-informed drivers can still get overly congested at peak periods. There's capacity to spare off-peak. Policies aimed to spread demand over time include flex-time work schedules and telecommuting. These policies are especially valuable at the morning commute because

Most business owners work relentlessly to get more productivity out of their investment in production infrastructure (i.e., plant and equipment); so should public transportation authorities.

commuters appear more time constrained in the morning than in the evening, and on days with bad weather, major pile-ups, or special events.

- **Traffic incidents (e.g., accidents, breakdowns).** Grenzeback and Woodle (1992) report that accidents and disabled vehicles contribute to about 60 percent of the vehicle hours lost to congestion.⁶ Appendix C provides some information on types of incidents and why they can be so expensive. Even seemingly expensive methods of clearing incidents might be worthwhile given their high costs (avoiding those high costs mean large benefits).
- **Toll plazas requiring cars to slow down or stop to pay tolls.** High-tech electronic tolling systems that read a bar code or detect a “smart card”, as described in Chapter 4, are increasingly being used to collect tolls on highways and bridges.
- **Bad drivers.** Speeding, cutting in, reducing safe driving distance, and shooting at people in other vehicles all disturb smooth traffic flow. TSM measures include more and more sophisticated law enforcement: speed monitoring signs, automatic photographing of vehicles, and ticketing of drivers for running red lights. Given the ubiquity of the system and the cost of labor for conventional methods of enforcement, automated systems have advantages.

We lack the space or will to discuss all the methods transportation agencies use or are experimenting with to reduce turbulence and smooth traffic flow. The Federal Highway Administration’s recent publication, “Traffic Congestion and Reliability: Trends and Advanced Strategies for Congestion Mitigation,” provides details and direction for further reading.⁷

More useful is an evaluation of TSM efforts as congestion mitigation strategies. Our conclusions:

- Many TSM strategies make economic sense. Most TSM strategies are analogous to the strategies private business owners use to “drive costs out of the system.” Thinking in terms of full costs might change obsolescent practices. When a fender bender occurs on a congested freeway, should drivers leave cars where they are (in travel lanes) until the police come? That lost time for so many other travelers is expensive.⁸ The lack of competition among transportation system suppliers (compared to that in many industries) has probably allowed too little emphasis on TSM relative to increases in capacity.
- Incident management is a particularly promising technique for driving costs out of the system. Downs (2004a, 99) cites multiple evaluations showing that the benefits of congestion relief and reduced accidents exceed the cost of Traffic Management Centers.⁹ Such centers coordinate response activities across multiple agencies, monitor the transportation flow (increasingly, electronically), dispatch vehicles to respond to incidents, divert (with information or barriers) traffic to alternative routes, and collect information about incidents that allows improvements in response.
- Improving efficiency requires more than increasing the numbers of vehicle trips. The reason is that trips vary in their value. Ramp metering provides a good example. It works to smooth the flow of cars entering a highway, so turbulence falls and the number of trips increases. But ramp metering benefits most those drivers who are already on the highway (e.g., those who have long commutes from suburbia) at the expense of those drivers entering the highway on the closer-in metered ramps. It’s not clear that the value suburbanites place on their trip is higher generally than that of people who live closer in. Ramp metering may get more throughput, but it tends to reward and encourage suburban sprawl.



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Ramp metering works to smooth the flow of cars entering a highway, so turbulence falls and the number of trips increases. But metering most benefits those already on the highway at the expense of those entering.

- TSM will not cure traffic congestion. TSM effectively works like adding lanes; it increases the capacity of the highway. But triple convergence almost guarantees that highway space will remain scarce.
- TSM has land-use impacts. The impacts are similar to those from adding capacity. The TSM improves access to both the existing commercial centers served by the highway and the suburban areas from which commuters and shoppers come. Thus, land in the commercial centers becomes more intensely developed as does land in the urban fringe.
- TSM should be used in conjunction with demand management. The objective of TSM is sensible: get the most value out of investment in infrastructure. But driver, business, and household reactions to lower travel times in the short run lead back to inefficient levels of congestion, though at higher volumes. Preventing this requires rationing. There are a variety of ways to ration, each with advantages, as we discuss in the next major section.

TABLE 6-2. SUMMARY OF TSM POLICIES

	SUPPLY-SIDE POLICIES	EFFECTIVENESS		COST		IMPLEMENTATION		POLITICAL ACCEPTABILITY
		Extent	Impact	Direct to commuters	To all society	Need for new institution	Ease of administration	
HIGHWAY	Building new roads without HOV lanes	Variable	Moderate	None	Great	Cooperative	Moderate	Poor
	Building added HOV lanes	Variable	Moderate	None	Great	Cooperative	Hard	Moderate
	Rapidly removing accidents	Variable	Great	None	Minor	None	Easy	Good
	Improving highway maintenance	Broad	Moderate	None	Moderate	None	Moderate	Moderate
	Upgrading city streets	Variable	Moderate	None	Moderate	None	Easy	Moderate
TRANSIT	Building new and expanding existing off-road transit systems	Narrow	Moderate	Minor	Great	Cooperative	Hard	Poor
	Increasing public transit usage by improving service, amenities	Narrow	Minor	None	Moderate	None	Hard	Moderate
ITS	Coordinating signals, traffic monitoring, ramp metering, electronic signs	Narrow	Minor	None	Minor	None	Moderate	Good

Sources: Adapted from Downs 2004a and Ferguson 1998 and 2001

In sum, a variety of TSM projects effectively increase VMT and travel-time reliability without converting more urban land to roadway, at least in the short run. (See Table 6-2 for a matrix of TSM policies and their effects.) That additional trip-making capacity is a benefit for both existing drivers and new drivers that now find conditions attractive enough to make a trip that they previously would have skipped, postponed, or made by a less convenient route or mode. In the long run TSM has effects similar to those from increasing capacity by adding lanes: the new capacity reduces the time cost of travel (its objective), which encourages more and longer trips, route switching, low-density land development at the urban fringe, and higher-intensity development in existing centers. With time, congestion returns. Improving long-term efficiency requires pricing or some other form of rationing (discussed later) in conjunction with TSM.

Better rail-transit efficiency

Here's a scenario with a basis in reality. The political leaders in a growing metropolitan area feel pressure to do something about traffic congestion. Pricing access to highways looks Kevorkian. Federal money is available to fund 80 percent of the cost of a rail transit system. That federal construction money looks great to local businesses, and rail looks a lot better than buses for attracting commuters out of their cars.

But ridership fails to meet expectations, in part because the system takes at least a decade to complete, and highway congestion appears unaffected. In addition to traffic congestion, the metropolitan area now has an expensive and underused rail system.

Pricing car travel would cure both of these problems, but that option is a political swan song, not a theme song. The burning question for the transit agency becomes: how do we increase ridership?

Dropping fares usually isn't a realistic option. Moreover, fares rarely cover 50 percent of *operating* costs (25 to 30 percent is probably a closer average), so operating subsidies are piled onto capital subsidies. The evidence suggests that, in general, dropping fares altogether would not have much effect: commuters who have the choice of driving are more sensitive to the nonmonetary costs of transit travel than to fare. A likely exception to that generalization is fare-free transit zones in dense downtowns, though there the issue is probably still more one of the convenience of the transit system (coverage, frequency, and no fares to mess with) than the fare itself.

What are these nonmonetary costs of transit travel? Chief among them are the time costs of starting from an origin (e.g., home), waiting for the transit vehicle, riding to the stop nearest the destination, perhaps transferring and waiting again for that transfer, then getting to the final destination (perhaps with a longish walk). The time costs can easily be double those associated with the same trip by car. In fact, that's what national averages show: average time for a commute trip by car, 23 minutes; by transit, 46 minutes. Then there are costs of reduced comfort, security, and flexibility, and the empirical evidence that people find waiting time more expensive than traveling time.

Transit operators realize that attracting drivers out of their cars requires performance. The dedicated rail right-of-way provides some of this performance because the train doesn't get stuck in congestion. The main benefit is decreased travel time relative to alternatives. Additional performance benefits, many of which ultimately make a difference because of effects on the amount and variability of travel time, include the following:

- Quick and easy access to secure and attractive stations
- Better coverage and service that reduces transfers
- Reliable service

- Easy to board, attractive, comfortable, and secure vehicles
- Efficient fare collection systems.

The last two of these are relatively easy to provide. Most rail commuter systems offer a variety of methods to pay the fare, including daily, weekly, or monthly passes. Most train cars are easy to board and reasonably attractive. The level of comfort varies, however, and is often inferior to that of a car. Climate control is especially difficult. And security has its costs: its better when the train is full.

Reasonable line-haul travel times and reliability are also reasonably easy to provide on exclusive or priority rights of way. It is a little trickier but solvable to keep LRT or BRT on time because they operate like a streetcar when in the CBD. It is essential to set traffic signals to give LRT priority over cars at intersections. Without separate rights-of-way, however, buses are subject to the same traffic and delay as regular traffic.

Accessibility is a critical characteristic, and is more difficult to provide. People who have options will be put off if it takes an unreasonably long time to access the platform at the station, and, once there, the train is nowhere in sight.

There are several options for improving access to rail transit, each with differing land-use implications:

- Improving car access to suburban rail stations. Usually this means park-and-ride lots and kiss-and-ride drop-off ramps. The disadvantage is that this improves access to the urban fringe, encouraging more sprawling suburban development.
- Providing a local bus system to feed rail stations. This feeder system has to be good enough so that the trip by transit competes with the trip by car.
- Increasing the number of commuters who live within walking distance of transit stations. A 15-minute walk to the transit station may appeal almost as much as a 15-minute drive. To be likely to make this walk, commuters

BRT IN EUGENE-SPRINGFIELD, OREGON

In 2007 the Lane Transit District began operating its BRT system—EmX. The first line connects downtown Eugene and downtown Springfield along the four-mile Franklin Corridor. Buses travel every 10-20 minutes and provide service to two high-transit-ridership generators: the Sacred Heart Medical Center and the University of Oregon. The system has several features to help reduce travel time:

- **Transitways.** Approximately 60 percent of the route is designed with exclusive lanes (transitways) for the EmX vehicle. They allow EmX vehicles to bypass traffic congestion and maintain constant travel times as congestion increases.
- **Stations.** Station spacing along this first route is approximately every half-mile, for a total of eight facilities. Raised platforms are designed to accommodate level boarding, and amenities include real-time information, lighting, furniture, and bike storage.
- **Intelligent Transportation Systems (ITS).** Computer, communication, and sensor technologies allow EmX to use signal priority and queue jumpers, which give the EmX vehicle



priority through intersections, and give passengers real-time information updates.

EmX uses a diesel hybrid 60-foot articulated vehicle with wide doors on both sides to allow for boarding on both sides of the bus in stations along the medians. Travel on EmX is currently free, although the system is designed to accommodate pre-boarding fare collection in the future. One month after beginning operations, daily average new ridership on the EmX Franklin Corridor is approximately 4,000 passengers, almost a 50 percent increase over previous transit ridership on the Franklin Corridor route.

ECONorthwest, summarized from (1) Wright, Jeff. "EmX rolls with strong ridership." *The Register-Guard*, February 14, 2007, and (2) Lane Transit District www.ltd.org (accessed 26 February 2007), including photos.

have to live well within a mile of the station. Quite a bit of housing in this short radius is needed to make much of an impact on ridership.

Transit operators often become advocates for the second and third options. Both work better where densities are higher. Indeed, rail transit worldwide works best when development densities are high. But high-density housing requires relaxation of suburban density and parking requirements.

Is low-density zoning the cause of low-density development? It might be. As described in Chapter 2, high-density housing can have negative fiscal and congestion effects, so neighbors oppose it, and zoning prevents the market from supplying it. There may be households, such as the growing cohort of empty nesters, who would be happy to live in higher-density housing and commute to work and play on a good rail transit system. Allowing high-density housing near transit stations seems a happy scenario from a metropolitan standpoint. Careful design and financing could minimize negative external impacts.

In some places, however, high-density development may be wishful thinking. Commuter rail lines have to usually be built to serve suburbanites (actually skipping past poorer inner-city neighborhoods) because these are the people who need to drive less to reduce congestion. Relaxing zoning in these areas may simply have no effect. The tendency then is to try to encourage high-density development through even more subsidy, this time of land development.

Urban containment can address the problems associated with each of the tactics above. An urban containment boundary prevents some of the sprawl induced by park-and-ride transit. The reduced supply of land reduces ridership, but the park-and-ride lot still attracts riders from nearby suburban areas.

The urban containment boundary also increases the demand for high-density development near transit stations. This may especially be needed if, as is usually the case, suburban highway capacity is being increased at the same time as rail transit. The reduction in supply of accessible and developable land raises suburban land prices encouraging more intense development. Relaxing zoning restrictions to allow it makes sense around transit stations.

Another alternative is BRT. It offers many of the advantages of rail transit, but potentially with more route flexibility that can reduce transfers. BRT routes can take advantage of separate right-of-ways, making them faster, more reliable, and safer than cars along congested corridors. Such rail-like right-of-way investments can provide land developers and their customers signals of commitment to an area that make it attractive for economic development. The buses can operate more flexibly than LRT on city streets. Successful examples include Curitiba, Brazil, and Bogota, Colombia.

Relatively quick, convenient, and reliable rail and bus feeder service, or BRT, could lead to long-term economic development benefits in terms of land-use, design, and travel patterns. But capturing those benefits requires a well-integrated and efficient system, which takes time to build (potentially decades) and more time to generate its full effects. To capture the long-run benefits of rail transit, which might well be thought of as a “legacy investment” (think about New York City’s subways), decision makers should probably use a 50- or 100-year rather than the typical 20-year planning horizon.

Toronto provides an example. Toronto invested heavily in rail transit in the 1950s, prior to highway-induced suburbanization. Rail transit has since provided a variety of benefits to urban form, the economy, and the livability of that city—impacts that a conventional benefit-cost analysis done in 1950 would likely have missed. These impacts have depended on implementation of a variety of regional land-use and transportation policies that have made rail transit competitive with the auto in Toronto.

Relatively quick, convenient, and reliable rail and bus feeder service, or BRT, could lead to long-term economic development benefits in terms of land-use, design, and travel patterns. But capturing those benefits requires a well-integrated and efficient system.

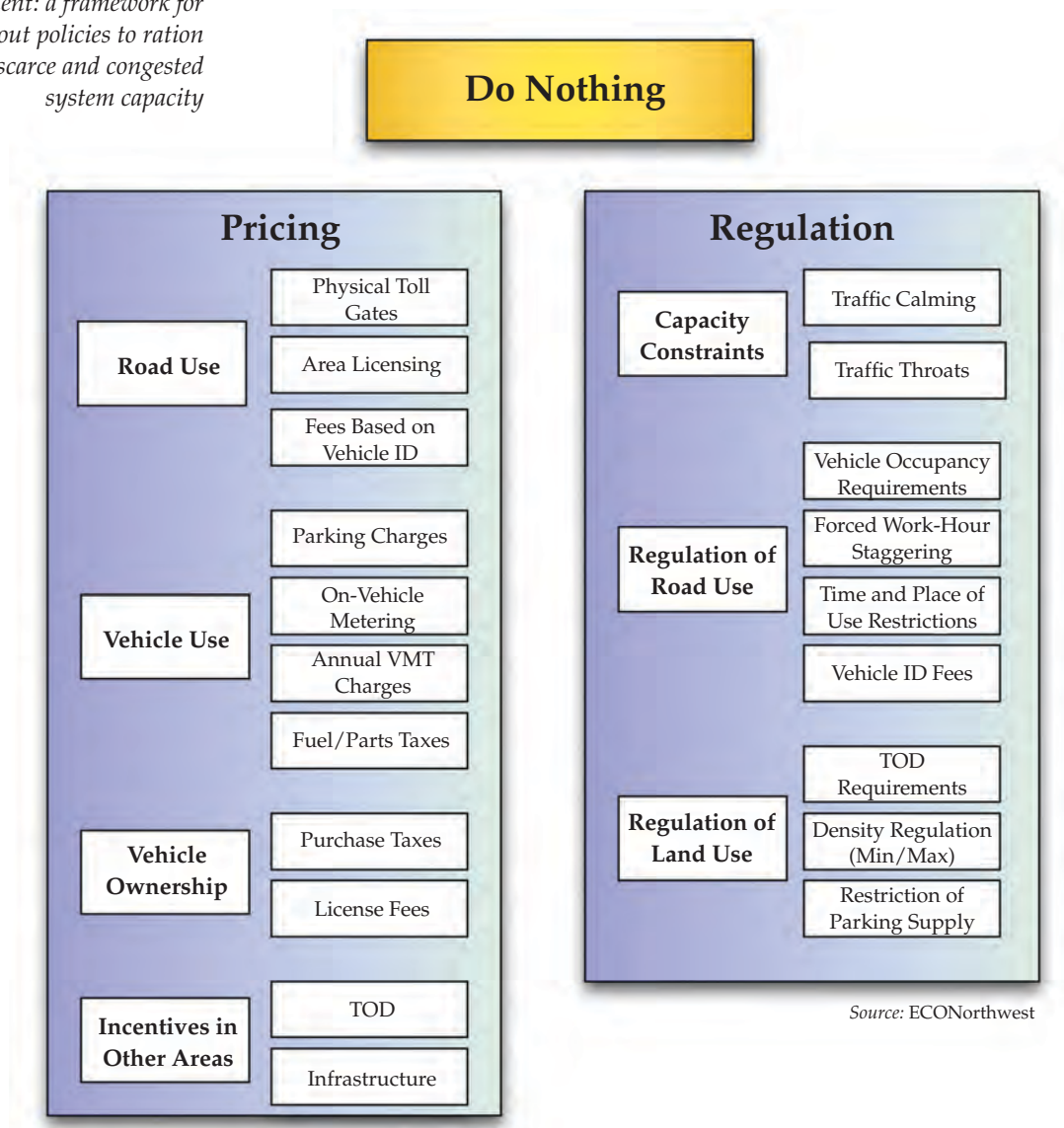
POLICIES TO IMPROVE THE USE OF VEHICLES

The public sector supplies most transportation infrastructure—the paths. The private sector supplies (builds, purchases, owns, and operates) most of the vehicles that use that infrastructure. But public policy can affect the way those vehicles are used.

The benchmark policy is peak-period pricing of highways and parking (i.e., congestion pricing). That policy prices vehicle trips, not person trips. Vehicle trips can, in concept, fall by a large proportion (and treating congestion requires only a small proportionate drop) without much if any decrease in person trips. A lot of vehicle capacity goes unused a lot of the time: passenger and back seats of cars and vans, empty seats in public buses and trains, bikes collecting dust in the garage, and legs that spend too much time crossed in front of a chair. Peak-period pricing of congestion and parking provides strong incentives to use that capacity more efficiently.

But pricing policies, and others that substitute for them, are difficult to implement. They affect behavior, and that's "social engineering" (which is apparently even worse than planning). The fundamental beliefs that planners are up against are: (1) I've paid for this system with gas taxes and my

Figure 6-1. Transportation demand management: a framework for thinking about policies to ration increasingly scarce and congested system capacity



purchase of a motor vehicle, (2) the fact that I've traveled this way for several months (or years or decades) means that I have a right to travel this way forever, and (3) I know what's good for me better than the government does.

That last point requires careful attention because it is so often true. Planners need a good understanding of where travel demand comes from and how people get benefits from the transportation system before they start slowing traffic, raising auto costs, and telling people that they are better off on transit. Appendices C and D contribute to that understanding—we provide an abbreviated discussion here.

A variety of factors influence the demand for a good or service. Especially important are the price of the good or service itself, the prices of substitute goods or services, and the purchasing power (i.e., wealth or income) of consumers. This section focuses on the first two of these in the context of demand for trips by modes other than the SOV:

- Policies that decrease the cost of modes alternative to SOV. These include subsidies to HOV travel on the highway (e.g., carpools, jitneys, and buses); HOVs that travel in other rights of way (e.g., rail transit or bus-ways); and travel on foot or bicycle on sidewalks or in bike lanes. Reducing the costs associated with travel by these modes (all else the same) increases travel overall, and shifts some person trips out of SOVs.
- Policies other than congestion pricing that ration travel by SOV. Congestion pricing rations space on the highway to those most willing and able to pay. While we sang its praises, the back beat was its political impediments. But there are other ways to ration space, each with its advantages and disadvantages. A big advantage of these policies, however, is that they attack the problem of congestion directly, and thereby encourage person trips by models alternative to SOV. In the parlance of transportation, such rationing is done with transportation demand management (TDM). Figure 6-1 illustrates three broad ways to ration demand:
 - Do nothing. Congestion will grow and drivers will respond to the price of that congestion
 - Change prices. This is the standard economic solution to scarcity.
 - Regulate use. This is a more typical government solution to scarcity.

A big advantage of TDM policies is that they attack the problem of congestion directly, and encourage person trips by models alternative to SOV.

The organization of this section could follow that of the previous one: first more, then more-efficient, vehicles. But there are already plenty, really too many, private motor vehicles. If highways, parking, and vehicle emissions were all properly priced, there would be demand for additional private or public vans, buses, train cars, and bikes. Industry can supply this demand. Thus, we skip the “more-vehicles” category of action for SOVs. (We acknowledge that a policy of “having more HOVs” is a reasonable one, but we deal with much of that topic above in the section on better rail and transit service. The main purpose of more vehicles in most cities is to reduce headways, not supply more capacity for excess passengers.) We focus on policies that encourage more efficient use of the vehicle capacity we already have, or could easily produce.

We organize the discussion based on how a policy affects the demand for vehicles: Does it reduce the cost of using an alternative to SOV (i.e., more HOVs)? or Does it ration the use of SOV, encouraging the use of other modes?

Increasing HOV

Congestion, parking, and emissions pricing (the benchmarks) work by increasing the total cost (out-of-pocket money plus time) to the driver of an

SOV trip, even though they decrease private time cost and total *social* cost. This higher total private cost encourages travelers to seek alternatives to travel by SOVs, primarily HOVs, which spread the higher out-of-pocket costs over more travelers. The greater demand for HOVs encourages improvements to HOV infrastructure and rolling stock. Travel by HOVs increases and highway congestion decreases.

Might things work the other way? Will reductions in the time and out-of-pocket costs and improvement in the other characteristics of travel by HOV encourage more travel by HOV and less congestion? Yes, the demand for travel by HOV will increase. But no, this will not reduce congestion, at least in the long run. The result is usually more travel and continued congestion. Nevertheless, improving the characteristics of alternatives to SOV travel are likely to improve the political prospects for congestion pricing—there's a decent alternative waiting to be used when the tolling starts—and can have desirable effects on land development.

In this subsection we first discuss the effect of dedicating existing highway space to HOV travel (i.e., carpools, vans and jitneys, and buses). Then we discuss subsidies of rail transit or BRT. We deal with the alternative modes of walking and biking (or whatever rolling mode gets used on sidewalks, bike lanes, and roads) in the later section on controlling neighborhood externalities.

HOV lanes

HOV lanes improve car commuters' incentive to increase the vehicle occupancy rates of their cars or vans. The incentive comes in the form of faster travel speeds. As long as there are relatively few multiple-occupant vehicles, dedicating one or more lanes to their use allows them to cruise past SOVs stuck in congestion.

HOV lanes similarly improve the demand for trips on standard public buses. A big cost, usually bigger than the fare, of riding a bus is the value of the time it takes walking and waiting to catch it. That point is critical to understanding how people make the choice between SOV and HOV trips. Transit trips are more likely to have multiple modes, and transferring modes has a time and convenience penalty. One has to walk (sometimes encumbered by bags, foul-weather gear, and weather), and one has to wait. Because there is variability around transit arrival-departure time, one must arrive early: wait time is inevitable if one is risk averse about missing a connection. Unless you live near a rail stop and work near another on the same line, there is a lot of access and waiting time.

An HOV lane can help compensate riders for the cost of time by providing faster and more reliable travel than that by SOV. An HOV lane makes a regular bus perform on the line-haul portion of the trip like BRT.

Dedicating a lane to HOV use has the short-run effect of speeding HOV travel, which may improve the overall efficiency of the transportation system: the time saved by the relatively many people in the HOVs may more than make up for the now even slower travel in the now fewer SOV lanes.

The faster HOV travel encourages more drivers over time to organize carpools, join van pools, pick-up respectable-looking hitchhikers, or buy blow-up dolls to serve as passengers. (There is evidence for all these responses: innovation emerges in a market economy—tell me the rules and I'll find creative ways to obey them.) The consequent drop in the number of SOVs, if significant, should increase travel speeds in the unrestricted lanes (at least temporarily: triple convergence again), and the increase in the number of HOVs should slow travel in the restricted lane.

One might expect that travel in all lanes will converge to the same speed on average. This is like the lines at the check-out counters at the grocery



Thania Navarro, istockphoto.com

Pricing congestion, parking, and emissions encourages SOV drivers to seek alternatives, primarily HOVs, which spread the higher out-of-pocket costs among more riders.

store: in equilibrium they all move on average at about the same pace. But travel in HOV lanes is likely to remain speedier on most days to compensate for the hassle of organizing the group travel. The result is that the drivers in SOVs are likely to see the HOV lane as underused.

Many moderately disgruntled commuters can produce quite a bit of political pressure to drop the restriction on the HOV lane(s). Roark (1981) notes that commuters tend to accept the building of an additional lane for HOV use (rather than restricting an existing lane) or converting a lane from the low-flow side of the highway. Commuters better accept HOV lanes that are physically separated from other lanes by a permanent barrier. Carpool lanes that require two occupants tend to work better and are more readily accepted better than those that require three occupants (due to lower organizing costs).

There are also ways to reduce the hassle of organizing group travel. An example strategy is “rideshare matching.” Employers or the public sector can use web sites or other methods to help those interested in carpooling to match up with others in their area who travel at similar times to similar locations. They can help provide locations for informal carpools, which skip the organizational hassles and can often operate on demand.

“Car sharing” is another clever example of a way to reduce a disadvantage of car-pooling. Commuting by transit or carpool means lack of access to a car during the work day. With car-sharing, participants purchase hourly or monthly plans to use a “share” of a pool of vehicles, and are able to use any vehicle in the program any time they like. Vehicles are ideally located throughout a downtown and once enrolled, participation in the program is simple.

More ways to improve mass transit

HOV lanes help, but both public rail and rubber-tire mass transit face serious challenges in suburbia. Buses and rail cars running mostly empty are common sights in many metro areas. Filling these vehicles, and hopefully additional ones, is a priority for transit agencies.

Consider the challenges for rail transit. Simple geometry dictates that very little of suburbia will be within walking or even reasonable cycling distance of a rail transit station. To see this, think of a city with five radial transit lines that radiate from its CBD. Transit stations are located every mile along every line.

To start, let’s go out five miles from the CBD—not far by suburban standards. The circumference of the urban area at this point, with a radius of five miles, is larger than 30 miles around. If there are five radial transit lines, each with a station at the five-mile market (i.e., right on our line), only one-sixth of the distance on that ring is within a half mile of a transit station. Move this out to 10 miles and the ratio falls to one-twelfth.

Now think in terms of land area. There are roughly 75 square miles of land in a ring of radius five miles. Given our assumptions, there are about 26 transit stations that service this area. If the bulk of their riders are within one-half mile of each station, each station serves a primary market area of about three-quarters of a square mile; 26 stations cover about 19 square miles; about 25 percent of the area has good access to the rail stations. Now double the radius to 10 miles. The land area quadruples to over 300 square miles. The number of transit stations only doubles to 51, so now stations cover about 38 square miles; and about 13 percent of the area has good access to the rail stations: double the radius, and the coverage drops to half. Suburbia is hard to service with radial transit.

The challenge for commuter transit in suburbia is finding efficient ways to feed rail transit stations and fill the buses that use HOV lanes on highways.

The elasticities reported in Appendix C suggest that the biggest impediments to bus ridership are:

- time spent walking;
- time spent waiting, especially with uncertainty about how long the wait will be due to poor reliability; and
- the comfort and security of the transit vehicle.

The last of these can be treated directly with an inflow of cash. Unless the alternatives are unacceptably bad, transit vehicles have to meet a standard of comfort not much lower than that offered by cars. They have to be clean, the seats have to be comfortable, and the commuter has to feel reasonably secure. If you have ridden in crowded buses or subways at rush hour, nose-to-pit with your fellow travelers as you jostle through the humid darkness, you see that is a hard test to consistently pass. Oh, and your cell phone doesn't work underground (although that's changing).

Reliability is especially important for trips that feed rail transit. More reliability means less waiting in uncertainty at both the bus stop and train station. Reliability is a tough problem for fixed-route buses operating on congested suburban streets. A brute-force option is to increase the number of vehicles on each route; more vehicles means shorter headways, and less waiting on average (though it's hard to guarantee that actual waits are close to the now shorter average).

A high-tech treatment is to communicate to riders electronically, perhaps over the Internet via home computer or cell phone, the location of the bus in real time. Commuters can then time their departure from home. Similar communication between the bus and the train helps ensure that the train doesn't pull out just as the bus arrives at the station.

An alternative to a fixed-route system on a fixed (but often unreliable) schedule is a flexible jitney service. The operator takes advantage of logistics software to pick up commuters in a van or small bus at their house. This eliminates walking. Electronic communication might eliminate uncertain waiting. Similar communication between the van and train helps coordinate the transfer. This system could be private or public, but it clearly represents a leap from that typically available at present.

In short, the performance characteristics of rubber-tire mass transit in most suburban areas require considerable improvement to draw many travelers out of their cars. That improvement requires cash, but also flexibility and entrepreneurship. As we noted in Chapter 4, the private sector might be in a good position to provide these services, given the opportunity to do so by relaxing restrictions against private jitney services.

The performance characteristics of rubber-tire mass transit in most suburban areas require considerable improvement to draw many travelers out of their cars. That improvement requires cash, but also flexibility and entrepreneurship.

Rationing vehicle trips

Treating congestion on highways and in parking lots, and reducing smog requires rationing. Pricing has several hard-to-beat advantages: (1) it reduces congestion and pollution; (2) it discourages the low-value trips (those trips whose drivers are unwilling or unable to pay the congestion, parking, and emissions tolls); and (3) it generates revenue that both signals the net benefits of and pays for highway and parking capacity.

Proper pricing will not be implemented soon in a big way in most metropolitan areas. But other rationing techniques and policies have been identified, analyzed, tested, and implemented. This section starts with a discussion in the abstract of the various mechanisms available to ration the limited supply to the huge demand that exists at low prices. We then turn to description and evaluation of strategies used in practice. We finish with an evaluation of planned congestion as a demand-management option.

The need and the possibilities

We discussed marginal-cost pricing as a method for rationing demand in Chapter 4. There are alternative methods of rationing that warrant consideration:

- *First-come, first-served (queuing)*. This is essentially the current method for allocating highway space. There is an efficiency rationale for queuing: those who most value the trip are the most willing to wait in line. Limiting the quantity of cars allowed on the highway, however, would improve travel speeds. For example, cars could line up at each interchange, with the first X number allowed onto the highway over any given period of time. This is a common mechanism for rationing access to popular rides at amusement parks, popular concerts, and popular sporting events. The time and energy spent queuing is itself, of course, inefficient.
- *Regulation*. A local government can choose how many trips to allow and to whom to allocate those trips. A simple example of a regulatory mechanism is allowing only certain license plate numbers: even numbers one day, odd the next. More likely, some sort of priority system would be developed. Health care is often rationed this way in public health systems. For example, the value of a vehicle trip could be rated by a point system according to the social value of each characteristic of the trip. Drivers with highly rated trip characteristics could apply on-line for a trip pass, print it, and take it to the control booth at a highway entrance point at the appointed time. (That should be a good defense against those claims of social engineering: try it at your next public meeting.)
- *Tournament*. Rather than setting up a priority points system ahead of time, the government could, in concept, allow drivers to compete or make their case for access to a permit. Though impractical given the large number of drivers, a tournament format could be, and sometimes is, used on the land-use side of the connection: a city restricts its number of residential building permits (justified based on its limited capacity to provide adequate public facilities), and allocates those permits to developers with the “best” proposals.
- *Lottery*. Again, the transportation authority chooses how many trips to allow, but the trips are ranked using some random mechanism, such as coin flips or a random-number generator. The lucky drivers get a pass to take the trip. Though seemingly egalitarian, fortune may not smile on people with extremely high-value trips. And revenues are not generated to pay for capacity. Lotteries are commonly used to allocate permits to float backcountry rivers, whose capacity cannot be further expanded without changing the experience that recreationists value and endangering the environment that is the source of that experience.
- *Tradable permits*. The government distributes the chosen number of trips to vehicles somehow equitably (e.g., by lottery or in proportion to the current use of the highway). The permits are transferable across vehicles, so drivers could buy and sell permits for trips at specific times. The government could create an on-line market or allow private market makers (e.g., Ebay) to match buyers and sellers. Trading establishes a price for permits that varies by time of day, which allocates trips to those most willing to pay. The allocation of trips is efficient. But the revenues generated by sales of permits go to those to whom the permits were initially allocated rather than to the transportation authority to pay for highway capacity. Tradable permit systems are in use to control some forms of air pollution; in this case, revenues cannot be used to increase the capacity of the environment to absorb pollutants.

- *Auction.* The government can choose the maximum allowable number of vehicle trips and let the participants in auctions set the price. Drivers, for example, could submit bids online for a permit to enter the highway over any short time period, and the available number of permits would be allocated to the highest bidders, with all bidders paying the lowest winning bid. That lowest winning bid is the price of access to the highway and would vary with demand; bidding establishes the congestion price. This kind of mechanism is used for auctioning federal government bonds.

At an abstract level, these, along with congestion pricing, are the mechanisms available for rationing space on a highway. Let's make it less abstract.

Rationing tactics in practice

Politics precludes full-cost pricing of car trips. But the results of nonpricing have motivated thinking to identify politically palatable policies that push the costs of vehicle trips in the right direction. We organize these by how they affect vehicle-trip demand.

- **Raise the out-of-pocket cost of a vehicle trip.** Congestion pricing, which raises the cost of private vehicle trips, is long run and not part of our list here. What's left, and what's right?
 - Raising the fuel tax. Among the best of the second best policies. Chapter 4 discusses its pros and cons in detail. Still a political challenge, but a surmountable one: it is not a new tax, it has an administrative structure in place, people understand it, and it has a relationship to the cost of driving. The big limitation: it is not sensitive to the key elements that define congestion: place and time.
 - Cashing-out free parking. Employers who typically provide employees with free parking could instead offer them the choice of free parking or a cash payment equal to the cost of the parking to the employer. It appears that no one loses, and some drivers will respond to the cash incentive to commute by an alternative mode. Shoup (2005b) estimates that such strategies could reduce automobile commuting by anywhere between 10 to 30 percent.
 - Converting some of the fixed costs associated with car travel to variable costs. Car owners already pay insurance premiums and fees. Neither usually varies with the miles the vehicle is driven. But the probability of an insurance claim rises with miles traveled (all else the same), as does the likelihood of incurring various public enforcement and administrative costs (pay-at-the-pump insurance premiums). A VMT tax may be a more appropriate method for charging for these costs. Though harder to collect, the vast majority of drivers may be no worse off, and many drivers might be better off driving less and saving on their insurance and registration payments.
 - Establishing tradable travel permits. Maybe clunky in practice, but the idea is simple: choose the desired level of traffic and distribute that number of travel permits freely to the everyday users of the corridor. Set up a trading website and let the market work. Commodities exchanges and futures markets are well established and efficient, why not add vehicle trips to the trading? The government doesn't get the revenue (that's part of the appeal, after all), but it still gets the price signals.
 - Returning congestion charge payments in a lump sum to drivers. This sounds odd: take their money a little at a time, and give it back all at once. But it could work to ration demand. On any given day, the

congestion toll has its usual effect of discouraging trips of low value.

- **Raise the time cost of an SOV trip.** Increasing travel time is a bad way to state the objective. The right way to say it is either or both (1) increase the degree to which travelers, as travelers (not as employees or taxpayers), pay for the full costs of their travel, and (2) pursue certain policies for the benefits they yield, and, in doing so, accept increases in the costs of SOV travel as one of the costs to pay for those benefits. Possible policies that meet this second condition:
 - Access management. Reducing or restricting the number of entrances to and exits from a highway or arterial street serves TSM purposes. Traffic moves more smoothly. But fewer access points means slight or substantial increases in the time costs of some car trips.
 - Traffic calming. Traffic on local streets generates negative noise, safety, and social externalities. Traffic calming involves street designs (e.g., speed bumps) that slow vehicular travel, which increases travel times and discourages trips (more on this in the next major section).
- **Place regulatory restrictions on car travel.** As a practical matter, many people are more sensitive to cash outlays than to inconvenience. In concept, time is money: a person who spends less time in traffic could be working more hours for pay. Converting time to money isn't always easy in practice; many hourly wage earners have no control over hours worked, and many salaried workers have no control over their salary. Regulations that serve a reasonable purpose, but that impose cost in terms of time or inconvenience, may generate less opposition than pricing. These include:
 - Relaxation of minimum parking requirements or imposition of maximum parking requirements. If parking has a beauty, it is the kind only an engineer can love, and it takes up a lot of space. A rational concern about not finding parking motivates trips by transit, foot, or bike.
 - Restricting car use in some areas (car-free zones). Proceed with caution. Shopping malls are car-free areas that are surrounded by parking lots. Cities around the country in the 1970s and 1980s experimented with downtown pedestrian malls, with some disastrous results. Nevertheless, it can work: witness Santa Monica, California, Boulder, Colorado, Charlottesville, Virginia, and any number of pedestrian areas in European cities. Prerequisites are worthwhile destinations and good access to the area either by car or alternative modes.

Table 6-3 shows the path by which these and other policies affect the demand for auto trips. There are probably other tactics and strategies that could both work and pass political muster. Indeed, there must be because too much vehicle travel generates a social loss. Society as a whole is obviously better off with better management of its scarce resources: land, infrastructure, and environment. The trick is finding ways to distribute the net gains in politically acceptable ways.

Doing nothing: congestion as rationing

The underpriced congestion and pollution costs of travel by auto have probably resulted in current highway capacity exceeding the optimal capacity in some areas, perhaps substantially. A paradox?: How can we have congestion and too much capacity at the same time? If you've read this far, you should be able to answer that question.¹⁰

Some planners have proposed that congestion simply be allowed to increase. As congestion increases, the cost of a trip as perceived by the driver

Society as a whole is obviously better off with better management of its scarce resources: land, infrastructure, and environment. The trick is finding ways to distribute the net gains in politically acceptable ways.

TABLE 6-3. HOW TDM POLICIES AFFECT THE DEMAND FOR AUTO TRIPS

TDM POLICY	DETERMINANT OF DEMAND AFFECTED	EXPECTED STRENGTH OF EFFECT ON AUTO TRIPS
Promote HOV use	Tastes	Weak
Coordinate carpools	Waiting time	Weak
Decrease fares or HOV costs	Transit fare	Moderate
Improve access to transit	Walking time	Moderate
Improve transit service	Waiting and Riding time	Moderate
Provide HOV parking	Riding time	Moderate
Guarantee a ride home	Waiting time	Moderate
Increase parking fees	Auto trip cost	Strong
Parking Cash Out	Transit trip cost	Strong
Increase parking fees	Auto trip cost	Strong

Sources: Compiled by ECONorthwest from Orski 1991, Ferguson 1990, Kuzmyak 1990, Wachs 1991, and Shoup 2005a, 2005b

increases, which encourages some drivers to switch to other modes. Halting new investment in an already overbuilt highway system allows local governments to direct funds toward improving access to other modes or toward projects that reduce the negative effects of the auto on urban environments. Though highway traffic volumes and travel times would, under this policy of planned congestion, remain higher than optimal, the efficiency of the entire transportation system would improve relative to what it would be if highway capacity had been expanded.

Doing nothing might also ease the transition from the current policy environment to a more efficient system of prices in the future. Congestion pricing would increase the demand for high-occupancy modes. Using funds diverted from capacity expansion to improve the quality of these modes might increase the political acceptability of pricing highway access because good alternatives would already be available. Second, a high level of congestion by itself increases the acceptability of pricing. Allowing congestion to build on existing highways may be an effective short-run approach to smoothing the transition in the long-run to a more efficient system of prices.

To make the argument more concrete, consider a congested bridge. An efficient peak-period congestion toll could reduce travel on the bridge sufficiently to eliminate the need for bridge expansion. In other words, the bridge might be the right size, but in the absence of efficient pricing (tolling) it remains overly congested. Moreover, money from state and federal fuel tax revenues is available to increase the capacity of the bridge. Nobody in the community would pay noticeably more for a new bridge: if this community chooses not to use the money, some other community will. Thus, the new capacity not only improves travel conditions (at least temporarily), but it also funnels money into the community in the form of construction contracts.¹¹

The proponents of planned congestion would argue that the bridge should not be expanded, but neither should the funds available for construction be turned down. Congestion on the bridge exists not only because access to it is underpriced, but because of the inefficiently high cost of using other modes. The funds earmarked for adding capacity to the bridge would be better spent improving infrastructure that serves other modes within the corridor.¹²

There is sense in all this, but there is a downside: the mammoth cost of the overly congested highway system. Millions of commuters spending perhaps 10 to 30 (or more) extra minutes every day commuting to work until pricing is implemented. Millions of dollars worth of lost time daily.

The following analogy may make the issues more clear by stepping away from transportation temporarily.

Yosemite and Yellowstone National Parks are congested during the summer peak. The logic goes like this: these parks are a national treasure for all; it would not be fair to price out lower-income visitors; we should have low entrance fees. The entrance fee is \$20–\$25 per car, good for seven days of use: for a family of four, that's less than a dollar a day per person; the weekly pass costs about what it would cost two people to spend two hours in a movie theatre. So, yeah, for these parks the price perceived by the user is low—very low.

Underpricing leads to over use: congestion. Among the costs of congestion are negative externalities: air and water pollution; loss of scenic value; loss of the experience value. The Park Service tries supply-side solutions: builds more camp grounds; puts in a bus system; adds other attractions to keep people occupied and satisfied. But eventually, expanding supply just seems too stupid. Nobody really wants all those people there, including the people who are there.

Some people shift to the off-peak season: it's less desirable than an uncrowded summer visit, but more desirable than a crowded one. Some people endure the congested entry and exit because it is a small part of the cost of their trip: they are passing through the crowded tourist areas on their way to a week alone in the high country.

But the majestic and accessible main valleys and attractions are very crowded. One solution: forget about it; do nothing and let people keep coming until it is too unpleasant to come anymore.

That solution seems unlikely to be efficient in any economic sense (though it has some populist appeal). Reducing the congestion by rationing the demand, even crudely, is likely to increase the total net benefits of visitors. Access to scenic rivers, such as the Colorado and Rogue, for example, is rationed according to a lottery. A poor person is as likely as a wealthy one to get a permit. This system does not maximize the *dollar value* of economic surplus, but it's both much better than doing nothing and retains the populist, egalitarian appeal.

So is do nothing the right solution? Definitely not if that remains the primary method of rationing demand. The cost in terms of lost travel time is too high. But it might be a better short-run policy than some inefficient capacity expansion, particularly if it speeds implementation of congestion pricing and the shift to higher-occupancy modes that pricing encourages. In short, there are better policies for addressing highway congestion than ignoring it, but there are not many that are easier.

POLICIES TO MITIGATE TRAFFIC EXTERNALITIES

Chapters 3 and 4 explain external costs as a form of market failure, why that failure is inefficient, and how public policy might address external costs via pricing or standards. The examples used were the standard ones from environmental economics: air and water pollution.

The construction and use of transportation clearly has negative impacts on air and water resources, and much of that impact is an external cost in the sense that those imposing the cost do not pay its full cost directly. But Chapters 3 and 4 have covered the theory and practical techniques for dealing with these types of external costs.

TABLE 6-4. SUMMARY OF TDM POLICIES

STRATEGY	DEMAND-SIDE POLICIES	CONGESTION RELIEF EFFECTIVENESS		SOV REDUCTION*	DIRECT COST		IMPLEMENTATION		POLITICAL ACCEPTABILITY
		Extent	Impact		Direct to commuters	To all society	Required institution	Ease of administration	
LAND USE	Adopting local growth limits	Narrow	Minor	Unknown	None	Minor	None	Easy	Good
	Clustering high-density housing near transit station stops (TOD)	Narrow	Minor	Moderate	None	Minor	Cooperative	Hard	Moderate
	Keeping densities in new growth areas above minimal levels	Broad	Moderate	Moderate	None	Minor	Regional	Hard	Poor
	Improving the job-housing balance	Broad	Minor	NA/ Moderate** to Great	None	Moderate	Regional	Hard	Poor
	Concentrating jobs in big clusters in areas of new growth	Narrow	Minor	Minor, depends on J/H Balance**	None	Great	Regional	Hard	Poor
ALTERNATIVE HOURS	Encouraging people to work at home	Broad	Minor	Positive	None	None	None	Moderate	Good
	Changing federal work laws that discourage working at home	Broad	Minor	Positive	None	Minor	None	Moderate	Moderate
	Staggered work hours	Moderate	Minor	None	None	None	Cooperative	Moderate	Moderate

ALTERNATIVE MODES	Encouraging formation of TMAs, promoting ride sharing	Narrow	Moderate	Moderate	None	Minor	Cooperative	Hard	Moderate
	Build comprehensive bicycle/transit network**	Variable	Minor	Moderate	None	Moderate	Nothing new	Moderate	Moderate
PRICING	Providing income tax deductibility for commuting allowances	Variable	Great	Great	None	Minor	None	Easy	Poor
	"Cashing out" free parking provided by employers	Broad	Great	NA/Great**	None	Minor	None	Hard	Moderate
	Substantially Increase gasoline taxes	Broad	Moderate	NA/Great**	Great	Moderate	None	Easy	Poor
	Increasing automobile license fees, sales taxes	Broad	Minor	NA/Great**	Moderate	Minor	None	Easy	Poor
	Eliminating income tax deductibility of free employee parking	Broad	Great	NA/Great**	Great	None	Cooperative	Moderate	Poor
	Instituting peak-hour tolls on main roads	Broad	Great	Moderate	Great	None	Regional	Moderate	Poor
	Parking tax on peak-hour arrivals	Broad	Great	Moderate	Great	None	Regional	Hard	Poor

TMA = transportation management association

Adapted from Downs 2004a; Ferguson 1998, 2001; and Zupan 1992

* Based on Zupan 1992. If "Not Addressed NA" then Authors' Opinion **

** Author's opinion

In this section we limit our discussion to a different type of external effect, though we admit the differences are hard to separate: the effects that occur to the social and economic environment along a street as it gets “improved” to facilitate traffic flow. More and faster traffic on a street increases noise, inconvenience to pedestrians and bicyclists, accident risk, and, ironically, congestion.

Congestion on highways and major corridors increases the demand for travel on less-congested local streets. Congestion pricing, by raising the cost of highway travel further, will further increase demand for travel on unpriced local streets. That diversion of trips to local roads is a strong source of political opposition to congestion pricing.

With or without congestion pricing, the negative externalities of vehicular traffic on the local street system merit attention. In this section we investigate policies to trade off the transportation capacity that the local grid offers with the desire for safe and pleasant neighborhoods.

Consistent with our views in the rest of this report, we see the options for dealing with these types of external costs as broadly: pricing (markets) or standards (government regulations). We have already exhausted ourselves, our readers, and the literature with our previous description of pricing policies. Clearly, any of the pricing policies described early in this chapter can have a beneficial effect on creating livable streets. For example, parking pricing can reduce travel, encourage mode shift, and help finance better design for livable streets.

Moreover, some of the nonpricing policies aimed at travel performance can also help create livable streets: with careful consideration and planning, a bypass may be able to divert faster, through traffic and leave a local main street with a traffic flow that permits and encourages walking and livable design. Another example: depending on how it is implemented, TSM measures to increase roadway performance may enhance or detract from livability. Appendix C shows that two roads of three lanes are more effective at carrying traffic than one road of six lanes; but TSM measures like one-way couplets will speed up traffic and, other things being equal, decrease walkability.

Thus, in this section we limit ourselves to a brief discussion of policies and designs aimed directly at dealing with the negative effects car traffic has on urban areas as social places. Another way to define the topic of this policy investigation would be “policies for creating livable streets.”

The street environment is a place where drivers, pedestrians, and bicyclists continually engage each other in an informal negotiation over who commands the street. Streets engineered for fast-moving cars generate noise, exhaust, and threat of injury that cause people to retreat from the street environment—the most accessible public space to any home for socializing and physical activity.

Appleyard (1981) championed the idea of livable streets a quarter-century ago: slowing traffic and lowering volumes enhance the street’s sense of comfort. Some recent research:

- Slowing traffic makes a street much safer. A pedestrian struck by a vehicle going 30 mph is eight times as likely to be killed as one hit at 20 mph, who will likely survive the crash 95 percent of the time.¹³ And the likelihood of such an accident happening is lower at 20 than at 30 mph.
- Jacobsen (2003) presents evidence that collision rates decline as the number of pedestrians and cyclists present increases. Drivers apparently travel with more care when they expect people to be around the street, which indicates that sharing the street between cars and pedestrians also makes them safer.

More and faster traffic on a street increases noise, inconvenience to pedestrians and bicyclists, accident risk, and, ironically, congestion.

- Narrow streets make the street safer. Swift (1998) found a high correlation between street width and accident rates.
- Dumbaugh (2005) found that beautifying streets with “livable streetscape” elements, such as buildings with complex facades close to the street and street trees, which are often viewed by traffic engineers as decreasing vehicle safety, actually cause drivers to slow down and drive more carefully, thereby increasing a street’s overall safety for all users.¹⁴
- Residents who live on pedestrian- or bicycle-friendly streets are more likely to use those modes of travel. Krizek and Johnson (2006) found that on-street bicycle lanes significantly increase the odds of bicycle use by people who live within 400 meters.
- Sampson, Radenbush, and Earls (1997) find that sense of community, more than social or economic class, strongly affects neighborhood crime rates.

The main policies for dealing with the external costs that highways and traffic impose on the livability of streets are (1) an agreement among multiple institutions that they want to deal with the livability issues as part of the evaluation of transportation solutions (i.e., an agreement to do the kind of integrated planning recommended throughout this report), and (2) design guidelines (softer) and standards (harder).

The next sections discuss some of those design guidelines in two parts: policies aimed at calming traffic, and policies aimed at making places more walkable. The two clearly are related, but they have a different emphasis.

Calming traffic

Implementing new design standards requires education, engineering, encouragement, and enforcement. Traffic calming is primarily an engineering strategy. Slowing traffic, not hindering access, should be the goal. Typical measures for slowing traffic are speed humps and raised crosswalks; narrower lanes, bends, and center islands; roundabouts; and partial and full street closures.¹⁵

The tension between efficiently moving traffic and keeping streets livable is obvious: if you try to maximize on one you lose a lot on the other. Figure 5-3 illustrates the trade-offs. Traffic engineers occasionally find themselves in what for them is a bizarre world: they have built a wide street and are now being asked to re-engineer it so that it does not work as well (for traffic). Appendix C explains, however, that the trade-off in travel time may not be as severe as some policy makers believe: on main streets with signalization, it may be that throughput can be maintained or even increased at slightly slower speeds.

For main streets, some on-street parking: can be critical for merchants; can be compatible with, and perhaps even enhance, livability by creating a buffer between the traffic and the pedestrians; and can calm traffic and (in some cases) improve flow. Among the many design ideas for mainstreets are the following:

- A distinct gateway announcing entry to a shared street (one shared by motorists with pedestrians and bicyclists; by passers-by with residents, shoppers, and workers).
- Sidewalk bulb-outs at intersections to reduce cross distance and clear time for pedestrians.
- Building design that has a strong relationship to the street.
- Wider sidewalks. Where space is tight, on-street parking (parallel or diagonal) can be punctuated by mid-block sidewalk bulb-outs for seating and gathering.

A HISTORY OF TRAFFIC CALMING AND DUTCH WOONERFS

Traffic Calming began as a grassroots movement in late 1960s in the Dutch city of Delft, when residents upset about cut-through traffic tore up their brick streets to create serpentine paths in order to slow the cars. These shared streets came to be known as Woonerfs, translated literally as “residential yards.” (They have also been referred to as “living yards,” “streets for living,” “living streets,” and finally in Britain as “Home Zones,” which additionally refers to the network of streets surrounding and leading to the core of the neighborhood.)

A Woonerf is a shared street incorporating a range of modes and activities, with provisions for moving traffic, parking, socializing among neighbors, and allowing children to play safely in front of their homes. Traffic is allowed access (ideally between 100 to 300 vehicles per hour during the afternoon peak period, when pedestrian and vehicle conflicts are more likely to occur), but speeds are slowed by both physical and visual measures. The Dutch originally specified the ideal speed as being the pace of walking horse (about 10 mph). Most people walk at about two to three mph. One of the key design features to slow traffic is to narrow the travel lanes and break up the visual continuity for the drivers, emphasizing the street as a residential place, rather

than a channel for traffic. According to Donald Appleyard, “The design philosophy of the Woonerf is to create a kind of ‘gestalt’ message that the streets belong to the residents and that drivers must attend ‘incessantly to the fact that the car is only one of the users and a guest.’” Drivers are made to feel it is natural to drive slowly via physical and visual measures, which include the following:

- Clear and distinct gateways to enhance the neighborhood’s identity and sense of place
- Curves in the travel lane that break up the driver’s sight line
- Visual features that serve both the needs of the residents and to slow traffic (e.g., trees, planters, landscaping, furniture and play equipment)
- Placing vehicles and pedestrian space on the same level, and thus eliminating continuous curbs (demarcated sometimes by bollards)
- Use of varied pavement with special treatments
- Parking spaced occasionally throughout the woonerf

Driver education courses and testing, as well as grassroots campaigns within neighborhoods, will improve driver awareness and show how to use the traffic calming features appropriately. Enforcement strategies include defining vehicle liability laws so that pedestrians and residents have priority on traffic calmed and, particularly, on shared streets. For example, many countries—like the Netherlands—support giving legal priority to pedestrians and cyclists in shared streets, making the driver responsible for the burden of proof in any dispute.

The sidebar gives more information on the history and practice of traffic calming; this endnote¹⁶ gives yet more sources.

A common concern about traffic calming, especially when it includes narrow streets, is access for emergency vehicles. Indeed, design requirements that make things easy for emergency vehicles may preclude good designs for livability. Nevertheless, on a shared street, maximizing for emergency response may make no more sense than maximizing for traffic throughput.

The reasonable response is compromise. It may take some aggressive lobbying to get the ball rolling, but it may be possible to find mutually agreeable accommodations. The response a planner might want from the providers of emergency services is smaller, more maneuverable vehicles. That may be costly, but it could be worth the cost. What a planner might have to give in return includes:

- parking restrictions that allow fire trucks to make turns even when they’re tight;
- staging areas for emergency vehicles near fire hydrants to help fire trucks maneuver;
- design changes (e.g., connecting local streets to form a network, rather than relying on cul-de-sacs); and

- alley access, undergrounding of electric lines so they do not get tangled in ladders, reflective visible street signs, and fire hydrants placed at intersections.

Creating walkable communities

If we had to select a single measure of livable streets and public space, it would probably be “walkability.” Yes, there is drive-through and electronic commerce, but walking is basic to any economic and social activity that requires people to meet in places, which essentially means nearly everywhere (work, shopping, education, recreation). Walking is part of almost every trip. In most cases, if there are a lot of people walking on streets, it is because those streets are attractive places. Causality might go either way: the attractive streets attract walkers, or the volume of walkers encourage businesses and governments to make streets attractive.

The previous parts of this chapter have been about policies to deal with motor transportation, and most of those policies are about system-level issues, not neighborhood issues. Chapter 5 and the sidebar below talk more about design.¹⁷

Livable communities have to accommodate non-motorized rolling stock too: primarily bikes. Yes, also roller blades, skateboards, strollers, Segways, and so on, but put in the context of this report, those are a primary source of transportation for only an unmeasurable percent of non-recreational person-miles of travel.

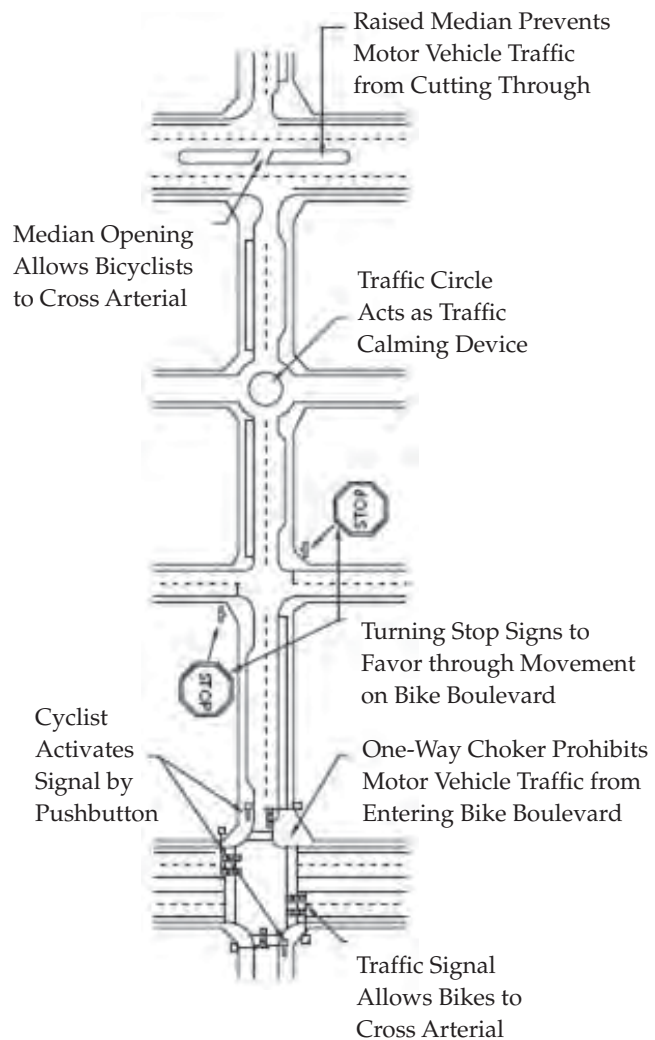
A bicycle in motion requires space to allow for the balancing and related weaving required to keep a bicycle upright and moving forward, and to give the bicyclist some margin for passing cars (left) and obstacles (right). That means about a five-foot width to ride comfortably. Figure 6-2 shows how bike lanes can be accommodated on the streets, as well as how to make bicycle boulevards. Placing bike racks on buses and allowing them on rail cars is an effective way to make cycling competitive timewise with a regional-level auto trip, even if the origin and destination are located in low-density suburban areas with poor transit service.

SUMMARY OF POLICY OPTIONS

We’ve covered a lot in this chapter, and done so using a non-traditional organization. There is much more to be said, and many other articles and books say it. Tables 6-2 and 6-4 (above) crib from some of the best works we have seen to give you a quick way to remember the many policy possibilities. They return to the supply-side (Table 6-2) and demand-side (Table 6-4) organization. For a more detailed matrix (more evaluation criteria) that assesses all the policy options that focus on urban congestion, see Downs’s (2004a) final chapter and mega-matrix.

So many policies, many of which move in different and potentially canceling directions. How does one assemble these into a comprehensive package of complementary, reinforcing policies that address transportation issues in the context of land use? Turn the page to Section III and our conclusions.

Figure 6-2. Typical bicycle boulevard features



Source: Oregon Department of Transportation 1995.

WALKABLE COMMUNITIES

By Paul Zytkofsky and Dan Burden

Walkability requires mixing important destinations (what we called Diversity in Chapter 5; one of the five “Ds”) so they are in close proximity to each other (two more Ds: Density and Distance), and doing so in a way that relates to the street network in an attractive way (Design, yet another D). Pedestrians are much more likely than motorists to notice the details on a building’s façade, items in a display window, the slope of a sidewalk, or the quality of a bench. A walkable community is a place that residents of all ages and abilities feel is safe, comfortable, and convenient.

Walking speeds and distances

The average adult walks three to four feet per second (about two to three mph); children, seniors, and people with disabilities move slower. The current standard used for pedestrian street crossing is four feet per second.

Untermann (1984) found that most residents will typically walk to destinations that are five minutes from their homes. At three feet per second, a person can walk from one-sixth to one-third of a mile in five to 10 minutes. Using the quarter-mile walk as the standard, a walkable neighborhood will roughly cover 125 acres.

Spatial needs

An average-size person takes up almost two feet of width while standing; that same person walking needs approximately four feet of width to allow for swaying or carrying a bag or briefcase. Since walking can often be a social activity, there should be at least six feet for two individuals to walk side-by-side.

Sidewalk design

Sidewalks should be buffered from the street. In commercial areas the buffer is often the “furniture zone” where utilities, signs, benches, transit shelters, planters, and trees should be placed.

In a low-density commercial zone the zone should be at least four feet: five to eight feet is most common. In a residential area, a continuous landscape strip distances and buffers pedestrians from the road, accommodates curb cuts without sloping the sidewalk, and can create a tree canopy to shade the sidewalk: it should be at least six feet wide to allow for healthy growth of trees. If space is tight and the sidewalk must be attached to the curb, an extra foot or two should be provided as an added buffer from the street. Avoid rolled curbs: they encourage drivers to park in the pedestrian realm.

Street design

Noise and safety risk cause people to distance themselves from streets with high speeds and high volumes. To encourage walking, residential streets should be designed to keep speeds at or below 25 mph, while larger, busier avenues should be kept at or below 35 mph.

Crosswalk design

Slower vehicle speeds and shorter crossing distances improve safety. In streets with on-street parking, curb extensions (bulb-outs) at intersections and mid-block crossing locations slow traffic entering and exiting the street, reduce the crossing distance, and make it easier for pedestrians to see and be seen by motorists.

Crosswalks in commercial areas should be at least 12 feet wide to allow people to flow in both directions. On wide streets, median refuges or islands with a median nose provide added protection. The clear path for pedestrians through the median island should be six feet. Countdown signals that let pedestrians know how much time they have left to cross reduce stress and accidents. At mid-block crossing locations, add refuge islands. An innovative design is to angle the crossing area by 45 degrees, forcing pedestrians and cyclist to look in the direction of oncoming motorists, and engage them visually. ■

CHAPTER 6 NOTES

1. We take you back to the Cartesian mind-body duality: the body (legs) are the vehicle to get the mind (spirit) where it wants to go.
2. We are not counting the growing expense of refitting old bridges and overpasses to meet higher maintenance and earthquake standards. We assert, by definition, that those are not about traffic safety and accidents.
3. “Anecdotal” because we know of no analysis that tries to control for other factors that affect development in the area, especially market forces.
4. The windfalls are smaller with congestion pricing, but there still are some: the value that commuters place on faster travel above the congestion fee still capitalizes into land values.
5. *Kelo vs. New London* (U.S. Supreme Court); Measure 37 in Oregon; Proposition 207 in Arizona.
6. The large proportion of congestion associated with disabled vehicles seems to suggest that policy should concentrate on ways to limit the impact of disabled vehicles rather than on pricing congestion. The problem is that disabled vehicles could not have this large effect in the absence of high traffic volumes. Moreover, the volume of traffic would increase with improvements in clearing obstacles;

experienced commuters implicitly take into account the probability that a disabled vehicle will slow their trip when they make decisions about when and where to travel.

7. http://www.ops.fhwa.dot.gov/congestion_report/index.htm (accessed February 2007).
8. We've seen modest accidents (no major injuries) block a lane and a half on a crowded two-lane (one-direction) interstate. Traffic backed up for at least three miles. The lanes probably held almost 200 vehicles per mile at the speed of less than 5 miles per hour. Average delay was probably 30 minutes per vehicle on one side. There was a more modest backup in the other direction as motorists slowed to take their turn looking. Doing some assuming and multiplication, that modest accident probably caused around 700 hours of vehicle delay, with a value of \$5,000 to \$20,000, depending on assumptions about persons per vehicle, value of time, and value of freight. Downs (2004a, 64) cites data suggesting that the average delay of a lane-blocking accident could be in the range of 1,000 to 5,000 vehicle hours. Cost of wasted gasoline might add another 10 percent. There is a good reason to move these accidents off the travel lanes quickly.
9. One study cited found a benefit-to-cost ratio of 23:1.
10. Hint: it has to do with the pricing of road use.
11. We refer to this method as benefit-benefit analysis: the highway creates travel benefits, and the cost of building it provides an economic benefit. The benefit of the benefit-benefit approach is its simplicity; its drawback is its irrelevance.
12. One of the major impediments to low-cost transit service is the fact that transit often gets stuck in auto traffic. Infrastructure that allows transit to avoid congestion delays would decrease the perceived cost of using transit relative to using a car. The congestion that remains on the bridge would further encourage travelers to use the alternative mode. Fitzroy and Smith (1993) note that even if congestion is priced, thus encouraging travel by other modes, the cars that remain on the highway continue to impede the progress of transit vehicles. They describe the impacts of policies implemented in Zurich in the late 1970s designed to give transit vehicles priority over autos. Between 1985 and 1990, transit trips increased by 33 percent. SAFETEA-LU allows some of this substitution to occur; it should encourage MPOs to be more explicit about, and take responsibility for, decisions about the relative amount of investment in different modes.
13. One theory is that since human's top sprinting speed is about 25 mph, we have evolved to survive head trauma at that speed.
14. This is an example of a general phenomenon: risk compensation. Reducing risk has lower effects, in this case on traffic accidents, than might be expected because drivers naturally exercise less caution as the risk of an accident falls. The economics are identical to those of latent demand: lower time costs encourage more travel, less risk encourages less caution. Car safety features, such as seat belts and anti-lock brakes, have had similar effects on driver behavior.
15. "Traffic Calming Measures." <http://www.trafficcalming.org/measures2.html>, accessed on June 18, 2006
16. For more on traffic calming see Ewing (1999); Appleyard (2006); Federal Highway Administration (1994); Ben-Joseph (1995); and <http://www.homezones.org.uk/public/guidance/index.cfm>
17. We could theorize about pricing policies for pedestrians, but we'll spare you.

The best way to predict the future is to invent it.

—Alan Kay



In preparing for battle I have always found that plans are useless, but planning is indispensable.

—Dwight D. Eisenhower

SECTION III.

INTEGRATING TRANSPORTATION INTO A COMPREHENSIVE REGIONAL PLAN

Section II provided a relatively complete listing and evaluation of individual transportation and land-use policies. More is required, however, to get regional adoption and implementation of a coordinated package of policies. Section III addresses how to assemble subsets of those policies into complementary packages of transportation and land-use policies, and provides advice on how to run a regional planning process to develop and get agreement on a policy package (i.e., on a regional plan).

Chapter 7. Coordinated Land-Use and Transportation Policy. Chapters 4, 5, and 6 (and several appendices) provide the details for evaluating individual policies. Chapter 5 makes the case that land-use and transportation policies should be considered and adopted together so that they reinforce one another. But how should they be assembled? What makes sense technically and politically? Chapter 7 gives some direction. It describes two policy packages: (1) Do what you do a little bit better (marginal improvements to existing policies): the likely policy future; and (2) Substantial improvements in the efficiency of pricing for urban transportation: the desirable policy future. The second package builds on the first. A final section discusses ways to transition from the first policy package to the second.

Chapter 8. A Process for Integrated Land-Use and Transportation Planning. We went back and forth on whether this chapter should precede the previous one. The argument for that alternative organization is that talking about a process for selecting policies (for developing a plan) *after* we had already discussed what those policies should be might interpreted as disingenuous and arrogant. The argument for the organization that we settled on is that it is clearer to follow the policy chapters (Chapters 4, 5, and 6) with a chapter that packages those policies (Chapter 7), and then follow the four policy chapters with a process chapter. So, Chapter 8 provides principles for political engagement and public participation in policy evaluation and adoption, but it presumes that such a process subsumes an evaluation of policies and decisions about a policy package. It draws on recent examples of attempts to do integrated transportation and land-use planning at the regional level.

Coordinated Land-Use and Transportation Policy

Organization: Chapters 4, 5, and 6 evaluate individual land-use and transportation policies. There are many, and many overlap: some are complements, some substitutes, and some in conflict. Which ones should a city or metropolitan region adopt?

The correct answer for any specific jurisdiction is: It depends, and we do not know enough about your circumstances to give you a definitive answer. But correctness, here and elsewhere, is often not very satisfying, so we are willing to speculate about some general answers that provide a place to start local debate and analysis.

This chapter has three main sections: one for each of two packages of policies, and one describing how to get from the basic Package 1 to the more-advanced and effective Package 2.

Basics: We accept that a systematic and broad effort at pricing road use will not happen soon. Other policies will be used to try to bring congestion to efficient levels and to reduce the negative externalities of urban transportation systems and travel.

- **Package 1: Do what you do a little bit better (marginal improvements to existing policies): the likely policy future.** This is the package we expect most metropolitan areas making a serious attempt to rethink and deal with urban congestion will adopt. It contains elements from many transportation policies: adding highway capacity where it provides big benefits, TSM, making travel safer, adding transit (especially if the federal government continues to pay most of the cost of rail), increasing emissions standards, and adding easier demand management policies (such as increasing the fuel tax). Land-use policies include organizing stronger regional institutions or ad hoc groups, reforming land-use and design policies to better accommodate multimodal transportation operations, and using user fees to get closer towards the efficient pricing of infrastructure.
- **Package 2: Substantial improvements in the efficiency of pricing for urban transportation: the desirable policy future.** This package is the one we think metropolitan areas should adopt. It builds on the first; in most cases it does not introduce new policies, but takes existing ones farther. Package 2 is a pumped-up version of Package 1. It focuses on proper pricing for highway trips. Transportation elements include evaluating projects based on full costs and benefits, properly pricing roads and parking, adding capacity where revenue from pricing pays for it, making better use of transit (including encouraging the provision of private transit), and properly pricing environmental externalities like emissions and noise. Land-use policies include strengthening the regional institutions, expanding land-use and design regulation (e.g., controlling development along bypasses), and using pricing policies for infrastructure.
- **Steps toward a transition to efficient transportation and land-use policies.** What can metropolitan areas do in the short run to increase the likelihood of

a transition from Package 1 to Package 2 in the long run? The most difficult part of getting from Package 1 to Package 2 is beginning to properly price road use. Second-best policies that approximate marginal-cost road pricing, such as fuel taxes and parking fees, can build a bridge from the short run to the long run.

Shortcuts: This chapter is the shortcut for Chapters 5 and 6. If you have already read and understood the policies in Chapters 5 and 6, you'll be able to read Chapter 7 more quickly; you can skim the discussion of how the policies work. If you skipped or skimmed Chapters 5 and 6, you can treat Chapter 7 as a practical, hands-on summary of and partial substitute for those chapters, though you'll miss a lot of the details of the policy evaluation.

Economists are easier to condemn than construe. On the one hand, as the successors to Malthus, they are the flag bearers for the Dismal Science. On the other hand, in the debate about global limits, when sooth needs saying, most are more likely to turn to Pollyanna than Cassandra. They are technological optimists. They acknowledge environmental problems as bad but believe them tractable. They point to steady increases in standards of living and believe it is more defensible to project that trend forward than to predict its collapse.

Thus, it is unexpected to find many economists who have studied the problem of congestion so pessimistic about the ability of government to effectively address it. Certainly government has had to deal with growth in demand for other public services. Why do many economists believe that government policy can't handle the demand for trips?

Chapter 2 gave part of the answer: economists believe that congestion is a necessary condition in an efficiently functioning urban area. If prices are right, some level of congestion is almost certainly efficient—congestion indicates expensive highway capacity is not idle; people want to be in urban areas. Chapters 4, 5, and 6 gave another part of the answer: government policy does not price trips to account for their full cost or to ration the trip-making capacity of its highways. If you are going to sell tickets to the Super Bowl for 25 cents each, you will never be able to build a big enough stadium to accommodate all demand and eliminate congestion. Some utilities, like power, telephone, and some water utilities, understand this point and use price to ration supply. For many reasons, some of which we've covered in this report, transportation is only beginning to be viewed as a utility; pricing is farther off yet.

The nature of transportation infrastructure, the institutional arrangements that provide it, and the expectations of the public and interest groups that use it suggest that a systematic and broad effort at pricing road use will not happen soon. Even if relatively strong pricing policies were adopted, the growth of population and real incomes will be pushing the demand curve out at all prices. There is increasing agreement that supply-side solutions to congestion are at the point of decreasing marginal returns. Almost everyone—including engineers and policy makers—gives at least lip service to the idea that metropolitan areas cannot build their way out of congestion. Until there is costless transportation, there will be congestion. The challenge for policy is not to eliminate congestion (an impossible task), but to bring it to efficient levels.

The evidence we have reviewed points to a preferred direction for changes in land-use and transportation policy to address urban transportation and quality of life: identify the magnitude of the externalities and use pricing mechanisms to correct them. Though that recommendation is technically defensible, it has both theoretical and practical problems for implementation we have already discussed. Given the intricacies of the interactions among key variables, how can one have any confidence that any individual policy or



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A preferred direction for changes in land-use and transportation policy to address urban transportation and quality of life is to identify the magnitude of the externalities and to use pricing mechanisms to correct them. The toll on the Golden Gate Bridge, for instance, has been adjusted for time of day.

set of policies actually makes society better off? Given that such confidence, if achieved, is as much a matter of faith as of fact, how can policy makers organize the faithful to lead the agnostics?

This chapter outlines two policy packages. The first describes what we think most metropolitan areas are *likely* to do. The second describes what we think they *should* do, but are unlikely to do in the short run, and perhaps not even in the long run.

- **Package 1. Do What You Do a Little Better.** This is the package we expect metropolitan areas that are making a serious attempt to rethink and deal with urban congestion will adopt. That certainly includes most metropolitan areas, and probably all of them (though some will go farther than others in the scope of policies they adopt to deal with congestion).

Unlike environmental impact statements, we do not include a pure Do-Nothing (No-Build) package. The problems and planning of transportation and land use are changing; metropolitan areas will not stand still as the changes sweep by. Thus, our first package assumes that metropolitan areas will do something (that is, that they will adopt some new policies) to deal with increasing congestion problems and their related land-use problems. It is not business as usual, but business as we expect it to be if policy makers are unable to adopt the more significant policies contained in the second policy package. Policy stays within the current policy envelope, making marginal changes to existing policy in step with marginal changes in technology and public attitude.

Package 1 is where we think policy will end up if metropolitan areas follow the trends, put in their time on planning, do the standard process, and make the marginal adjustments to existing policy that do not require putting a heroic amount of political capital at risk. In particular, this policy package does not include policies that would price congestion.

- **Package 2. Substantial Improvements in the Efficiency of Pricing for Automobile Trips and Other Public Services.** This package is the one we think metropolitan areas should adopt. It builds on the first; in most cases it does not introduce new policies, but takes existing ones farther. Package 2 is a pumped up version of Package 1. It focuses on proper pricing for highway trips. Chapter 4 described why pricing policies—especially policies that increase the price of driving on roads, and especially during peak hours—are likely to be the most efficient solution to congestion, pollution, and other externality problems that result from the failure of market activities to reflect the full social costs of the resources they consume (including environmental quality and travel time). Despite the difficulties of adopting this package, we believe it is necessary if metropolitan areas are going to make headway on congestion problems. Package 1 may slow your slide toward gridlock, but slide you will. Package 2 could also include policies that push transit now so it will pull later. Once auto prices more closely approximate true costs, additional investments in transportation capacity may be warranted.

We describe the policies and their likely effects as a package.¹ For the most part, the effects we describe are ones that we think could occur within 10 years if the packages were adopted today, with two notable exceptions. First, some of the packages include technologies that, though demonstrated as possible, are not fully field tested or mass produced. Second, land use and its impacts on transportation change slowly in response to public policy. Some of the impacts we describe, therefore, may be 15 to 20 years off.

Plenty of people have tried to assemble packages of policies for regions. Dozens of regional planning efforts in the last 10 years have done that. The sidebar gives a recent example from a project in the Chicago metropolitan

For the most part, the effects we describe are ones that we think could occur within 10 years if the packages were adopted today.

EXAMPLE OF COMPREHENSIVE AND COMPLEMENTARY POLICIES FOR A REGION

1. Encourage Redevelopment, Re-use, and Infill
2. Achieve a Balance Between Jobs and Housing
3. Provide Affordable Housing Opportunities
4. Promote Livable Communities
5. Promote Economic Vitality
6. Promote Diversity and Inclusiveness
7. Promote Compact, Mixed-Use Development
8. Promote Telecommunications Infrastructure and Technology
9. Promote Walking and Bicycling as Alternative Modes of Travel
10. Ensure that Transportation Facilities Are Context-Sensitive with Natural Resources, Historic Resources, and Community Values
11. Coordinate Land Use and Transportation
12. Maximize Use of Existing Infrastructure
13. Protect Water Resources
14. Plan for Sustainable Water Supply from Ground, Surface, and Lake Michigan Sources
15. Protect and Enhance Biodiversity
16. Enhance and Connect Green Areas
17. Preserve Farmland

Source: The Northeastern Illinois Planning Commission, <http://www.nipcc.org/2040/>

area that was selected by the American Planning Association as the national plan of the year in 2006. But that list mixes (to some degree) policies with goal statements and leaves out a lot of detail on implementation. We try to provide some of that detail in this chapter.

A big question is, What can metropolitan areas do in the short run to increase the likelihood of a transition from Package 1 to Package 2 in the long run? We conclude this chapter by providing some ideas about that transition.

PACKAGE 1. DO WHAT YOU DO A LITTLE BIT BETTER (MARGINAL IMPROVEMENTS TO EXISTING POLICIES): THE LIKELY POLICY FUTURE

Why would a metropolitan area pick this package? Because it can. Most Metropolitan Planning Organizations (MPOs) believe (and the evidence we have presented supports this belief) that metropolitan transportation investments need to shift from additional highways to Transportation Systems Management (TSM), Transportation Demand Management (TDM), and transit. But congestion pricing is hard to implement and big transit systems are hard to fund. Choosing this package allows MPOs to show a response to a pressing problem, to make some progress on solving it, and to avoid the controversy of more radical solutions.

The policies are primarily marginal adjustments to existing policies, with a leaning toward some politically doable demand management (including land-use policy) and alternative modes, which can be justified by the federal requirements in the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU), the Clean Air Act, and the preferences of a majority of the people who get involved in the regional planning processes.

Transportation Policies This Package Might Contain

Squeeze more out of existing highway capacity without squeezing demand (Transportation System Management)

The order of the day is to work smarter—there is little debate about the desirability of getting more efficient (though some attempts at efficiency can cost more than they save). The idea of getting more performance from existing investments should pass political muster if the technical justification is unequivocal. Metropolitan areas will phase-in more TSM: basic signalization (smart and coordinated) and some high-tech. Ramp metering can smooth flows, but it can have the disadvantage of penalizing shorter, central trips. Message boards on highways, highway cameras, Internet sites, and call-in sites (over half the states in the U.S. have 511 calling or clicking) provide instantaneous information about traffic conditions. The evidence for the net benefits of improved incident response is compelling, and no technological obstacles exist. Congestion monitors on freeways that transmit information to computers installed on vehicles are feasible. A big part of the program (possibly the most expensive part) will be more HOV lanes, created by either squeezing existing roadway or by adding more lanes, with coordination of HOV lanes and transit (e.g., park-and-ride lots near entrances to HOV lanes).²

Add highway capacity

The era of freeway building is over, but ample opportunity still exists for adding highway capacity. Metropolitan areas will certainly add small pieces of highway where they are (a) possible, and (b) likely to provide big benefit. Big metropolitan areas growing at the urban fringe will be pressed to add limited-access highways. In many metropolitan areas, opportunities remain for completing connections: thus, we expect marginal additions to

the proliferation of major state highways, interstate spurs, and bypasses that punctuate metropolitan areas. Most of those additions will be for routes planned long ago; many already have secured the right-of-way. Most of the new arterial capacity will come from construction of suburb-to-suburb and bypass highways.

Make travel safer for all travelers

Every metropolitan area can certainly make cost-effective investments to improve safety. The benefit-cost calculations here, though controversial, are not difficult. Considerations of bicycle and pedestrian safety, coupled with what is typically a stated policy objective of increasing trips by those modes, should increase investments in traffic calming. Those investments will include one at the intersection of transportation and land use: converting some sections of arterial strips into main streets.

Help transit

The impetus for this report was the amount of attention that planners and the public are giving the interaction of transportation and land use and the failure of building road capacity to reduce congestion. That realization has found its way into metropolitan, state, and federal policy—SAFETEA-LU formally acknowledges the need for nonhighway solutions to the problem of urban congestion and frees funds to try those solutions.

The most obvious nonhighway solution is improvements to traditional transit. We expect those improvements to focus on improving bus service for most metropolitan areas; the final section of Appendix C gives some idea of the relative effectiveness of ways such improvements might be made. More money into transit will decrease the perceived total cost to a traveler of a transit trip relative to an auto trip.

While some large metropolitan areas may bust for bus, many will rail for rail. That push comes only partially from technical considerations. Most of the evidence suggests that, in the short run, the benefits of rail are not worth the cost: it is a very expensive way to induce a small increase in transit ridership. The strongest technical argument is theoretically plausible but empirically contentious; namely, in the long run, land will develop and redevelop in such a way that rail ridership will take a substantially larger share of total trips, congestion will be substantially less than it otherwise would have been, and urban areas will be better places to live in so many ways. Part of the argument is that a fixed guideway system is necessary if developers are to have confidence that the transit capacity to support dense development will always be there; rather than having cars chasing development, development will chase rails. But the upside of the lack of a fixed rail is flexibility: roads are everywhere, so buses can be also; they can respond to changing demand.

There are other points relevant in the debate about rubber or rail. First, trains are cool and buses aren't. Or, more rigorously, rail has the potential to offer greater convenience and amenity to riders than buses, so (other things equal) more people will ride. This argument is the one heard most often from light rail and streetcar advocates: riding rail is more pleasant and social than riding a bus, so a lot more people will do it.

Those benefits for rail come at a cost. Buses can use the existing street network. Heavy rail certainly cannot; light rail can, but there are additional construction costs, including the substantial costs in delay, inconvenience, and lost business during the construction.

Even if the net benefits (in the rigorous sense of benefit-cost analysis, as elaborated in Appendix D) of rail were less than a rubber-tire alternative, the benefit of a rail system may exceed the cost *from the perspective of metropolitan*

While some large metropolitan areas may bust for bus, many will rail for rail.

areas because those areas usually pay only a small part of the costs of building such a system. Most systems built in recent years had about 80 percent or more of the project funded federally. The incentives have been to go after rail. One of the benefits of the flexible funding that SAFETEA-LU provides is that regions will now be spending their own money on these projects rather than use-it-or-lose-it funds that forced a particular transportation solution. Seattle, for example, has been working for at least 15 years on rail investments originally scoped to cost on the order of \$10 billion to \$15 billion. That full system is unlikely if local voters must either raise those funds or transfer them from funds used for highway improvements.

Slowly increase emissions standards

Transportation problems are more than the lost travel time from congestion. Chief among other externalities is air pollution. Much of the improvement in vehicle performance will come at the federal level. But state and metropolitan governments can and have pushed even faster for cleaner, quieter cars. The history of policy in this area shows a heavy reliance on regulatory standards to control these externalities. That trend is probably the best prediction of the future: a majority of metropolitan areas will not push to directly price vehicle emissions.

Incrementally introduce demand management policies

Though big changes in pricing are farther down the road, we expect to see some small change in most metropolitan areas. At the federal level, the last increase³ in the fuel tax of any size (4.3 cents per gallon) was enacted in the early 1990s. Removing federal tax incentives that encourage employers to offer free parking has been talked about for two decades: recent changes (2002) now allow employers to give employees the benefit of subsidized transit service as an option to the free parking that most provide.⁴ This is, however, an employer option: those that own excess and capitalized parking capacity will see the benefit as an extra cost with no offsetting savings.

As states try to deal with the substantial difference between their estimates of revenues and needed highway investments, the fuel tax is a likely place for them to turn. Even if they do so while considering only political acceptability and ignoring economic efficiency, they could do much worse: there is at least a rough correlation between some of the external costs of the auto and the amount of driving that occurs, which is highly correlated with fuel consumption in the short run.

Some states have discussed (it's all talk at this point) approximating a highway-use fee by taxing based on vehicle-miles traveled (by reading odometers). Per our earlier arguments, this is mainly just a refinement to the gas tax that charges for general use. It lacks the two key components of congestion pricing: pricing by time and location of travel.

We expect a few experiments with the non-pricing-based TDM policies (e.g., trip-reduction ordinances), especially in suburban commercial and light-industrial areas. The temptation to push the cost on to the private sector will be too great for government to resist. Most of the programs will be voluntary, but some will provide economic incentives and disincentives to get the desired shift in travel behavior.

Land-Use and Public Service Policies This Package Might Contain

Organize regional institutions for coordinated solutions to regional problems

We could just have easily put this class of policies under transportation. Most regional agencies that exist today were created for the single or primary purpose of dealing with transportation problems. Given that the money supporting this planning is federal and state, and no comparable federal and state

One of the benefits of the flexible funding that SAFETEA-LU provides is that regions will now be spending their own money on these projects rather than use-it-or-lose-it funds that forced a particular transportation solution.

funding is available for regional land-use planning, transportation is the likely Trojan horse for integrated regional planning. In fact, this observation is one more piece of evidence about the correct answer to the planners' query, What leads: land use or transportation? Answer: follow the funding.

But if integrated policy is what regions are after, regional policy is more logically integrated under the heading of land use than transportation. Metro in Portland has justified using various sources of funds from transportation-related sources to support Region 2040, a long-run regional planning effort with land-use roots. It is possible that a few more regional organizations with powers like those in Portland and Minneapolis/St. Paul might form, but we expect that regional government will be primarily advisory or ad hoc (i.e., created to address a single issue, like a utility district). Some examples of new institutional arrangements: Envision Utah, and the Chicago Metropolitan Agency for Planning, a merger of the Northeastern Illinois Planning Commission and the Chicago Area Transportation Study (the former MPO).⁵

The role of land use in metropolitan transportation planning is so obvious as to be inevitable. MPOs need to allocate population and employment to the small-area zones in their transportation models so that the models can run. Those allocations happened outside of ("are exogenous to") those models. The connection between trips, people, and the buildings necessary to allow people to do anything in a transportation zone (live, work, shop, recreate) is clear. MPOs are, by choice or by force, having to consider a long-run development pattern to do their transportation planning and are finding themselves involved, at some level, with regional land-use planning (the visioning at least, but little on the regulation). Tying land use into the MPO long-range transportation plan (LRTP) process is probably the best single hope for the coordination of planning for land use and transportation in metropolitan areas.

Even without a strong regional organization, an obvious need (and required by the federal laws governing the operations of MPOs) exists for regional transportation and land use to integrate with local efforts. The sidebar gives one example of how this might occur.

Tying land use into the MPO long-range transportation plan (LRTP) process is probably the best single hope for the coordination of planning for land use and transportation in metropolitan areas.

INTEGRATING PLANNING FOR LAND USE AND TRANSPORTATION, REGIONALLY AND LOCALLY

To effectively integrate transportation and land-use planning, analysis and funding should occur in the right order:

- *Identify and rank areas for growth.* Define boundaries and policies for the areas to be protected from development. Rank the remaining areas for development potential based on factors like location, capacity, infrastructure cost, and market demand. MPOs should work with local governments to plan for these targeted growth areas, making sure that transportation projects contained in the Long-Range Transportation Plan (LRTP) support this primary goal of focusing development.
- *Develop land-use and transportation plans and funding programs that support a 4-D development strategy (density, diversity, design, destinations).*
- *Implement the development strategy.* Those working on the MPO's funding program for the LRTP and Transportation Improvement Program (TIP) should work with those responsible for land-use planning in the constituent localities and with the various stakeholders. The TIP should be coordinated with local Capital Improvement Plans (CIPs).
- *Evaluate road projects.* Consistent with a theme of this report, a vision for community development, articulated in a comprehensive plan, should lead the development of a transportation system, and alternatives to new roads (e.g., access management, demand management, transit) should be considered at the same time as road projects to encourage a comparison of relative benefits, costs, and trade-offs. If a new road is to be built, it should link into a greater network; if the road extends outside the targeted growth area, access management policies and possibly conservation easements should be established in those places before moving ahead with the road project.

Incrementally reform land-use and design policies

The extent of changes in land-use policy will depend in part on what policies a jurisdiction is now using. For example, Oregon law already requires cities and metropolitan areas to establish urban growth boundaries (UGBs), a policy that many planners trying to manage growth in other states view as highly desirable and barely achievable. In Oregon, growth management policies will go farther; in other states, many metropolitan areas are unlikely to adopt the policies in place in Oregon now.

Besides the standard list of growth management policies, we expect many metropolitan areas to push for land-use policies that reinforce a rational and mutually supportive connection between land-use policy and transportation policy. As we just noted, MPOs will almost certainly be moving in this direction for several reasons.

Planning is one thing; regulation is another. Jurisdictions will vary in the forcefulness of their regulations. Some will do planning studies that will provide information to developers about how they could change design and make their projects more marketable without substantially increasing their costs. Others will require certain design features (e.g., bus turnouts). They will implement design standards that attempt to provide good vehicle access to areas planned for relatively high-density development and maintain a pleasant environment for pedestrians (e.g., the neotraditional development models; traffic calming to reduce and slow traffic in residential areas and direct traffic to arterial streets; and safe crossing of arterials for pedestrians and bicyclists).

The metropolitan areas at the forefront of growth management regulations will continue experimentation with policies like transit-overlay zones that allow, or even require, development at higher intensities with designs that improve access to transit. Most unconventional policies are likely to be non-binding, either intentionally (e.g., policies that “encourage,” “promote,” and so on) or by default (e.g., maximum lot sizes applied in small subareas where market conditions are already building at the required densities or greater).

Our guess is that policies relating to a jobs/housing balance will get lip service, but the difficulties in defining, implementing, and enforcing them means they will get little more. Concerns about having housing near employment opportunities will probably get addressed through mixed-use zoning or performance standards, which would be more sensible than numerical quotas for a specific mix of jobs and residences.

Planning is one thing; regulation is another. Jurisdictions will vary in the forcefulness of their regulations.

Use more pricing policies for infrastructure

Our observation is that during the second half of the twentieth century, moral and political sensibilities shifted slowly. People who grew up in the Depression and fought in World War II had at least a tolerance, and probably a respect, for all levels of government. They paid taxes and assumed their public servants would use them well. Many of their children learned to distrust government in the 1960s and to enjoy the benefits of capitalism as middle-class consumers in the 1980s. The ascendancy of Republicans in the federal legislative and executive branches reinforced the rhetoric of big government, inefficiency, and tax cuts.

The direction from taxpayers to local government has pointed toward working smaller, working smarter, outsourcing, and—the point here—toward user fees. The logic of charging true cost is supported by the long-run and incremental political shifts just described: people should pay for the services they use. The problem is getting from the rhetoric to the reality. Plenty of evidence supports the theory of full-cost pricing, but that theory is not enthusiastically embraced when various special interests see what it means to them. We have given examples in previous chapters.

We are relatively confident about the direction of change (toward more pricing and higher user fees) but not about the magnitude (what percentage of full cost will jurisdictions be able to charge). Most types of utilities (water, sewer, electricity) are well aware of the importance and effects of pricing; any reluctance to go to full-cost pricing is more political than technical. Not only will public utilities increase their efforts to charge full fare for the obvious direct costs (capital, operation, maintenance), but some of them will try to incorporate some estimate of environmental externalities into their costs and rates.⁶ These efforts by other utilities will lend support to the idea that perhaps transportation is a utility also and should be priced as other utilities are.

Likely Effects of These Policies on Travel and Land-Use Patterns

The details of our answer to questions about the likely effects of these policies are found in Chapters 4, 5, and 6. We just summarize the key points here:

- There will be short-run improvements in travel times on some routes resulting from increases in capacity. For growing metropolitan areas (or attractive areas within them), the improvements will almost certainly be short run: growth will come (in part because of the transportation improvements) and move congestion back toward its previous levels or beyond.
- Transit ridership will increase, but not enough to reduce congestion or operating subsidies. Transit agencies will increase bus services and, if they have rail, build park-and-ride lots around suburban rail stations to attract ridership. The concern about increasing congestion will cause more money to be shifted from gas taxes and other sources into transit: the percent of transportation funding spent on transit will increase. The short-run benefits will be less than the costs; the bet is that the transit network will get fused to the highway system to create a backbone that better urban development can conform to in the future. We discuss that point more in Package 2.
- TDM, combined with congestion and HOV lanes, will provide a small increase in the average number of riders per vehicle. Those increases will not be enough to have a big effect on congestion because it will be difficult to design them in a way to keep other trip makers from shifting to highways where congestion is reduced, unless they include policies that increase the price of auto trips.
- The effects of the previous transportation policies on land use will depend on land-use policies. In the absence of regulation, new highway capacity (and, we believe, new transit capacity) will continue to provide economic incentives for low-density development in suburbs. Radial transit systems encourage development in the urban center (though the effect might not be very strong) and low-density development in the suburbs (to the extent they attract riders). The few bypass highways that can still be built in metropolitan areas will cause instability in the development of suburban land, with the ultimate outcome again depending on land-use policy (but the likely outcome being the one that history suggests: strip and subcenter development along the bypass and, hence, at the urban fringe).

Thus, the key question is whether the transportation policies that encourage decentralization will be accompanied by countervailing land-use policies that will curb suburbanization and encourage (or attempt to require) density. Even if such a balance can be achieved (which is unlikely in many places), the effects on the public that the planners are trying to serve are ambiguous, though we suspect they will net to a negative. Our suspicion is based on a simple observation: trying to achieve a social optimum by adopting policies that often attempt to cancel each others'



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The key question is whether the transportation policies that encourage decentralization will be accompanied by countervailing land-use policies that will curb suburbanization and encourage density (e.g., with mixed-use developments).

effects is inherently inefficient, and, more importantly, these policies leave drivers and developers still trying to optimize based on prices that send the wrong signals about the value of resources.

- The previous points notwithstanding, many new developments of the type discussed in Chapter 5 and supported by advocates of smart growth will be built. On this point, however, we tend to agree with Downs (2004a): that amount of desired development will be small relative to the total development. We are probably more optimistic, however, that such development can be not only exemplary (encouraging by its success more such development elsewhere) but also catalytic (e.g., getting as little as 30 acres developed properly as a main-street, mixed-use center may be enough to influence the form of development on 10 or even 100 times that acreage).
- Moderate improvements to reduce emissions of pollution and noise per vehicle will continue. Whether the per-vehicle improvements lead to a net improvement in ambient conditions is hard to predict. Our guess is that decision makers will be satisfied if they can maintain current ambient levels without having to resort to more stringent standards or pricing.

PACKAGE 2: SUBSTANTIAL IMPROVEMENTS IN THE EFFICIENCY OF PRICING FOR URBAN TRANSPORTATION: THE DESIRABLE POLICY FUTURE

Package 2 is not a substitute for Package 1—it is an enhancement. In general, it assumes that everything in Package 1 gets done, and then goes farther: Package 1 with an attitude. It does not, for example, ignore the expansion of highway capacity; rather, it assumes that the expansions done as part of Package 1 cover (perhaps more than cover) what are cost-effective investments in highway capacity (the main thrust, by far, of almost all MPO investment plans).

To pick this package, decision makers would have to be convinced that congestion is a crisis for the health of the metropolitan economy and its quality of life, and that the prescription we have articulated in this report is the correct one. They would do so knowing they might be risking their political futures on technically correct policies opposed by many powerful interest groups. In other words, it's not likely that a significant amount of Package 2 will be adopted by any metropolitan area in the short run. The technical arguments are compelling; our doubts stem from uncertainties about the desire and ability of decision makers to convince the public to accept pricing policies. In that context, this package serves two purposes: it provides an ideal to inform and motivate public policy, and it describes where we think policy should end up in the long run.

The package consists of stronger policies to price the auto to reduce peak-hour congestion. It generally assumes that transit plays an important supporting role, but that the region is shaped more by its existing highways than by its future bus routes or rail lines.⁷ Land uses are regulated to make transit work better, but investments in transit (especially rail) do not reflect an assumption that a transit grid is an organizing framework for future land use.

A second part of Package 2 could include substantial investment in transit if metropolitan decision makers believe that investment in a transit grid, though not cost-effective when evaluated with a conventional analysis and current market conditions, will pay off in ways not easily measured in the long run. We consider this to be "the Toronto model," in part because it may be what Toronto had in mind in the 1950s and in part because it's what the planners who point to Toronto as a model of transit planning believe happened in Toronto. For reasons we described in Chapter 5, we believe a necessary condition for the long-run success of this kind of commitment to and

Package 2 is not a substitute for Package 1—it is an enhancement. In general, it assumes that everything in Package 1 gets done, and then goes farther: Package 1 with an attitude.

investment in transit is the implementation of many of the highway pricing policies contained in Package 2. Unfortunately, the likely future is that decision makers will invest in big-ticket transit solutions before they will adopt pricing schemes that charge highway users the full marginal costs of their trips. We are certain such investments will be less efficient in the absence of better pricing of highway trips; we are less certain about whether the long-run net benefits of such investments will occur in the absence of highway pricing.

We assume that Package 2 contains all the policies we described in Package 1. Thus, we describe only new policies and additional effects in this section. The primary new effects concern demand for auto trips (because of increasing price), how future transportation investments would get evaluated, and the amount of investment in transit.

Transportation Policies This Package Might Contain

Improve the evaluation of projects and policies to provide a sounder basis for investment decisions

This could happen, and will to some extent, as part of Package 1. We put it in Package 2 because we are talking about a big shift in thinking for most MPO evaluation processes.

A lot of what we have discussed in this book relates to making decisions about the desirability of transportation investments (i.e., about the expected benefits and costs) based less on intuition and more on analysis. The components of such analysis are not arcane. They are well articulated in the planning of other public utilities. The idea of a comparison of full benefits and costs is articulated in a publication of the American Association of State Highway and Transportation Officials, *User Benefit Analysis for Highways* (2003), and in Appendix D of this report.

Those methods, however, are rarely applied with technical rigor or political commitment to investment decisions. We are not saying that technical arguments can or should carry the day: technical analysis is only part of what elected and appointed officials look at when making decisions on public policy. Rather, we believe that such analysis should be better than it is; that though technical analysis may eventually defer to political considerations, it should not be influenced too strongly by an analyst's guesses about political feasibility. Let the planners present a best estimate of a range of benefits and costs based on the information available; let the decision makers decide on how to use and weigh that information. Planning limited by planners' assumptions about political practicality will "have no magic to stir men's souls." Moreover, let those best estimates be based on the kind of information road pricing provides about how much and where people believe it is worth paying to have congestion reduced.⁸

Though technical analysis may eventually defer to political considerations, it should not be influenced too strongly by an analyst's guesses about political feasibility. Let the planners present a best estimate of a range of benefits and costs based on the information available; let the decision makers decide on how to use and weigh that information.

Push harder on demand management (TDM, especially pricing)

We listed this category of policies at the tail end of Package 1; squeezing demand with efficient pricing is at the heart of Package 2. It is the policies in this category that most clearly distinguish Package 2 from Package 1. Moreover, we believe that moving in this direction will be strongly helped by the previous policy: more rigorous evaluation of the benefits and costs of alternative transportation investments. Diminishing returns to capacity expansions (increasing marginal cost; decreasing marginal benefits) combined with increasing constraints on funding suggest a shift toward more strategic, targeted investments, including ones in policy instead of structures.

The theme of road pricing is to charge users for the full (marginal) costs of their automobile trips; these costs change depending on the time and location of those trips. In the economically ideal solution, vehicles are tracked by technology now available and pay for road use by amount, location, and time of day, a pricing concept not radically different from the ones used by

many public utilities (e.g., telephone and electricity) and by other transportation modes (e.g., airlines). Short of that solution, government could charge congestion prices on major limited-access highways, a policy consistent with the kinds of improvements in electronic communication discussed in SAFETEA-LU. Short of that solution, a simpler implementation of the pricing of congested roads is the pricing of vehicle access to high-intensity areas (e.g., the pay-for-a-permit system in Singapore or the London cordon pricing system). Short of that solution, increasing the fuel tax significantly could be a rough proxy for congestion pricing (inferior on the demand side for all the reasons we have already discussed).

Parking should also be priced properly. Chapter 4 showed that parking is a big component of the total cost of an auto trip. At the federal level, the tax incentives for providing subsidized parking to employees should be removed. The problem of pricing parking has many similarities to the problem of pricing congestion: it should be done at a regional level for all facilities. Such implementation is unlikely; the best one can hope for in the foreseeable future are price increases in the most congested areas. Subareas where the demand for access is inelastic (perhaps downtowns or suburban office centers) might be able to implement parking taxes without significant impacts on location decisions; for retail centers, the problems seem less tractable. At a local level, municipalities will have to make some unpopular choices about whether to abet congestion with cheap parking or risk the further decay of downtowns by charging full costs that push businesses to suburban locations. Shoup (2005a; 2005b) provides most of the detail any jurisdiction needs on the reasons and methods for better parking pricing and policy.



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If bypasses are built, they should include strong land-use controls that do not allow the development of new subcenters at all access points to the expressway. The strongest control would be to purchase the property or the development rights along the new highway.

Add highway capacity

We considered leaving this policy out of Package 2 on the assumption that all of the cost-effective new highway capacity, and probably more, would be built as part of Package 1. But given the emphasis in Package 2 on better evaluation, new highway projects might get built and different ones almost certainly would.

Given the thrust of evaluation and pricing in Package 2, increases in highway capacity should be limited to those fundable, at least theoretically, by revenues from congestion tolls (or equally rigorous benefit-cost criteria). Our prediction is that such a policy would lead to fewer and perhaps few new lane miles being added to existing highway systems in developed areas. Both the theory and examples we presented in Chapter 4 suggest that if the peak-hour drivers causing government to build more highways were charged the full cost of that building, they would find more efficient ways to satisfy their travel needs. Some building on key corridors may still be justified; express lanes for through-traffic might be built, but circumferential bypasses probably would not be. If bypasses are built, they should include strong land-use controls that do not allow the development of new subcenters at all access points to the expressway. The strongest control, and one consistent with our other prescriptions about market-based policy, would be to purchase the property or the development rights along the new highway as part of construction (but at pre-improvement land prices).

Many of the TSM policies we described in Package 1 would still apply: they might prove to be the least-cost ways of reducing congestion at places where revenues from congestion pricing suggest that decreases in congestion are most valued.

Partner with the private sector

Public-private partnerships, at one level, are old news. The great majority of urban highways, roads, and rails get built by the private sector, with public

money. That's a partnership. But the special and rising use of the term is for arrangements that include not only private construction, but also private operation, maintenance, and even ownership.

Given the market orientation of this report, it will be no surprise that we think introducing private-sector incentives for efficiency into the design-and-build process can save money. That savings, however, is in part a result of a different process: one that has a more focused bottom line and less concern about involving everyone until they are done wanting to be involved. The focus is return on investment, subject to compliance with regulatory requirements. Part of the efficiency comes by cutting out the process and project extras for which interest groups have acquired an appetite. Process is no longer our most important product.

Moreover, the big theme—that private efficiency and financing will refloat the boat—is, at best, debatable. Financing is not funding. Ultimately, someone has to pay, and the someones are all of us in our various capacities as travelers, homeowners, workers, business owners, taxpayers, and so on. *Maybe* experienced management and strong incentives allow the private sector to build, design, and operate 10 percent (even 20 percent) cheaper than a nimble, professional, quasi-autonomous government institution created for the same purpose. But it does not allow project costs to be cut in half. Private firms are only interested in running and owning the public transportation system if they can run it at a profit. Either they are going to find a way to charge users (e.g., tolls and fares), or they are going to require government to guarantee, in one way or another, a rate of return (which will have to be backed from other sources, like the general fund). Probably the bigger advantage of the public-private partnership is the pricing that it will require.

There are plenty of examples to suggest that turning over public assets to the private sector can meet short-run development objectives. But the long-run costs are harder to assess. Railroads opened the west, but gave away millions of acres of forests, minerals, and watershed in a checkerboard pattern. We suggest caution.⁹

Push harder on transit

It is better to narrow the gap between the price of auto trips and transit trips by decreasing the subsidy to the auto than by increasing the subsidy to transit. We prefer that the pricing policies of this package be in place before moving on to the strong policies for transit development that follow. Our expectation is that if the pricing of automobile trips were corrected, transit advocates would see the increase in the demand for transit that they want.

We tend to be conservative about big investments in transit in the absence of improved pricing for auto trips. Pricing policies address congestion by increasing the price of travel, which yields revenues but also decreases the demand for trips and the problems those trips create. Investments in new transit systems, on the other hand, address congestion by decreasing the price of travel (by supplying better alternatives), which increases the demand for public funds and increases the demand for trips and the pressure for decentralization. In short, other things being equal, correcting price distortions (which cause inefficient decisions about the use of scarce resources) by removing them is preferable to introducing new distortions intended to compensate for the original ones. At a minimum, the major investments in alternative modes should not precede the pricing policies—we allow they might occur in tandem.

An alternative view is that the gap between the price of a trip by auto and a trip by an alternative mode should be reduced however that reduction is possible: if it is easier to get more money for transit than to charge more for the auto, so be it. Moreover, the argument continues, one cannot expect



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auto drivers to shift modes if those modes are nonexistent or inconvenient relative to the auto (which, to us as economists, is another way of saying that nonauto trips cost too much more).

Even if one accepts that argument, one still needs to make some important choices about desirable transit investments: expensive rail systems may not be the best way to spend transit dollars if the objective is to reduce auto trips. With road pricing, the cost of each vehicle trip increases—the more people per vehicle, the lower the cost of each person trip. In other words, HOV looks better. That calculus, combined with a reduction in some of the red tape associated with trying to provide local, private transit, could result in a much greater provision of private transit. Jitney service gets transportation economists almost as excited as road pricing. Such service springs up spontaneously in developing countries, though the lack of regulation (or full pricing) results in substantial externalities in terms of emissions and safety. Those problems, however, are already well covered by other regulations in the U.S.¹⁰ We believe that opening the market for the private provision of rubber-tired transit could further increase the competitiveness of transit (lower price and better service). The public sector does not have substantial scale economies when it provides rubber-tire transit; the big-scale economies are in the production of the highway system, not in the operation of transit vehicles.

For rail, we still lean toward public provision. Even more important than the argument about natural monopoly is the one about the problems of coordinating the development of a corridor through a multijurisdictional region: a role for public involvement in the planning and decision making of urban rail corridors clearly exists (construction and operation could be easily privatized).

But the empirical evidence is insufficient to allow an unequivocally technical judgment about the long-run net benefits of investments in big transit systems. The weight of the evidence of the rigorous short-run evaluations is that the investments in big regional rail systems don't pay for themselves. The long-run analyses are few. An evaluation of BART, the rail transit in the San Francisco region (Landis and Cervero 1999) found office development in downtown San Francisco grew strongly around stations, but grew only modestly around other stations, and that BART had little effect on the location of population or employment.

The theoretical argument for rail is that, despite its poor performance in the short run, in the long run it provides an infrastructure for design and location decisions whose net benefits more than offset the short-run costs. It is certainly true that in a world of higher prices for auto trips, rail transit would work better than it does now. Handy (2005, 163) summarizes the literature on the transportation/land use connection and comes to four conclusions, one of which is "LRT can encourage higher densities under certain conditions." Those conditions include a healthy economy, economic forces and attitudes that support higher density, and public-sector involvement in planning, regulation, and investment.

Our conclusion is that the experience of U.S. cities with light and heavy rail in the last 30 years supports our arguments about the importance of both economics and the land use/transportation connection. Rail tends to be more successful where (1) economics predicts it will be—that is, where land values and density are both high; and (2) land-use policies both allow (with appropriate zoning) and encourage (with public investment and partnerships with the private sector) the kind of transit-oriented development desired.

Whether justified by revenues from congestion pricing or by expectations about unmeasurable but substantial public benefit in the long run, heavy investment in public transit would mean going beyond the marginal,

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cost-effective system improvements we discussed in Package 1. The key idea is to get serious about improving transit service both within and between communities. These are very big investments that can only be made with committed leadership supported by a vision of regional development that is generally shared by stakeholders and the public.

Control the environmental externalities of trip making with pricing

In Package 1, we discussed these changes as optional and marginal. In Package 2, they are required and substantial. Metropolitan areas must price auto emissions, and probably should price noise as well. Vehicle emissions fees are probably the most promising opportunity for policy that would price externalities: the emissions are measurable; the costs can be calculated; the regulatory system is tested and in many cases in place; and no new technology needs to be installed.

One strategy is to require that all cars and trucks get checked once a year and pay fees based on the type, condition, and mileage of the vehicle. A more high-tech (and less likely) solution, which would be tied to the implementation of a comprehensive program of road pricing based on electronic tracking of vehicle use, would be to add a noise and emissions charge to the road-use bill.

Another strategy—one that demonstrates that not all solutions have to look like taxes or fees—is to subsidize the removal of polluting vehicles from the highway by paying people to junk their junkers. Such a policy could be economically efficient (and would also address equity issues) if the marginal damage of operating a junker is great, and the costs of reducing air pollution and noise through other policies is high.

Easiest of all, and a lot better than nothing, is to use the fuel tax as a proxy for an emissions fee. In its favor, it is already in place and politically acceptable, and is reasonably related to the damage emissions cause. Its problems are that not every mile driven in every place at every time of day or year is equally damaging.

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Land-Use and Public Services Policies This Package Might Contain

Organize regional institutions for regional problems

Our forecast for Package 1 applies here, though the likelihood of regional coordination and integration increases because of the demands for integration a regional system of road pricing will make.

Expand the scope of land-use and design regulation; make a greater effort to control externalities through pricing and performance measures

We offer nothing new here: planners have long had the full list of policies available for influencing land development patterns, and we have already discussed many of them. The problem has been uncertainty about their likely benefits and full costs, and a lack of will to go forward with anything but marginal changes in the face of that uncertainty. Many of the land-use and design policies in Package 2 are the same as the ones in Package 1: the difference is that in Package 2 many of the more difficult and expensive ones get implemented.

Some of the policies under this category are the obvious correlates of the road-pricing policies that drive this package. Getting the road prices right should be accompanied by getting land development prices right also. Such pricing policies could include emissions fees for industries to protect air and water quality, the public purchase of property and development rights, and a shift of more of the property tax burden from improvements to land.

Consistent with policies that try to correct for external costs by making people pay those costs would be the relaxation of some of the command-and-control

policies that, in the extreme, dictate the use of every parcel to attempt to ensure compatibility. Chief among those policies would be replacing ordinances that segregate uses with flexible zoning ordinances to allow mixed use and greater density, providing that performance standards are used to control externalities. Form-based zoning is a variation of performance zoning: it specifies some parameters for a consistent pattern and look, but lets uses vary within the constraints of design and performance criteria.

We discussed the general thrust of the design policies of this package when we discussed policies to help transit (e.g., controlling development along new bypasses).

Unlike some economists, we do not have big problems recommending some command-and-control land-use policies that orient development toward transit. As we described in Chapter 3, government was heavily involved in the promulgation and implementation of policies that supported development oriented toward the automobile. We encourage jurisdictions, however, to try policies that provide economic incentives and disincentives for the type of development desired. Among the policies jurisdictions might consider:

- Remove land-development policies that decrease development densities (e.g., minimum lot size zoning, minimum parking requirements).
- Adopt policies that shift the design focus and standards in residential areas from trips by auto to those by foot and by bike: fewer cul-de-sacs or at least more pathways to eliminate dead-ends for bicyclists and pedestrians; smaller, calmer streets; reduced requirements for space devoted to parking; parking oriented so as to not restrict access by modes other than the auto; reduced setbacks to allow buildings to be located at the corner of a parcel instead of at its center (which will allow parking now and redevelopment with parking structures when market conditions grow to merit it).
- Use pricing policies to encourage higher-intensity development (e.g., reduce property taxes on improvements, reduce development fees for projects that have lower impacts on the transportation system).
- Use design and pricing standards to encourage mixed-use development and limit its negative impacts.
- Control access on urban arterials (e.g., minimum access distances by street classification); prohibit or strongly regulate development near access points to unlimited-access highways in rural areas; prohibit development along unlimited-access highways that connect communities (or, better, buy the land within a certain distance of the highway to prevent development and compensate landowners).

Unlike some economists, we do not have big problems recommending some command-and-control land-use policies that orient development toward transit. We encourage jurisdictions, however, to try policies that provide economic incentives and disincentives for the type of development desired.

Increase the use of pricing policies for infrastructure

In addition to the changes we expect will occur anyway, Package 2 should make more progress in the following areas:

- Increase land-development fees to equal cost, considering the long-run marginal costs of service provision and of use.
- Price industrial emissions.
- Provide financial incentives to encourage developers to choose designs that improve the character of the public aspects of development (e.g., provide cash awards for good designs; provide low-interest loans).

Likely Effects of These Policies on Travel and Land-Use Patterns

The details of our answer are found in Chapters 4 and 5. We just summarize the key points here and describe most impacts relative to Package 1 (e.g., more or less, increase or reduced, compared to Package 1):

- Reduction of congestion on limited-access highways to efficient levels; congestion on alternative unpriced routes possibly made worse
- Large shift from the auto to other modes, and higher occupancy of autos
- Reduction in the rate of growth in per-capita vehicle miles traveled (VMT), with a corresponding decrease in death and injuries caused by vehicles (no prediction about whether per-capita VMT will be reduced absolutely, however)
- More stability in central business districts and suburban subcenters
- Slower development at urban fringe
- Greater density and more mixed-use in residential areas
- Higher-intensity development, especially near transit stations
- Reduction in the rate of growth in noise, air emissions, and other pollution
- Improvements in the character of public spaces (due to stability, better development-design standards, and financial incentives)

In short, given the dimensions we are concerned with in this PAS Report (the efficiency of transportation and land use, and the quality of the urban environment), urban areas are likely to be better places under Package 2 than they would be under Package 1.

STEPS TOWARD A TRANSITION TO EFFICIENT TRANSPORTATION AND LAND-USE POLICIES

In the long run, the policies of Package 1 cannot effectively cope with congestion; in the short run, the policies of Package 2 will not be adopted. Package 1 maintains an emphasis on the supply-side solutions that have dominated U.S. transportation policy, though it begins to work on the demand side of trip choice. Package 2 focuses on the demand side, which means increasing travel costs to trip makers and increasing political costs to policy makers. How can metropolitan areas make the transition from the short run to the long run? How can they increase the possibility of coping with congestion problems in the long run without having to make difficult political decisions about pricing in the short run?¹¹

A fundamental problem between developers and planners, between markets and visions, is a different definition of the relevant time horizon. Developers look more at the short run, and, in the short run, they are right. The planners' visions don't pencil out. Not many people driving cars will shift to transit. Given the huge number of highways and vehicles already available, the private cost of auto travel will remain much less than the private cost of transit trips for most travelers: people want housing that facilitates the mode of travel they use, which is demonstrably the automobile; most people won't pay more money for less housing in transit-oriented developments (TODs). But planners look at the long run, and in the long run *they* may be right. They *assume* that policies, preferences, and prices have changed and led to a future where the cost advantage of the auto is substantially diminished, where land use and multimodal transportation are integrated. In that world, density, TOD, and light-rail to serve it pencil out.

Pure economic theory would argue that both parties cannot be right. If planners are right about the long-run changes, developers can see those changes as well as planners and can calculate a present value for each potential type of development, incorporate some analysis of risk and uncertainty, and make the correct economic decision. Arguably, they are doing that now and either discounting future returns heavily or assessing the risk as high. The more likely reality is that such calculations are not made formally, but implicitly. The long-run future is too uncertain: too many things can change. The prudent business decision is to build for today's market.

A fundamental problem between developers and planners, between markets and visions, is a different definition of the relevant time horizon.

Planners point to the problem that decision creates: it perpetuates a development pattern that depends on the automobile and makes the transition to the[ir] desired development pattern more difficult. If cities keep building for the auto, only the auto can serve the transportation demand effectively, at least until the congestion problems planners predict become a reality that forces new solutions.

This problem is well illustrated in planning for the areas around proposed stations for a new light-rail transit (LRT) line. If the LRT were operating, developers would be more likely to see an opportunity for the kind of high-density, mixed-use development that station-area plans typically call for. But the LRT is not operating. In fact, from the time discussions start about an LRT to the time it operates, 10 to 20 years have passed. Is the developer supposed to build now what can only be served well later. If not now, when? When the corridor is selected? When the station areas are selected? When the funding is secured? When the ground is broken? When the train is running? When its functionality is demonstrated? Our experience suggests that developers discount the potential rail heavily *at least* until track is being laid.

That perspective allows the kind of proposal that occurred in the Portland metropolitan area in 1993, years after its LRT line had been touted as a success. Metro, the regional government, had secured funding for an extension of the LRT to the west side of Portland: its EIS was completed, engineering was under way. A developer wanted to buy property at one of the station sites and develop a golf course. Maybe the golf course penciled out. But from a land-use and transportation perspective, in light of a billion-dollar transit investment, it was not the type of TOD planners had in mind (though it would have encouraged some internal walking trips and trips by electric cars).

Planners have to look for the bridge from the short run to the long run. It is not too hard to think of principles that would assist with that transition. The public sector could control the land by buying it. If planners are right, they are making a great investment, and the public may someday acknowledge their foresight. Or public policy could facilitate or require development that acknowledges a future change in use: shadow platting and changing design standards (e.g., setbacks) can facilitate future infill and redevelopment.

The policies that would facilitate a transition between Package 1 and Package 2 are bounded by the policies in those two packages. Transitional policies would contain all or most of the policies we described in Package 1. They would contain as many of the policies described in Package 2 as possible, pushed as far as political friction allows them to slide. For many policies, the transition would do no more than Package 1; for others, it would take the policies of Package 1 farther. The thrust of the transition is to implement *second-best policies that approximate marginal-cost road pricing*. For examples, we repeat briefly some of the second-best policies that start to get toward pricing:

- An *efficient fuel tax* is probably the easiest of the pricing policies because people are already familiar with it. Data in Chapter 4 suggests substantial underpricing of average and peak-period auto travel. One way—the easiest though not necessarily the most efficient—to start to correct that pricing is to increase fuel taxes that are already in place. In Package 1, we said we expected a small increase in the tax. In Package 2, we encourage states and metropolitan areas to go farther. We also strongly recommend that they do not make the old mistake of the federal policy of using the gas tax increase to fund budget shortfalls (in the federal case, the deficit) not directly related to transportation: the gas tax should approximate a user fee. Policy makers can use the fuel tax as a surrogate for an *emission fee*: our estimate of the cost of emissions in Chapter 4 is low enough that small increases in the fuel tax could cover it.

Planners have to look for the bridge from the short run to the long run. It is not too hard to think of principles that would assist with that transition.

- Another second-best pricing idea is a *variable vehicle registration rate based on miles traveled*. As with the fuel tax, such a price would (1) be highly correlated to highway use, which is correlated (less perfectly) to highway cost, and (2) not address well that component of trip cost that relates to the time and location of a trip (i.e., the aspect of the trip that contributes to peak-hour congestion). While less than perfect as a pricing mechanism, it has the additional advantage of reinforcing the idea of user fees and beginning the transition to road pricing.
- Parking fees are an obvious target for pricing, but a hard one to hit. For reasons we discussed, we do not recommend using parking as a second-best means of pricing congested roads (i.e., charging at all parking lots time-of-day fees calculated to cover the estimated cost of highway congestion). Rather, the policy should aim at charging the price of parking per se. Parking pricing could be tied to other TDM policies like the removal of parking subsidies or offering cash payments to workers that use a mode other than the auto.

Many of the land-use policies for the transition are of the same type but a different magnitude as to those in Package 2; namely, do as much as possible to price industrial emissions, replace ordinances that segregate uses with flexible zoning ordinances, allow higher-density development in residential areas, increase development fees to cost, prevent development along bypasses, and so on.

For both technical and political reasons, pushing for full pricing of urban services other than highways can probably move faster than road pricing. Consumers are accustomed to metering for water, electricity, and telephones; to paying for the local roads of new development; and to impact fees (system development charges). Local governments, however, are schizophrenic about these charges. In Oregon, for example, a law describes an economically correct method (as far as it goes) of calculating and allocating the *direct* cost (forget the externalities) of providing public services to new development and requires municipalities to charge no more than the fee that would cover those costs (i.e., an impact fee). The reality, however, is that many jurisdictions found the calculation would allow them to justify fees two or three times higher than what they charge now. Their councilors and commissioners opted instead for small increases, partially because of pressure from developers, partially because they feared the impacts on their objectives for economic development, and partially because they convinced themselves that local residents should pay for some of the growth.

Perhaps as important as the specific policies is a change in the way policies get evaluated. Emphasizing cost-effectiveness is a logical transition to pricing. SAFETEA-LU will help encourage regions to consider cost-effectiveness if it in fact gives them greater discretion over how to use a single pot of un-earmarked funds. It will force trade-offs. If an MPO goes heavily into transit systems, it will be leaning toward a do-less policy on highways. Or it may decide it cannot afford the big LRT *and* the key highway improvements it needs, so it goes for expanded buses instead. In either case, these trade-offs lead to a more effective use of resources—a distinct change from a policy of “build whatever the federal government will pay for.”

For the transition, state and local governments do not have to set tolls—rather, they start presenting analyses that support transportation investment decisions in terms of the toll that would have to be charged. We argued in Chapter 4 that calculating such tolls is relatively straightforward; implementation, however, is difficult. Imagine how decision making might change if, instead of an EIS saying, “The highway expansion will cost \$50 million,” it said, “Under the assumption that users should pay and that peak users are

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the ones requiring the highway expansion, peak trips would have to pay a toll of \$3 per trip to pay for the expansion.” We think the debate about the alternatives would sharpen. If everybody says, “Come on—nobody would pay \$3. They’d travel at different times or take Route 5. They might pay 25 cents,” then: that’s the answer. In short, the additional capacity isn’t worth its cost. The debate on highway expansion would get focused where it should be: on high-volume, highly congested corridors with few good alternative routes.

That reasoning supports a conclusion that is probably obvious without it: MPOs should only add new transportation capacity if it’s worth it. That conclusion leads to the possibility of a policy to Do Nothing. Though the Do-Nothing policy (i.e., hold the current capacity fixed and let congestion and land use adjust to the constraint) is at odds with our general penchant for proactive public policy, an argument can be made for it as part of the transition to Package 2. If policy makers believe that (1) the cause of congestion is the substantial underpricing of auto trips on major corridors during peak hours, and (2) that, politically, they cannot charge a toll for those trips, then they may choose to allow an increase the price of auto trips by letting congestion get worse. This deterioration of level of service will increase the public demand for government to do something, and the public willingness to pay the price of a solution. All of these conditions make road pricing more attractive. Not a great policy, but better than one of building high-cost/low-value transportation facilities.

No metropolitan area will get to road pricing quickly: all will take tiny steps. The more of those steps it can take, the more those steps are subsidized, the more they produce successful results—the faster they will move the transition to road pricing. This point supports the idea of adopting many incremental but complementary policies (what Downs (2004a) refers to as “one hundred small cuts”).

That point, together with the one we made about the different perspectives developers and planners have on economic efficiency, suggests that demonstration projects and programs are a critical part of the transition. Compared to many of the policies on the table, demonstration projects are cheap. For example, if policy makers are convinced that developers are myopic about the market for high-quality, moderate-density urban development (the urban village idea), they should put money into demonstrating that fact. If government subsidizes a product and it sells as well as the planners predict, the government recovers its costs and the developers see that a market exists. If it doesn’t sell, everyone learns something at a price much less than what would result from large-scale regulations for minimum densities.

A final point about the transition—one at the foundation of this report. Planners, policy makers, and the public need to think more about integrating and bundling policies. MPOs cannot develop a transportation plan without considering a land-use plan. Neither can they develop a land-use plan without considering the transportation system that makes development possible. And, for most jurisdictions, land-use patterns and transportation systems cannot be evaluated without reference to policies or (yes, we’ll say it) visions: about the economy, the environment, social intercourse, and justice. We predict that transportation planning in metropolitan areas will fail to the extent that it does not open the scope to include the entire, messy metropolis.

What are the likely effects of these policies on travel and land-use patterns? In the case of transition policies, the simple answer is also the complete answer: the effects are somewhere between those we described for Package 1 and those we described for Package 2. How much closer a metropolitan area gets toward Package 2 depends on the type, number, and strength of second-best policies it adopts.

Planners, policy makers, and the public need to think more about integrating and bundling policies. MPOs cannot develop a transportation plan without considering a land-use plan. Neither can they develop a land-use plan without considering the transportation system that makes development possible.

CONCLUDING THOUGHTS

The policy packages make our main themes clear: (1) pricing policies can strongly influence travel behavior and land development in ways the public wants while still allowing individual choice, and (2) without some notion that the resources society is consuming to accommodate trips are priced at full cost, it would be only good luck if most of the policies now being considered led to a more efficient transportation system (one in which the costs of the investments society is making are less than the benefits those investments produce), much less something approaching an optimal one.

All of the regional planning efforts now in process in metropolitan areas across the nation are fundamentally about improving the quality of life for the people that live in those regions. They all want to preserve environmental quality, provide economic opportunity, and re-create a sense of community. These are the right goals to have—the problem is that, even if they do not conflict with each other, they cost something to achieve. Visionaries and planners are often not explicit about those costs; the public often is. A fundamental assumption underlying our analysis has been that being explicit about those costs—about the costs of the policies that lead to the vision—will become increasingly important as resources become more scarce and the debate about transportation policy becomes more heated (if hell can get any hotter).

Contributing to the difficulty of any debate about public policy is a fuzziness about efficiency and equity. Are transportation and land-use policies supposed to increase economic efficiency (e.g., by providing trip benefits greater than costs or by increasing land values), or are they to achieve equity objectives (e.g., by subsidizing travel for special groups or by putting jobs near where people who need them live), or both? If both, there is no technically correct answer about transportation policy: inefficient investments are fine because they comport with an idea of fairness held by society (as represented by elected decision makers). If both, trade-offs must be made. Either metropolitan areas should make decisions about rights (e.g., the right to equal mobility regardless of income) and then evaluate policies to maximize efficiency subject to the constraints of rights, or they should describe an efficient system of land use and transportation and the policies that would achieve it, and then implement policies that explicitly deal with issues of equity and social justice.

Our emphasis on pricing is not a polemic about the ineffectiveness of government and the efficiency of the market. Our cities and transportation systems are a result of private and public investments. Government has been involved, will be involved, and, for many good reasons, should be involved in the planning and development of transportation systems. We didn't start this book with that point as our destination; we just followed the signs and ended up there.

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CHAPTER 7 NOTES

1. We do not discuss individual policies in the packages: that analysis, as far as we have taken it, is in Chapters 4, 5, and 6. We do not discuss the political problems of getting the packages adopted: discussion of political process is in Chapter 8.
2. HOV could be considered new capacity, but we treat it as TSM.
3. That increase is still small in comparison to (1) the increases that occurred during the energy crisis of the 1970s (increases the, despite their size, had only temporary effects), and (2) other estimates of the amount of underpricing of auto trips (on the order of \$1 per gallon or more). Even though the tax has not increased since then, the allocation to transportation has increased slightly as the amount of the tax going to the general fund for "deficit reduction" has decreased.
4. http://www.fta.dot.gov/news/colleague/news_events_4664.html accessed 18 February 2007.
5. www.envisionutah.org/ and www.chicagoareaplanning.org/

6. The concept of *environmental adders* (adding the costs of external environmental impacts to the costs of production) is like congestion pricing in transportation. How it would work and its benefits have been understood by economists and occasionally debated politically for 50 years.
7. Several arguments support this position: (1) transportation corridors respond to natural features that, in most cases, are a binding constraint on location; (2) in most metropolitan areas, a network of corridors has already been built in response to those constraints, meaning the opportunities for new corridors are few, especially relative to the number and extent of existing corridors; (3) the amount of extant development in metropolitan areas strongly suggests that, from a regional perspective (the view from a altitude of about 20,000 feet), in 20 years they will look a lot like they do now; and (4) the geometry of radii and area (area growing at the square of distance from the center) means that a lot of the growth in the next 20 years can be accommodated by small expansion at the fringe, whose traffic would continue to use existing corridors as it approaches the core. Even with increases in the price of auto trips, auto trips will continue to carry the largest portion of trips for a long time; many developed countries charge much higher prices for auto trips, yet most trips are still made by auto.
8. We want to be clear that we are recommending a level of analysis substantially different than what shows up in a state-of-the-practice environmental impact statement (EIS). The idea of EIS was a good one (describe all impacts; i.e., benefits and costs). The reality of the EIS for highway projects is a disaster: a routine, legalistic recitation of sanitized phrases having less to do with informing decisions than appeasing stakeholders. The kind of simple cost-benefit calculations we advocate are not only not done, they are not allowed. EISs give broad lists of impacts to allow all parties to the decision a maximum of wiggle room; they do not present any rigorous analysis of the trade-off among competing objectives and alternatives.
9. See *Mother Jones* magazine (January/February 2007), "The Highwaymen," for one version of a deal to transfer the 157-mile Indiana Toll Road to a private consortium for the next 75 years, and a discussion of similar types of public-private partnerships under consideration. Governor Daniels, a proponent of the Indiana partnership, wrote "any businessperson will recognize our decision here as the freeing of trapped value from an underperforming asset, to be redeployed into a better use with higher returns." At a hearing of the House Subcommittee on Highways, Transit, and Pipelines, Daniels testified about the benefits of the partnership and Oregon's Peter DeFazio said "So you're saying that there's no political will to raise the tolls, but if you enter into a binding contract which gives a private entity the right to infinitely raise tolls, then that'll happen—but politically you couldn't say we're going to go out and raise the tolls . . . Are we outsourcing political will to a private entity here?"
10. These issues may be covered so well by other regulations that private transit service might never be possible without subsidy. We are thinking of the Americans with Disabilities Act (ADA) requirements, which now encourage large buses with equipment to assist individuals with disabilities. If a larger fleet of smaller vehicles were required to have similar equipment on all vehicles, the costs might be prohibitive. The benefits of ADA notwithstanding, we expect it will increase the cost and slow the development of future innovations in the supply of peak-hour transit service.
11. Before trying to answer those questions, we'll take a paragraph to explain why we bothered asking them. After all, our analysis in this report puts us firmly on the side of pricing solutions. Why should we even consider what we have denigrated as second-best solutions? The answer is that our enthusiasm for pricing solutions to transportation problems is bridled by our experience with the reality of public policy. We recognize that adopting full pricing is politically very difficult (probably impossible) in the short run. But we are not ready to adopt the pessimism of many analysts, economists included, who believe that pricing will never happen, choking peak-hour congestion is therefore inevitable in metropolitan areas, and commuters' most cost-effective response to the congestion is the acquisition of a good CD player, companionable ridesharers, and an attitude that commuting is really leisure time (though that response is probably the best one for individual trip makers in congested metropolitan areas).

A Process for Integrated Land-Use and Transportation Planning

Organization: Chapter 7 might appear to have gotten ahead of itself by coming to conclusions about reasonable packages of land-use and transportation policies for metropolitan regions. Since a pervasive theme of this report has been complexity, uncertainty, and the need for political debate as part of decision making, how could we presume to specify a preferred policy package? Surely such preferences get established through a public process.

They do. The way a region gets to agreement about a package of policies (i.e., about a regional vision for land use and transportation) is to go through a planning process that involves the public, stakeholders, and elected officials. We jumped ahead to policy packages in Chapter 7 because it seemed like the logical conclusion to the discussion of individual policies in Chapters 4, 5, and 6.

In this chapter, we get to the obvious and important question: If there is agreement among technicians on the most effective policies, why don't governments adopt them? For most readers, the obstacles we describe will be familiar. We spend most of the chapter discussing principles for the public process and decision making that is a necessary part of any attempt to create regional, integrated land-use and transportation planning.

This chapter has two sections:

1. *Overview of Public Decision Making.* The problems of integrated land-use and transportation planning at the regional level; options for process.
2. *Practical Principles for Integrated, Regional Planning For Transportation and Land Use.* A baker's dozen.

Basics: Big change is difficult but not impossible; a planning process that is both comprehensive and incremental helps. An efficient process will take on some of the characteristics of a "visioning" process: stakeholders will be involved early and throughout a process that looks comprehensively at desires and possibilities over the long run. Policies most likely to be correct technically are most likely to be unpopular politically; even if preferred policies could be adopted, the assumptions about their technical superiority are untested. Some advice about running an integrated process for regional planning:

- Subregional planning in the absence of regional planning is not likely to be successful; neither is regional planning in the absence of an empowered regional organization.
- Regional government may be a necessary condition for successful regional planning, but it is not a sufficient one.
- The goals for regional growth do not vary across metropolitan areas. They generally fall into the categories of: economic prosperity, environmental quality, urban amenity, and cost of living.
- The possibilities for future urban form are few: options at the regional level are limited.

- Public involvement is necessary, expensive, and incapable of satisfying all stakeholders; regional planning efforts will be simultaneously criticized as too little and too costly.
- Measurement in a multiobjective world is always faulty; regional planning led by technicians for technicians will be interesting to technicians only.
- Focus on direction, not destination; the best way for most people to evaluate a long-run vision is to focus on the short-run policies that are the first steps towards it.
- Work with the market to change behavior: change prices. Pricing is a policy direction, not a destination for urban form.
- Evaluation of regional policies focus on efficiency; interest groups and the public also care about equity—what will this mean for me?
- Bundle complementary policies. Integrated planning for land-use and transportation means integrated policies.
- Preserve long-run opportunities. Existing neighborhoods should be planned to enhance redevelopment and transit opportunities down the road.
- Integrated regional planning needs champions.
- If you want to affect the long-run, take a long-run attitude toward change.

Shortcuts: If you are a technician focused on policy evaluation and can have someone else bother with the process or politics, you can skip this chapter. If you are going to be developing or running a regional planning process, or if you are a sucker for good stories about the politics of planning, read this chapter.

OVERVIEW OF PUBLIC DECISION MAKING

Problems for Integrated, Regional Land-Use and Transportation Planning

On the one hand, our perspective—that land-use and transportation decisions are fundamentally economic decisions—simplifies the technical analysis. It leads us to conclude that:

- automobile travel, especially during peak hours in single-occupant vehicles, is underpriced;
- the underpricing leads to suboptimal amounts or patterns of highways, trips, land use, and congestion; and
- policies should aim at reducing the inappropriate differential between a consumer's cost of an auto trip (especially during peak hours on congested routes) and the cost of a trip by another mode or route, or at a different time.

On the other hand, that perspective highlights at least two fundamental problems for public decision making. First, the policies most likely to be correct technically (increasing the relative price of auto trips) are most likely to be unpopular politically. Second, even if the preferred policies could be adopted, their technical superiority ultimately rests on untested and untestable assumptions. There is no assurance that they lead unambiguously to a better world, or, in the jargon of economics, that they lead to an increase in net social welfare.

We address the second problem first. Most of the chapters and appendices of this report provide some evidence that the complexity of the interactions among land use and transportation, and among markets and government, means it is hard to make a quantified and incontrovertible case for the superior performance of a particular pattern of urban land use or transportation. There is no way to unambiguously add up the net welfare of all people in the region, now and out into the future, to get to estimates of net social welfare of alternative policy packages. Rather, we argue (especially in Appendices A and D) that, in the best case, policy gets developed more or less the way this report is organized: a lot of information is gathered, reorganized, and

discussed; the discussion leads to some level of agreement about diagnosis, prognosis, and, ultimately, prescription.

To these limitations on technical analysis we will add another. The previous few paragraphs, and all the analysis that precedes this chapter, have addressed the *efficiency* of public policy. It is difficult to imagine a system that could estimate the effect on each individual firm and household in a region, and simultaneously establish the relative importance of each effect on each firm or household so that the preferences could be aggregated for the relevant society.¹ Efficiency as an objective of public policy has no meaning in the absence of agreement about the relative prices of resources (which reflect the weights people place on them).²

Politicians must implicitly determine the weights that the technicians cannot measure. Technicians might be able to give general ideas about who are the gainers and losers, but it is the public and their elected officials who must decide how that distribution of benefits and costs should be valued. Is short-run sacrifice by some worth the long-run benefits to them and others? Do the institutions necessary for the implementation of the policies exist (or can they be created), or should policies be revised to fit with the existing institutional structure? Moreover, even if individual preferences could be aggregated, there remain the intractable problems of *equity*: should the needs and preferences of special groups of people (the rich, the poor, the disadvantaged, the informed, the interested) be given special weight?³

Rather than detour through the extensive planning literature on how to define the public interest, we simply restate our conclusion: no amount of technical information will lead to a unique and unambiguously objective definition.⁴ This point is true not only in general, but also in particular for configurations of land use and transportation: no research has (or ever will) specify an optimal system of urban development and infrastructure in any absolute sense. Decision makers can expect no more from technicians than the kind of partial analysis contained in this report: “if your assumptions are *j* and your objectives *k*, policy *x* will probably lead to your objectives more effectively than policies *y* and *z*.”

Technical analysis does not make decisions; rather, it informs them. Decisions about public policy are made by elected and appointed decision makers, which returns us to the first problem we listed; namely, optimal decisions in that context are inhibited not only by the uncertainty of the technical information about the issues of interest, but also by the realities of politics in a pluralist democracy.

Politicians can make big changes in policy only if the public or powerful special interests believe such change is desirable and supportable. That belief typically emerges on big and difficult issues (transportation and land development qualify) only in response to a crisis—a problem that is palpable, broad, large, and immediate. With rare exception, in the absence of a clear crisis (like a war or natural disaster), government at all levels in the U.S. has generally shown itself unable to make significant and comprehensive policy changes in the face of powerful minority opposition. Examples of failure or extensive delay include the siting of a nuclear waste repository, health care reform, and deficit reduction. Traffic congestion has reached crisis status in only a few metropolitan areas.

Even with agreement on the existence of a crisis, agreement on solutions is unlikely. Technicians cannot say with certainty what policies maximize net social welfare; interest groups take a narrow view that maximizes their interest; services once considered special benefits harden by familiarity into rights. Politicians must make decisions in the face of substantial technical and political uncertainty.

Even with agreement on the existence of a crisis, agreement on solutions is unlikely. Technicians cannot say with certainty what policies maximize net social welfare; interest groups take a narrow view that maximizes their interest; services once considered special benefits harden by familiarity into rights.

Our conclusion is that big change is very difficult, but not impossible. The combination of a perceived crisis and a political champion can lead to significant legislation. Both the Oregon land-use program and the Washington Growth Management Act were passed under these conditions. But, when the crisis evaporates or is averted, and as the champions move on, the idealistic goals mature into regulations and bureaucracies, the rights of the right sweep out what's left of the left (or vice versa). A sustained change requires public support. Support for the kinds of policies that would have real effects on land-use and travel patterns is tenuous in most places.⁵ The treatment is worse than the illness for most citizens.

Options for a Planning Process

If the public interest for urban policy is unknowable (or, at best, ambiguous) given the size and complexity of metropolitan areas, it is not surprising that local governments have been unable to specify an optimal land use/transportation configuration. But planners do not, and probably should not, let this complexity justify inaction.⁶ On the one hand, they cannot assume away the complexity and adopt simpleminded, ineffective solutions. On the other hand, they cannot give up; the arguments we have developed so far convince us that a lack of public policy in a system that already depends on it will lead to even greater inefficiencies.

A key role for planners in the development of integrated land-use and transportation policy is to make the complexity of the interactions intelligible to decision makers and their constituents so that decisions are better informed. The inability to prove a best solution does not mean that better ones cannot be described and technically defended. Thus, our focus in this section is on the characteristics of a process for getting better decisions about integrated land-use and transportation policy.

Hundreds of books and articles have been written on how to improve the process of public decision making (planning). As we did above, they tend to focus on the technical and political aspects of decision making. (We include public process—for the general public, and private and public stakeholders—under the heading of political.) A fair generalization is that the bulk of that literature recommends a process that encourages the participation of all stakeholders (including not only the powerful, but also representatives of groups who are unorganized or unempowered) in a discussion of objective technical information that leads to a consensus about optimal policy for the planning jurisdiction. Though a lot of the literature talks about the pragmatism of short-run incrementalism, our observation is that planners in planning agencies tend to take a long-run, comprehensive view of the planning process, especially for planning aimed at developing integrated, regional policy for land use and transportation.

The propensity toward comprehensiveness among planners has a distinguished lineage: from Burnham's big plans, to the master planning of the Model Cities program, to comprehensive land-use planning and statewide programs, to strategic planning, to visioning. The same basic body of thought gets dressed by new generations of planners in the vernacular of the day. The steps are the same no matter what the title and are well known to planners: define the problems (goals); describe and evaluate alternative policies for solving the problem (policies that are comprehensive—all relevant geographies, long-term, and multidisciplinary); adopt, monitor, and revise the policies. In the comprehensive planning model, those technical steps must occur as part of a public process in which all opinions are sought and considered. The planners and the planned learn from each other, evaluate trade-offs, and eventually arrive at an agreement on policies that, by definition, make things better than they were.

A key role for planners in the development of integrated land-use and transportation policy is to make the complexity of the interactions intelligible to decision makers and their constituents so that decisions are better informed. The inability to prove a best solution does not mean that better ones cannot be described and technically defended.

This planning model dominates public decision making for a good reason; namely, it is a logical (and therefore understandable and defensible) process for making decisions. It is a good description of the way society would like to make decisions given the knowledge, money, time, and community spirit. In practice, however, all of those ingredients are in short supply. In practice, planners must fit their desire for the all-by-all matrix of alternatives and impacts to the boundaries described by these constraints. Whether they admit it or not, their planning reflects a bounded comprehensiveness; they look ahead and around as far as the constraints of the problem and the process will let them, and then focus on the step in front of them.

This brief restatement of the debate about comprehensive versus incremental planning is old news to planners. It gets pulled out and polished only in academic articles and books like this one because most practitioners accepted long ago the necessity of both comprehensiveness and incrementalism, strategy and tactics, a long view and an action plan. A blend of comprehensiveness and incrementalism is at the heart of strategic planning, whereas a comprehensive environmental scan provides a context for a short-run, specific (incremental) action plan.

Though the debate about stereotypical theories of planning may be artificial, the choice among methods for implementing a planning process is real. How much citizen involvement should occur? When? Who gets involved? How should opinions be gathered? What weights should those opinions have? Should technicians provide information in advance to inform the public debate, or after citizens have told them what information they want? We look at some examples of integrated regional planning for answers.

Likely Characteristics of an Efficient Process Given the Difficulties of Adopting Policy

Planners and decision makers across the U.S. differ in the planning process they choose for a given problem; the choice depends on the characteristics of the problem, the agency, the technicians, and the stakeholders. But the scope of the planning efforts that must occur to integrate land-use and transportation planning suggests they all should have some common features. They should tend to be:

- long range and comprehensive (looking across time, geography, type of effect, and professional discipline);
- broad-brush (a consequence of their comprehensiveness, and the limits of budget, time, and methods);
- pro-involvement (a consequence of the need for regional solutions—myriad stakeholders will be involved; both the reality and appearance of substantive involvement are necessary); and
- pro-visioning (a consequence of all the above).

Our evaluation of regional planning efforts shows they share these characteristics. We acknowledge that our conclusions here are based on a nonrandom sample of brief case studies. The cases we studied tend to be in western states, where the kind of large, regional planning effort of interest to us was occurring in many major metropolitan areas. We found the planning efforts we reviewed to have very similar processes for decision making, public and agency involvement, and technical analysis. We leave it to readers of this report to determine for themselves whether our generalizations comport with their experience with long-range regional planning efforts.

Since all of the processes we reviewed were long range and comprehensive, we were not surprised that all of them referred directly or indirectly to their efforts as visioning, the 1990s' name for long-range comprehensive planning. Klein (1993) summarizes what he feels to be visioning's differences in emphasis, which include:

Planning efforts to integrate land-use and transportation planning have some common features. They should be:

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- *broad brush;*
- *pro-involvement; and*
- *pro-visioning.*

THIRTEEN PRACTICAL PRINCIPLES FOR INTEGRATED, REGIONAL PLANNING FOR TRANSPORTATION AND LAND USE

1. Subregional planning in the absence of regional planning is not likely to be successful; neither is regional planning in the absence of an empowered regional organization.
2. Regional government may be a necessary condition for successful regional planning, but it is not a sufficient one.
3. The goals for regional growth do not vary across metropolitan areas.
4. The possibilities for future urban form are few.
5. Public involvement is necessary, expensive, and incapable of satisfying all stakeholders; regional planning efforts will be simultaneously criticized as too little or too costly.
6. Measurement in a multiobjective world is always faulty; regional planning led by technicians for technicians will be interesting to technicians only.
7. Focus on direction, not destination; the best way for most people to evaluate a long-run vision is to focus on the short-run policies that are the first steps toward it.
8. Work with the market to change behavior: change prices.
9. Evaluations of regional policies focus on efficiency; interest groups and the public also care about equity—what does this mean for me?
10. Bundle complementary policies.
11. Preserve long-run opportunities.
12. Integrated regional planning needs champions.
13. If you want to affect the long run, take a long-run attitude toward change. ■

- getting stakeholders (including the public) involved early, including in the definition of the problem;
- getting them substantively involved (often through the use of facilitators);
- communicating technical information to the public; and
- consideration of the very long run.

We reinterpret the first three points to mean that visioning is finally doing the kind of agency and stakeholder involvement the standard planning model always envisioned. Regarding the last point, our previous critique applies: time spent on the very long run comes at the expense of time spent on immediate problems and the difficulties of implementing actions. Given the importance of the long-run evaluation to long-run investments, some attention to the long run is likely to be cost-effective, if not essential. But given the limits of data, methods, and comprehension, the benefits of looking at the long run probably diminish quickly. Visioning, if done well and cost-effectively, looks a lot like bounded comprehensiveness as a framework for incrementalism (action plan).

We won't belabor the definitions. If you like visioning as a label for your ideal process for integrated planning for land use and transportation, use it. We will move on our conclusions about such integrated planning efforts.

PRACTICAL PRINCIPLES FOR INTEGRATED, REGIONAL PLANNING FOR TRANSPORTATION AND LAND USE

Principle 1. Subregional planning in the absence of regional planning is not likely to be successful; neither is regional planning in the absence of an empowered regional organization.

Subregional planning cannot handle the problems of growth, land use, and transportation that do not respect jurisdictional boundaries. For the reasons we gave in Chapter 4, the solution to problems that derive from externalities is internalization: all the geography that comprises activities that contribute significantly to the problem must be included in the solution. That internalization can be approximated, in theory, without regional government if individual jurisdictions pursue their self-interest and make bargains among themselves. The problem with that solution is that transaction costs are extremely high (what is the process and forum that allows these deals among many parties to be struck?) and the free-rider motivations still exist.

The other type of solution is a regional government or coalition. The Bay Area Council in San Francisco, a business-sponsored organization involved in research and advocacy on regionwide public policy issues, came to similar conclusions:

... the Bay Area needs a new form of regional governance. . . . The Council strongly urges that a regional growth management agency be created for the nine-county Bay Area. . . . (from policy adopted by the Bay Area Council Board of Directors, March 18, 1991)

Kaplan and Cuciti (1993), Downs (2004a), and Katz and Puentes (2005) provide good descriptions of several institutional arrangements for regionally coordinated infrastructure and congestion-relief policies, including voluntary cooperation among local governments, regional agencies, public-private organizations, and state planning. Downs (2004a, 298–320) recommends public-private associations like the San Francisco Bay Area Council. Envision Utah (Salt Lake area) is a good example of a privately instigated private-public partnership that has had a lot to say and a big impact on regional land-use and transportation planning.

Regional solutions can be encouraged by state sticks and carrots. In Oregon, an administrative rule adopted by the state's land-use planning agency

requires local governments to prepare regional solutions to transportation problems.⁷ For larger jurisdictions, the obvious regional organization is the designated Metropolitan Planning Organization (MPO). The Sacramento Area Council of Governments (SACOG), an association of local governments in the six-county Sacramento Region, used its status and obligations as the MPO to create an integrated, regional land-use and transportation plan (the “Blueprint”) that pushes many of the policies we described in Chapter 7. The Northeastern Illinois Planning Commission merged with the Chicago Area Transportation Study to become the new Chicago Metropolitan Agency for Planning and prepared the 2040 Regional Framework Plan (the American Planning Association’s 2006 national plan of the year)—a vision for land and transportation development. The plan is advisory: implementation occurs by convincing local governments it is in their interest to adopt policies that are consistent with the regional vision.⁸

In the final analysis, however, well-funded organizations with real power to adopt the kind of policies we have discussed will be essential if land-use and transportation systems are to be amended in any significant way. Such organizations could be coalitions of existing institutions (e.g., local governments, special service districts, state agencies), and could be expanded to include private sector and nongovernment organization advocacy groups. But they must have the discipline and power to enforce conformance with a regional plan. That is a tough charge without state-delegated power to a regional government, but so is creating such a regional government. Unlike some advocates of market solutions to societal problems, we believe that such government intervention is essential to creating or facilitating the markets that will allow pricing solutions to occur.

Principle 2. Regional government may be a necessary condition for successful regional planning, but it is not a sufficient one.

Metro, the regional planning agency in Portland, Oregon, is one of the most powerful regional agencies in the nation. It has responsibility for land-use and transportation planning. It has an elected council with legislative power to require local governments (cities and counties) to adjust their plans to comply with regionally adopted plans for land use and transportation. It is the MPO that influences expenditure of state and federal transportation money. Despite these nominal powers, integrated land-use and transportation planning has been an incremental process at Metro, as it has in most other areas of the country.

The reality is that Metro needs the support of local governments. Though Metro has power now, it must be careful if it wants to maintain it. If enough local governments or public interest groups dislike what Metro is doing, they can try to curb its powers through the state legislature or referendum. Moreover, Metro councilors are elected and must respond to their constituencies. For all the obvious political reasons, Metro has moved cautiously and incrementally on developing its long-range vision for land-use and transportation. The Oregon land-use program found its powers effectively reduced as the state continues (as of 2007) to sort out the effects of a statewide ballot measure that required compensation for land-use regulations that reduced property value or a waiver of those regulations. In the Seattle area, the Puget Sound Council of Governments was disbanded and reorganized in part because of the dissatisfaction of local governments that constituted it.

Having the power on paper is not enough. The regional agency must have not just authorized authority, but also legitimate authority (Benevise 1989). It must meet a need perceived by those who allow it to exercise that authority. As we have noted, in the absence of meeting a clear need of resolving a clear problem—in the absence of a public outcry about the form

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of urban development and traffic congestion—even regional government will have difficulty implementing the more effective of the policies we described in Chapter 7.

Principle 3. The goals for regional growth do not vary across metropolitan areas.

All the planning efforts we reviewed used the standard planning model. Proposed public policies (also known as strategies, scenarios, alternatives, prototypes, choices) get evaluated against criteria (also known as goals, objectives, principles, factors, impacts measures, benchmarks) in an effort to estimate their relative performance in achieving those criteria. Most planning efforts distinguish a hierarchy of objectives from the general to the specific: (1) for impacts or outcomes, the hierarchy is typically goals, objectives, measurements; (2) for doing something to achieve those outcomes the hierarchy is typically policies to actions (regulations, incentives, investments). Appendix A provides more detail.

All regional planning processes start with goals because it is logical to argue that the region should know where it is going before adopting policies to get there, and goals are a lot easier to agree to than specific policies.

Our impression based on the projects we reviewed is that every regional planning effort feels an obligation to develop anew its goals. The reasons are academic, substantive, and procedural. The pedagogy of planning is based on the logic of making decisions to achieve goals; therefore, goals must lead. Many planners believe that citizens have ideas about goals (have visions) that will not be aired without extensive public involvement at the beginning of the planning process. Finally, the appearance of public involvement is as important as the reality. Planners and policy makers feel they cannot propose solutions until all parties have had a chance to comment on the problems.

Our research convinces us that long-range planning processes end up with essentially the same goals, independent of the methods or amount of time used to develop them. Appendix A shows typical goals and offers several conclusions.

First, one is not likely to uncover previously unknown and broadly held goals during the public involvement process (though one might learn about the relative importance of the goals). Goal-setters typically list a clean environment, a healthy economy, affordable housing with good access to jobs, and efficient public services (including transportation) among main goal headings. Some efforts may generate general social goals (about opportunities for diversity and choice, investing in people, and fairness). In a few instances, proffered goals are specific enough to be considered policies (e.g., Bay Area 2020 included as part of its land-use goals the idea of an urban containment boundary for the Bay Area). We do not consider goals like “increase density” or “improve the jobs-housing balance” (which often show up in these regional plans) to have the specificity of policies (e.g., changes in land-use plans or zoning).

Second, how goals are organized varies little across regional plans. A region will not learn much by reviewing the goals of other regions: there are no surprises. There is agreement about the usefulness or necessity of a hierarchy of goals—from the general to the specific; from principles to goals to objectives to measurements to actions; from high-order values to the conditions necessary for actualizing those values. A Toronto plan started with eight high-order factors (e.g., urban form, environmental quality), divided these into 16 goals, and divided these again into 42 measurements. Metro Portland did something similar but used different terms: six fundamentals were further specified by dozens of goals, which were further specified by dozens of measures.

Our research convinces us that long-range planning processes end up with essentially the same goals, independent of the methods or amount of time used to develop them.

Finally, while the more detailed studies have been able to specify ways to measure goal achievement, they have been less successful in aggregating these measurements into a few high-order goals. A big technical problem derives from the fact that the goals must eventually get refined into specific, comprehensive, and mutually exclusive evaluation criteria (measurements). Consider the problems:

- *Goals overlap.* First, they overlap as they move from the general to the specific: from principles to goals to objectives to measurements. For example, an economic plan for the state of Oregon had three main goals (to develop and maintain exceptional people, outstanding quality of life, and a diverse, robust economy) that got subdivided into 18 subcriteria, 55 sub-subcriteria, and, ultimately 159 “benchmarks” (measures). Second, goals overlap across categories. Is good access a livability issue or an economic one? Does environmental quality go in its own category or under livability? To what extent does it affect economic development? As one moves from goals to more specific criteria, this kind of overlap means criteria are not mutually exclusive; many factors get counted in more than one criterion. That problem exacerbates the next one.
- *Goals and the criteria based on them do not add up.* All long-run regional planning studies are examples of the multicriterion problem; public policy always has multiple objectives, which, even if quantifiable, cannot be easily expressed in a way that allows aggregation and summary. Quantitative solutions to this problem (e.g., benefit-cost analysis, multi-attribute utility analysis) are typically beyond the scopes of these projects even if such techniques were appropriate (which, for the reasons we gave at the beginning of this chapter, we doubt).
- *If criteria cannot be added, a logical gap almost always remains between criteria and prescription, and is usually bridged by rhetoric.* For example, some regional studies describe a vision of a metropolis (i.e., the criteria) and then jump to the policies believed to be most conducive to achieving that vision.
- *If an evaluation goes beyond five to 10 criteria, even the technicians will get lost.* Since the criteria cannot be added (because they are all in different units), they cannot be summarized using spreadsheets. The performance of each alternative on each criterion must be described in numbers, text, or graphics. Go to a typical Environmental Impact Statement to see how bad things can get when there are too many criteria and no way to reduce them.
- *The need for detail is as important as the need for simplification.* A typical comment by participants in large-scale decision-making processes like these is, How can we decide whether this is good for X (the environment, the economy, the disadvantaged, social equity, etc.) unless you give us some information about a, b, c, . . . ? A tension will always exist between those making decisions, who want detailed information across policy alternatives for all criteria, and those supplying that information, facing schedules and budgets that demand alternatives and criteria be reduced to only a few to make detailed investigation feasible. The dilemma for a regional planning process (in fact, for any planning process) is how to proceed efficiently in the face of constraints on time and resources; how to get from scores of policy alternatives to a few best alternatives in the absence of the kind of detailed information that will later be collected about those few alternatives.
- *Criteria usually end up getting weighted implicitly.* The conclusions drawn from the performance of the criteria can easily be skewed unintentionally or intentionally. In most of the many studies we reviewed, there was no formal process to weight the criteria, which means that, implicitly, each

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criterion has equal weight. Thus, the more criteria one uses, the less important any criterion is, and the more ways one can measure some objective, the more important it is.

No obvious best solution to these problems exists; one must optimize. The optimal solution starts with a rough idea about higher-order objectives (e.g., Livability, Choice), works down to the detailed information about measurements, and then rolls-up that information into summary tables for public debate (with the detailed documentation available for those who want it).

This generalization leaves some specific problems. First, there is the question of consistency across categories. For example, should the high-order categories be organized from the perspective of human values and needs (shelter, education, employment, social interaction), the physical systems that provide for those needs (the ambient environment, natural resources, the built environment, infrastructure), or the policies that can provide for those needs (urban containment, transit development, social services)? In any of these organizations, where do criteria like social justice and vertical, horizontal, and intergenerational equity go? And what about process criteria like open, accessible, democratic decision making, all of which are important components of the quality of life in the future?

Second, to what level of abstraction should the specific measurements be summarized? For example, a measurement of the amount of peak-hour transit ridership could be summarized to increasingly general categories: quality of transit, quality of transportation, quality of infrastructure, or quality of life. The answer depends on what the process wants to emphasize and the discretion, whims, and bias of those running the process.

Those complications are real, but one cannot let them cause paralysis. We offer a hierarchy of goals as good as any we came across, and better than most. In concept, at the top of a hierarchy of goals or criteria for evaluating alternatives should be an implicit, highest-order criterion—something like: Is it good; does it work? That criterion is too general to be useful: it begs the question, What does good mean? Our answer, and the one that most economists would give, is “good” means society gets what it wants in a way that is efficient and fair (equitable). Again, this is the correct answer in concept, but goals of efficiency and equity are still too general for measurement and don’t really reflect the kinds of things people think about when they think about the benefits and costs of a long-run plan.⁹

Appendix A (Figure A-3) suggests the following set of high-order goals for urban areas, all of which contribute to the higher goal of Quality of Life:

- Economic Welfare (Prosperity): jobs (amount, type) and income
- Environmental Quality: air, water, habitat, open space
- Urban Amenity: public facilities and services (significantly for this PAS Report, transportation), housing, shopping, etc.
- Cost of Living: provide as much of the benefits in the three previous categories as possible while still controlling the cost of living

A slightly different alternative for high-order goal categories:

- Environment and Natural Resources (or, the Natural Environment)
- Development and Built Resources (or, the Built Environment)
- Economic and Cultural Resources (or, the Socioeconomic Environment)
- Government and Institutional Resources (or, the Institutional Environment)

In that second organization, “amenities” are part of the Built Environment (e.g., recreation facilities and shopping centers) and the Cultural Environment (e.g., educational and health services).

In any of these organizations, where do criteria like social justice and vertical, horizontal, and intergenerational equity go? And what about process criteria like open, accessible, democratic decision making, all of which are important components of the quality of life in the future?

These sets of categories come as close as any we have seen to meeting the requirements for comprehensiveness (they cover everything) and mutual exclusivity (they cover everything only once). The assumption is that a region needs all categories of resources functioning efficiently to be able to efficiently provide its citizens what they want. Equity issues could also be addressed as subsets of these categories (i.e., as the distribution of impacts among selected areas or groups). Each of these high-level categories could be further divided and detailed, down to the level of specific measurements. Appendix A gives more examples.

For some readers, the last couple pages will have been tedious. Unfortunately, that is part of our point—that's how it plays to the public as well. We reject the academic reasons for extensive goal setting; real planning does not have to happen that way. We reject the substantive reasons for extensive goal setting; all the evidence suggests that the months or even years of setting goals (Portland Metro's Regional Urban Growth Goals and Objectives took more than two years from start to adoption) yield goals that look the same across regions. What is left is the procedural argument; namely, people need to be involved in defining the problem and establishing goals if policies of any substance are ever to be implemented.

While we agree with that point in general, the interesting questions for planning is, What public process gives the best probability of agreement on policy? We believe strong arguments can be made for a much more abbreviated problem/goal/scoping process than the one used by most of the efforts we reviewed (Moore 1988). To state the conclusion more bluntly: a six- to 24-month goal-setting process is unlikely to lead to the adoption of goals any different than those contained here or in Appendix A. The length of time one takes getting to those goals is predominantly a political decision about process, not a technical decision about substance.¹⁰

The interesting questions for planning is, What public process gives the best probability of agreement on policy?

Principle 4. The possibilities for future urban form are few.

Almost all of the integrated regional planning efforts we are familiar with focus on physical planning: what should a map of land uses and transportation facilities look like in 20 or 50 years? The point here is similar to the one we made about goals: no matter how much research one does to find new regional land-use patterns to address regional quality in general, and traffic congestion in particular, the options are few. All of the regional planning efforts we have reviewed arrive at the same alternatives. Save time by getting the public debate quickly to those alternatives (or better, as we argue below, past them to discuss the policies that would be required to make them a reality rather than a vision).

Some clarifications. First, saying the future patterns of regional development are limited is not the same as saying the factors contributing to growth and development are few: they are not. Scenario planning, for example, might explore all manner of potential events or long-run trends—national and international, natural and man-made, economic and demographic, technological and political—that could change the amount, composition, and requirements of development. But in the end, for the foreseeable (50-year) future, the bulk of the U.S. growth of population and economic activity is going to occur in metropolitan areas, those people and activities are going to be accommodated in buildings connected by surface transportation systems, and that development is going to occur in the context of already defined urbanization patterns. The factors that may cause the amount and timing of growth to vary are many; the way that growth will be accommodated in metropolitan areas are not.

Second, we are addressing regional urban form, not neighborhood site development. Regional planning is eight miles high: its building blocks

are cities, major arterials, and square miles. Its purpose is to define a fuzzy context for more detailed plans for subregional development. Thus, our comments here are not about the many possible options for site-specific design, which would logically get addressed after a regional planning effort has given some general guidelines about the intensity of use expected or desired in different subareas of the region. It is at the regional level that options for urban form are limited.

Regional form is limited by the fact that what you see is not only what you've got, but what you'll get. A high growth rate for an established metropolitan population and employment might be an average of 2 percent per year: population and employment would double in 35 years. Much of that growth will be accommodated by infill and redevelopment in existing residential and commercial areas. Thus, any planning for future urban form must acknowledge as given much more than half of the form.

Moreover, the geometry and economics of urban development mean that even less of that new growth can really be directed to contribute to a new regional development pattern. The geometry of a circle, where area grows at the square of the radius, means that as metropolitan areas get established and big, they expand outward less rapidly (even though they might be consuming land at an increasing rate). More important, the historical form of their infrastructure has already placed some substantial limitations on the development of any significant new urban centers. Metropolitan areas today inherit highway corridors established 50 or 100 years ago: any significant new development will occur at existing high-access crossroads or new ones that can be created. In the case of highways, new crossroads are increasingly limited to new beltways, which explains much of the phenomenon of Edge Cities.

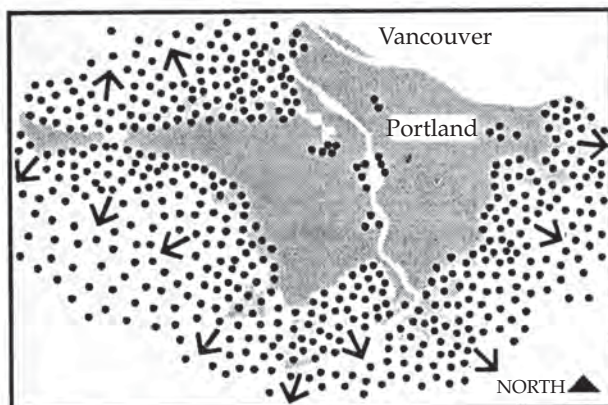
These considerations help explain why only a few distinct ways exist to describe regional growth patterns and urban form, though certainly a large number of those patterns that could develop at the subregional level remain (e.g., one could use many ways to allocate growth to subareas of the region that would be consistent with a concept of with the concepts that follow). Figure 8-1 shows the options for regional land-use and transportation patterns (from Metro's Region 2040). We classify them in the following ways:

- *Expansion.* For most metropolitan areas, this is synonymous with "Let the current trends continue" (or base case, status quo, business as usual, no change). Such a growth option would emphasize growth in the suburbs, the urban fringe, and adjacent to the urban fringe. This option might have two variations: in both, the location of growth would be similar, but one would emphasize housing patterns found in current markets, while the other would assume different patterns (higher-density, better orientation to transit).
- *Concentration.* Emphasize growth in the existing urbanized area. An obvious policy to encourage this concentration is an urban growth boundary. With a fixed supply of land, growth can occur only by using land inside the boundary more intensively (e.g., by increasing densities of new developments, infill development, and redevelopment).¹¹ Another way to increase density (and one more in keeping with our emphasis on proper pricing) is to increase the costs of service extensions, including transportation. While concentration could, in theory, occur uniformly throughout a region, all of the regional plans we reviewed called for pockets of increased density along transportation corridors and at crossroads. Most of them also envisioned high-capacity transit (usually light or heavy rail) as a key to achieving concentration. Concentration is usually described in a regional plan as achievable by directing growth to (increasing density in) a hierarchy of *centers and corridors*.

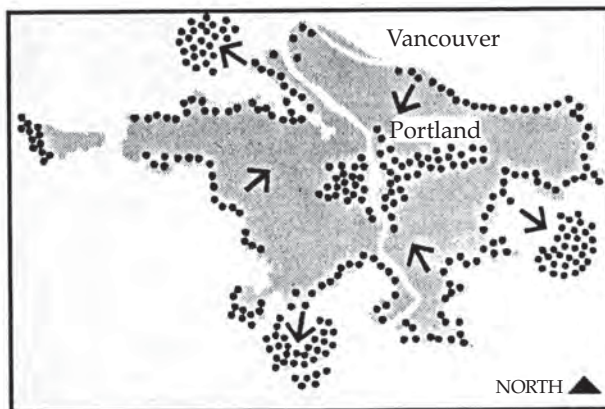
Regional form is limited by the fact that what you see is not only what you've got, but what you'll get.

- Concentration of the expansion.* This concept takes a little from both of the previous ones. It could also be called “expansion of the concentration.” It relieves some pressure inside the urban area by allowing concentrated development in limited areas outside the urban area. In other words, it creates new urban areas in and beyond the urban fringe. Concentration would still be encouraged in existing urban areas, and sprawl at the fringe would be reduced. In the jargon we have used in this report, the concept is to create new, large urban subcenters in the polycentric metropolitan area. These cities might be entirely new, greenfield development, or, more likely, expansions of towns or cities in the metropolitan hinterland. Edge Cities like Tysons Corner, Virginia, are one version of the concept; more distant new cities like Reston, Virginia, and Columbia, Maryland, another.
- Deflection.* In every regional effort to plan for growth someone asks, Why do we have to plan for growth? Isn’t growth the cause of the problems we are now trying to solve? Why not reduce or stop growth? Why not an alternative that doesn’t accommodate the expected growth? This concept would emphasize growth outside the region, either proactively or reactively. An example of proactive policies to achieve this end is joint planning with outlying areas in the metropolitan hinterland or adjacent regions in the hope that those areas would become larger as policies in the metropolitan area forced growth elsewhere. An example of reactive policies to achieve the same end might be a variation of the concentration concept (i.e., by limiting land supply and imposing other constraints on development, the forecasted growth just does not occur). In either case, this concept is a nonconcept with respect to alternatives for a regional urban form to accommodate growth: it doesn’t.

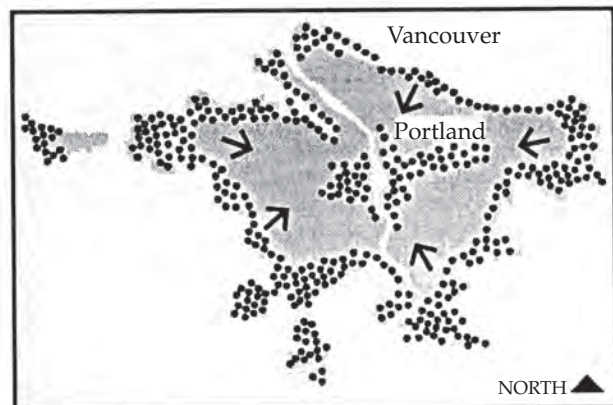
Figure 8-1. Options for regional land-use and transportation patterns



Source: Metro, Portland, Oregon



Expansion. Letting current trends continue will result in single-family dwellings being sited in the urban fringe.



Concentration. Concentrating growth within the urbanized area means more intensive use of land, more multifamily uses, smaller lots, and growth along existing highway and transit corridors.

Concentration of the expansion. Encouraging more intensive growth within the urbanized area but also promoting growth at the urban fringe will mean new or larger communities outside the urbanized area.

In summary, the concepts differ in where they accommodate growth: Will growth expand the region at its urban edge, increase the density of development in its existing urbanized area, or shift to existing or new cities outside the urban area? Since “where” is also about “how much” at some location, the question of density is implied, and with it, more detailed questions about urban form. At the regional level of analysis in all the planning processes we reviewed, those concepts cover the range of options being considered.

Moreover, in *all* cases, citizens never pick “Trend” or “Expansion.” They *always* conclude (by consensus or majority) something is wrong with the way growth is occurring, outward expansion should be reduced, and a regional pattern of denser centers and corridors is the way to do that.

At the level of neighborhood or site design, more alternatives clearly remain: mixed-use subcenters, neotraditional development, transit-oriented development, traffic calming, and so on. In fact, it is only this level—where people can see the implications of a particular regional growth concept on the places where they might live or work—that most citizens can become engaged in the planning process.

Principle 5. Public involvement is necessary, expensive, and incapable of satisfying all stakeholders; regional planning efforts will be simultaneously criticized as too little and too costly

The dual purposes of public involvement are information exchange and credibility. We stated previously our conclusion that the extensive goal setting of regional planning efforts is not justified on technical grounds. Most of the information technicians will gain is predictable in advance. Even if this were not true, the public pulse can be taken more economically and accurately by polling and focus groups than by blue-ribbon panels and open houses. Why doesn’t it happen that way?

Our answer is that most public involvement is about public education and credibility. When the hard decisions have to get made, when neighborhoods and interest groups pack that final public meeting, the decision makers must be able to say, *You all had a chance; we listened; we can’t please everyone; here is our decision.* Thus, the appearance is as important as the reality of public involvement. Since many people either misunderstand or mistrust statistical generalizations based on samples, and random sampling does not allow the participation of anyone who chooses, it seems unlikely the surveys will replace meetings, workshops, and open houses. We believe, however, surveys could be used more frequently. All the regional planning efforts we reviewed used random sampling, some extensively.

Whether the process of soliciting public opinions about problems and goals lasts two weeks or two years, at decision time, policy makers will face the same set of policy choices and the same uncertainty about their effects. What may be different is the amount of agreement by powerful interest groups about the preferred policy. The sidebar gives an idea of how a process shifts from one that is driven by highway engineering considerations to one that tries to be more thoughtful about the connection of transportation and land use.

Neither we nor anyone else can detail a planning process that can be applied without amendment to any metropolitan area. Yes, stakeholders and the general public must be involved, but the weight of evidence is that such involvement should be concentrated. Citizens and even interest groups have trouble sustaining interest for a year, much less five: people move, problems change, politicians and CEOs are replaced. A common strategy for local jurisdictions and special interest is to spend a little time monitoring but not weigh-in until there is a draft policy document to respond to.

Get public agreement on the need to evaluate regional policy, and on the scope and schedule of that evaluation and of the public involvement that

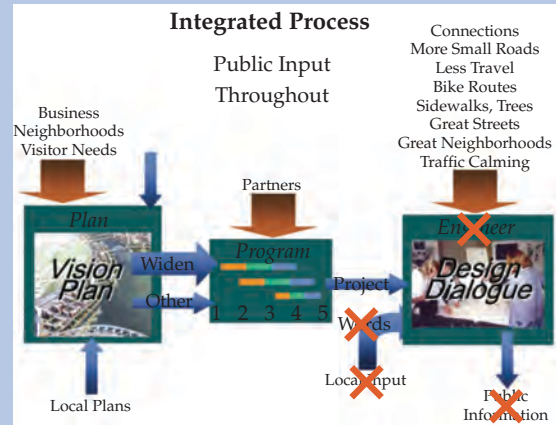
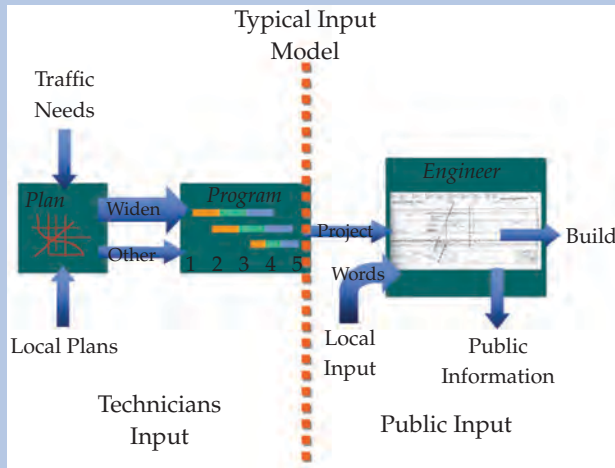
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COMMUNITY ENGAGEMENT FOR INTEGRATED LAND-USE AND TRANSPORTATION SOLUTIONS

By Troy Russ, Principal, Glatting Jackson Kecher Anglin

There was a time, and it may occasionally be with us still, when transportation planners identified problems, modeled solutions, and then asked for public comment on the solutions.

That process can predetermine a roadway's design solution, such as four lanes changing to six lanes, without a real public engagement process. Transportation and land-use are evaluated separately and not at the appropriate scale until the design decision has already been



established. Public input can be late (during final design phase) and focused on the location of the widening (one side of the other), not whether there are alternatives to it.

A integrated land-use and transportation process is driven by an engaging visioning effort that allows and creates community dialog; from concept development to programming and final design. Transportation solutions are developed in unison with the land use vision.

must integrate with it. Document the agreement and move ahead. Move from broad goals to specific policies. Document and publicize interim decisions to protect against attacks at the end that try to go back to challenge fundamental assumptions: your decision makers need that documentation if they want to be able to stand up to those attacks and keep the project on schedule. Don't lose momentum. Set a decision point and get to it. If the decision is to do nothing, accept it and move on to something else, which will almost certainly include returning to the ripening problem a few years later.

Principle 6. Measurement in a multiobjective world is always faulty; regional planning led by technicians for technicians will be interesting to technicians only.

No model of land use and transportation will ever be able to predict the effects of policy with such accuracy and certainty that a set of policies will be demonstrated to be unambiguously superior to all others. We obviously believe, however, that technical analysis can be useful to decision makers. We concur with Beneveniste (1989) that technical arguments are "... rarely sufficient, on their own, to convince stakeholders. ... [Rather they] provide the legitimacy for organizing coalitions of supporters."

The technical analyses that we reviewed for evaluating regional growth options were rife with methodological problems they themselves acknowledged. But Toronto did its study in six months and presented the results in a matrix that facilitated comparison while acknowledging the approximate nature of its results. The lesson is one we all know but forget for fear of being thought softheaded—decision makers need an intuitively reasonable description of problems, interactions, and policy directions, not a box of numbers.¹²

We have two suggestions about how to make the complexity of regional transportation and land use (and, hence, the complexity of the policy choices)

real to the public. First, describe alternative futures; use scenarios.¹³ Scenarios may be a combination of possible or desired futures, and the policies that would be consistent with their realization. Scenarios need not be desirable or even probably futures: they may be defined to clearly illustrate the impact of some external event or local action. Describe the following clearly:

- A limited number of economic, demographic, and policy variables likely to have the biggest influence on regional goals and possible futures. Some of the variables may not be able to be affected much by local policy (e.g., global warming, the price of oil, interest rates).
- Some assumptions about the future behavior of these variables.
- The likely future form of the region and its neighborhoods given these assumptions.

These scenarios may have models and simulations underneath them, but the public and policy makers will want to work from text, pictures, and some summary tables.

Second, a good way to illustrate a future is to reference other regions that, 20 years ago, looked like your region does today. How did they grow? What is the quality of life in these areas? What policies appear to have contributed? Poor Toronto and Vancouver: the U.S. Federal Transit Authority should have had to fund their development as a demonstration project given the extensive use transit planners in U.S. metropolitan areas make of it as a model. Nonetheless, it supplies images of what life could be like in a large metropolitan area that can be contrasted to U.S. metropolitan areas of similar size that have focused on highways.

The caveat to using these techniques is that they must be tied to the costs of achieving them. Some regional visioning efforts have surveyed citizens about what kind of urban environments they like by showing them pictures of different neighborhoods and commercial areas. Not surprisingly, they prefer delis to drive-ins, perking lattes to parking lots, landscape to moonscape. That preference is not very useful for public policy unless people have simultaneously considered questions like: What does it cost to develop given the market forces? What are the costs of public policy (either direct subsidies or lost opportunities) to make it happen? Who is going to pay?

Principle 7. Focus on direction, not destination; the best way for most people to evaluate a long-run vision is to focus on the short-run policies that are the first steps towards it.

We are pessimistic about the prospects for end-state planning. The world changes too rapidly to be able to predict with confidence the behavior of the forces driving urbanization for even 10 years, much less 50. The rate of technological change has grown exponentially. The number of people and firms with the economic resources to make location decisions has grown. The order of the day, and the likely future, is “just-in-time” manufacturing. As the world becomes smaller and the economies more interdependent (a process facilitated by changes in transportation and communication technologies), migration of people and capital becomes more fluid.

Could planners 50 years ago have specified the regional form we have today? Or if that form is a bad one, could they have specified all the technological, demographic, social, economic, and political changes that have occurred and, based on their certitude about those changes, have specified a better regional form? All of our research tells us that not only can planners not specify an optimum urban form, but they cannot model myriad and complex interactions to allow them to predict how that form or any other would be achieved.

From our previous discussion about comprehensive versus regional planning, one can infer our feelings about end-state planning: pop-up for

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the occasional long look to make sure you're generally headed in the direction of the distant mountain, but spend most of your time looking down to hack the blackberries off the path and to make sure you don't walk off the cliff. Big change, when it's possible, is most often a result of a series of small changes.¹⁴

A problem with some of the regional planning processes we reviewed was their failure to address clearly the problems of implementation. They started by asking people what they wanted without telling them what they would have to pay. The costs are not just the tax dollars needed to build highways, light-rail transit, and sewage treatment plants. They are also the policies one must adopt to get to the end-state: changes in government structure, requirements for increased density, restrictions on automobile driving, changes in property rights, and so on.

The lack of emphasis on implementation has several explanations. First, because no one can model regional land-use and transportation patterns for 50 years, it's hard to convey any confidence when trying to relate policies that would require sacrifices today to a better but distant future. Second, talking about change and sacrifice is not politically popular; the debate can be kept alive as long as it is about a better and distant future. Third, the problem may not be bad enough for most people yet; no strong, grassroots movement to create a political will for implementation exists.

Whether by chance or choice, attempts at integrated planning for land use and transportation have moved very slowly toward a discussion of the policies that have to be adopted now to move in the right direction. Until they do, such planning will continue to be primarily a debate among technicians. Citizens are savvy consumers. Give them a product and give them a price; give them a vision and the policies they must endure to achieve that vision. Let them evaluate the trade-offs and choose. Who would make a decision to buy something as big as a refrigerator, much less a region, without knowing the price?

Who would make a decision to buy something as big as a refrigerator, much less a region, without knowing the price?

Principle 8. Work with the market to change behavior: change prices.

If one cannot know the end-state, how can one choose a direction? You can think of lots of examples to illustrate the idea that one can make movements toward better without knowing best. The research we report in previous chapters convinces us that better quality of life in metropolitan areas depends, among other things, on better pricing, and that better pricing means charging something closer to the full cost of land development and driving. Proper pricing is a policy direction, not a destination for urban form. We don't have to know what we cannot know (what is the optimum urban form); all we have to know is that pricing resources closer to their true costs will lead to a more efficient form, whatever it is, than the one we have now.¹⁵ That assumption makes changes in policy possible.

Congestion pricing shows how, in theory, pricing might work to reduce traffic problems. It has not been used much; many analysts believe it is a good idea in theory, but that public agencies will never be able to implement it for technical and political reasons. We agree that implementation is difficult, but it is not impossible. Moreover, compared to the suggestions of many planners of rearranging urban land uses by building transitways on a massive scale, the costs and effectiveness of pricing looks good. In the rest of this section, we discuss two ideas that might make congestion pricing more feasible technically and politically.

One simple way to use the market to achieve public purposes is to increase fees for services now believed to be underpriced. In the case of congestion pricing, the technical issues are manageable. The technology for pricing exists. Setting the price is not an intractable calculation, but it is made even

easier by three circumstances. First, it does not have to be very exact to achieve a lot of the desired effect; if the marginal costs to society are increasing rapidly relative to traffic volume, even charging half of the efficient toll will capture most of the net savings from lost time (Decorla-Souza and Kane 1992, 301). Second, once a toll is set, it is easy to measure its effects and adjust it. Third, it is our judgment that the toll that could be justified technically far exceeds what decision makers can defend politically.

In discussing the fee approach to externalities, however, Baumol and Oates (1988) say this iterative process of setting a fee, observing, and adjusting the fee may not work because: (1) it still requires a lot of information because one must be able to measure changes in performance to know how to adjust the fee; and (2) one cannot, as we have argued throughout this report, know what the optimum is that policy should be moving toward. For example, we showed in Chapter 4 that auto trips were underpriced, not only because of emissions and delays, but because of other social costs as well; measuring the full social costs of auto trips to set a proper trip tax would be a very difficult task.

Thus, Baumol and Oates suggest what they call the *standards and charges approach*. Though they share the economists' penchant for pricing to reduce externalities and increase efficiency (as opposed to direct controls, which are almost always less efficient in theory and practice), they argue that, in the real world, estimating the marginal damage of the externalities requires too much information. They suggest that a region select standards (e.g., targets for emissions or travel time) through a typical political process (which includes some technical analysis, individual judgment, and compromise), and then use charges (pricing) to achieve those standards. They say, ". . . under appropriate conditions [pricing in the form of taxes or subsidies] is the least-cost method for the achievement of these targets" (Baumol and Oates 1988, 164).

As with the previous method, policy makers could then observe the effects of the charges and make adjustments. The advantage is that the adjustments would be pegged to achievement of the standard, rather than to a measure of marginal social cost. (Policy makers must be careful about being too cavalier about adjusting rates to get to the standard since households and firms will be making investment and location decisions in response to these new rates.)

The standards-and-charges approach is easier than pure pricing because it makes one additional assumption, and it's a big one: the standards that ultimately get adopted are consistent with achievement of net social welfare. In other words, society is better off if the standards are met. As Baumol and Oates note, in the presence of externalities and public goods, where market signals are inadequate, it is necessary to use the political process to determine the proper level of an activity, and that process can only estimate optimal results. "The very ignorance that serves as the rationale for the adoption of such standards implies that we can hardly be sure of their consequences" (Baumol and Oates 1988, 174). Nonetheless, when it is likely the social costs are high and they could be significantly and feasibly reduced by reducing the level of activity that causes them, standards and charges have a good chance of being efficient.

A key point is that, in most cases, charges will almost certainly achieve desired standards more efficiently than direct regulation. If a region determines it must reduce peak-hour traffic by 20 percent, does it make sense for everybody to drive 20 percent less or 20 percent fewer miles? No. But that is the way direct regulations are often written. Part of the unpopularity of direct regulation may result from an intuitive understanding on the part of the public of these inefficiencies.¹⁶

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Many regions are considering standards, though most have not made movements to achieve those standards with charges. In Oregon, for example, the state economic plan includes Oregon Benchmarks, more than 150 measurable standards relating to human, natural, and economic resources. They include the percentage of people who commute in less than 30 minutes (2010 target: 88 percent), the percentage who make that commute by a means other than single-occupant vehicle (2010 target: 60 percent), and percentage of miles of limited access highways in Oregon in urban areas that are not heavily congested during peak hours (2010 target: 60 percent). We are not commenting here on the likelihood that these particular measures, if achieved, would lead to net social benefits. What we can say is that what society chooses to measure has a strong bearing on what it will achieve. In concept, there is no reason one could not use charges to try to achieve these standards.

Principle 9. Evaluations of regional policies focus on efficiency; interest groups and the public also care about equity—what will this mean for me?

Our discussion so far has focused on the efficiency of policies. Also important to the ability to adopt policy changes is an evaluation of equity (the distribution of impacts). The techniques here are necessarily more qualitative. They all, however, are trying to answer the same question: Who benefits and who pays? Evaluation of policies should provide a description of likely winners and gainers by type (e.g., public vs. private, developers vs. residents, current residents vs. future residents, high-income vs. low-income households) and by geography (which neighborhoods, by city, by county).¹⁷

The tension between efficiency and equity is apparent throughout transportation policy. In 1993, a proposed federal gas tax, which we have argued is a reasonable second-best policy for addressing the underpricing and over consumption of auto trips, was criticized by members of Congress from oil-producing states and low-density western states (e.g., Wyoming) where people drive greater distances. They argued it would be an unfair burden to their constituents. But a back-of-the-envelope estimate is that the burden would have been on the order of 10 to 15 cents per day per driver.¹⁸

Another example of that tension is found in public transit. Many analysts have shown with numbers what casual observers criticize: big buses carrying a few passengers, and multiples buses with expensive hydraulic equipment for wheel-chair access that gets used only occasionally. An obvious solution to this inefficiency is smaller buses, less frequent service, and limited service to low-density areas. The concerns about the economically or physically disadvantaged could be addressed more efficiently with special service (e.g., dial-a-ride buses) or even vouchers for private transportation (e.g., taxis). But the decision of what service to provide is not made on efficiency criteria. Advocates for the disadvantaged argue that they should have equal opportunity and equal access; they have a right to enjoy the services that others enjoy, as part of the mainstream, without the stigma of special programs. The federal Americans With Disabilities Act supports this position.

Our discussion so far has focused on the efficiency of policies. Also important to the ability to adopt policy changes is an evaluation of equity (the distribution of impacts).

Principle 10. Bundle complementary policies.

Bundling changes in complementary policies can have an effect greater than the sum of the effects of changing any of these policies in isolation. For example, the effect of changing a single transit policy (e.g., lowering fares) in an environment dominated by designs that ease auto travel probably differs significantly from the effect of a similar change in an environment more supportive of transit. Even providing comfortable and fast transit from the suburbs to the Central Business District will have limited effects in an otherwise auto-dominated environment: witness BART in the Bay Area.

The current conventional wisdom in transit planning is that increasing transit ridership, especially among travelers who can choose between transit and auto, requires increasing the reliability, accessibility, and overall quality of transit service. Increasing reliability requires more than increasing the number of buses: it requires redesigning bus corridors so that buses can avoid delays and stay on schedule. Increasing accessibility means more than building nicer shelters: it requires increasing the number of people willing to walk to a transit station, which requires less space devoted to surface parking lots, improvements to the quality of the pedestrian environment, and so on. Encouraging people to ride transit requires a package of complementary changes in land-use and transportation policies.

Similarly, increasing the price of an auto trip without improving transit service reduces discretionary auto trips less than it might have because many people have few competitive transportation alternatives. For example, higher gas prices in the later 1970s had only a temporary impact on transit ridership. Transit service remained of generally poor quality relative to the convenience of the auto, and people adjusted to higher fuel prices by purchasing more fuel-efficient cars. Increasing the quality of transit service (by changing a wide variety of land-use, building-design, and transportation policies), while at the same time forcing drivers to pay the full cost of their trip, would probably have much stronger effects on transit ridership and per-capita VMT than any of even the best new rail transit systems. All those new systems, despite their cost, face the impossible task of competing with the auto in a policy environment dedicated to easing auto travel.

We have been consistent on this obvious theme throughout this report: integrated planning for land use and transportation means integrated policies. Chapter 7 shows how we would try to do that integration. The need for such integration is at the core of any rejoinder to the many analysts who argue that construction of extensive rail transit systems is a poor investment of public funds. They argue that rail transit simply cannot compete cost-effectively given the land-development patterns and travel habits typical in all but a few of the biggest, densest cities in the U.S. We agree: the experience of transit agencies across the nation provides ample support for the position. The question remains open, however, about how well transit would do if many policies—all aimed at charging full cost to autos and increasing the effectiveness of transit—were adopted simultaneously.

In a different vein, a potential advantage of integrated policy is related to the issue we just raised about equity: it may facilitate the siting of regional facilities that have net benefits to the region as a whole but cause net losses to the neighborhoods in which they are located (Locally Unwanted Land Uses—LULUs—and Not In My Back Yard uses—NIMBYs). There are many facilities like this: landfills, solid waste transfer stations, power generators, bus terminals, low-income housing, and so on. Part (we admit, only part) of the problem in siting these facilities is that they are attempted one by one. Cities, under pressure from neighborhoods around a possible site, have every incentive to say no: why should they be the one city of two dozen in the region to take the facility? But, if all facilities were sited simultaneously (not necessarily to be built immediately, but sites selected for future building), the possibility exists that city decision makers could argue with their aggrieved neighborhoods that all they got was their fair share.¹⁹

Principle 11. Preserve long-run opportunities.

As a region grows, development should occur both in existing neighborhoods and in the undeveloped fringe of the region. Neighborhoods redeveloping at higher densities require changes, especially in transportation infrastructure.

The current conventional wisdom in transit planning is that increasing transit ridership, especially among travelers who can choose between transit and auto, requires increasing the reliability, accessibility, and overall quality of transit service.

Existing low-density neighborhoods should be planned to enhance redevelopment at higher densities and to allow increased transit service.

Improving transit service in areas developed under auto-oriented policies requires ingenuity. Public agencies need to be flexible enough to respond to opportunities that present themselves, such as purchasing land for rights-of-way or station areas. If government truly believes its investments in transit will make land more valuable in the future, it should find ways to convince the public it serves that it would be a good idea to purchase that property.²⁰ Its risk is slightly diminished to the extent it has control over the policies that allow and encourage land to develop.

Ironically, one of the biggest obstacles to preserving opportunities for change may be public policy itself. Levine (2005) makes the case that the main obstacle to the higher-density vision of transit-oriented development that shows up in all regional planning efforts is intractable single-family zoning. He presents evidence to show that market forces would move toward such development patterns but are prohibited by policy. A first step for public policy: get rid of the policies keeping you from doing what you want and not providing at least offsetting benefits.

Principle 12. Integrated regional planning needs champions.

This is an obvious point made by many experts on planning process. Our experience is consistent with their recommendations. Without some policy makers and opinion setters committed to keeping integrated planning pumped up, it will collapse under its own weight.

Principle 13. If you want to affect the long run, take a long-run attitude toward change.

One way to have a long-run attitude is to work for small successes with the hope they will eventually add to big change. In this context, and consistent with our advice to work with market forces, demonstration projects make sense. For example, a lot of money in the last few years has gone to studies for planning neotraditional neighborhoods. Many developers claim there is no market; planners counter that no market is seen because there's nothing to buy.

This situation is an obvious candidate for a demonstration project, with the public underwriting some of the risk. If the planners are proved correct, the cost is small (because the project will pay for itself either directly or by encouraging more of the kind of development public policy may desire). Demonstration projects for development around transit stations, franchising jitney service on main corridors, congestion pricing, and other market-oriented policies are possible. In addition to the advantage of a relatively small cost, a demonstration project allows policy makers plausible deniability; it is overtly an experiment—if it fails, society benefits from the knowledge gained, and politicians can claim small credit for having the foresight to test before investing.

A second long-run technique is to focus on education. Much of the critique of planners' plans by the private sector is that they don't fit with market conditions: people won't pay for the high density, or the open space, or the transit that planners want developers to build—there is no effective demand. But in economists' basic formulation of a demand function, one of its key determinants is preferences. Preferences change. Among the things that can change them is better information (e.g., education or advertising). They can sometimes change rapidly. Not many people would have predicted in the 1980s that smoking in public transportation, public buildings, restaurants, offices, and even parks would be prohibited or restricted to the extent it has been the last decade of the twentieth century (or, maybe more to the point for this report, taxed, and the companies who make the cigarettes sued by public representatives for public damages and costs). Attitudes changed.

Ironically, one of the biggest obstacles to preserving opportunities for change may be public policy itself. Levine (2005) makes the case that the main obstacle to the higher-density vision of transit-oriented development that shows up in all regional planning efforts is intractable single-family zoning.

American attitudes about property are anchored deep. In the long-run, the kind of integrated planning for land use and transportation that planners want may be best served by increasing the number of people who share their beliefs about property: namely, that private property rights be exercised in the context of public purposes and the rules to achieve them.

Thirty years after its state land-use program was adopted, the debates on the proper scope of planning still rage in Oregon. In 2007, at the time of this writing, Oregon has a new property-rights law (a backlash to the regulations of the state land-use requirements) and a task force in the middle of a three-year effort to evaluate and adjust the program. But among the hooks that keep the program hanging on, however tenuously, is a tacit acknowledgment by a majority of voters that that's how things are done in Oregon—quality of life and the environment matters more there, and public debate about public policy in the context of statewide goals makes sense. Our prediction is that the Oregon program will not be thrown out, but it will be adjusted. Some argue that the failure to adjust earlier caused the property-rights ballot measure, which is unlikely to play out as a measured adjustment to the program.

If planners want change, they should spend some time developing a constituency that will look at their attempts at integrated regional planning and say, Of course, how else would you do it? Now let's get the policies ironed out. If the profession takes a long view of change, it should spend some resources educating the next generation of leaders and the next generation of voters. Elementary schools are already teaching environmental science. Civics classes should be teaching what's at the root of the three branches of government: Why government? Why not just markets? Why is it that government intervention can lead to a better world? And, yes, What should we be doing collectively to make our metropolitan areas develop in ways that increase our quality of life as individuals?

CHAPTER 8 NOTES

1. The only system that comes close is the extant market system: individuals register preferences via their purchases. The problem with this system is one we already identified: if the prices are wrong, the preferences based on them are likely to be wrong.
2. Many of the standard texts on welfare economics make this point and proceed to describe how policy will be evaluated under the assumptions that: (1) efficiency gains were all that mattered (based on the further assumption that those gains could be redistributed to compensate losses); (2) relative prices reflect the true value of resources (the aggregate of individual preferences as reflected by their willingness to pay for those resource); (3) the current distribution of wealth (resources) is socially optimal; and (4) the impacts of alternative policies can be identified, measured, and added to estimate their relative performance (benefit-cost analysis). There are several notable exceptions to the standard, including Baumol and Oats (1988), who state that the presence of externalities and the uncertainties they introduce into any efficiency analysis means that "we may not even know in which direction to modify the level of an externality-generating activity if we want to move toward an optimum" (page 159) and the "efficiency prices are robbed of much of their normative usefulness" (page 131).
3. See Banerjee (1993) for a discussion of why planning is required to deal with issues relating to the distribution of resources, power, rights, and opportunities.
4. For an economist's proof of this assertion, see Baumol and Oats (1988), Chapter 8.
5. Even Oregon, known for its aggressive policies on land use and its connection to transportation, is now (2007) backing off (still short of full retreat) in response to a property-rights ballot measure (Measure 37).
6. We say "probably should not" to allow the possibility of the conclusion that: (1) the inertia of our political system and the magnitude of the change needed make meaningful change impossible in the absence of a crisis; (2) any efforts by planners to do anything less than meaningful change is just postponing the crisis necessary for meaningful change to occur; so (3) planners should quit postponing the collapse, which will only be more traumatic for their efforts.

7. An often-used political strategy is for a higher level of government to promulgate policies that require local governments to do something that their decision makers may have wanted to do anyway but could not accomplish without someone else to blame it on. State requirements for local land-use planning work this way, at least in part. To a large extent, however, the political problem of regional action still exists, but at a higher level of government for a larger region: How does one get agreement at the state level about the kind of policies we have been discussing? By making the federal government require them? The problems about process are different in scope but similar in kind at all levels of government.
8. <http://www.sacregionblueprint.org/sacregionblueprint/> and <http://www.nipc.org/2040/> (accessed December 2006).
9. We've already discussed the problems of how or whether the preferences of individuals can be aggregated into preferences for society. The concept of fairness is only occasionally addressed in technical studies, probably because it is so difficult to measure. The best way to handle it for most regional planning efforts is by evaluating alternatives from different perspectives: How does the alternative affect different groups (e.g., city/suburban, east/west, low income/high income)?
10. We vacillate on this issue. On the one hand, as technicians, we support Moore (1988): just do it. Get some provisional ideas out quickly; start the discussion; be responsive to critique; find the hot buttons; get adequate technical information quickly to address issues of concern; decide before the public runs out of interest or moves. On the other hand, we've seen slow process work to create, step by baby step, a base of political support—take a long time on goals to keep the conversation going and the controversy low; don't introduce actions until people have had a lot of time to process that actions will probably need to be taken. Like the grieving process or settlement in litigation: some decisions require more time, not more information.
11. Note that this conclusion is based entirely on supply-side analysis. It assumes demand will not be reduced (in other words, the growth of population and employment would not be less than predicted) despite restrictions of land supply that would increase the price of land. At its extreme, this alternative begins to approximate the kind of regulation—unprecedented at a regional scale—that would be required to slow growth to rates substantially below those expected.
12. We can't resist quoting Beneveniste (1989, 128) on ineffective planners. "They often tend to write trivial plans—lists of decisions that have already been taken. They defend their professional turf by inventing esoteric languages. They spend their energy in developing planning methodologies that are arcane and difficult to interpret. They often avoid complex choices, preferring to list alternatives and to provide large volumes of data without facing the need for choices. . . . They invent names and functions for planning and often protect themselves by routinizing planning. Planning becomes bureaucratic, data are collected, analysis takes pace routinely, and no one seems to have much to say about the usefulness of these planning efforts except those who carry them out. . . . [T]hey overemphasize the professional sanctity of their craft."
13. For a thorough exposition of how scenarios get used in a public planning process, see Hopkins and Zapata (forthcoming 2007, Lincoln Institute of Land Policy).
14. These ideas are a restatement of the mixed-mode scanning of Etzioni and consistent with more recent work of Lew Hopkins on "a system of plans." Cities have lots of different plans (different topics, time horizons, purposes, authoring departments or agencies, quality) that collectively define its vision for development and the tools to achieve it. The end-state, if there is one, is fuzzy: the individual plans need to be flexible and capable of relatively quick adjustment to keep them consistent with market and political forces, and with each other.
15. We will not enter into the extensive economic literature about the qualifications of this statement. We will summarize by saying it can be shown to the satisfaction of most economists that changing prices to more accurately reflect their value (opportunity cost) does not necessarily lead to improvements in social welfare if the prices of any other goods do not reflect their value (see Baumol and Oats 1988). That conclusion, if acted on, leaves policy makers paralyzed in the real world: no policy can be provided to contribute to the net welfare of the community for which it is being adopted. One escapes that paralysis only by making some assumptions. We prefer to make as few as possible about the value of planning and direct regulation, but we do assume that the level of externalities from driving the auto during peak hours is so great that increases in the price of an auto trip will improve efficiency.
16. Baumol and Oats (1988, 172), commenting on studies of pollution policies, have generally found that "the least-cost solution [charges] entails costs that are only

a modest fraction of those under the current direct control system: the estimates range from a high of roughly 50 percent of current costs to less than 10 percent.”

17. For an overview of the techniques available and references to the professional literature on this topic, see Patton and Sawicki (1993) and Krumholz and Forester (1990).
18. If, on average, people in these burden states drive 50 miles more per day than people in other states, and do so 300 days per year, they drive 15,000 miles more per year. At 15 miles per gallon, they use an additional 1,000 gallons of gas, which, with a tax of 4.3 cents per gallon, would cost them \$43/year—about 10 to 15 cents per day.
19. We won’t detour into other techniques for siting NIMBYs. We will note, however, that here also market mechanisms seem to offer the best hope: somehow the local areas being damaged must be compensated with some of the net benefit that accrues to the region.
20. Because of the significant federal role in funding transit, and the limitation on the use of those federal funds, this policy requires either a change in federal policy or local funding.

EPILOGUE

Big Questions, Answers, Equivocations, and Directions

We hope the previous chapters have addressed the big questions. This section forces us to be sure and helps meet the demands of the readers who wish that we had limited the whole report to this size in the first place.

WHAT LEADS: TRANSPORTATION OR LAND USE?

There should be no doubt that transportation and land development interact. So the question is one of dominance: Is the dominant force of transportation causing land development, vice versa, equal, or indeterminate?

If the question is historical and long run, transportation dominates. Prior to the Industrial Revolution, cities grew predominantly in places that people and goods could get to most easily: ports, crossroads, and so on. Even after the Industrial Revolution, cities in the newly urbanizing U.S. grew around transportation hubs: ports, rail terminals, and, later, highway interchanges.

If the question is about growth now, in modern metropolitan areas, the answer is equivocal because of the inherent feedbacks between transportation and land use. New highways get built, and people use them. The improved access encourages land development. The new land development generates trips, which generate fuel-tax revenues that flow into transportation agencies charged mainly with improving traffic flows. New highways, and sometimes mass transit systems, get built to serve the places with the most trips, which are the places where land has developed. The cycle continues, sometimes viciously.

If the question is about what *should* lead (transportation or land use) our answer is still equivocal. Ideally, they should happen together. In many cases they do not, and even when they do, the integration is more superficial than fundamental. But in a metropolitan area where the responsibility for the planning resides in multiple agencies, land use is the more logical nexus than transportation for an integrated view of regional development.

Chapters 5 and 7 and Appendix G make that case. Planning for quality of life in metropolitan areas is multidisciplinary and long run. It is typically a land-use planning process that draws together issues relating to land development, supporting infrastructure (including transportation), environmental quality, economic development, and

other public services in the context of developing a long-run vision for a high-quality urban environment and policies that make achieving that vision more likely.

But because of funding, transportation agencies may be leading the development of the urban vision instead of accommodating it through the transportation system. In those cases, transportation agencies should not view land-use controls primarily as congestion management tools. Creating good urban and metropolitan areas is complicated and multidimensional. Everything interacts. There are trade-offs. One cannot find optimal solutions to complex urban problems by maximizing on transportation variables (VMT, peak travel speeds, and so on) alone. Transportation must help support a larger vision for quality of life in a metropolitan area.

WHY CAN'T WE BUILD OUR WAY OUT OF URBAN CONGESTION?

We phrase the question immediately above in the way we do instead of, Can we build our way out of congestion?, because even highway departments now say, We can't build our way out of congestion. But why not?

Many terms describe the same phenomenon: latent travel demand, induced travel demand, triple convergence. Travelers are not charged for their use of congested roads by location or time of use (too many vehicles in the same place at the same time is what defines congestion), and demand for travel at low prices substantially exceeds the supply of facilities that would allow those facilities to be uncongested. One cannot limit the use of the highway to the drivers that were using it before it was expanded. The inevitable result: in congested areas drivers quickly find the new highway capacity and fill it. Moreover, the core highway system in most metropolitan areas is primarily built: adding new capacity is increasingly expensive and decreasingly effective per lane mile.

Unless you started reading this report at the end, you already know our theme: pricing matters. Scarce and valuable transportation capacity should be rationed, and probably to the users who are willing to pay to show how much they value it. Better pricing (which means, in most cases of congestion, higher prices) of all aspects of transportation is required to get to efficient levels of congestion. Apropos of the question this section poses, note that proper pricing does not eliminate congestion: proper pricing reduces congestion to the point where reducing it further costs more than the value of the additional transportation benefits (primarily travel-time savings, travel-time reliability, and increased safety).

WHAT POLICIES ARE BEST FOR . . . ?

Pricing matters, but it won't happen quickly: it has technical and political obstacles. Even if it did happen instantaneously, the rest of the transportation and land-use systems are not ready for it: policy needs to prepare for a transition. Thus, metropolitan areas will have to work through less-than-ideal policies to create, incrementally, a better transportation system. Our opinion (which we acknowledge as such) is that such "second-best" policies are likely to include more emphasis on:

- reducing vehicle trips without reducing person trips by using demand management techniques;
- encouraging higher-density and mixed-use development; and
- limiting urban sprawl.

This opinion implies another: one works toward "better" because "best" is unknowable. The public interest in a metropolitan area cannot be defined in any rigorous technical sense that simultaneously gains broad political acceptance. Policy evaluation is contingent; it depends on goals and as-

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sumptions. Would travelers prefer to shift toward public buses and trains operating in separate rights-of-way, or would they preferred private jitneys and carpools operating in HOV or HOT lanes? Should cities limit urban sprawl with growth boundaries or with public-services requirements and development impact fees? Should cities seek higher densities by simply relaxing minimum lot size, parking, and use requirements? By paying for infrastructure upgrades to promote infill and redevelopment in older areas? Or by charging for parking and requiring a mix of uses?

To say (correctly) that “it depends” is defensible but unsatisfying for readers and authors alike. We’ll risk going a little farther:

- Working on highway capacity without working on demand is not going to solve congestion. Less obviously, expanding or improving alternative modes will reduce the costs of using those modes relative to the automobile, but without demand management on the highway/auto side of the equation, improving other modes gives only temporary relief from highway congestion (triple convergence rules)—too many resources get channeled away from more valued activities.
- That said, the purpose of transportation policy should not be to make auto trips arbitrarily expensive. Stating the goal that way (as many advocates do) is technically incorrect and politically explosive. The purpose of policy should be to ration capacity efficiently and to encourage innovation that reduces the full costs of transportation. Imposing the full marginal cost of each trip (or a better approximation of it) on drivers does both. And it signals where transportation improvements are most valued.
- In the absence of better pricing (and it will be absent in most metropolitan areas for years to come), land-use and transportation planners should work to adopt policies that approximate the kinds of transportation and land-use patterns that would exist if prices were more in line with marginal costs. The direction of the shift likely to be more efficient is toward higher densities and more mixing of uses, especially in centers and corridors.
- These policies are neither technically nor politically easy. By eschewing pricing solutions, policy makers must choose regulatory solutions, which require people and businesses to act in a specific way estimated to be better for society in some sense. But the people who have studied the growth of metropolitan economies and development are among the first to admit they have only a crude understanding of how that growth works. A cadre of PhDs and PCs can’t predict with certainty or in detail how proposed policies will ripple through the natural, built, and social environments of a metropolitan area.

Policy makers are talking the talk, but can they and their constituents walk the walk (or, failing that, buss the bus)? If they can, they haven’t. They still drive in droves.

WHAT CAN BE DONE TO GET THOSE POLICIES ADOPTED?

Is the new (or renewed) emphasis on the connection between transportation and land use likely to lead to any significant changes in policy? Our response to this question—yes and no—summarizes the points we just made.

On the positive side is the demonstrable fact that integrated planning for transportation and land use is getting a lot more attention from policy makers and technicians. It has moved from an idea rediscovered in the 1990s to mainstream today. That attention results from and leads to more technical work on the interrelationships between transportation and land-use policy.

Policy makers are talking the talk, but can they and their constituents walk the walk (or, failing that, buss the bus)? If they can, they haven’t. They still drive in droves. The evidence is overwhelming that technical solutions to transportation problems require better pricing as a cornerstone. But full marginal-cost pricing is usually the last policy tried because the public and their policy makers believe that they should be able to get more for less. That

belief underlies movements to reinvent government; to privatize government services; to cut property taxes; and to fund new facilities and services with creative financing (i.e., to get those facilities without having local taxpayers fully pay for them or know that they have fully paid for them).

What expectations should one have about the reliability of the intuitions of policy makers or the results of a public process where responses to a questionnaire are based on five minutes of thinking during TV commercials or five hours of thinking in a workshop? Most citizens do not understand the metropolis in the way we have described it in this report. They care about and understand some part of it—the part that directly affects their interest—but not the entire system.¹

It is no surprise, then, that the public processes for integrated regional planning we review focus on questions that everybody can have an opinion about: goals, values, and visions. Regional planning for land use and transportation, as we predicted in the first edition of this report, is ascendant—lots of metropolitan areas are thinking regionally. But most of these efforts use a safe public involvement process more likely to generate predictable generalizations about values and policy directions than substantive debate about trade-offs and tough policies.

Some wasted effort is an inevitable part of the friction of collective decision making and action. Accountability and involvement have a cost. But even if you find yourself bound to the typical process of public involvement, consider the changes we have suggested that may make the process more productive by getting it focused on real choices:

We suggest these changes to make the planning process more productive:

1. *Get quickly past general goals to specific policies.*
2. *Talk about directions rather than end-states.*
3. *Consider policies that work by changing incentives.*
4. *Think about how to redistribute gains.*
5. *Avoid subsidies on top of subsidies.*
6. *Create and work from a regional forum.*
7. *Focus on quality of life, not just transportation.*
8. *Recruit and support champions.*

- Get quickly past general goals to specific policies. Confront policy makers and the public with real trade-offs requiring real choices. A big part of the cost of some of the visions for future urban form is the regulations and investments that would be required to achieve them. Citizens want to debate these policies.
- Talk about directions rather than end-states. Visions of future development patterns help sell changes in policy. But such visions must have some link to attainable changes in policy. Moreover, too many potential futures exist to be able to predict with any confidence the one that will unfold. Look for big themes and then for policy directions consistent with many of them: such policies allow quicker and smarter responses as the future presents itself.
- Consider policies that work by changing incentives. Policy in all areas has been moving from regulation toward market-based incentives and disincentives. Pricing is a simple policy that allows options for innovative responses: do what you want—find your own best solutions—as long as you pay the full costs (and suffer the full consequences) of your actions. Think more in terms of improving incentives (matching individual objectives with social objectives) than of forcing specific outcomes.
- Think about how to redistribute gains: policy changes that improve efficiency make the pie bigger, but they usually generate losers as well as winners. Policy makers often have trouble adopting efficient policies because some powerful interest group fears that it will be made worse off. Such fear may be justified (e.g., if that group has been enjoying implicit subsidies), but the politics may work better if efficiency gains are spread and used to reduce the pain of the transition.
- Avoid subsidies on top of subsidies. Subsidizing an activity to counteract the effect of an existing subsidy for another activity may improve relative prices, but the combined subsidy further distorts incentives. And subsidies are hard to take back once given.

- Create and work from a regional forum. The biggest gap in government structure is at the regional, metropolitan level. State boundaries rarely correspond to a relevant physical or economic geography (e.g., a watershed, a commute shed, a trade area). City boundaries have the advantage of providing consumers (residents and businesses) choice, but their boundaries are usually too small to deal with all the externalities and transactions costs of regional land-use and transportation issues. Thus, there is a need for a regional body charged with thinking about regional, cross-boundary issues, and across subject areas. Debate is not enough: binding, enforceable decisions are desirable, but compliance through moral suasion (e.g., through the convening of an ad hoc group of public and private opinion setters) may have to suffice.
- Focus on quality of life, not just transportation. Metropolitan Planning Organizations exist, but their focus is on transportation. They tend to think of land-use regulation as a way to treat highway congestion: we built the highway to serve your local land-use plans, then you local governments allowed development to get out of hand. The regional planning body has to think more broadly about the interactions between land use and transportation.
- Recruit and support champions. The kinds of policies necessary to significantly affect metropolitan transportation problems will not be easy to adopt. Technical analysis can support committed political leadership, but it cannot substitute for it: you need champions from the beginning²

Enough calisthenics: it's game time. And playing is the thing, because the game never ends and a win or loss is never final. To travel happily is better than to arrive.

EPILOGUE NOTES

1. This statement is not a condemnation of the ignorance or apathy of the average citizen. As Richardson and Gordon (1993) have noted, citizens have a "rational ignorance" about public policy they perceive as having little effect on their interest. They leave long-run regional policies to their elected officials or the interest groups they fund, or they leave it alone entirely.
2. We do not have space to discuss the details of how technical analysis works in a political process. We subscribe to the views of Beneveniste (1989).

Framework for Evaluating Public Policy¹

Policy makers and analysts agree, in theory, that good evaluation of public policy² requires identifying and measuring significant benefits and costs. In practice, however, they disagree about how and how well those benefits and costs can be identified, measured, and summed.

Just identifying the potential effects of a large project is difficult. Estimating the *direction* of an effect (positive or negative) gets harder. Estimating the *magnitude* of an effect (how big is that positive or negative impact?) harder yet. Consolidating the many effects into a summary measure of net impacts is beyond the capabilities of almost any impact analysis—although that is often not recognized in the analysis.

The point is not that public policy decisions cannot be made without a thorough and reliable evaluation: such decisions are made daily. Rather, the point is that if a community (local, regional, state, or national) is serious about basing a decision on an assessment of likely impacts, it should accept some well-supported principles about the structure of such an impact analysis.

This appendix discusses principles fundamental to the evaluation of *any* large public investment decision (not just decisions about transportation projects and programs). It has three sections, plus a summary at the end:

1. **Overview of Policy Evaluation** describes the broad goals of any evaluation of the full costs of large public investment decisions.
2. **Principles for Evaluation** defines terms and gives some guidance on how to identify and measure what matters to the public and elected officials.
3. **Using Technical Work in a Political Process** describes how multiple measures of desired objectives can get combined in a way that is technically defensible and makes sense to policy makers and the public.

OVERVIEW OF POLICY EVALUATION

Public policy gets adopted to achieve public objectives. Thus, at the most basic level, an evaluation of public policy must answer two questions: What do we want to achieve? and What do we do to increase the likelihood that we will achieve it? Figure A-1 illustrates the questions.

The two questions come together when possible policy actions get evaluated against measures of desired outcomes. Below is a description of the ideas and terms used in Figure A-1:

- *Things a community wants to achieve (desired outcomes)*. These things may be stated broadly or specifically. Positives (what we want; desired outcomes; benefits) are mirror images of negatives (what we want to reduce or avoid; undesired outcomes; costs).
 1. *Goals* are broad statements of desired outcomes. Examples: increase environmental quality, economic prosperity, transportation choice, social justice. If the high-level goals get parsed into sub-goals, they are often called *objectives*. Logically, since goals and objectives are the categories of things people care about, they are roughly synonymous with the term *impacts*: the objectives are about good impacts that a community wants to increase, and bad impacts that it wants to reduce.

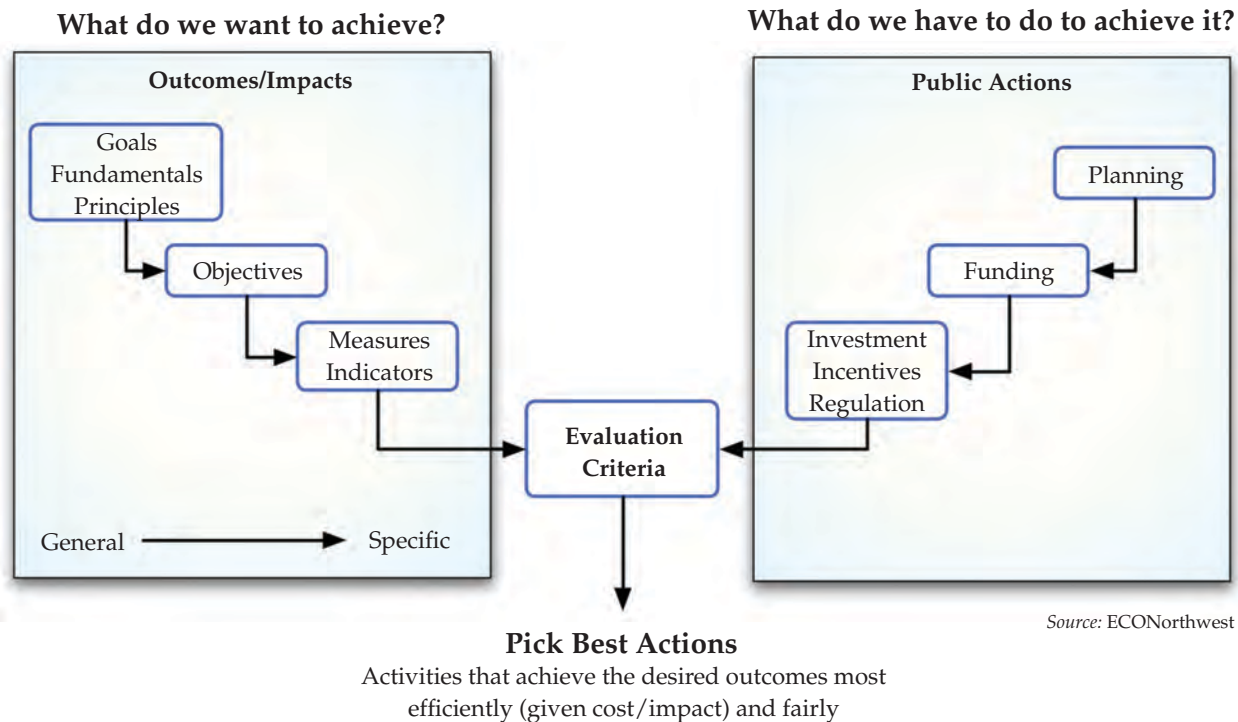


Figure A-1. The standard planning model for most regional planning efforts

2. *Measures* are specific statements of outcomes. They should fall under (“nest within”) one of the higher-level goals or objectives. Examples: number of new jobs by type; reported crimes per capita; amount of new investment in lower-income neighborhoods. Measures are also called *indicators*.
 3. The measures—which should be consistent with, related to, and more specific than the goals—are *evaluation criteria*: logically, policy choices should be evaluated based on how effectively they achieve desired outcomes as specified by the measures deemed to best reflect those outcomes.
- *Things a community is willing to do to increase its chances of achieving what it wants to achieve.* The things that the public sector (a local, regional, or state government) can do are *actions* or *policies*. There are several ways actions can be classified (by where they get applied; by who implements them; by the area of development they affect). The following is a list of the techniques that can be used to achieve desired outcomes. The taxonomy here is as close as we can get to one that is comprehensive and mutually exclusive:
 1. *Planning*: identifying efficient opportunities for collective action
 2. *Funding*: getting the public to agree to pay for some of those actions; getting the money
 3. *Investment*: building public facilities; providing public programs (follows Planning and Funding; there should be agreement on what to do and on how much can be paid for before building can occur)
 4. *Incentives*: giving financial incentives (direct or in-kind) to the private sector to provide the desired public facilities or programs (an alternative to direct public Investment)
 5. *Regulation*: requiring the private sector, as a condition of development, to preserve or provide certain public facilities, amenities, or services (an alternative to Incentives)
 6. *Coordination*: getting everyone to cooperate, and to do so efficiently: not just public-private partnerships, but public-public partnerships.

Getting agreement on definitions is essential to any intelligent discussion. Fuzzy language risks fuzzy policy. Some related points:

- The distinction between what we want to achieve and what we should do is often not made. Goals get confused with actions. They are related, but different: Figure A-1 shows that.
- There are many terms that cover more or less the same idea, though they get used differently by different jurisdictions and even by different people in the same jurisdiction. Some examples:
 1. Terms related to outcomes: Goals, principles, fundamentals, objectives, impacts, measures, indicators, evaluation criteria
 2. Terms related to actions: strategies, policies, implementation tools, programs, regulations, investments

There are lots of ways to define and combine these terms. The essential point is to pick one and enforce its consistent use. Figure A-1 and the text describing it is an example.

Long-run urban planning presumes that thinking and taking collective action now can lead to a better future than failing to do so. When making decisions, long-run planning is trying to describe and evaluate alternative futures. Those futures are called alternatives or scenarios and are defined by the measures and the actions deemed relevant to achieving them.

There is a useful distinction to be made between *likely*, *desirable*, and *possible/hypothetical* futures. Consistent with the professional literature on scenario evaluation, we recommend the following use of terms:

- Reserve the term *scenarios* for hypothetical futures, independent of whether they are desirable or likely. A scenario might be, for example, “Very high energy costs” based on a combination of reduced supply from the Middle East and new, high gas taxes. That may not be likely, and it is hardly desirable, but it is possible, and a scenario might explore the implications of that future for travel demand and spatial development.
- Reserve the term *alternatives* to mean “potentially desirable and somewhat likely futures.” Alternatives are real choices about policy direction—the typical choices in any planning or policy evaluation process. In the context of long-run planning for growth and development (the focus of a city’s comprehensive planning), typical alternatives are defined spatially: “trend or status quo,” “focus more growth in centers and corridors,” “expansion at the urban fringe,” and so on.

We now examine the left-hand side of Figure A-1 in a little more details: outcomes and impacts. The goal of public policy (of government action) is, in broad terms, to make the people government serves better off. The presumption is that collective action in some areas will yield superior results to a lack of collective action. That is the justification for taxing people: government will provide some desirable services individuals might otherwise be unable to provide by themselves (e.g., certain aspects of environmental quality) or would not provide very efficiently (e.g., a regional highway system).

Government usually operates at the base of psychology’s hierarchy of needs, trying to make sure certain biological, physiological, and safety needs are met (e.g., clean air and water; shelter; personal and property security). Individuals must supply their higher needs (belongingness, esteem, self-actualization) themselves. The assumption is that adequately meeting the base needs provide the time, security, and economic resources for pursuing the higher needs.³

In economic terms, the goal of government policy is to increase *welfare*: the economic and social well-being of the citizens it serves. Ideally, policy makers and the public they represent would like to know how alternative investments perform relative to one another. They want to know what they get (benefits) for what they give up (costs). Adding benefits (positives) to costs (negatives) yields, in theory, a

measure of *net benefits*. In broad terms, decision makers should be choosing among public investments with the highest net benefits.

A decision to change public policy or to make a public investment is a decision to try to change the future—an assumption of change is implicit in all such decisions. Thus, good evaluation requires a comparison of a proposed policy's benefits and costs to whatever the benefits and costs would otherwise be without the policy. That "different future" is often referred to as the "base-case," "trend," or "status-quo" scenario or alternative. The base case represents how the world relevant to the policy decision is expected to look if policy does not change.

Figure A-2 illustrates the challenge for policy evaluation: to measure all types of relevant impacts on all people at all times. The literature of policy evaluation and benefit-cost analysis sometimes refers to this goal as full-cost evaluation—usually synonymous with a framework that attempts to identify and quantify all impacts.

Figure A-2. Goals of policy evaluation: what, who, and when

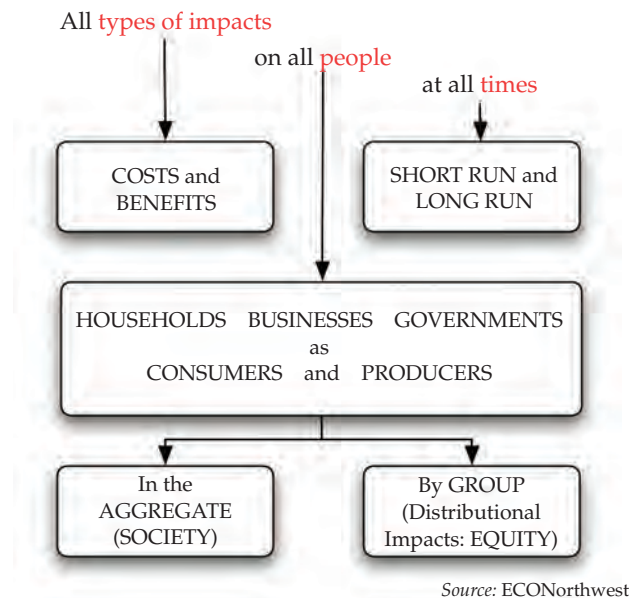
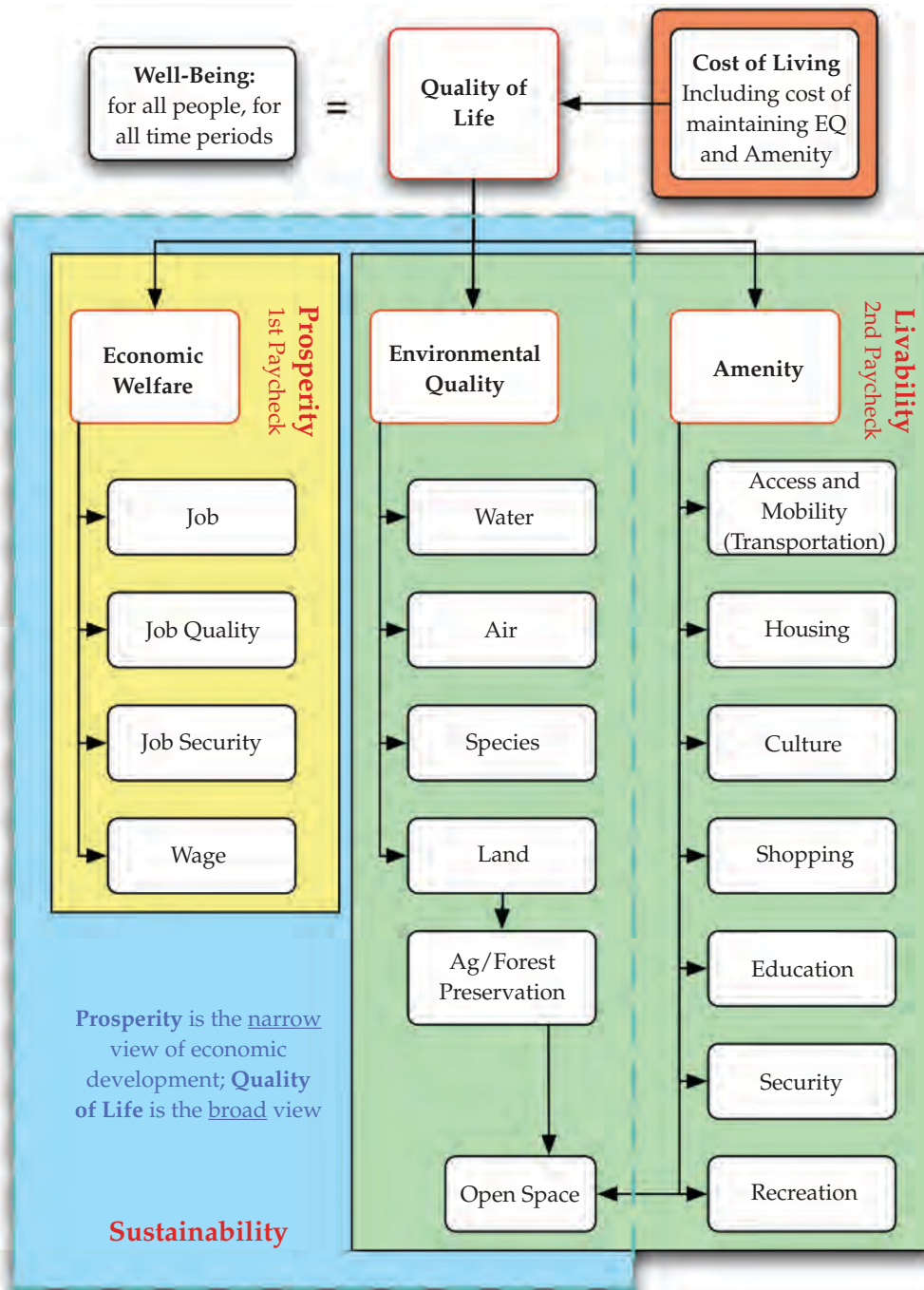


Figure A-2 appears simple, but each of its boxes contains a lot of complexity. Consider that there are:

- potentially dozens of categories of impacts, many with multiple measures of impacts;
- potentially millions of people, thousands of businesses, and scores of jurisdictions and interest groups in a metropolitan area, with overlapping affiliations; and
- changes to both impacts and affected people that occur over time. At the most basic level, communities choose to incur costs now (e.g., to build transportation projects) because they expect an acceptable return from future benefits.

Simple statements of broad objectives reveal their artificial simplicity when they get poked. "We are building this new road because it will make citizens better off." Really? In what way? What citizens in what roles: as families, workers, businesses owners, or property owners? Does "citizens" mean voters or residents? What about people that work in the city but don't live here? What about people that don't live here now but will move here within the next five years? And so on, and on, and on: specifying objectives in a measurable way is a very big deal.

These kinds of questions illustrate that a demonstration of net benefits in the aggregate (e.g., for an entire metropolitan region on average) says nothing about the net impacts to subsets of the region (e.g., geographic subareas, special groups). In other words, an efficient policy (one with likely net benefits) is not necessarily a fair one.⁴



Source: ECONorthwest

Figure A-3. Types of
impacts to be evaluated

The challenge is to somehow make all people in some geography, now and in the future (not just the people that live in those boundaries now and vote, but unknown people who will live in those boundaries in the future), better off.

Figure A-3 considers the impacts examined in a good evaluation, dividing them into categories that try to be both *comprehensive* and *mutually exclusive*. If not comprehensive, important impacts might not be considered; if not mutually exclusive, some impacts may be counted more than once.

Figure A-3 shows that all types of impacts can be generally defined as the things affecting a community's quality of life.⁵ The quality of life is a function of four main forces: economic welfare (or prosperity), the quality of the natural environment, the quality of the built and social environment (amenities), and the cost

of achieving quality in those three dimensions (cost of living). The boxes under the first three forces show some of their components, and the boxes around the forces show their main connections with terms often used when people talk about broader objectives for public policy (e.g., the term *livability* usually emphasizes environmental quality and amenity; *sustainability* usually emphasizes the environment and the economy).

The factors identified in Figure A-3 are all the different ways that new programs and policies can affect different groups of people over time. Some factors are not too hard to quantify with existing data: typically those under the “economic welfare” category. But an evaluation of a program or policy should consider all these impacts, not just those that can be counted.

Figure A-3 not only presents what planners and policy makers are typically trying to achieve in urban areas (i.e., the goals of public policy); it is simultaneously a relatively good model of regional growth and migration. People are attracted to an area because of economic opportunity (their “first paycheck”). But everything else influences the overall quality of that area: the natural features, built environment, and social and cultural services. The level of quality-of-life factors is a “second paycheck.” People make location decisions based on their assessment of the size and value of both paychecks. Offsetting the first and second paychecks is the cost of living.

These three variables—the first and second paychecks, and the cost of living—have dampening effects on each other. If economic opportunity and wages are high, new residents will migrate to that region. The region might either invest to maintain livability (which may increase the cost of living) or defer maintenance on livability (the second paycheck goes down). Different regions equilibrate at different levels of the three variables.

Figure A-3 greatly generalizes the many concerns people have about regional development and policies that affect land use, transportation, economic development, and environmental quality. To generalize further, for each of these and other key impacts, citizens and their elected representatives should want policy decisions to consider these questions in the areas of efficiency and fairness:

- *Efficiency*: Can a reasonable case be made that a new policy will make enough of the larger collective enough better off? and Is that policy worth the disadvantages created for the rest of the collective (the net-benefits criterion)?⁶
- *Fairness*: Is there some reason (moral, legal) that some subset of the population should pay less than the estimated full cost of benefits, and another subset should pay, independent of the efficiency effects?

Public policy ultimately seeks to improve the overall quality of life for a community. But any policy or program cannot simultaneously maximize all these components of quality of life. Instead, it must *optimize*; that is, it must find the right combination measures to enhance quality-of-life so that it is maximized overall. In other words, good policy balances competing objectives—not only across components of quality of life, but across classes of people, subareas of the affected area, and existing and future residents.

In the context of regional planning, it is common to think about desired outcomes not in terms of efficiency and fairness, but in terms of what people most often think makes a region livable: economic opportunity and security, environmental quality, urban amenity and services (including, significantly, the pattern and quality of built space and transportation), and the cost of living.

PRINCIPLES FOR EVALUATION

This section describes an analytical framework to work towards the goals described above. The framework assumes that better information about the new policies citizens want and are willing to pay for is essential to the creation of sound public policy. An evaluation should try to do the following three things:

1. *Identify what matters to citizens.* Depending on the program, things that matter are called goals, objectives, impacts, outcome, priorities, and so on.
2. *Measure what matters.* Different disciplines use different terms, but basically the evaluation should measure positive and negative impacts: the pros and cons, the pluses and minuses, the benefits and costs, the values and prices, the causes and effects, the outcomes. That measurement can and should include a qualitative assessment of impacts (e.g., public opinion) as well as a quantitative one (e.g., the outputs of travel-demand models) while being careful not to double-count either a benefit or a cost.
3. *Let the technical work in the previous step inform the choice of public policies or investments.* Though the choices are ultimately political (made by a small group of decision makers elected or appointed to represent a larger public), their choices should roughly conform to a ranking of projects based on net benefits (or cost-effectiveness), subject to constraints imposed by goals for the distribution of net benefits (fairness and equity).

Identifying What Matters

A lot has been written about this step: though important, it is not central to our interest here, which is a framework for technical evaluation (the next two steps). We cannot resist, however, making a few points:

- *List the relevant impacts.* The impacts are essentially evaluation criteria, and evaluation criteria are measures. Evaluation criteria are similar to and derive from goals. They differ not in kind but in degree: evaluation criteria are more detailed and measurable versions of goals. Sources for the “relevant impacts” are the professional literature, similar studies by other jurisdictions, and public opinion (policy makers, stakeholders, the public).
- *Consider short-run and long-run impacts.* In the short-run, conditions are relatively constant. In the long-run, conditions will change with or without the action being considered, and changes will occur in reaction to the project.
- *Consider the incidence of benefits and costs.* The distribution of those benefits and costs can make a difference among alternative actions. For example, an action may provide net benefits for a city by creating large benefits for a small group of people and small benefits for everyone else. The question of who pays is also important—the costs may fall disproportionately on groups who are least able to pay or on groups that do not receive a proportionate level of benefits from the action.
- *Don't waste time reinventing the obvious:* high-level goals tend to be the same for all urban and regional land-use and transportation projects. Those goals look a lot like the top boxes in Figure A-3. In any long-run evaluation of transportation investments or policies, the top goals will be some approximation of (1) transportation system performance (access/travel time, safety); (2) urban form and amenity; (3) economic development; (4) environmental quality; (5) equity (social justice: the distribution of impacts); and (6) cost-effectiveness. Everything anyone cares about measuring can probably fit in one of those categories.

Any hope of rolling up dozens of measures in some technically defensible way into some small group of summary measures requires that the measures nest within each other. Since (1) the literature of psychology and common experience suggest that people have a hard time focusing on lists with more than five to seven items,, and (2) an evaluation of the impacts of a regional transportation system may require consideration of dozens or scores of measures of things people care about, the measures must get nested under the goals. Each goal may need subgoals (objectives) to achieve this seven-item limit, which means any goal could have up to 49 (7×7) items—a clear indication how cumbersome this can quickly get if not done carefully. We discuss this important point in more detail below.

Measuring What Matters

In the context of policy evaluation, specific and measurable outcomes are often referred to as *performance measures*. Developing performance measures for evaluating transportation policies and investments requires decisions about the following:

- *The nature of the measure.* What, precisely, is the measure measuring and what form will the measure take? One must consider the following:
 1. What level of detail is being addressed in the measure? Will the measure be applied to the entire subject or only a subset of the subject?
 2. How will the impact be specified? Assume, heroically, that policy makers can agree on a type of measure. How should it then be reported: as a total, as a change, as a percent change, as a ratio, by the rank of the region against another metropolitan area, or by some other method? Dozens of impacts, each with dozens of potential measures, each with a dozen ways they could be specified and reported.
- *The geographic focus of the measure.* Will the measurement of impact be applied to the entire metropolitan area?; a region within the metropolitan area?; a city?; a neighborhood?; or some other area?
- *The length of time the measure will be applied.* Will it be applied for a year after the end of the project, a five-year period, or 10 years? Does it consider activity before and after the project?
- *The link between action being considered and the measure of the impact.* In the context of transportation, does a clear and significant link exist between a certain type of transportation action and the measure of its impact? If so, what is the direction of the link? How strong is the relationship? What is the magnitude of the relationship? For example, does the change in the transportation system (e.g., a policy, backed by funding, to make transit stations safer for and more accessible to elderly travelers) have a positive, strong relationship with the nature of the measure (e.g., the percentage of elderly that use public transportation) that can result in a large change (e.g., overall increased ridership, more equity in the various modes of transportation)?
- *Bang for the buck.* What is the impact of the project per dollar spent? If the project provides a net benefit, does it provide enough benefit or does it provide as much benefit as a different project?
- *Winners and losers.* Who are the beneficiaries of the project? Are they the “right” beneficiaries? Equity considerations, in essence, expand the reporting of measurements as a multiple of the number of sub-groups of interest. If, for example, the measure was “change in travel time,” that single measure may need to be calculated in a dozen ways: by city, by corridor, by mode, by income, and so on.
- *Does the measure work at the margins?* In the context of transportation, for example, does the measure cover all transportation projects or does it focus on specific projects or specific types of projects?

No single measure of any category of goals and objectives can cover all these issues. What seems like a simple item to measure might require many measurements to cover everything people care about.

For example, one measure of a healthy economy is the change in the number of jobs in the region. Jobs and transportation are related because firms depend on the transportation system to move goods and people (clients and employees). Once the relationship is clear, developing the performance measure involves answering a number of questions. Does the measurement include all jobs, jobs by sector, or only traded-sector jobs? If the region gets jobs, is that enough?, or Does one care about their distribution to subareas? What time period is being measured: for example, the year after the project or after 10 years with the project? Can one link the creation of jobs via existing models to transportation improvements to make it possible to forecast changes in measures of jobs based on assumptions about changes in transportation?

Can local models do that consistently and reliably for the different combinations of transportation actions (investments, programs, policies) under investigation?

A lot more can be said about measures, but lots of literature already says it. Every regional long-term transportation plan done by a Metropolitan Planning Organization (MPO) must address the measurement issue, and many have tried to do it well. Our advice: network. Subscribe to APA's Planning Advisory Service (they'll point you to the good examples); make calls to the bigger MPOs. You should be able to have plenty of information in a few days.

USING TECHNICAL WORK IN A POLITICAL PROCESS

These technical steps—defining goals and objectives (i.e., impacts you care about) and measuring them—have been central to rigorous and effective policy evaluation for decades. They show up in transportation benefit-cost analysis, cost-effectiveness analysis, least-cost planning, choosing by advantages, budgeting for outcomes, and processes of other names.

The technical evaluation is a *decision-aiding* tool, not a *decision-making* one. The presumption and hope of most policy analysts is for policy to be based on good information (i.e., understandable and accurate, with assumptions and variability clearly documented), resulting in better decisions.

The use of measures to aid public decision making usually follows some approximation of the following steps:

1. *Define alternatives* (desired alternative futures and packages of actions intended to increase the probabilities of arriving at those futures)
2. *Describe the important attributes of those alternatives* (the impacts, which become measures and evaluation criteria)
3. *Compare the alternatives* across attributes
4. *Pick the alternative* with the best array of attributes

Principles for Comparing Measures

A proper analysis employs several principles of structure and logic to correctly identify and estimate the impacts related to a particular project. Some examples:

- *Frame the analysis with and without the action under consideration.* The impact of a project or planning alternative is the difference between what the world would be *with* the project and what it would be *without* the project. Framing the analysis in this way forces one to consider future changes likely to happen without the project—impacts from these changes are not impacts of the project because they will happen anyway. Framing an analysis as “before and after” often causes analysts to incorrectly attribute impacts to a project that would happen without the project. In the context of an Regional Transportation Plan (RTP), such an evaluation could theoretically be done (1) at the project level (e.g., what are the changes in outcomes—in impacts—when the project gets built and operated?) or (2) at the system level (e.g., with packages of projects organized around some theme; say, a focus on highway expansion versus a focus on transit expansion).
- *“No Action” is not “No Change.”* The “Without Action” alternative referred to in the previous bullet should almost certainly not be defined as “nothing changes.” The economy, politics, and other policies may change the future even in the absence of the adoption of the policy being evaluated. A No-Action alternative represents the world without the project. The environment is likely to change under the No-Action alternative, and these changes must be considered to accurately assess the impacts unique to the project. Changes will occur even without the project because of other planned or likely projects, population growth, economic shifts, increased travel, and many other factors not directly attributable to or even related to the proposed project.
- *Focus on differences among alternatives at the margin.* Many analyses report the total impact of various alternatives, without reporting the marginal differences

between the alternatives. For example, “health of the economy” is always listed as one of the top five considerations in any regional planning process. But if the transportation policy options under consideration cannot be shown to have any measurable difference on the health of the economy, then that variable is (technically, and in theory) irrelevant to the decision making. There are no *differences*.

Aggregating Measures into a Summary Evaluation

Public policy making always has multiple goals, objectives, and, therefore, criteria. An implication is that public policy is trying to *optimize*, not *maximize*. Technically, one can only maximize a single criterion. Hence, goals like “minimize environmental impacts” are, practically and technically, an impossibility in the real world: pollution can be reduced from what it might have been in the absence of some policy choice or perhaps even reduced in an absolute sense, but it cannot be minimized in any scenario that keeps the economy and government running about like they do now. The evaluation of any region’s transportation alternatives is unavoidably in the world of multicriterion (multi-impact) analysis.

Because there are multiple criteria, one cannot avoid the issue of *weighting*: what is the *relative importance* of each measure/evaluation criterion? If no weights are specified, criteria implicitly get weighted equally. Social scientists have been working on developing a method for estimating the relative values of different objectives for decades without finding an ultimate solution. They never will. Among the difficulties:

- A city or region may consist of tens or hundreds of thousands of people, all of whom have slightly different values, preferences, and circumstances, and many will be affected somewhat differently by a change in policy.
- Regional economies, ecosystems, and public policies are complex and interrelated; many effects occur only over a long period; and outside market, social, and natural forces affect those systems. Thus, the net impact of a policy change on all significant aspects of those systems is impossible to predict.
- Even if one could somehow add all the different types of impacts for all individuals to get some estimate of the total net impact, and even if that impact were positive, policy makers might still decide that negative impacts on some people are too great to justify the total net benefits to society as a whole.

Only a small percentage of policy makers, and a smaller percentage of citizens, have the desire or patience to wade through the complexity of the interrelationships in an urban economy and ecosystem. Even if technical experts attempt to deal with that complexity, it ultimately must be simplified substantially. That simplification cannot occur without value judgments.

Benefit-cost analysis is one way to deal with the multicriterion problem. First, in the context of evaluating transportation actions, it collapses many of the possible measures of transportation performance into a single measure of net benefits: a measure of the efficiency of the project. Second, it attempts to convert other, nontransportation impacts to dollars so they can be added to or subtracted from estimates of transportation net benefits. Third, it can, if an analyst chooses, address issues related to equity by showing how those impacts are distributed across different areas or different groups.

This PAS Report takes an economic perspective and argues that transportation systems are so complicated that estimating the net benefits of packages of transportation investments cannot be evaluated without reference to demand-and-supply interactions and the *consumer and producer surplus* they generate. Thus, it should be no surprise that we argue for the application of benefit-cost techniques to the evaluation of long-run regional transportation investment choices. Anyone who wants to explore that advice can find an introduction in Appendix D, and the state-of-the-practice manuals (as of 2006) at the sites in this endnote.⁷

But benefit-cost analysis, in its full glory, is not commonly used. A more common approach to the multicriterion problem involves defining categories of impacts

(outcomes) and creating measures of how different transportation projects or packages of projects perform on those measures. As a practical matter, that method usually means that (a) the final public decision on a region's growth alternatives (for the development of land and infrastructure, including transportation) must be about no more than five alternatives (two or three is better), with five to 10 criteria to evaluate those alternatives, and (b) the performance of the alternatives based on the criteria must get summarized in some kind of matrix to facilitate comparison of the alternatives. EISs sometimes miss on both points.

We call this type of evaluation a *matrix display* because the final product is usually a matrix whose cells show how different actions (e.g., transportation project investments: the rows) are expected to perform on outcomes the planning process has defined as important (the columns). To get from the matrix of actions/impacts to a decision about which alternative to choose, one must either score the alternative actions or use a less quantitative means of comparison. If scoring is chosen, one must either (a) put all the criteria in the same units so they can be added (usually dollars, when this method is used, in which case it is a cost-benefit analysis), or (b) give each measurement a score, which implies a formal weighting scheme.

Constructing a matrix of impacts can be accomplished using methods that usually contain some close variation of the following steps:

1. Work with a task force, technical advisory committee, or study team to agree on alternatives and general categories of criteria.
2. Propose ways to measure the criteria (quantitative preferred; qualitative where measurement is not reasonable).
3. Prepare a table and text discussing, for each criterion, the relative performance of the alternatives on that single criterion (the problem of weighting does not occur until one tries to compare alternatives *across* criteria). Each row of the table may be summarizing a somewhat or much larger technical analysis: a section, chapter, or technical appendix of a final report.
4. Summarize all that information into a single table. The table shows a summary measure, score, or text description of impacts/performance for each alternative for each criterion, but does *not* try to add up performance measures across criteria to get a score for each alternative.
5. Facilitate a discussion among technicians and policy makers that allows them to identify (a) clearly inferior alternatives, (b) clearly superior alternatives, and (c) alternatives in between, in a process that leads ultimately to a choice between two or three alternatives. Note that criteria are not weighted upfront. Such upfront weighting seems to protect fairness (people can not later change weights to favor the projects they want), but it can be a mistake both technically and politically.
6. Conduct additional research to supply information policy makers think they need to make a final decision.

The technical analysis to support the decision making should be aimed at a summary evaluation that can be reported in a matrix that looks (in concept) something like Table A-1.⁸

What makes Table A-1 different from a standard matrix display of the performance of policy choices on decision-making criteria is its focus on *relative advantages* when comparing alternatives across criteria. The early steps in the evaluation are the same ones that would be needed to construct Table A-1:

- *Define the alternatives* (see below). In this project, the alternatives are primarily transportation ones, but they have a land use/development component.
- *Define the criteria* (see below). At the broadest level, criteria are goals. At the most specific level, criteria are performance measures. Both will be used—goal-based criteria will be the categories within which analysts can consider and “roll-up” more specific performance-based criteria depending on the required level of detail.

TABLE A-1. TYPICAL EVALUATION MATRIX

CRITERIA	ALTERNATIVES			
	A1 (base case)	A2		An
C1				
C2	All of the cells in the matrix get filled in with (1) facts about the performance of each A on each C, (2) an evaluation of the relative performance of ALL A's across each C, and (3) an identification and relative weighting of the most important advantages. Each row (criterion) may be supported by a full report.			
....				
Cn				

Source: ECONorthwest

- Fill in the facts relating to each criterion, for each alternative. A fact may be a single number, or it may be an entire report with many numbers.
- Add criteria as necessary. While an initial set of criteria and performance measures will be established, it may become appropriate to add criteria or performance measures later in response to the findings of the study as such measures occur.

Table A-2 illustrates the next step in the evaluation, assuming all the data (facts) have been collected and summarized into the matrix. It illustrates how one would compare facts *across each criterion* (work across the rows). (Note: Tables A-2 and following are illustrative only: there would be many more criteria for a real evaluation.) The comparison can and must happen across criteria because, for each criterion, the facts for each alternative are measured in the same units. One cannot work across an alternative (by column) because all the criteria for a given alternative are measured in different units and are, therefore, not easily comparable.

In summary, Table A-2 illustrates these steps:

- Describe the difference in advantage for each criterion.
- Highlight the best alternative for each criterion. For illustration, we assumed that Alternative 3 performs best on Criterion 1, and Alternative 1 performs best on Criterion 2.

Many evaluation exercises stop here. If one alternative has the most highlighted cells, the decision might be relatively easy. Even if it does not, it may be that the display of facts and relative advantages is enough of a base for an informed discussion and consensus decision.

TABLE A-2. ILLUSTRATIVE EVALUATION MATRIX, PHASE 1, RELATIVE ADVANTAGES

CRITERIA	ALTERNATIVE 1 (BASE CASE)		ALTERNATIVE 2		ALTERNATIVE 3	
Format	Describe Facts for Alt 1		Describe Facts for Alt 2		Describe Facts for Alt 3	
	Describe difference in advantage	Score	Describe difference in advantage	Score	Describe difference in advantage	Score
Criterion 1 Air Pollution	10 PPM of X		20 PPM of X		5 PPM of X	
	5 PPM more than Alt 3	Score	15 PPM more than Alt 3	Score	Lowest pollution	Score
Criterion 2 Cost	\$1,500,000		\$2,000,000		\$2,500,000	
	Lowest price	Score	\$.5M more than Alt 1	Score	\$1M more than Alt 1	Score

Source: ECONorthwest

If the analysts and decision makers want to go farther into quantification of the relative importance of the advantage (i.e., into weighting and scoring), the matrix facilitates the process. Table A-3 shows the next steps. The fact-finding part is over. Now some group (technicians, stakeholders, citizens, policy makers: it can happen at any or all levels) are trying to make quantitative judgments (i.e., by using scoring) about value—about the relative importance of the advantages.

There are five steps illustrated in Table A-3:

1. Look at all the highlighted boxes. Make a judgment (through whatever process) about which cell contains the most important advantage and highlight that cell (green box). This is not necessarily the most important criterion, but the most important relative advantage.
2. Arbitrarily give that cell a score of 100.
3. Look at the remaining boxes highlighted in yellow. Give them scores relative to the most important advantage (for illustration, we gave a score of 90).
4. For each criterion, rank each alternative relative to the score for the best alternative on that criterion.
5. Ultimately, the whole matrix of cells gets a score, and the scores can be added by alternative (column).

TABLE A-3. ILLUSTRATIVE EVALUATION MATRIX, PHASE 2, SCORING

CRITERIA	ALTERNATIVE 1 (BASE CASE)		ALTERNATIVE 2		ALTERNATIVE 3	
Criterion 1 <i>xxx</i>	Describe Facts for Alt 1		Describe Facts for Alt 2		Describe Facts for Alt 3	
	Describe difference in advantage	Score	Describe difference in advantage	Score	Describe difference in advantage	Score 50
Criterion 2 Air Pollution	10 PPM of X		20 PPM of X		5 PPM of X	
	5 PPM more than Alt 3	Score	15 PPM more than Alt 3	Score	Lowest pollution	Score 90
Criterion # Cost	\$1,500,000		\$2,000,000		\$2,500,000	
	Lowest price	Score 100	\$.5M more than Alt 1	Score	\$1M more than Alt 1	Score

Source: ECONorthwest

We said above that we would return to the idea of a *nested hierarchy* of goals, objectives, and measures. Figure A-4 illustrates the problems and the solution.

The problem is that as one moves from broad goals to narrower goals (objectives), to impacts of concern (positive and negative), to measures of those impacts, the number of categories expands. Figure A-4 suggests that six broad goal categories⁹ might turn into 50 to 100 different measurements. All those measurements are in different units. They cannot be added. Decision makers and the public cannot deal effectively with 10 different measures, much less 100. Planners know well the standard rule from above “If it’s not summarized on a single page, Councilor X won’t read it.” What does a planner do?

We’ve already given one solution: work through the details of Tables A-1 through A-3. But for 50 measures?

The likely reality is that technicians will have to systematically roll-up multiple measures into some kind of rating for the category that comprises them. In Figure A-4, for example, that means that various measures of congestion for different transportation alternatives (proposed actions) would get rolled into a single score (e.g., better or worse; + or 1; 1 to 5) on the objective called *Congestion Relief*. Then the scores on congestion relief, safety, choice, and so on would get rolled-up into a score for the goal *Transportation Choices*.

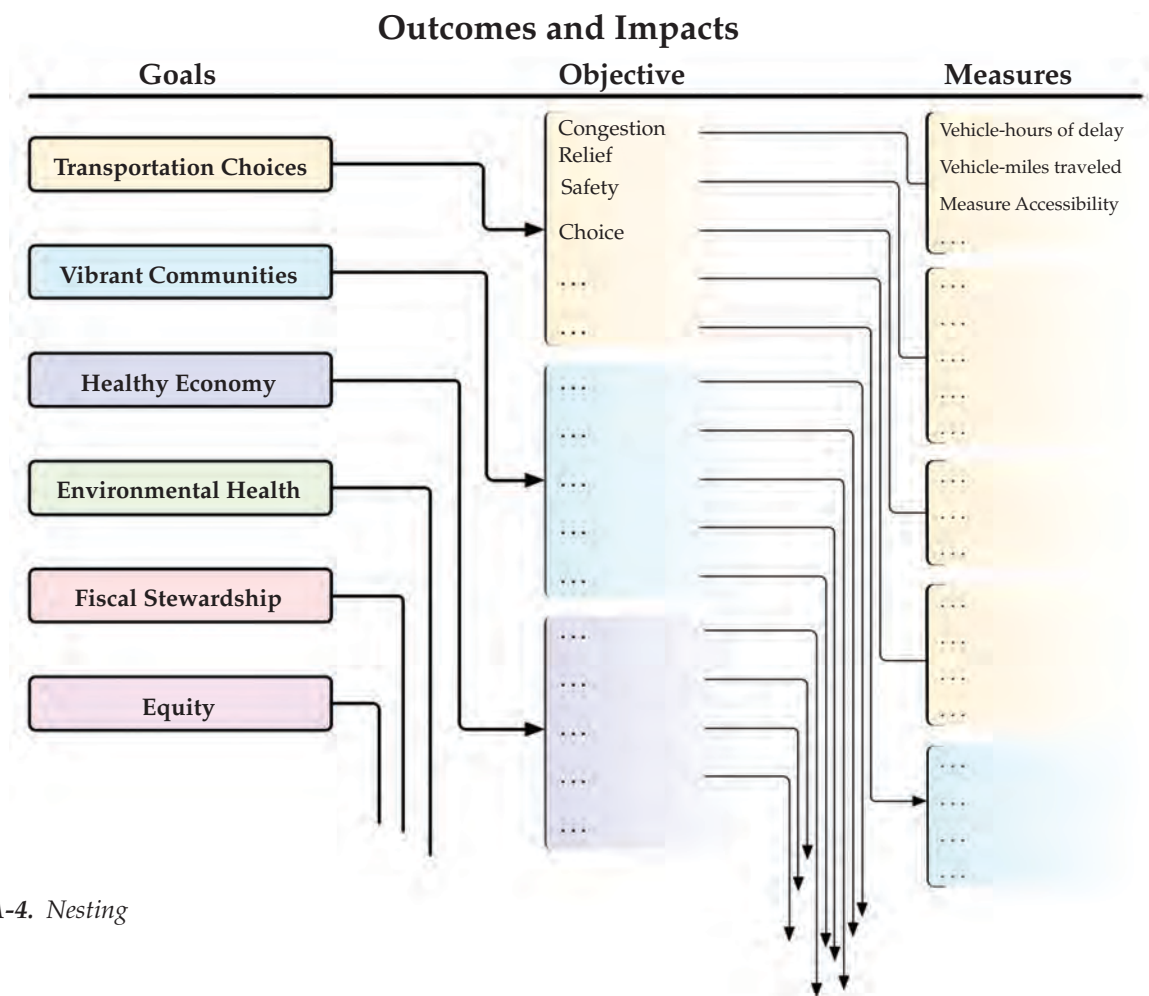


Figure A-4. Nesting

Source: ECONorthwest

Clearly, weighting has to occur to do that. It may happen formally and explicitly; it may happen implicitly without anyone knowing it. But it will happen. If it is explicit, it may happen by scoring (e.g., in a process like the one described for Tables A-1 through A-3), but more likely it will happen by visual inspection, expert judgment, voting, and consensus.

However it happens, it will happen more easily and more defensibly if the process makes an effort from the beginning to nest objectives in goals, and measures in objectives. That is a practical way of approximating the principles of comprehensiveness (every covered) and mutual exclusion (everything covered only once).

The following example makes the theory more concrete. Assume that four alternative transportation investment packages are being considered (the *policy alternatives*), all specified to meet the same financial constraint (i.e., all have the same budget of “reasonably available” funds). Assume a public/political process has led to agreement on the use of several *measures* that relate to achieving a broad goal of “Better Transportation System Performance.” These measures will be in different units (e.g., vehicle-miles traveled, person-hours of delay, accidents per million vehicle-miles), so they cannot be added. Technical staff (MPO staff, consultants, local government advisory group, a formal task force) could look at these multiple measures of Transportation System Performance and then work through some qualitative process to arrive at a relative ranking of the four alternatives on the broad goal of “Better Transportation System Performance.” That process, if applied to all the broad goals (e.g., land use, the economy, the environment) would lead to a summary matrix of

relative performance of each of the four alternatives on each of four to seven broad goals that could be presented on a single page.

Note that the method just described assumes an answer to an important question: Should weights be assigned before measuring impacts, after, or not at all? Tables A-1 through A-3 assume weights are assigned after impacts have been measured. The political wisdom of that order is that policy makers (1) may not have a good idea about weighting criteria in the abstract, and (2) may want the flexibility to adjust weights to try to select their project preferences after impacts have been measured. An additional downside of weighting before impacts are measured: it takes a lot of time to do it properly and can slow down the front-end of a planning process.

The second point in the previous paragraph, however, is why most technicians tend to recommend weighting *before* impacts have been measured: so that policy makers cannot “game” the evaluation process by assigning higher weights to criteria that makes their preferred project look better. We are in the minority in believing that to be potentially as much an advantage as a liability. Policy makers need some time and flexibility to find their best interest. Abstract weights will not have much meaning to many until they see how they affect policy and project rankings.

We do not think there is an unambiguous right answer to the question of when to weight criteria, but the results of the analyses will probably be different depending on when weighting is done.

SUMMARY

Much more could and has been written about the topics covered in this appendix. Our purpose here was to provide a framework for thinking about how to evaluate public policy, not a users manual. The main points:

1. In the context of regional planning, there are many doors, but all lead to the same room. Whether you enter because of a primary concern about land use, transportation, economic development, or environmental quality, you will arrive at the same place: one in which all those concerns, and more, have to be considered. Citizens and policy makers will not accept, in response to their questions about the impacts of a transportation project on the achievement of a land use plan, the response, This is a transportation plan, so we didn't deal with land use. In short, you have to deal with multiple objectives, and ones not bounded by any single element or discipline.
2. If you accept the previous point, you are confronted by the task of trying to develop the all-by-all, eye-of-God matrix for a regional economic and ecological system. Everything affects everything else in complicated ways that cannot be modeled well individually, much less collectively. Computer models are suggestive, but not definitive. Moreover, policy makers and the public are suspicious of “black boxes” (a process that does not explain itself; things go into the box and come out, but what happens inside is not observable). They want something transparent. They want planners to make something inherently complex into something simple and intuitive (i.e., something they can understand according to their individual intuitions, which, of course, can vary widely across individuals). Good luck.
3. In the context of regional transportation, an application of the principles and techniques of benefit-cost analysis has the best chance *technically* of dealing with the multiobjective/multicriterion problem. *Politically*, however, such an analysis is complicated and too much of a black box for most policy makers. In some cases, a benefit-cost measure may be only *one criterion* of a multicriterion evaluation. In short, you are very likely going to do your evaluation with some version of *matrix display*.
4. If you use a matrix display (its limitations notwithstanding), your work will be better both technically and politically if you follow the principles described in this appendix. Those include nesting measures under broad goals; having solid analytical work sitting behind the summary evaluation in the summary matrix;

nesting evaluation criteria and measures inside of objectives and goals; evaluating the relative advantages of each policy alternative on each measure/evaluation criterion; and scoring (if you choose to do scoring) based on relative advantages, with scores anchored to the primary advantage.

APPENDIX A NOTES

1. The material in this appendix draws heavily on other work by Terry Moore and ECONorthwest, published in part in several other reports and books. Thanks in particular to the Transportation Research Board (TCRP and NCHRP), the Lincoln Institute of Land Policy, and Metro (Portland, Oregon).
2. In the rest of this appendix, we use the term *policy* broadly to include any type of public decision, including budget allocations (investments) for infrastructure or programs, or changes in regulatory requirements.
3. Government does provide certain social programs (especially education) that help people meet the higher needs, but those policies are not the focus of land-use and transportation agencies, and are beyond the scope of this book.
4. “Social justice” is the current jargon for referring to issues of equity and fairness.
5. Quality of life is how planners tend to refer to what economists refer to as economic welfare or well being.
6. Some texts on public policy distinguish between *efficiency* (do things right) and *effectiveness* (doing the right things). It is not a benefit to society if public actions do the wrong things efficiently. In this appendix, we use the term efficiency to mean net benefits, which implies a consideration of effectiveness.
7. ECONorthwest for the National Cooperative Highway Research Program and American Association of State Highway and Transportation Officials (AASHTO). 2003. *A Manual of User Benefit Analysis for Highways, 2nd Edition*. Washington, D.C.: AASHTO; and ECONorthwest/Parsons Brinkerhoff for the Transportation Research Board. 2002. *Report R-78. Estimating the Benefits and Costs of Public Transit Projects: A Guidebook for Practitioners*. Washington, D.C.: TRB.
8. The method illustrated in Table A-1 and following is called “Choosing by Advantages” and is described in: Suhr (1999).
9. These six are from the Long-Range Transportation Plan of Metro (Portland, Oregon), 2007.

Microeconomic Analysis: Basic Concepts and Tools¹

By Arthur M. O'Sullivan, Lewis and Clark College

Transportation and land-use policies operate in the context of a market economy. Land-use policies consist mainly of (1) regulatory requirements with which the private participants in land markets must comply (e.g., constraints on lot sizes and building materials), and (2) incentives intended to influence the market behavior of the suppliers and consumers in the market (e.g., tax abatements). Transportation agencies both regulate and influence incentives, but they play a much larger role as the sole supplier of transportation infrastructure.

Critical to identifying and evaluating the effects of policy is an understanding of how the market system operates in the absence of government influence. This appendix reviews the most basic tools economists have developed to understand and evaluate market performance and the effects of public policies on markets (especially markets for land development and transportation services).

We introduce these tools in general; we do not focus on land use and transportation in this appendix. We apply the tools to the special case of transportation in Chapter 4 and Appendix C, and to a wide range of land-use and transportation policies in Chapters 5 and 6.

This appendix describes tools and concepts fundamental to an economic evaluation of the allocation of resources in any market. It is organized as follows:

1. **Individual decision making.** Markets consist of individual suppliers and consumers (those who demand products made by suppliers), whose decisions to maximize their benefits, as they perceive them, define markets.
2. How markets function:
 - a. **The mechanics of market allocation.** The interaction of suppliers and demanders in markets establishes the prices at which goods or services are exchanged. A market *equilibrium* price clears the market.
 - b. **Evaluation of market allocation.** The objective is to maximize the net benefits from consumption. We show how, in concept, to determine the net benefits from market exchange and describe the conditions under which they are maximized.
 - c. **Market response to changing conditions.** The situations faced by suppliers and consumers change over time, and markets respond to those changes.
 - d. **Rationale for public policy and its effects on markets.** We look at the rationale for public policies in a private market economy and the market effects of both regulatory policies and policies that influence incentives.

A summary at the end restates the main points important for understanding market forces that affect land-use patterns and surface-transportation systems in metropolitan area.

INDIVIDUAL DECISION MAKING

Markets consist of groups, sometimes large groups, of individual buyers and sellers. An understanding of group behavior builds on an understanding of how the individuals they comprise make consumption and production decisions. We start, as any analysis of this sort must, with some basic assumptions. We then describe what we mean by “rational” decision making, show how to measure the benefits from rational decision making, and then identify several common pitfalls in decision making.

Assumptions

Our analysis begins with several critical assumptions. Decision makers (consumers or business operators in the private sector; policy makers in the public sector) are characterized by the following:

- They have well-defined objectives. Consumers come to the market to purchase products that increase their well-being, happiness, or (in economic jargon) utility. Businesses bring products to market seeking to earn profits.
- Their preferences and circumstances vary, and, thus, the value they place on products traded in markets also varies.
- They pursue their objectives effectively. Given enough experience and information, decision makers usually make reasonably “rational” decisions. They use their brains, not just their hearts or instincts, when deciding what and how much to buy.
- They have limited resources (i.e., at least some resources are “scarce”). There are not enough resources for all people to have as much as they want of everything they want. Choices have to be made.

The assumption of scarce resources seems uncontroversial. But many people have trouble accepting the assumption of “rationality.” Everyone makes mistakes, and we all know people whose behavior, at least on occasion, can hardly be described as rational. Nevertheless, most people behave reasonably rationally when it comes to the activities they care about and engage in day in and day out.

We are unwilling to go down the path of assuming that most people most of the time do dumb things or are ridiculously easy to persuade. We may not approve of or even immediately understand some people’s choices. But that does not mean they are in general behaving irrationally. Most people have a better understanding of their situation than do casual observers. We discuss some caveats shortly.

Rational Decision Making: The Marginal Principle

So, what do we mean by “rational behavior” in the context of economic decision making. Essentially, it means following the “marginal principle”:

If the marginal benefit of an activity exceeds its marginal cost, do more of it. If possible, pick the level of the activity at which the marginal benefit equals the marginal cost.

The marginal benefit of an activity is the extra benefit from a small (one-unit) increase in the activity; for example, the extra satisfaction from eating one more piece of pizza. The marginal cost is the additional cost from a small increase in the activity (i.e., the additional expense incurred purchasing that additional piece of pizza).

Applying the marginal principle to pizza consumption: eat another slice as long as the dollar value of the additional satisfaction exceeds the dollar cost of the slice.

To more fully illustrate the marginal principle, consider a movie producer who must decide how many sequels to produce. Suppose the original version of a movie is successful enough that the producer expects a sequel to be profitable too. If the sequel turns out to be profitable, the producer then has to decide whether to make a third movie, then a fourth, and so on. A profit-maximizing producer uses the marginal principle to figure out when to stop making sequels.

We can plot the movie producer’s marginal benefits and costs on a graph. Figure B-1 has dollars on the vertical axis and the number of movies on the horizontal axis. The marginal benefit is the revenue generated by the original movie (\$300 million, point MR1) and then each additional sequel to the original movie: \$210 million for the second movie, point MR2 (the first sequel), and \$125 million for the third movie, point MR3. This pattern follows the general rule of thumb in the movie business that a sequel generates about 30 percent less revenue than the original, and revenue continues to drop at the same rate with additional sequels.

Next consider marginal costs. Suppose that the typical blockbuster movie costs about \$65 million to produce and \$60 million to promote. The marginal cost of the

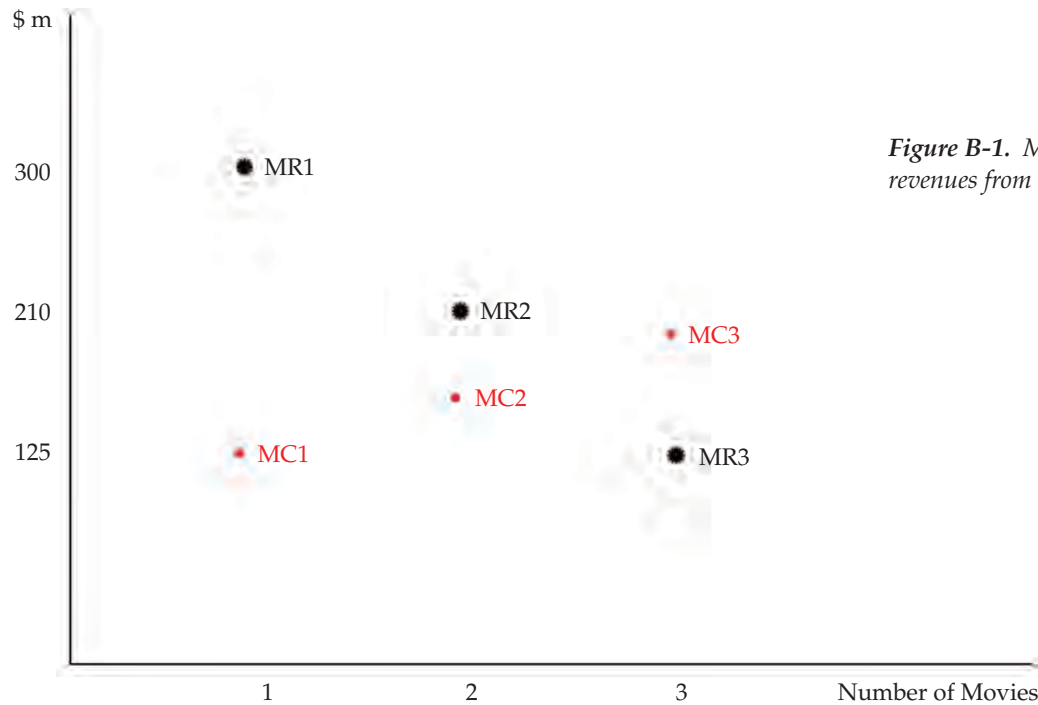


Figure B-1. Marginal costs and revenues from movie sequels

first movie is \$125 million, point MC1. The marginal cost increases with the number of movies because film stars typically demand higher salaries to appear in sequels in part because the sequel depends on their participation and in part because they have other interesting offers. Starring in a successful movie raises the opportunity cost of a movie star's time. Angelina Jolie, for example was paid more for *Tomb Raider 2* than for *Tomb Raider*. The marginal cost of the second movie is \$160 million, point MC2, and \$195 million for the third, point MC3, reflecting the rising cost of the movie stars' time.

In this example, the first two movies are profitable, but the third is not. For the first movie, the marginal benefit (\$300 million) exceeds the marginal cost (\$125 million), generating a profit of \$175 million. Although the second movie has a smaller marginal benefit and a bigger marginal cost, it is profitable because the marginal benefit still exceeds the marginal cost by \$50 million (\$210 million–\$160 million). In contrast, the marginal cost of the third movie (\$195 million at point MC3) exceeds the marginal benefit (\$125 million at point MR3), so the third movie is a losing proposition. In this example, the producer should stop after the second movie.

Although this example shows that only two movies in a series are profitable, other outcomes are possible. If the revenue from the third movie were higher or the cost were lower, the marginal benefit could exceed the marginal cost, and making a third movie would be profitable. Indeed, there are many examples of movies with multiple sequels and prequels, including *The Pink Panther*, *Star Wars*, and *Rocky*. Conversely, there are many examples of profitable movies that didn't generate any sequels. In these cases, the expected drop-off in revenues and run-up in costs were large enough to make producers believe a sequel would be unprofitable.

Measuring Net Benefit (Economic Surplus)

The marginal approach allows a calculation of *net benefit*, also known as “economic surplus,” from a particular activity. Marginal *net benefit* equals marginal benefit minus marginal cost. In the movie example, the marginal net benefit is the profit from each additional movie.

In figure B-1, this profit is shown by the vertical distance between the marginal benefit and the marginal cost. The vertical distance is \$175 million for the first movie (\$300 million–\$125 million) and \$50 million for the second (\$210 million–\$160 million).

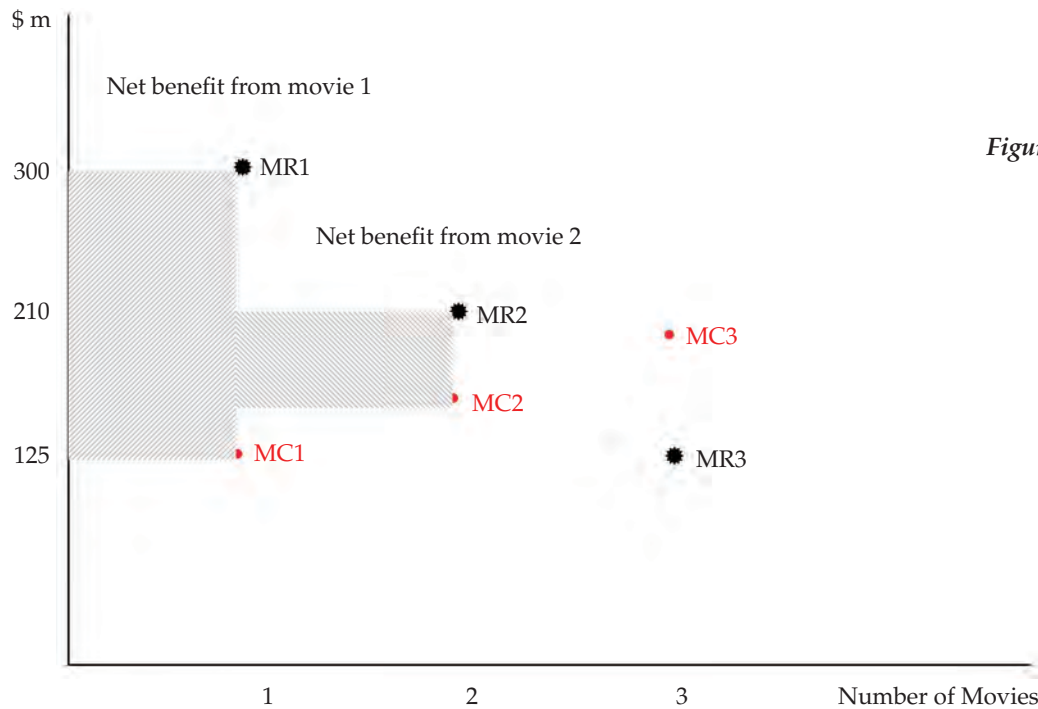


Figure B-2. The net benefit from movie sequels

Note that one can view the marginal net benefit as an *area* on the graph rather than as a vertical distance. Recall that the area of a rectangle is $\text{base} \times \text{height}$. The height is the vertical distance just described. The base is equal to one additional movie. Thus $\text{base} \times \text{height} = \text{height}$ because base equals one. This may seem trivial, but as we show shortly, viewing net benefit as an area is often useful.

The *total* net benefit then is the sum of the marginal net benefits. The marginal principle indicates that two movies are worth producing. The shaded areas in Figure B-2 show the profits from each of the first two movies: \$175 million from the first movie, and \$50 million from the second, or \$225 million total. This is equal to the combined areas of the two different sized rectangles shown in Figure B-2.

Pitfalls in Decision Making

Most people most of the time behave rationally (i.e., in accordance with the marginal principle). That said, accomplishing rational behavior isn't always easy. Indeed, Frank and Bernanke (2006) identify several common pitfalls in decision making.² People figure out how to overcome these pitfalls in most day-to-day activities because failing to do so imposes costs. But it is useful for our purposes to discuss them. We do so in the context of the movie production example:

1. Ignoring opportunity costs. The cost of a sequel is often higher than the cost of the original movie. The reason we cited is that the opportunity cost of the actors' time has increased, so they require greater compensation to do the sequel. It is often easy to focus on the explicit (i.e., out-of-pocket) costs, but implicit costs, such as that of time, are often at least as important.
2. Failing to ignore sunk costs. Suppose that making the original movie required special sets, training, and equipment that cost \$20 million. Once these investments have been made, the costs are "sunk" (i.e., gone forever) and irrelevant to marginal decisions. For example, suppose that the same sets, training, and equipment can be used for the second and subsequent sequels. These costs should not be included, either fully or on average, as part of the marginal costs of sequels.
3. Failing to understand the distinction between marginal and average. Let's do a quick calculation. The first three movies generate a total of \$635 million, or

roughly \$212 million on average. The total cost of the first three movies is \$480 million dollars, or \$160 million on average. Average revenue greatly exceeds average cost, so the producer should make all three movies. Right? Wrong! The third movie is a loser as we have already demonstrated. Total net benefit is higher (and average benefit much higher) after two movies than after three. Averages have their uses, but, as a practical matter, are generally not useful when the objective is maximization.

HOW THE MARKET WORKS

Market Allocation of Resources

We took the revenues generated as given in the movie example of individual decision making. How are those revenues determined in the market for movies? Competing with Angelina Jolie's *Tomb Raider* and *Tomb Raider 2* are other movies distributed by a variety of companies. The revenues from the *Tomb Raider* series depend not only Jolie's star power, but also on the general conditions in the *market* for movies.

Our objective in this section is to reveal how competitive markets work. Stigler and Sherwin (1985) offer an apt definition of a market: that set of suppliers and demanders whose trading establishes the price of a good.³ We first look at the behavior of demanders, then of suppliers, and then how they interact in markets. We are especially interested in demonstrating the critical role of market prices and profits in the market allocation of resources.

The demand for a product

The market demand curve shows how the quantity of a product that consumers are willing and able to purchase varies with price. Demand curves slope downward: as price falls, quantity demanded rises. In other words, consumers buy more at lower prices, all else the same. This is referred to as the "law of demand" in introductory economics textbooks and hardly seems controversial.

But it's worthwhile looking more closely at the source of this downward slope. Suppose that you have been working hard in the yard all morning. You go for lunch to a pizza place that sells pizza by the slice. Suppose that the pizza place charges \$3 for a slice. How many slices do you buy?

The first slice is a no-brainer; it's good pizza, and you're hungry. Indeed, you would have bought the first slice even if it cost \$5. That \$5 is your *reservation price* for the first slice, it represents your valuation (i.e., your willingness to forego whatever else you would have bought with the \$5) of that first slice. That \$5 is also the *value* you place on the slice (even though you only had to pay \$3).

You purchase the second slice but only after a little more thought because your valuation of that second slice is less than that of the first one; you're not as hungry, and the taste isn't as novel. This is an application of the *law of diminishing marginal returns*, the closest thing to a physical law of nature in economics: over any short time period, the return from additional units of any activity eventually falls. It turns out that your valuation of that second slice is close to but still greater than \$3, so you decide to buy it. You choose not to buy a third slice, given your declining marginal benefit and the price of \$3 per slice.

What would happen if the price of a slice were lower, \$2 say. The decision to purchase the second slice becomes easy, and the third difficult. There's a good chance, however, that you'll buy the third slice. Your "quantity demanded" increased as the price fell.

Market demand comes from all the people who, like you, are having pizza for lunch. They vary in their valuation of each consecutive slice of pizza. But each person buys slices as long as the value they place on an additional slice (i.e., the marginal value or benefit) exceeds the price (i.e., each applies the marginal principle). Marginal benefit falls as a result of the law of diminishing marginal returns. So most consumers increase their consumption as price falls: quantity demanded in the market increases as price falls, so the market demand curve slopes downward.

The supply of a product

The market supply curve shows how the quantity of a product sellers are willing and able to bring to market varies with price. Supply curves usually slope upward: as price rises, quantity supplied rises. In other words, a higher price encourages businesses to bring more of their product to market, all else the same. This is the “law of supply” in introductory economics textbooks.

The supply curve slopes upward for essentially the same reason as the demand curve slopes downward: the law of diminishing marginal returns.

To see this, consider the situation faced by a seller of pizza slices. In the short run, the seller has to operate with a fixed amount of production equipment (e.g., pizza ovens), space for customer seating, and so on. As the amount of raw materials and labor increases to increase pizza production at periods of peak demand, congestion begins to increase production cost per slice. The pizza ovens become more congested with raw materials, which cook more slowly, and the additional workers waste effort trying to avoid bumping into each other. There are diminishing returns from these resources given the scarcity of work space. In addition, customers have to queue to place orders and some have to stand and eat. So, the cost of producing and consuming an additional slice (the marginal cost) rises as production increases due to the scarcity of a key resource: production facilities.

The marginal production cost is the supplier’s reservation price for supplying each unit.

The profit-maximizing pizza supplier applies the marginal principle. He increases production only as long as marginal production cost is less than price. Since costs are higher at peak times, he may raise price (or, as is often the case, stops offering the price he offers at off-peak times, say, from 3 to 5 in the afternoon). Or he may simply stop letting additional people in or require them to wait when marginal cost threatens to rise above the menu price.

The individual pizza seller can treat this problem of rising marginal cost in the long run by adding onto the shop. If this expansion has no significant effect on the prices of land, labor, or materials (the likely case), the pizza seller’s long-term supply curve is very flat (i.e., the long-term supply is highly elastic). But if many pizza suppliers are expanding at the same time, their combined activity might bid up the prices of land, labor, or anchovies. Marginal costs rise with increased industry production, and the long-term market supply curve slopes up.

The general idea is that the market supply curve slopes upward because some input is scarce. That scarce input might be plant and equipment, a natural resource (e.g., oil), or labor. Competition among suppliers for the scarce input bids up its price, increasing marginal production costs.

The interaction of suppliers and demanders in the market

The quantities of a product consumers are willing to buy and the quantities suppliers are willing to sell each depend on price. There is a price at which these quantities are the same, a price that “clears” the market.

If price differs from the market-clearing price (if it is higher or lower), market forces push price toward the market-clearing price. This market-clearing price is, therefore, an *equilibrium* price. The market system establishes equilibrium prices for the range of goods and services traded in markets.

One can see how market forces work by superimposing market supply and demand curves, as in Figure B-3. The market demand curve shows the quantity of product consumers are willing to buy at each price, and the supply curve shows the quantity suppliers are willing to sell at each price.

Where the curves cross is a special point. At the corresponding price, labeled P^{eq} , quantity demanded equals quantity supplied. The market clears. At the end of the trading period, all of the suppliers and demanders are in an important sense “satisfied” with the day’s trading, given the market price.

To understand this sense of satisfaction, suppose that the market price were higher than P^{eq} . At this higher price, quantity demanded is lower, and quantity supplied is

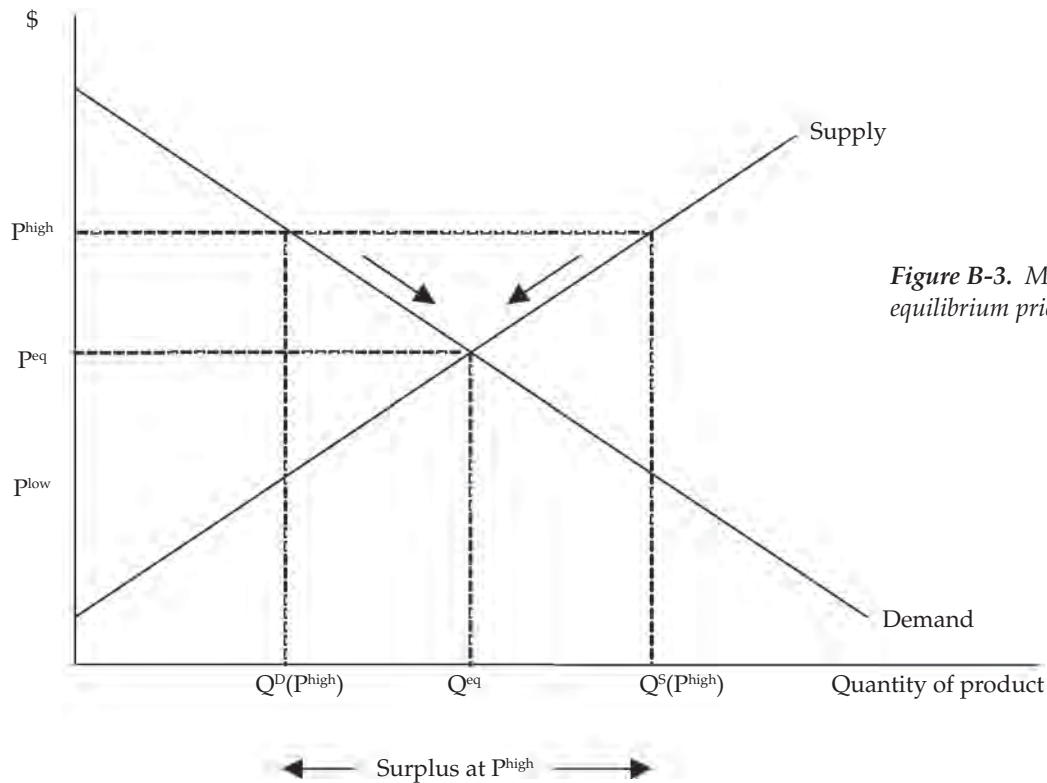


Figure B-3. Market adjustment to equilibrium price and quantity

higher. So at P^{high} consumers want to purchase only $Q^D(P^{\text{high}})$ units, while suppliers would like to sell $Q^S(P^{\text{high}})$ units: there is an *excess supply*, or *surplus*, of product on the market at P^{high} . Some sellers are left with unsold product at the end of the trading period, and are, therefore, unsatisfied. Each of these sellers has incentive to drop his price next period to encourage higher sales. Other suppliers will have to follow suit to avoid losing sales. So price falls, causing movement downward along the demand and supply curves until quantity supplied equals quantity demanded at Q^{eq} .

There is an analogous story of market adjustment if the market price starts lower than P^{eq} . At the lower price, quantity demanded is higher and quantity supplied lower. At P^{low} , the quantity suppliers make available for sale is much lower than the quantity customers would like to buy at that price: there is an *excess demand*, or *shortage*, of product on the market. Suppliers notice that some customers go away empty-handed and unsatisfied, and most of these customers have incentive to compete for the available supply by offering to pay more than P^{low} . This encourages suppliers to increase prices and bring more product to market. So price rises causing movement upward along the demand and supply curves until quantity supplied equals quantity demanded at Q^{eq} .

The action stops when the price settles at the market-clearing level. This price is an *equilibrium* price because market participants feel no incentive to change their behavior next period.

Evaluating the Market Equilibrium

Demand and supply curves show the quantities consumers are willing to buy and suppliers are willing to sell at each price. That is the “horizontal interpretation” of what these curves tell us. Give me a *price*, I look it up on the vertical axis, then slide horizontally to the demand and supply curves, and read off *quantities* demanded and supplied at that price.

There is also a “vertical interpretation” of demand and supply curves. We described individual demand and supply curves in terms of reservation prices. An individual consumer purchases a unit of product if his reservation price/marginal benefit exceeds the market price. Marginal benefit falls as the individual consumes more of

the product within a given time period (obeying the law of diminishing marginal returns). The individual consumer's demand curve shows the reservation price for each additional unit consumed. Give me a *quantity*, I look it up on the horizontal axis, then slide vertically to the demand curve and read off the reservation *price*. This is the vertical interpretation: it shows reservation price/marginal benefit as a function of quantity consumed.

The market demand curve arranges units of output in order of decreasing marginal benefit. It is as if each unit of output is auctioned to consumers one by one. The first unit goes to the consumer who values that unit most highly because that person has to outbid everyone else to lay claim to the first unit. Because there are many consumers of this product, the individual consumer has to bid very close to his or her reservation price. The second unit goes to the person with the second highest valuation. And so on.

The vertical interpretation of the demand curve provides a marginal *benefit* curve. Give me any *quantity*, I look it up on the horizontal axis, then slide vertically up to the demand curve, which reveals the highest reservation *price* given that everyone with a higher reservation price has already been allocated one unit of a product.

The supply curve is analogously a marginal *cost* curve. An individual business supplies a unit of product if its reservation price/marginal production cost is less than the market price. The market supply curve arranges each unit of output by all suppliers in order of increasing marginal cost. So, give me a quantity, I look it up on the horizontal axis, then slide vertically up to the supply curve, which reveals the lowest reservation price/marginal cost that all suppliers with lower reservation prices have already been supplied the product.

In short, the market demand and supply curves can be interpreted as marginal benefit and marginal cost curves.

This allows us to calculate the *gains from trade*. Consider the situation depicted in Figure B-4. The equilibrium market price is \$3—the market clears at \$3—and the equilibrium quantity traded is 1,000 units.

The demand curve shows that consumers are willing to pay more than \$3 for most of the units traded. For example, 500 units would still sell if price rose to \$4. Because the market price is only \$3, the people who buy each of these units of output at \$4 now get to purchase those units of output at \$3 and have a dollar left over with which to

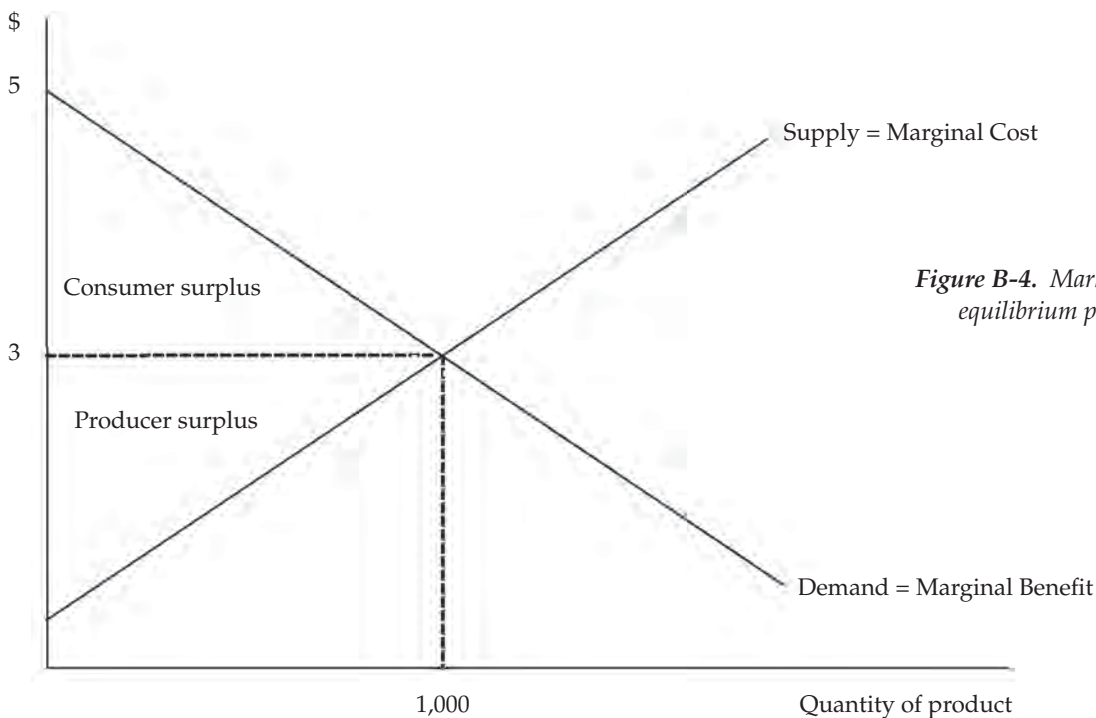


Figure B-4. Market adjustment to equilibrium price and quantity

buy something else. The relatively low market price provides these consumers with “surplus,” referred to as *consumer surplus*.

We can calculate the consumer surplus on each unit traded. Consumer surplus equals how much the consumer would have been willing to pay (his or her reservation price) minus the market price. This is just the vertical distance between the demand curve and the price line in Figure B-4. If we draw that line for each unit of output using a marker that is one unit of output wide, total consumer surplus very closely equals the *area* under the demand curve and above the price line. As drawn, that area equals \$1,000.⁴ As a group, the consumers in this market benefited by \$1,000 from trading at \$3.

We can tell an analogous story on the supply side. The supply curve shows the reservation price/marginal cost of supplying each unit of output. Suppliers earn surplus on every unit but the marginal (i.e., 1,000th) unit. The total *producer surplus* equals the area above the supply curve and below the price line.

The total gains from trade (i.e., total surplus) equals consumer surplus plus producer surplus, which equals the *area* between the demand/marginal-benefit and supply/marginal-cost curves to the left of the quantity traded.

Note that total surplus is maximized at 1,000 units of output. At 900 units of output, marginal benefit, MB, is greater than marginal cost, MC; there remain consumers who value the product more highly than additional units of the product would cost to produce. There’s, in a sense, money left on the table. At 1,100 units, MC is greater than MB; the cost of producing the marginal unit exceeds its value to consumers. The resources that went into production of that unit would have generated greater surplus in production of some other good or service. Surplus is maximized where MB equals MC (in accordance with the marginal principle).

Critical to note is that the surplus maximizing (i.e., *optimal*) output of 1,000 units is the market *equilibrium* output. A large number of consumers and suppliers interacting independently in markets generate an equilibrium price of \$3, which results in trades of 1,000 units. This mechanical and decentralized process generates a highly desirable result.

Economists refer to this happy result as market *efficiency*. It has three components:

1. *The right quantity of output is produced.* The market obeys the marginal principle: MB equals MC at Q^{eq} .
2. *Output is produced cost-effectively.* The suppliers of the product are those that can produce the product at a cost not more than the market price.
3. *Output is allocated to the consumers who value it most highly.* Only those consumers who are willing to pay at least the market price purchase and consume the product.

That’s the magic of the market system: pursuit of individual self-interest *under the right conditions* leads to social gain. Adam Smith famously put it this way in 1776:⁵

It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest. Every individual . . . neither intends to promote the public interest, nor knows how much he is promoting it. . . . He intends only his own security, only his own gain. And he is in this led by an invisible hand to promote an end that was no part of his intention. . . . By pursuing his own interest he frequently promotes that of the society more effectively than when he really intends to promote it.

Given this plug for private market production and allocation of resources, one might wonder about the role of public policy. We’re getting to that. But before we do we need to look at how markets adjust to changes in conditions.

Market Response to Changing Conditions

A practical problem concerns the measurement of economic surplus. We can observe prices and quantities traded, but these might vary only a little bit around their equilibrium values. How can we calculate surplus if we lack information about the marginal benefits and costs along the full range of output?

We can't, but fortunately it often doesn't matter in practice. What typically matters is not the total surplus, but the *change* in surplus from some change in market conditions. The market demand and supply curves show the relationship between price and the quantities demanded and supplied, *all else held constant*. In this section, we identify the variables that affect quantities demanded and supplied *at each price*, analyze how the market reacts to changes in those variables, and then discuss the extent to which we can evaluate those effects.

Variables other than price that affect supply and demand

Changes in variables other than price shift the entire supply or demand curve, throwing the market out of equilibrium at the original equilibrium price. The market responds by working toward a new equilibrium.

To see this, first consider the variables other than the current product price that affect the quantity of product demanded by the consumers in the market. These variables include all of the following:

- *Consumer incomes.* An increase in income increases the quantity demanded of all "normal" goods, shifting the demand curve to the right. An increase in income decreases the quantity demanded of all "inferior" goods, shifting the demand curve to the left. Note that, from the perspective of the market, a consumer with a higher income *values* normal goods more highly than another *otherwise identical* consumer. Preferences may be the same, but willingness to pay is higher with higher income.
- *Prices of substitute products.* An increase in the price of a substitute good, such as an increase in the price of natural gas relative to that of electricity, decreases the *relative price* of electricity, increasing its demand for use in home heating.
- *Prices of complementary products.* Two goods are complements if they tend to be consumed together: scooters and sunglasses, computers and software, and so on. An increase in the price of a good, such as gasoline, reduces demand for its complementary good, large cars. Sales of large cars tend to fall as gasoline prices rise, all else (e.g., car prices) being the same, because the demand curve for large cars has shifted to the left.
- *Expectations of future prices.* If consumers expect the price of a good to be lower in the future, some consumers choose to delay their purchase, current demand falls, and the demand curve shifts to the left.
- *Number of consumers.* An increase in the number of consumers in the market shifts the demand curve to the right: at every price, more people are willing to buy a good. For example, an increase in a city's population increases the demand for housing.
- *Preferences or tastes.* Consumers' tastes for certain items, such as fashion items, can change over time, changing demand. A warning is in order here, however. Many commentators explain a change in market outcomes as an effect of changing tastes when in fact they reflect changes in incomes or relative prices or expectations. Beware this cheap, easy, and often incorrect and unhelpful explanation.
- *Other variables specific to particular markets.* Changes in the weather, for example, affect demand for seasonal clothing.

A change in any of these variables changes the quantity demanded *at every price*; that is, the entire demand curve shifts. For example, an increase in consumer incomes increases the quantity demanded of a normal good, such as restaurant meals, at every price; the demand curve shifts to the right.

The short-run effects of a change in a nonprice variable

We can see the effects of an increase in demand on market outcomes in Figure B-5. On a typical Saturday night, the 20 restaurants in a particular city sell 2,000 meals at a price per meal of \$15. Now suppose that over the next year the prices of the city's export products rise, increasing household incomes in the city. The demand curve in the market for meals shifts to the right, creating excess demand (a shortage) at the original price of \$15: the quantity of meals demanded at \$15 is now 2,400.

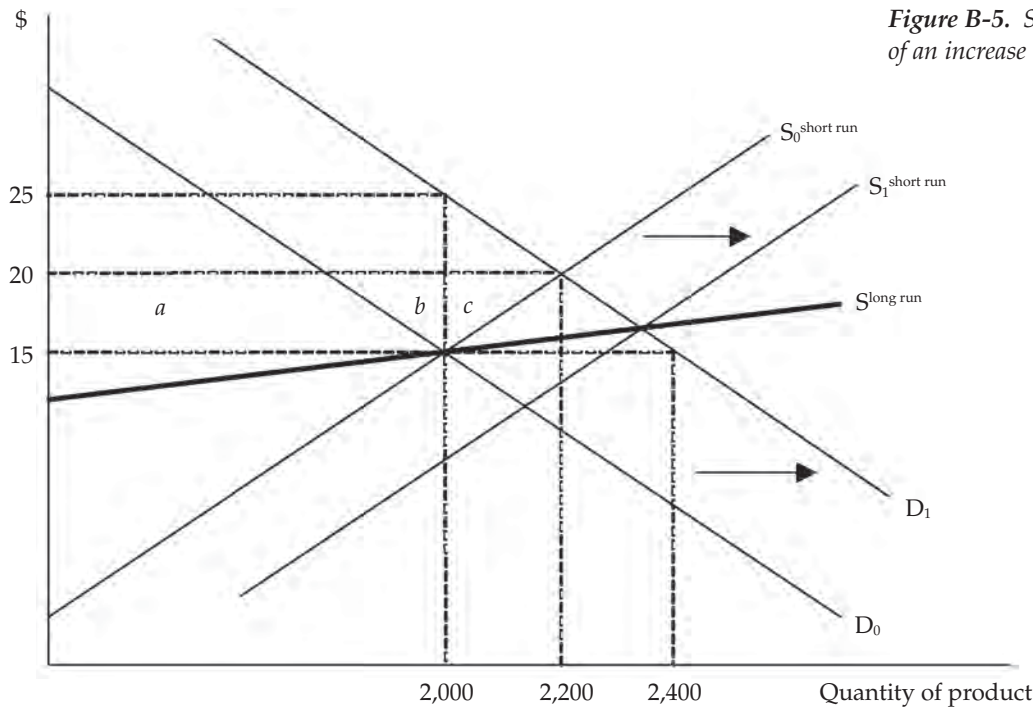


Figure B-5. Short-run market effects of an increase in demand

How does the market respond? Initially, restaurateurs may have to allocate the available supply by some means other than price (e.g., first-come, first served, or some form of lottery). In this case, quantity supplied does not change, and some consumers go away unsatisfied. Or the restaurateurs could use price to clear the market by raising the price of a meal to \$25.

Given some time, suppliers can increase output. They can, for example, order additional raw materials for delivery on Saturdays and schedule more waiters and cooks to work on Saturday nights. Unfortunately, given the capacity of their kitchens and the amount of space for seating, marginal costs rise. So they raise prices on Saturday night meals.

The higher price clears the market. Suppliers move upward along the supply curve, and consumers move upward along the demand curve; some consumers decide the price is too high even with their higher incomes—they are priced out of the market. The market equilibrates at a new equilibrium price of \$20 per meal, and the number of meals traded increases to 2,200.

Before moving on, note with care how the increase in income works through the system. The rising incomes directly affect demand: the demand curve shifts to the right. This increase in demand, and the consequent opportunity to raise price, induces a supply response. The indirect effect of the rise in incomes is an increase in quantity supplied. Increases in consumer incomes do not in the short run affect the supply curve.

We are not finished with our analysis of market adjustment to the increase in incomes. But to carry on, we need to look at the variables that directly affect the supply curve:

- **Input prices.** Anything that decreases the cost of producing a given quantity of output allows suppliers to bring more product to market at every price, shifting the supply curve to the right. Inputs include raw materials, labor, and physical capital (i.e., plant and equipment).
- **Expectations about future product prices.** If suppliers expect the price of their product to fall over time, some would prefer to sell now while the price is high, current supply increases, and the supply curve shifts to the right.
- **Technology.** Technical innovations cut production costs by improving the productivity of labor and other inputs, cutting marginal production costs even though unit input prices remain unchanged. The supply curve shifts to the right.

- *The number of suppliers.* An increase in the number of suppliers (or production facilities, generally) allows suppliers collectively to supply more product at each price, shifting the supply curve to the right.
- *Other variables specific to particular markets.* Changes in the weather, for example, affect the supply of agricultural products.

A change in any of these variables changes the quantity supplied *at every price*; in other words, the entire supply curve shifts. For example, a decrease in the wage rate reduces production costs and increases the quantity the supplier is willing and able to supply at every price; the supply curve shifts to the right.

The long-run effects of a change in a nonprice variable

Now return to the restaurant example shown in Figure B-5. Consumer incomes have increased, shifting the demand curve to the right, inducing suppliers to increase output by increasing raw materials and labor. The bad news for consumers is that marginal production costs have increased (due to the limited kitchen and seating space), pushing the market equilibrium price up to \$20 per meal.

The good news for suppliers is that the higher price has increased supplier profits. We can see this by noting the increase in producer surplus. Producer surplus, remember, is the area above the supply curve and below the price line. That amount has increased by the sum of the areas labeled *a*, *b*, and *c* in Figure B-5. Restaurateurs are taking home more money.

That increased profitability has long-run effects. To investigate this, we need to distinguish among three types of “profit”:

- *Accounting profit.* Each restaurant has one or more owners. These owners may or may not work at the restaurant. They may hire managers who then hire waiters and cooks, order raw materials, and make many day-to-day decisions. Or the owners may work as managers. In addition, the owners provide the financial capital necessary to get the restaurant working and to keep it in operation. *Accounting profit* is the amount of money left over after all the bills/explicit costs are paid.
- *Economic profit.* The checks that the owners write to pay their bills cover the costs of providing the inputs—raw materials, labor, floor space, equipment—used to produce the firm’s product. But the owners have more costs: the *opportunity costs* of the owners’ time and financial investment in the restaurant. The owners could spend their time in some other activity and invest their savings somewhere else. The foregone returns from these alternative activities are real economic costs, but they are implicit inasmuch as no checks are written to pay them. *Economic profit* is the amount of money left over after all the costs, both explicit and implicit, are subtracted from total revenue.
- *Normal profit.* Economists consider zero economic profit to be a “normal” profit. A firm that earns a normal profit is just covering all of its costs. In particular, the firm’s owners are earning only as much as they could if they invested their time and savings in the next best activities. A normal profit is like an equilibrium price: it provides the incentive to keep things as they are, to make no changes. Conversely, greater-than-normal or less-than-normal profits provide suppliers incentive to alter the allocation of resources.

Suppose that prior to the increase in demand restaurants were earning normal profits: enough to just compensate owners for their direct/explicit costs and their implicit/opportunity costs. The increase in the equilibrium price of meals from the increase in demand also increases profits. The owners of these restaurants are earning higher returns than those being earned in other activities, which provides the incentive for change.

The change comes in the form of added capacity: more restaurants. The relatively high returns attract additional investment from both inside and outside the local industry. The increase in the number of restaurants *shifts the (short-run) supply curve to the right*.

By how much will the supply curve shift? Suppose that four new restaurants open, each with average capacity: each can produce 100 meals at a marginal cost of \$15 and 110 meals at a marginal cost of \$20. With these four additional restaurants, the equilibrium price falls back to \$15: quantity demanded equals quantity supplied at \$15.

Is this the likely outcome? It is only if the increased demand for variable inputs—raw materials and labor—has no effect on the prices of these inputs. Another possibility is that the new restaurants will have to compete for a limited supply of waiters and cooks by paying better wages. The incumbent restaurants will similarly have to offer higher wages to retain staff. So maybe only three new restaurants open, and the new equilibrium price is \$16, up from the original equilibrium price of \$15, but down from the short-run equilibrium price of \$20.

One important lesson here is that more time allows suppliers to respond more aggressively to changes in price. We can see this by drawing a line through the original and new long-run equilibrium price/quantity combinations. This line is the *long-run* supply curve. It shows the quantity of the good suppliers, as a group, bring to the market at each price given that they have enough time to adjust all of their inputs. It is flatter than the short-run supply curve because enough time has elapsed to add new capacity to the industry. It still slopes upward, however, due to some scarcity in labor.

The bigger lesson concerns how market participants respond to the changes in consumer incomes. The increase in income shifts the demand curve to the right, pushing product price up in the short run. The higher price generates greater-than-normal profits for the suppliers originally in the market. Those profits attract additional resources to the market, shifting the short-run supply curve to the right. The market, *in the absence of central control or guidance*, reallocates resources to the now more highly valued activities. Prices and profits are the signals that guide this reallocation.

Evaluating the market response to change

In principle, economists can put a dollar value on the increase in surplus from the rise in incomes and the consequent market adjustments. They can identify the changes in consumer and producer surplus as areas on the simple supply/demand graph, measure these areas, and add them up to get the total change in surplus.

But problems occur in practice. A big one is that information about consumer benefits (the demand curve) and supplier costs (the supply curve) is often limited. We begin in this section by showing what one can and cannot measure from observations on equilibrium prices and quantities before and after a change in market conditions. We find that one typically needs additional information about supply or demand to make reasonably accurate estimates of the total change economic surplus. Fortunately, situations exist in which additional information is available, and it allows for more accurate estimates.

Start by evaluating the effect in the long run of the increase in income depicted in Figure B-5. Figure B-6 simplifies the diagram by including only the long-run supply curve. Comparing the total economic surplus before with the surplus after the increase in demand, one can see that the increase in total surplus corresponds to the sum of the areas labeled *a*, *b*, and *c* in Figure B-6. Area *c* is easy to calculate as the area of a right triangle: $1/2 \times 360 \times \$1$ equals \$180. The shift in demand essentially traces out, at least approximately, the supply curve.⁶

Estimating areas *a* and *b* is more difficult. Each requires knowledge of the slopes of the demand curves before and after the shift. We drew the shift as a parallel one: the increase in income has the same effect on the value of each meal. This may occur, but it does not seem particularly likely, and the observations about the before-and-after prices and quantities do not help. We have to work with the information available.

The situation is similar if supply shifts. Suppose, for example, that the local government pays for an improvement to local infrastructure. That improvement reduces production costs for local businesses, shifting the private supply curve to the right. Equilibrium price falls and quantity produced and traded rises. In this case, the old and new equilibrium points trace out the demand curve but not the supply curve.

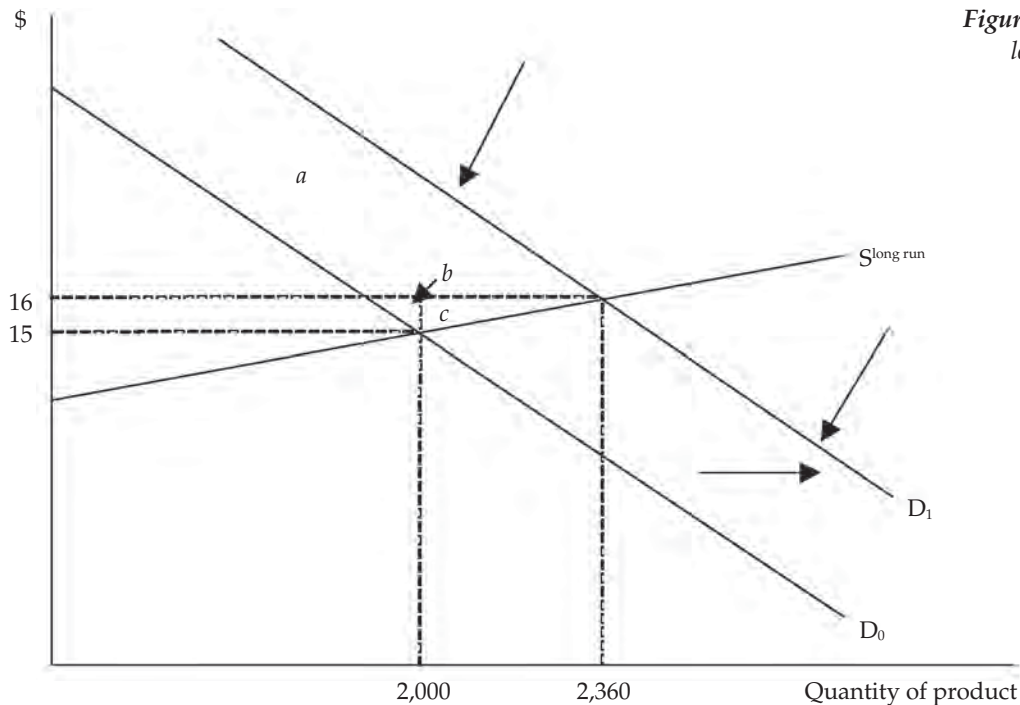


Figure B-6. Evaluation of the long-run market effects of an increase in demand

The good news is cases exist in which we can achieve a more complete evaluation. First, we can do better if a government agency is the supplier of the good or service. In that case, the books are open, and we have relatively good information about the supplier's costs.

We can also learn about the supply curve if the demand curve shifts periodically. Demand for restaurant meals and for trips on highways and transit varies over the course of the day, week, and sometimes seasonally. Demand for a variety of products varies with cycles in general economic conditions, such as interest rates and employment. In these cases, the shifting of the demand curve traces out the (short-run) supply curve, providing more information from which to calculate changes in producer surplus from public policies that affect production costs.

Similarly, we can sometimes learn about demand curves by looking at how similar sets of consumers react to differing conditions across metro areas, regions, or countries. The idea is that supply/marginal production costs vary over space a lot more than consumer preferences do. The variation in supply curves therefore traces out the demand curves of representative groups of consumers. Knowledge of the demand curves allows more accurate measurements of the changes in consumer surplus from public policies that affect consumers directly, such as policies that reduce local or regional air pollution levels.

The short version: It helps greatly that most of the time we are interested only in measuring the change in economic surplus that results from a change in policy rather than in measuring total surplus. Unfortunately, observing only prices and quantities before and after the policy is generally not sufficient. But additional information is frequently available to allow at least a roughly accurate estimate of the net benefits from the policy.⁷

Rationale for Public Policy and Its Effects on Markets

Let's review. We've shown that market trading leads to the establishment of an equilibrium price. That price clears the market. Under the conditions specified, this equilibrium price maximizes the gains from trade—the total economic surplus. We know this because the market satisfies the marginal principle: the equilibrium price equals marginal consumer benefit, which equals marginal production costs. The right amount of goods and services are produced in the most cost-effective way

and allocated to the consumers who value them most highly. All this without public oversight or intervention.

So what is the role of government? From a strictly market-oriented perspective, there are four types of roles:

1. *Defining and enforcing the rules of the game.* The market system works only if individuals, households, and businesses have well-defined and well-enforced rights. Perhaps the most important of these rights are property rights. And, indeed, a great deal of legislation, enforcement, and litigation concerns property rights. Laws and public policies with respect to contracts and tort, for example, reflect the government's effort to define and enforce property rights.
2. *Stabilizing the macro market economy.* Key characteristics of market economies, especially unemployment and inflation rates, tend to cycle over time. Indeed, market swings were often sudden and severe through the 1930s. Though economies still suffer recessions, a variety of public policies developed and implemented over the last 50 years has greatly reduced the frequency and severity of economic downturns.
3. *Redistributing income.* Efficient market performance *requires* variation in incomes. Wages and salaries are prices in labor markets. An increase in demand for a particular good results in an increase in demand for workers to produce that good, which drives up wages in the short run. The higher wages attract workers away from less-valued activities. The promise of higher wages and salaries induces individual workers to acquire valuable, but costly, skills and training. Good returns on investment of savings encourage more saving and investment. And people vary in how much they want to save and in how much of their time they want to spend at work. The market system cannot operate efficiently without variation in incomes. The world, however, is a risky place, and some people suffer periods of low income through no fault of their own. They may be born with disabilities or suffer a debilitating disease or accident. Old age typically reduces productivity. Workers become unemployed due to changes in market conditions. So, governments use a variety of techniques, including direct redistribution of income and a variety of subsidies and regulations, to protect people from poverty.
4. *Counteracting factors that inhibit efficient market performance.* Our analysis earlier implicitly assumes that several conditions hold:
 - a. Market participants have good information about market prices and product characteristics. They know the characteristics of the goods they are buying, and they know who is selling the goods they want at the lowest prices.
 - b. Resources can easily move among the various market activities. Consumers have access to goods in low-priced markets, and resource owners can direct their raw materials, labor, and physical capital to production of goods in which market prices and profits are relatively high.
 - c. Market participants bear the full costs and enjoy the full benefits of their activities. Neither production nor consumption of a good by an individual firm or household bestows additional benefits or imposes costs on others not party to the transaction.

It is this fourth role of government—treating market “failures”—that we elaborate on in this section. Our objective is to provide some insight into the nature of each of these three sources of market failure and to briefly outline the types of policies governments use to improve market efficiency. These policies are discussed and elaborated on throughout this PAS Report.

The key feature of these market failures is that the full marginal benefit differs from the full marginal cost at the market equilibrium. If marginal benefit exceeds marginal cost in equilibrium, market activity generates inefficiently little of the good or service. If marginal cost exceeds marginal benefit in equilibrium, market activity generates inefficiently much of the good or service. In either case, economic surplus

is not maximized. The objective of government policy is, or should be, to help market participants capture some of that lost surplus.

Note that inefficiency in markets implies “money left on the table.” Unrealized surplus means that some goods and cash that should change hands don’t. This by itself represents a profit opportunity: market participants are willing to pay to break down market inhibitors. A wide variety of private businesses specialize in doing just this. The government steps in when the problem is somehow too big or too difficult for market participants or other private companies to handle on their own.

Limited information about price or product quality

Our supply-demand analysis of a competitive market implicitly requires that consumers know the characteristics of what they are buying and that all suppliers charge the same market-clearing price because any other price makes no sense. Things are often not so neat in reality.

Limited consumer information about product price

Our idealized model of a competitive market corresponds most closely in practice to a few highly organized and centralized markets, such as the financial securities markets in New York and commodities markets in Chicago. At any time, all shares of General Motors stock or contracts for delivery of pork bellies sell for the same market price.

But things are not always so neat in typical retail markets. Suppose that I want to buy some Wilson tennis balls. A tin of three balls might on average cost about \$5, but the price could easily vary by 50 cents either way depending on the retailer I visit. Given time and experience, I may discover the retailer that generally has the best price. But finding that retailer takes some effort, and, frankly, saving 50 cents might not be a high priority for me. But making an extra 10 percent on tennis balls and many other products that retailer sells might make quite a bit of difference to the retailer. The upshot is that there is no single market price for many retail items.

This multitude of prices persists in large part because of search costs. Learning of the best price and exploiting it requires expenditure of both money and time. Each retailer may learn that bumping prices up a little bit hardly hurts sales because many customers realize that searching for the best price is not worthwhile. If all retailers take this approach, the slightly higher price becomes standard, creating the incentive to creep prices up a bit more. So prices rise generally above strictly competitive levels.

By how much might prices rise? The top end depends on consumer sensitivity to price. As the price of tennis balls rises, I tend to use a set of tennis balls a little longer, and I play less tennis and more racquetball. Higher prices mean fewer sales, as shown by the demand curve.

So retailers as a group face a trade-off. A higher price means more revenue per sale. But a higher price also means fewer sales. It makes sense to raise price only so long as the price effect dominates the sales effect. We can investigate this further by using the concept of *elasticity*.

Elasticity is a measure of sensitivity or responsiveness. The price elasticity of demand, for example, measures how sensitive consumers as a group are to changes in price. The price elasticity of supply correspondingly measures how sensitive suppliers as a group are to changes in price.

Elasticity is based on proportional (i.e., measured by percentage) changes.⁸ The price elasticity of demand is the percentage decrease in sales from each 1 percent increase in price. Suppose, for example, that the price elasticity of demand for tennis balls is 0.5. Then a 10 percent price increase, from \$5.00 to \$5.50, results in a 5 percent decrease in sales.

Now we can see an advantage of using elasticity. If the price elasticity of demand is less than one, then a 10 percent increase in price leads to decrease in sales of less than 10 percent. So the price effect on revenue dominates the sales effect. Increasing price when elasticity is less than one (i.e., when demand is inelastic) clearly increases

supplier profits; revenues increase and variable costs decrease due to the lower sales and production levels.

Conversely, a high demand elasticity discourages price increases. If the price elasticity of demand is greater than one, then a 10 percent increase in price results in a sales drop greater than 10 percent, and revenue falls. The sales effect dominates. Costs fall as well, but raising price is certainly not the no-brainer it is when consumers are relatively insensitive to price.

What determines the price elasticity of demand? Mainly the availability of substitutes. If the price of Wilson tennis balls increases, many people will switch to Dunlop. The price elasticity of a particular brand of tennis ball is probably high because substitute brands are readily available. If the prices of all brands of tennis balls rise, some consumers will substitute older balls or play other sports. But the price elasticity of demand for tennis balls is generally much lower than that for a particular brand of tennis balls (unless there's something very special about one brand).

But the demand for tennis balls may be relatively *inelastic* for another reason: they are not a big part of most consumers' budget. Suppose that I buy one tin of tennis balls per month for \$5. Then a 10 percent increase in price means I spend 50 cents more per month, or \$6 more per year, to buy the same number of tins. I may simply not respond much to that extra 50 cents per month.

This helps explain both the relatively high retail mark-ups and the dispersion in prices across retailers for goods like tennis balls, which cost relatively little to produce. The elasticity of demand is probably low at prices around the actual production cost of tennis balls. Even with a healthy mark-up, many consumers will not bother to shop around to look for a lower price. It is hard to imagine that retailers do not take advantage of this low consumer sensitivity to price that arises ultimately from the costs of searching for relatively good prices.

How do retailers fare from the relatively high mark-ups? The answer depends on the reaction of other suppliers to the higher profits the higher mark-ups generate. In our basic supply-demand analysis, we saw that positive economic profits attract competitors. New competitors *either* push prices down toward competitive levels *or* take some of the incumbent suppliers' market share. Either way, profits fall. In the second case, prices remain above competitive levels even though profits fall; consumers suffer, and suppliers do not prosper.

Limited consumer information about product quality

In our base analysis, we assumed that consumers had essentially full information about the characteristics of the products they were buying. For many products, this is essentially true in practice: many food products at grocery stores, fuels, and so on. But the quality of a large variety of products and services can be difficult for consumers to discern: cars and appliances, legal and financial services, etc. Consumers have to take a bit of a gamble when purchasing these goods and services.

What are the effects on market outcomes of this limited information about product quality? The basic answer was first provided in 1970 by George Akerlof, who won a share of the 2001 Nobel prize in economics in large part for this work.⁹ In short, markets in high-quality product may not just fail to allocate efficiently—they may not exist at all! We start in this section by explaining this, and then, more importantly, discuss why and how private market participants respond to this potentially serious type of market failure.

Akerlof thought about the problem in terms of the market in second-hand cars. In this case, information about product quality is “asymmetric”: through experience, the seller knows more about the car's quality than does the potential buyer. But the buyer knows that the quality of second-hand cars varies. So how does the buyer react to the asymmetric information? Suppose he assumes that the car is of median quality. Then he knows that half the time he'll be lucky and the car will be of relatively good quality and half the time he'll be unlucky, but if he offers a price reflecting the average quality, he'll on average get a tolerable deal.

This sounds simple and reasonable, but it has surprisingly ill effects on the market for second-hand cars. A price that reflects the quality of a typical car will seem low to the potential seller of a high-quality second-hand car. That seller will therefore be inclined not to sell his car at that price. So the cars offered for sale will tend to be those of relatively low quality. This is referred to as *adverse selection*: the quality of the cars brought to market is not representative of the quality of used cars generally—it is lower.

Consumers reasonably respond to the lower quality by reducing their offers for cars. But this just exacerbates the adverse selection. Low-quality cars effectively drive high-quality cars off the market. All of the potential economic surplus from trade in high-quality cars is then lost. This is a rather extreme form of market failure.

The problem can arise in a variety of important markets. Workers applying for jobs know more about their productivity than do the firms looking for workers. The people purchasing insurance policies know more about the likelihood of their making a claim than do the insurance companies. People applying for a loan know more about the likelihood of default than do the lenders. The potential loss of economic surplus due to adverse selection could be rather high.

It is this potentially high cost that provides the incentive to try to overcome the problem of asymmetry. Market participants, both sellers and buyers, have found ways to provide incentives to other participants. Consider the following examples.

- *Reduce the variation in product quality.* Adverse selection does not arise because cars vary in quality; it arises because consumers cannot distinguish the quality of a particular car. But if quality does not vary, the problem goes away. Some car makers work hard to reduce variation in quality.
- *Supply information.* Potential car buyers are willing to pay a mechanic for expert evaluation. And car makers assist in this process by installing systems that make it easier to assess the condition of the car—systems that make sense anyway (e.g., sensors that monitor engine performance and feed information to an onboard computer).
- *Establish a reputation for quality.* New car dealers are in a relatively good position to sell good second-hand cars. Many consumers of new cars bring their car in for service at the dealer. The dealer then has a record of maintenance and repair for the vehicle. When the owner of the car decides to trade it in for a new model, the dealer has unusually accurate information about the quality of the car. The dealer can offer a good trade-in price if they can then sell these good used cars at premium prices by establishing a good reputation among used-car buyers.
- *Offer a warranty.* Reputable sellers are willing to guarantee the quality of their product, which reduces the risk to consumers and encourages them to offer a higher price.
- *Support independent rating agencies.* Many types of service providers are willing to contribute to the support of private or public groups that rate and accredit their operations.
- *Signal.* In his 1974 book, Michael Spence (who shared the 2001 Nobel Prize with Akerlof) showed that in some markets sellers can send buyers signals that convey information about a product's quality.¹⁰ Education, for example, is often used as a signal from job applicants to employers. Successfully completing a tough major at a good school with high grades sends a strong signal to potential employers.

There are surely additional ways that private market participants respond to the inefficiency of adverse selection. The big point here is that market inefficiency implies lost surplus, which provides incentive for market participants themselves to treat the market failure. Government policy (i.e., intervention) makes sense only when the government can supply the needed fix more cost-effectively than can private suppliers.

Immobile resources

In the basic competitive market model, prices and profits guide the allocation of resources. An increase in demand for restaurant meals, for example, increases prices

and profits in the short run (i.e., the period of time too short to build new capacity), which attract additional resources—buildings and equipment, labor, and raw materials—to the production of additional meals. Expansion continues until prices and profits have dropped to “normal” levels (i.e., until the return to deployment of resources in this activity is similar to the return in other activities). As we noted, this decentralized and mechanical process results in highly desirable results: the efficient allocation of resources across activities.

But the process requires that resources are mobile. Entrepreneurs have to be able to shift building materials, workers, and raw materials to the markets in which prices and profits are relatively high. In many cases, resources are in fact fairly mobile, at least given enough time. This resource mobility is reflected in a concept called “the law of one price”: in long-run equilibrium, a given product trades for roughly the same price across the locations in which it is sold. The reason is that suppliers and demanders respond to price differentials in ways that tend to eliminate them.

In practice, the law of one price appears poorly enforced. One reason is that market participants lack perfect information about product prices and characteristics. Gathering and correctly analyzing information has costs. Similarly, moving physical resources has costs. These “adjustment” costs act as a wedge between actual and theoretically efficient prices.

Monopoly pricing results from resource immobility. A patent, for example, is a legal barrier that prevents nonpatent holders from entering a market. The idea is that the greater-than-normal profits the patent gives the patent holder provide strong incentives to innovate. In other cases, firms gain monopoly power because efficient production requires a very large upfront investment. The extreme case is termed a “natural” monopoly because one supplier can produce at lower average cost than can two or more suppliers operating in the same market. In general, maintaining monopoly prices and profits requires barriers against resources entering the monopolized market.

Some important resources that appear mobile are less than perfectly so. Labor, for example, appears mobile: workers chase relatively high wages. Legal barriers, however, (e.g., national borders) limit labor mobility. And workers care about more than wages; some workers are willing (though not keen) to accept lower wages to remain near extended families or social networks. This results in wages (the prices of labor) to vary across otherwise similar individuals.

Other important resources are inherently highly immobile. Land is the defining example, and land price can vary tremendously across even relatively small distances. In practice, these price differentials often induce at least a limited supply response. For example, many of the shallow areas around good ports have been filled to create new land (increasing its supply) in highly valued locations. But well-located land remains almost by definition in limited supply.

To the extent that land is immobile, its price differs in concept from that of mobile resources. The price of a mobile resource cannot rise much beyond its value in the next best activity. The price of well-located land can, in contrast, rise far above its value in the next best activity because it’s very tough for the owners of low-priced land to simply move their land to chase high land prices. This gives rise to an important concept: economic rent. Economic rent is any payment for a resource greater than the minimum necessary to attract the resource to the activity.

The concept of economic rent, and specifically land rent, has generated a great deal of debate.¹¹ Much of the debate concerns taxes. Taxes typically have effects on the allocation of resources. Taxes on labor income, for example, affect individuals’ decisions about how many hours to work and even in which state or city to work. Taxes on retail goods affect how much of the goods people want to buy, and therefore on the amounts produced and traded. Thus, many taxes reduce the efficiency of resource allocation (known as the “excess burden” or “deadweight loss” from the tax). But to the extent that land is in strictly fixed supply, a tax on land has no effect at least on the quantity of land supplied to the market; the land owner has little choice

but to accept the tax and carry on as before. The tax affects the land owner's wealth and income, but not the quantity of land supplied.

High land prices, however, can induce a supply response that does not involve moving or creating new land. Johann Heinrich von Thünen in 1826, in his book, *The Isolated State*, described why agricultural land prices rise with proximity to a city or port: lower land prices farther from the port compensate for higher transport costs. The owners of more-distant, and therefore lower-value, land would of course like to move their land closer to the port. Moving land is usually not possible. But one can, through changes in the characteristics of the transportation system, move the port closer to the lower-priced land. High land prices do in reality affect the allocation of resources by encouraging transportation system changes that effectively increase the supply of "well-located" land.

Changes in land prices can be used to fund the transportation improvement. Privately supplied infrastructure can be funded through land sales and publicly supplied infrastructure through land taxes.

Note that resources in addition to land can have similar characteristics. Once built, houses, commercial buildings, and a variety of public infrastructure are difficult to move. Similarly, workers with specialized training have trouble changing employers or industries. House prices and the wages of workers with specialized training can fall below "replacement" cost in areas suffering an economic downturn. The owners of rental housing, for example, have little choice but to continue to supply housing at lower rents. This resource immobility can assist an area to weather the downturn because lower housing prices and wages allow local employers to remain in business and keep workers employed. It also can result in urban blight if the economic downturn is sufficiently long and severe.

External costs or benefits

An externality is usually described as a cost or benefit that spills over onto people not party to the transaction (or activity) that generates the cost or benefit. Air or water pollution are the standard examples. A manufacturer, say, disposes of waste cost effectively—from his perspective—by dumping the waste into the air, water, or ground. The pollution has costs, however, because it adversely affects people downstream or downwind.

An activity can also generate positive externalities. A well-educated person is not only privately more productive, but is more likely to contribute more to public issues and ventures. A large retailer generates traffic that benefits the smaller retailers nearby. Of course, the traffic that the large retailer generates also may impose costs on nearby residential areas. One need not look hard to find externalities.

An alternative definition of an externality offers some advantages: a cost or benefit *not priced* in the market. Markets allocate resources efficiently when the equilibrium market price equals both marginal production cost and marginal consumption benefit. Markets, therefore, allocate inefficiently when some aspect of the transaction, such as disposal of waste into the atmosphere, isn't priced; the *full* marginal production cost exceeds both price and marginal private cost.

This begs the question of why some costs or benefits aren't priced. The reason often is that property rights to some resource aren't well defined. Who owns the airshed? Because no one owns the airshed, no one has a strong incentive to look after the resource and to charge for its use.

One could reasonably argue that allowing some individual or business to own the airshed is at best impractical. We agree. But the airshed is a scarce resource, using it has costs, and for the market system to work well the users of the resource have to pay a price equal to marginal cost.

To analyse the effects of externalities on markets, consider the case of the polluting industry. Waste generation is part of the production process. The problem is not that the industry generates waste, but that waste disposal in the river or air is unpriced.

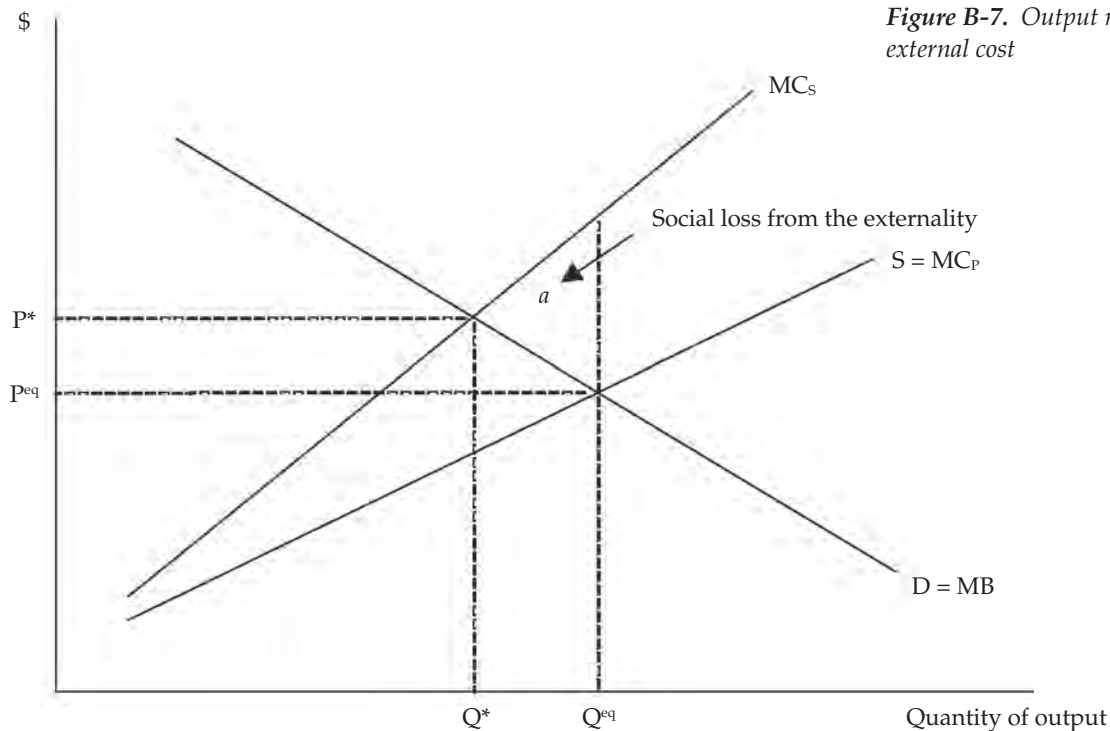


Figure B-7. Output market with an external cost

Figure B-7 shows the source and characteristics of the market failure. The curve labelled MC_p shows the industry's marginal production costs when it dumps its waste into the river. This curve is the standard industry supply curve; it consists of the vertical sum of the marginal costs of all the inputs—labor, materials, energy, equipment, and so on—that the firms in the industry have to pay for in the respective input markets. Given these costs, and the demand curve for the industry's product, the market equilibrium price-quantity combination is P_{eq} and Q_{eq} .

The curve labelled MC_s shows the marginal *social* cost of producing the industry's output. The subscript s stands for "social" in contrast to the subscript p , which stands for "private." The MC_s curve is the vertical sum of the costs that each firm in the industry pays for, as shown by the MC_p curve, plus the costs incurred by people downstream as a result of the industry's pollution. The vertical difference between the two marginal cost curves is the *marginal external cost*—the additional external cost associated with each one-unit increase in production.

If firms had to pay for their waste disposal, the MC_s curve would be the industry supply curve. The market price would rise to P^* and the market equilibrium quantity produced and traded would fall to Q^* . An external/unpriced cost results in output prices that are too low and output that is too high.

Note that pricing an unpriced input does not eliminate pollution. It does, however, encourage firms to economize on their polluting, and it generates revenues with which to compensate those harmed by the pollution. In other words, the market in access to the environment would work like the markets in other scarce resources. Production would be efficient in that the price of the final product would include the full social cost of production.

Society in sum would be better off by the dollar amount corresponding to the area of the triangle labelled a . For each unit of output in excess of Q^* , the full marginal cost of production exceeds the marginal benefit from consumption; this represents a net social loss. The total loss is the sum of all those marginal losses (i.e., the area of the triangle labelled a , which is referred to as the "social loss triangle").

What are the potential policy responses?

- *Government sets a price on emissions.* One could think of this as government claiming ownership of the resource on behalf of the citizens. The ideal emissions charge equals the marginal external cost at the optimal output level. Each firm then decides how much to pollute, just as it decides how many workers to hire.
- *Government sets a ceiling on the quantity of emissions.* This is the conventional practice in the U.S. and other developed countries. The government then has to allocate emissions across the various polluters. It can do this in two ways:
 1. Centrally, through quotas on each firm based on available recognized control technologies, which is the conventional approach, or
 2. Through the market by allocating emissions permits to firms and allow trade in those permits. The market in permits establishes a price for emissions, and each firm makes decisions based on that price. Markets in a variety of pollutants are in operation in the U.S. and other countries.
- *Government tries to facilitate negotiations between producers and people affected by externalities of production.* Note that the social loss associated with excessive emissions provides incentives, collectively, to look for a fix. Polluting firms, for example, save money by dumping waste freely into the environment, but workers demand higher wages to live in polluted cities. Total costs would be lower if polluters collectively reduced emissions. The government could try to facilitate those negotiations.

Facilitation is most likely to work when the numbers of participants in the negotiating process is small. One could view zoning boards and commissions as facilitators between land owners. Zoning requirements provide the zoning board with leverage—they have the power to make a decision. But they often have quite a bit of discretion. Given the threat of an uninformed decision, participants have the incentive to participate in negotiations.

SUMMARY

This long appendix is a short course in microeconomics. In the rest of this PAS Report, we use implicitly or explicitly most of the economic concepts it introduces. The ones we call on most frequently are the following:

- People usually make economically sensible decisions in their day-to-day activities. This means they behave roughly in accord with the *marginal principle*.
- Marginal benefit usually falls and marginal cost usually rises with increases in an activity over a given amount of time in accord with the *law of diminishing marginal returns*.
- A market consists of all those demanders and suppliers whose interactions determine the market price of a good or service.
- The market demand curve can be interpreted as a social marginal benefit (MB) curve and the supply curve as a social marginal cost (MC) curve.
- Market interactions push the market price toward the price that “clears” the market. That price and the quantity of goods or services traded at that price are referred to as the market equilibrium. Important to remember is that the market equilibrium is established through an essentially “mechanical” process.
- Under favorable conditions, the market equilibrium is “optimal” or “efficient” inasmuch as the right amount of the good or service is produced by those who can supply it most cost-effectively and is allocated to those consumers who value it most highly. Put another way, the market equilibrium maximizes net social benefit.
- The dollar value of the net social benefit from production and trade (also known as *economic surplus* or the *gains from trade*) can be measured as the area between the demand (i.e., MB) and supply (i.e., MC) curves.
- Changes in market conditions generate short-run changes in market prices and profits that guide longer-run reallocations of resources.

- *Elasticity* is a useful measure of the sensitivity of market participants to changes in price. It measures the percent change in quantity supplied or demanded from a 1 percent change in price.
- One can often measure the dollar value of the change in net social benefit from a change in market conditions (due, for example, to a public investment or change in public policy) with reasonable accuracy.
- Unfavorable conditions—poor information about prices or product quality, constraints on the movement of people or goods, or poorly defined property rights—inhibit efficient market allocation, which reduces the net gains from trade.
- This loss of economic surplus provides incentives for market participants to reduce or eliminate the unfavorable conditions. Private, nongovernmental solutions are frequently successful.
- Sometimes, private solutions are or appear infeasible or cost-ineffective, and the public sector can use its special powers to more effectively treat the market failure. This is the rationale for the public land-use and transportation policy alternatives that we adopt in this PAS Report.

APPENDIX B NOTES

1. The draft of this appendix was written by Arthur O'Sullivan and drawn, with his publisher's permission from his text, *Urban Economics* (6th edition, Prentice Hall, 2006). Paul Thorsnes built from Art's base to create the final version of this appendix; Art liked the improvements.
2. Discussed in O'Sullivan (2006, Chapter 1).
3. Stigler and Sherwin (1985, 555–85).
4. The area of a right (i.e., a 90-degree) triangle = $1/2 \text{ base} \times \text{height} = 1/2 \times 1,000 \times (5-3) = 1,000$.
5. "An Inquiry into the Nature and Causes of the Wealth of Nations," 1776.
6. This is approximate because we have assumed that the long-run supply curve is linear between the original and new equilibrium points.
7. An additional complication is that changes in one market spill over to affect other markets. In some cases, these spillovers may be fairly small, but in others they may be important and need to be recognized and measured using the same kind of methods.
8. Specifically, the price elasticity of demand is the *percent* change in quantity demanded divided by the *percent* change in price. This is in contrast to *absolute* changes; for example, the absolute drop in sales with, say, a \$1 increase in price (measured by the slope of the demand curve).
9. Akerlof (1970).
10. Spence (1974).
11. This debate is often associated with Henry George and his book, *Progress and Poverty*, published in 1879.

Surface Transportation: How It Works

This appendix discusses principles of surface transportation from both a physical and a social perspective. It is a crash course in surface transportation planning and the connection of transportation to land use in urban areas. It covers topics familiar to engineers and economists (but probably not to both at the same time) and helps urban planners with a technical bent get some familiarity with what those engineers and economists are talking about.

The majority of the appendix is an application to highway transportation of the economic concepts described in Appendix B. We show that traffic congestion on a toll-free highway can be thought of as a form of market failure resulting from the decision to fund roadway construction and maintenance with a tax on fuel rather than with a highway toll that varies with demand. The analysis provides a useful framework for identifying and evaluating a variety of common policy responses to traffic congestion.

The appendix is organized into two main sections:

1. **The Physics of Traffic** describes the constraints on travel imposed by the physical characteristics of cars and highways.
2. **The Economics of Traffic** adds traveler decision making to the physics. It has two parts: (1) *Traffic Externalities* works through the standard economic framework for evaluating congestion, emissions, and other transportation-related externalities; and (2) *Travel Demand Elasticities* reports estimates of traveler sensitivity to changes in factors that influence travel choices.

THE PHYSICS OF TRAFFIC

Analogies use the familiar to explain the new. They can be helpful or misleading. For example, the Heisenberg Uncertainty Principle, arguably correct at the subatomic level, has been used as an analogy to explain all manner of physical, social, and spiritual inquiries—uses that may be inappropriate.

While we recognize the trap that analogies present, some are too seductive to resist. We think it helps to start an understanding of urban traffic by an analogy to physics: traffic as water flowing through a system of streams or pipes. Water in a pipe is probably the better analogy for traffic because it implies some ultimate capacity.

If a pipe is straight and smooth, water flows smoothly in what physicists call laminar flow. If there is no pressure—if the pipe is not full and not constricted at its downstream end—water moves along at the pace gravity allows. Any of several conditions, however, can cause the flow to become turbulent: increased pressure (trying to force more water through the pipe); imperfections in the inner surface of the pipe, or at joints, that become more important as the pipe fills and the pressure increases; turns and constrictions in the pipe; and (look out) something getting stuck in the pipe and constricting flow.

Turbulence is a problem for engineers trying to optimize the functionality and life of the water-delivery system. First, turbulence means that less water is getting through in a given period of time. When the pipe is not full, more throughput (e.g., gallons per minute) can be achieved by forcing more water into the pipe (opening a valve; turning on a pump). At some point, when the pipe is full but before turbulence kicks in, the maximum throughput is achieved. Past that point, trying to push more water into the upstream end of the pipe not only does not get more water out of the

downstream end: it actually reduces the output because of the turbulence. Water molecules collide, the system falls into disorder, and less water gets down the pipe in a given amount of time

Second, turbulence means that the insides of the pipes are getting banged by cavitation (pockets of air in the water). Over time that air banging around causes pipes and joints to wear, which will then require new investment and, in the meantime, will lead to even more turbulence.

A related analogy for understanding the phenomenon of traffic congestion is “sticks in a stream.” A bunch of sticks make their way, with occasional rest stops in eddies, down a creek. They slow down at the curves. But if too many sticks arrive at the same time at an otherwise negotiable constriction, they can pile up. If that pile-up is not released, the flow behind backs up, slows, and looks for other paths to follow downstream.

Another analogy is sand through an hourglass. Imagine a tall hourglass with all sand speedily in freefall until it hits the waist of the hourglass. At that constriction sand continues to flow, but at a rate determined by the capacity of the constriction. More pressure from the sand stacking up above makes little difference on the rate of flow at the constriction.¹ Demand exceeds capacity and upstream backups continue to occur.

The analogies to traffic flow are too obvious to belabor. If a highway is uncongested, without hazards or accidents, the flow is smooth and consistent. Highway geometrics cause turbulence as more traffic is allowed into the pipe; accidents create new and more acute constrictions.

All of us have been slowed to a standstill in freeway traffic and have assumed an accident lays ahead—a constriction that explains the back-up. Eventually we move, and we see no evidence of an accident. If no accident, why were we stuck?

Plenty of research has attempted to answer the question of how mere congestion can cause a jam. Traffic physicists (e.g., Kerner 1999) define three types of traffic flow: free flow, synchronized flow, and traffic jam (no flow). In free flow, each lane flows at a different average speed. People pass one another readily, and no one is preventing anyone else from driving as fast or slow as they want.

The flow becomes synchronized as density increases. Lanes fill with cars and room to pass disappears. Each car is going slower than in free flow, but at a relatively high speed considering the short following distance. This traffic level is tenuous; a jam may be looming. Any anomaly in the system, even random fluctuations in speed by the momentary inattention of drivers may cause a jam. If someone switches lanes and causes the person behind him to brake, or if someone turns to pick up a cell phone, change the radio station, or reprimand a child, a jam could occur.

The instability of synchronized flow results from the limited spacing on roads during synchronized flow. As following distances decrease during periods of high volume and flow becomes synchronized, the reduction in distance is not met by a proportional cut in speed. Drivers slow down, but not enough to maintain a safe following distance.

Engineers, doctors, and psychologists have spent decades studying response time—the results do not change because the laws of physics and metal and motor do not change: a safe driving distance is considered to be about two seconds behind the car in front of you—the amount of time needed to observe, respond, and slow your vehicle in response. At 10 mph, that is about 30 feet; at 65 mph, that is about 200 feet. It is rare to the point of unobservable to find drivers adhering to these suggested standards at peak travel periods in urban areas. In congested urban areas, one is lucky to find 1.5-second spacing; tailgating at a car-length or two is cutting it to a half a second.² Downs (2004a, 360) reports an extensive California study that found maximum traffic flow on freeways at speeds of 60 mph and average following distances of between 1.6 and 1.1 seconds.³ In those conditions, any fast stop has a relatively high probability of causing an accident, and that accident really creates upstream congestion: miles of it.

Driving habits in response to high traffic volumes stretch the slack out of the system. Drivers ride on the edge, saving a few seconds or minutes a day, but substantially increasing the risk of much longer delays on some days.

Synchronized high-density flow becomes a traffic jam as individual decisions ripple throughout the entire system. As drivers approach an area where they anticipate a slowing of traffic (that anticipation may be pretty short: Oh, brake lights ahead; I'll start braking now), they may brake harder and longer than they need to. A person following may also err on the side of caution and brake even harder. This collective reaction propagates itself backward in a wave of congestion that flows upstream through traffic, compressing the concentration of cars until people sit bumper to bumper, curse to curse.

Schewe and Stein (1997) explain this congestion, common at peak times in urban areas, in the context of our physics analogy as a *phase transition* from a liquid to a solid. Traffic in free flow resembles a liquid flowing uninhibited down the path of least resistance to a destination. As traffic density increases from free flow to synchronized flow, everyone moves at approximately the same speed. This flow level could turn to a jam at any moment: it could reach its freezing point. The traffic will move smoothly, but if there are small perturbations (small chunks of ice) caused by momentary inattentions, "rubber necking," lane changing, or sudden braking, other drivers react and the solid chunk of ice grows as more liquid water molecules are trapped by the ice and crystallize. Traffic moves glacially as part of a large iceberg.

But drivers, for better and for worse, are not water molecules. They cannot flout the laws of physics, but they can ignore the laws of traffic. They aren't as predictable as water molecules; they act independently and can cause congestion all on their own. Thus, the conversion from synchronized flow to a traffic jam can be more random than the conversion of water from a liquid to a solid at 32 degrees Fahrenheit.

At low traffic levels, the system has enough capacity to deal with erratic driver behavior. Drivers have more distance on the road, and one person's decisions have less immediate effects on other drivers. At low traffic levels, if someone is driving erratically, he can simply be passed. At high traffic levels, however, with no room to pass, good drivers (the rest of us) must respond to the erratic driver by braking more often; if they choose to pass, cars in the passing lane will be forced to slow to accommodate the extra space the lane change requires.

Put basic physics facts about today's urban freeways into the context of *yesterday's* urban freeways (special exercise for baby boomers). You know—the ones that had wider lanes, a shoulder on each side, relatively little congestion, and nice people driving. Some of today's freeways (we've seen examples in Los Angeles) have expanded capacity by creating lanes out of what used to be shoulders. We now drive shoulder to shoulder, and shoulder to shoulder, with only our false faith and low-but-rising probabilities of an accident protecting us from gridlock.

Traffic engineers break the causes of congestion into two classes (Downs 2004a, 355). Recurrent congestion is what our discussion so far has focused on: having too many vehicles in the same place at the same time. Even if everything goes reasonably well, the flow of vehicles decreases for all the reasons we have given. We have also alluded to a second type of congestion: congestion caused by "incidents" (i.e., an accident or any other event that perturbs the smooth flow). There may be a vicious cycle: the worse the former, the worse the latter.

On a congested road, when an accident (or overheated engine, or flat tire) does occur, there is nowhere to go. With so many more cars traveling, the probabilities of an incident are much higher, as are its consequences. Drivers have traded higher average speeds on most days for the risk (the certainty, it appears) of very much lower speeds on some days.

Our example in this appendix has been a limited access highway. The same principles (and water analogy) applies to arterials with traffic signals. If the volume is low and the lights are synchronized, flow is unimpeded (the pipe is smooth and open) and the traffic on the arterial behaves similarly to traffic on a freeway, though at a

lower optimum speed (probably 30 to 40 miles per hour) because it is impossible to synchronize traffic perfectly in two directions on a two-way street. But if the lights are not synchronized or cannot be because of the density of the traffic flows, the lights are equivalent to short-term incidents: traffic backs up. Thus, congestion can happen on roads with lower travel speeds. If traffic backs up too much, one gets what traffic engineers refer to as *cycle failure*: some vehicles fail to clear the intersection in a single cycle—you've been there; we all hate it.

Let's turn off the water and tap into the real concerns of traffic engineers: volume, capacity, speed, density, and flow of traffic.

Volume (v) is usually measured as the number of vehicles on a road segment at a given time. Typical measures are (1) Average Daily Traffic (ADT) for a roadway, all lanes, in both directions (e.g., 60,000 vehicles ADT); (2) peak-hour traffic in one direction (e.g., 6,000 vehicles northbound in the morning peak hour); or (3) vehicles per lane per hour (e.g., 1,800). When volume is measured in vehicles per lane per hour, it is identical to *flow*.

Capacity (c) is defined by The *Highway Capacity Manual* (2000) as "the maximum sustained 15-minute rate of flow, expressed in passenger cars per hour per lane (pcphpl), that can be accommodated by a uniform freeway segment under prevailing traffic and roadway conditions in a specified direction."⁴

Engineers use the ratio of volume to capacity (v/c) to describe the amount of congestion on the road. This ratio ranges from 0 to 1.0 (or more). At a ratio of 0, there is no volume: no cars are on the roadway. At a ratio of 1.0, volume equals capacity: the road is carrying the maximum volume of vehicles it was designed to handle. At values below 0.5 (or even higher), speed is not affected by volume. But well before the ratio reaches 1.0, the roadway becomes congested. As it closes in on 1.0 it gets very congested: as with water and pipes, if too much volume is trying to get through a stretch of road, turbulence can reduce the ability of the roadway to handle traffic, and less traffic makes it through.

Speed (S) is just what you would expect: distance traveled in some amount of time, usually measured for traffic in the U.S. in miles per hour, but sometimes feet per second.

Density (k) is the amount of cars in a given distance. This measurement tells how tightly the roadway is packed. During periods of congestion, average speed is inversely related to density. Intuitively this makes sense: at peak traffic hours, as more and more cars enter the highway, density increases. To accommodate this density, drivers must (eventually) slow down to account for the reduced braking distance between cars.

Flow is the amount of traffic moving past a point in a given amount of time. Traffic planners and engineers often see maximizing flow as a socially desirable goal because at the maximum flow rate, the greatest number of vehicles are getting where their drivers and occupants want to go in a given amount of time.⁵ Flow is equal to density times speed ($F = D \times S$). As both speed and density increase, more cars pass through a roadway in a given time. But for traffic, density and speed are inversely related: more density means less speed. Thus, the maximum flow occurs at a balance between the two where a lot of cars (but not too many) are traveling at relatively high (but not maximum) speed (maybe 55 mph on an urban freeway).

Level of Service (LOS) describes how well traffic is moving, from level A (great) to F (failure). It is not a measure of flow (many different flows can all have LOS A); it is highly correlated to the v/c ratio and density for limited access highways like freeways.⁶ In fact, LOS is often illustrated by photographs or movies of a strip of highway with a different number of vehicles on it. For limited-access highways, a v/c of 0.2 is going to be LOS A; a v/c of 1.0 will be LOS F.

Think about a stretch of road with cars bumper to bumper. The only safe speed for them to drive without colliding is zero. If slightly fewer cars have a foot or two of space between them, they may be able to go five miles per hour. While traffic density may have been reduced, the increase in speed more than offsets the density

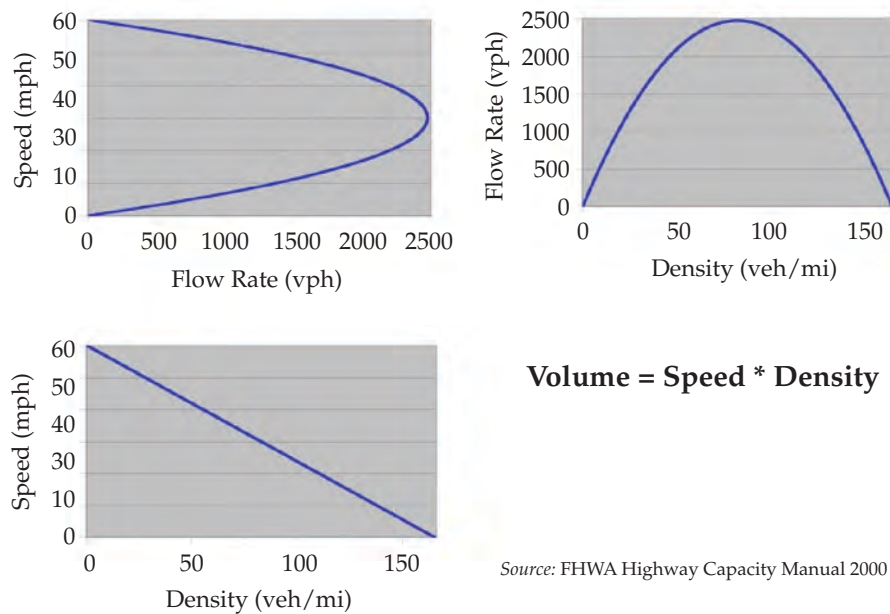


Figure C-1. Relationship between speed, density, and volume/flow

reduction, and the flow is higher. At the opposite extreme, one car may drive through an empty road at 100 miles per hour. If one more speeder is added to the road, there is still plenty of space, and neither will have to slow much to accommodate the other. Flow will increase since the increase in density more than offsets the decrease in speed. These marginal changes to extreme conditions point to the correct conclusion: the maximum flow is achieved somewhere in between, at a compromise between density and speed.

Figure C-1 illustrates the relationship between speed, density, and volume/flow. It assumes that traffic speed depends on and is highly correlated with traffic density, which is in turn highly correlated with traffic volume: thus, volume is the implicit determinant of speed. At high speed and low flows (top left graph), flow can be increased by increasing density and decreasing speed. Transportation planners do not need to worry about that condition much: the flow is low because nobody wants to travel on that road at that time—their choice. At low speeds and high density (traffic congestion), flow can be increased by decreasing traffic density and increasing speed. In other words, when the road gets too congested, the only safe way to move toward a higher flow that day is to reduce the traffic density (over a longer period, planners think about increasing capacity in any of several ways).

The maximum flow is achieved at the cusp on the right of the top-left graph, where the balance between speed and density is achieved. If you care about efficiency (a lot of traffic engineers do),⁷ this is the sweet spot and where you want the system operating. That spot is probably around a .85–.90 v/c ratio, which gives enough buffer capacity for the flow to recover from the perturbations introduced by less-than-perfect driver behavior.

The sidebar summarizes how the physics of traffic relates land-use and the goal of having livable, walkable communities.

At least two factors break the analogy between vehicles and water molecules when trying to understand flow. First, vehicles are much less uniform, and that variability leads to variability in flow. There are “incidents.” Downs (2004a, 63) cites research showing that about 80 percent of them are vehicle breakdowns, 20 percent of which block lanes (estimated delay of 15–30 minutes and 500–1,000 vehicle-hours each). The other 80 percent get to the shoulder (estimated delay of 15–30 minutes and 100–200 vehicle-hours each). Accidents account for only 10 percent of the incidents, and 40 percent of them block lanes (estimated delay of 45–90 minutes and 1,200–5,000 vehicle-hours

TRAFFIC FACTS THAT MATTER FOR LAND USE

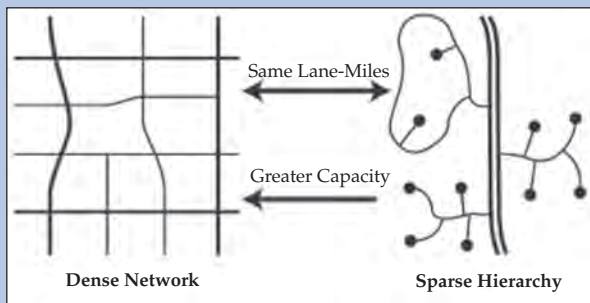
By Troy Russ, AICP, Principal, Glatting Jackson Kercher Anglin, Inc., and Walter Kulash, Principal, Glatting Jackson Kercher Anglin, Inc.

Connectivity

Many subdivision regulations do not require street connectivity. They force traffic to fewer roadways which then require widening. The bulk of arterials have been designed for mobility: as high-speed, single-purpose roadways, which then limit land-use choices, discourage buildings from fronting the street, and reduce the quality of the walking environment.

For a given number of lanes, a connected network of smaller streets will perform better than a sparse network of larger streets for two reasons:

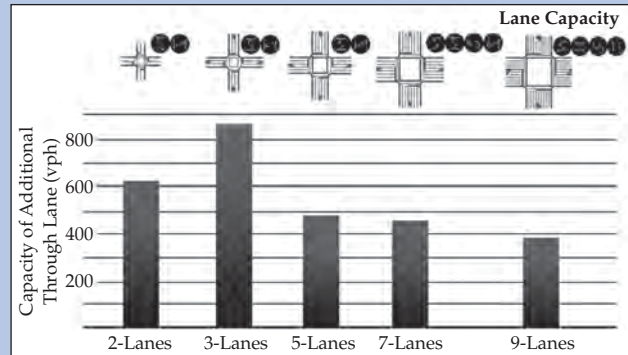
- In a sparse network, left turns are gathered up from multiple locations and focused at a single location, requiring protected left-turn signals and robbing intersections of “green time” and capacity. A connected network of disperses left turns and may not even require protected left-turn lanes.
- Signalized intersections of larger roads require longer clearance (yellow) phases for crossing pedestrians, increasing signal times and further reducing capacity. Beyond three lanes (one lane in each direction and a left turn lane at intersections), there is a diminishing rate of return on pavement width.



Street Size

This point relates to the one about connectivity. As roads add lanes they become less efficient: more lanes do provide more capacity, but each new lane adds less new capacity than its predecessor.

Going from two lanes to three adds capacity because it allows protected left turns that increase the volume of through traffic.



Then volume per lane falls. Two three-lane streets carry more cars than a single seven-lane street.

In addition to transportation benefits, there are land-use benefits. Three-lane streets do not limit land-use choices, they do not discourage building from fronting a street, and they do not damage the quality of walking environment. Residential, retail, and commercial land uses can all succeed; pedestrians and cyclists feel comfortable.

Travel Speeds

A free-flowing arterial will carry more cars per lane between 25-30 miles per hour than at any other speed. Land uses work at these speeds. Posted speed limits should 30 and 35 mph in urban areas in which the use calls for a slower street; in many situations signal progression dictates speeds lower than 30 mph.

Roadway Safety

Design speeds control the level of roadside improvements a community can make: higher speeds require longer “clear zones.” Minor reductions in design speed can leverage large gains for street trees, lighting, and other pedestrian amenities. Increasing speeds from 20 mph to 40 mph will not simply double stopping sight distance, it will increase it threefold.

Two different and reasonable approaches to roadway safety are used in the U.S. One increases sight distances, which then leads to increased design speed. On average, 15 percent of drivers exceed the intended speeds limits. The severity of accidents increase with speed. The other approach matches design speeds with existing sight distances. Often referred to as traffic calming, this approach trades off design speed for greater land-use amenity and decreased severity of accidents.

each). The other 80 percent get to the shoulder (estimated delay of 45–60 minutes and 500–1,000 vehicle-hours each). Other delay is mostly from construction.

Since accidents contribute so heavily to congestion delay, they are worth understanding better. Again, Downs (2004a) does a great job of summarizing the literature and drawing conclusions. He shows accidents rates are down, and that reduction is correlated with lower alcohol use, anti-crash restraints, more law enforcement, better road design, and, ironically, increased congestion (the number of accidents probably increases, but the severity drops).

The second factor that breaks the analogy is a different contributor to variability: human choice. How does one make all those independent drivers adhere to the production function? Time for the economists.

THE ECONOMICS OF TRAFFIC CONGESTION

Water molecules lack a system view. They are compelled by the laws of physics to choose the fastest immediate path, even though that path may lead to an eddy that causes them an extended delay in their journeys downstream. The analogy to some extent applies to drivers: though they have the capacity to look ahead, many do not. But experienced drivers, consciously or unconsciously, consider many possible routes to a destination and try to select the best, given their information about roadway conditions. If one part of the system is experiencing a lot of pressure, they choose an alternative. As congestion builds, some drivers select less direct but quicker routes.

Molecules of water do not make conscious choices as drivers do: they don't listen to radio reports to decide which route to take or eddy out to let the log jam clear. In this section, we use the tools of economics to understand the decisions of drivers, even if drivers do not. This section puts the basic microeconomic principles from Appendix B into the context of transportation:

- Drivers' decisions establish an equilibrium in the transportation system in which, during peak periods, the expected time and operating costs are roughly equal across alternative routes.
- This peak-period equilibrium is inefficient in the absence of some constraint on the numbers of cars entering the system (i.e., market failure occurs). The total net benefits from the system (i.e., economic surplus) would increase, potentially by quite a bit, if trips of low value could be discouraged during peak periods.
- The inefficiency results because each individual driver experiences less than the full social cost of his or her trip. Highway capacity is a scarce resource during peak travel periods: an additional driver slows all other drivers. The perceived price of a trip to each individual driver is less than the full cost of that trip to society.
- What may seem an obvious solution to congestion—reducing scarcity by increasing capacity—often fails to work in the long run. Expanding a road reduces travel time, which encourages more driving, and the congestion that follows.
- An effective method for treating overcongestion is to charge a price for the use of scarce roadway space. The optimal road toll varies with demand and consists primarily of the dollar value of the time cost an additional car imposes on all other travelers when it enters the highway.
- A congestion toll not only discourages marginal trips; it also signals the need for capacity expansion. Tolls are high on high-value routes. Revenues from peak-period tolls cover the cost of providing additional capacity on those routes.
- A key cause of currently overcongested routes, therefore, is the practice of paying for roads with a tax on fuel.

Factors That Affect Travel Choices

Drivers make travel decisions every day about where to go, when to go, how to go, and by which routes to go. Economists begin by assuming these decisions are made by people who behave rationally: they competently act to maximize their personal satisfaction by applying the marginal principle (Appendix B) given the information available.

Those assumptions are not perfect, but they are not unsupportable: most travelers do think about the costs and benefits of travel choices; in other words, they make what they think is the best trip for them. Even if they are not making precise computations, they generally behave as if they were. When thinking about travel costs, people consider the out-of-pocket costs of fuel, upkeep, or transit fare, and the time cost of travel. They also consider difficult-to-quantify costs and benefits, such as the

comfort of a seat on the bus or the prestige or opprobrium of humming down the passing lane. These costs and benefits are difficult to put dollar values on, and many are subjective. One commuter may value time spent driving higher than another commuter; one traveler sees walking as a cost, while another sees it as a benefit.

Travel choice also affects choices about where to live and where to work. People weigh the larger lots and more desirable schools and other amenities in the suburbs against the longer commutes they have; they could choose smaller dwellings and yards and a different mix of urban amenities in the city to cut those commutes. Some people consider their transit choices before deciding where to live. If someone does not want to drive to work, he locates near a transit line or within walking distance from his workplace. Transportation choices have long-run impacts on location decisions, and, conversely, location decisions have short-run impacts on transportation choices.

For present purposes, assume choices about home and work location and about vehicle ownership are made. On a day-to-day basis, people still have choices about how to get to work, school, or shopping. They have choices to make about the mode of travel (car, car pool, transit, bike, walking), the route, and the time. How do people make that choice? Back to microeconomics: they consider the costs and benefits of their alternatives.

At first blush, the categories of benefits and costs seem pretty clear: get there as quickly and as cheaply as possible. Quickness is easy to measure: door-to-door travel time. Costs seem easy as well, but things get tricky in a hurry. When driving, fuel is a cost ($[\text{miles} \div \text{miles per gallon}] * [\text{fuel price per gallon}]$). Don't forget oil and tire wear. And what about periodic maintenance, car insurance, and the cost and depreciation of the vehicle itself? The American Automobile Association estimates that, in 2006, the costs of gas, oil, tires, and maintenance sum to about 15 cents per mile, on average, and total out-of-pocket costs, which include vehicle depreciation, insurance, and licensing sum to about 52 cents per mile for a typical car driven 15,000 miles per year.⁸

But neither of these numbers really captures all the things people consider when deciding whether to make a particular trip. The cost and benefit calculation gets more complicated when one considers the time cost of travel and other nonmonetary attributes of the travel. Table C-1 gives a feel for the variety of factors that influence how many trips a person takes and by what mode. Service attributes and time costs can easily be as important as out-of-pocket costs.

One additional critical distinction exists when thinking about costs: the one between *fixed costs* and *variable costs*. Fixed costs do not vary with the decision at hand. For example, a particular trip does not affect insurance costs, license costs, or the portion of vehicle depreciation that comes with age rather than miles. These costs are all "sunk" when it comes to deciding whether to take a particular trip and therefore have, or at least should have, no impact on decision making.

One can find estimates of "the full cost" of auto travel (including the depreciated annual value of auto ownership and insurance) coupled with a discussion of daily trip making. These full-cost estimates are relevant to long-run decisions about where to live and work, and whether to own a car; they are not particularly relevant to daily decisions about mode choice or number and length of trips.

Relevant to these day-to-day decisions are variable costs. Most travelers consider the time the trip is likely to take, and the cost of fuel and parking (if they have to pay for parking that day, but not if they pay on a monthly basis for an "all-you-can-park" pass). They may also be aware of the marginal costs of oil, tire wear, and mileage related maintenance and depreciation. And most travelers consider comfort, safety, and reliability, though these costs are more difficult to pin a number to.

Sometimes, analysts are not careful about these short-run/long-run, fixed-cost/variable-cost, monetary/nonmonetary differences. To see this, consider the choice of whether to take a trip from home to a mall five miles away. According to the estimates of the American Automobile Association, the full amortized cost to the individual of the 10-mile round trip is about \$5.20. Not a fortune, but enough to give many people

TABLE C-1. FACTORS THAT AFFECT THE AMOUNT OF TRAVEL AND MODAL CHOICE

	AUTOMOBILE	TRANSIT	CYCLING AND WALKING
Out-of-pocket costs	Fuel and maintenance Roadway tolls Parking fees	Transit fare	Bike maintenance
Time costs	Driving Parking search Walking	Walking Waiting Riding	Riding Walking
Service attributes	Effort Safety Comfort Reliability Flexibility	Effort Safety Comfort Reliability Flexibility	Effort Safety Comfort Reliability Flexibility
Other factors	Environmental concerns Health	Environmental Health	Environmental Health

Source: ECONorthwest

pause. But much of this cost is sunk; a better number to use for the out-of-pocket cost of this trip would be 15 cents per mile, or \$1.50 for the round trip. As a rough approximation, drivers may be paying attention to less than one-third of their out-of-pocket costs when they make a trip: their *perceived* marginal cost is significantly less than their full average cost. That is a key point for some of the analysis that follows.

But that \$1.50 is nearly \$1.50 more than the out-of-pocket expense of a trip by bicycle. The problem is that the bike trip usually takes more time, certainly involves more effort, and often involves more risk of injury. It doesn't take much concern for these nonmonetary factors to outweigh the savings of \$1.50. The walking and waiting associated with transit can have the same effect.

In addition to the multiple categories of costs and benefits inherent in the travel itself, travelers must consider the multiple benefits of being at the places they want to be when they want to be there. Consequently, they construct complex itineraries (shopping after work on Monday; picking up Cody after lacrosse on Tuesday, Wednesday, and Friday; aren't they re-surfacing Main Street this week? And so on).

Rational travelers make trips until the perceived marginal benefit from the trip equals its perceived marginal cost (where costs and benefits are thought of broadly as in Table C-1). Some decisions are easy. Drivers decide to make the trips that have high net benefits—because the benefits are high (a trip to the dentist to fix an aching tooth), the costs are low (walking half a block to pick up milk at the local market), or both. Having made their priority trips, drivers then decide whether to make trips that have lower benefits or higher costs or both. As long as the marginal benefit exceeds the marginal cost, more travel will bring additional net benefit to the traveler.

As they make trips of lower and lower priority, the additional net benefits decrease. When marginal travel benefit equals marginal travel cost, every trip up to the last has generated net value, and every trip after the last will be made at a loss. It is time to stop traveling (or, as in the movie example of Appendix B, stop making sequels).

Though people do not dress their decisions up in the dry, bookish language of economics, they do make reasonably good decisions about when a trip is not worth the effort.

Travel Budget

In a day of 24 hours, people must sleep, eat, work, and engage in other required and elective activities that take time. That time is not available for trip making. Some boundaries surround the amount of time people are willing to spend traveling in a

day. One can think of the amount of time that people are willing to spend on travel as their travel budget. That budget obviously varies by person, but the great majority of working adults probably have a travel budget of 30 to 150 minutes per weekday (most probably from 40 to 90 minutes per day). Downs (2004a, 274) cites an international study of cities and villages in Europe, Asia, Latin America, and Africa that found a consistent average of just under 1.0 to just over 1.5 hours per day on travel. He says “these are remarkably similar results considering the variety of income levels, travel modes, cultures, ethnic groups, and stages of economic development.”

The concept of a travel budget helps explain travel behavior. For example, suppose a commuter now spends two hours each day for the round trip from work to home. Suppose that her job and the capacities of computers and telecommunications are such that she determines telecommuting can work. She now has to go to the office only two days per week; she has cut six hours out of trip making.

Suppose she continues to include those six hours in her travel budget. She may make other trips to allow new activities. If telecommuting works and her job looks stable, she may move farther out to enjoy a rural lifestyle, commute three hours round trip, and still have time for other activities. The concept of a travel budget implies that people are rather rigid (in the economic jargon that we introduce at the end of the appendix, their demand for travel is relatively *inelastic*) when allocating time to different types of activities. If people are this rigid, improvements to the transportation system that reduce trip times result in people increasing the number of trips because they see “travel” as something they’ve already budgeted: if I spent only six of my budgeted 10 hours, I can afford to spend four more hours on travel.

This suggests the limitations of the travel-budget concept. The fact that a person spends 10 hours a week commuting now does not mean that he will spend 10 hours under a wide range of circumstances relating to congestion, location, and job characteristics. People make trade-offs. In the short run, with home and job locations fixed, they expand or contract their travel budgets as necessary to get to and from work. They may take actions to reduce travel time (e.g., traveling earlier to avoid peak traffic). In the longer run, when travel time gets insupportable, they can change the location of their home or work.

These considerations lead to the second limitation of the concept of travel budget when one tries to apply it in detail: namely, travel budgets are widely distributed among trip makers. Some people spend less than 30 minutes a day on all trips; some people spend more than four hours. The distribution of average weekday travel budgets among trip makers probably looks like a normal distribution with a mean somewhere between 30 and 60 minutes with a long, thin tail to the right (created by a few people who spend up to four or even five hours on trip making).⁹ The reason for that distribution: (1) people face different travel times to the locations of their required and optional activities; (2) they place different values on travel time; and (3) they are free to make trade-offs between the time spent on travel and the time spent on other activities.

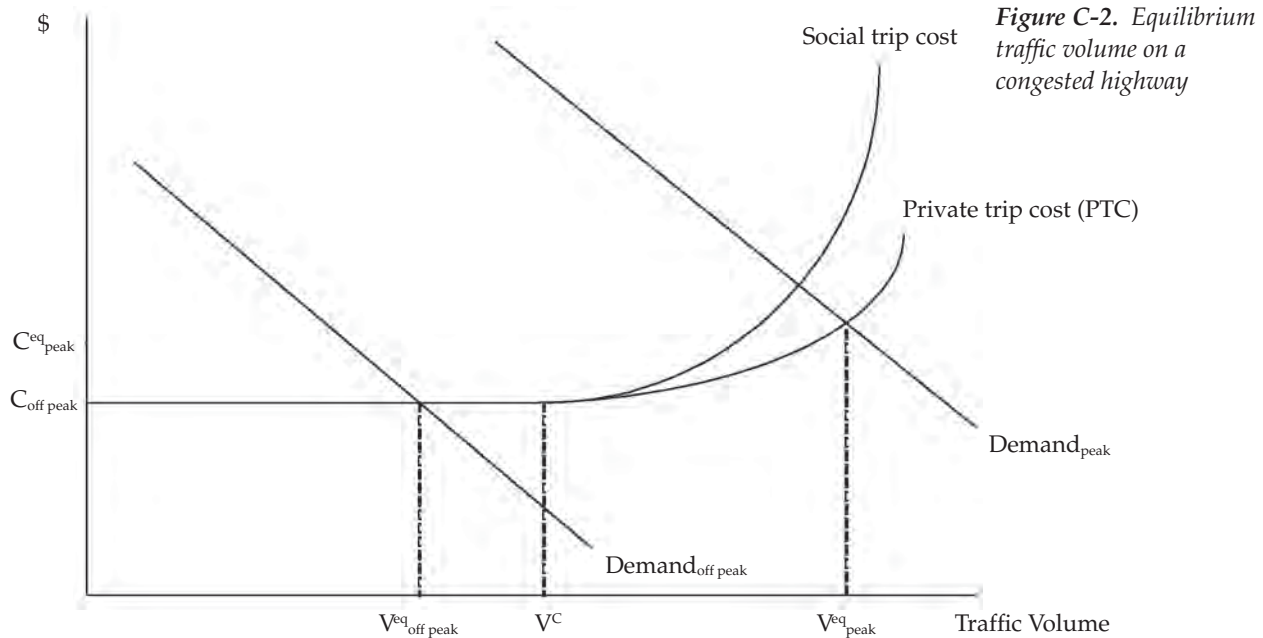
Thus, the concept of a travel budget provides a broad concept for thinking about travel but limited insight into predicting individual responses to changes in the factors that contribute to the cost of a trip.

Explaining Congestion with Demand and Supply Curves

From a planning perspective, individual choices, while foundational to the workings of the transportation system, offer little to guide public policy. For a more practical and useful picture, we zoom out and apply the marginal principle to a larger unit of analysis: a group of drivers on a roadway that suffers peak-hour congestion.

The basics

Figure C-2 shows the standard economic diagram of the costs and benefits from travel on a highway segment that suffers peak-period congestion. The vertical axis measures prices, costs, and benefits in dollars. The horizontal axis measures the



Source: Standard diagram rendered by ECONorthwest

quantity of traffic on the highway segment in some suitable metric (e.g., cars per lane). We describe each of the four curves in the diagram in turn.

We begin with the curve labeled *private trip cost*. The horizontal portion of the curve on the left corresponds to uncongested travel. At low traffic volumes, all cars travel at the speed limit (or higher). That point is consistent with the speed/flow relationships shown in Figure C-1. The cost of their trip, labeled $C_{\text{off-peak}}$, is the sum of out-of-pocket and time costs. Out-of-pocket costs include fuel and mileage-related maintenance and vehicle depreciation. The time cost depends on travel speed.

PTC rises as the highway gets congested. Cars begin to slow down to maintain safe travel distances at traffic volume V^c . Slower travel translates to higher time costs; the PTC curve begins to slope upward. It gets steeper as more cars enter the already-congested roadway. This curve gets its shape from the speed/flow diagram in Figure C-1. In fact, the PTC curve is simply the speed/flow curve inverted because lower speeds translate directly into higher time costs.

Next look at the *demand* curves. The left-hand curve shows the quantity of trips demanded at each “price” during some *off-peak* hour, 10:00 to 11:00 a.m., say. The right-hand demand curve shows the quantity demanded at each price during the *peak* period.

The demand curves slope downward, as usual: quantity demanded rises as price falls. Why?

- *Substitution*: As the cost of a trip increases, drivers change their behavior. As congestion increases the time cost of travel or higher fuel prices increase out-of-pocket expense, some drivers choose not to drive alone and instead ride public transit, telecommute, shift their working hours, or carpool.
- *Income constraints*: As the price of a trip increases, some drivers can no longer afford to make them. Some cut back on the number of trips they take; others simply give up car travel. As congestion increases the time cost of driving, fewer people can afford to sit in traffic.

The demand curves show the marginal benefit of travel—the monetary equivalent of the value or benefit drivers derive from a trip on this highway segment at this time, all else constant. The demand curve implicitly arranges trips from high to low benefit. The trips on the left side of the graph provide high benefit—priority trips with few or

no substitutes. As the time or out-of-pocket expense of the trip declines, the quantity of total travel increases. The additional trips are lower priority trips—ones with more substitutes (e.g., go at a different time, or by a different mode) and ones previously too expensive for many drivers. The travel demand curves are in concept no different from the demand curves for pizza slices described in Appendix B.

But the PTC curve *does* differ from the supply-of-pizzas curve described in Appendix B. The supply-of-pizzas curve orders pizzas from low production cost to high production cost, just as the pizza demand curve orders pizzas consumed from high value to low value. The ordinary supply curve is thus a marginal-cost curve, just as the ordinary demand curve is a marginal benefit curve. One can do this ordering on both the supply and demand curves because the suppliers and demanders differ from one another. Most pizza consumers are not pizza suppliers, and vice versa.

That is not the case in Figure C-2. In drawing the demand curves, we assumed that the benefit a traveler derives from the trip varies across travelers. So we arranged travelers left to right in decreasing order of benefit. But the travelers are also the “suppliers” of trips, and the cost of the trips varies across travelers. Some drive fuel-efficient cars, others don’t. Some value their time highly, others less so. We could order travelers from low to high cost, as we usually do with supply curves. But we cannot order travelers *both* from high to low benefit *and* from low to high cost unless benefit and cost are perfectly negatively correlated—an extremely unlikely circumstance. This problem arises because the travelers are both suppliers and demanders in this analysis.

The solution to this problem—one that allows us to continue with the exposition—is to assume the cost of all trips for all travelers is equal, but the benefits of each trip vary for each traveler.

The PTC curve differs in another way from an ordinary supply curve. An ordinary supply curve is a *marginal-cost* curve, but the PTC curve is an *average-cost* curve. Some pizza sellers enjoy lower production costs than other pizza sellers; that some suppliers suffer higher costs does not mean all suppliers suffer higher costs. This is not the case on the highway; if congestion slows anyone down, it slows everyone down. At $V^{\text{eq}}_{\text{peak}}$ in Figure C-2, for example, all drivers endure total trip cost of $C^{\text{eq}}_{\text{peak}}$.

The supply (i.e., marginal-cost) curve in Figure C-2 is the curve labeled *social trip cost*. When the highway flows freely at low traffic volumes, social costs equal private costs: the social and private cost curves coincide. But social costs rise faster than private costs as congestion builds because an additional car entering the highway slows everyone down a little, which causes everyone’s PTC to increase. The social cost of that one additional trip (i.e., the *marginal* cost of the trip) is the PTC of that single trip *plus* the value of the additional time everyone else has to spend on the highway when the additional car enters and slows everyone down. The social trip cost curve is the marginal-cost curve—it shows the social cost of each additional trip on the highway segment.

Equilibrium versus optimal traffic volumes

Figure C-2 provides cost and benefit curves for travel on a highway segment that experiences peak-period congestion. We can use the curves to identify *equilibrium* traffic volumes, and then *evaluate* that equilibrium.

Drivers will enter the highway as long as the benefit they receive from the trip, as shown by the demand curve, exceeds the cost they experience. The cost each driver experiences is shown by the PTC curve. Thus, during peak demand, the equilibrium traffic volume is $V^{\text{eq}}_{\text{peak}}$. The market “price” of that trip is $C^{\text{eq}}_{\text{peak}}$. This price is paid in the form of expenditure on fuel, maintenance, and vehicle depreciation plus the time spent traversing the highway segment.

Is this equilibrium traffic volume economically efficient? No! The marginal cost of a trip is shown by the social trip cost curve: at $V^{\text{eq}}_{\text{peak}}$, marginal social cost exceeds marginal benefit. The inefficiency results because each driver entering the congested highway fails to consider the effect his trip has on all the other drivers. The last

(i.e., marginal) driver is nearly indifferent about making the trip—she wouldn't make the trip if she had to consider the additional time costs her trip imposes on other drivers.

The usual way economists look at this problem is as an *externality*. Each driver entering a congested highway imposes external costs on the other drivers: he slows everyone else down. The additional time cost might not be much per driver, but it adds up if there are a lot of drivers.

Appendix B explained that externalities often result from unpriced access to a scarce resource. What is the scarce resource in this context? Answer: the highway itself. The problem is that space on the highway becomes scarce at high traffic volumes; not everyone can enter the highway without imposing costs on others. In the absence of a charge that reflects the cost of using the scarce resource, the resource becomes “exploited,” which in this case means overly congested.¹⁰

Appendix B also explained that the absence of a price often results from a lack of property rights. If no one owns a scarce resource, no one has the incentive to look after it or has the ability to charge for its use. This doesn't seem to apply in the case of highways. State and local transportation agencies “look after” highways on behalf of taxpayers. These agencies effectively own the property rights to highways, and they do charge for the use of the highway system, collecting taxes on fuel and tires and on the weight of trucks.

The problem with these charges, however, is that they do not vary with demand for the use of the scarce resource (i.e., with the extent of the scarcity). These taxes are a form of what economists refer to as *average-cost pricing*.

The source of inefficient levels of congestion is the practice of financing highway infrastructure through charges that do not vary with demand for the use of the highway infrastructure. In particular, the charges miss two key components of demand: people want to be in a certain *place* at a certain *time*. “Too much of something in the same place at the same time” is the fundamental definition of congestion. This important idea warrants further discussion.

Congestion tolls versus the fuel tax

Compare the peak-period equilibrium in Figure C-2 with the off-peak equilibrium. In peak-period equilibrium, the costs generated by the marginal trip exceed the benefit it generates. The reason for this inefficiency is that the marginal driver fails to internalize the cost his trip imposes on other drivers in the form of longer travel times. The market inefficiency results not because of external costs but because they are unpriced. The obvious solution (at least to economists) is to price peak-period use of the highway.

At off-peak times, in contrast, the external time costs are zero in the diagram (private costs coincide with social costs, which, by definition, means there are no external costs) and are very low in practice. So the equilibrium traffic volume, $V_{\text{off-peak}}^{\text{eq}}$, is economically efficient. This suggests (perhaps surprisingly for some readers) that drivers should pay nothing for the use of a rather expensive highway during uncongested periods.

How can this make sense? The answer is that most of the highway capacity is there to accommodate peak users; the capacity of the highway is determined by peak-period demand. That capacity is excessive during off-peak periods. Why should off-peak users have to pay a full share for a highway they as a group use relatively little of? Answer: they shouldn't.

This reasoning is applied routinely in the private sector.¹¹ Ever wonder why the price of lunch at many restaurants is so much lower than the price of dinner at the same restaurant, even though there's little difference in the meals? There are two reasons: (1) demand is lower at lunch (off peak) than it is at dinner (peak); and (2) the restaurant owner recognizes that the costs of the restaurant itself, the fixed costs, are sunk costs at lunch hour.

Put yourself in the restaurant owner's position. Suppose that the owner estimates the weekly total cost of his fixed inputs (i.e., the building and equipment, insurance,

the value of the owner's time and financial investment, etc.) and divides this by the total number of meals he expects to serve per week. Suppose this comes to \$5 per meal. Then he computes his variable costs (labor, materials, energy, etc.), which come to \$7 per meal. So the average total cost of a meal (lunch or dinner) is \$12.

Suppose he charges \$12 per meal regardless of time of day or time of week; that is, he charges average cost regardless of demand. What happens? Dinner (i.e., peak) crowds are great, in fact they're too big, but lunches are slow—very few people want to pay \$12 for lunch. The natural inclination would be to reduce lunch prices and raise dinner prices. This makes sense to an owner who understands that the capacity of the restaurant is chosen on the basis of dinner traffic. Any lunch price over \$7 contributes to profit, given that the restaurant itself is paid for by the dinner crowd. Dinner prices have to rise above \$12 to cover fixed costs, which reduces dinner sales.

But the net effect is positive. We can modify Figure C-2 to show the gains from marginal-cost (i.e., peak-period) versus average-cost pricing. A highway is conceptually identical to the fixed restaurant capacity. Figure C-3 differs from Figure C-2 in that it distinguishes three PTC curves (the other curves are the same as in Figure C-2).

Consider each PTC curve in turn. The lowest PTC curve shows PTC as a function of traffic volume assuming no taxes: PTC is simply the sum of fuel and time costs assuming no tax on fuel (or tires, etc). The intermediate PTC curve adds the cost of a tax on fuel (or of a road toll that does not vary by time of day). It corresponds to the PTC curve that drivers currently face on most highways. The highest PTC curve substitutes an optimal congestion toll for the fuel tax. The optimal congestion toll equals the external time cost imposed on other drivers by the marginal driver. The toll internalizes this external cost.

The fuel tax represents average-cost pricing. The fuel tax spreads over everyone who drives, regardless of time of day, the cost of the highway, or more precisely, of the highway system. Given that each driver pays this fuel tax, the equilibrium *peak-hour* traffic volume on this highway segment is $V^{\text{eq}}_{\text{peak}}$. As before, this traffic volume is inefficiently high, and travel speeds inefficiently low, because the fuel tax is lower than the external cost imposed by the marginal driver: the social cost of the marginal trip exceeds its benefit to the traveler (i.e., $MC > MB$ at $V^{\text{eq}}_{\text{peak}}$).

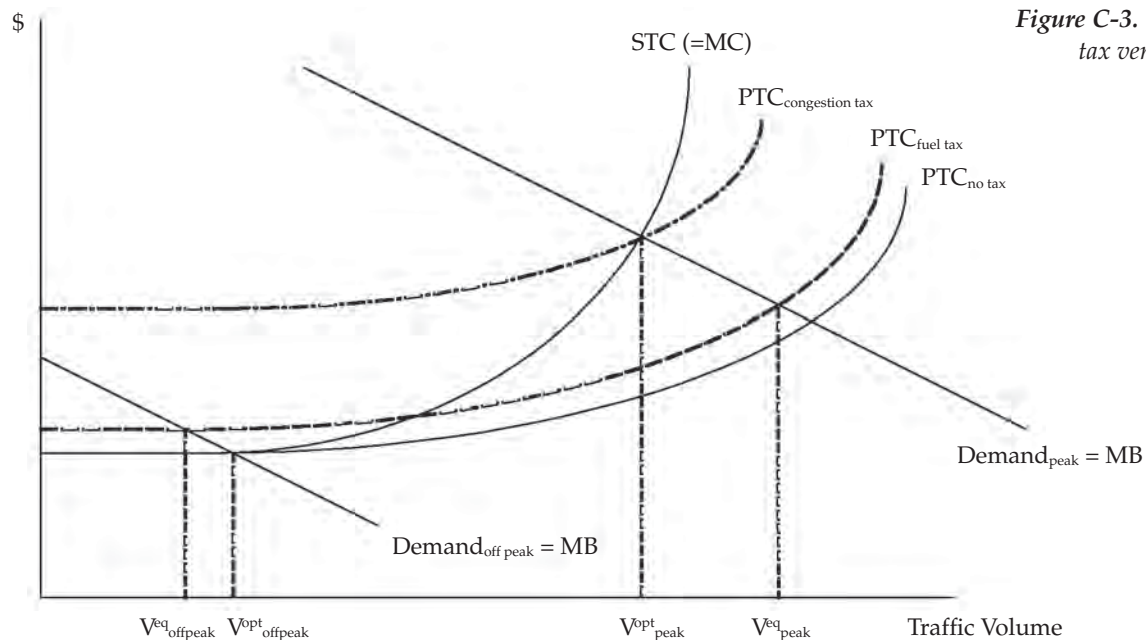


Figure C-3. Congestion tax versus fuel tax

Source: Standard diagram rendered by ECONorthwest

The opposite is the case during off-peak hours. Equilibrium *off-peak* traffic volume is $V^{\text{eq}}_{\text{off peak}}$. At this traffic volume, marginal benefit is greater than marginal cost; a driver has to pay the tax even though she imposes no time cost on other drivers. The highway capacity is to serve peak users, so its cost is sunk off peak, and external time costs are small or nonexistent when the highway is not congested.

The congestion tax represents marginal-cost pricing. The toll is set so that PTC equals marginal social cost. During peak periods, the congestion toll will be higher than the fuel tax, so equilibrium travel volumes will be lower and travel speeds higher. Correspondingly, the congestion toll will be lower than the fuel tax during off-peak periods, so equilibrium travel volumes will be higher off peak.

The revenues from the congestion toll may not differ much from those taken in by the fuel tax. But the size and timing of the toll has valuable and potentially large effects on travel speeds and volumes. The higher congestion toll during peak periods discourages marginal trips, so travel speeds increase and the chance of an incident decreases. Travel costs decrease during off-peak periods, so travel at these times increases, allowing more intensive use of expensive infrastructure. And, importantly, revenues flow from highway segments instead of from gasoline retailers.

The revenue flows from a well-designed congestion toll provide more accurate signals about the value of capacity improvements than does congestion itself. Return to Figure C-3. Suppose that two similar highways experience similar levels of congestion under the current average-cost pricing (traffic volume equals $V^{\text{eq}}_{\text{peak}}$ on both highways). But suppose that the demand curve is flatter on one highway than on the other. The flatter demand curve means that more of the marginal trips are of low net value; a small toll will encourage relatively many drivers to get off the highway. Smaller revenues signal no need for new capacity, *even though current congestion levels are high*. Accurate signals are extremely important for making good infrastructure investments.

The congestion toll is also an important signal for highway users. Figures C-2 and C-3 provide strong support for an uncontroversial proposition: equilibrium traffic congestion is too high under the current average-cost pricing system. Some of these drivers should not be on the highway during peak periods. But how do we identify which ones to discourage? The toll does that job for us: it discourages those drivers who place relatively low value on the trip.¹²

Capacity Expansion and Induced Demand

The previous analysis makes it clear that *how* we pay for transportation facilities can have a big impact on how and when they get used (i.e., on transportation system performance). It should also make clear that just because a road is congested, the appropriate policy response may *not* be to build more capacity. Average-cost pricing is the big cause of overly congested highways, usually not undercapacity. The gas tax helps little to discourage the marginal (i.e., low-value) trips that cause congestion.

In this section, we build on Figure C-2 to analyze the effect of expanding capacity on a congested highway in the absence of congestion pricing. The analysis takes much of the mystery out of the concept of *induced demand*, the tendency for new capacity to encourage additional trips until congestion is restored.¹³ The primary lesson of this section is that a policy to build one's way out of congestion is misguided inasmuch as it denies the reality of resource scarcity and is pernicious inasmuch as it magnifies the inefficiency caused by average-cost pricing.

Figure C-4 extends the analysis of Figure C-2 to consider the effects of capacity expansion in the short run (weeks and months) to those in the long run (years). The curves subscripted with a "0" reflect the original conditions on the highway. Peak demand is high, and the highway is congested. The curves subscripted with a "1" reflect conditions after the capacity expansion. In particular, congestion builds, and drivers have to slow down only at higher traffic volumes.

In the *short term*, the new capacity provides benefits of congestion relief. The locations of households and businesses are fixed in the short run. Drivers whose relatively

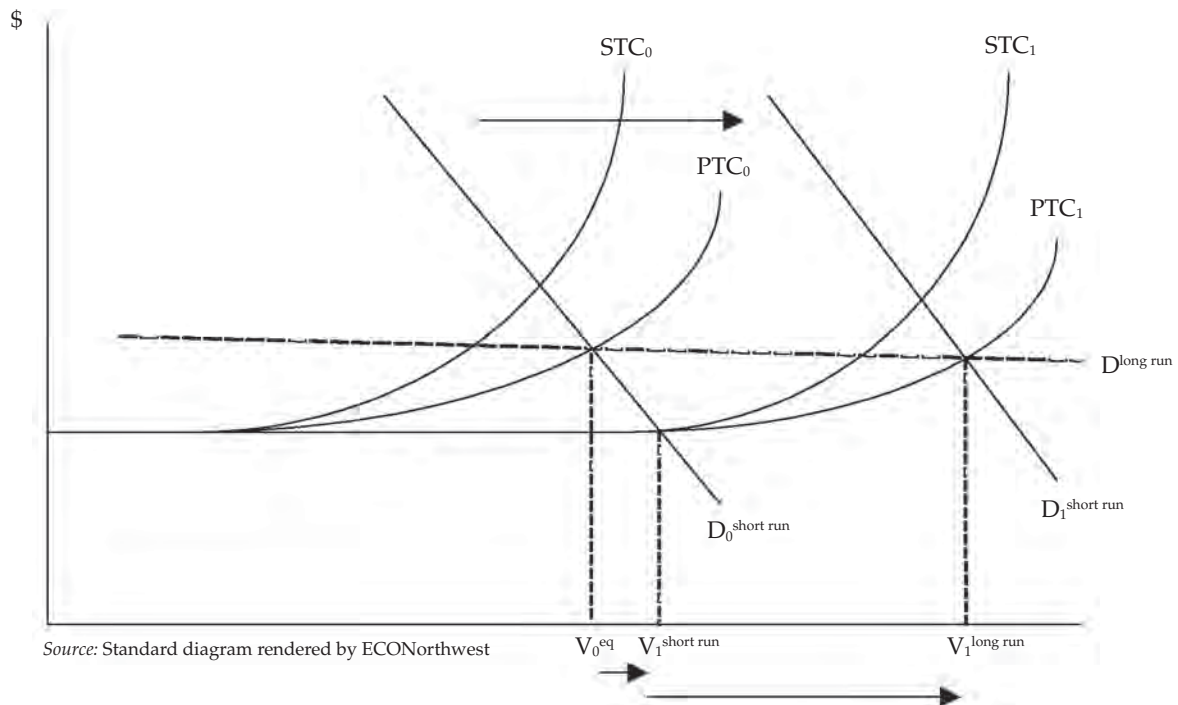


Figure C-4. Short-run versus long-run effects of capacity expansion

low marginal benefits kept them from driving before the expansion because of the relatively high costs now see that costs have decreased. They are “induced” to make the trip they previously did not make or made on another mode, on another route, or at another time. Downs (2004a) refers to this as a *triple convergence*: new trips converge on the new capacity from other routes, times, and modes. Like water, the traffic flow soon finds its equilibrium level, where travelers see no big benefits in further changes. But the effects on the new capacity are relatively small—trip volumes rise from V_0^{eq} to $V_1^{short\ run}$ —and congestion and trip times diminish.

The *long-term* effects are primarily driven by shifts in land use. Cervero (2003), for example, reports that faster travel times encourage development along newly expanded highway corridors, which consequently generates more traffic and reduces the initial congestion relief of the added capacity. That finding is consistent with all of the theory of urban development we have described in this book: the theory is so fundamental that we would be surprised by and suspicious of any empirical work with contrary findings. Both households and businesses take advantage of the new capacity to move to lower-price land farther from developed centers, and the businesses that serve them follow.

The short-run demand curve shifts as these relocations occur. This shift is shown as that from $D_0^{short\ run}$ to $D_1^{short\ run}$ in Figure C-4. The increase in demand builds congestion on the expanded roadway. The slower travel speeds on the congested highway discourage further business and household relocations that shift the short-run demand curve; the new capacity has reached its equilibrium in the larger metropolitan transportation system. As drawn, equilibrium travel times are only a little bit faster than they were before the capacity increase.

Does this restoration of congestion eliminate the benefits of the new capacity? No. Commuters who used the highway before the highway expansion get relief in the short run. But that relief gets eroded over time by new trips. The conclusion that they and others draw is the highway expansion provided no benefit: trips for them take just about as long as before the expansion. What they miss is that a lot more people are taking trips: those who did not use the highway in its original congested state now do. These trips have value to those people, so travel benefits have increased.

The critical question is whether those additional benefits were worth their cost. That question is very difficult to answer in the absence of congestion pricing.

Does the new capacity generate new economic development? The answer depends on the boundaries of the inquiry. Several studies show that the added road capacity and faster travel times encourage shifts in the locations of housing, retail, and employment opportunities from one part of a community to another (Boarnet 1995, 1997; Forkenbrock 1990). Only part of the new development represents net growth to the city. Not surprisingly, the percentage effect on vehicle miles traveled (VMT) of a 10 percent increase in metropolitanwide capacity is less than the percentage effect on VMT of a 10 percent increase in capacity on a single highway link; the new capacity on the single highway link attracts relocations from other parts of the city.

Taxes on Fuel, VMT, and Axle Weight

The analysis above leaves little doubt about our view of the fuel tax as a mechanism to expand the capacity of congested highways. Congestion is fundamentally a pricing problem; too many drivers enter a congested highway because those on the margin fail to consider the time costs they impose on other drivers. A congestion toll addresses this problem directly—it internalizes the external cost, and it provides funds for cost-effective capacity expansion. A fuel tax falls short.

Is there a role for the fuel tax? The answers are (1) “yes” on political grounds because the gas tax is the bedrock of transportation finance in the U.S. (but we are not making political arguments in this section), and (2) a qualified “yes” on efficiency grounds because burning gasoline or other fossil fuels generates air pollution. Just as drivers generally fail to consider the time costs they impose on the other drivers on a congested highway, they also tend not to think much about the costs of their contribution to smog or climate change. A fuel tax internalizes this cost and provides revenues for its treatment.

The answer is a qualified one because the emissions generated from burning a gallon of gasoline vary with the characteristics of the gasoline and the car. Some states require cleaner fuels than others; some engines more thoroughly burn fuels than others. Ideally, one would tax the emissions that each car generates. But taxing fuel might be more cost-effective.

A fuel tax can be used sensibly to fund construction and maintenance of uncongested highways. The majority of roadway mileage around the country does not experience significant congestion; one or two lanes is enough almost all of the time. In principle, tolls that do not vary by time of day would be the most direct method of paying for these roads. But a fuel tax might in practice be a more cost-effective way to raise the funds.

Some states and municipalities are considering a tax on VMT. The argument is that some auto externalities depend mainly on the number of miles the car is driven. For example, the chance that the car is involved in an accident or the visual disamenity cars create vary with how much each car is driven. A VMT tax does not, however, work well as a congestion toll.

Many states charge heavy trucks for use of the highway system. Heavy trucks do far more damage to road surfaces than do cars. And the damage depends on axle weight and often on air temperature. Fees that vary with axle weight encourage freight companies to distribute weight over potentially many axles. And if the fee increases with temperature, freight companies would avoid putting their trucks on the road when pavements are especially vulnerable. The revenues from such fees would be used to contribute to the funding for roadway resurfacing.

Other transportation-related externalities, such as those from noise and disposal of vehicles, clearly need to be considered as well. The point here is that externality taxes work best when targeted rather precisely at the externality-generating activity. A congestion toll works when applied to “fixing” congested highways because it targets congestion.

Travel Demand Elasticities

Table C-1 shows factors that influence the demand for travel by various modes. To the extent that public policies are going to be successful in influencing travelers and travel, they must work through their influence on these factors. This raises a critical question for policy analysis: How sensitive are travelers to changes in these factors?

A focus of quantitative research over the past several decades has been on obtaining reasonable estimates of various *elasticities* of the demand for transportation. Elasticity, recall, is the percent change in one variable (e.g., VMT, transit ridership) that results from a 1 percent change in one of the factors that influences that variable, holding all other influencing factors constant.¹⁴ Small elasticities (near 0 in absolute value) mean people are relatively unresponsive to the change in the factor, while elasticities around 1.0 in absolute value indicate responsiveness.

Travel-demand elasticities are helpful in evaluating the changes in policy we describe in Chapters 5 and 6. We begin by briefly describing some of the most important problems researchers face when estimating elasticities. We then report elasticities that measure the effects on car travel and transit ridership of changes in the various components of the cost of a car trip. We are not aware of any efforts to measure the effects of changes in prices on walking and bicycling.

Problems in estimating elasticities

The elasticity estimates we report were obtained using methods designed to sort out statistically the effect of changes in each variable as if all other variables were held constant (i.e., multivariate regression analysis). In other words, the researchers used statistical methods in an attempt to estimate the effect that would be observed in a controlled experiment.

An example shows the importance of this point. Suppose that a transit agency raises fare by 10 percent while simultaneously decreasing the time spent waiting for the bus by 10 percent (e.g., by increasing the number of buses on the route by 10 percent). The agency observes that ridership increases after raising the fare and concludes that out-of-pocket expense has little effect on transit riders. A more reasonable explanation is that the elasticity of ridership with respect to time spent waiting exceeds the elasticity of ridership with respect to fare. In general, many important factors vary between observations of behavior across both time and space. In lieu of controlled experiments, statistical regression analysis offers researchers the best way to isolate the effect of changes in individual factors.

To untangle the various effects at work in urban transportation and land markets, researchers require sophisticated empirical models, large quantities of accurately measured data, and a great deal of luck and insight. Researchers are gradually increasing the sophistication of the empirical models, helped by better understanding of the interactions in an urban economy and the increase in the power of computers. Changes in methods have led to changes in estimates of important elasticities. The problem: as with any scientific endeavor, the starting point—good and abundant data—takes time and money to collect.

In fact, it may require more than funding to collect the kind of data needed for accurate analyses. Since the 1950s, policy in the U.S. and Europe (the places from which most data are collected) has been directed toward decreasing the price of travel by auto. The roadway and building designs that improve auto travel often increase the cost of using other modes. The effect of changing a single transit policy (e.g., lowering a fare) in an environment dominated by designs to ease auto travel probably differs significantly from the effect of a similar change in an environment more supportive of transit.

This lack of impact from a change to a single transit policy suggests that changing a number of complementary transportation policies may cumulatively exceed the sum of the effects from changing one policy at a time. Put another way, estimates of specific elasticities (e.g., the price elasticity of demand for auto trips) could change substantially if other policies changed simultaneously in a reinforcing way (e.g., if the quality of

transit service was much higher than it had been). Unfortunately, the general thrust of policy is quite uniform throughout the developed world (though some differences do, of course, exist), which inhibits exploration of these synergistic relationships.

Thus, we were not surprised to find that many interesting elasticities remain without estimates, and sometimes the elasticities that have been estimated are the subjects of great disagreement. The results of empirical studies should be interpreted with care. Some limitations of even high-quality research:

- The reported estimates are averages; the elasticity applicable to a particular project may differ considerably from the point estimate reported.
- Elasticities assume other things equal—they usually aren't, and controlling effectively for the influence of other factors is difficult.
- Researchers may neglect to model short-run versus long-run changes explicitly (though interest has recently focused on that distinction). Investments in transportation infrastructure and most transportation policies have long-run effects. Both theory and the few empirical estimates we found of long-run elasticities suggest that elasticities can change substantially with time.
- Elasticity estimates show the response in a dependent variable (e.g., transit ridership) from a small change (usually 1 percent; occasionally as much as 10 percent) in the independent variable (e.g., transit fare), other independent variables (e.g., travel time) held constant. A larger change in the independent variable may result in a different percentage change in ridership. The percentage change in the dependent variable may also vary with the magnitude of that variable before the change. A decrease in fare from \$3.00 to \$1.50, for example, may have a different (probably larger) percentage effect on transit ridership than a decrease in fare from \$1.00 to 50 cents, the same percentage change. Moreover, the elasticity estimate may vary with differences in the values of the other variables held constant.

These considerations suggest that we should (1) focus on the relative magnitudes of estimates (e.g., the percent change in VMT from a 1 percent increase in the variable cost of auto travel is roughly twice that of a 1 percent decrease in transit fare), and (2) use theory to help understand the effects of large changes in one variable and simultaneous changes in more than one variable.

Effects on auto travel of changes in the components of price

We expect that drivers respond to changes in the price components of a car trip in proportion to the effect of those changes on the overall cost of a trip. The extent to which drivers can make changes other than in the amount they drive (by buying more fuel-efficient cars, for example) reduces the magnitude of the effect. Over time—as changes in location occur—the sensitivity to changes in costs increases (i.e., elasticity increases).

Table C-2 summarizes empirical estimates of the effects on VMT of changes in each of the components of the cost of a trip. The first column lists those components. The second column summarizes measurements of the average effect on VMT of a change in each component of trip cost that occurs *in the short run*, usually within a year or two of the change in price. The “+” or “–” indicates the direction of the change. For example, if the price of fuel increases, VMT decreases, other things constant (therefore the “–” sign). The third column shows the strength of the effect *in the long run*, typically from two to 10 years after the change in price.

The second and third columns of Table C-2 do not contain numbers, though most empirical studies do yield numerical estimates. Low, Med, and High is our summary of the strength of the effect based on the literature we reviewed (Low means relatively inelastic). Elasticity estimates should not be regarded as highly precise. To dispel any illusions of accuracy, we chose to report the range into which most of the estimates we found fell. The ranges are described in the footnotes to the table.

An increase in the *price of fuel* tends to have small effects on VMT in the short run and also relatively small effects in the long run. The short-run effects are small for

TABLE C-2. MEASUREMENTS OF THE EFFECT ON VMT OF CHANGES IN THE COST COMPONENTS OF AUTO TRAVEL

COMPONENT OF THE PRICE OF A TRIP	MAGNITUDE OF EFFECT IN THE SHORT RUN	MAGNITUDE OF EFFECT IN THE LONG RUN
Out-of-pocket cost		
Fuel (work)	– Low	–Low-Med
Fuel (nonwork)	– Med	– Med*
Congestion tolls	– Low	– Low-Med
Parking fees	– Low	– Low-Med*
Time cost		
Riding time/parking search	– Low-Med	– Med-High*
Cost of alternative modes		
Transit fare	+ Low	+ Low*
Transit access time	+ Low*	+ Low*
Nearby development		
Jobs-housing balance	NA	Low-Med
Nearby retail/services		Low

Sources: Acutt and Dodgson 1995; Altshuler 1979; Cervero and Duncan 2006; Goodwin 1992; Olszowski and Xie 2005; Oum, Waters, and Yong 1992; Pucher and Rothenberg 1976; Schiffer et Al. 2005; Shoup 2005a; and Winston 1985.

Low: elasticity between 0 and 0.3

Medium: elasticity between 0.3 and 0.7

High: elasticity greater than 0.7

*We could find no numerical measurement of this elasticity. We report the effect suggested by theory and other measurements (e.g. surveys).

two reasons. First, fuel is not a large component of the total cost of a trip. Suppose a 10-mile commute takes 20 minutes by car each way, and the car gets 20 mpg. Suppose the price of a gallon of gasoline rises by 50 percent, from \$1.80 to \$2.70. The out-of-pocket cost of commuting increases by 90 cents per day, or \$4.50 per week. For most commuters, the time cost of the trips *each day* exceeds this change, possibly by a lot; a big percentage increase in fuel price is a smaller percentage increase in total trip cost. The second reason for the small fuel-price elasticity is that many people have few good alternatives to driving in the short run. Is it worth saving 90 cents to access transit from a location not well oriented to transit? Doing so could easily double the travel time.

Not surprisingly, the effect of an increase in fuel prices on discretionary (nonwork) trips is higher than on commuter work trips. But the long-run effects are not much higher than those in the short run because people can respond to higher fuel prices by purchasing more fuel-efficient cars. This long-run effect showed itself clearly in the aftermath of the oil shortages of the 1970s. And as fuel prices dropped in the 1990s, many people switched back to less fuel-efficient vehicles. Thus the effect of an increase in the price of fuel on fuel consumption is stronger than the effect on VMT.

Of special interest is the effect on VMT of congestion tolls. A study of the sensitivity of travelers in Singapore to changes in congestion tolls is revealing (Olszewski and Xie 2005). Though congestion pricing has been implemented elsewhere (e.g., Norwegian toll rings and the cordon pricing in London), only Singapore has experience with congestion pricing over a long time period: since 1975. Singapore implemented an electronic road pricing (ERP) system in 1998. Tolls are deducted electronically from a pre-paid smart card as autos pass under a gantry. Importantly, the congestion tolls vary by time of day, type of vehicle, and type of road.

Here are some details. There are a total of 45 gantries, 28 control access to the central area (cordon pricing), 12 collect tolls on expressway segments, and five collect tolls on radial arterial streets. Tolls rise by type of vehicle, from motorcycle, to car, small truck, and large truck. And tolls vary by time of day. For example, in August 2004, the toll (all tolls provided here are in Singapore dollars) on cars traveling on an

expressway segment rose from \$1.00 from 7:00 to 8:00 in the morning, to \$2.50 from 8:00 to 8:30, to \$3.00 from 8:30 to 9:00, and back down to \$1.00 from 9:00 to 9:30.¹⁵ The charges are reviewed every three months and are set to achieve travel speed targets (which is simpler and more easily appreciated by the public than setting charges equal to some estimate of external time cost). The relatively frequent variation in prices can be used to estimate elasticities.

The elasticity estimates are relatively low on average. For cars they range from -0.1 into the central area to -0.2 on expressways. They are low for a variety of reasons. First, the initial prices are not terribly high, and the changes are correspondingly modest. Second, since prices are adjusted frequently, the elasticity estimates are probably short run. Third, congestion prices target the relatively high-value commute trips; elasticities are generally lower for work trips than for discretionary travel.

But even small elasticities matter on congested highways. Synchronized flow allows reasonable travel speeds even at high traffic volumes. As the speed flow relationship graphed in Figure C-1 demonstrates, a small proportionate increase in the number of cars on a crowded highway can have a large impact on travel speeds. Thus the 2 percent drop in trips caused by a 10 percent (i.e., 30 cent) increase in the toll can have a big impact on travel speeds.

Estimated elasticities vary by both time of day and type of vehicle. Car drivers are more sensitive during the evening than during the morning rush hour; elasticity rises to -0.3. Commuters may have more flexibility in time of travel in the evening than in the morning. Patterns of elasticities are similar across types of vehicle, but light delivery vehicles are especially less sensitive (lower elasticities) compared with others.

One might expect an increase in parking fees to have an effect similar to that of an increase in a congestion toll. But there is a difference: a driver may be able to use another route to avoid a toll, even in the short run. To the extent other routes are available, we suspect that VMT *on the tolled highway* drops more from an increase in the toll than from a similar dollar increase in fuel or parking prices. An increase in the price of parking is effectively similar to that of an increase in the price of entry to the central area in Singapore.

Shoup (2005a) points out that the impact of an increase in the price of on-street parking in central business districts (CBDs) might be especially large. On-street parking is usually hard to find because cities tend to price parking too low; higher prices would mean more open parking spaces. The result is that people cruise for parking. This wastes their time, but it also has potentially large external costs on other drivers; cruising for parking increases congestion. The impact of higher charges on parking meters is less cruising.

The effect of an increase in the price of fuel, tolls, or parking probably increases with the availability of other travel modes. For example, given current patterns of transit service, the effect on trips to the CBD of a regionwide increase in parking fees probably exceeds, perhaps substantially, the effect on trips between suburban areas. Because transit usually serves trips to the CBD better than trips to suburban employment centers (in large part because the low density and auto-oriented design of development in suburban employment centers usually prevents effective transit service), travelers to the CBD can switch to transit in response to higher parking prices.

Travel time (riding, searching for parking, moving slowly in congestion) is a relatively large component of the cost of travel. Time spent traveling replaces time spent working or at leisure. This suggests that people should value time spent traveling at about their after-tax wage. Small (1992) summarizes the results of empirical studies of the value of time spent traveling. Commuters value time spent *in the vehicle* at between 20 percent and 100 percent of their *gross* wage. He suggests using half of gross wage as an appropriate rule of thumb for valuing time spent commuting. Time spent traveling for business is probably valued at closer to gross wage. One would expect changes in travel times to have strong effects on VMT.

Highway capacity expansions in congested corridors provide large short-run improvements in travel speeds and trip times. There consequently has been a growing

interest in estimating the short-run versus long-run elasticities of travel time on traffic flows. Schiffer et al. (2005) review 31 studies, most published between 1996 and 2003. Estimated short-run elasticities vary from near 0.0 to about -0.4—traffic builds quickly as travel times fall on new capacity, but not by enough in the short run to significantly slow travel. The long-run elasticity is higher, ranging from -0.5 to -1.0. Business and household relocations cause congestion to build on some expanded corridors. Estimated elasticities decrease with the size of the project; congestion is likely to return with more of a vengeance on expansions of individual links or corridors compared to systemwide expansions. Expansion of capacity in corridors that experience little congestion is likely to have relatively little effect in the short run or long run. Schiffer et al. conclude that induced-demand effects are important.

Table C-2 also shows the effect on auto travel of a change in the price of transit. Marginal changes in the characteristics of transit will have little effect on VMT, holding constant the characteristics of the roadway system. Acutt and Dodgson (1995), for example, estimate the elasticity of transit fare in the UK to VMT is 0.02. A big reason is that the percentage effect on auto travel of a shift from transit to car is necessarily smaller than the percentage effect on transit. Trips by car account for more than 90 percent of all vehicle trips (in suburban areas that number often increases to 99 percent). A large percent change in transit riders necessarily translates into a small percentage affect on highways. In many suburban areas, eliminating transit would have only a very small effect on VMT.

The long-run effect is harder to predict. On the one hand, decreasing the price of transit trips, especially by improving access to the transit system, will encourage more people to use transit. On the other hand, as more people switch from auto travel to transit, the decrease in congestion will encourage new auto trips. We suspect that the effect on VMT of improved transit service will not be large in the long run, but it may be larger than in the short run. Enough commuters use transit in some cities that a disruption in transit service (e.g., by striking transit employees) causes serious increases in congestion on highways. We suspect that large improvements in transit service, combined with moderate increases in the price of auto travel, could encourage strong long-run reductions in VMT.

Cervero and Duncan (2006) compare empirically the effects of mixed-use development to that of jobs-housing balance on VMT. Proximity of households to their typical destinations should decrease overall VMT. Travel survey data from the San Francisco Bay area indicate that a 10 percent increase in the availability of jobs that generally match the travelers type of job within four miles of the traveler's house decreases VMT by about 3.3 percent, an elasticity of 0.33. The analogous estimated elasticity of VMT with respect to proximity to shopping is about 0.17. Cervero and Duncan conclude that jobs-housing balance, at least when the jobs match the preferences of the householders, has a bigger effect on VMT than nearby retail and service development, but both matter.

Effects on transit ridership of changes in components of price

Table C-3 summarizes empirical estimates of the effects on transit ridership of changes in the price of a trip on transit. The format of Table C-3 follows that of Table C-2.

Table C-3 shows that a change in *fare* has a moderate impact on ridership in the short run, especially with respect to work trips (elasticity estimates range from about -0.2 to -0.6). Most people who travel to work on transit probably have difficulty adjusting quickly to an increase in transit fare. The fare elasticity of nonwork trips exceeds that for work trips because people can make changes more easily. They can, for example, simply take fewer shopping trips if the price goes up.

The low response of transit riders to increases in fare has led many researchers to conclude that transit agencies could decrease operating subsidies by increasing fares. The increase in revenues from the higher fares would more than offset the loss in revenues from the loss of riders. Recent studies, however, cast doubt on this assessment, especially for rail. Goodwin's (1992) review reports average fare elasticities

for rail of -0.8. Studies that explicitly account for long-run effects frequently estimate long-run elasticities as greater than -1.0. Voith (1991a) estimates short- and long-run fare elasticities for rail transit in Philadelphia. His estimate of -0.6 for the short-run elasticity is consistent with, though on the high side of, most existing short-run estimates. Over the course of eight years, however, the magnitude of the fare elasticity increases to -1.6. Goodwin reports that both short- and long-run fare elasticities for bus travel are typically less than those for rail. Perhaps people that ride rail have higher incomes than people who ride the bus and, therefore, have better access to an auto should fares rise. Gilbert and Jalilian (1991), however, estimate long-run fare elasticities for bus travel in London at between -1.2 and -1.3.¹⁶

Recent research indicates that long-run fare elasticities vary significantly across cities and over time. For example, Bresson et al. (2004) find that long-run fare elasticities vary from -0.1 to -1.0 across 62 French urban areas, with a mean long-run elasticity estimate of about -0.6. In an earlier paper (Bresson et al. 2003), they report long-run fare elasticities that vary from about -0.2 to -1.2 across English counties (mean about -0.7). They also report that transit travelers become more sensitive to increases in fares as the starting fare gets higher. Thus the long-run responsiveness to fares is higher than in the short run, but there appears to be variability across cities and over time.

Bresson et al. (2004) also report interesting results about the impact of changes in income on ridership. Income elasticities of ridership are usually negative; ridership falls as incomes rise—transit is considered an “inferior good.” Using a standard specification, they estimate the income elasticity at about -0.2. Adding car-ownership to the regression equation, however, eliminates the negative effect of income. Higher incomes lead to higher car ownership rates, which reduces ridership.

O’Sullivan (1993) reports that travelers value time spent walking to and waiting at a transit station more highly than they value the time spent riding in the vehicle, perhaps equal to their after-tax wage. Decreasing walking and waiting times has

TABLE C-3. MEASUREMENTS OF THE EFFECT ON TRANSIT RIDERSHIP OF CHANGES IN THE PRICE OF TRANSIT

COMPONENT OF THE PRICE OF A TRIP	MAGNITUDE OF EFFECT IN THE SHORT RUN	MAGNITUDE OF EFFECT IN THE LONG RUN
Out-of-pocket cost		
Fare (work)	– Low-Med	– Med-High
Fare (non-work)	– Med	– High*
Time cost		
Riding time	– Low	– Low
Frequency (peak)	+ Med	+ Med
Frequency (non-peak)	+ Med	+ High
Access time	– Med	– High*
Reliability	+ Med-High*	+ High*
Cost of alternatives		
Variable cost of auto	+ Med-High	+ High
Fixed cost of auto	+ Low	+ High
Service characteristics		
Climate control	+ Med-High*	+ High*
Luxuries	+ Low*	+ Low*
Safety	+ High*	+ High*

Sources: Bresson et al. 2003, 2004; Meyer and Miller 1984; Wachs 1991; Voith 1991a.

Low: elasticity between 0 and 0.3

Medium: elasticity between 0.3 and 0.7

High: elasticity greater than 0.7

*We could find no numerical measurement of this elasticity. We report the effect suggested by theory and other measurements (e.g. surveys).

a correspondingly large effect on ridership relative to a similar decrease in riding times. Voith (1991a) reports that an increase in the number of transit vehicles (which decreases waiting time) has a strong effect on ridership in the long run, especially at nonpeak times (elasticity equals 1.9).

Cervero (2006) estimates the effect of residential densities on transit ridership. Controlling for the relatively strong effect of more frequent transit service in denser areas (elasticity of about 0.6), a 10 percent increase in density increases transit ridership by about 3 percent (elasticity about 0.3). This effect comes in part from the higher cost in relatively dense areas of owning and operating a car, and in part because more people live within easy walking distance of transit stops.

Wachs (1991) suggests that the *reliability* of transit has relatively strong effects on ridership, though we found no measured elasticities. A person can spend less time waiting if she knows when the vehicle will arrive at the station, and when it will arrive at its destination. Transit vehicles that run in their own rights-of-way have greater control over reliability than ones that must compete for highway capacity and respond to the vagaries of congestion. A person who can choose between taking a car and taking a form of transit that can avoid congestion may choose transit to avoid the uncertainties of driving in congestion.

The measurements of the effects on transit ridership of changes in the *cost of a trip by car* support this notion. Voith (1991a) reports that in the short run, a 10 percent increase in the price of a trip by car increases rail ridership in Philadelphia by about 10 percent (cross-elasticity equals 1.0). The cross-elasticity in the long run increases to about 2.7. These estimates seem quite high, and cross-elasticities may be smaller for buses or for transit in other cities. But the estimates do support the intuitively plausible notion that transit ridership is far more sensitive to changes in the cost of an auto trip than the other way around (in no small part because a small percentage shift away from car travel implies a large percentage increase in transit travel).

Casual observation supports the importance of the variable cost of auto travel on transit ridership. For example, in cities in the Northwest, where significant snowfall is infrequent, drivers and road crews cannot respond adequately to the occasionally strong winter storm. After such a storm, the price of auto travel (in dollars, time, safety, and convenience) increases substantially, and so does transit ridership. Transit districts in both Portland and Eugene, Oregon, estimate that ridership increases by about 25 percent on days with snowy or icy road conditions.

Wachs (1991) reports that bringing all aspects of transit service up to a reasonable level can have significant effects on ridership. Surveys show that people have little concern for luxuries on transit, but they want control over climate. Too hot or too cold buses have trouble attracting riders. Wachs also reports that people expect security on transit; decreases in security frighten riders away.

We know of no study that measures the combined effect of complementary changes in more than one component of price. We suspect, however, that given the current orientation of public infrastructure toward easing travel by auto, improving transit service in several ways simultaneously could have a strong effect on transit ridership. For example, holding fare constant (or even raising it considerably) while improving the comfort, accessibility, and reliability of transit could greatly increase ridership.

Summary of elasticities

To aid in our analysis of policy in Chapters 5 and 6, we draw the following conclusions (though we are aware that the measurements reported are insufficient to make these conclusions uncontroversial):

- The response of travelers to changes in the cost of travel is probably larger than that conventionally believed. We do not believe that people make most trips by car simply because they love their cars. They travel by car because the cost of most trips by other modes exceeds the cost of the trip by car when all components of cost, especially time cost, are taken into account. If the price of other modes decreases relative to the car, some people will change travel behavior.

- The effects in the long run of changes in the factors that individuals consider when making travel choices usually differ from the effects in the short run. Over time, people can adjust to changes first by changing the timing or routing of trips, then by buying cars with different characteristics, and eventually by changing the location of their origins and destinations. Whether the long-run effect is stronger or weaker than the short-run effect depends on the specific situation.
- The effects of a change in the user cost of a trip by a particular mode depend on the user cost of alternative modes. Responses to changes in cost differ across routes, modes, and cities in part because the costs of alternatives differ across routes, modes, and cities. For example, an increase in the price of a trip by car has a larger effect on suburb-to-CBD trips than on suburb-to-suburb trips because transit systems typically serve suburb-to-CBD trips relatively well.
- The effect of simultaneous changes in the factors that affect travel demand in complementary ways may exceed the sum of the effects of those changes made in isolation. For example, improving the quality of transit service while simultaneously increasing the price of an auto trip probably has a greater effect on ridership than the sum of the effects from doing either in isolation.
- More research is needed. The methods to estimate empirically the affects of changes in policy continue to evolve. Recent research suggests that people respond more strongly to changes than previously recognized. Our review of the literature suggests that researchers remain far from a consensus regarding many elasticities important to policy makers.

APPENDIX C NOTES

1. Thanks to Wayne Kittelson for this analogy.
2. We got a good laugh out of a publication we received from our automobile insurance company, telling all its clients to keep “three or four seconds” behind the car in front in good weather, and to double that time at night or in rain. At 65 mph, six seconds of travel time is about 600 feet, a spacing of less than 10 cars per mile; a flow of about 600 vehicles per hour. When did you ever see that on your rainy commute home in a metropolitan area? Exercise for the interested student: why do insurance companies recommend that traveling distance?
3. See Downs (2004a), Appendix A, for much more details about the numbers.
4. It would be defined differently for different types of transportation facilities.
5. Transportation economists are fussier. That specification of social maximization implies that every trip is as valuable as every other trip. Economists know that isn’t true, and so do you. So their preference is to maximize the sum of the net value (benefits less costs) of the individual trips. That criterion, however, has some big measurement problems, and some big traffic engineering problems. Tolling, the economists’ tool of choice, helps solve both of those problems and is discussed extensively in Chapter 4.
6. To keep the story simple, we are talking about freeways, which are analogous to long, uniform sections of pipe. The v/c ratio is not relevant for evaluating LOS at intersections, which can fail even at low v/c ratios or work fine at high ones.
7. Traffic planners and engineers often see maximizing flow as a socially desirable goal because at the maximum flow rate the greatest number of vehicles are getting where their drivers and occupants want to go in a given amount of time. Economists, however, disagree. Among the reasons, as we demonstrate later, is that it is not just numbers of cars that matters, but also how long it takes those cars to get through and the chances of an “incident” stopping traffic altogether. The economically optimal (i.e., efficient) flow is less than capacity.
8. From *Your Driving Costs*, American Automobile Association, 2006.
9. Downs (2004a) reports that the average commuting trip takes 22 minutes, but that 30 percent of all commutes require more than 30 minutes, and 12 percent require more than 45 minutes. The total amount of time people spend driving daily probably varies as much as the time people spend commuting.
10. An overly congested highway is an example of a “tragedy of the commons,” a term coined by William Forster Lloyd in 1833 and popularized by Garret Hardin in his article of the same name published in the journal *Science* in 1968. Though not cited by Hardin, the economics of the situation was analyzed in the context of an open-access fishery by H. Scott Gordon in 1954 (“The Economics

- of a Common-Property Resource: The Fishery." *Journal of Political Economy* 62 (April): 124–42).
11. Average-cost pricing, in contrast, is common when goods are supplied or regulated by government agencies. Electricity, for example, typically sells at a fixed retail price due primarily to regulatory requirements despite wide swings in wholesale prices. Charges for water and sewer services are often constant across a city or metro area despite the fact that costs vary geographically. Average-cost pricing may seem fair, but it has serious negative consequences.
 12. That is an answer about *efficiency*, not *fairness*. Chapter 4 discusses fairness: should lower-income drivers who, on average, will have lower-value trips (because their ability to pay is less) be the ones to get bumped off the highways by tolls?
 13. Other terms sometimes used for induced demand are latent demand and triple convergence (Downs 2004). The professional literature on the transportation/land use connection also confuses things by using "induced demand" to refer to the tendency of new transportation facilities to induce land-use development (the construction of buildings) beyond what would have been expected in the absence of the provision of new capacity.
 14. Analysts prefer elasticities to other measures of absolute or percent changes because elasticities measure the size of the effect on travel relative both to the size of the change in the price and to the original amount of travel. To see the point, consider the following statement: a \$1 increase in the price of parking results in 5,000 fewer cars entering the city during rush hour. One has trouble applying that observation to other situations without knowing the initial price of parking and volume of traffic.
 15. At the time, one Singapore dollar was worth 58 U.S. cents.
 16. The London example reminds us of another point worth paying attention to: elasticities of demand may change as one moves to different places on the demand curve. Thus, the new charges for congestion in London (see Appendix F) have increased cost and moved the point where demand and supply intersect: elasticities may have changed.

Benefit-Cost Analysis and Project Selection¹

Appendix A developed a framework for evaluating any type of public policy. This appendix discusses a specific evaluation technique consistent with that framework: benefit-cost analysis. It provides an overview, discusses some techniques and limitations, and comments on issues relating to its application to transportation as part of a regional planning effort.

Benefit-cost analysis is often criticized as biased or impractical. Like any tool for evaluating policy, one must understand its limitations and assumptions to use it properly. We present it here not because we think it can be applied exclusively or well in all transportation evaluations. Rather, we think it provides a logical framework for thinking about transportation investments and a point of departure for any systematic evaluation. We describe benefit-cost analysis as a decision-*aiding* technique, not a decision-*making* technique. It is “not a substitute for political decisions, but it makes their implications more transparent” (Small 2003).

OVERVIEW OF BENEFIT-COST ANALYSIS

If all of the principles of evaluation identified in Appendix A were applied to an evaluation of a transportation investment, the evaluation would have the following steps:

1. *Consider all relevant project, program, and policy alternatives.* Policy makers should not arbitrarily restrict themselves to a particular portfolio of project alternatives. All reasonable alternatives and combinations of alternatives (as appropriate) should be considered.² Project alternatives have to be specified in such a way that their benefits and costs are unrelated (mutually exclusive).
2. *Define a relevant base case.* Projects can be evaluated only relative to some characterization of what would happen without the project in place. Typically, the base case tries to represent the conditions that will prevail if no change (no new transportation project, program, and policy) is initiated. Developing data about base case conditions is just as complex as the measurements required for the project alternatives.
3. *Take a networkwide, multimodal perspective.* Any nontrivial benefits produced (or costs imposed) anywhere on the regional transportation network, on any affected mode, should be measured.
4. *Measure all types of impacts.* A full analysis should measure not just transportation user benefits and transportation system costs, but also the impacts on nonusers, other public agencies, environmental quality, and economic development. Economists serious about a technically correct evaluation of choices see benefit-cost analysis as inclusive. They do not limit the analysis to a small subset of impacts traded in the market and measured in dollars, which is often the claim of those who oppose benefit-cost analysis (based on how it is often, but incorrectly, done). This appendix uses benefits and costs to mean all significant impacts, whether they are readily quantifiable (in dollars or other units) or not. The measurement of benefits and costs for transportation projects has these subtasks:
 - Identify all the significant categories of benefits and costs, and the relevant geographic extent of each.
 - Measure all the direct transportation costs (construction, operation, capital maintenance and replacement) over a reasonable assumed life for the investment (20 to 50 years).

- Estimate monetary value of the user benefits of the investments by looking at the values of all individual trips, which would vary by person, mode, time, and location.
 - Describe and, where possible, quantify and monetize, all other benefits and costs that do not strictly fall on transportation users (e.g., environmental, economic development, social).
5. *Consolidate the stream of future benefits and costs into a single measure of aggregate or average benefits and costs, and subtract the cost measure from the benefit measure to estimate net benefit.* This step requires the proper analytical treatment of benefits and costs that occur in different amounts over different years. Annual benefits (in both type and magnitude) and costs (both capital and operating) change over the life of the project. The timing of the occurrence of these costs affects the attractiveness of a project because of the time value of money.
 6. *Measure impacts on important subgroups.* A consideration of subgroups rather than society as a whole helps policy makers consider how to evaluate the *distribution* of impacts and the *equity* (fairness) of that distribution. Policy makers might be interested not only in aggregate net benefits, but in net benefits (or costs) by travel mode or income.
 7. *Compare all the information above across the different investment options.*
 8. *Make a decision in the context of budgetary and financing constraints and considerations of risk and uncertainty, and select one or more projects to implement.* Although all projects with benefits that exceed costs (properly measured) are projects worth pursuing, budgetary constraints may limit what can be pursued. This restriction is usually the result of policy, legal, or technological considerations.

Though easy to describe, to implement these steps in the real world is technically complex. Typical questions that must be addressed:

- How does one measure the benefits of something that does not yet exist, especially when it interacts in a complex way with other products or services?
- What does one do if some benefits or costs are not susceptible, at all, to measurement? What if the saving or loss of human life is potentially involved?
- What if the benefits or costs play out over time? How should these delays be incorporated in the analysis?
- How does one treat uncertainty and risk?
- What if many projects have positive net benefits, but budgets are limited? Which projects should be selected for implementation?
- What if a project has negative net benefits but is particularly effective at helping a targeted or protected class of user (such as the poor)?

The rest of this appendix examines some of the technical issues relevant to the application of benefit-cost analysis to the evaluation of public investments in transportation.

TECHNIQUES OF BENEFIT-COST ANALYSIS

Measuring Benefits and Costs

Principles for evaluating the impacts of transportation investments on transportation performance

The focus (not the full picture, but the focus) of performance measurement in a transportation plan should be on measures of transportation system performance for the following reasons:

- The primary purpose of transportation investments should be improving the performance of the transportation system. If other objectives are more important (e.g., economic development, environmental quality, social justice), other, nontransportation policies probably address those objectives more directly and efficiently.
- Some of the other categories of potential objectives/benefits may be largely double-counts of improvements to transportation system performance. For example,

increases in the intensity or value of land development resulting from a transportation improvement are largely a double-count of the capitalization of the improved access and mobility into land values.

An important (arguably the most important) criterion in evaluating a transportation investment (actually, any investment decision) is *efficiency*: how much bang (benefits to users and nonusers of the transportation system) do you get for the buck (construction cost, and other secondary costs that construction and operation cause)?³

The majority of the benefits produced by transportation projects come from reductions in user costs.

User benefits are determined by travel costs in three distinct areas: travel time costs, operating costs, and accident costs. Taken together, the total of these costs is essentially the price that travelers must pay to travel. When a comparison is made between the costs of traveling and the number of trips taken at each price level, a relationship is determined between the cost of travel and the demand for trips. When all users are aggregated together, the difference between the travel “price” that travelers are required to pay and what they would have been willing to pay is the user benefit affiliated with the trip. Any reduction in travel costs (i.e., trip price), then, will result in a benefit to the traveler. (ECONorthwest 2003)

We skip over the detail of how to measure user benefits to some conclusions:

- Any rigorous evaluation of alternative transportation projects has to focus on user benefits.
- The theory and procedures for correct measurement are neither intuitive nor simple to apply. It is not the case that the components of user benefit can be measured individually and simply added. Travelers evaluate as a package all the components of a trip (e.g., travel time, out-of-pocket cost, accident risk). Recommended procedures for estimating user benefits are to construct a travel demand curve (see Appendix B) to estimate willingness to pay and consumer surplus. Again, we skip the details and go to the conclusions: by using a travel-demand model and benefit-cost techniques, one can get to an estimate of the *efficiency* of some proposed transportation improvement: its net user benefits and the net benefits relative to the cost. Efficient investing in transportation projects is fundamentally a search for projects that have positive net benefits.
- Without that kind of analysis, one must deal even more with the multiattribute problems we describe below in this appendix. Various measures of transportation performance do not add in any obvious or rigorous way to some measure of net benefits.

Consider, as just one example, the reduction of vehicle miles traveled (VMT) as a desired outcome for transportation investment. It is clear to transportation economists that reducing VMT is not necessarily a good idea (it might be, but it might not be). People travel to get access to many things that enhance their well-being: work, health care, shopping, recreation, and so on. Other things being equal, people are benefited by lower cost travel, which (again, other things being equal) allows them to travel more (increasing VMT). So VMT (or VMT per capita) might be a measure of travel *benefit*, not travel dysfunction.

Yet, many transportation and land-use planners argue that decreases in VMT are the desired direction for the transportation system. Those making the argument will often justify it by saying that *access* (the ability to get to places we care about) is what matters more, and *mobility* (the ability to travel anywhere quickly) matters less. With that logic, what they hope to achieve is getting everyone where they want to go but with the smallest increase in VMT. The notion, not illogical, is that government policy (probably more about land use than transportation) can rearrange origins and destinations so that fewer, shorter, and nonauto trips provide the same access.

But trips are not a homogeneous commodity: one trip is not as good as any other trip. Going to the same place but at a different time, in a different mode, and at a different

speed, is not as good as going quickly when and where you want. And the cost of rearranging land uses so that access is improved must also be considered. It's complicated.

Transportation economists are clear on the theory and even on the techniques: to measure the efficiency of a transportation project, what needs to be measured is the change in *consumer surplus*, which is the difference between a traveler's willingness to pay for a trip (the benefit or value of the trip) and his perceived cost of the trip, summed across all travelers across all the relevant time periods, and discounted to a present value (see Appendix B). Our guess is that few regional agencies do that in their transportation planning.

Thus, our conclusion is that the matrix display described in Appendix A (showing various measures of transportation performance) is good as far as it goes, and probably goes as far as it can, but it cannot answer unambiguously the question of whether a particular project is more efficient than another. It can point in the right direction: for example, spending a lot of money to expand a highway with relatively light congestion is probably not efficient, so measures of congestion and congestion relief are probably correlated with consumer surplus and efficiency. VMT, however, is probably not.

Measuring the direct transportation benefits and cost to users of transportation facilities

Transportation improvements alter the characteristics of a region's transportation network. In response, travelers may change their choices of the time, mode, path, and frequency of travel. The impact of an improvement on travel behavior is the primary source of societal benefits. The impacts on users of the transportation system constitute the major share of the benefits; they are the primary reason to incur the costs of constructing a transportation facility or implementing a transportation program.

The benefits (and costs) of a transportation improvement redound not only to *users* of the transportation system but to *nonusers* as well. A new road, for example, clearly benefits those who use the facility itself (users). If the new road also reduces air pollution (say, because of reduced stop-and-go driving), it may also benefit nonusers (anyone who breathes the affected air, whether they use the facility or not).⁴ The section after this one discusses those nonuser (indirect; secondary) benefits and costs.

At the heart of benefit-cost analysis is the estimation of benefits and costs to transportation users. The costs to users include the following:

- Direct, out-of-pocket costs—the costs of purchasing, maintaining, and operating an automobile, including fuel, fuel taxes, insurance, and so on.
- Other costs—the most important of these are *time* and *risk of accident*. Time includes all time needed to complete a trip: not just in-vehicle time, but time to walk, wait, transfer buses, find parking, and so on. Many studies have shown that the activity that a person is spending time on influences the value he or she puts on that time: for example, people perceive waiting time as more costly than riding time. Other costs include perceptions of comfort, convenience, crime risk, and so on.

Users care only about the travel costs they *perceive*, and those costs will not include all direct costs when public funds are used to make large transportation investments. If highway improvements are funded by general property taxes, a transportation user does not perceive them as relevant to the marginal cost of a trip, those costs will not influence the trip decision (whether to make the trip, when, where, and by what mode). The fact that travelers do not consider such costs, at the margin, however, does not mean they do not exist. They do: they are real and significant, and must be measured as part of a benefit-cost analysis.

Project evaluation requires an estimate of any change in direct costs specifically associated with implementation of a transportation improvement. Thus, an analyst should be interested in the dollar value of costs:

- incurred for the planning, construction, operation, or maintenance of transportation facilities or equipment; some of these costs may be ancillary but necessary (e.g., increased costs for parking and policing associated with increased bus service);

- typically paid by the nonuser sector (as opposed to many user costs paid directly by travelers, such as gas, insurance, travel times); and
- measured *relative* to the base (nonimprovement) case; hence, the analyst need be interested only in *changes* in costs associated with the improvement, as opposed to the circumstances that will prevail without the improvement.

Revenues are relevant here also. The source of funding makes a difference to the analysis of overall distributional impacts later. But while revenue sources are almost always of interest to policy makers, in the context of benefit-cost analysis, they are transfers, not benefits, and should not be added to other benefits.

Costs and direct revenues are not hard to identify in concept, and even their measurement is relatively straightforward. The measurement of user benefits is more complex. The project doesn't yet exist: how can user benefits be measured? The economist's answer in the context of transportation benefit-cost analysis is that one must estimate (forecast) users' aggregate *willingness to pay* for transportation services at various quantities of those services.

The willingness-to-pay relationship for trips between an origin and a destination is a schedule of the aggregate quantity of the trips users would be willing to make at various levels of cost per trip. (Economists call the willingness-to-pay relationship the *demand relationship*.) Measurements of willingness to pay for transportation services allow transportation analysts to convert changes in travel activity to changes in economic benefits to travelers. In the overall picture of benefit-cost analysis, this focus on willingness to pay makes sense, too. If users of a service are collectively unwilling to pay what it takes to provide or improve that service (conceptually, at least), it is harder to argue that these users receive net benefits from the service enhancement and that new facilities are needed (unless large benefits accrue to nonusers).⁵

The difference between what users (in the aggregate) would have been *willing* to pay, and what they are *required* to pay, is called *consumer surplus*. In economic terms, willingness to pay is defined by a demand curve, which also can be viewed as a marginal benefits curve. When a transportation improvement reduces the users' cost of a trip between two points (which means a change in the supply / marginal-cost curve), the willingness to pay remains the same, but since users' perceived cost of travel is less, consumer surplus will increase. Users who were already making the trip get to make the trip at lower cost, and new users (those for whose the willingness to pay was previously less than the old cost of the trip) are induced to travel.

Figure D-1 shows the basic relationship.

The "demand curve" shows the relationship between the volume of trips (x axis) and the user cost per trip that travelers must bear (y axis). The less the trip is perceived to cost, the more trips users will make, everything else being equal. In the case depicted in Figure D-1, a certain number of users (volume V_0) are willing to incur costs of U_0 , but additional users would be willing travel if the costs are reduced to U_i . The notion of consumer surplus recognizes that, at virtually any point on the demand curve (such as the level of demand associated with cost U_0), most of the people who are using transportation services would be willing to pay more than they are actually paying.

This point is so fundamental that it bears restatement: the main reason for doing a transportation project is almost always that its proponents believe that *users will be better off* (will enjoy consumer surplus, which is the measure of their increased net benefits).

This conclusion suggests how to measure the new user benefits of a transportation project: subtract consumer surplus *before* the improvement from the consumer surplus *after* the improvement. To do so, one must estimate two things:

1. The willingness-to-pay (demand) relationship.
2. The effect of the improvement on the users' perception of travel cost.

To calculate changes in consumer surplus for new trips, however, one must estimate how many additional users there will be after the improvement: in economists'

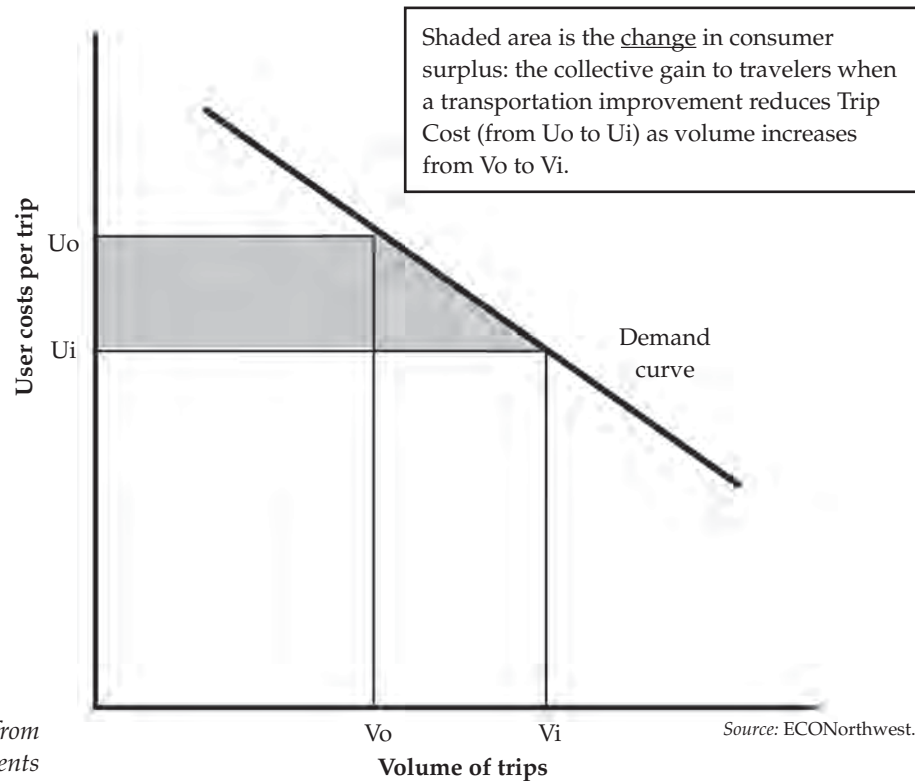


Figure D-1. User benefits from transportation improvements

terms, how *elastic* is the response of demand to the travel cost reduction associated with the improvement. Economists measure the elasticity of travel-demand relationships numerically as the percent change in the quantity of travel that results from a 1 percent change in the perceived unit cost of travel. The proper measurement of *elasticity of demand with respect to travel cost* is one of the key informational needs of transportation project selection methodologies. Modern techniques for measuring demand elasticities are accurate, but information intensive (Hensher and King 1998).

Measurement of the effect of the improvement on users' perception of travel cost is the other piece of information needed to measure traveler benefits. An unusual aspect of transportation activity (relative to many other goods and services in the economy) is that users commit their personal *time* to transportation activities. Consequently, the perceived cost of travel includes not only expenses like gasoline or transit fares, but also the value of the time spent traveling. This makes the *value of time* a crucial factor in benefit-cost analysis in transportation, in addition to the various cash or *out-of-pocket* costs. The derivation of the appropriate value of time for users of various types, income classes, and trip purposes requires the same detailed information on the demography of actual travelers and the pattern of their travel behavior used to estimate demand relationships (Calfee 1998; Wardman 1998).

Measuring the secondary benefits and cost to nonusers of transportation facilities

Many people who would not directly use the proposed transportation improvement may experience benefits or costs as a consequence of its construction or operation. They can be categorized as:

- suppliers of transportation services (transit companies, road authorities or departments, etc.);
- users of facilities elsewhere in the transportation network on the same, or other transportation modes; and
- other businesses and households.

Suppliers (producers) of transportation services are affected by changes in use of their facilities. Analogous to the *consumer surplus* enjoyed by users, suppliers enjoy

a *producer surplus* if they are able to charge more for a service than they are willing to charge to simply provide it. When the cost of a service declines from a user's perspective (thereby increasing consumer surplus), this may cause the supplier to experience a *reduction* in producer surplus. From a cost-benefit accounting standpoint, declines in producer surplus offset gains in consumer surplus. Calculating changes in producer surplus requires knowing the effect of the proposed project on:

- the producer's *willingness-to-supply* relationship (usually referred to simply as the *supply* relationship); and
- the revenues enjoyed by the supplier (typically, the projected toll or fare revenue).

In practice, producer surplus effects are seldom calculated, even when they may be important, as in the case where the supply relationship slopes down with additional load (i.e., when there are economies of scale or scope, as with bus transit).

Users of facilities elsewhere in the transportation network are usually affected by transportation projects that affect a select portion of the network. It is very common for a positive benefit generated in one portion of the network to be at least partially offset by a disbenefit elsewhere on the network as traffic levels adjust to improvements. The principle of measuring the benefits on these indirectly affected portions of the network are the same as for the primary users' benefits. The challenge here is primarily in quantifying these *network effects*. Although the state of transportation demand and network modeling has advanced significantly, the linkages between these modeling technologies and benefit-cost analysis is virtually nonexistent.

Other businesses and households can be affected apart from their use of some portion of the network. In economics parlance, these effects are *externalities* the transportation system engenders. Externalities can result in benefits or (more commonly) costs in many categories:

- *Environment*. A transportation improvement may either improve or degrade environmental conditions. Transportation activity has significant impacts on air, noise, and water resources; some argue that transportation activity contributes significantly to the apparent global warming trends. Much work has been done on this topic over the last 15 years, but what value to assign to some externalities is still debated, and only a few meta-analyses of transportation externalities have been done (Murphy and Delucchi 1998).
- *Safety*. Some of these costs may already be included in the perceived user cost; others may not. Costs not perceived by the user, such as traffic delays due to an accident, are external costs.
- *Economy*. Transportation improvements can affect the health of a regional economy by increasing the overall production possibilities of the economy.
- *Cost of public services*. The use of transportation facilities affects the activity of courts, police departments, emergency service departments, etc.
- *Land use*. These impacts have become a big concern in the U.S. in the last 15 years. From an economic perspective, however, these impacts look primarily like intermediate impacts in the sense that changes to land use may yield changes in impacts already being measured: on travel, the environment, the economy, and public services.

Estimating values when benefits and costs cannot be easily quantified

Specific estimates for the value of time, demand elasticity, the value of human life, impacts on regional economic growth and the environment are either contested or only partially developed. Nonetheless, decision making must and does proceed. By default, the values for these parameters are set implicitly. For example, when decision makers choose not to do something (say, not spend \$10 million on a road improvement that would save one life every year), they are adopting *de facto* a value of life (or at least an upper bound on the value) as surely as if they had asserted it directly.

If enough decisions were observed from this perspective, it would be possible to infer the values of decision makers. For explicit estimates of decision-maker values,

however, the most rigorous procedures generally involve what are known as *multicriterion* decision processes (Nijkamp and Blaas 1994). Decision makers (who may either be officials or the public) are asked to rank the alternative transportation projects on a number of dimensions or criteria, such as net economic benefits, environmental soundness, and so on. In so doing, they reveal the weight they apply to various performance attributes of the projects which, in turn, implies something about the value of the underlying benefit-cost parameters.

The problem with these techniques, from an economist's point of view, is the presumption that, as a group, decision makers somehow harbor knowledge about the "true" value of these parameters and that, once extracted, these values can be usefully and repeatedly applied. It is unlikely that decision-maker intuition is able inherently to solve complex scientific questions. In the worst case, if the decision makers happen to be completely wrong, consistent reliance on the results of multicriterion exercises results in a large number of wrong decisions. The issue here is the same one described in Appendix A as "weighting," and some of the solutions there overlap the ones discussed here.

Discounting and Present Value

Most transportation projects involving capital improvements provide benefits over an extended period of time. For as long as the project is operational, it is providing transportation services. Therefore, some benefits (and costs) will be associated with it throughout its life. The savings (or increases) in user travel time and expense, relative to what they would have been without the project, extend far into the future. Similarly, the development and operating costs associated with the improvements play out over an extended period of time. Though the bulk of the capital, or development, costs usually occur at the front-end of the implementation process, operating costs and maintenance costs occur over the life of the project.

The process of telescoping the stream of benefits and costs that occur over time into an equivalent single figure in today's dollars is the process of *present valuation*. Present valuation is not simply a matter of adding up each year's benefits net of costs in a simple, arithmetic fashion. An adjustment needs to be made for society's perception of what a dollar is worth when it is received or spent in the future, as opposed to today.

An example illustrates the reasoning. Given the choice of \$100 today or a note redeemable for \$100 one year from now, most people would choose the \$100 today. But if that note were worth \$1,000 in one year, most people would choose the note over the immediate \$100; that is, they would accept the postponement of gratification, the erosion of inflation, and the risk that, for whatever reasons, the payment in a year will end up being less than \$1,000. At some future payment amount more than \$100 and less than \$1,000, people are indifferent between \$100 today and some larger future payment.

In other words, a dollar received today is worth more now than it would be if received later because one always has the option of investing it and turning it into more than a dollar later. In yet other words, individuals *discount* future dollars: a dollar next year is worth less than a dollar today, even if there were no inflation. Likewise, society as a whole is indifferent to receiving a dollar's worth of benefits in the future or some lesser amount today. This lesser, discounted amount is called *the present value of the future benefit* (or cost). The rate at which those future benefits (or costs) are brought back to present value is called *the discount rate*.

A complicating factor in evaluating public transportation investments is the choice of an appropriate discount rate for discounting future costs and benefits. The rate of return required for public investments is generally lower than that required for private investments due to the lower level of risk for public investments. The public rate is sometimes referred to as the *social rate of time preference* when it is used as a discount rate. If private market considerations dominate financial markets, and private markets care only about the consumption prospects of current generation of

market participants (and thus discount future benefits at a higher rate), it may be the case that society is *underinvesting*.⁶ Many transportation projects involve high up-front costs, with benefits that play out only over long periods of time. If the social rate of time preference is lower than the private rate, projects discounted at the social rate will have net present values higher than if they were discounted at the private rate.

Economists have debated whether very low discount rates should be used on public projects despite considerations like the *social* underinvestment hypothesis (Feldstein 1988). One reason for the debate is that through the operation of futures markets and by virtue of bequest behavior, the consumption prospects of future generations may be adequately considered in present-day, private decision making.

Given these considerations, the choice of a discount rate (the types are described in this bulleted list) depends on the particular project being evaluated:

- *Riskless Nominal Rate.* If the analyst has structured the analysis so that no or insignificant uncertainty or risk is involved, a riskless, nominal (not adjusted for inflation) discount rate should be used. The best source for this rate is the current yield on government securities with a term similar to the project's term.
- *Riskless Real Rate.* If no significant risk is involved in the project and costs and benefits for future years have been estimated in *real* (inflation-removed) terms, the use of a riskless real discount rate is appropriate. Government securities are also an appropriate place to look for determining the riskless real rate. The U.S. government issues inflation-protected bonds ("Treasury Inflation-Indexed Bonds") for terms of five, 10, and 30 years. The difference between the yield of these bonds and conventional bonds of the same term is the market's current estimate of inflation for future years. At the time of this report (2007), the implicit, riskless real rate was approximately 3.5 percent. In general, economists expect this rate to be in the range of 3 percent to 4 percent.
- *Risk Premiums.* A risk premium can be added to the two discount rates described above to obtain the risk-adjusted discount rate if the analyst wishes to adjust for risk as part of the discounting process. The private sector provides guidance for determining the size of the risk premium, as private entities providing transportation services to the public often fund these investments with issuance of debt (bonds) and equity. The yield offered on these private securities (with a similar term to the project being analyzed) minus the yield of riskless government securities provides an approximation of the size of the risk premium. By its nature, project risk is very idiosyncratic and conservatism argues for a larger, rather than smaller, risk premium.

To summarize and simplify, here is a rule of thumb probably adequate for a base-case evaluation. The effect of different discount rates could (and should) be tested quickly with sensitivity analysis:

- If all the costs and benefits have been measured in real dollars, use a discount rate of 3 percent.
- If the costs and benefits have been measured in nominal dollars, use a discount rate equal to 3 percent plus the annual, future inflation rate that was used in the analysis. In 2002, in the U.S., the consensus opinion for future inflation was 2.5 percent annually, so the discount rate would be $3.0 + 2.5 = 5.5$ percent.⁷ As discussed above, if risk or uncertainty is a consideration, the analyst might add a risk premium to this amount but must be careful about double-counting: that risk may already and more appropriately be accounted for in the initial estimates of benefits and costs, or in sensitivity analysis.

Small (2003, 156) points out an additional problem related to timing: dealing with large-scale, long-lived projects that are much more than marginal adjustments to an existing system (e.g., a new light-rail system; the "Big Dig" in Boston). We share his conclusion: forecasting and valuation in this situation is even more daunting than it otherwise always is, and the focus should be on exploring possible futures (not

picking one) and on developing investment strategies that allow incremental response to an uncertain and unfolding future.

Ranking Projects Based on Efficiency

Benefit-cost analysis is about bang for the buck: benefits for costs incurred. In that sense, it is about *efficiency*. A transportation investment is efficient if benefits exceed costs. More specifically, it is efficient if the presented discounted value of the estimated future stream of beneficial impacts is greater than the presented discounted value of the estimated future stream of negative impacts (costs). A slightly different way of expressing the efficiency relationship is as the ratio of benefits to costs: a project is efficient if the benefit/cost ratio is greater than 1.0 (benefits are greater than costs).

Appendix A talked about the “matrix display” method of evaluation: criteria are selected, and measurements related to those criteria are displayed for each relevant policy choice. It noted that an inherent problem with multiple criteria and multiple measures is that the measures are in different units and cannot be added. Thus, they must be scored or weighted. Sometimes, formal scoring or weighting occurs, but such methods are inherently flawed and, in addition, often poorly implemented.

In benefit-cost analysis, that problem is theoretically eliminated and certainly reduced. All the weighting is already done implicitly by stating all impacts in terms of dollars. Those, different impacts are, on that basis, additive, and have already been weighted (in dollars).

Though true in theory, the reality is usually that not all the benefits and costs of alternative projects or groups of projects are included in the benefit-cost calculation. Some can probably never be included in any rigorous way. For example, “social justice” is a concern not about the efficiency of the transportation investment, but about the distribution of its benefits and costs. “Compliance with the regional land-use vision” might be an important criterion for evaluating transportation investments, but one that does not lend itself easily to a measurement in dollars. And though economists have done a lot of work over the last 40 years to try to estimate the value of environmental quality in dollars, many interest groups and policy makers are not comfortable with the use of those techniques.

Thus, it seems likely that in many cases the best one can hope for from benefit-cost analysis is that it be one criterion among several for selecting transportation improvements. But even that more limited role would potentially be a big technical improvement over current more qualitative methods capable of being manipulated.

For the rest of this discussion, assume that some of the key impacts of transportation performance have been included in a measure of *net benefits*. Those impacts that could reasonably be monetized (specified in dollars) include the value of:

- time savings (or losses);
- decreased (or increased) operating costs;
- reduced (or increased) accidents (property, injury, mortality); and
- decreased (or increased) air pollution.

The primary rule for selecting a project is that net present value should be positive (i.e., the present discounted value of benefits should be greater than the present discounted value of costs).

But society cannot pursue all projects with positive net present value for a lot of reasons, including the following:

- *Projects are not necessarily mutually exclusive.* Candidate projects may overlap in benefits and costs: building a set of interrelated projects is likely to amount to different net benefits than the sum of the net benefits estimated for all the projects built individually.
- *Budgets are limited.* Decision makers may not have budget authority sufficient to build all of the projects that provide positive net benefits. This circumstance is probably rare; it is more likely that the list of transportation project “needs”

includes many projects with negative net benefits. Moreover, for projects with positive net benefits, the budget constraint could be resolved in many cases through user charges: indeed, if the projects are beneficial to users, properly structured user charges should be able to “capture” part of these benefits to provide the financing. Nonetheless, the perception of budget limitations is common, and in some cases, institutional rigidities exist that truly impose arbitrary budget constraints.

That project benefits and costs are not mutually exclusive means the project selection process should focus on alternative menus and configurations of projects in some cases, rather than on individual projects. The issue of budget limitations means a decision rule other than the simple benefits-greater-than-costs rule must be evoked. The solution is to devise project configurations (the mutual exclusivity issue) and affordable subsets of these configurations (the limited budget issue) that maximize net present value. In particular, consider dealing with the following issues:

- *Nonmutual exclusivity.* Analysts should include projects in project configurations if those projects maximize the sum of own-project and other-project net present value. By calculating beforehand the interactions of each project with all others (most of which will be zero), devising net present-value maximizing configurations is a manageable task.
- *Limited budgets.* Analysts should rank projects and project configurations in descending order, based on the ratio of the present value of benefits to costs. Candidate projects should be added to the implementation list until the budget is exhausted, starting with the project with the highest ratio.

If budgets for transportation system improvements were unlimited, (1) all projects that produce benefits in excess of their costs could be built, and (2) the transportation system would be as good as it could be, given technological and physical constraints. Budgets are limited, however, so a process must be put in place to select those projects that will provide the most benefit from the limited budget.

Several factors complicate the process of identifying and selecting the transportation projects that deserve implementation:

- *Competing political interests and community needs.* Though many sources of transportation funding can be used only for transportation, clear limits exist as to how much people are willing to assess themselves in taxes and fees to pay for all the things that might be collectively provided. Education, parks, environmental management, and so on all compete for tax- and rate-payer dollars.
- *Project interdependence.* The feasibility or efficiency of one transportation project may depend on the existence of another.
- *Competing public policy goals to produce transportation outcomes.* For example, the transportation project that does the most to reduce travel time may give most of the benefits to a small group, conflicting with goals for fairness.

Dealing with Distributional Issues

Note that the project ranking we’ve discussed in this appendix was based only on *aggregate estimates of benefits and costs that can be quantified and monetized*. But, not only do some costs and benefits resist quantification, it is also the case that *disaggregated* estimates of impacts (e.g., by subarea or subgroup) are important to the final ranking. Assume that one has done a thorough job of evaluating a transportation investment along the lines outlined above: all user and nonuser, direct and secondary, benefits and costs have been estimated for all people in the relevant geography, over an extended time period, and future benefits and costs have been discounted to present value. More must still be done before one can make a technically and politically defensible decision.

Benefit-cost analysis is sometimes criticized as being blind to the *distribution* or *equity* of benefits and costs. Distributional problems are not a conceptual fatal flaw in benefit-cost analysis from the Hicks-Kaldor perspective. If a project generates

net economic benefits, it should be possible to use tax or pricing schemes to capture enough of the benefits from the winning group and redistribute them (in an effective way) to the losing group to nullify their loss. The issue from the Hicks-Kaldor perspective, therefore, is whether it is possible to derive effective redistribution schemes.⁸

In the 1980s, economists raised the stakes on distributional issues in benefit-cost analysis by presenting ways that project-selection processes could be modified to support the more general income redistributional aims of society, if so desired. They proposed that the selection of *even inefficient projects* (i.e., those with negative net present values) might be justified under some circumstances if those projects have the effect of improving the economic status of certain targeted or protected classes of individuals.

The logic of this argument is that when society decides to assist the less fortunate, it is implicitly saying it is willing to trade off a certain amount of *efficiency* for improvement in societal *equity*. The only issue is the selection of the method for effecting the income transfers; it should, of course, have the smallest possible adverse efficiency effects. By this reasoning, it may be worthwhile under some circumstances to choose projects not strictly on the basic benefit-cost criterion (net present value of benefits greater than net present value of costs) if a project has sufficiently large, positive effects on the well-being of the targeted groups relative to the efficiency penalty.

One way for conventional benefit-cost analysis to address equity issues computationally is to apply weights to the benefits received or costs paid. If, for example, a targeted class is deemed to merit special benefits, the estimate of those benefits could be multiplied by a number greater than 1. As a result, a project is made to appear better or worse than it otherwise would in the normal, aggregated estimate of net benefits. The weights to use in this recalculation can come from decision makers via methods noted previously, or they could be derived analytically (e.g., by observing the relative treatment of various income groups in other aspects of policy—say, the tax or welfare systems—or by calculating the marginal value of additional income from other income and consumption data) (Weisbrod 1968; Gramlich 1979).

The introduction of distributional considerations (and any other qualitative assessments of impacts, for that matter) raises some problems and must be treated carefully. For example:

- To what extent does a project actually benefit or harm the targeted class? Existing transportation travel data and demand models are not particularly good at identifying the income class of beneficiaries of transportation projects. Modelers know very little about precisely who travels where in metropolitan areas or inter-regionally. In addition to this problem (and partly because of it), travel demand parameters (such as demand elasticities) are not typically differentiated by income class. Hence, the measurement of the likely effects of a transportation project on any particular identified class of beneficiaries is highly speculative.
- Who is keeping track of the aggregate effect of such concessions to income distribution? If efficiency concessions are made, over time, to select projects that provide services to certain classes of travelers, how will decision makers know when enough projects have been adopted? Measured income statistics, of course, might well remain unchanged even if many transportation projects friendly to the targeted class have been built.
- Is the development of inefficient transportation services the best way to raise incomes? This is the central question that, to date, has been begged in decision-maker acceptance or rejection of transportation projects or policies because of equity considerations. Policy toward transit, older polluting vehicles, uninsured motorists, and congestion pricing are often influenced by the implicit weighting based on concerns about the impacts on specific groups (e.g., the efficiency benefits of congestion pricing are offset disproportionately⁹ by concerns about impacts on lower-income drivers).

LIMITATIONS OF BENEFIT-COST ANALYSIS

A typical critique of benefit-cost analysis is that it measures everything in dollars or only what can be measured in dollars. In fact, though practitioners of benefit-cost analysis try to measure more things in dollars (e.g., putting a dollar value on, say, a measured increase in air pollution), they acknowledge it would stretch credulity if some impacts were measured in dollars (e.g., the benefit of civic pride or the value of political acceptability). Their recommendation, and one endorsed here, is that some impacts be measured in dollars, some be measured in natural units (e.g., parts per million, number of people affected), and some be only described (i.e., not quantified).

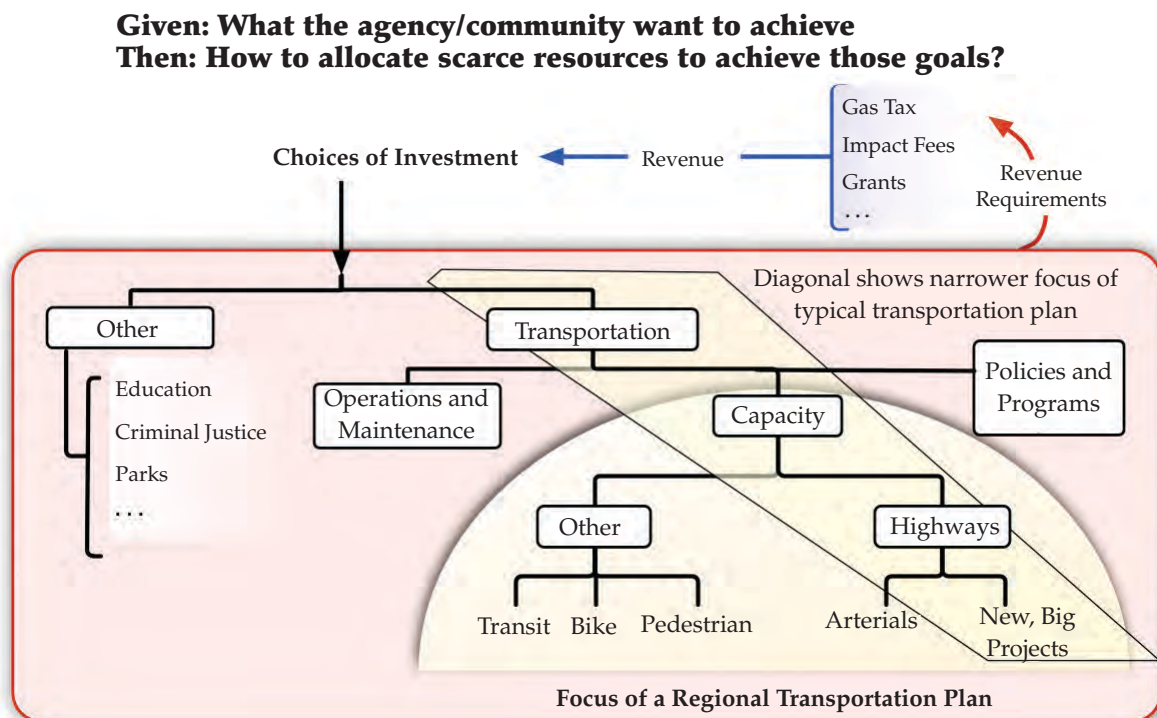
A related critique is that even if technicians understand the limitations of benefit-cost analysis, the public and their elected officials may not, with the result that some people will argue that all decisions about transportation can be made mechanistically based on, say, the benefit-cost ratio. A variation of that argument is the quantification of benefits and costs leads to a false sense of confidence in the validity and reliability of the estimates. All of the calculations rest on some or many assumptions, alternative assumptions are always possible and often justified, and final results may vary (perhaps substantially) when different assumptions are made.

The critique is not fatal to benefit-cost analysis as we have defined it. If analysts and policy makers generally accept the idea that better identification and measurement (where possible) of transportation's full benefits and costs can be useful in public debates about transportation policy and investment, the methodological debate is not about the benefit-cost framework, but about the details of the measures and measurement.

SOME IMPLICATIONS FOR APPLICATION TO TRANSPORTATION

We provided ample reasons for transportation analysts to be pessimistic about their ability to measure all the impacts (benefits and costs) of transportation projects and policies in a way that is comprehensive, mutually exclusive, and amenable to consolidation. Figure D-2 piles on. It illustrates in a stylized way a hierarchy of choices a region must make when it tries to select transportation projects (typically as part of federal requirements for a Long-Range Transportation Plan).

Figure D-2. Types of investment choices in a regional transportation plan



Source: ECONorthwest

The main points in Figure D-2 are:

- *Transportation investments compete with other public investments.* Though much of the funding is dedicated (e.g., revenues federal and state highway trust funds), much is not (e.g., allocations from state or local general funds). All agencies usually ignore that fact, in part because their missions assume they will and require them to. If you are a planner in a transportation agency, your job is to serve the public by making transportation improvements, not by building schools or providing health care. Yet, at the margin, investing in education and health care may give the public a bigger bang for the buck. You don't know, and it's not your job as a transportation planner to care. But for elected officials running a region or a state, and for the public they represent, that comparison is not only relevant—it is critical.
- *Transportation projects and programs compete with each other.* Once an amount is dedicated to transportation, the first big choice is how much of the total funding to allocate to operations and maintenance. That choice is first based on the logic that it is likely to be more efficient (bang for the buck) to keep existing capacity functioning properly now and into the future than it is to build new capacity. The decision clearly can be informed by benefit-cost techniques (in engineering terms, life-cycle costing). No elected official will disagree with the truism that it is often more effective to maintain the infrastructure we have than to build new. That makes engineering and economic sense. But decisions these officials make about regional and local transportation investments rarely implement that principle rigorously. Our work throughout the northwestern U.S. suggests that, depending on the place and the calculation, local and state governments are funding between 30 and 70 percent of what engineers and economists would estimate as cost-effective maintenance. In the context of the techniques of benefit-cost analysis, investment in the maintenance of existing capacity may yield a better return (present value of benefits over costs) than investment in new capacity. Many regional transportation plans give only perfunctory attention to that important point. Cost-effective maintenance has less voter appeal than a new highway or new lane.
- *The revenue remaining is available for other things: primarily for projects to increase capacity.* The diagonal in the figure shows where the bulk of the transportation money has gone historically: to capacity, to highways, to new and big projects. In the jargon of transportation planners, these are “modernization” projects. But a region can also get net benefits by investing in *policies and programs*. In fact, at the heart of the argument for better transportation/land use connections is the belief that investments in land policy (e.g., in the rearrangement of origins and destinations) can have big impacts on the effectiveness of transportation. Also in this category are transportation demand management policies (e.g., pricing, carpool matching). Clearly we believe these are important. Figure D-2 shows them outside the hemisphere to illustrate that they are usually not the primary focus of regional transportation planning: they should at least share the spotlight.
- *Coming back to capacity, those modernization projects could be for highways or for alternative modes.* How does that choice get made? At yet a finer level, there are choices about the types of projects. For example, in the context of highways, should a region invest in big, new projects (new beltways or bridges or big digs), or in more modest expansions of existing arterials? More difficult yet, should it seek to solve congestion problems by investing in transport improvements of any type or by funding land-use policies and programs that change origins and destinations?

The reality is that technical analysis cannot answer these critical questions unequivocally even if it were allowed to try, and it almost never is. Some of these big allocation decisions are made quickly and politically. Some jurisdictions use flexible funding from federal and state highway trust funds to support transit; others do not. How do they decide? Almost certainly not by using benefit-cost analysis or any

other rigorous evaluation technique to try to get to some estimate of net benefits by investment alternative.

Much, much more could be said about specific techniques for project evaluation. We are not going to say it here. Those interested should look in the professional literature: multiattribute or multicriterion utility analysis (e.g., Merrick and Garcia 2004); analytical hierarchy process (e.g., Saaty 1994); FHWA and TRB websites; websites for large MPOs.

Our goals for this appendix are less grand. Our point, made at several places in this report, is that while technical analysis cannot be definitive, it is not irrelevant. We believe that thinking about transportation investments in the context of benefits and costs is the right way to approach a technical evaluation of the investment alternatives. The kind of thinking that benefit-cost analysis requires is probably more important to solid decision making than the formal techniques for estimating benefits and costs. Trying to add some rigor to the evaluation of projects—the kind of rigor suggested by Appendices A, C, and D—improves the debate about these complicated investments and, by doing so, improves the final decisions.

In the context of the framework we established in Appendix A, benefit-cost analysis and a more rigorous estimate of net benefits (efficiency) can feed into a matrix-display evaluation format as one of a few key criteria. In our opinion, it should probably be the most important one: regions should be making transportation investments to get the biggest bang they can in transportation performance (improvements in travel time, safety relative to a base case) for a given amount of investment. We think that wedding the concepts and some of the techniques of benefit-cost analysis to a multicriterion matrix display is a practical direction for system and project evaluation.

An explicit benefit-cost framework for evaluating transportation options has the important additional advantage of explicitly dealing with the pricing of transportation and the demand/supply relationships, which are part of the evaluation of user benefits. At a minimum, some basic benefit-cost analysis would show some of the worst boondoggles for what they are and would help regions focus on what they are giving up when they allocate scarce public resources to one project or another. Small (2003, 168–70) says it well:

... just as accounting rules curtail the tendency of corporations to manipulate financial statistics in their favor, professional standards for project evaluation limit the extent to deception that [sic] can pass for objective analysis. ... The best method of presentation is one that makes it possible to understand and justify political decisions that are in the interests of the citizenry at large, while embarrassing those who would make decisions favoring only narrow interest groups.

APPENDIX D NOTES

1. This appendix draws on many related reports done by ECONorthwest (primarily by Terry Moore and Randy Pozdena), including ECONorthwest/Parsons Brinckerhoff (2002) and ECONorthwest (2003). That work was done independently from, but is consistent with, Small (2003).
2. Making a decision in advance about what is “reasonable” will be difficult. The point is to find the balance between: (1) going into the evaluation with a very narrow focus (perhaps for institutional or political reasons), and (2) burdening the evaluation with many fringe ideas that have little hope of surviving the screening process. The standard, and best, solution is for the evaluation to screen more than once, at increasing levels of detail as alternatives are narrowed.
3. In economic jargon, this is benefit-cost analysis, and the resulting measure is either net benefits (benefits less costs) or a benefit/cost ratio (benefits divided by costs).
4. In a benefit-cost framework, *benefits* can be positive or negative: a negative benefit is the same as a *cost*.
5. Induced demand is what Downs (2004a) describes more graphically as triple convergence: users change time, route, and mode to converge on the new capacity provided.
6. Arrow and Kurz (1970) and others argued that the *social rate of time preference* should be lower than the private market rate of time preference (i.e., the private market discount rates is higher than the one that should be used in public projects).

7. The correct arithmetic is more complicated, but this approximation works.
8. Even if it is not possible to implement a redistribution scheme within the context of the project itself, the adverse effects could be nullified through tax or other more general redistribution schemes. Hence, from a Hicks-Kaldor perspective, distributional issues are not a fatal flaw to implementation of efficient projects.
9. "Disproportionately" in the sense that the Hicks-Kaldor criterion might show that the disbenefited parties could be, theoretically, compensated for their losses and there would still be net social benefit, but the proposed project dies because of concerns about low-income drivers.

Models Used in Transportation Planning¹

By Joel Freedman, PB, Portland, Oregon

This appendix describes the general characteristics of the mathematical models most used by agencies charged with planning and managing transportation systems. It starts by describing the four-step urban transportation modeling system (UTMS)², which has been in widespread use since the 1960s. The four-step paradigm is somewhat flexible and has allowed for variation in particular model formulations across regions. It then describes the weaknesses of the four-step modeling paradigm, and the ways in which models have evolved to address these concerns. Readers wanting a parallel description, also at an introductory level, should see Beimborn (1996).³

HOW TRANSPORTATION AGENCIES TYPICALLY MODEL TRANSPORTATION AND LAND-USE SYSTEMS

Since the 1970s most big cities and metropolitan areas have used some form of UTMS. UTMS is a sequential process for organizing a series of computer models to simulate urban travel patterns. Figure E-1 shows the typical process.

The analyst inputs data (almost always estimated outside the transportation model) that describe the pattern of land use (the locations of population and employment, also known as the demand), and the characteristics of the transportation system (or supply). That information feeds into a sequence of transportation models. The results from one step feed into the models in the next step (shown by the solid arrows) and may feedback into previous steps (shown by the dashed arrows). The entire model system describes the demand for travel in a region for a given set of inputs, commonly referred to as a scenario.

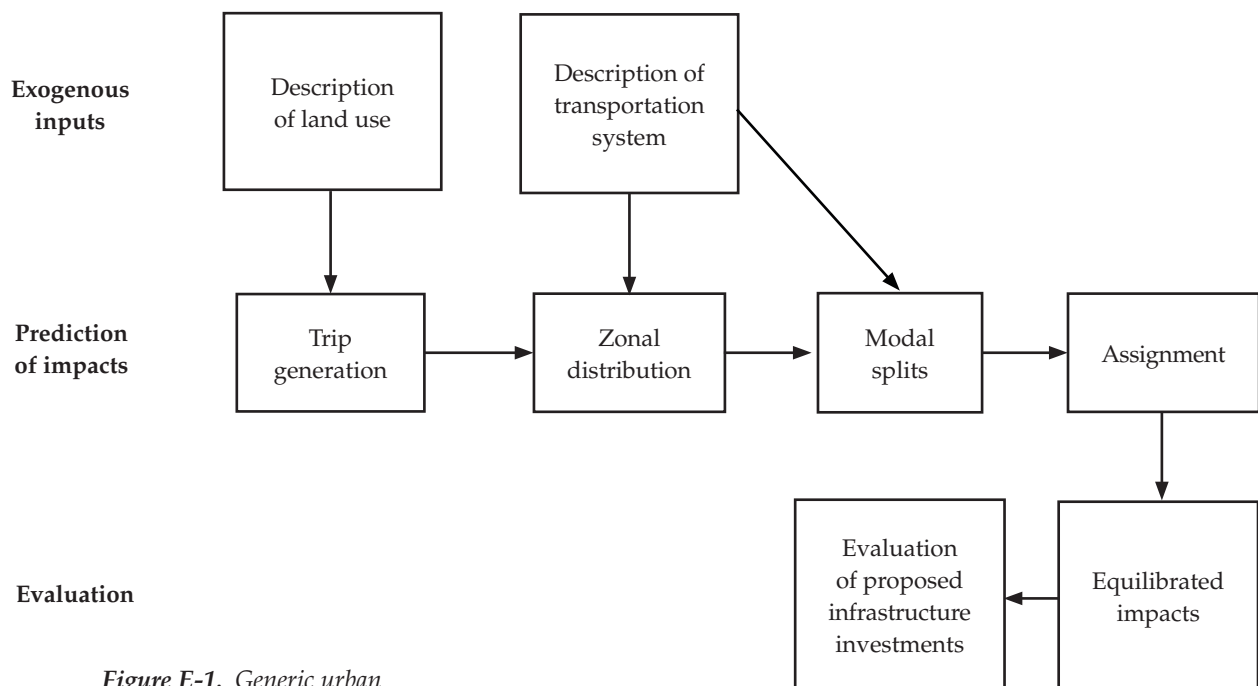


Figure E-1. Generic urban transportation modeling system

Source: Adapted by ECONorthwest from Beesley and Kemp 1987

Inputting the characteristics of proposed changes in the transportation system allows evaluation of those changes. Typical outputs include total trips, flows of person trips between different parts of the region, trips by mode, vehicles/passengers on the highway and transit system, and resulting congestion. Trips may be summarized by different purposes and socioeconomic market segments, to the extent that the particular model system considers such segmentation. UTMS outputs may be fed into other post-processing steps to determine air quality impacts of particular scenarios, noise, more detailed traffic characteristics, or optimum tolls.

This four-step UTMS approach was developed in the 1960s to help regions plan many of the major highway projects constructed in the 1960s and 1970s. The methods used within the four-step model process have evolved to meet changing analytical needs, such as the introduction of more sophisticated mode choice models within many model systems in the 1980s and 1990s to more realistically model transit demand. The basic structure of the sequential forecasting process has remained largely constant. The four-step model paradigm has its roots in the days of mainframe computer systems whose processing power is easily surpassed by desktop computers today. New computing power has been consumed by more sophisticated model components and/or more explicit treatment of time and space in the model system.

The particular form of each model within the UTMS is mostly independent of the particular software used by an agency, with the possible exception of the last step, trip assignment (see below). The commercial software employed by the agency is simply a tool for implementing the model system; it is up to the analyst to build the model using those tools. In many cases, consultants with national experience in developing models are employed by the agency to either provide expert advice or to take on model development activities for agencies.

The transportation analyst starts by dividing the urban area into Transportation Analysis Zones (TAZs). UTMS analyzes travel *between* these zones, so the analyst defines as many zones as the data allow, anywhere from several hundred for small cities to several thousand for large metropolitan areas. The number of TAZs in a model is a trade-off between computational burden (more zones means more calculations) and the desire for a good spatial representation of the model area.

Data requirements include (1) information about the number and characteristics of households and businesses within each zone, often referred to as transport demand, and (2) information about the characteristics of travel between zones (e.g. modes available, cost of a trip on each mode), commonly referred to as transport supply. Transport supply is represented in the model system by one or more interrelated transportation networks, which include arterial roads and highways at a minimum, and often represent facilities at the level of minor arterial or collector. Transit networks are also coded to represent bus and rail routes and their characteristics. These exogenous inputs are shown in the top row of Figure E-1. Systems that provide more detailed results require larger amounts of data.

The specification of land use comes from outside the four-step transportation model. Most states, for example, have computer models that forecast population and employment growth by region. The urban area can determine its share of regional growth and allocate that growth to the TAZs used in transportation modeling in any number of ways (e.g., manual allocation, computer models, expert guesses). For example, a growth allocation process was recently conducted in the San Francisco Bay Area, using a Geographic Information System (GIS)-based planning tool called PLACES. Public workshops were held in which participants allocated regional growth to TAZs using assumptions about zoning and development patterns. That information was then refined by the regional agency responsible for growth forecasts for the region. Land-use allocation and modeling tools are also available in which population and employment forecasts are allocated to TAZs according to mathematical equations or are explicitly represented within a modeling system (see below). Such tools tend to be more expensive to develop and maintain than manual or GIS-based allocation methods, and are therefore less common.

The four-step UTMS process (the second row, labeled “prediction of impacts” in Figure E-1) first calculates the number of trips into and out of each zone (*trip generation*). Trip generation models are composed of two submodels: (1) trip productions, which are generated largely from households, and (2) trip attractions, which are generated largely by employment. Cross-classification methods are commonly employed for trip production models, wherein households are stratified by key socioeconomic variables, such as household size, income, and/or number of cars owned. The total number of households within each socioeconomic category is estimated for every TAZ, and a separate trip rate, estimated from home-interview survey data, is applied to each household type. This method is used to reflect observed relationships between the socioeconomic variables and the numbers of trips generated; for example, households with more persons generate more trips on average. Regression models are often used to determine the rate of trips attracted by different types of employment, such as basic, service, and retail. This stage of the four-step process estimates *trip ends*, with one production and one attraction representing each end of one trip, identified separately by purpose (work, shop, etc).

The four-step process then distributes trips between zones (*trip distribution*), linking the trip ends produced and attracted in trip generation into trip tables. Though many types of trip distribution models exist, the most common is called the gravity model. The gravity model distributes trips produced in zone A to attractions in zone B based on the relative attractiveness of zone B compared to all other possible attraction zones. The relative attractiveness of zone B is inversely related to the spatial separation between zone A and zone B, and directly related to the size (number of attractions) of zone B. The gravity model is calibrated by iteratively adjusting a set of factors that vary by spatial separation (distance, travel time, or generalized cost), until the gravity model results in a trip-length distribution that matches observed trip-length data from a home-interview survey.

Although gravity-based trip distribution models attempt to accurately simulate the observed distribution of trips between zones, the models are not based on behavioral theory and do not operate well under certain conditions. An alternative form of trip distribution model, based on utility-maximization choice theory, is becoming more common. This model form is referred to as a logit-based destination choice model, which has some advantageous mathematical properties compared to the gravity model (Ben-Akiva and Lerman 1985).

The third stage in the process is referred to as *mode choice*. The mode-choice model estimates the probability that an individual traveler with certain characteristics will choose a particular mode of travel given the costs of taking the trip by each mode.⁴ This model is also based on utility-maximization theory, which states that individuals evaluate all alternatives available to them and then choose the best alternative (the one with the highest utility). To determine the utility of each alternative, the model uses information about:

- the travel times and costs associated with each available mode (collectively, all these costs are sometimes referred to as the generalized travel cost);
- the characteristics of the members of households that travel;
- the type of trips individuals are making;
- the relative weights that individuals attach to each component of time and cost; and
- the relative effects of person, household, and trip characteristics in determining mode. In the four-step model, the probabilities for each mode are multiplied by the number of trips estimated by the trip-distribution model to determine the number of trips by mode.

The *trip assignment* model assigns trips to specific routes between zones for each mode. For auto assignments, most agencies use an assignment technique referred to as *equilibrium capacity-restraint*. This algorithm recognizes that the cost of travel on a

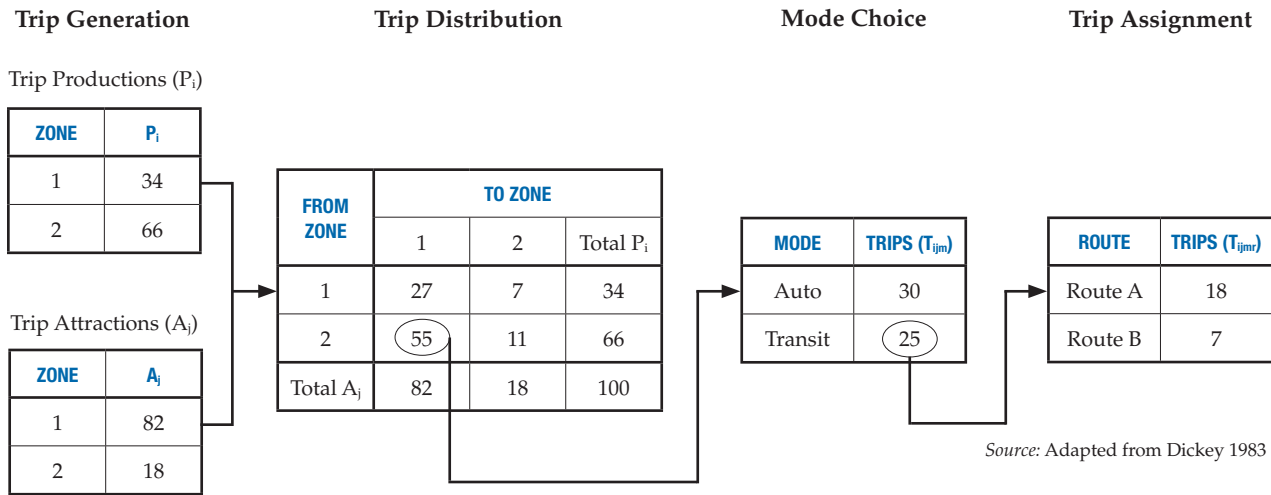


Figure E-2. Steps in a travel-demand model

particular route increases as the volume of traffic increases and is based on the theory that travelers will try to find the least-cost route, taking into account congestion imposed by all other travelers. The algorithm iteratively assigns trip tables from mode choice to the auto network until it finds an equilibrium state for the assignment process where no traveler can benefit from changing routes. Many different algorithms are used for assigning transit trips to transit routes, taking into account in-vehicle time, out-of-vehicle time, fare, and other factors. A description of the various approaches is beyond the scope of this appendix.

Figure E-2 shows an example of a four-step travel-demand model for a simple two-zone region. The first step is trip generation; trip productions (P_i) and trip attractions (A_i) are estimated for each TAZ. The outputs from this step are fed into the trip distribution model, which creates a trip table (T_{ij}). The trip table is then fed into a mode choice model which estimates trips by mode for each i - j zone pair (T_{ijm}). Finally, the outputs from mode choice are fed to a trip assignment model which assigns a route to every trip (T_{ijmr}).

ADVANCES IN TRAVEL-DEMAND MODELING

The four-step model approach has been criticized for a number of reasons. The most common are that: (1) there is significant aggregation bias within the four-step framework; (2) certain model components are weakly supported by underlying theory regarding travel behavior; and (3) the basic model takes land use as given: within the model, changes in the transportation system affect travel demand but not land use.

Aggregation bias occurs in a number of dimensions within the four-step model paradigm.

- *Market segmentation bias.* Because TAZs are the basic unit of demand in four-step models, it is not possible to introduce person-level variables in models that significantly affect travel, such as age of traveler, gender, and occupation. Such bias limits the sensitivity of models to policies that have differential effects across these person segments (e.g., parking pricing) and limits the ability of the models to provide information for equity analysis/environmental justice.
- *Temporal aggregation bias.* Most four-step model systems consider only two aggregate time periods (peak and off-peak) at all model stages except for assignment. This limits the sensitivity of four-step models to policies that vary by time of day (e.g., variable pricing, extension of transit service into the evening, and so on.).
- *Spatial aggregation bias.* The geographic division of the model area into TAZs and the generalization of the transportation network limit the sensitivity of four-step models to policies that effect specific sites/parcels, (e.g., pedestrian-friendly

design). The division also reduces the ability of models to reflect network characteristics that affect travel (e.g., lane configuration, signal timing, vehicle queuing, and so on).

That the four-step model paradigm limits behavioral fidelity is apparent both from the sequential structure of the four-step process and the forms of models used for each component. Clearly the number of trips, their destinations, and the modes used for trips are highly interrelated for any individual traveler and across travelers within a household in the real world. Those interrelationships are only weakly represented within the four-step paradigm, if at all. The gravity model is an example of a model component with very little underlying behavioral theory.

All of these criticisms have resulted in improvements to the four-step modeling paradigm. Many agencies have adopted models with more market segmentation (through the consideration of many different types of household segments), added zonal detail to overcome spatial aggregation bias, and attempted to model multiple time periods. Agencies have also sought to improve the behavioral fidelity of models, in part by adopting destination choice models and implementing feedback loops to overcome the sequential nature of the model system. The four-step model paradigm, however, is limited in its ability to accommodate increased market segments and time periods, and the essential framework of modeling trips as separate entities is a major limitation in improvements to the behavioral fidelity of models.

Tour-Based Models

A new paradigm that addresses many of these criticisms, often referred to as *activity-based* or *tour-based modeling*, is beginning to move from research into practice. Tour-based models are different from trip-based models in three fundamental ways:

1. Individual travelers are the basic decision-making unit in the model system, as opposed to TAZs. This allows the consideration of person characteristics and interpersonal relationships to be explicitly represented;
2. The tour (a series of trips from home to one or more destinations and back to home) is the basic unit of travel as opposed to the trip. This allows trips within a tour to be modeled consistently, with respect to destinations, modes, and times of day, increasing the behavioral fidelity of the model system.
3. Tour-based models are implemented within a microsimulation framework, in which a synthetic population is created, representing every household and person within the study area. Then models are applied to each person that result in discrete choices made which condition the set of choices for the next set of decisions. For example, suppose an auto ownership model is applied to a two-worker household and the number of autos chosen is 1. The first worker goes to work, and chooses drive-alone as the mode for the work tour. Next, the second worker travels to work at the same time as the first worker. But since there is no auto available, the mode choice is constrained to either share a ride with the first worker, or take transit. Tour-based models implemented with micro-simulation allow the introduction of many more explanatory variables than are possible within a trip-based model, overcoming many of the aggregation biases inherent in the trip-based model paradigm. Note that the actual form of models used in the tour-based model system can be very similar (e.g., logit models) as those used in the four-step model paradigm.

Tour-based models have been adopted by at least three transportation agencies in the U.S. The San Francisco County Transportation tour-based model was adopted in 2000 and models San Francisco County residents (Bradley et al. 2001). The New York Metropolitan Transportation Council Best Practice Model was adopted in 2001 and considers the entire New York metropolitan region (Parsons Brinckerhoff 2005). The Mid-Ohio Regional Planning Commission model was developed in 2004 and covers the Columbus, Ohio, planning area (Parsons Brinckerhoff 2006). There are

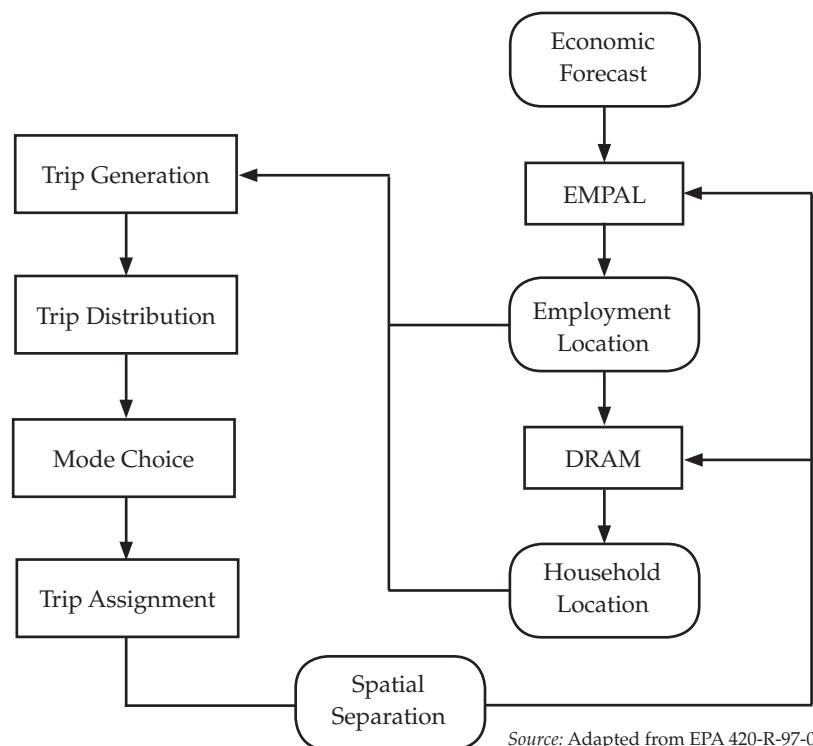
tour-based models under development for at least four MPOs in the U.S.: the Atlanta Regional Commission, the Denver Regional Council of Governments, the Metropolitan Transportation Commission located in the Bay Area, and the Sacramento Council of Governments. Many more MPOs are considering moving to the tour-based paradigm in their next major model update.

Integrated Land-Use and Transportation Models

The third criticism of the four-step model paradigm—that it takes land-use inputs as a given without reflecting the effects of transport accessibility on the type and location of development—is being addressed by a growing number of integrated land-use and transportation model systems. The key improvement of these models is that they allow land use to be affected by changes in travel accessibility reflected by the transportation model, instead of assuming that land use will remain fixed regardless of transportation system changes. Most of these model systems iterate a land-use model with a four-step travel-demand model. That interaction helps their predictive power, probably substantially. They fall into two categories: (1) systems that allow land use to interact with transportation but contain no explicit land-use economics; and (2) systems that make more use of land-use economics.⁵

Planning agencies in several urban areas already have implemented models of the first type largely because these models require data similar to that required by the four-step model.⁶ ITLUP (Integrated Transportation and Land Use Planning) has been particularly popular. ITLUP is actually composed of three components; DRAM (Disaggregated Residential Allocation Model), EMPAL (Employment Allocation Model), and a travel-demand model. ITLUP relieves the analyst of predicting future land-use patterns. The analyst inputs expected aggregate growth in employment and population, the locations of existing employment and population, and characteristics of the transportation system. ITLUP uses a gravity model to allocate new employment and population to zones. This calculated pattern of land uses feeds into the four-step transportation model to determine traffic flows. This information feeds back to the land-use model, which updates land-use patterns based on the effects on accessibility. The model iterates until little change occurs in each iteration (i.e., until the system converges). A schematic of the model system is shown in Figure E-3.

Figure E-3. *Integrated Transportation and Land Use Planning (ITLUP) Model*



Source: Adapted from EPA 420-R-97-007

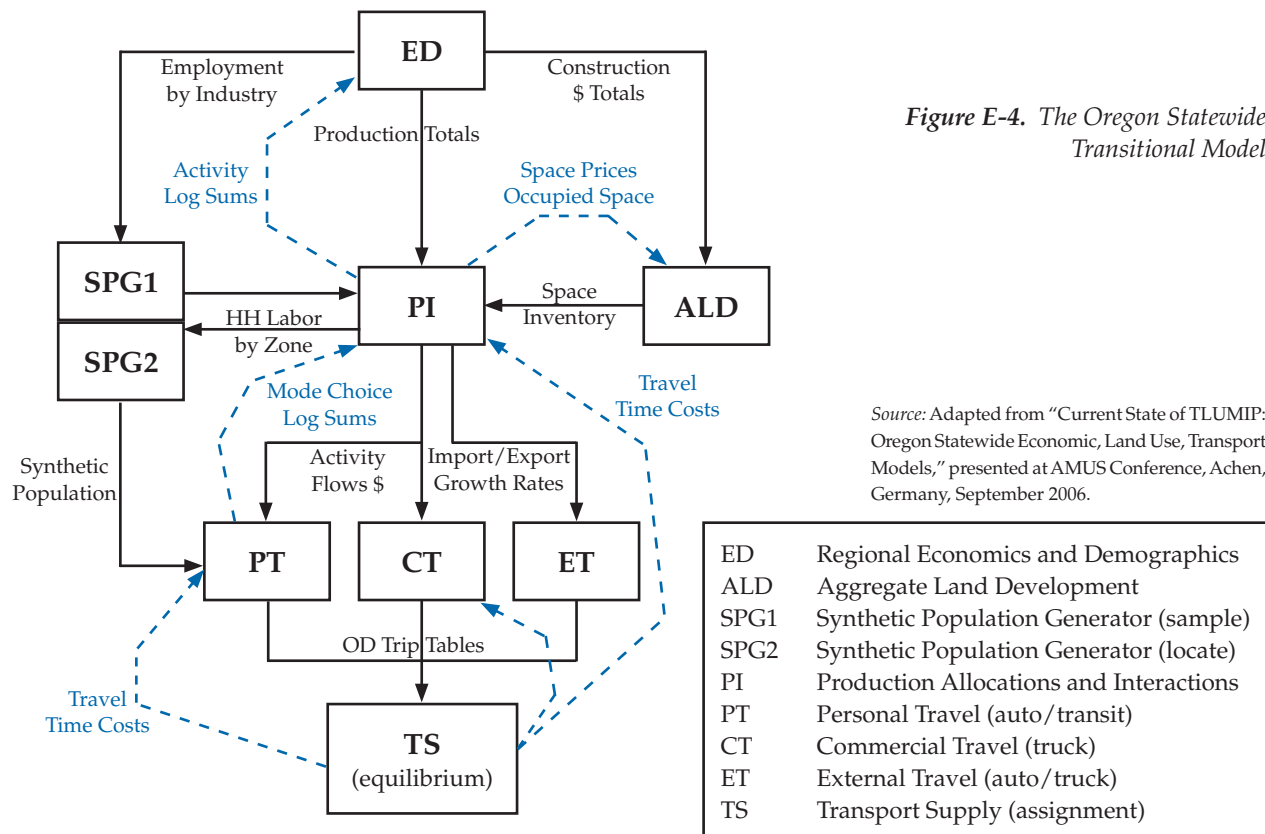
The attractions of ITLUP are that it simulates current conditions well, incorporates changes in the transportation system which affect land use, and requires no more data than that typically collected by transportation planning agencies. But it requires assumptions about overall land-use growth as an input and shares the weaknesses of the gravity model at its core. Gravity models have the flexibility to simulate current conditions well (either land-use or travel patterns) but perform less well for forecasting, especially in cases where significant changes in underlying economic conditions are occurring. Without explicit consideration of the variables driving travel and development patterns, currently estimated gravity-model parameters will likely not yield good predictions.

Models of the second type offer a more theory-driven approach to specifying land use and including economic methods to help forecast growth in employment and population. MEPLAN for example, is an integrated land use/transport model that relies on the production and consumption relationships among business and households embedded within an economic spatial input-output model, to drive the location of land use activities. Thus businesses locate in order to minimize cost to reach raw materials and end markets, while household locate where they can sell their labor and buy necessities, a more economically driven approach than the gravity model of the ITLUP model. Once located, the flow of goods and services among producers and consumers (including labor produced by household and consumed by industry) is translated into travel and assigned to the transport network. MEPLAN still requires external forecasts for the “basic” drivers of the economy (imports/exports or basic industry), as well as zonal measures of land capacity. Within the model, floor space rental rates by zone signal the location of desirable locations for future land-use change, thereby reflecting economic market forces in land-use decisions.

These models should improve the accuracy of forecasts. However, they require more detailed data about land uses and the transportation system than transportation agencies typically collect. MEPLAN has been tested against reality (validated) in Bilbao and Sao Paulo. 1000 Friends of Oregon (1991, 27) reported, “The Sao Paulo results showed good aggregate forecasts, especially in the transportation sectors, though zonal forecasts were less satisfactory. The forecasts in the Bilbao study were mostly fairly accurate.” TRANUS, a similar land-use transport model, has been successfully applied and calibrated as a statewide model for Oregon under the Oregon Transportation and Land-Use Improvement Program (TLUMIP) (Upton 2003). Figure E-4 shows the structure of the TLUMIP model.

Since the 1990s, two other notable model systems have been developed and applied to metropolitan areas as well as statewide within the U.S. The UrbanSim model system is a logit-based system of models that seeks to represent the decisions of households, businesses, and developers that give rise to development. A set of logit models (there is one package of models for each of the actors described above) is used to allocate exogenous regional population and employment projections to parcels. UrbanSim is typically run in five-year increments; at each increment, the outputs from the land-use model are fed into a transportation model, which then feeds travel accessibilities into the next land-use simulation year. The model requires information about existing land-use, zoning and growth controls, population, and employment. UrbanSim has been applied and successfully calibrated in a number of regions in the United States, including Salt Lake City, Houston, and Honolulu (Waddell 2002).

The PECAS model is also an integrated land use/transport system, that builds upon the MEPLAN and TRANUS economic framework. In PECAS, the relationships among industries and households are expanded to explicitly represent the exchange of goods and service commodities and internally calculate market-clearing commodity prices that inform location and land-use decisions. The resulting commodity flows in economic units (dollars) provide a valuable starting point for freight modeling and travel demand, where they can be converted to vehicle trips and assigned to a transport network. PECAS also includes a more detailed microsimulation of the land-use system at a small grid or parcel level. As in UrbanSim, land use in PECAS is modeled as the



outcome of developer actions driven by expected revenues (from model-calculated floor space prices) and costs of developing various land use changes, given constraints imposed by zoning and other growth management policies. A key advantage of the PECAS framework is that it is capable of linking to a regional economic model for its forecasts and a variety of transport model frameworks. Feedback to an economic model allows internally-calculated changes in transport and land-use conditions to affect the overall growth or contraction of the regional economy. PECAS has been applied in several regions, including Sacramento, and Baltimore. PECAS is being calibrated at a statewide level in both Oregon and Ohio; in both cases, a tour-based microsimulation model has been developed in concert with and integrated with the land-use model system (Weidner et al. 2007).

A 1991 survey of transportation planners in the 20 largest cities in the nation (1000 Friends 1991) found that eight of 19 respondents used no formal model to allocate land use to TAZs; five used a formal land-use model, but their transportation model did not provide feedback to land use; four were in transition, moving from a traditional approach to the implementation of a system that allows land-use and transportation models to interact; and two had implemented a computer modeling process that interacts land use and transportation. As of 2005, there were at least 80 worldwide applications of integrated land-use/transportation models as described in an examination of six such models (Hunt, Krieger, and Miller 2005).

Undoubtedly, more detailed evaluations of these advanced models will appear as more transportation agencies apply them. Our view is that they represent movement in the right direction. The data they require and the analyses they provide shed needed light on many aspects of urban development. But decision makers should not count on these models to provide the accurate forecasts of land development that they would like to have. These models have not advanced to the point that they can replace informed professional judgment. As we discuss in Chapter 7, the failure of land-use, transportation, and growth management policy results more from an

inability of government to get consensus about any significant policy change than from an inability to predict which changes are most likely to lead to more efficient land-use and transportation systems.

Alternatives to Large-Scale Models

The models described here are big, complicated, and data hungry. They often cannot deliver answers quickly for all the variations on policy questions that get asked during evaluation of transportation options, especially questions about local impacts. Cervero (2006) points out that many of the concerns of supporters of smart growth—concerns about things like transit-oriented development and the ability of land-use patterns and design to influence travel demand at the neighborhood level—are not things that big models can deal with quickly or well (if at all). He describes the state-of-the art for post-processing models (which pivot off the data from a larger travel-demand model) and direct models (stand-alone models that estimate local travel demand as a function, in part, of land-use, urban and transit facility design, and transit service parameters).

APPENDIX E NOTES

1. This appendix, originally written by Joel Freedman for the first edition, has been substantially updated by the authors for this edition.
2. The process is also often referred to as the urban transportation planning (UTP) system. Our description follows that in Meyer and Miller (1984).
3. Also available at www.environmentaldefence.org/documents/1859_InsideBlackBox.pdf (accessed October 2006).
4. Note, in the context of discussions about the full social cost of a trip in Chapter 4, that the discrete-choice analysis looks at the cost of the trip to the individual, not to society. That is the correct perspective for predicting individual and aggregate travel behavior, but not for determining optimal system configuration.
5. This section follows LUTRAQ, Volume 1.
6. Systems of this type include ITLUP from S.H. Putman and Associates, TRANSTEP from R.J. Nairn & Partners, and TRACKS from Gabites Porter, Ltd.

Congestion Pricing: Case Studies¹

Chapter 4 makes the case that a big part of urban congestion problems result from improper pricing of transportation facilities and that those congestion problems are best solved, at least in theory, by pricing policies.

The first edition of this book predicted that pricing would become more prevalent: congestion would increase, supply-side solutions would become increasingly expensive, pricing technology would improve and its costs would decrease, and all these factors would chip away at the conventional wisdom that pricing was politically infeasible. This appendix provides some evidence that transportation policy and investment are moving in the predicted direction.

Cordon pricing could also be called area pricing: some sub-area of a metropolitan area gets priced while other areas do not. The cordoned area is usually a center rather than a corridor, and always the central area of the biggest city of the region for starters, because that's where the most congestion is. The technology for cordon pricing can be simple (e.g., paper permits on the windshield with visual inspecting and ticketing by a patrolling staff) or more sophisticated (e.g., electronic reading, billing, and ticketing of the license plates of vehicles crossing a cordon line). London, Stockholm, Singapore, and Norway all use some variation of cordon pricing.

Corridor pricing could also be called facility pricing: some facility (usually a limited-access, high-speed, high-capacity highway) gets priced while other facilities do not. Again, pricing can be low-tech (pay a person at a toll booth) or higher-tech (prepaid smart cards read electronic while the vehicle that passes without stopping). The examples from California, Canada, Texas, and Florida all use some variation of corridor pricing.

Not described in these case studies is a very-high-tech implementation: GPS-based tracking and pricing, which we will call *area pricing*. That technology makes the distinction between cordon pricing and area pricing less relevant: it can be applied to either or both simultaneously. It allows “ubiquitous pricing” with the constraints being mainly storage capacity and bandwidth. There are no examples in this appendix, but a sidebar in Chapter 4 talks about an on-going experiment with GPS pricing in Seattle.

CORDON PRICING

London

Program

To date, the most ambitious congestion pricing program is London's. This program debuted in February 2003 to a skeptical public but has since preformed extremely well, providing a visible success story to congestion pricing advocates.

The London system is based on cordon pricing. Drivers are charged electronic tolls when they cross a line into the heart of the central business district (CBD). This type of program is designed to remedy congestion in a specific geographical area and works well for a city such as London that has no room for road expansion and sees limited car traffic—less than 10 percent of total travel in Central London was via automobile prior to the program according to Litman (2006).

In 2000, Ken Livingstone, London's liberal candidate for mayor was elected having run with a congestion pricing proposal as a major piece of his platform. While the proposal was highly criticized and seemed too radical for many, the mayor pushed forward capitalizing on popular support to find solutions to heavy congestion.

SOURCES FOR LONDON CASE STUDY

Leape, J. 2006. "The London Congestion Charge." *Journal of Economic Perspectives* 20(4).

Litman, Todd. 2006. "London Congestion Pricing." Victoria Transport Policy Institute. Victoria, B.C. <http://www.vtpi.org/london.pdf>. (accessed July 2006).

Transportation for London. 2006. "Central London Congestion Pricing: Forth Annual Report." <http://www.tfl.gov.uk/tfl/cclondon/pdfs/FourthAnnualReportFinal.pdf>. (accessed July 2006).

Livingstone's program was eventually implemented as a response to years of crippling congestion in central London. The program aimed to reduce congestion, enhance bus service, decrease time travel for car users, and create a more efficient distribution system for the city's goods and services.

Stakeholders' primary concerns were: (1) public transit would be unable to handle the increase in ridership; and (2) congestion would simply be transferred to areas outside the city center.

The program debuted with a £5 charge to any vehicle driving in the city center (about 22 square miles) during the hours of 7:00 a.m. and 6:30 p.m. during weekdays. (This charge was raised to £8 in 2005.) Drivers pay tolls by internet, at retail outlets, in booths, and via cell phone text message.

The toll is enforced through a system of 400 video cameras around the city that record auto license plates and cross-reference them against a database of people who have paid the toll. Those who have not paid the toll are fined £80.

Certain vehicles (e.g., taxis, motorcycles, buses, vehicles used by disabled persons, and alternative fuel vehicles) are exempt from the charge. Additionally, people living inside the cordon receive a 90 percent discount on the toll.

Results

The independent research group, Transportation for London (TfL), was commissioned to issue annual reports monitoring the program's successes and failures.

It reported that though expectations for the program were difficult to quantify prior to implementation due to the unprecedented size and scope of the program, the goals of the mayor and TfL were to reduce traffic levels inside the zone by 10 to 15 percent and to reduce vehicle delays by 20 to 30 percent.

After just one year of operation, TfL found that their goals had been met. Traffic levels (measured by "flow") fell by 15 to 18 percent and continued to moderately decrease through 2005. Congestion (measured by speed of travel) fell by 30 percent but since has increased slightly, stabilizing at a 22 percent reduction through 2005.

Though traffic increased slightly after the first year of the program (4 percent) on the road just outside of the cordon, this was lower than expected and has decreased slightly through 2005.

Fears about public transit's ability to handle the increase in ridership were alleviated by an increase in reliability and availability of bus service. With less congestion and more funding from congestion tolls, excess waiting time for bus service fell 30 percent in the first year and 18 percent in the second year of the program as the revenue from congestion tolls enhanced the public transit system. Despite the 37 percent increase in ridership, the system met demand.

The program was less immediately successful for businesses; a general economic slowdown combined with the implementation of congestion pricing led to poor business performance following the charges in 2003. However, according to TfL, congestion pricing was not a significant contributor to poor economic performance. Since the initial dip in business, the economy has turned, and many businesses have recognized the benefits of the pricing scheme, including increased pedestrian traffic and improved transportation efficiency.

According to Litman (2006), the program is projected to earn £190 million in total revenue and £97 million in net revenue in 2004/05 budget year. This money is to be funneled back into transportation infrastructure.

The program has been widely recognized as a success, even by people initially opposed, and mayor Livingstone was reelected in 2004 largely on the perceived success of the program. A more recent evaluation (Leape 2006) concludes that congestion pricing in London "has been both a practical success in reducing congestion and a popular success."

Stockholm

Program

Using London's cordon-based pricing system as an example, Stockholm began charging drivers as part of a congestion pricing pilot study in January 2006 to determine if they should implement their own cordon-based pricing system. The trial ran through July 2006.

Similar to London, Stockholm's trials were designed to curtail growing congestion in the nation's largest city. The Stockholm program emphasized environmental concerns more than London's. "It is important to me for Stockholm to become an exciting region in Europe," said Mayor of Stockholm Annika Billström. "From an international perspective, it is important to not only have economic growth, but environmental growth. Many cities have serious environmental issues. We are now doing this trial with a modern, exciting, new system which the rest of Europe and the world can learn from." (<http://www.govtech.net/news/news.php?id=98651>, accessed 7/19/06).

The city wants to improve accessibility and environmental quality by reducing traffic on the busiest roads by 10 to 15 percent, reduce emissions, improve the urban environment, and provide more money for public transportation.

The program charges drivers that cross a cordon surrounding the central city on weekdays between 6:30 a.m. and 6:29 p.m. Drivers are charged between 10 and 20 Krona (\$1.39–\$2.79 in US\$ (2006)) on a sliding scale depending on the time of day they cross the cordon, with a maximum daily fee of 60 Krona (\$8.37 in US\$ (2006)).

Traffic flow is not affected by fee collection as cameras record vehicle license plates. Some vehicles are also equipped with tags that record their movement and automatically deduct congestion fees.

London's exemptions for taxis, emergency vehicles, motorcycles, alternative fuel vehicles, and vehicles used by disabled persons exist in Stockholm as well. Additionally, drivers passing through the city are allowed to drive for free.

Similar concerns existed for Stockholm to those of London. How would people adapt? Could the public transportation system handle increased capacity? Would roads outside the cordon see an increase in congestion?

Results

Results as of June 2006 have been promising. According to the city of Stockholm's Expert Group Summary of the program, the goal of 15 percent less traffic in and out of the central area was met and exceeded, with an actual observed reduction in traffic of 20 to 25 percent. Queue times (amount of time spent waiting in traffic) fell 30 to 50 percent, and emissions decreased 14 percent in the inner city.

Drivers experienced significantly smaller increases in traffic than had been predicted on the roads immediately surrounding the cordon. Traffic also decreased in areas further outside of the cordon.

During the trial program, buses operated at higher speeds, and rider satisfaction is high. Public opinion has changed since implementing the program; prior to implementation only 44 percent of the population thought that the program was good idea. As of June 2006, 54 percent believe it is a good idea; in September 2006, nearly the same percentage (53 percent) voted in favor of continuing the system (<http://www.streetsblog.org/2006/09/19/stockholm-votes-for-congestion-charge-and-its-opponents/>).

Projected annual net revenue for the system is 760 million Krona/year. At an initial cost of 3.1 billion Krona/year, the system is expected to pay for itself in four years. This is a much faster payback than that for rail or road systems (usually 15 to 25 years).

Singapore

Program

Singapore, an island city-state with a total population of nearly 4 million, has long used pricing to control congestion in its central area. The Singapore Area License

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SOURCES FOR SINGAPORE CASE STUDY

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Willoughby, Christopher. 2000. "Singapore's Experience in Managing Motorization and Its Relevance to Other Countries." Discussion Paper TWU-43. Washington, D.C.: The World Bank, April. http://www.worldbank.org/transport/publicat/twu_43.pdf (accessed 25 January 2007).

The World Bank. 2000. *World Transport Development*. Washington, D.C., August.

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Scheme (ALS) was originally established in 1975 when the population was closer to 2 million and residents owned about one car for every 16 persons.

Singapore's windshield-license-based ALS was very effective in reducing traffic, and over the years was enhanced and modified. In 1998, the paper-based system was converted into the fully automated Electronic Road Pricing (ERP) system, and Singapore added a series of radial and peripheral highways in the metropolitan region into the pricing scheme.

The ERP uses in-vehicle transponder units (IUs) that accept stored-value smart cards for payment. The stored-value-card system was preferred over a standard billing system because it offered better privacy protection. Enforcement is handled using video cameras.

The new electronic pricing system allows motorists much more flexibility in choosing whether and when to enter the restricted zone. Monthly commitments are no longer involved, and lower tolls are charged at off-peak times. Under ERP in 2000, restricted zone entry tolls apply on weekdays from 7:30 a.m. to 7:00 p.m. They range, for private autos, from free (10:00 a.m. to 12:00 p.m.) to S\$2.50 (US \$1.66 (2007)) (8:30 to 9:00 a.m. at most entry points), according to a fixed schedule keyed to the time of day. Previously the ALS applied tolls of S \$3.00 (US \$1.95 (2007)) throughout the entire morning and evening peak periods. Pricing under ERP covers not only more routes, with different prices along expressways according to location, but also more hours for restricted zone entry, than under the ALS.

As of 2000, residents own about one car for every nine persons, but the pricing system still effectively mitigates congestion. Singapore's Land Transport Authority intends the ERP system to be a traffic management tool rather than a revenue generator. They review tolls every three months to ensure the program objectives are being met. Traffic flows on selected routes are reviewed quarterly and may be adjusted up or down depending upon degree of congestion. As long as speeds do not drop below a 20 to 30 kilometers per hour range, pricing is not increased.

Results

The key analyses were based on before-and-after observations conducted in accordance with a comprehensive study design. After the new traffic patterns stabilized, measurements showed that the weekday traffic volume entering the Restricted Zone had dropped by 20 to 24 percent, from 271,000 vehicles per day in 1997, to between 206,000 and 216,000 in 2000. With the lower traffic volume, average traffic speeds in the zone increased from 30 to 35 kilometers per hour to 40 to 45 kilometers per hour.

These changes are relative to conditions under the previous ALS pricing scheme. Some peak spreading within the former peak periods has occurred as motorists take advantage of lower tolls on either side of the peak hour. The authorities have enforced a no-stopping rule near the toll gantries to prevent motorists from waiting on the shoulder for scheduled toll reductions.

On the technical side, the system works even when vehicles pass at 120 kilometers per hour. The CashCard, which is used to store the prepaid value used for tolls, is also accepted as a means of payment at stores, government agencies, and Internet sites.

The Singapore government views it as an important step in becoming a cash free society. The card can be recharged at automatic teller machines, special kiosks, and gas stations. With violations recorded by video cameras for later enforcement, the ERP violation rate is only 0.3 percent.

The ERP system costs are high, but still much less than the cost of building new capacity. The Singapore system cost S \$200 million (about US \$110 million on the basis of 2002 exchange rates) to implement, including equipment for up to 60 overhead gantry points and nearly 1.1 million in-car sensors. Half of the S \$200 million cost was for fitting of sensors, provided free to most vehicles.

Norway

Program

Several Norwegian cities have established “toll rings.” The first such Norwegian system was established in 1986 in Bergen, Norway’s second most populous city (population 300,000). The idea subsequently spread to Oslo, Norway’s capital (population 700,000) in 1990, and to Trondheim, Norway’s third largest city (population 140,000) in 1991 (Gomez-Ibanez and Small 1994). At least one additional Scandinavian installation, in Stavanger, Norway, has been added since.

The primary motivation behind the Norwegian toll rings has been revenue generation rather than congestion mitigation, and they feature relatively low tolls that do not vary much by time of day. They are significant, though, in being the first citywide applications of road pricing outside Singapore (Gomez-Ibanez and Small 1994).

Bergen’s system charged motorists about NOK 5 as of 1994 (1 Norway Kroner = US \$0.80 in 1994) to enter the central area except during night hours and on weekends. The small number of access routes to Bergen facilitated toll-ring implementation. Just seven toll stations handle about 70,000 vehicles per day. Nearly 60 percent of car crossings are by vehicles that have passes and are thereby allowed to traverse dedicated lanes.

Oslo charges a toll of about NOK 11 for crossing the cordon at all times of the day. Surrounding mountains and water concentrate traffic entering the city into three corridors. The system uses electronic toll collection to ensure minimal delays. Consistent with its intent to simply raise revenue, the toll ring has had a small impact on traffic.

The Trondheim toll ring opened in 1991 and features toll rates that vary somewhat based on the time of day. During the morning peak period, the toll is around NOK 10, and during the midday, it is around NOK 8. After 5:00 p.m. and on weekends, there is no charge.

Results

Though not intended to limit congestion, the Norwegian tolls have affected drivers. Despite the low charge, Bergen’s tolls appear to have restrained weekday traffic growth slightly and to have lowered vehicle crossings by perhaps 6 to 7 percent during hours of operation. Traffic during hours of free passage increased by about 10 percent (Gomez-Ibanez and Small 1994).

In Oslo, estimates of the reduction in cordon crossings in response to the toll average 5 percent. Most of the drop was during off-peak periods, implying that mainly nonwork trips were eliminated or diverted. Many peak-period trips are subsidized. Slightly more than 60 percent are covered by monthly subscriptions, and employers pay 60 percent of those. Effects of the toll on business and land use could not be discerned (Gomez-Ibanez and Small 1994).

In Trondheim, surveys before implementation indicated that 91 percent of the area’s residents had some alternative to paying the toll—riding the bus, changing destinations, or changing work hours. Respondents indicated a willingness to shift a trip 10 minutes earlier or later in order to save the toll charge.

Shopping trips appeared to be the most influenced by the toll. Shopkeepers contributed to this by lengthening hours to enable free access after 5:00 p.m., and about 19 percent of inbound shopping trips shifted as a result. No decrease in downtown retail revenues was observed, though costs probably increased with the extended shopping hours. Work trips exhibited a lesser shift. Some 3 percent of inbound home-to-work trips and 13 percent of work-to-home trips shifted in time to cross the cordon after 5:00 p.m. (Gomez-Ibanez and Small 1994).

CORRIDOR PRICING

San Diego (I-15)

Program

The two-lane exclusive high-occupancy-vehicle (HOV) facility on I-15 in the northeast side of San Diego is approximately eight miles long. It was opened in 1988 with a

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San Diego Association of Governments. 1997. "I-15 Congestion Pricing Project." San Diego, Calif. ■

two-person-per-vehicle occupancy requirement. The lanes operate in the southbound direction from 5:45 to 9:15 a.m., in the northbound direction from 3:00 to 7:00 p.m., and are closed at other times. There is one entry point and one exit point.

In light of unused available capacity in the HOV lanes, a value pricing demonstration was considered as one potential transportation control measure for the regional air quality plan and emerged as one of the preferred alternatives for the I-15 corridor.

The ultimate primary goals of the demonstration, as per the "Phase II Year Three Overall Report," were: 1) to maximize use of the existing express lanes; 2) to fund new I-15 corridor transit and HOV improvements; 3) to test whether allowing priced single-occupant vehicles (SOVs) to use the HOV lanes can help freeway mainline congestion; and 4) to set tolls on the basis of a market-based approach.

The I-15 Freeway HOV Pricing project was a congestion-pricing demonstration funded through the ISTEA of 1991, with the FHWA as the primary funding agency. The two-phase project provided a test of allowing SOVs to use the I-15 HOV lanes for a fee. Funding for express bus service in the corridor and a freeway service patrol was provided locally. Private enterprise was used for operations and marketing, and the California Highway Patrol (CHP) was paid for enforcement during the project.

ExpressPass, the Interim Operations phase of the demonstration, was in operation from December 1996 to March 1998. During this phase, a limited number of monthly permits were sold to motorists on a first-come, first-served basis. Drivers with permits could use the HOV lanes during all operating hours without meeting the vehicle occupancy requirement.

Carpools and vanpools with two or more persons continued to use the lanes for free. Initially, the monthly fee was set at \$50, and permits sold were held to 500. In February 1997, the limit was increased to 700 monthly passes, and in March the monthly fee was raised to \$70. The number of available monthly permits was increased to 1,000 by the end of the Interim Operations Phase in March 1998.

The full implementation phase, or FasTrak, started on March 30, 1998. This second phase ended December 1999, but FasTrak was still being used in 2002. FasTrak tested variable fees with electronic toll collection for SOV use of the HOV lanes. The fees were based on congestion levels in the general-purpose lanes and ranged from \$0.50 to \$4.00 under normal traffic conditions. During severe traffic, the fees could go as high as \$8.00.

Variable message signs are used to display the fees in effect to motorists approaching the lanes. As of July 1998, all 5,000 of the initial transponders had been distributed to 3,700 customers, and a waiting list was being maintained. By December 1999, 11,091 transponders had been issued.

New bus service, called the Inland Breeze, was implemented in March 1997 as part of the demonstration. The route connected to the San Diego LRT system at the Fashion Valley Mall Transit Center. During the demonstration peak period, service headways were 30 minutes, and midday and off-peak service was provided every 60 minutes.

Results

A monitoring and evaluation program continued throughout the demonstration phases. The "Traffic Study" collected and analyzed data on traffic volumes and speeds, travel modes and times, violations of the exclusive lanes, and associated quantitative information.

The "Attitudinal Panel Study," covering three different commuter groups, surveyed 1,500 persons every six months. The overall evaluation also addressed air quality, delay, business impacts, bus ridership, bus and carpool park-and-ride lot occupancy, media response, public acceptance, marketing, and institutional issues. For certain analyses, the nearby I-8 corridor was selected as a control corridor. Comparisons were hampered by that corridor's lack of freeway express lanes and differing geographic orientation and socioeconomic makeup, but it did serve

a similar travel function and offered an indication of which effects were external to the demonstration project.

During Phase I, the total vehicles using the HOV lane increased by approximately 27 percent. Carpools comprised a large share of this increase in vehicles. Use of the HOV lanes by ExpressPass permit holders was less than expected. Fewer than 55 percent of the permits were used on a regular basis. The remainder was used periodically or infrequently.

In March 1998, the last month of Phase I operation, pass holders comprised approximately 10 percent of vehicles using the HOV lanes on any given day. This periodic use by paying customers continued in Phase II, with the majority of FasTrak participants becoming occasional users. From April through September 1998, 53 percent of the FasTrak transponders were used one to five times a month, 18 percent were used six to 10 times, 11 percent were used 11 to 15 times, and 19 percent were used 16 to 40 times.

Revenues have been lower than projected, but they successfully funded operating expenses and the Inland Breeze bus service. During the ExpressPass phase, monthly revenues started at \$25,000 and plateaued—after five months—at about \$65,000. Revenues during the FasTrak phase at first quickly stabilized at roughly \$80,000 a month, and then in 1999, rose to more than \$100,000 per month.

More use of the Express Lanes was achieved as desired, without detriment to HOV-lane level of service. Total I-15 peak period vehicular volumes increased correspondingly, while volumes and traffic conditions on the main general purpose freeway lanes held fairly steady throughout, meeting additional project objectives.

An Attitudinal Panel survey covering Phase I of 500 ExpressPass participants and 1,000 other I-15 commuters indicated general support for the demonstration. Seventy percent of I-15 commuters responded that the program was fair to travelers in the HOV lanes and the general-purpose lanes. Approximately 95 percent of the ExpressPass respondents drove alone before they purchased a permit, while 56 percent were carpoolers.

In the fall of 1999 final panel survey, more than 90 percent of FasTrak customers, more than 50 percent of carpoolers, and almost 40 percent of I-15 solo drivers judged the program a success. Customers tended to be middle-aged (more women than men), to be from two-vehicle households, and to have higher incomes, more years of work experience, more education, and higher levels of home ownership than the general population.

In September 1998, reducing per-trip fees during the shoulders of the peak periods further refined the FasTrak pricing structure. This was done to create an incentive for users to travel outside of the peak of the peak, and thereby distribute traffic volumes more evenly throughout the peak periods. FasTrak customers responded to this change. The average toll paid for inbound morning trips dropped from \$2.25 in the spring of 1998 to \$1.77 in the fall of 1999. FasTrak's variable pricing was able to redistribute some of the paid component of HOV lane traffic from the middle of the peak into the shoulders.

A post-demonstration public opinion survey of users and nonusers of FasTrak conducted in late 2001 revealed overwhelming support for the HOT lanes, including operating policies, toll rates, the technology, use of revenues, and proposals for facility extension. Carpoolers reported satisfaction throughout with travel conditions, and Level of Service C was indeed maintained throughout virtually the entire demonstration project. These results pertained across users and nonusers, and all socioeconomic groups.

Los Angeles (SR 91)

Program

The SR 91 Express (91X) lanes are 10 miles of four-lane divided toll highway built in the median of the Riverside Freeway (SR 91) in Orange County, California, in the

SOURCES FOR LOS ANGELES CASE STUDY

California Private Transportation Company. 2001. "Eastbound Toll Schedule/Westbound Toll Schedule." <http://www.91expresslanes.com> (Webpages dated November 1, 2001).

Orange County Transportation Authority. 2002. "Orange County Transportation Authority Reaches Agreement with CPTC to Purchase 91 Express Lanes Toll Road and Franchise Agreement." <http://www.octa.net/news/late/042302.asp> (webpage dated April 23, 2002).

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Small, K. A. 2001. "The Value of Value Pricing." *Access*, no. 18 (Spring): 17–20. <http://www.uctc.net/access/access18lighter.pdf> (accessed 25 January 2007).

Sullivan, E. C. 2002. "Updated Observations from the State Route 91 Value Priced Express Lanes." Transportation Research Board 81st Annual Meeting, Preprint Paper No. 02-2554. Washington, D.C.: Transportation Research Board.

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———. 1998. "Evaluating the Impacts of the SR 91 Variable-Toll Express Lane Facility: Final Report." Sacramento: California Department of Transportation, May.

Yan, J., Small, K. A., and Sullivan, E. C. 2002. "Choice Models of Route, Occupancy and Time-of-Day with Value Priced Tolls." Transportation Research Board 81st Annual Meeting, Preprint Paper No. 02-2617. Washington, D.C.

form of barrier-separated express lanes. There are no on or off ramps on the express facility. The location, which includes passage through suburbs of Anaheim and Yorba Linda, California, is within the westerly half of the only freeway connector across the topographic barrier between heavily suburbanized Orange and Riverside Counties. This is a rapidly growing area. The affected portion of SR 91 was formerly one of the most heavily congested freeway sections in the state, with typical peak-period delays of 20 to 40 minutes.

The \$134 million project is one of four private, for-profit investments of the type authorized by the California legislature during the 1990s, selected for this location because of an inability to fund improvements quickly through the public sector. The lanes give motorists on the parallel freeway lanes, still significantly congested, the option to pay to use dedicated lanes that normally provide a faster trip. In addition, HOV 3+ (vehicles with three or more passengers) can use the lanes at a 50 percent discount (at the outset of the program, the fee for such HOVs had been waived).

The 91X lanes opened December 27, 1995, and feature tolls that vary by time of day and day of week. The private operator, free to set tolls within the constraints of a maximum financial return, elected to employ a published schedule. The toll schedule has been adjusted and has become more complex over time. Since the 91X toll collection system is fully automated, it is not difficult to have the tolls changing throughout the day. The range of tolls in 1996 was from \$0.25 off-peak to \$2.50 during the peak. As of 2001, the tolls ranged from \$1.00 off-peak to a high of \$4.75 during the evening peak.

In 1996, only five different toll amounts were charged. They were based on time of day and whether the trip was made Monday through Thursday, Friday, or on the weekend. In 2001, for example, the Eastbound toll at 6:00 p.m. was different each weekday, ranging from \$3.50 on Monday to \$4.25 on Friday, and 16 different toll amounts were charged based on time of day and day of week.

Results

An extensive data set was collected as part of federally funded evaluations of the SR 91 project. California Polytechnic State University has provided the principal investigator for these evaluation efforts and produced two major reports, which, along with several papers on the subject, describe the observed effects of 91X lanes. Much of the data is from traffic observations, including classification counts collected both before and after 91X lanes were built, along with results of telephone and other surveys conducted between 1995 and 1999 of highway users, randomly sampled households, and panel samples.

Two separate modeling efforts have provided estimates of elasticities. The "corridor," for which selected findings are reported, is not a broad swath. The corridor, as defined, consists of only the SR 91 Riverside Freeway, inclusive of the 91X lanes located in its median.

The 91X lanes provided two additional highway lanes of capacity in each direction, priced to maintain uninterrupted traffic flow. Motorists report being attracted to the toll lanes by the travel-time savings, improved driving comfort, and perceptions of improved safety.

The result has been improved travel conditions overall, albeit dissipating as traffic grows. The 91X tolls have been maintained at levels sufficient to preserve good levels of service during even the peak periods. In the first few months of operation, SOV traffic increased because of mode shifts among prior users of SR 91, traffic moving over to the freeway from city streets and local roads, induced demand—primarily discretionary nonwork purpose travel, and continuing development growth in the area.

HOV traffic also grew significantly, in part due to direct toll incentives for HOV 3+ motorists, and the opportunity to split tolls for HOV 2 commuters. The initial overall change in average vehicle occupancy (AVO) was downward because of the large number of new SOV users in the narrowly defined travel corridor. Over time, the new lower AVO has not changed much, and not much traffic has returned to

local roads, but the peak-period nonwork travel has declined toward prior levels as congestion has gradually returned to the SR 91 free lanes.

The commute trip mode shares on SR 91 (free and toll) in 1999 were 73 percent SOV, 19 percent HOV 2, 7 percent HOV 3+, and 1 percent motorcycle and other. Five general toll increases took place on 91X lanes between the time they opened and the summer of 2000. The first general toll increase, which occurred in January 1997, was followed by a significant and permanent reduction in the rate of traffic growth on 91X lanes. Subsequent toll increases, which were less substantial and focused on fine-tuning the hour-by-hour toll schedule, were not followed by perceptible changes in traffic trends. The opening of an alternative route for some through Orange County, the Eastern Toll Corridor, initiated a decline in 91X traffic that lasted about nine months.

Use of the priced lanes has been found to be highly selective. Half of the toll-lane users choose to do so once per week or less. The decision to use the lanes is largely related to travel conditions, but other factors contribute. Motorists on business or traveling to or from work are more likely to use the lanes than motorists traveling for nonwork purposes.

Travelers with higher incomes and education use the lanes more frequently than those with lower incomes and education. Female commuters are about one-third more likely than men to make frequent use of the express lane option. The youngest and oldest travelers are less likely than others to be frequent toll facility users. Demographic information obtained through the project evaluation surveys allows findings concerning the distribution of use across different traveler groups.

In this relatively affluent area, the 1999 income distribution of 91X users was 19 percent users from households with less than \$40,000 annual income; 23 percent \$40,000 to \$60,000; 37 percent 60,000 to \$100,000; and 21 percent more than \$100,000. Commuters in the high-income group were 2.5 times as likely as commuters in the “low” income group to have used the 91X toll lanes for their most recent SR 91 corridor trip in 1996 or to have used a toll facility (91X or the new Eastern Toll Road) for their most recent comparable trip in 1999. Nevertheless, roughly 20 percent of the lowest-income travelers chose a tolled route for their most recent trip in both years.

The basic associations between income and use of the tolled lanes remained stable between 1996 and 1999, with one notable change: a decline from 40 to 25 percent in the number of trips by travelers in the \$40,000 to \$60,000 middle-income category made on toll facilities. This suggests a particular sensitivity to the toll increases among these middle-income travelers. Commuters in the corridor overall are more affluent than the overall urban population of Southern California, and thus caution should be applied in generalizing the results to areas with different demographics.

Toronto (Hwy 407)

Program

Toronto’s Route 407 ETR is a 108-kilometer (67-mile) toll road project, built and operated as a public-private partnership that runs through Metropolitan Toronto. The highway, opened in several phases, features electronic road pricing technology enabling it to operate without toll booths—an approach known as “open road tolling.” Most cars carry transponders, which allow use of the road without incurring an added fixed charge per trip. Cars without transponders are tracked using digital-video-based technology that matches the license plate against the Motor Vehicle Registry’s database and sends a monthly bill.

Until a change implemented at the start of 2002, the pricing on the road varied by time of day. Under the variable pricing schedule, road use by passenger cars and other light vehicles cost CAD \$0.10 per kilometer (US \$0.19 (2007) per mile) during peak weekday periods, CAD \$0.07 (US \$0.13 (2007) per mile) per kilometer during nonpeak and weekend daytimes, and CAD \$0.04 per kilometer (US \$0.08 (2007) per mile) at nighttime (Shiller 1998).

SOURCE FOR TORONTO CASE STUDY

Shiller, E. 1998. *The Road Ahead: The Economic and Environmental Benefits of Congestion Pricing*. Santa Cruz, Calif.: Pacific Research Institute, February.

On January 1, 2002, it cost a flat CAD \$0.115 per kilometer (US \$0.22 (2007) per mile) at all times, with an added CAD \$2.65 (US \$3.10) per-trip processing charge for use without a transponder.

Results

Traffic volume has risen steadily since the road opened, and as it has been lengthened, trips per weekday now stand at more than 300,000. When first opened, the highway operated without tolls for about four months, attracting a daily volume of about 300,000 vehicles to the open segment. When the tolls were first introduced, the traffic declined by more than one-third to less than 200,000 per day.

The news is not all good, according to Ontario Department of Transportation's "Review of Customer Service—Highway 407 ETR." Many users complain of excessive toll charges, problems with billing, and flaws in the system's technology. Dissatisfaction with poor customer service provided by the company operating Highway 407 is also widespread.

Houston (I-10)

Program

The I-10W Katy Freeway HOV facility in Houston is a single, barrier-separated, reversible lane located in the freeway median. The facility was opened in stages between 1984 and 1990, and is now 13 miles in length. The vehicle occupancy requirement on the lane was gradually liberalized, stabilizing in 1986 at buses, vanpools, and HOV 2+ carpools. However, in October 1988, a HOV 3+ occupancy requirement was reinstated for the peak of the peak period in response to high volumes and a corresponding decline in speeds and travel time reliability. By late 1991, after fine-tuning, the HOV 3+ requirement applied from 6:45 to 8:00 a.m. and from 5:00 to 6:00 p.m. The I-10W Katy Freeway HOV lane facility is the only HOV lane in the country that uses occupancy requirements that vary by hour (Stockton, McFarland, and Ogden 1998).

The reinstated HOV 3+ restriction resulted in excess lane capacity, and as traffic on the regular lanes of the facility worsened, pricing was considered as a way of making more effective use of the HOV facility. A demonstration project was implemented in January 1998 that allowed a limited number of two-person carpools to use the HOV lane for a fee during the HOV 3+ restricted hours. The system uses electronic toll collection and charges a \$2.00 fee for each HOV +2 trip allowed (Shin and Hickman 1999a).

Results

In little more than a year, 650 transponders were issued and between 150 and 200 tolled trips were made on the facility daily during both peak periods combined. Most people use the facility only occasionally. About 25 percent of the registered QuickRide participants used their tag on a given day and only 6.5 percent of enrolled tags produced five or more trips per week (out of a maximum of 10). Six months into the program, approximately 25 percent of the registered tags had never been used at all, but many of these (40 percent) were second tags owned by a single household.

Original concerns that the number of potential QuickRide participants might need to be capped at 600 were not borne out as a result of the low number of trips per registrant. While it may be inappropriate to draw strong conclusions from a low-key demonstration without strong marketing, it appears that many enrollees view having an electronic tag as insurance for the occasional need and opportunity for a quick trip (Stockton, McFarland, and Ogden 1998; Stockton and Smith 1998; Shin and Hickman 1999a and b; Federal Highway Administration 2000b).

Morning QuickRide usage is heavier than evening usage, and it builds during the week, with the lightest use on Mondays, and the heaviest use on Thursdays and Fridays. QuickRide appears to have had some effect in encouraging carpools, but the small number of users has caused observed changes in person throughput to be statistically insignificant and rendered its effect on congestion in the corridor negligible (Shin and Hickman 1999a). The program continues and consideration has

SOURCES FOR HOUSTON CASE STUDY

Federal Highway Administration. 2000. "Value Pricing Notes No. 5" (Fall).

LKC Consulting Services, Inc., and Texas Transportation Institute. 1998. "Public Acceptance of QuickRide: Draft Final Report." Prepared for Houston Metro, Houston, Texas, December.

Shin, S. W., and Hickman, M. D. 1999a. "Effectiveness of the Katy Freeway HOV-Lane Pricing Project: Preliminary Assessment." *Transportation Research Record* 1659.

Shin, S. W., and M.D. Hickman, M. D. 1999b. "The Effectiveness of the Katy Freeway HOV Lane Pricing Project." *Session 94: Hot News on Hot Lanes*. Oral Presentation. Transportation Research Board 78th Annual Meeting, Washington, D.C.

Stockton, B., F. McFarland, and M. Ogden. 1998. *Feasibility of Priority Lane Pricing On the Katy HOV Lane: Feasibility Assessment*. College Station: Texas Transportation Institute.

been given to expanding the scope to include other HOV facilities in the area (Federal Highway Administration 2000b).

A survey of travelers on the mixed (free) traffic lanes of Houston's Katy Freeway (I-10W) revealed low knowledge of the QuickRide HOT lane demonstration program, but 55 percent thought the approach was fair, 67 percent viewed it as effective for the HOV lanes, and 85 percent perceived a benefit for the regular lanes. In actuality, the low QuickRide usage has not resulted in any significant changes in person throughput on the freeway (LKC Consulting Services, Inc., and Texas Transportation Institute 1998; Shin and Hickman 1999a).

The QuickRide program on Houston's I-10W Katy Freeway HOV facility appears to have had some effect in encouraging carpools, and negligible adverse impact on usage of HOVs, including public transit. QuickRide registrants—persons “buying in” for two person carpool use of the lane during periods of 3+ occupant eligibility requirement—were surveyed in the initial months of the pricing demonstration (Shin and Hickman 1999a and b; LKC Consulting Services, Inc., and Texas Transportation Institute 1998).

The small number of QuickRide trips involved raises questions about transferability of findings, but the exploration of prior modes of travel provided is the only such information presently available for HOV lanes having a value pricing application differentiated by HOV occupancy level.

Complex travel changes among QuickRide participants, involving time, spatial and mode shifts, are indicated by the survey results. Some two-person QuickRide carpoolers went from using the HOV lane in the shoulders of the peak (2+ occupant eligibility requirement) to the peak (3+ occupant normal requirement). This and other options open to QuickRide trip makers apparently contributed to roughly a 20 percent increase in peak trips by QuickRide participants. There was also significant movement of two-person carpools from the general-purpose lanes to the HOV lane. New carpools were also formed.

Survey responses suggest that a quarter to a third of QuickRide trips on any given day were previously drive alone trips on the freeway. In contrast, diversion of bus, vanpool and 3+ occupant carpoolers to QuickRide usage appeared to be limited in total to approximately 5 percent, and certainly no more than 8 percent, of the small number of QuickRide trips (Shin and Hickman 1999a and b; LKC Consulting Services, Inc., and Texas Transportation Institute 1998).

Lee County, Florida

Program

From 1980 to 1996, Lee County, Florida, experienced a population growth of 82 percent and now has a population approaching 400,000. Located in southwestern Florida, the county includes the cities of Fort Myers and Cape Coral. Fort Myers is an employment center, and Cape Coral more of a residential community. Separating the cities is the Caloosahatchee River, which most residents cross using either the Midpoint or Cape Coral toll bridges. Free alternatives over the river are US 41 and I-75, but both are north of the cities and not as convenient for most residents (Burris and Swenson 1998).

In 1998, a demonstration program was implemented on both the Midpoint and Cape Coral bridges to encourage users to avoid crossing during the peak period. The bridges carry an average of 75,000 vehicles per day. The preexisting toll structure encouraged frequent users to purchase a discount pass (\$40 per year) with a toll of \$0.50 per crossing. The majority of users with discount passes were commuters who traveled during peak periods.

Thus, planners faced the dilemma that peak-period users were paying the least to use the bridge. Getting them to shift trips from the peak without raising tolls would be difficult (Burris and Swenson 1998; Center for Urban Transportation Research 2000).

Planners decided to offer a 50 percent savings for traveling during specific discount periods focused on the shoulders of the peaks. The toll discounts were underwritten

SOURCES FOR LEE COUNTY CASE STUDY

Burris, M.W., and C. Swenson, C. 1998. “Planning Lee County’s Variable-Pricing Program.” *Transportation Research Record* 1617.

Cain, A., M.W. Burris, and R.M. Pendyala. 2001. “Impact of Variable Pricing On Temporal Distribution of Travel Demand.” *Transportation Research Record* 1747.

Center for Urban Transportation Research. 2000. “The Lee County Variable Pricing Project.” Prepared for U.S. Federal Highway Administration. Tampa: University of Southern Florida.

by a federal grant and limited to participants in the LeeWay electronic toll account program.

Approximately 25 percent of total bridge traffic became eligible for the variable pricing discount. Of the eligible travelers, 93 percent pay for the annual discount pass. Therefore, the vast majority of eligible users save only \$0.25 by traveling in the discount periods (Cain, Burris, and Pendyala 2001).

Results

Looking at the same six-month period before and after implementation, traffic-volume data provided little evidence of any active peak-spreading effects at the bridge and the basic patterns of the daily demand profiles remained unchanged. This is best explained by the fact that the number of commuters subject to the discount was small relative to the total number of vehicles crossing the spans.

However, at the disaggregate level, among users eligible for the discount, positive shifts in demand were observed within all discount periods, and negative shifts in demand were observed during peak periods. The strongest response occurred in the morning at the Midpoint Bridge, where an 18 percent increase in eligible traffic was observed during the 6:30 to 7:00 a.m. discount period, associated with a 7 percent decrease during the nondiscount period from 7:00 to 9:00 a.m.

In surveys, approximately 38 percent of eligible bridge travelers indicated that they had altered their travel since the program began to take advantage of the discounts, and 50 percent indicated they took the discounts into consideration when planning trips across the bridges. Of those modifying their travel, 84 percent changed their time of travel, 9 percent changed their route, and 6 percent changed their number of trips (Cain, Burris and Pendyala 2001; Center for Urban Transportation Research 2000).

APPENDIX F NOTE

1. The majority of this appendix used the following as its source: Transportation Cooperative Research Program. "Road Value Pricing: Traveler Response to Transportation System Changes." Report 95. Washington D.C. 2003. Case studies have been edited and reformatted for this appendix.

Institutions and Agents of the Transportation/Land Use Connection

Public policy is made by government institutions. Shifting the intersection of transportation and land use toward land use (our argument in Chapter 5) requires an understanding of the institutional framework.

Transportation institutions tend to be large (federal and state), consolidated, well funded (from dedicated revenues, principally the gas tax), and less dependent on the private sector to complete their missions. Land-use institutions are smaller (local), more diffuse, have less funding, and require more interaction (regulation and partnership) with the private sector (land developers, builders, banks, insurance agencies, Realtors).

We have argued that large transportation projects have done more historically in metropolitan areas to influence land development than large development projects have done to influence transportation systems. If that is to change, institutional change will be required.

This appendix provides an overview of the U.S. institutions central to the transportation/land use connection. It has three main sections:

1. **Institutions and Agents Responsible for Transportation Planning and Development.** This section discusses important historical and organizational factors that shape and constrain the behavior of the major institutions and agents in transportation planning and development.
2. **Land-Use Planning and Development: Local Governments and Developers.** This section shows how, in contrast to transportation infrastructure, the planning and development of urban buildings is done primarily by the private sector and has many more agents whose decision making relies more on a decentralized interpretation of market information than on command-and-control from a centralized agency.
3. **Implications for Coordinating and Balancing Transportation and Land Use.** This section discusses problems and directions for change.

INSTITUTIONS AND AGENTS RESPONSIBLE FOR TRANSPORTATION PLANNING AND DEVELOPMENT Historical Development of Transportation Institutions

The roadway system in the U.S. is primarily owned and operated by the public sector. While freeways, highways, and streets function as a single system, building and maintaining that system is the joint responsibility of federal, state, and local Departments of Transportation (DOTs).

Road systems in urban areas are extensive and cross many jurisdictions. Efficiently building and maintaining such a complex system requires planning to coordinate the investments of multiple jurisdictions. Large urban areas are required by federal and state law to coordinate plans for transportation improvements at a regional level.

The Federal-Aid Highway Act of 1956 authorized billions of dollars for the construction of the National System of Interstate and Defense Highways, the world's largest public works program at the time.¹ In 1962, Congress responded to concerns over the implementation of the interstate highway system by passing the Federal-Aid Highway Act. The Act required the establishment of regional agencies that were to follow the "3C" planning process—Comprehensive, Cooperative, and Continuing. This requirement still governs transportation planning today.

THE THREE-C PROCESS

Comprehensive means specifically considering all transportation modes, economic development, financial resources, demographics, land use, travel patterns, terminal and transfer facilities, traffic control, and social- and community-value factors, such as environmental issues, preservation of open space and historical sites, and aesthetics.

Cooperative refers to cooperation (working together to achieve a common goal or objective) between the various levels of government and among agencies at the same level of government.

Continuing refers to the need to continually reevaluate and update the transportation plan and program. ■

Starting in the 1960s, after Congress created the U.S. DOT, a series of federal funding laws gradually required additional coordination between highway and transit planning programs and the many jurisdictions involved. Eventually, metropolitan highway and transit planning were unified under Metropolitan Planning Organizations (MPOs) (Wachs 2004). Often the MPOs evolved out of the organizations that carried out the 3C planning process for federal transportation funding.

MPOs were formally established by federal mandate in the late 1960s and early 1970s to help manage the 3C planning process. MPOs were required for urbanized area with a population greater than 50,000 people. Their oversight extended to transit projects resulting from successive Urban Mass Transportation Acts starting in the mid-1960s. The 1970 Act financially empowered the Urban Mass Transportation Agency (the precursor to the Federal Transit Administration) to help fund local and regional transit agencies. It also gave the agency the power to certify regional planning agencies to administer these funds to these transit operators.

Ultimately, an MPO's structure and operation get determined by the state government legislation (often with significant influence from the state DOTs and local communities). The resultant organizational structures vary from state to state, and even across metropolitan areas within states. For example, the Metropolitan Transportation Authority, the MPO in the Los Angeles region, is also the state-designated Regional Transportation Planning Agency for Los Angeles County: it plans and builds transportation systems; does multimodal funding and programming; and operates the transit system. In contrast, the Metropolitan Transportation Commission in the San Francisco Bay Area is an MPO that manages the funds for independent, potentially competing, transit service providers. Transportation planning, therefore, is mostly driven either at the regional or state level, depending on the political powerbase and legacy in each state and even in each region.

The Surface Transportation and Uniform Relocation Assistance Act of 1987 was, effectively, the last authorization bill of the Interstate era. In 1991, the federal government passed the Intermodal Surface Transportation Efficiency Act (ISTEA). ISTEA was innovative in the several ways. Specifically, it:

1. increased money for surface transportation (from \$87 billion in 1987 to \$150 billion);
2. streamlined administration by reducing the number of categorical programs from more than 40 to 12 and allowing more flexibility to shift money among programs;
3. broadened goals and objectives to include air quality (working in concert with 1990 Clean Air Act Amendment); enhancement of urban areas; and environmental consequences;
4. emphasized maintenance and management of the system, not just new construction;
5. gave more decision-making authority to local and regional agencies (especially MPOs);
6. required better evaluation for decision making (e.g., Congestion Management System, Pavement Management System);
7. encouraged fiscal realism by requiring that metropolitan transportation plans match planned projects to reasonable expectations about funding;
8. promoted intermodalism and multimodalism;
9. required state transportation planning; and
10. increased the requirements for citizen participation.

Since ISTEA, two more Transportation Authorization bills have built upon the ISTEA legacy: the Transportation Equity Act for the 21st Century (TEA-21) in 1998 and the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)² in 2005. Federal planning factors for transportation planning have changed only slightly; in TEA-21 they were to:

1. support the economic vitality of the metropolitan area, especially by enabling global competitiveness, productivity, and efficiency;
2. increase the safety and security of the transportation system for motorized and nonmotorized users;
3. increase the accessibility and mobility of people and for freight;
4. protect and enhance the environment, promote energy conservation, and improve quality of life;
5. enhance the integration and connectivity of the transportation system, across and between modes, for people and freight;
6. promote efficient system management and operation; and
7. emphasize the preservation of the existing transportation system.

The only changes to those goals in the SAFETEA-LU reauthorization were to (1) increase the security of the transportation system (after September 11, 2001), and (2) promote consistency between transportation improvements and state and local planned growth and economic development patterns.

Following the Money

Most of what will happen in metropolitan transportation and land-use is determined by what households and businesses, as private consumers and taxpayers, are willing to pay for. Institutional arrangements and policies can strongly influence the flow of funding. A key to figuring out how to effectively coordinate transportation and land-use planning is to understand where the money comes from.

Land-use development depends primarily on dispersed sources of mostly private funding from developers and real-estate consumers, influenced by financial institutions and factors. Transportation development, in contrast, is driven by public allocations of taxes and user fees. Figure G-1 illustrates how funds flow to transportation projects in a typical metropolitan area. It shows where the funding comes from (from businesses and households, primarily via fuel taxes), and where it goes (back to the MPO and local governments, often after traveling to and through federal and state DOTs).

The financing provisions for federal funds come from the Federal Transportation Authorization bills such as ISTEA, TEA-21, and most recently SAFETEA-LU. The main source of funds is Federal Highway Trust Fund, supplemented by general fund appropriations. User fees and taxes (e.g., motor fuel taxes; motor vehicle excise taxes; registration and license fees; air, transit, rail, or ferry fares; and toll proceeds) also provide a large source of funds for transportation, particularly for highway projects. The federal gas tax (as of January 2007) is 18.4 cents a gallon (last raised in 1993).

State gas taxes range from as high as 32 cents per gallon (New York, Wisconsin) to as low as 8 cents per gallon in Alaska. Combined, federal, state, and local gas taxes average about 46 cents per gallon (2005) across all states. In general, the tax rate has grown more slowly than inflation. Paradoxically, as fuel economy has improved, tax revenue has not kept pace with increased vehicle miles traveled.

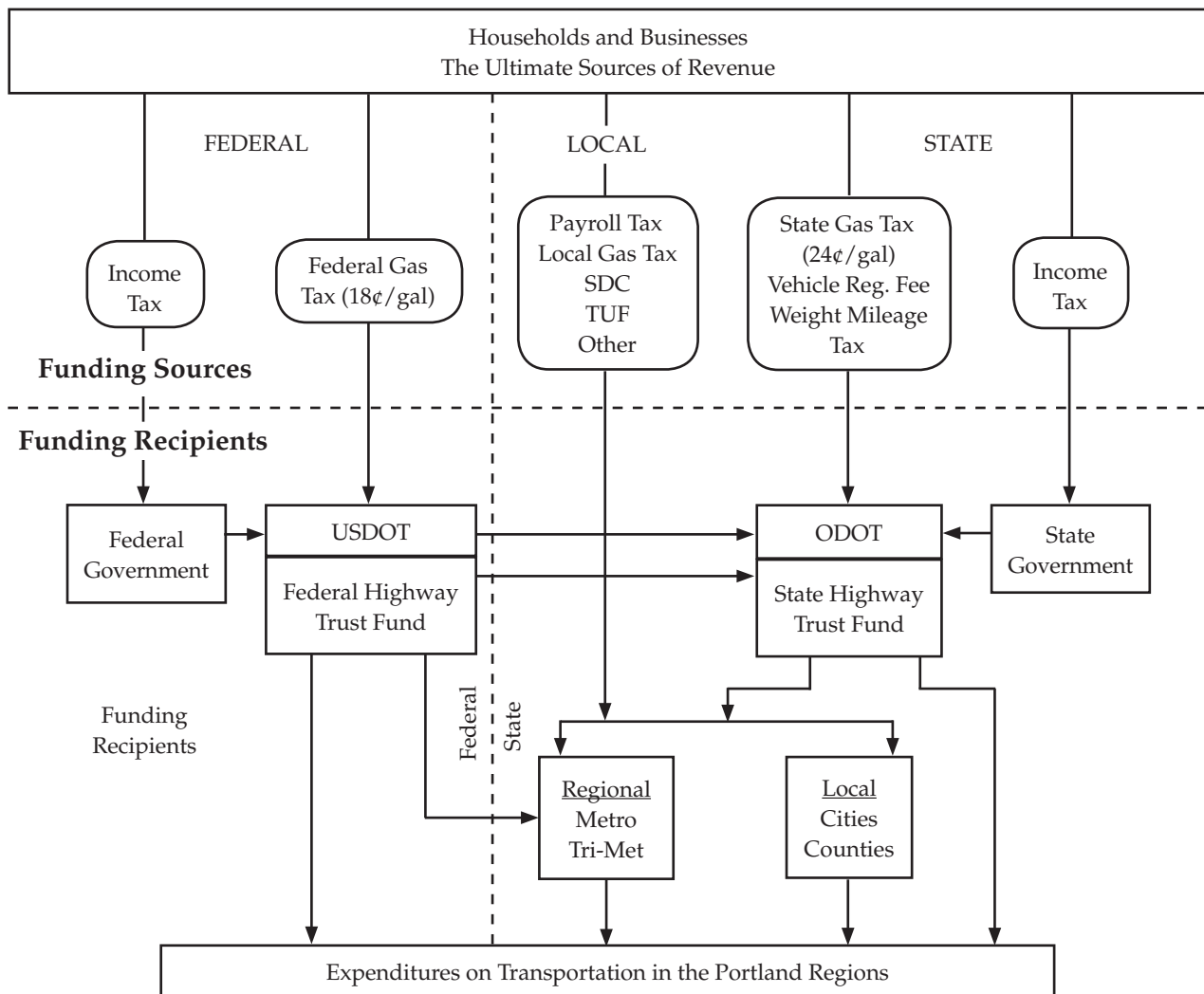
In addition, a variety of state-level user fees and tax revenues fund transportation projects, such as state sales tax. General fund appropriations for transportation purposes are also an option, as are public/private funding arrangements.

Funds from the Federal Highway Administration (FHWA) road program are administered in large part by the individual state DOTs, which then pass them on to the individual MPOs. Minor MPOs (50,000 to 200,000) rely mostly on state DOT decisions for their allocations, while major MPOs get a portion of surface transportation funds directly from the federal government.

Most transit funds are distributed directly from the FTA to regional transit operators through an MPOs with state DOT oversight. An MPO, through its certification of the metropolitan transportation planning process, ensures the federal government that existing and future transportation expenditures are planned and prioritized

ABOUT MPOS

- The designation of the boundary and structure of an MPO is made jointly by the governor and affected local governments in accordance with federal regulations.
- The area that an MPO serves may be contained in one county or may span several counties or even multiple states.
- An urbanized area with more than 50,000 people in a contiguous urban area must have an MPO.
- Metropolitan areas between 50,000 and 200,000 are labeled “minor” MPOs and rely almost exclusively on the state DOTs for final allocation of funding.
- Urban areas of 200,000 or more in population are designated as “major” MPOs as well as Transportation Management Areas (TMAs). They have more specific and stringent responsibilities than minor MPOs, and receive some funds directly from the federal government.



Source: ECONorthwest, 2006

Figure G-1. Typical funding for metropolitan transportation (Portland, Oregon)

based on a certified 3C planning process. This is required for the distribution of federal transportation funds.

MPO Requirements for Planning and Coordination with Federal and State DOTs

As a condition of funding, the federal government—through U.S. DOT and FHWA—requires MPOs to convene the local governments of the region, to follow the 3C process, and to produce three documents:

1. A Unified Planning Work Program (UPWP) outlines what planning studies and tasks the MPO will undertake over the subsequent one to two years. It must be updated every year.
2. A Long-Range Transportation Plan (LRTP) establishes transportation needs at a regional level and identifies policies, programs, and projects to meet these needs. For a project to be in a Transportation Improvement Program (TIP), it must be in the LRTP. The LRTP project list is typically input into a transportation model and an air quality model, where it is checked for compliance with federal regulations associated with air quality (the Clean Air Act (CAA)) and social equity (the Civil Rights Act). If a metropolitan area is considered noncompliant with CAA air quality standards, for example, the LRTP is required to be updated every three years instead of five.

3. A TIP is updated at least every two years with a three-year horizon. A region's TIP is required to comply with the Statewide Transportation Improvement Program (STIP), as determined by a state's DOT. There is at least one case where a state DOT threatened to refuse to include an MPO's TIP into the STIP to ensure that a bypass project was included that the MPO did not want. While the law (23 U.S.C. and 49 U.S.C.) appears to be on the side of the MPO, there was enough concern that the state DOT could take such an action and thus keep the MPO from receiving any federal funds that the MPO policy makers changed the TIP.
4. The projects in the TIP that involve the local jurisdictions responsible for the transportation system in the region must be consistent with policies, programs, and projects identified in the LRTP. Additionally, because projects must be in the LRTP to be in the TIP, they are eligible for most federal and state funding programs.

While a LRTP can include policies, strategies, and programs, the focus of a LRTP is usually on *capital investments* to improve existing roadways, construct new roadways, and improve transit service. A key requirement for regional transportation plans is that they be *fiscally constrained*. In short, the cost of activities identified in the LRTP cannot exceed the level of funding considered reasonably available in the region.

The cost of all projects in a region that could contribute to improvements in travel time and safety almost always exceeds the financial resources considered *reasonably available* to pay for them. For this reason, the defining task of an LRTP is to prioritize projects and select ones that are within the constraint of available funding. Determining what funding is "reasonably available" in a metropolitan region is complicated and, in practice, somewhat discretionary.³

LAND-USE PLANNING AND DEVELOPMENT: LOCAL GOVERNMENTS AND DEVELOPERS

In contrast to transportation infrastructure, the planning and development of urban buildings is done primarily by the private sector and has many more agents (property owners, developers, builders, bankers, Realtors) whose decision making relies more on a decentralized interpretation of market information than on command-and-control from a centralized agency like a state DOT.

Local Governments

The role of government is still very important in the development and planning of buildings, but the importance shifts from federal and state government to local government. The common development tools used by almost all local governments are subdivision and zoning ordinances, comprehensive plans, and building codes. Most probably also use at least some additional tools: impact fees, design and environmental regulations, and incentives (e.g., for downtown development, and commercial and industrial development in certain sectors).

In the early twentieth century, planning commissions were established in an effort to protect the planning function by placing it outside the influence of public officials, and to ensure that planning decisions were made based on objectivity and science as opposed to political power and influence. The first planning commissions in the U.S. were established in Hartford in 1907, in Milwaukee in 1908, and in Chicago, Detroit, and Baltimore in 1909. In 1913, Massachusetts led the way for many states by making it mandatory for cities of more than 10,000 people to create official planning boards.

In 1926, the U.S. Department of Commerce published the Standard State Zoning Enabling Act, encouraging states to adopt laws that would allow local governments to form planning commissions responsible for overseeing the creation and enforcement of the zoning codes. In 1928, it published the Standard City Planning Enabling Act, which promoted city planning. But the Act created several problems:

1. It confused the comprehensive plan with the zoning code, causing some communities to make zoning decisions without reference to long-range plans.
2. It recommended piecemeal adoption of the comprehensive plan, undermining the purpose of comprehensive planning.

FINANCING VERSUS FUNDING

These terms often are used interchangeably, but preserving a distinction can clarify thinking.

Providing transportation facilities and services costs money, and somebody has to pay for these costs. The ultimate source of revenue for these costs is *funding*. When the funds for transportation costs are borrowed and paid back over time, these costs have been *financed*. Public agencies finance costs for the same reasons as households and businesses—to reduce the current out-of-pocket costs by spreading out payments over time. But the ultimate source of funding for financed costs is not the financing instrument itself—bonds, for example—but rather the revenue sources used to repay the borrowed funds.

GARVEE bonds, for example, get money for transportation improvements now by pledging future revenues from an existing funding source (gas tax) to pay off the debt in the future. Thus, a transportation agency gives up some of its ability to make improvements in the future so it can make them now. Some public agencies have not made this fact transparent in public debate.

The public is probably, or should be, more concerned about funding than financing. Saying that a project will be funded by bonds misses the point, which is: What revenue source will be used to pay back those bonds? Tolls, gas tax, sales tax, property tax, . . . ? Each case distributes the burden a little differently, and people care. ■

THE DILLON RULE AND THE POWER OF LOCAL GOVERNMENTS

John Forest Dillon was the chief justice of the Iowa Supreme Court in the late 1800s and an authority on municipal law. He distrusted local governments and local officials. He articulated his famous rule in 1873, which grants power to localities very narrowly. It was supported by state supreme courts around the nation.

The Dillon Rule says that if a question exists about the extent of a local government's power or authority, the local government does not receive the benefit of the doubt, and it is assumed that the local government does not have the power in question.

A Dillon-Rule state is one in which local governments have only the authority explicitly granted them by the General Assembly. In contrast, "home rule" states grant localities authority unless it is explicitly removed by the state legislature. Home rule is frequently granted by revising the state constitution to authorize local governments to enact home-rule charters. More than 40 states have granted some measure of home rule. The scope and style of home rule varies considerably from state to state.

The Dillon Rule is still in effect in every state in some form or another. It is the strongest in the southeastern states. Not only are the local governments weaker in these states, but their state DOTs tend to be large and strong, which moves in the direction of the kind of institutional imbalance we describe in this appendix. ■

3. It may have weakened the emerging role of planning in local government through its recommendation that an appointed planning commission oversee planning operations rather than a directly elected legislative body.

During the 1930s and 1940s, planning commissions evolved as the primary clients of zoning codes and master plans, shifting influence away from the business and development interests. They gained responsibility for the budgeting and planning of roads. In the 1950s, they were the local authority for the surveys, studies, and plans undertaken to determine eligibility for increased federal funds generated from Section 701 of the Housing Act of 1954.

In recent decades, the need for planning commissions has been questioned: corruption has been largely eliminated from local government; they create an extra layer of bureaucracy between the planning department staff; and, ultimately is it the locally elected legislative body (city councils, county commissions, and so on) where the final decisions on controversial development issues will get made.

Though the federal government and most state governments⁴ do not get directly involved in the regulatory or comprehensive planning aspects of local development, they still play a large role in metropolitan urban form. An obvious influence is the theme of this report: federal and state decisions about the type and location of transportation improvements have had a very large effect on urban patterns. The tax-deductibility of interest on home mortgages (including second homes) has encouraged home ownership.

Developers

Developers (the term we use here for the private-sector building industry in general) design and construct buildings and local roads, according to local standards. They are involved in multiple aspects of project development and construction (financing, design, market research, entitlements, and so on). In the capitalist economy of the U.S., the historical precedent and current assumption is that the private sector will build most of what people commonly view as urban areas. By our rough estimates, probably two-thirds to four-fifths of the infrastructure and buildings in a city or metropolitan area are owned by the private sector.

Many large, master-planned communities have been developed by the private sector. But they amount to a small percentage of the total annual investment by developers. Most of the development occurs in subdivisions or lot by lot by independent developers with little interest, incentive, or ability to create a plan for regional development into which their development fits. That role is largely left to local government.

What most developers want from local government are clear planning and design guidelines that increase certainty and lead to quick entitlement review and approval.

IMPLICATIONS FOR COORDINATING AND BALANCING TRANSPORTATION AND LAND USE Problems

We provide evidence in Chapters 2 through 5 to support our conclusion about the connection between transportation and land use:

1. In many U.S. metropolitan areas, in much of the 20th century, *transportation led land use*. Urban land developed around transportation infrastructure, especially in western states that grew mainly after the development of the automobile, and especially in suburban areas at the urban fringe (where most of the growth was occurring).
2. In U.S. metropolitan areas in the twenty-first century, *land use should lead transportation*. Comprehensive land-use planning is the logical vehicle for making decisions about the dozens of attributes of quality of life in an urban area (see Appendix A). Urban form is how most people can relate to urban growth in public discourse: What do we want this region to look like in 10, 20, or 50 years? Which areas are protected; which ones are developed, and at what intensity? What pattern will

balance our concerns for economic opportunity, environmental quality, urban amenity, and equal access to all of them? Having defined and agreed on some possible urban forms consistent with these and other local goals, the public and its elected officials and professional staff can then evaluate infrastructure and public facilities (including transportation) can facilitate those urban forms. *Transportation should follow land use.*

A theme of this report is that land-use and transportation planning have to happen together: they interact; mutual adjustment is required. Thus, when we say “transportation should follow land use” we mean that the required interaction should probably take place in the context of long-run land-use planning.

Thus, our conclusion is that policy needs to shift in metropolitan areas to having comprehensive (and, ideally, regional) plans define a long-run vision for urban development and having transportation serve that vision. Our term “imbalance” is another way of saying “a shift is needed.” Making that shift requires understanding and changing institutions.

The institutional arrangements for urban transportation and land development help explain why the imbalance exists and give some insight into the kinds of needed changes to shift the emphasis toward land use:

1. Government transportation and land-use agencies operate in separate silos, at different scales, with different planning horizons, and with different policy and funding tools.
2. Transportation projects use primarily public funds, generated from gas tax revenue and other taxes and fees. Land-use development relies on private funding and financing involving investors, banks, and individual consumers.
3. Compared to land-use actions, transportation projects are relatively large in scale, executed in a more coordinated fashion by public agencies (e.g., DOTs and MPOs), are more expensive, and are funded with public dollars. In contrast, land-use projects, while numerous, are mostly relatively small in comparison, implemented on individual and incremental timeframes, and involve relatively complex relationships (e.g., local governments, developers, and consumers) and actions (obtaining zoning approvals, financing).
4. Local governments rely on the developers and market forces to implement their land-use plans. Not only is land-use implementation in a weaker position relative to transportation, but regional governance (for transportation, let alone land use) is in a weaker position relative to the influence of federal, state, and local governance.
5. At the local level, developers build local roads according to local standards. Those standards are often the result of formulation by federal and state agencies, endorsement by financial institutions, and adoption by the local governments. The formulation of these street standards may lack any consideration of local community or environmental contexts (Southworth and Ben-Joseph 2003). Given that historical development, it should not be surprising to find local street standards emphasizing vehicle throughput and safety, not livability or walkability.

Solutions

Cervero (2001) provides guidance on what needs to be done to get coordinated and balanced planning and development of transportation and land use:

1. Embrace urban planning and articulate a vision for the future that can guide all subsequent planning decisions.
2. Use infrastructure investments to leverage, focus, and shape the location and intensity of development, including water, sewer, as well as transportation.
3. Engage in regional cooperation, including local land use as well as transportation decisions guaranteeing regional accessibility to important job and retail opportunities. A key issue to deal with is revenue sharing, which will foster a shift toward

long-run optimization of development and investment from regional perspective instead of short-run fiscal zoning (which leads localities to overzone for retail and office and underzone for affordable housing and certain basic industries).

4. Balance the twin aims of urban design: form and function.

We concur with Cervero and believe the quality of life in a metropolitan area, and the functions, goods, and services that must be addressed to have some hope of optimizing it, are regional in scope and must be addressed by regional policy.

MPOs are a logical place to look for regional coordination of multiple governments, nongovernmental organizations (NGOs), and private interests. MPOs are a good start, but as structured they are in a coordinating role, not a decision-making one. They are conveners of local government interests in the MPO region (cities and counties) that determine regional transportation priorities. With a few notable exceptions (Metro in Portland, Oregon), they have virtually no control over land-use decisions.

Nonetheless, the role of land use in metropolitan transportation planning is so obvious as to be inevitable. MPOs need to allocate population and employment to the small-area zones in their transportation models so that the models can run. Those allocations are exogenous to those models. The connection between trips, people, and the buildings necessary to allow people to do anything in a transportation zone (live, work, shop, recreate) is clear. So who decides what is going to get developed in those zones, and how? Is that development inevitable (historical growth trends, when extrapolated, dictate development?), or can its type, intensity, and location be influenced by public policy?

Our answer, of course, is that it can be influenced by public policy. We think that is what most people believe. Thus, MPOs are, by choice or by force, having to consider a long-run development pattern to do their transportation planning and are finding themselves involved, at some level, with regional land-use planning (the visioning at least, but little on the regulation). Tying land use into the MPO LRTP process is probably the best single hope for the coordination and planning of land use and transportation in metropolitan areas.

Many of the successful efforts to coordinate transportation and land-use planning have come from state-level action (oftentimes at the request from local and regional entities) that strengthen the role of regional governance and land use. Oregon has provided one of the best examples of consistent and supportive planning and growth management legislation, spanning three decades. With the passing of the property rights initiative, Measure 37, Oregon's innovative growth management policies are facing their most serious adjustment to date. Maryland and Massachusetts provide other examples for how growth management policies can flourish and recede under different political administrations. Bureaucracies have inertia, but they can be moved if executive and legislative leadership are clear and in agreement about desired change.⁵

APPENDIX G NOTES

1. The 1956 Act was the culmination of two decades of studies and negotiation. As a result of the *Interregional Highways* report, Congress had adopted a National System of Interstate Highways not to exceed 40,000 miles in the Federal-Aid Highway Act of 1944. However, money was not authorized for construction of the system. Based on the recommendations of the U.S. Bureau of Public Roads and the Department of Defense, a 37,700-mile system was adopted in 1947. This network consisted primarily of the most heavily traveled routes of the Federal-Aid Primary System. The remaining 2,300 miles were reserved for additional radials, bypass-loops, and circumferential routes in and adjacent to urban areas. Studies of urban area needs were made by the states with the cooperation and aid of city officials. The urban connections were formally designated in 1955 (U.S. Dept. of Commerce 1957).
2. We had a hard time with the LU in SAFETEA-LU. The first time we heard the acronym, we assumed it stood for Land Use. But Legacy for Users: what's that all about? The answer, which we have not corroborated but sounds right, is that the wife of one the principal Congressional sponsors of the bill is named

Lu. That anecdote supports a theme of this book for planners: understand the technical stuff, but don't forget that good analysis alone is not going to get your ideas over the bar.

3. That such discretion exists is not unreasonable, if MPOs are not unreasonable in exercising it. The future is uncertain; future funding is not dictated by past funding. Technicians do not "know" what future revenues will be with any precision. Thus, the amount of funding that is "reasonably available" is a policy decision based on technical information.
4. About 20 to 25 percent of the states have something that might be considered a statewide authority for regulating or monitoring land development.
5. The administration of Governor Romney in Massachusetts consolidated several state departments, including transportation, under the Department of the Commonwealth to get large development projects coordinated on objectives for economic development, environmental quality, transportation, affordable housing, and smart growth, and restructured funding to local governments to support smart growth principles. The state used its discretionary funding authority (grants) to encourage local governments to meet state objectives for smart growth by consolidating the multiple grant programs of various agencies and establishing clear guidelines that it would use to evaluate grant applications. We wouldn't be surprised to see that program reduced or dismantled under a new administration.

Glossary

This glossary is compiled from others. Please consult these resources for more transportation- and land-use-related terms.

- American Planning Association. *A Planners Dictionary*, edited by Michael Davidson and Fay Dolnick, Planning Advisory Service Report No. 521/522. 2004.
- Federal Highway Administration (FHWA). *Planning Glossary*. <http://www.fhwa.dot.gov/planning/glossary/index.cfm>
- Mineta Transportation Institute. *Glossary of Terms*. <http://transweb.sjsu.edu/comglos.htm>
- Sacramento, California, Transportation and Air Quality Collaborative. *Glossary of Terms: Transportation Funding*. http://www.sactaqc.org/Resources/primers/Glossary_Trans_Funding.htm
- Transportation Planning Capacity Building Program. *The Metropolitan Transportation Planning Process Key Issues, A Briefing Notebook for Transportation Decision Makers, Officials, and Staff*. <http://www.planning.dot.gov/metro.asp#briefing>
- Victoria Transport Policy Institute. *Online TDM Encyclopedia—Glossary*. <http://www.vtpi.org/tdm/tdm61.htm>

access management Measures regulating access to streets, roads, and highways from public roads and private driveways. Measures may include, but are not limited to, restrictions on the siting of interchanges, restrictions on the type and amount of access to roadways, and use of physical controls (such as signals and channelization, including raised medians) to reduce impacts of approach road traffic on the main facility.

accessibility The ability to obtain desired goods, services and activities.

Advanced Traffic Management Systems (ATMS) A high-tech version of the old-fashion traffic cop, ATMS uses a variety of means to more efficiently manage traffic. It can include roadside sensors, ramp metering, HOV lanes, and synchronized traffic signals that respond to traffic flows.

Americans with Disabilities Act of 1990 (ADA) Civil rights legislation enacted by Congress that mandates the development of a plan to address discrimination and equal opportunity for disabled persons in employment transportation, public accommodation, public services, and telecommunications.

area sources Small stationary and non-transportation pollution sources that are too small and/or numerous to be included as point sources but may collectively contribute significantly to air pollution (e.g., dry cleaners).

arterial street A class of street serving major traffic movements (high-speed, high volume) for travel between major points.

attainment area An area considered to have air quality that meets or exceeds the U.S. Environmental Protection Agency (EPA) health standards used in the Clean Air Act. Nonattainment areas are areas considered not to have met these standards for designated pollutants. An area may be an attainment area for one pollutant and a nonattainment area for others.

automated toll collection (ATC) Sensors at toll booths that pick up billing information from devices installed in cars, allowing motorists to drive through without stopping; the amount of the toll is automatically deducted from a prepaid account or billed to the vehicle's owner.

automobile dependency Transportation and land-use patterns that result in high levels of automobile use and limited transportation alternatives. In this case, "automobile" includes cars, vans, light trucks, sport utility vehicles (SUVs), and motorcycles.

average vehicle occupancy (AVO) The number of people traveling by private passenger vehicles divided by the number of vehicles used. The AVO during commute hours for the Bay Area in 1990 was 1.097.

capacity A transportation facility's ability to accommodate a moving stream of people or vehicles in a given time period.

Capital Improvement Program (CIP) A ranked list of capital projects, such as transportation improvements, expected to be completed during the project timeline (typically five years).

capital program funds Financial assistance from the Capital Program of 49 U.S.C. This program enables the Secretary of Transportation to make discretionary capital grants and loans to finance public transportation projects divided among fixed guideway (rail) modernization; construction of new fixed guideway systems and extensions to fixed guideway systems; and replacement, rehabilitation, and purchase of buses and rented equipment, and construction of bus-related facilities.

carbon monoxide A colorless, odorless, tasteless gas formed in large part by incomplete combustion of fuel. Human activities (i.e., transportation or industrial processes) are largely the source for CO contamination.

Central Business District (CBD) The downtown retail trade and commercial area of a city or an area of very high land valuation, traffic flow, and concentration of retail business offices, theaters, hotels and services.

Clean Air Act (CAA) The original Clean Air Act was passed in 1963, but the national air pollution control program is actually based on the 1970 version of the law. The 1990 Clean Air Act Amendments (CAAA) are the most far-reaching revisions of the 1970 law. The 1990 Clean Air Act Amendment is the most recent version of the 1970 version of the law. The 1990 amendments made major changes in the Clean Air Act.

Congestion Management System (CMS) Systematic process for managing congestion. Provides information on transportation system performance and finds alternative ways to alleviate congestion and enhance the mobility of people and goods, to levels that meet state and local needs.

Congestion Mitigation and Air Quality Program (CMAQ) The Intermodal Surface Transportation Efficiency Act created the CMAQ program to deal with transportation related air pollution. The program is continued under TEA-21. States with areas that are designed as nonattainment for ozone or carbon monoxide (CO) must use their CMAQ funds in these nonattainment areas. A state may use its CMAQ funds in any part of its particulate matter (PM) nonattainment areas, if certain criteria are met. Funds are directed towards projects and programs in certain non-attainment areas that meet standards contained in the Clean Air Act amendments of 1990.

congestion pricing Road pricing that varies with the level of traffic on a congested roadway. Congestion pricing is intended to allocate roadway space efficiently.

cordon pricing Tolls charged for entering a particular area (a "cordon"), such as a downtown.

corridor A broad geographical band that follows a general directional flow connecting major sources of trips that may contain a number of streets, highways and transit route alignments.

cost Resources used to produce a good or service, which may include money, time, materials, land or even risk and discomfort. Costs can be categorized in several ways: 1) **fixed cost**: A cost that is not related to consumption, such as vehicle registration fees and residential parking; 2) **variable cost**: A cost that directly increases with consumption, such as fuel and road tolls; 3) **internal cost**: A cost borne directly by the user of a good, such as transit fares and vehicle operating costs; 4) **external cost**: A cost that is borne by others, or by users indirectly and not related to their consumption, such as pollution emissions and general taxes used to fund transportation services; 5) **market cost**: The cost of a good that is commonly traded with money in a competitive market, such as vehicles and fuel; 6) **nonmarket cost**: The cost of a good not commonly traded in a market, such as crash risk and air quality; and 7) **social cost**: The sum of all costs.

demand responsive transit Non-fixed-route transit service utilizing vans or buses with passengers boarding and alighting at prearranged times at any location within the system's service area. Also called Dial-a-Ride.

Department of Transportation (DOT) At the federal level, a cabinet agency with responsibility for highways, mass transit, aviation and ports; headed by the secretary of transportation. The DOT includes the FHWA, the FTA, and the FAA, among others. There are also state DOTs.

differential pricing (*also known as variable pricing*) User charges that vary by factors, such as vehicle type, vehicle occupancy, time of use, congestion level, facility location, and air quality impact.

distance-based charges Fees charged for using a vehicle based on the vehicle miles traveled (VMT) in a jurisdiction. They can include distance-based road user charges, emission charges, and insurance charges.

dynamic pricing Congestion pricing that reflects real-time congestion levels, as opposed to variable pricing that follows a fixed schedule.

environmental assessment (EA) An environmental analysis prepared pursuant to the National Environmental Policy Act (NEPA) to determine whether a federal action should significantly affect the environment and thus require a more detailed environmental impact statement.

Environmental Impact Statement (EIS) A document required of federal agencies by the National Environmental Policy Act (NEPA) for major projects or legislative proposals significantly affecting the environment. A tool for decision making, it describes the positive and negative affects of the undertaking and cites alternative actions.

employment areas Areas of mixed employment that include various types of manufacturing, distribution, and warehousing uses, commercial and retail development, and some residential development. Retail uses should primarily serve the needs of the people working or living in the immediate employment area. Exceptions to this general policy can be made only for certain areas indicated in a functional plan.

elasticity of demand A measure of the sensitivity of demand for a commodity to a change in its price. It equals the percentage change in consumption of the commodity that results from a 1 percent change in its price. The greater the elasticity, the more price-sensitive the demand for the commodity. Price elasticities are an important indicator of the effects of pricing strategies.

emissions budget The part of the State Implementation Plan (SIP) that identifies the allowable emissions levels, mandated by the National Ambient Air Quality Standards

(NAAQS), for certain pollutants emitted from mobile, stationary, and area sources. The emissions levels are used for meeting emission reduction milestones, attainment, or maintenance demonstrations.

emission charges Vehicle charges that vary based on emission rates.

environmental justice Consideration of the views of all people affected by the outcomes of a plan or project and guarantees that any negative outcomes will not fall disproportionately on one segment of a population.

Environmental Protection Agency (EPA) The federal regulatory agency responsible for administering and enforcing federal environmental laws, including the Clean Air Act, the Clean Water Act, the Endangered Species Act, and others.

externality An action by one or more persons, firms, or households that affects third parties and for which the third parties do not pay (in the case where the effect is a benefit) or are not reimbursed (in the case where the effect is a cost).

fare elasticity The extent to which ridership responds to fare increases or decreases.

Federal Highway Administration (FHWA) A branch of the U.S. Department of Transportation that administers the federal-aid Highway Program, providing financial assistance to states to construct and improve highways, urban and rural roads, and bridges.

Federal Transit Administration (FTA) A branch of the U.S. Department of Transportation that is the principal source of federal financial assistance to America's communities for planning, development, and improvement of public or mass transportation systems.

financial planning The process of defining and evaluating funding sources, sharing the information, and deciding how to allocate the funds.

financial programming A short-term commitment of funds to specific projects identified in the regional Transportation Improvement Program (TIP).

fine particulates Particulate matter less than 2.5 microns in size (PM-2.5). A micron is one-millionth of a meter. See "Particulate Matter" below.

fiscal constraint Making sure that a given program or project can reasonably expect to receive funding within the time allotted for its implementation.

formula capital grants Federal transit funds for transit operators; allocation of funds overseen by the Federal Transit Administration (FTA).

freight intermodal facility An intercity facility where freight is transferred between two or more modes (e.g., truck to rail, rail to ship, etc.)

fringe parking An area for parking usually located outside the Central Business District (CBD) and most often used by suburban residents who work or shop downtown.

full cost accounting Quantification of transportation costs, both public and private, that extends beyond the "direct costs" or "market price." The term implies that "externalities" are included in the cost estimate.

generated traffic Additional vehicle trips on a particular roadway or area that occur when roadway capacity is increased or travel conditions are improved due to latent demand (additional trips that travelers would make if traffic conditions were improved). A portion of generated traffic often consists of *induced travel*; that is, an increase in total vehicle mileage (which excludes travel shifted from other times and routes).

Geographic Information System (GIS) Computerized data management system designed to capture, store, retrieve, analyze, and display geographically referenced information.

Global-Positioning-Based (GPS-based) pricing Road pricing based on the location and time that a vehicle is driven, using GPS (Global Positioning System) technology. It is a subcategory of distance-based charges.

High-Occupancy Vehicle (HOV) a passenger vehicle carrying more than a specified minimum number of passengers. HOVs include carpools, vanpools, and buses.

High-Occupancy-Vehicle Lane Exclusive road or traffic lane limited to buses, vanpools, carpools, and emergency vehicles.

High Occupancy Toll Lanes (HOT Lanes) HOV facilities that allow lower-occupancy vehicles, such as solo drivers, to use the facility if they pay a toll. This offers users three options: 1) drive alone on an unpriced but congested general purpose lane, 2) drive alone and pay to use a less congested lane, or 3) rideshare (carpool, vanpool or ride transit) to use a less congested lane without any additional fee.

Intelligent Transportation Systems (ITS) The application of advanced technologies to improve the efficiency and safety of transportation systems.

intermodal The term “mode” is used to refer to and to distinguish from each other the various forms of transportation, such as automobile, transit, ship, bicycle and walking.

Intermodal Surface Transportation Efficiency Act (ISTEA) This landmark federal legislation signed into law in 1991 made broad changes in the way transportation decisions are made. ISTEA emphasized diversity and balance of modes, as well as the preservation of existing systems before construction of new facilities. ISTEA expired in 1997, but much of its program structure is carried forward in new federal legislation.

Interstate Highway System (IHS) The system of highways that connects the principal metropolitan areas, cities, and industrial centers of the U.S.

jitney Privately owned, small, or medium-size vehicles usually operated on a fixed route but not on a fixed schedule.

land use Refers to the manner in which portions of land or the structures on them are used (e.g., commercial, residential, retail, industrial, etc.)

Level of Service (LOS) A qualitative assessment of a road’s operating conditions by transportation officials that reflects the relative ease of traffic flow on a scale of A to F, with free-flow being rated LOS-A and congested conditions rated as LOS-F.

linked/unlinked trip An unlinked trip is a passenger trip made on a single vehicle, such as a single automobile or bus ride. A linked trip is a person’s entire trip between an origin and destination, which may involve transferring between vehicles (e.g., Park and Ride, or bus and rail transit), or multiple stops.

local improvement district A method that allows a group of property owners to share the cost and benefits of public improvements. The local jurisdiction finances the construction. Costs are paid back by property owners through a special assessment.

Long Range Transportation Plan (LRTP) A document resulting from regional or statewide collaboration and consensus on a region or state’s transportation system, and serving as the defining vision for the region’s or state’s transportation systems and services. In metropolitan areas, the plan indicates all of the transportation improvements scheduled for funding over the next 20 years.

maintenance area Any geographic region of the U.S. previously designated non-attainment pursuant to the Clean Air Act Amendments of 1990 and subsequently redesignated to attainment subject to the requirement to develop a maintenance plan under section 175A of the amended Act.

managed lane A lane restricted to particular types of vehicles providing improved travel conditions to eligible users. Eligibility can be based on the number of people in

the vehicle, the type of vehicle, users who are willing to pay a toll, or other criteria. Examples include HOV lanes, HOT lanes, freight lanes, and value-priced lanes.

market distortions Market conditions that violate basic market principles, such as consumer choice, cost-based pricing, or economic neutrality.

Metropolitan Planning Organization (MPO) A federally required transportation planning body responsible for the Regional Transportation Plan and the Transportation Implementation Plan in its region; the governor designates an MPO in every urbanized area with a population larger than 50,000.

Metropolitan Transportation Plan (MTP) The official intermodal transportation plan developed and adopted through the metropolitan transportation planning process for the metropolitan planning area, in accordance with 23 U.S.C.134, 23 U.S.C.135, and 49 U.S.C.5303.

mode A specific form of transportation, such as automobile, subway, bus, rail, or air.

National Ambient Air Quality Standards (NAAQS) Federal standards that set allowable concentrations and exposure limits for various pollutants. EPA developed the standards in response to a requirement of the Clean Air Act. Air quality standards have been established for the following six criteria pollutants: ozone (or smog), carbon monoxide, particulate matter, nitrogen dioxide, lead, and sulfur dioxide.

National Environmental Policy Act of 1969 (NEPA) Established a national environmental policy requiring that any project using federal funding or requiring federal approval, including transportation projects, examine the effects of proposed and alternative choices on the environment before a federal decision is made.

national Intelligent Transportation System (ITS) architecture A systems framework to guide the planning and deployment of Intelligent Transportation Systems infrastructure. The national ITS architecture is a blueprint for the coordinated development of ITS technologies in the U.S. It is unlikely that any single metropolitan area or state would plan to implement the entire national ITS architecture.

nonattainment A geographic region of the U.S. that the EPA has designated as not meeting the air quality standards.

ozone (O₃) A colorless gas with a sweet odor. Ozone is not a direct emission from transportation sources. It is a secondary pollutant formed when Volatile Organic Compounds (VOCs) and nitrous oxides (NO_x) combine in the presence of sunlight. Ozone is associated with smog or haze conditions. Although the ozone in the upper atmosphere protects us from harmful ultraviolet rays, ground-level ozone produces an unhealthy environment in which to live. Ozone is created by human and natural sources.

parking management Strategies aimed at making better use of available parking supply. Parking management strategies include preferential parking or price discounts for carpools and/or short-term parkers, and disincentives, prohibitions, and price supplements for those contributing more to congestion.

parking cash out A program under which those offered a free parking space are also offered the cash equivalent of the value of their parking space.

particulate matter Airborne solid particles and liquid droplets. Particulate matter may be in the form of fly ash, soot, dust, fog, fumes, etc. Fine particles (those smaller than 2.5 microns) that can be a byproduct of fossil fuel combustion can easily reach remote lung areas, and their presence in the lungs is linked to serious respiratory ailments such as asthma, chronic bronchitis, and aggravated coughing. Exposure to these particles may aggravate other medical conditions such as heart disease and emphysema and may cause premature death. In the environment, particulate matter contributes to diminished visibility and particle deposition (soiling).

peak period Morning and afternoon time periods when all transportation modes are most heavily used; usually referred to as “rush hour.”

peak-period surcharge An extra transportation fee imposed during peak periods (rush hours).

ped-shed Also called a walkable area. The area within walking distance of a destination (the term is derived from “watershed,” the area from which stormwater flows into a stream or lake).

performance measures Indicators of how well the transportation system is performing with regard to such things as average speed, reliability of travel, and accident rates. Used as feedback in the decision-making process.

person trip A trip taken by an individual. For example, if three persons from the same household travel together, the trip is counted as one household trip and three person trips.

person miles The aggregate distances traveled by all persons on a given trip.

public transit Passenger transportation services, usually local in scope, available to any person who pays a prescribed fare. It operates on established schedules along designated routes or lines with specific stops and is designed to move relatively large numbers of people at one time.

rail, commuter Urban passenger train service for short-distance travel between a central city and adjacent suburb. Does not include rapid rail transit or light rail service.

rail, heavy A railway with the capacity for a “heavy volume” of traffic and characterized by exclusive rights-of-way, multi-car trains, high speed, rapid acceleration, sophisticated signaling, and high platform loading. Also known as “rapid rail,” “subway,” “elevated (railway),” or “metropolitan railway (metro).”

rail, light A railway with a “light volume” traffic capacity compared to heavy rail. Light rail may use shared or exclusive rights-of-way, high or low platform loading, and multi-car trains, or single cars. Also known as “streetcar,” “trolley car,” and “tramway.”

Regional Planning Organization (RPO) An organization that performs planning for multijurisdictional areas. MPOs, regional councils, economic development associations, rural transportation associations are examples of RPOs.

road pricing The general term for any charge for use of a roadway. It is sometimes limited to direct charges, such as tolls, or may include other vehicle fees, including fuel taxes, license fees, and parking charges.

right-of-way (ROW) A public or private area that allows for the passage of people or goods. Right-of-way includes passageways, such as freeways, streets, bicycle and pedestrian off-street paths, and alleys. A public right-of-way is one dedicated or deeded to the public for public use and under the control of a public agency.

single occupant vehicle (SOV) A vehicle with one occupant, the driver.

smart growth A set of policies and programs design to protect, preserve, and economically develop established communities and valuable natural and cultural resources.

sources Refers to the origin of air contaminants. Can be point (coming from a defined site) or nonpoint (coming from many diffuse sources). **Stationary sources** include relatively large, fixed facilities such as power plants, chemical process industries, and petroleum refineries. **Area sources** are small, stationary, nontransportation sources that collectively contribute to air pollution, including dry cleaners, bakeries, surface coating operations, home furnaces, and crop burning. **Mobile sources** include on-road vehicles, such as cars, trucks, and buses, and off-road sources, such as rains, ships, airplanes, boats, lawnmowers, and construction equipment.

sprawl The movement of people from the central city to the suburbs, characterized by dispersed, low-density, single-use, automobile-dependent land-use patterns.

stakeholders Individuals and organizations involved in or affected by a planning process.

State Implementation Plan (SIP) Produced by the state environmental agency, not the MPO. A plan mandated by the CAA that contains procedures to monitor, control, maintain, and enforce compliance with the NAAQS. Must be taken into account in the transportation planning process.

State Infrastructure Bank (SIB) A revolving fund mechanism for financing a wide variety of highway and transit projects through loans and credit enhancement. SIBs are designed to complement traditional federal highway and transit grants by providing states increased flexibility for financing infrastructure investments.

state planning and research (SPR) funds The primary source of funding for state-wide long-range planning.

State Transportation Improvement Program (STIP) A staged, multiyear, statewide, intermodal program of transportation projects, consistent with the statewide transportation plan and planning processes as well as metropolitan plans and Transportation Implementation Plans (TIPs).

Surface Transportation Program (STP) A federal highway funding program that funds a broad range of surface transportation capital needs, including many roads, transit, sea and airport access, vanpool, bike, and pedestrian facilities.

system development charge (SDC) A fee assessed to pay for increases in transportation needs caused by a project.

Transportation Equity Act for the 21st Century (TEA-21) Enacted June 9, 1998, as Public Law 105-178. TEA-21 authorized the federal surface transportation programs for highways, highway safety, and transit for the six-year period 1998-2003. The TEA 21 Restoration Act, enacted July 22, 1998, provided technical corrections to the original law. The combination of these two laws is referred to as TEA-21.

telecommuting Communicating electronically (by telephone, computer, fax, etc.) with an office, either from home or from another site, instead of traveling to it physically.

time-variable pricing Any user charge that varies by time period as a way to manage demand and reduce congestion. This is the basis of congestion pricing. Many services (telephone, electric utilities, airlines) use time-variable pricing to encourage more efficient use of system capacity and allow users to save by shifting their consumption to off-peak periods. Some road tolls and parking charges are time-variable pricing.

Title VI Title VI of the Civil Rights Act of 1964. Prohibits discrimination in any program receiving federal assistance.

Transportation Management Area (TMA) As defined in federal regulations, “an urbanized area with population over 200,000” and “applies to the entire metropolitan planning area.” These areas must meet certain standards, and nonattainment TMAs must meet additional planning requirements.

toll road A section of road where motorists are charged a direct user fee (or toll).

transit-oriented development (TOD) A type of development that links land use with transit facilities to support the transit system and to help reduce sprawl, traffic congestion, and air pollution. It includes housing, along with complementary public uses (jobs, retail and services), located at a strategic point along a regional transit system, such as a rail hub.

transponder An electronic tag installed in a vehicle for electronic road pricing. The tag is read by a tolling device that automatically assesses the amount of the user fee.

transportation conformity A process to assess the compliance of any transportation plan, program, or project with air quality implementation plans. The conformity process is defined by the Clean Air Act.

Transportation Control Measures (TCM) Transportation strategies that affect traffic patterns or reduce vehicle use to reduce air pollutant emissions. These may include HOV lanes, provision of bicycle facilities, ridesharing, telecommuting, etc. Such actions may be included in a State Implementation Plan if needed to demonstrate attainment of the NAAQS.

transportation demand management (TDM) Various strategies that change travel behavior (how, when, and where people travel) in order to increase transport system efficiency and achieve specific objectives such as reduced traffic congestion, road and parking cost savings, increased safety, improved mobility for nondrivers, energy conservation, and pollution emission reductions.

Transportation Improvement Program (TIP) A document prepared by a metropolitan planning organization (MPO) that lists projects to be funded with federal funds.

Transportation Infrastructure Finance and Innovation Act of 1998 (TIFIA) A federal credit program under which the DOT may provide three forms of credit assistance—secured (direct) loans, loan guarantees, and standby lines of credit—for surface transportation projects of national or regional significance. The fundamental goal is to leverage federal funds by attracting substantial private and nonfederal co-investment in critical improvements to the nation's surface transportation system.

Transportation Management Area (TMA) All urbanized areas greater than 200,000 in population, and any other area that requests such designation.

Transportation System Management (TSM) Strategies and techniques for increasing the efficiency, safety, or level of service of a transportation facility without increasing its size. Examples include, but are not limited to, traffic signal improvements, traffic control devices (including installing medians, channelization, access management, and ramp metering), incident response, targeted traffic enforcement, preferential transit measures, and restriping for high-occupancy vehicle (HOV) lanes.

trust fund A fund credited with receipts that are held in trust by the government and earmarked by law for use in carrying out specific purposes and programs in accordance with an agreement or a statute.

Unified Planning Work Program (UPWP) The management plan for the (metropolitan) planning program. Its purpose is to coordinate the planning activities of all participants in the planning process.

urbanized area An area that contains a city of 50,000 or more population plus incorporated surrounding areas meeting size or density criteria as defined by the U.S. Census.

value pricing A synonym for congestion pricing that applies when motorists are offered a choice between an unpriced but congested facility and a value-priced facility that provides a superior level of service. It emphasizes the extra benefits that users receive in exchange for paying a premium.

Vehicle Miles Traveled (VMT) The sum of all miles traveled by automobile. Reducing VMT helps ease traffic congestion and improve air quality.

List of References

The standard list of references follows. There is much more good material than what we have cited in this text. To help you find it, we provide these brief instructions on how to use *Google Scholar* to find more detailed information on the transportation and land use connection literature.

- Go to <http://scholar.google.com/>. You can also get here by first going to <http://www.google.com/> then click on “More,” then “Even More,” and then scroll down to “Scholar.”
- Type in an author’s name and follow your instincts.
- Here are *some* of the most important and cited authors on the transportation/land-use connection. The same technique could be applied to any author in the bibliography that follows.

Boarnet, Marlon
Cervero, Robert
Crane, Randall
Deakin, Elizabeth
Downs, Anthony
Ewing, Reid
Giuliano, Genevieve
Handy, Susan
Jacobs, Allan
Kockelman, Kara
Shoup, Donald
Small, Kenneth
Southworth, Michael
Wachs, Martin

There are many good websites: USDOT, Transportation Research Board, MPOs across the county. Standard search techniques will get you there. A couple of sites with lots of material:

- Access, a magazine from the University of California Transportation Center: <http://www.uctc.net/access>
- The Victoria Transport Policy Institute: <http://www.vtpi.org/>

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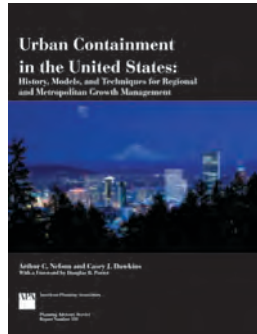
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of special interest



Urban Containment in the United States

PAS 520. Arthur C. Nelson and Casey J. Dawkins. 2004. 130 pp.

Growth boundaries, service limits, and priority growth areas are meaningless without effective methods of implementation. Urban containment is a set of tools that can turn a line on a map into actual sustainable development. This report explores urban containment as a framework that regions and communities can use for managing growth. The authors review the history and central characteristics of urban containment, evaluate more than 100 urban containment plans, and present a four-framework typology. They also offer case studies of each of the four frameworks and summarize some of the implications for planners.



Parking Cash Out

PAS 532. Donald C. Shoup. 2005. 119 pp.

Free parking is the most common fringe benefit offered to workers in the U.S. Is it any wonder, then, that 91 percent of them drive to work—or that most of them drive solo? The cost of this parking subsidy is about 1 percent of the gross national product and four times the amount of funding for public transit. This report, a complement to Shoup's *The High Cost of Free Parking*, shows how employers who offer their employees the option to cash out their parking subsidies can discourage solo driving and its attendant social, environmental, and infrastructure costs. It also suggests ways planners can bring this option to their communities.

Planning Active Communities

PAS 543/544. Marya Morris, ed. 2006. 116 pp.

Medical professionals have spent years telling overweight Americans to eat right and exercise—but obesity rates are still climbing. This may be, in part, because, as we've become more efficient in how we work, eat, and get around, we have engineered physical activity out of our daily routines.

This report looks at how planning processes, development regulations, and community participation can be used to ensure that development patterns facilitate everyday physical activity. Includes information about safe routes to school programs and accessible schools, along with case studies of planning active communities in Denver, San Diego, Portland, Nashville, Minneapolis, and more.



